

Competitive Programming Notebook

ICPC team reference • August 13, 2026

HOW TO FIND SOMETHING (in order of speed). 1. You know the routine's name → the alphabetical *Index of every routine* at the very back gives the page. 2. You know the topic → the running head on every page names the section (left) and the sub-topic (right); thumb until it matches. 3. You know neither → section **12 Problem Solving** maps a problem's *shape* and its *constraints* onto the technique, and docs/PLAYBOOK.md does the same at more length.

BEFORE YOU PASTE. Line 1 of every file says what it does; a // Needs: line names the templates it depends on; a // NOTE: line warns when another file declares the same symbol. Vertices and array positions are **0-based** unless the header says otherwise. Geometry: P<ll> for every decision, P<ld> only when the answer is a real number.

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1 Miscellaneous

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1.1 Template

```
#include <bits/stdc++.h>
using namespace std;

// #define int long long
#define ll long long
#define lll __int128 // used unnamed by
    ↳ CHT, BigInt, PrimeCounting
#define ld long double
#define fi first
#define se second
#define pb push_back
#define eb emplace_back
#define all(x) x.begin(), x.end()
#define rall(x) x.rbegin(), x.rend()
#define sz(x) (int)(x).size()
#define ones(n) __builtin_popcountll(n)
#define msb(n) (63 - __builtin_clzll(n))
#define lsb(n) __builtin_ctzll(n)
typedef vector<int> vi;
typedef pair<int, int> pii;

const int N = 2e5 + 5, M = 1e3 + 5, LOG = 31;
const int inf = 0x3f3f3f3f;
const ll llinf = 0x3f3f3f3f3f3f3f3f;
const int MOD = 1e9 + 7;
const ld eps = 1e-9, PI = acos(-1.0L); // ld, not double:
    ↳ geometry needs the digits

#ifdef LOCAL // compile with
    ↳ -DLOCAL
template <class T> void _p(const T& x);
void _p(const string& s) { cerr << "''" << s << "'"; }
void _p(char c) { cerr << "'" << c << "'"; }
template <class A, class B> void _p(const pair<A, B>& p) {
    cerr << '('; _p(p.fi); cerr << ", "; _p(p.se); cerr << ')';
}
template <class T> void _p(const T& x) {
    if constexpr (is_arithmetic_v<T>) cerr << x;
    else { cerr << '{'; int f = 1;
        for (auto& e : x) { if (!f) cerr << ", "; f = 0; _p(e); }
        ↳ cerr << '}' ; }
}
void _pr() { cerr << "\n"; }
template <class H, class... T> void _pr(const H& h, const T&... t) {
    _p(h); if (sizeof...(t)) cerr << " | "; _pr(t...);
}
#define dbg(...) (cerr << "[" << #__VA_ARGS__ << "] = ",
    ↳ _pr(__VA_ARGS__))
#else
#define dbg(...)
#endif

void testCase() {
}

void preCompute() {
}

int32_t main() {
    ios_base::sync_with_stdio(false), cin.tie(nullptr),
    ↳ cout.tie(nullptr);
    cout << fixed << setprecision(10);
    preCompute();

    int tc = 1;
    cin >> tc; // DELETE THIS LINE
    ↳ for single-test problems
    for (int TC = 1; TC <= tc; TC++) {
        // cout << "Case #" << TC << ": ";
        testCase();
    }
}

/*

*/
```

1.2 fastIO

HAND-ROLLED FAST IO (fread/fwrite buffered). Use only when cin with sync off is measurably too

```
// slow: >1e6 numbers to read, or heavy output. Otherwise prefer the
//   ↳ template's
// ios::sync_with_stdio.
const int MOD = 1e9 + 7;
const int BUF_SZ = 1 << 15;

inline namespace Input {
    char buf[BUF_SZ];
    int pos;
    int len;
    char next_char() {
        if (pos == len) {
            pos = 0;
            len = (int)fread(buf, 1, BUF_SZ, stdin);
            if (!len) { return EOF; }
        }
        return buf[pos++];
    }

    int read_int() {
        int x;
        char ch;
        int sgn = 1;
        while (!isdigit(ch = next_char())) {
            if (ch == '-') { sgn *= -1; }
        }
        x = ch - '0';
        while (isdigit(ch = next_char())) { x = x * 10 + (ch - '0'); }
        return x * sgn;
    }
} // namespace Input

inline namespace Output {
    char buf[BUF_SZ];
    int pos;

    void flush_out() {
        fwrite(buf, 1, pos, stdout);
        pos = 0;
    }

    void write_char(char c) {
        if (pos == BUF_SZ) { flush_out(); }
        buf[pos++] = c;
    }

    void write_int(int x) {
        static char num_buf[100];
        if (x < 0) {
            write_char('-');
            x *= -1;
        }
        int len = 0;
        for (; x >= 10; x /= 10) { num_buf[len++] = (char)('0' + (x %
        ↳ 10)); }
        write_char((char)('0' + x));
        while (len) { write_char(num_buf[--len]); }
        write_char('\n');
    }

    // auto-flush output when program exits
    void init_output() { assert(atexit(flush_out) == 0); }
} // namespace Output
```

1.3 grid

GRID DIRECTION VECTORS. First 4 entries = 4-neighbourhood (L,R,U,D), all 8 = king moves.

```
// Knight moves: dx {1,1,-1,-1,2,2,-2,-2}, dy {2,-2,2,-2,1,-1,1,-1}.
//   ↳ Guard with 0<=x<n && 0<=y<m.
int dx[] = {0, 0, -1, 1, -1, -1, 1, 1};
int dy[] = {-1, 1, 0, 0, -1, 1, -1, 1};
char di[] = {'L', 'R', 'U', 'D'};

// KNIGHT DISTANCE ON AN INFINITE BOARD, O(1). BFS is fine up to a
//   ↳ 1000x1000 board; past that, or
// when the coordinates are 1e9, you need this closed form. Reduce by
//   ↳ symmetry to x, y >= 0, then
// the answer is driven by three lower bounds - you cover at most 2
//   ↳ in either axis per move and at
// most 3 of (x + y) per move - and parity, since every move changes
//   ↳ x + y by an odd amount. Three
// positions beat the formula and must be special-cased; they are the
//   ↳ classic reason a hand-rolled
// version WAS. Exact integer arithmetic, any magnitude that fits in
//   ↳ ll.
ll knightDist(ll x, ll y) {
    x = abs(x), y = abs(y);
    if (x < y) swap(x, y); // now x >= y
    ↳ >= 0
    if (x == 1 && y == 0) return 3; // the three
    ↳ exceptions, all near the
    if (x == 2 && y == 2) return 4; // origin,
    ↳ where the board "runs out"
    ll d = max((x + 1) / 2, (y + 1) / 2, (x + y + 2) / 3);
    while ((d & 1) != ((x + y) & 1)) d++; // parity: x
    ↳ + y flips every move
    return d;
}

// ON A BOUNDED BOARD the formula is wrong near the edges and corners
//   ↳ - BFS, or use the formula
// and repair the O(1) corner cases by hand. On a HALF-PLANE (x, y >=
//   ↳ 0) the same three exceptions
// are exactly the corrections needed, which is what the code above
//   ↳ assumes.
// KING distance is max(|dx|, |dy|); ROOK is 0/1/2; BISHOP is 0/1/2
//   ↳ when the colours match and
// unreachable otherwise. Those need no code, only the reminder that
//   ↳ they are not Manhattan.
```

1.4 random

RANDOMNESS. Seed from the clock so tests cannot be prepared against you.

```
mt19937_64
↳ rng(chrono::steady_clock::now().time_since_epoch().count());
ll rnd(ll l, ll r) { return uniform_int_distribution<ll>(l, r)(rng);
↳ } // inclusive

vector<int> randPerm(int n) { // random
    ↳ permutation of 0..n-1
    vector<int> v(n);
    iota(all(v), 0);
    shuffle(all(v), rng);
    return v;
}

// Shuffling the INPUT is a real algorithm, not just a utility:
// - minimum enclosing circle is O(n) only after a shuffle
// - quickselect / random pivots
// - defeats anti-hash and anti-quicksort tests
// - random rotation of points removes degenerate ties in a
↳ geometry sweep
```

1.5 coordinateCompression

Coordinate Compression - $O(N \log N)$

```
template<typename T>
struct Compress {
    vector<T> v;

    Compress(vector<T>& a) : v(a) {
        sort(all(v));
        v.erase(unique(all(v)), v.end());
        for (auto &x : a) {
            x = lower_bound(all(v), x) - v.begin();
        }
    }

    T operator[](int i) const {
        return v[i];
    }
};
```

1.6 grayCode

GRAY CODE - orders $0..2^n-1$ so consecutive values differ in exactly ONE bit. Use it to walk all

```
// subsets while changing one element at a time (incremental DP,
// ↳ XOR-basis sweeps, Tower of Hanoi).
// Gray Code Generator -  $O(2^N)$ 
// g(n) =  $n \oplus (n \gg 1)$ 
vector<int> grayCode(int n) {
    vector<int> res(1 << n);
    for (int i = 0; i < (1 << n); i++) {
        res[i] = i ^ (i >> 1);
    }
    return res;
}

int rev_g (int g) {
    int n = 0;
    for (; g; g >>= 1)
        n ^= g;
    return n;
}
```

1.7 customHashHashTable

UNORDERED_MAP THAT CANNOT BE HACKED. The default `std::hash` for integers is the identity, so an

```
// adversary can force every key into one bucket and turn  $O(1)$  into
// ↳  $O(n)$ . splitmix64 + a clock seed
// fixes that. gp_hash_table is ~3x faster than unordered_map;
// ↳ reserve() and max_load_factor
// help more.
#include <ext/pb_ds/assoc.container.hpp>
using namespace __gnu_pbds;

// Custom Hash to prevent anti-hash tests in unordered_map /
// ↳ gp_hash_table
struct custom_hash {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }

    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM =
        chrono::steady_clock::now().time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};

// Usage: gp_hash_table<ll, int, custom_hash> table;
// unordered_map<ll, int, custom_hash> safe_map;
```

1.8 pragmas

COMPILER PRAGMAS. O3 + unrolling, and AVX2 for auto-vectorised loops (bitset, simple array math).

```
// Put them at the very top, before any include. Risk: 'target'
// ↳ crashes judges without those CPU
// features - never use avx2 on an unknown judge, and never on
// ↳ Codeforces problems with old
// machines.
// GCC Optimizations
#pragma GCC optimize("O3,unroll-loops")
#pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")
```

1.9 int128

__int128 Stream Operators Helper

```
typedef __int128_t int128;
typedef __uint128_t uint128;

ostream &operator<<(ostream &os, int128 n) {
    if (n == 0) return os << "0";
    if (n < 0) { os << "-"; n = -n; }
    string s;
    while (n > 0) { s += (char)('0' + (n % 10)); n /= 10; }
    reverse(all(s));
    return os << s;
}

istream &operator>>(istream &is, int128 &n) {
    string s; is >> s;
    n = 0; int i = 0, sign = 1;
    if (!s.empty() && s[0] == '-') { sign = -1; i = 1; }
    for (; i < (int)s.size(); i++) n = n * 10 + (s[i] - '0');
    n *= sign;
    return is;
}
```

1.10 SearchIdioms

The four searches. Getting the boundary right here saves more contest time than any template.

```
// BINARY SEARCH ON A PREDICATE. pred must be monotone: F F F T T T.
// firstTrue returns the first x in [lo, hi] with pred(x), or hi+1 if
    ↪ none.
template <class F> ll firstTrue(ll lo, ll hi, F pred) {
    hi++;
    while (lo < hi) { ll m = lo + (hi - lo) / 2; pred(m) ? hi = m :
        ↪ lo = m + 1; }
    return lo;
}
// lastTrue returns the last x in [lo, hi] with pred(x), or lo-1 if
    ↪ none. T T T F F F
template <class F> ll lastTrue(ll lo, ll hi, F pred) {
    lo--;
    while (lo < hi) { ll m = lo + (hi - lo + 1) / 2; pred(m) ? lo = m
        ↪ : hi = m - 1; }
    return lo;
}
// Reals: fixed iteration count beats an epsilon loop (no infinite
    ↪ loops, predictable time).
// 100 iterations halve the range 2^100 times - always enough.
// Returns the boundary: pred is false on [lo, x) and true on [x,
    ↪ hi]. ('lo' and 'hi' converge.)
template <class F> ld binReal(ld lo, ld hi, F pred, int it = 100) {
    while (it--) { ld m = (lo + hi) / 2; pred(m) ? hi = m : lo = m; }
    return hi; //
    ↪ first TRUE, matching firstTrue
}

// TERNARY SEARCH on a UNIMODAL function (strictly decreasing then
    ↪ increasing => minimum).
// Integer version: stop when the window is tiny, then scan. Never
    ↪ use 'l < r' alone.
template <class F> ll ternaryInt(ll lo, ll hi, F f) { //
    ↪ returns argmin
    while (hi - lo > 2) {
        ll m1 = lo + (hi - lo) / 3, m2 = hi - (hi - lo) / 3;
        f(m1) <= f(m2) ? hi = m2 : lo = m1; // <=
    }
    ↪ keeps the LEFTmost min
    ll best = lo;
    for (ll x = lo; x <= hi; x++) if (f(x) < f(best)) best = x;
    return best;
}
template <class F> ld ternaryReal(ld lo, ld hi, F f, int it = 200) {
    while (it--) { ld m1 = lo + (hi - lo) / 3, m2 = hi - (hi - lo) /
        ↪ 3;
        f(m1) < f(m2) ? hi = m2 : lo = m1; }
    return (lo + hi) / 2;
}
// If f is only WEAKLY unimodal (has flat regions), ternary search is
    ↪ WRONG: one comparison
// f(m1) == f(m2) inside a plateau discards the side holding the
    ↪ optimum. For any DISCRETELY
// CONVEX f (non-decreasing difference sequence, plateaus allowed)
    ↪ use the derivative instead -
// this is strictly safer AND uses half the evaluations:
template <class F> ll convexArgmin(ll lo, ll hi, F f) {
    return firstTrue(lo, hi - 1, [&](ll x) { return f(x + 1) >= f(x);
        ↪ });
}

// NEWTON'S METHOD for roots of a smooth f (quadratic convergence;
    ↪ needs a decent start).
template <class F, class DF> ld newton(ld x, F f, DF df, int it = 60)
    ↪ {
    while (it--) { ld d = f(x) / df(x); x -= d;
        if (fabsl(d) < 1e-15L * max((ld)1, fabsl(x)))
    }
    ↪ break; }
    return x;
}
// Integer sqrt / cbrt with correction - NEVER trust sqrt() alone
    ↪ near 1e18.
ll isqrt(ll n) {
    ll r = (ll)sqrtl((ld)n);
    while (r > 0 && r * r > n) r--;
    while ((r + 1) * (r + 1) <= n) r++;
    return r;
}
ll icbrt(ll n) {
    ll r = (ll)cbrtl((ld)n);
    while (r > 0 && r * r * r > n) r--;
    while ((r + 1) * (r + 1) * (r + 1) <= n) r++;
    return r;
}
```

```
}
// BINARY SEARCH ON THE ANSWER - the pattern, not the code:
// "minimise the maximum X" / "maximise the minimum X" / "can we
    ↪ finish within T"
// -> guess the answer, write a GREEDY feasibility check, binary
    ↪ search on it.
// PARALLEL BINARY SEARCH - when you need this for MANY queries at
    ↪ once: bucket the queries by
// their current midpoint, sweep the shared structure once per level,
    ↪ O(log) sweeps total.
```

1.11 BitTricks

BIT MANIPULATION CHEAT SHEET =====

```

/* ===== BIT MANIPULATION CHEAT SHEET
↳ =====
BUILTINS (add ll suffix for 64-bit: __builtin_popcountll, ...)
__builtin_popcount(x) #set bits      __builtin_ctz(x)
↳ #trailing zeros (x != 0)
__builtin_clz(x)      #leading zeros  __builtin_parity(x)
↳ popcount & 1
__lg(x) = 63 - clzll(x) = floor(log2 x)    (GCC only; write it
↳ yourself elsewhere)
lowest set bit: x & -x      clear it: x &= x - 1
↳ set/flip: x |= b, x ^= b
is power of two: x && !(x & (x - 1))
round UP to a power of two (x >= 1): x == 1 ? 1ULL : 2ULL << (63 -
↳ __builtin_clzll(x - 1))
- clzll(0) is UNDEFINED, so x = 1 must be special-cased, and the
↳ shift must be 64-bit

SUBMASK ENUMERATION - iterate every subset of m:
for (int s = m; ; s = (s - 1) & m) { ...; if (!s) break; }
Over ALL masks this is O(3^n) total (each element is
↳ in/out/neither).

SUPERMASK enumeration (every s with m subset of s, within n bits):
for (int s = m; s < (1 << n); s = (s + 1) | m) { ... }

K-SUBSETS in increasing order (Gosper's hack):
for (int s = (1 << k) - 1; s < (1 << n); ) { ...;
    int c = s & -s, r = s + c; s = (((r ^ s) >> 2) / c) | r; }

GRAY CODE  g(i) = i ^ (i >> 1)      inverse: x ^= x >> 1 repeatedly
↳ (see 06 - grayCode.cpp)
consecutive gray codes differ in exactly one bit - use for "change
↳ one thing at a time".

XOR FACTS
x ^ x = 0, x ^ 0 = x, xor is its own inverse -> prefix xor answers
↳ range xor
xor of 0..n = [n, 1, n+1, 0][n % 4]
a + b = (a ^ b) + 2*(a & b)      a | b = (a ^ b) + (a & b)      a +
↳ b = (a | b) + (a & b)
"OR of a range equals its SUM" <=> the values are pairwise
↳ BIT-DISJOINT
(so at most 30 non-zero values fit in any such window -> two
↳ pointers)

COUNTING PER BIT - the standard decomposition: handle each of the
↳ 30/60 bits independently.
sum over pairs of (a_i ^ a_j) = sum over bits b of (#with bit) *
↳ (#without bit) * 2^b

BITSET
std::bitset<N> b; b._Find_first(), b._Find_next(i) (GCC) -
↳ iterate set bits fast
b <= k, b |= other are O(N/64) - this is what makes bitset DP
↳ and reachability work.
=====
↳ */

// Iterate all submasks of m (including m and 0).
template <class F> void forEachSubmask(int m, F f) {
    for (int s = m; s = (s - 1) & m) { f(s); if (!s) break; }
}
// All n-bit masks with exactly k bits set, in increasing order
↳ (Gosper's hack).
vector<int> kSubsets(int n, int k) {
    vector<int> r;
    if (k == 0) return {0};
    for (int s = (1 << k) - 1; s < (1 << n); ) {
        r.push_back(s);
        int c = s & -s, x = s + c;
        s = (((x ^ s) >> 2) / c) | x;
    }
    return r;
}
// Sum over all pairs (i<j) of (a_i XOR a_j), by handling each bit
↳ independently.
ll pairXorSum(const vector<ll>& a, int B = 62) {
    ll n = a.size(), r = 0;
    for (int b = 0; b < B; b++) {
        ll c = 0;
        for (ll x : a) c += (x >> b) & 1;
        r += c * (n - c) % MOD * ((1LL << b) % MOD) % MOD;
    }
    return r % MOD;
}
}

```

1.12 StressTest

STRESS TESTING =====

```

/* ===== STRESS TESTING =====
The single highest-value 20 lines in this notebook. Use it the
↳ moment a solution WAS on a
test you cannot see. Write gen.cpp + brute.cpp, then loop until they
↳ disagree.

---- run.sh
↳ -----
g++ -O2 -o sol sol.cpp && g++ -O2 -o brute brute.cpp && g++ -O2 -o
↳ gen gen.cpp
for i in $(seq 1 100000); do
    ./gen $i > in.txt
    ./sol < in.txt > out1.txt
    ./brute < in.txt > out2.txt
    if ! diff -qb out1.txt out2.txt > /dev/null; then echo "WA on
↳ test $i"; cat in.txt; break; fi
done

↳ -----
Windows cmd: replace the loop with 'for /L %i in (1,1,100000) do (
↳ ... fc out1.txt out2.txt )'

RULES THAT MAKE IT ACTUALLY WORK
1. Generate SMALL cases (n <= 8, values <= 5). Bugs hide in tiny
↳ corner cases, not big ones.
2. Seed the generator from argv[1] so every iteration differs and
↳ is reproducible.
3. If the answer is not unique, do not diff - write a CHECKER that
↳ validates sol's output.
4. If there is no brute force, stress against yourself: compare two
↳ different solutions, or
    compare an O(n^2) version of your own idea against the O(n log
↳ n) one.
5. For "find any valid X" problems, verify the constraints instead
↳ of comparing.
===== */

// ---- gen.cpp ----
// #include <bits/stdc++.h>
// using namespace std;
// int main(int argc, char** argv) {
//     mt19937 rng(atoi(argv[1]));
//     auto rnd = [&](int l, int r) { return
↳ uniform_int_distribution<int>(l, r)(rng); };
//     int n = rnd(1, 8);
//     printf("%d\n", n);
//     for (int i = 0; i < n; i++) printf("%d ", rnd(1, 5));
//     puts("");
// }

// ---- in-process version: no files, no scripts. Put brute + fast in
↳ one binary. ----
// Fastest way to start; switch to the shell loop only when you need
↳ the failing input on disk.
template <class Gen, class Fast, class Brute>
void stress(Gen gen, Fast fast, Brute brute, int iters = 100000) {
    for (int it = 1; it <= iters; it++) {
        auto in = gen(it);
        auto a = fast(in), b = brute(in);
        if (!(a == b)) {
            printf("MISMATCH on iteration %d: fast=", it);
            cout << a << " brute=" << b << "\n";
            return;
            // print 'in'
        }
        // here too
    }
    puts("all tests passed");
}

```

1.13 Fraction

Exact rational. Always normalised (gcd reduced, denominator > 0).

```
// Comparison uses __int128 so it is exact for |num|, |den| up to
//  ~9e18.
// Use whenever comparing ratios: slopes, "maximum average",
//  ~ scheduling exchange arguments.
struct Frac {
    ll n = 0, d = 1;
    Frac(ll n = 0, ll d = 1) : n(n), d(d) { norm(); }
    void norm() {
        if (d < 0) n = -n, d = -d;
        ll g = gcd(n < 0 ? -n : n, d);
        if (g) n /= g, d /= g;
        if (!d) d = 1; // 1/0 -> treat
    }
    // as 0/1; guard your input
    Frac operator+(Frac o) const { return Frac(n * o.d + o.n * d, d *
    o.d); }
    Frac operator-(Frac o) const { return Frac(n * o.d - o.n * d, d *
    o.d); }
    Frac operator*(Frac o) const { return Frac(n * o.n, d * o.d); }
    Frac operator/(Frac o) const { return Frac(n * o.d, d * o.n); }
    bool operator<(Frac o) const { return (__int128)n * o.d <
    (__int128)o.n * d; }
    bool operator==(Frac o) const { return n == o.n && d == o.d; }
    bool operator<=(Frac o) const { return !(o < *this); }
    ld val() const { return (ld)n / d; }
    friend ostream& operator<<(ostream& os, Frac f) { return os <<
    f.n << '/' << f.d; }
};
// OVERFLOW: + and * multiply denominators, so chains of operations
//  ~ blow up fast. If you only
//  ~ need COMPARISONS, keep the raw pair and cross-multiply in __int128
//  ~ - do not normalise.
// For unbounded exactness use Python/BigInt, or the Stern-Brocot
//  ~ search (ContinuedFractions).
```

1.14 Inversions

INVERSIONS: #pairs $i < j$ with $a[i] > a[j]$. $O(n \log n)$ by merge sort (also sorts a).

```
ll inversions(vector<int>& a) {
    int n = a.size();
    if (n < 2) return 0;
    vector<int> buf(n);
    function<ll(int, int)> rec = [&](int l, int r) -> ll {
    // [l, r)
        if (r - l < 2) return 0;
        int m = (l + r) / 2;
        ll res = rec(l, m) + rec(m, r);
        int i = l, j = m, k = l;
        while (i < m && j < r) {
            if (a[i] <= a[j]) buf[k++] = a[i++];
            else buf[k++] = a[j++], res += m - i;
        }
        // a[i..m) all beat a[j]
        while (i < m) buf[k++] = a[i++];
        while (j < r) buf[k++] = a[j++];
        copy(buf.begin() + l, buf.begin() + r, a.begin() + l);
        return res;
    };
    return rec(0, n);
}
// WHY IT KEEPS APPEARING
// - minimum number of ADJACENT SWAPS to sort = inversion count
// - minimum adjacent swaps to turn a into b: relabel b's positions,
//   ~ then count inversions
// - "how far is this permutation from sorted" / bubble-sort distance
// - inversions modulo 2 = permutation PARITY (= sign, used in
//   ~ determinants and 15-puzzle)
// BIT VERSION (needed when you must also support updates, or count
//   ~ online):
// compress values, sweep left to right, ans += (#inserted so far)
//   ~ query(a[i]), then add a[i].
```

1.15 ExpressionParser

EXPRESSION PARSER (shunting-yard). Handles $+$ $-$ $*$ $/$ $\%$ $\wedge$, unary minus, parentheses.

```
// Left-associative except  $\wedge$ . Extend by editing prec() / applyOp().
struct Parser {
    static int prec(char op) {
        switch (op) {
            case '+': case '-': return 1;
            case '*': case '/': case '%': return 2;
            case '^': return 3;
            case '~': return 4; // unary minus
        }
        // (internal symbol)
        default: return -1;
    }
    static bool rightAssoc(char op) { return op == '^' || op == '~'; }

    static ll apply(char op, ll a, ll b = 0) {
        switch (op) {
            case '+': return a + b;
            case '-': return a - b;
            case '*': return a * b;
            case '/': return a / b;
            case '%': return a % b;
            case '~': return -a;
            case '^': { ll r = 1; while (b > 0) { if (b & 1) r *= a;
            a *= a, b >>= 1; } return r; }
        }
        return 0;
    }
    static ll eval(const string& s) {
        vector<ll> val;
        vector<char> ops;
        bool wantOperand = true; //
        // distinguishes unary from binary minus
        auto pop = [&]() {
            char op = ops.back(); ops.pop_back();
            if (op == '~') { ll a = val.back(); val.pop_back();
            val.push_back(apply(op, a)); }
            else { ll b = val.back(); val.pop_back(); ll a =
            val.back(); val.pop_back();
            val.push_back(apply(op, a, b)); }
        };
        for (size_t i = 0; i < s.size(); i++) {
            char c = s[i];
            if (isspace((unsigned char)c)) continue;
            if (isdigit((unsigned char)c)) {
                ll x = 0;
                while (i < s.size() && isdigit((unsigned char)s[i]))
                x = x * 10 + (s[i++] - '0');
                i--;
                val.push_back(x), wantOperand = false;
            } else if (c == '(') ops.push_back(c), wantOperand = true;
            else if (c == ')') {
                while (!ops.empty() && ops.back() != '(') pop();
                if (!ops.empty()) ops.pop_back();
                wantOperand = false;
            } else {
                if (c == '~' && wantOperand) c = '~'; // unary
                while (!ops.empty() && ops.back() != '(' &&
                (prec(ops.back()) > prec(c) ||
                (prec(ops.back()) == prec(c) &&
                !rightAssoc(c))))
                    pop();
                ops.push_back(c), wantOperand = (c != '~') || true;
            }
        }
        while (!ops.empty()) pop();
        return val.empty() ? 0 : val.back();
    }
};
// To build an AST instead of evaluating, push node indices instead
//   ~ of values.
// For boolean/comparison operators just add them to prec() with
//   ~ lower precedence.
```


1.16 CycleFinding

CYCLE FINDING in a FUNCTIONAL GRAPH $x \rightarrow f(x)$. Every component is a "rho": a tail feeding

```
// into a cycle. Answers: where does x end up after k steps, cycle
  ↳ length, distance to cycle.

// Floyd (tortoise & hare), O(1) memory. Returns {start index of the
  ↳ cycle, cycle length}.
template <class F> pair<ll, ll> floydCycle(ll x0, F f) {
    ll t = f(x0), h = f(f(x0));
    while (t != h) t = f(t), h = f(f(h));
    ll mu = 0; // steps from
  ↳ x0 to the cycle entry
    t = x0;
    while (t != h) t = f(t), h = f(h), mu++;
    ll lam = 1; // cycle length
    h = f(t);
    while (t != h) h = f(h), lam++;
    return {mu, lam};
}

// f applied k times, using the rho structure (k can be 1e18).
template <class F> ll iterate(ll x0, ll k, F f) {
    auto [mu, lam] = floydCycle(x0, f);
    if (k < mu) { while (k--) x0 = f(x0); return x0; }
    for (ll i = 0; i < mu; i++) x0 = f(x0);
    ll rem = (k - mu) % lam;
    while (rem--> 0) x0 = f(x0);
    return x0;
}

// WHOLE-GRAPH version when f is given as an array over 0..n-1:
  ↳ colour DFS, O(n).
// state: 0 unvisited, 1 on the current path, 2 done. Fills
  ↳ cycId/cycPos/distToCycle.
struct FuncGraph {
    int n;
    vector<int> f, state, cyc, pos, dist, cycLen; // cyc[v] = id
  ↳ of the cycle v reaches
    FuncGraph(vector<int> f) : n(f.size()), f(f), state(n, 0), cyc(n,
  ↳ -1), pos(n, -1), dist(n, 0) {
        for (int i = 0; i < n; i++) if (!state[i]) {
            vector<int> path;
            int v = i;
            while (!state[v]) state[v] = 1, path.push_back(v), v =
  ↳ f[v];
            if (state[v] == 1) { // found a NEW
  ↳ cycle, starting at v
                int id = cycLen.size(), len = 0;
                for (int u = v; u = f[u];) { cyc[u] = id, pos[u] =
  ↳ len++; if (f[u] == v) break; }
                cycLen.push_back(len);
            }
            for (int j = path.size() - 1; j >= 0; j--) {
                int u = path[j];
                state[u] = 2;
                if (cyc[u] == -1) cyc[u] = cyc[f[u]], dist[u] =
  ↳ dist[f[u]] + 1;
            }
        }
    }
    int kth(int v, ll k) { // f^k(v)
        while (k > 0 && dist[v] > 0) v = f[v], k--;
        if (!k) return v;
        int L = cycLen[cyc[v]], p = (pos[v] + k % L) % L, u = v;
        for (int i = 0; i < (p - pos[v] + L) % L; i++) u = f[u];
        return u;
    }
};

// USES: "apply the permutation k times", Pollard's rho, "where does
  ↳ the ball stop",
// detecting a repeat in a modular sequence, linked-list loop
  ↳ detection.
```

1.17 ParallelBinarySearch

PARALLEL BINARY SEARCH - binary search the answer for MANY queries at once, sweeping the

```
// shared structure once per level instead of once per query.
// PRECONDITION: test(i) must be MONOTONE in time - once it is true
  ↳ it stays true. If updates
// can also undo things, parallel binary search does not apply; use
  ↳ offline divide & conquer over
// the timeline with a rollback DSU (02 - Data Structures/14) instead.
// Total: O(log(maxT) * (cost of one full sweep)).
// Shape: "after which update does query q first become satisfiable?"
// lo[q], hi[q] track each query's remaining range; every round,
  ↳ bucket the queries by their
// midpoint, replay the updates in order, and test each query when
  ↳ the sweep reaches its mid.
template <class Apply, class Reset, class Test>
vector<int> parallelBinarySearch(int q, int T, Apply applyUpdate,
  ↳ Reset reset, Test test) {
    // applyUpdate(t): apply update t to the shared structure
    // reset(): clear the structure
    // test(i): is query i satisfied by the current structure?
    vector<int> lo(q, 0), hi(q, T); // answer in
  ↳ [lo, hi]; hi == T means "never"
    while (true) {
        vector<vector<int>> at(T + 1);
        bool active = false;
        for (int i = 0; i < q; i++)
            if (lo[i] < hi[i]) at[(lo[i] + hi[i]) / 2].push_back(i),
  ↳ active = true;
        if (!active) break;
        reset();
        for (int t = 0; t <= T; t++) {
            if (t) applyUpdate(t - 1); // now updates
  ↳ 0..t-1 are applied
            for (int i : at[t]) { if (test(i)) hi[i] = t; else lo[i]
  ↳ = t + 1; }
        }
        return lo; // lo[i] == T
    }
    // CONCRETE USES
    // - "for each query, the first moment a path/edge set connects u
  ↳ and v" (structure = DSU)
    // - k-th smallest in a range, offline (structure = BIT over values,
  ↳ updates = insert a value)
    // - "smallest capacity so that a flow/matching of size k exists"
    // - anything where a single query's binary search would need its
  ↳ own O(T) replay.
    // If the structure supports ROLLBACK, prefer divide-and-conquer over
  ↳ the timeline instead
    // (see OfflineDynamicConnectivity) - it is one pass instead of log
  ↳ passes.
```


1.18 MeetInTheMiddle

MEET IN THE MIDDLE - $n \leq 40$. Split into halves, enumerate $2^{(n/2)}$ subsets each side,

```
// then recombine by sorting + binary search / two pointers.
    ↪  $O(2^{(n/2)} * n)$ .

// All subset sums of a half, as a sorted vector.
vector<ll> subsetSums(const vector<ll>& a) {
    int n = a.size();
    vector<ll> s(1 << n, 0);
    for (int m = 1; m < (1 << n); m++) {
        int b = __builtin_ctz(m);
        s[m] = s[m ^ (1 << b)] + a[b];
    }
    return s;
}

// #subsets with sum <= S. Works with NEGATIVE values too (no
    ↪ pre-filter on the left half).
ll countAtMost(vector<ll> a, ll S) {
    int n = a.size(), h = n / 2;
    vector<ll> L(a.begin(), a.begin() + h), R(a.begin() + h, a.end());
    auto ls = subsetSums(L), rs = subsetSums(R);
    sort(all(rs));
    ll c = 0;
    for (ll x : ls) c += upper_bound(all(rs), S - x) - rs.begin();
    return c;
}

// Largest achievable sum <= S (the classic "closest subset sum from
    ↪ below").
ll bestAtMost(vector<ll> a, ll S) {
    int n = a.size(), h = n / 2;
    vector<ll> L(a.begin(), a.begin() + h), R(a.begin() + h, a.end());
    auto ls = subsetSums(L), rs = subsetSums(R);
    sort(all(rs));
    ll best = LLONG_MIN;
    for (ll x : ls) {
        // no 'x <= S'
        ↪ filter: x may be > S
        auto it = upper_bound(all(rs), S - x);
        // and still
        ↪ pair with a negative right sum
        if (it != rs.begin()) best = max(best, x + *prev(it));
    }
    return best;
}

// OTHER SHAPES
// - EXACT target: put one half in a hash map, look up (target - x)
    ↪ from the other.
// - 4-SUM: split the four arrays 2+2, hash all pairwise sums.
// - "choose exactly k items": bucket each half's subsets BY
    ↪ POPCOUNT, then pair bucket i with
//   bucket k-i (sort each bucket once).
// - Optimising two objectives: sort one half by objective A and
    ↪ take a prefix maximum of
//   objective B, so each query is one binary search.
// If  $n \leq 20$  a plain  $2^n$  loop is simpler; meet in the middle earns
    ↪ its keep from  $n = 30$  upward.
```

1.19 Scheduling

SCHEDULING / EXCHANGE-ARGUMENT THEOREM CARD
=====

/* ===== SCHEDULING / EXCHANGE-ARGUMENT THEOREM CARD
 ↪ =====

Each of these turns a hard-looking problem into a sort. Prove any
 ↪ new one by the EXCHANGE
ARGUMENT: assume an optimal schedule, swap two ADJACENT jobs, and
 ↪ compare the two costs.

SMITH'S RULE ($1 \parallel \sum w_j C_j$: minimise the weighted sum of
 ↪ completion times)
Sort by p_j / w_j ASCENDING. Adjacent swap of h, k improves iff
 ↪ $w_k * p_h - w_h * p_k > 0$,
so ALWAYS compare by cross-multiplication ($w_k * p_h$ vs $w_h * p_k$),
 ↪ never by float division.

MOORE-HODGSON ($1 \parallel \sum U_j$: minimise the NUMBER of late jobs),
 ↪ $O(n \log n)$
Sort by deadline ascending; keep a max-heap of the processing
 ↪ times taken so far.
For each job: take it, push p_j , add p_j to the running time; if
 ↪ the running time exceeds
this job's deadline, pop the largest p and subtract it. The heap
 ↪ size is the answer
(the rejected jobs can go last in any order).

EARLIEST DEADLINE FIRST (preemptive, one machine)
If ANY schedule meets all deadlines, EDF does. Feasibility check
 ↪ for non-preemptive
unit-machine: sort by deadline and verify prefix sums of p_j never
 ↪ exceed d_j .

JOHNSON'S RULE ($F2 \parallel C_{\max}$: two-machine flow shop, minimise
 ↪ makespan)
Set A = jobs with $p_1 < p_2$, sorted by p_1 ASCENDING
Set B = jobs with $p_1 \geq p_2$, sorted by p_2 DESCENDING
Optimal order = A then B .

LAWLER ($1 \parallel \text{prec} \mid f_{\max}$: minimise the maximum penalty, with
 ↪ precedence), $O(n^2)$
Build the schedule BACK TO FRONT. With t = total remaining time,
 ↪ among the jobs that have
no unscheduled successor pick the one minimising $f_j(t)$, and place
 ↪ it last.

CLASSIC GREEDY COMPANIONS
Interval scheduling (max #non-overlapping): sort by RIGHT
 ↪ endpoint, take greedily.
Interval point cover (stab all intervals): sort by right endpoint,
 ↪ place a point at it.
Minimum #machines for overlapping intervals: max overlap = sweep
 ↪ $+1/-1$ over endpoints.
Deadline scheduling with profits (each job unit length): sort by
 ↪ profit DESC, place each
job as late as possible before its deadline (DSU "next free slot")
 ↪ makes it near-linear).
Fractional knapsack: sort by value/weight DESC.
Minimise sum of $|x - a_i|$: x = the MEDIAN. Minimise sum of $(x -$
 ↪ $a_i)^2$: x = the MEAN.

=====

↪ */

// Moore-Hodgson: maximum number of jobs finished on time. jobs =
 ↪ {processing time, deadline}.

```
int maxOnTime(vector<pair<ll, ll>> jobs) {
    sort(all(jobs), [](auto& a, auto& b) { return a.second <
        ↪ b.second; });
    priority_queue<ll> pq;
    ll t = 0;
    for (auto [p, d] : jobs) {
        t += p, pq.push(p);
        if (t > d) t -= pq.top(), pq.pop();
    }
    return pq.size();
}
```

// Johnson's rule: optimal two-machine flow-shop order (returns the
 ↪ job indices).

```
vector<int> johnson(const vector<pair<ll, ll>>& jobs) { // {p on
    ↪ machine 1, p on machine 2}
    vector<int> A, B;
    for (int i = 0; i < (int)jobs.size(); i++)
        (jobs[i].first < jobs[i].second ? A : B).push_back(i);
    sort(all(A), [](int x, int y) { return jobs[x].first <
        ↪ jobs[y].first; });
    sort(all(B), [](int x, int y) { return jobs[x].second >
        ↪ jobs[y].second; });
```

```

    A.insert(A.end(), all(B));
    return A;
}

```

1.20 Josephus

JOSEPHUS PROBLEM. n people in a circle $0..n-1$, every k -th is removed; who survives?

```

// Recurrence:  $J(1) = 0$ ,  $J(n) = (J(n-1) + k) \bmod n$ .  $O(n)$ .
ll josephus(ll n, ll k) { ll r = 0; for (ll i = 2; i <= n; i++) r =
    (r + k) % i; return r; }
//  $k = 2$  closed form: write  $n = 2^m + l$  with  $0 \leq l < 2^m$ , the
    survivor is  $2 * l$  ( $0$ -indexed).
ll josephus2(ll n) { return 2 * (n - (1LL << (63 -
    __builtin_clzll(n)))); }
// LARGE  $n$ , SMALL  $k$ : one full lap kills  $\text{floor}(n/k)$  people, so recurse
    on  $n - \text{floor}(n/k)$  and map the
// index back.  $O(k \log n)$ .
ll josephusFast(ll n, ll k) {
    if (n == 1) return 0;
    if (k == 1) return n - 1;
    if (k > n) return (josephusFast(n - 1, k) + k) % n;
    ll r = josephusFast(n - n / k, k) - n % k;
    return r < 0 ? r + n : r + r / (k - 1);
}
// ORDER OF ELIMINATION / the  $j$ -th person to die: keep a BIT over
    alive flags and binary-search the
//  $((\text{pos} + k - 1) \bmod \text{alive} + 1)$ -th alive person each step.  $O(n \log$ 
     $n)$  and handles " $k$  changes".
// LAST TWO SURVIVORS: run the recurrence starting from  $J(2) = \{0, 1\}$ 
    instead of  $J(1) = 0$ .

// OTHER ONE-LINE CLASSICS worth having on paper:
// GRAY CODE:  $g(i) = i \wedge (i \gg 1)$ ; inverse:  $x \wedge x \gg 1$ ,  $x \gg$ 
     $2$ , ... until  $0$ .
//  $n$ -th PERMUTATION (factorial number system): digit  $i = k /$ 
     $(n-1-i)!$ , then  $k \% = (n-1-i)!$ .
// 15-PUZZLE SOLVABLE iff (#inversions of the tiles, blank excluded)
    + (row of the blank counted
// from the BOTTOM, 1-based) is EVEN, for a  $4 \times 4$  board. For an odd
    width  $w$ , solvable iff the
// inversion count is even. General  $w \times h$ : inversions + (blank row
    from bottom) parity rule above
// when  $w$  is even, plain inversion parity when  $w$  is odd.
// TOWER OF HANOI:  $2^n - 1$  moves; on move  $t$  (1-based) move disk  $(t \&$ 
     $-t)$  counted by trailing zeros,
// and for odd  $n$  the smallest disk cycles  $A \rightarrow C \rightarrow B \rightarrow A$ , for even  $n$ 
     $A \rightarrow B \rightarrow C \rightarrow A$ .
// BULLS AND COWS / stable-marriage style pairings: Gale-Shapley is
     $O(n^2)$ , always terminates, and
// is optimal for the proposing side.
// CATALAN-STYLE BALLOT COUNTING: paths from  $(0,0)$  to  $(n,m)$  staying
    strictly above  $y = x$  are
//  $(n-m)/(n+m) * C(n+m, m)$  (Bertrand's ballot theorem).

```

1.21 BigInt

BIG INTEGERS, non-negative, base $1e9$. Enough for the usual "the answer does not fit in 64 bits"

```

// problems: exact factorials / Catalan numbers / determinants, huge
    Fibonacci, exact comparisons.
// Add/sub  $O(n)$ , schoolbook multiply  $O(n*m)$  (fine to  $\sim 1e4$  digits),
    divide by a small int  $O(n)$ .
// If you need big*big beyond that, convolve the limbs with FFT and
    carry.
struct Big {
    static const ll B = 1000000000;
    vector<ll> d;
    // little-endian limbs, base  $1e9$ 
    Big(ll v = 0) { for (; v > 0; v /= B) d.push_back(v % B); }
    Big(const string& s) {
        for (int i = s.size(); i > 0; i -= 9)
            d.push_back(stoll(s.substr(max(0, i - 9), min(9, i))));
        trim();
    }
    void trim() { while (!d.empty() && d.back() == 0) d.pop_back(); }
    bool operator<(const Big& o) const {
        if (d.size() != o.d.size()) return d.size() < o.d.size();
        for (int i = d.size() - 1; i >= 0; i--) if (d[i] != o.d[i])
            return d[i] < o.d[i];
        return false;
    }
    Big operator+(const Big& o) const {
        Big r; ll c = 0;
        for (size_t i = 0; i < max(d.size(), o.d.size()) || c; i++) {
            if (i < d.size()) c += d[i];
            if (i < o.d.size()) c += o.d[i];
            r.d.push_back(c % B), c /= B;
        }
        return r;
    }
    Big operator-(const Big& o) const {
        // requires *this >= o
        Big r = *this; ll c = 0;
        for (size_t i = 0; i < r.d.size(); i++) {
            r.d[i] -= c + (i < o.d.size() ? o.d[i] : 0);
            if ((c = r.d[i] < 0)) r.d[i] += B;
        }
        r.trim();
        return r;
    }
    Big operator*(const Big& o) const {
        if (d.empty() || o.d.empty()) return Big();
        vector<ll> t(d.size() + o.d.size(), 0);
        for (size_t i = 0; i < d.size(); i++)
            for (size_t j = 0; j < o.d.size(); j++) t[i + j] +=
            (ll)d[i] * o.d[j];
        Big r; ll c = 0;
        for (size_t i = 0; i < t.size(); i++) c += t[i],
        r.d.push_back((ll)(c % B)), c /= B;
        for (; c; c /= B) r.d.push_back((ll)(c % B));
        r.trim();
        return r;
    }
    Big operator*(ll v) const { return *this * Big(v); }
    Big operator/(ll v) const {
        // small divisor only
        Big r = *this; ll c = 0;
        for (int i = r.d.size() - 1; i >= 0; i--) c = c * B + r.d[i],
        r.d[i] = (ll)(c / v), c %= v;
        r.trim();
        return r;
    }
    ll operator%(ll v) const {
        ll c = 0;
        for (int i = d.size() - 1; i >= 0; i--) c = (c * B + d[i]) %
        v;
        return (ll)c;
    }
    string str() const {
        if (d.empty()) return "0";
        string s = to_string(d.back());
        for (int i = d.size() - 2; i >= 0; i--) {
            string t = to_string(d[i]);
            s += string(9 - t.size(), '0') + t;
        }
        return s;
    }
};
// SIGNED: keep a separate sign flag and dispatch +/- to add/sub on
    magnitudes; comparison first.
// BASE  $1e9$  packs 9 decimal digits per limb, so printing is trivial -
    that is the whole point.

```

```
// FAST DIVISION of two Bigs is long and almost never needed; if a
    ↪ problem really wants it, binary
// search the quotient limb by limb, or move to Python-style thinking
    ↪ and avoid it.
// Sqrt: binary search with the multiply above, or Newton on doubles
    ↪ then fix up by a few steps.
```

1.22 RankingAndDates

RANKING / UNRANKING COMBINATORIAL OBJECTS, and CALENDAR ARITHMETIC. Small, and each one costs

```
// 20 minutes to re-derive under pressure.

// --- FACTORIAL NUMBER SYSTEM: rank and unrank a PERMUTATION. O(n^2)
    ↪ as written; use an
// order-statistic tree or a BIT for O(n log n) when n is large.
// order(p) = how many permutations of the same multiset are
    ↪ lexicographically smaller.
ll permRank(vector<int> p) { // p
    ↪ is a permutation of 0..n-1
    int n = p.size();
    ll r = 0, f = 1;
    for (int i = n - 1; i >= 0; i--) {
        int c = 0;
        for (int j = i + 1; j < n; j++) if (p[j] < p[i]) c++; //
        ↪ the Lehmer code digit
        r += c * f, f *= (n - i);
    }
    return r;
}
vector<int> permUnrank(int n, ll k) { //
    ↪ k-th permutation, 0-indexed
    vector<int> avail(n), p(n);
    iota(all(avail), 0);
    vector<ll> f(n + 1, 1);
    for (int i = 1; i <= n; i++) f[i] = f[i - 1] * i;
    for (int i = 0; i < n; i++) {
        ll d = k / f[n - 1 - i];
        k %= f[n - 1 - i];
        p[i] = avail[d], avail.erase(avail.begin() + d);
    }
    return p;
}
// NEXT PERMUTATION in place is std::next_permutation, O(n) - use it
    ↪ unless you need the rank.
// COMBINATIONS: rank of a k-subset {c_1<...<c_k} of [n] is sum
    ↪ C(c_i, i); unrank greedily by
// finding the largest c with C(c, i) <= remaining. Same shape, C
    ↪ instead of factorials.
// GRAY CODE is g = i ^ (i >> 1) (06 - grayCode.cpp).

// --- BALANCED BRACKET SEQUENCES: rank, unrank, and the next one in
    ↪ lexicographic order, with
// '(' < ')''. The counting table is the BALLOT NUMBERS: f[i][j] =
    ↪ number of ways to finish when i
// characters remain and j brackets are still open. f[n][0] is the
    ↪ Catalan number C_{n/2}.
// USES: "the k-th valid bracket sequence", "how many valid sequences
    ↪ are smaller than this one",
// iterating all sequences of length 2m without generating and
    ↪ filtering. Same skeleton works for
// any object counted by a DP: the rank is a walk that adds the
    ↪ subtree sizes you skip over.
// O(n^2) to build the table, O(n) per rank / unrank / next.
    ↪ Overflows for n > 60 - use __int128
// or a modulus if you only need the count.
vector<vector<ll>> ballot(int n) { // n =
    ↪ total length, must be even
    vector<vector<ll>> f(n + 1, vector<ll>(n + 2, 0));
    f[0][0] = 1;
    for (int i = 1; i <= n; i++)
        for (int j = 0; j <= i; j++)
            f[i][j] = f[i - 1][j + 1] + (j ? f[i - 1][j - 1] : 0);
    ↪ // place '(' then ')'
    return f;
}
ll bracketRank(const string& s, const vector<vector<ll>>& f) { //
    ↪ 0-indexed rank, s must be valid
    int n = s.size(), j = 0;
    ll r = 0;
    for (int i = 0; i < n; i++) {
        if (s[i] == ')') r += f[n - i - 1][j + 1]; //
        ↪ skip every sequence with '(' here
        j += s[i] == '(' ? 1 : -1;
    }
    return r;
}
string bracketUnrank(int n, ll k, const vector<vector<ll>>& f) {
    string s;
    int j = 0;
    for (int i = 0; i < n; i++) {
        ll c = f[n - i - 1][j + 1]; //
        ↪ completions starting with '('
        if (k < c) s += '(', j++;
        else k -= c, s += ')', j--;
    }
    return s;
}
```

```

    }
    return s;
}
bool nextBracket(string& s) { // in
    ⇨ place; false if s is the last
    int n = s.size(), d = 0;
    for (int i = n - 1; i >= 0; i--) {
        d += s[i] == '(' ? -1 : 1;
        if (s[i] == '(' && d > 0) { //
            ⇨ flip it to ')' and reset the rest
            d--;
            int open = (n - i - 1 - d) / 2;
            s = s.substr(0, i) + ')' + string(open, '(') + string(n -
            ⇨ i - 1 - open, ')');
            return true;
        }
    }
    return false;
}

// --- JULIAN DAY NUMBER: date <=> integer, closed form, no loops,
    ⇨ correct for the Gregorian
// calendar including every leap rule. Date differences, "what
    ⇨ weekday", and "add 1000 days" all
// become integer arithmetic once you are here.
ll dateToInt(int y, int m, int d) { //
    ⇨ y-m-d Gregorian -> day number
    return 1461 * (y + 4800 + (m - 14) / 12) / 4
        + 367 * (m - 2 - (m - 14) / 12 * 12) / 12
        - 3 * ((y + 4900 + (m - 14) / 12) / 100) / 4 + d - 32075;
}
array<int, 3> intToDate(ll jd) { //
    ⇨ inverse: {y, m, d}
    ll x, n, i, j, dd;
    x = jd + 68569, n = 4 * x / 146097, x -= (146097 * n + 3) / 4;
    i = 4000 * (x + 1) / 1461001, x -= 1461 * i / 4 - 31;
    j = 80 * x / 2447, dd = x - 2447 * j / 80, x = j / 11;
    return {(int)(100 * (n - 49) + i + x), (int)(j + 2 - 12 * x),
    ⇨ (int)dd};
}
int weekday(ll jd) { return (int)((jd + 1) % 7); } // 0
    ⇨ = Sunday
// LEAP YEAR: y % 4 == 0 && (y % 100 != 0 || y % 400 == 0).
// ZELLER'S CONGRUENCE gives the weekday directly, but the day number
    ⇨ is strictly more useful.

// --- STERN-BROCOT SEARCH: the simplest fraction satisfying a
    ⇨ MONOTONE predicate. ---
// Descend the mediant tree, but take many steps in one direction at
    ⇨ once (double the step size)
// so a 1e18 denominator costs O(log^2) instead of O(answer).
// USES: "smallest denominator q with property P", recovering p/q
    ⇨ from a decimal prefix, the
// simplest fraction strictly inside an interval, and best rational
    ⇨ approximation with a bound.
// ok() must be MONOTONE in the value p/q: false on small fractions,
    ⇨ true from some point on.
// Returns the SMALLEST such fraction with denominator <= N. Each
    ⇨ iteration takes as many steps
// as possible in one direction (found by doubling, then binary
    ⇨ search), so it is O(log^2 N)
// rather than O(answer) - the difference between instant and
    ⇨ hopeless at N = 1e18.
template <class F>
pair<ll, ll> fracSearch(ll N, F ok) {
    ll lp = 0, lq = 1, rp = 1, rq = 0; //
    ⇨ 0/1 (not ok) .. 1/0 (ok)
    for (;;) {
        ll moved = 0;
        auto walk = [&](ll& ap, ll& aq, ll& bp, ll& bq, bool want) {
            ⇨ // steps from a toward b
            ll k = 0, s = 1;
            auto test = [&](ll t) {
                return aq + t * bq <= N && ok(ap + t * bp, aq + t *
            ⇨ bq) == want;
            };
            while (test(k + s)) k += s, s *= 2;
            for (s /= 2; s; s /= 2) if (test(k + s)) k += s;
            ap += k * bp, aq += k * bq, moved += k;
        };
        walk(lp, lq, rp, rq, false); //
        ⇨ push the FALSE end up
        if (lq + rq > N) return {rp, rq};
        walk(rp, rq, lp, lq, true); //
        ⇨ pull the TRUE end down
        if (lq + rq > N || !moved) return {rp, rq}; //
        ⇨ !moved: nothing left to refine
    }
}

```

```

    }
}
// The Stern-Brocot tree IS the continued fraction expansion: going
    ⇨ left/right k times is one
// partial quotient. See 04 - Number Theory/05 - Rational for the CF
    ⇨ machinery and semiconvergents.

```

1.23 MatroidIntersection

MATROID INTERSECTION - the largest set that is independent in TWO matroids at once, and the

```
// max-weight version. Ground set is {0..n-1}; each matroid arrives
// as an ORACLE object exposing
// build(S) prepare for the current independent set S (a
// sorted vector of element ids)
// canAdd(e) is S + e independent? (e not in S)
// canSwap(e, x) is S - x + e independent? (e not in S, x in S)
// build is called once per augmenting phase, so it may rebuild
// everything from scratch - that is
// exactly why the concrete oracles below can get away with |S| + 1
// rebuilt structures per phase.
// matroidIsect returns a maximum-CARDINALITY common independent set,
// sorted, as element ids.
// matroidIsectW returns a maximum-WEIGHT one over all cardinalities;
// force maximum cardinality
// first by adding C > sum|w| to every weight.
// WHEN: one selection problem carrying TWO independent structural
// constraints. Colourful spanning
// tree / spanning tree with at most c_i edges of colour i, and
// rainbow spanning tree (c_i = 1):
// graphic + partition. Arborescence: graphic on the underlying
// undirected graph + partition by
// head vertex with cap 1, so "one incoming edge per vertex, no
// cycle". Spanning tree with bounded
// degrees: partition the edges by an endpoint - legal ONLY if no
// edge joins two constrained
// vertices, otherwise an edge lands in two classes and the reduction
// is silently wrong. Bipartite
// matching is two partition matroids (but use Hopcroft-Karp). THREE
// matroids is NP-hard.
// MIN-MAX: max |I| = min over U of (rank1(U) + rank2(complement of
// U)), handy for proving a bound
// or for a "why is the answer stuck at k" argument.
// COMPLEXITY in oracle calls: one phase pops each element once and
// spends O(|S|) or O(n) calls on
// it, so O(r n) calls plus 2 builds per phase, and there are r + 1
// phases: O(r^2 n) calls total.
// O(r^1.5 n) is the best known bound for this scheme and needs
// Cunningham's phase batching
// (augment along many shortest paths per BFS, Hopcroft-Karp style) -
// rarely worth typing. Budget
// r * n * (cost of one oracle call) per phase and multiply by the
// number of phases. Weighted:
// Bellman-Ford over n nodes and O(r n) arcs, so O(r n^2) per phase
// on top of the same oracle work.
// Memory is the oracles': O(r V) ints for graphic, O(63 r) for xor,
// O(colours) for partition.
// DEGENERATE: n = 0 or every element a LOOP (self-loop edge, zero
// vector, colour with cap 0)
// returns {}; loops fail canAdd on an empty S and are never picked.
// Duplicate / parallel elements
// are fine. Ties are broken arbitrarily - no lexicographic
// guarantee. Weighted with all weights
// negative returns {}. Both routines start from S = {} and cost
// nothing extra if the answer is 0.
// EXACT: pure combinatorics, no floating point; weights are ll and
// one path cost sums at most n of
// them, so |w| <= 1e17 is safe. Two traps that fail SILENTLY: the
// unweighted search must use BFS
// (a shortest augmenting path, fewest arcs) and the weighted one
// must break cost ties by fewest
// arcs - swap in a DFS or drop the tiebreak and you get a set that
// is not independent.
template <class M1, class M2>
vector<int> matroidIsect(int n, M1& m1, M2& m2) {
    vector<char> in(n, 0);
    vector<int> S;
    for (;;) {
        m1.build(S), m2.build(S);
        vector<int> par(n, -2), q; // -2
        // unvisited, -1 source; q is the BFS order
        int t = -1;
        for (int e = 0; e < n; e++) if (!in[e] && m1.canAdd(e))
            par[e] = -1, q.pb(e);
        for (int h = 0; h < sz(q) && t < 0; h++) {
            int u = q[h];
            if (in[u]) { // leaving S: u
                // out, some v in, keep M1 happy
                for (int v = 0; v < n; v++)
                    if (par[v] == -2 && !in[v] && m1.canSwap(v, u))
                        par[v] = u, q.pb(v);
            } else if (m2.canAdd(u)) t = u; // reached the
            // sink side: augment
            else for (int v : S) if (par[v] == -2 && m2.canSwap(u,
                v)) par[v] = u, q.pb(v);
        }
    }
}
```

```
}
if (t < 0) return S; // no
// augmenting path: S is maximum
for (int v = t; v != -1; v = par[v]) in[v] ^= 1;
S.clear();
for (int e = 0; e < n; e++) if (in[e]) S.pb(e);
}
template <class M1, class M2>
vector<int> matroidIsectW(int n, vector<ll> w, M1& m1, M2& m2) {
    vector<char> in(n, 0);
    vector<int> S, best;
    ll cur = 0, bw = 0;
    for (;;) {
        m1.build(S), m2.build(S);
        vector<pair<ll, int>> d(n, {llinf, inf}); // (path cost,
        // arcs) - lexicographic
        vector<int> par(n, -1);
        vector<pii> ed;
        vector<char> snk(n, 0);
        for (int e = 0; e < n; e++) if (!in[e]) {
            if (m1.canAdd(e)) d[e] = {-w[e], 1}; // sources; a
            // node costs -w in, +w out
            if (m2.canAdd(e)) snk[e] = 1;
            for (int x : S) {
                if (m1.canSwap(e, x)) ed.pb({x, e});
                if (m2.canSwap(e, x)) ed.pb({e, x});
            }
        }
        for (int it = 0; it < n; it++) { //
            // Bellman-Ford: weights are negative on entry
            bool ch = false;
            for (auto& [u, v] : ed) if (d[u].fi < llinf) {
                pair<ll, int> nd = {d[u].fi + (in[v] ? w[v] : -w[v]),
                // d[u].se + 1};
                if (nd < d[v]) d[v] = nd, par[v] = u, ch = true;
            }
            if (!ch) break;
        }
        int t = -1;
        for (int e = 0; e < n; e++) if (snk[e] && d[e].fi < llinf &&
            // (t < 0 || d[e] < d[t])) t = e;
            if (t < 0) break;
        cur -= d[t].fi; // the path
        // cost is minus the weight gained
        for (int v = t; v != -1; v = par[v]) in[v] ^= 1;
        S.clear();
        for (int e = 0; e < n; e++) if (in[e]) S.pb(e);
        if (cur > bw) bw = cur, best = S; // S is now
        // optimal among sets of size |S|
    }
    return best;
}
// --- ORACLE 1: COLOURFUL / PARTITION matroid. Independent iff
// colour c is used <= cap[c] times.
struct ColourM {
    vector<int> col, cap, cnt; // col[e] in
    // [0, |cap|)
    ColourM(vector<int> col, vector<int> cap) : col(col), cap(cap),
    // cnt(cap.size()) {}
    void build(const vector<int>& S) {
        fill(all(cnt), 0);
        for (int e : S) cnt[col[e]]++;
    }
    bool canAdd(int e) { return cnt[col[e]] < cap[col[e]]; }
    bool canSwap(int e, int x) { return canAdd(e) || col[e] ==
    // col[x]; }
};
// --- ORACLE 2: GRAPHIC matroid. Ground set = edges over vertices
// 0..V-1, independent iff acyclic.
// One DSU for S and one for each S - x, all rebuilt per phase:
// O(r^2) per build, O(r V) memory.
struct GraphicM {
    int V; vector<pii> e; vector<int> pos; vector<vi> d; // d[0] =
    // S, d[i+1] = S without S[i]
    GraphicM(int V, vector<pii> e) : V(V), e(e) {}
    int f(vi& p, int x) { return p[x] == x ? x : p[x] = f(p, p[x]); }
    bool ok(vi& p, int i) { return f(p, e[i].fi) != f(p, e[i].se); }
    void build(const vector<int>& S) {
        int m = sz(S);
        pos.assign(sz(e), 0), d.assign(m+1, vi(V));
        for (auto& p : d) iota(all(p), 0);
        for (int i = 0; i < m; i++) pos[S[i]] = i;
        for (int i = 0; i < m; i++) for (int j = 0; j <= m; j++) if
            // (j != i+1) {
            vi& p = d[j];
            p[f(p, e[S[i]].fi)] = f(p, e[S[i]].se); // union; a
    }
}
```

```

    ↪ no-op when already joined
    }
}
bool canAdd(int x) { return ok(d[0], x); }
bool canSwap(int x, int y) { return ok(d[pos[y] + 1], x); }
};
// --- ORACLE 3: LINEAR matroid over GF(2). Ground set = masks,
    ↪ independent iff xor-independent.
// b[j][i] is the basis vector whose top bit is i. Masks must be in
    ↪ [0, 2^62); 0 is a loop.
struct XorM {
    vector<ll> v; vector<int> pos; vector<vector<ll>> b; // b[0] =
    ↪ S, b[i + 1] = S without S[i]
    XorM(vector<ll> v) : v(v) {}
    bool ok(vector<ll>& b, ll x, bool ins = false) {
        for (int i = 62; i >= 0; i--) if (x >> i & 1) {
            if (!b[i]) { if (ins) b[i] = x; return true; }
            x ^= b[i];
        }
        return false;
    }
    void build(const vector<int>& S) {
        int m = sz(S);
        pos.assign(sz(v), 0), b.assign(m + 1, vector<ll>(63, 0));
        for (int i = 0; i < m; i++) pos[S[i]] = i;
        for (int i = 0; i < m; i++)
            for (int j = 0; j <= m; j++) if (j != i + 1) ok(b[j],
    ↪ v[S[i]], true);
    }
    bool canAdd(int x) { return ok(b[0], v[x]); }
    bool canSwap(int x, int y) { return ok(b[pos[y] + 1], v[x]); }
};
// MORE ORACLES, all trivial: UNIFORM U(k, n) - canAdd = sz(S) < k,
    ↪ canSwap = true. FREE matroid
// (everything independent) - both true; intersecting with it
    ↪ degenerates to "greedy on one
// matroid". TRANSVERSAL matroid (elements matchable into a bipartite
    ↪ graph) - one Kuhn augmenting
// path per query, legal but slow. COGRAPHIC (complement stays
    ↪ connected).
// ONE matroid only, max weight: sort by weight descending and add
    ↪ while canAdd - that is Kruskal,
// and matroid intersection is the price you pay for the second
    ↪ constraint.
// MATROID UNION and "partition the ground set into k independent
    ↪ sets" both reduce to intersection.

```

1.24 SchreierSims

SCHREIER-SIMS - a strong generating set (a stabiliser chain) for the permutation group generated

```

// by the given permutations of {0..n-1}. Gives |G| exactly,
    ↪ membership testing in O(n^2), and an
// explicit word in the generators for any member. Permutations are
    ↪ 0-indexed images: p[i] is where
// i goes, and mul(a, b) means "apply a, THEN b", so (a*b)[i] =
    ↪ b[a[i]]. Level i of the chain holds
// coset representatives of the pointwise stabiliser of 0..i-1,
    ↪ indexed by the image of i, so
// |G| = product of the transversal sizes and every element factors
    ↪ uniquely as one rep per level.
// WHEN: "you may apply these shuffles to the array any number of
    ↪ times, in any order - how many
// distinct arrangements are reachable, and is arrangement X
    ↪ reachable". Model a shuffle as the
// permutation it applies, then the reachable arrangements are
    ↪ exactly the group elements: order()
// counts them and has() decides reachability. word() prints the
    ↪ moves that get you there. Also:
// the group generated by a puzzle's legal moves, the size of a
    ↪ symmetry group before a Burnside
// count, and "do these generators generate all of S_n" (compare
    ↪ order() with n!).
// CONVENTION TRAP: an arrangement is the permutation taking the
    ↪ START to it, so ask has(q) with
// q[i] = the index the item now at position i started at (build it
    ↪ once, consistently, then stay
// consistent). If the array has DUPLICATE values, the number of
    ↪ distinct arrangements is |G| over
// the size of the stabiliser of that multiset, NOT |G| - this file
    ↪ does not give you that.
// COMPLEXITY: level i holds at most n - i reps, so O(n^2) stored
    ↪ permutations and O(n^3) ints of
// memory (under 1 MB even at n = 60). The constructor sifts every
    ↪ rep at level i against every rep
// at a level >= i: O(n^4) sifts costing O(n^2) each, so O(n^6) when
    ↪ the group is all of S_n.
// MEASURED on three random generators of S_n: n = 40 takes 0.5 s, n
    ↪ = 50 takes 2 s, n = 60 takes
// 6 s, and that is the wall - past it you need a real Schreier-Sims
    ↪ (Schreier generators kept per
// level, plus Jerrum's filter) or the randomised variant. has() and
    ↪ word() are O(n^2) per query.
// DEGENERATE: no generators or only identities gives order() == 1
    ↪ and has(id) == true; n = 1 the
// same. Duplicate generators are free. word(identity) is the empty
    ↪ word.
// EXACT: integers only. OVERFLOW: |G| <= n! exceeds ll at n = 21
    ↪ (20! = 2.4e18 fits), so switch
// order() to __int128 (good to 33!) or accumulate modulo the
    ↪ answer's modulus for larger n.
struct PermGroup {
    int n; bool keepW;
    vector<vi> lk; // lk[i][j] =
    ↪ index of the level i rep i->j
    vector<vector<vi>> T, Ti, W; // reps, their
    ↪ inverses, and their words
    static vi mul(const vi& a, const vi& b) {
        vi c(sz(a));
        for (int i = 0; i < sz(a); i++) c[i] = b[a[i]];
        return c;
    }
    static vi inv(const vi& a) {
        vi c(sz(a));
        for (int i = 0; i < sz(a); i++) c[a[i]] = i;
        return c;
    }
    // Strip p through the chain. Returns the level where p is NEW
    ↪ (and inserts it there if add),
    // or -1 when p reduces to the identity, i.e. p is in the group.
    ↪ w accumulates the word of the
    // running residue, so it ends as the word of p^-1 when p sifts
    ↪ through.
    int sift(vi p, vi& w, bool add) {
        for (int i = 0; i < n; i++) {
            int k = lk[i][p[i]];
            if (k < 0) {
                if (!add) return i;
                lk[i][p[i]] = sz(T[i]), T[i].pb(p), Ti[i].pb(inv(p));
                if (keepW) W[i].pb(w);
                return i;
            }
            if (keepW) for (int j = sz(W[i][k]) - 1; j >= 0; j--)
    ↪ w.pb(~W[i][k][j]); // w *= rep^-1
            bool one = true; // p *= rep^-1

```



```

    ↪ in place; now p fixes 0..i, so
        for (int j = i; j < n; j++) // only j >= i
    ↪ can move and only they matter
        one &= (p[j] = Ti[i][k][p[j]]) == j;
        if (one) return -1; // already the
    ↪ identity: skip the rest
    }
    return -1;
}
PermGroup(int n, const vector<vi>& gen, bool keepW = false) :
    ↪ n(n), keepW(keepW) {
        lk.assign(n, vi(n, -1)), T.resize(n), Ti.resize(n),
    ↪ W.resize(n);
        vi id(n);
        iota(all(id), 0);
        for (int i = 0; i < n; i++) lk[i][i] = 0, Ti[i].pb(id),
    ↪ Ti[i].pb(id), W[i].pb(vi());
        vector<pii> el; // every
    ↪ non-identity rep, as (level, index)
        auto push = [&](vi p, vi w) {
            int i = sift(p, w, true);
            if (i >= 0) el.pb({i, sz(T[i]) - 1});
        };
        for (int g = 0; g < sz(gen); g++) push(gen[g], vi[g]);
        for (int i = 0; i < sz(el); i++) for (int j = 0; j <= i; j++)
    ↪ { // el grows as we go
            auto prod = [&](pii a, pii b) {
                vi w;
                if (keepW) w = W[a.fi][a.se], w.insert(w.end(),
    ↪ all(W[b.fi][b.se]));
                push(mul(T[a.fi][a.se], T[b.fi][b.se]), w);
            };
            if (el[i].fi <= el[j].fi) prod(el[i], el[j]); // a
    ↪ level i rep times a
            if (el[j].fi <= el[i].fi && i != j) prod(el[j], el[i]);
    ↪ // level >= i element is

    ↪ // a Schreier generator
    }
}
ll order() {
    ll r = 1;
    for (int i = 0; i < n; i++) r *= sz(T[i]);
    return r;
}
bool has(vi p) { vi w; return sift(p, w, false) < 0; }
vi word(vi p) { //
    ↪ PRECONDITION: has(p), and keepW was set.
        vi w; // token g >= 0
    ↪ is gen[g], ~g is its inverse
        sift(p, w, false);
        reverse(all(w));
        for (int& x : w) x = ~x; // w held p^-1;
    ↪ invert it
        return w;
    }
};
// Rebuild p from word(p) with res = mul(res, x >= 0 ? gen[x] :
    ↪ inv(gen[~x])) over the tokens in
// order. Word LENGTH IS NOT BOUNDED - it is a product of stored
    ↪ words and can blow up; use it only
// where the checker accepts any move sequence. If you need short
    ↪ words, use only for small groups.
// ORBIT of one point (which positions a single item can reach) is a
    ↪ plain BFS on the generators
// in O(n t) - do not build the whole chain for that. The level-0
    ↪ transversal IS the orbit of 0.
// The pointwise stabiliser of 0..i-1 is generated by the reps at
    ↪ levels >= i, which is how you
// answer "how many arrangements fix these positions" (= product of
    ↪ |T[j]| for j >= i).
// Related: Burnside / Polya for counting colourings up to the group,
    ↪ and the parity argument that
// solves the 15-puzzle style problems in one line when the group is
    ↪ the full alternating group.

```

1.25 RangeBitset

RANGE BITSET - a bitset with SET / RESET / FLIP of a whole range and POPCOUNT of a range, all

```

// in O((r - l) / 64 + 1). std::bitset cannot do range updates at all
    ↪ (its set()/reset()/flip()
// are either one bit or the whole thing), so anything that wants
    ↪ "turn on bits [l, r]" needs
// this. Bits are 0..n-1, ranges are INCLUSIVE on both ends, and l >
    ↪ r is a silent no-op.
// WHEN: a sweep that paints intervals into one bitset per colour
    ↪ ("recolour [l, r] to c" over
// 100 colours: AND out the range from the old bitset, OR it into the
    ↪ new one); interval DP
// reachability; "is any position in [l, r] still free" answered by
    ↪ findNext; knapsack-style DP
// where the transition is x |= x << k.
// COMPLEXITY: n/64 words. O((r-l)/64 + 1) per range op, O(1)
    ↪ test/set of one bit, O(n/64) for
// the word-parallel operators and for a shift. The point is the 64x
    ↪ constant, not the O.
// EXACT: pure unsigned integer, nothing can overflow.
// THE WORD-BOUNDARY TRAP - the only thing that ever goes wrong in
    ↪ this file:
// - a range touches THREE pieces: the tail of word l>>6 (bits
    ↪ l&63..63), the whole words
// strictly between, and the head of word r>>6 (bits 0..r&63).
// - if l>>6 == r>>6 you must do ONE masked write and return.
    ↪ Falling through to the three-piece
// path writes bits l&63..63 AND bits 0..r&63 of that same word,
    ↪ i.e. the entire word - the
// classic off-by-one that makes this silently wrong only when l
    ↪ and r are 64-aligned near
// each other. Test n = 64, 65, 128 and l, r in the same word.
// - never shift by 64: ~0ULL << 64 is UB (x86 shifts by 0 and hands
    ↪ you all-ones). mask(a, b)
// below shifts by a and by 63-b with 0 <= a <= b <= 63, so both
    ↪ counts are in [0, 63].
// - keep the bits >= n zero (trim()), or count() and findNext()
    ↪ over the whole bitset lie.

```

typedef unsigned long long ull;

```

struct RangeBitset {
    int n; vector<ull> w; // w[i]
    ↪ holds bits 64i .. 64i + 63
    RangeBitset(int n = 0) : n(n), w((n + 63) / 64, 0) {}
    static ull mask(int a, int b) { // bits
    ↪ a..b of ONE word, 0<=a<=b<=63
        return (~0ULL << a) & (~0ULL >> (63 - b));
    }
    template <class F> void each(int l, int r, F f) { //
    ↪ f(word, mask) over the range
        if (l > r) return;
        int a = l >> 6, b = r >> 6;
        if (a == b) { f(a, mask(l & 63, r & 63)); return; } // SAME
    ↪ word: one write, then STOP
        f(a, mask(l & 63, 63)), f(b, mask(0, r & 63)); // the
    ↪ two partial words
        for (int i = a + 1; i < b; i++) f(i, ~0ULL); // the
    ↪ full words between them
    }
    void set(int l, int r) { each(l, r, [&](int i, ull m) { w[i] |=
    ↪ m; }); }
    void reset(int l, int r) { each(l, r, [&](int i, ull m) { w[i] &=
    ↪ ~m; }); }
    void flip(int l, int r) { each(l, r, [&](int i, ull m) { w[i] ^=
    ↪ m; }); }
    int count(int l, int r) {
        int c = 0;
        each(l, r, [&](int i, ull m) { c += ones(w[i] & m); });
        return c;
    }
    int count() { return count(0, n - 1); } // n ==
    ↪ 0 gives l > r, so 0
    bool test(int i) const { return w[i >> 6] >> (i & 63) & 1; }
    void set(int i) { w[i >> 6] |= 1ULL << (i & 63); }
    void reset(int i) { w[i >> 6] &= ~(1ULL << (i & 63)); }
    void flip(int i) { w[i >> 6] ^= 1ULL << (i & 63); }
    int findNext(int i) { // first
    ↪ set bit >= i, or n if none
        if (i < 0) i = 0;
        if (i >= n) return n;
        int b = i >> 6;
        ull c = w[b] & (~0ULL << (i & 63)); // drop
    ↪ the bits before i
        while (!c) { if (++b == sz(w)) return n; c = w[b]; }
        return b * 64 + lsb(c);
    }
    void trim() { if (n & 63) w.back() &= mask(0, (n - 1) & 63); }
}

```



```

    ↪ // bits >= n must stay 0
    void operator|=(const RangeBitset& o) { for (int i = 0; i <
    ↪ sz(w); i++) w[i] |= o.w[i]; }
    void operator&=(const RangeBitset& o) { for (int i = 0; i <
    ↪ sz(w); i++) w[i] &= o.w[i]; }
    void operator^=(const RangeBitset& o) { for (int i = 0; i <
    ↪ sz(w); i++) w[i] ^= o.w[i]; }
    void orShl(int k) { // x |=
    ↪ x << k, the subset-sum step
        int q = k >> 6, s = k & 63;
        for (int i = sz(w) - 1; i >= q; i--) { //
    ↪ DESCENDING: every source word is
            ull v = w[i - q] << s; // below
    ↪ i, so still unmodified
            if (s && i - q) v |= w[i - q - 1] >> (64 - s); // s ==
    ↪ 0 would be a shift by 64
            w[i] |= v;
        }
        trim(); // bits
    ↪ that slid past n
    }
};
// VARIANTS / RELATED
// - Whole-bitset AND/OR/XOR plus _Find_first/_Find_next only: use
    ↪ std::bitset, it is faster and
// already written. Come here the moment a RANGE update appears.
// - Range ASSIGN with O(log) amortized per op and no n/64 term: the
    ↪ interval set ("Chtholly",
// a std::set of maximal constant runs). It does assign, not flip,
    ↪ and needs random data.
// - Range assign + range count with true polylog: a lazy segment
    ↪ tree. This struct wins up to
// n around 1e7 purely on constant factor, and is a third of the
    ↪ code.
// - Static "count ones in [l, r]": a prefix popcount array, no
    ↪ structure needed.
// - findPrev: mask with mask(0, i & 63) and scan words downward,
    ↪ taking msb().

```

1.26 TrygubNum

TRYGUB NUMBER - a big number that eats 1e5+ updates of the form "add $x * b^e$ " in O(1)

```

// amortized carries, and answers "what is digit k" at any time.
// WHAT: maintains  $V = \sum e \text{ of } d[e] * \text{base}^e$  exactly, where
    ↪ base =  $b^K$  packs K base-b digits
// into one machine digit. add(x, e) does  $V += x * b^e$  with x AND e
    ↪ of either sign. kthDigit(k)
// returns floor( $V / b^k$ ) mod b, i.e. the k-th digit of V in base b
    ↪ counting from the units digit
// at k = 0. sign() returns +1 / 0 / -1.
// THE TRICK: digits are NOT normalised. d[e] may be anywhere in
    ↪ (-base, base), the carry is
// pushed lazily one position at a time and stops as soon as it dies,
    ↪ and only NONZERO digits are
// stored, in a map - so a negative e (a fractional position) costs
    ↪ nothing and the number needs
// no length. Amortized it is O(1) carry steps per add
    ↪ (binary-counter argument), each step one
// map operation, so O(log D) with D = number of stored digits.
    ↪ Memory O(D).
// WHEN: 1e5 updates "add  $5 * 10^k$ " to a huge number with digit
    ↪ queries interleaved. A plain
// vector is O(len) per update; a segment tree over digits drowns in
    ↪ carry logic. Also the clean
// way to compare two such numbers: subtract, look at sign().
// NEGATIVE e: split  $e = q * K + j$  with FLOOR division, not C++
    ↪ truncation. For  $e = -1$ ,  $K = 30$  the
// language gives  $e/K == 0$  and  $e \% K == -1$ , which would multiply x by
    ↪  $b^{(-1)}$ . Take
//  $j = ((e \% K) + K) \% K$  first, then  $q = (e - j) / K$ , which is exact
    ↪ in both signs.
// NEGATIVE x: nothing special in add - integer division truncates
    ↪ toward zero, so after
//  $d[q] -= t * \text{base}$  the residue KEEPS the sign of d[q] and  $|d[q]| <$ 
    ↪ base. That is exactly how
// negative digits appear, and it is why kthDigit needs the borrow
    ↪ below.
// THE BORROW RULE in kthDigit (the whole difficulty): the digits
    ↪ below q may sum to a negative
// number. Their total is bounded by  $\text{base}^q$  in absolute value (sum of
    ↪  $(\text{base}-1) * \text{base}^e$  for  $e < q$ 
// is  $\text{base}^q - 1$ ), so the top stored digit below q decides the SIGN
    ↪ of the whole low part, and a
// negative low part eats exactly 1 from position q: 'if
    ↪ (prev(it)->se < 0) a--'. This is why
// zeros MUST be erased from the map - a stored 0 sitting just under
    ↪ q would answer "not
// negative" and lose the borrow.
// OVERFLOW BUDGET: a digit lives in (-base, base) and add pushes  $x *$ 
    ↪  $b^{(K-1)}$  into it, so you
// need  $\text{base} + |x| * b^{(K-1)} < 9.2e18$ .  $b = 2$ ,  $K = 30$  (base ~  $1.07e9$ )
    ↪ and  $b = 10$ ,  $K = 9$ 
// (base =  $1e9$ ) both leave room for  $|x| \leq 1e9$ . Bigger K = shorter
    ↪ carry chains, so take the
// largest one the budget allows.
struct TrygubNum {
    ll b, K, base; vector<ll> pw; // pw[i] =
    ↪  $b^i$ , base =  $b^K$  = one machine
    map<ll, ll> d; // digit. e ->
    ↪ digit, ZEROS ARE ERASED
    TrygubNum(ll b, ll K) : b(b), K(K), pw(K + 1, 1) {
        for (ll i = 1; i <= K; i++) pw[i] = pw[i - 1] * b;
        base = pw[K];
    }
    void add(ll x, ll e) { //  $V += x *$ 
    ↪  $b^e$ , both signs allowed
        ll j = ((e \% K) + K) \% K, q = (e - j) / K; // FLOOR split
    ↪  $e = q * K + j$ ,  $0 \leq j < K$ 
        d[q] += x * pw[j];
        ll t;
        do {
            t = d[q] / base; // truncates
    ↪ toward 0: residue keeps sign
            d[q] -= t * base, d[q + 1] += t;
            if (!d[q]) d.erase(q); // erasing
    ↪ zeros is load-bearing, see above
            q++;
        } while (t); // carry died
    ↪ -> stop, O(1) amortized
        if (!d[q]) d.erase(q); // the digit
    ↪ above may have been left at 0
    }
    int kthDigit(ll k) { // floor( $V /$ 
    ↪  $b^k$ ) mod b,  $k \geq 0$ 
        ll q = k / K, a = 0;
        auto it = d.lower_bound(q);

```

```

    if (it != d.end() && it->fi == q) a = it->se;
    if (it != d.begin() && prev(it)->se < 0) a--; // low part is
    ↪ negative: it borrows a 1
    a = ((a % base) + base) % base;
    return (int)(a / pw[k % K] % b); // the (k %
    ↪ K)-th base-b digit of a
}
int sign() { // top stored
    ↪ digit dominates all the rest
    return d.empty() ? 0 : (prev(d.end())->se > 0 ? 1 : -1);
}
};
// DEGENERATE: empty number -> sign() 0 and kthDigit() 0 everywhere.
    ↪ add(0, e) stores nothing.
// Adding and then subtracting the same term returns the map to
    ↪ empty. k above the top digit
// gives 0. kthDigit is meant for V >= 0; for V < 0 it still returns
    ↪ floor(V / b^k) mod b, which
// is the b-complement expansion (...bbb1 for -1), not the digits of
    ↪ |V| - negate first.
// VARIANTS / RELATED
// - Reading the whole number: from prev(d.end())->fi * K + K - 1
    ↪ downward through kthDigit.
// - Only nonnegative e and no queries until the end: just use a
    ↪ vector and normalise once.
// - Full arithmetic (multiply, divide, print): 21 - BigInt.cpp. This
    ↪ struct only does += x*b^e.
// - Same lazy-carry idea works for "add x at position e" in any
    ↪ positional structure; the map is
// the only reason it stays sparse.

```

1.27 PermutationRoots

CYCLE DECOMPOSITION AND WHAT IT ANSWERS: the k-th root of a permutation, the minimum number of

```

// transpositions (with the swaps), and the order.
// Convention: p is 0-indexed, p[i] is the image of i, (p . q)[i] =
    ↪ p[q[i]], p^k is p applied k
// times. All three routines are O(n) after the O(n) decomposition
    ↪ (the order pays a small
// factorisation on top).
vector<vector<int>> cycles(const vector<int>& p) { // each
    ↪ cycle as i, p[i], p[p[i]], ...
    int n = sz(p); vector<vector<int>> c; vector<char> vis(n, 0);
    for (int i = 0; i < n; i++) if (!vis[i]) {
        c.pb({});
        for (int j = i; !vis[j]; j = p[j]) vis[j] = 1, c.back().pb(j);
    }
    return c; // fixed
    ↪ points are cycles of length 1
}
vector<int> fromCycles(int n, const vector<vector<int>>& c) {
    vector<int> p(n);
    for (auto& v : c) for (int i = 0; i < sz(v); i++) p[v[i]] = v[(i
    ↪ + 1) % sz(v)];
    return p;
}
// --- K-TH ROOT: q with q^k = p, or {} if none exists. k >= 1 (k = 0
    ↪ is not a question).
// DERIVATION. Raising to the k-th power splits a q-cycle of length m
    ↪ into gcd(m, k) cycles of
// length m / gcd(m, k) - so the q-cycle behind a group of p-cycles
    ↪ of length L must have length
// m = g*L where g is the group size, and it must satisfy g = gcd(gL,
    ↪ k). Divide that out: g = k
// is not required, the condition is exactly
// g | k AND gcd(L, k / g) = 1.
// Let t = product over primes r dividing L of r^(exponent of r in k)
    ↪ - computed below by
// stripping gcd(L, .) off k until it is 1. Then k/t is coprime to L,
    ↪ every legal g is a multiple
// of t, and every g = t*z with z | (k/t) is legal. So the p-cycles
    ↪ of length L can be packed iff
// t divides their COUNT, and greedily taking groups of exactly t
    ↪ always works. Groups of size t
// also maximise the number of cycles of q, so this returns the root
    ↪ with the FEWEST
// transpositions. Existence therefore = "for every L, t(L, k)
    ↪ divides the number of cycles of
// length L". A cycle length that does not appear is vacuously fine.
// CONSTRUCTION: interleave the g cycles v_0..v_{g-1} into one array
    ↪ c of length m = g*L by
// c[(x + y*k) mod m] = v_x[y]. That is a bijection precisely because
    ↪ gcd(m, k) = g, and then
// stepping one place in c is q, so q^k moves v_x[y] to v_x[y+1] =
    ↪ p(v_x[y]).
vector<int> permRoot(const vector<int>& p, ll k) {
    int n = sz(p);
    vector<vector<vector<int>>> by(n + 1); //
    ↪ p-cycles bucketed by their length
    for (auto& v : cycles(p)) by[sz(v)].pb(v);
    vector<vector<int>> out;
    for (int L = 1; L <= n; L++) {
        if (by[L].empty()) continue;
        ll t = 1, rem = k, g;
        while ((g = gcd((ll)L, rem)) != 1) t *= g, rem /= g;
        ↪ // t = minimal legal group
        if (sz(by[L]) % t) return {}; // count
        ↪ not divisible: NO ROOT
        for (ll i = 0; i < sz(by[L]); i += t) {
            ll m = t * L, kk = k % m; vector<int> c(m); // reduce
            ↪ k first: y*k can overflow
            for (ll x = 0; x < t; x++) for (ll y = 0; y < L; y++)
                c[(x + y * kk) % m] = by[L][i + x][y]; // the
            ↪ interleave
            out.pb(c);
        }
    }
    return fromCycles(n, out); // n == 0
    ↪ returns {}, which is right
}
// --- MINIMUM TRANSPOSITIONS to sort p is n - (number of cycles),
    ↪ and no fewer: one swap changes
// the cycle count by exactly +-1 (inside a cycle it splits, across
    ↪ two it merges), and sorted
// means n cycles. To turn a into b it is n - cycles(b^-1 . a). Both
    ↪ are the SAME loop: drag the
// wanted value into place, which kills one cycle per swap.
vector<pii> swapsToMatch(vector<int> a, const vector<int>& b) {

```

```

    ↪ // b = 0,1,...,n-1 means "sort"
    int n = sz(a); vector<int> pos(n); vector<pii> s;
    for (int i = 0; i < n; i++) pos[a[i]] = i; // pos[v]
    ↪ = where value v sits now
    for (int i = 0; i < n; i++) if (a[i] != b[i]) {
        int j = pos[b[i]]; // j > i
    ↪ always, so one pass suffices
        s.pb({i, j}); pos[a[i]] = j, pos[a[j]] = i, swap(a[i], a[j]);
    }
    return s; // sz(s)
    ↪ == n - cycles(b^-1 . a)
}
// --- ORDER of p = lcm of the cycle lengths. It OVERFLOWS EARLY: the
    ↪ max over permutations of n
// (Landau's g(n) ~ e*sqrt(n ln n)) passes 1e18 around n = 130, so
    ↪ never store it in an ll. Keep
// it factored, or reduce it mod M. Returns prime -> exponent; order
    ↪ mod M is then the product of
// q^e mod M over the map, and the exponents sum to at most
    ↪ log2(order).
map<ll, int> permOrder(const vector<int>& p) {
    map<ll, int> f;
    for (auto& v : cycles(p)) {
        int m = sz(v);
        for (ll q = 2; q * q <= m; q++) if (m % q == 0) {
            int e = 0; while (m % q == 0) m /= q, e++;
            f[q] = max(f[q], e);
        }
        if (m > 1) f[m] = max(f[m], 1); // the
    ↪ leftover prime factor
    }
    return f; //
    ↪ identity / n == 0 gives {} = order 1
}
// VARIANTS / RELATED
// - p^k in O(n): inside each cycle of length L step by k, giving
    ↪ gcd(L, k) cycles of length
// L / gcd(L, k). That is the root loop read backwards, and the
    ↪ cheap way to check a root.
// - p^k for huge k: reduce k modulo the cycle length of each
    ↪ element, not modulo the order.
// - SQUARE ROOT (k = 2): t = 2 for even L and 1 for odd L, so a root
    ↪ exists iff the cycles of
// every even length come in pairs. Odd-length cycles always have a
    ↪ root alone.
// - Need a root with a prescribed PARITY of transposition count:
    ↪ once, merge 2t cycles of some
// length instead of t (legal iff k/t is even and that length has
    ↪ >= 2t cycles). It changes the
// cycle count by one, hence the parity.
// - MINIMUM ADJACENT swaps is a completely different quantity - the
    ↪ inversion count. See
// 14 - Inversions.cpp. Do not confuse it with n - cycles.
// - The permutation is even iff n - cycles is even; that is the
    ↪ sign, and it is what tells you
// whether the 15-puzzle position is solvable.

```

1.28 SubarrayAggregator

LOGARITHMIC SUBARRAY AGGREGATOR - every value of $op(a[l..r])$ over every subarray, in $O(n \log A)$.

```

// Fix r. As l walks from r down to 0 the aggregate  $op(a[l..r])$  only
    ↪ ever moves one way, and every
// strict move is a big jump, so only  $O(\log A)$  DISTINCT values occur
    ↪ and equal ones form contiguous
// runs of l. subarrayAgg calls cb(r, runs) for every r (0-indexed)
    ↪ with runs = {v, lo, hi} meaning
//  $op(a[l..r]) = v$  for exactly the l in [lo, hi]; the runs partition
    ↪ [0, r] and are listed from
// l = r downwards, so runs[0].hi == r and runs.back().lo == 0.
// WHEN: "count / find the subarrays whose gcd (or / and / min) is
    ↪ X", "how many subarrays have
// gcd > 1", "longest subarray with or <= k". Replaces a sparse table
    ↪ plus a binary search per l,
// and it is shorter and exact.
// PRECONDITION for correctness: op associative, and  $op(x, y)$ 
    ↪ monotone in the same direction for
// every x (so equal values are adjacent and the merge below is
    ↪ legal).
// PRECONDITION for the  $O(\log A)$  run count: every STRICT change must
    ↪ shrink a potential of size A -
// gcd at least halves, and / or turn at least one bit off / on, lcm
    ↪ capped at LIM at least doubles.
// gcd, and, or, capped lcm all qualify. min and max are monotone so
    ↪ the code is CORRECT for them,
// but their run count is  $O(n)$ , not  $O(\log A)$  ( $a[i] = i$  gives n runs
    ↪ at  $r = n-1$ ) - use a monotonic
// stack there. sum, xor and product do not qualify at all: nothing
    ↪ forces values to repeat.
// COMPLEXITY:  $O(n \log A)$  time,  $O(\log A)$  live memory plus whatever cb
    ↪ stores.
// DEGENERATE: empty a calls cb zero times. n == 1 gives one run
    ↪ {a[0], 0, 0}. All-equal gives one
// run per r. Zeros are fine for gcd ( $gcd(0, x) == x$ ), so an all-zero
    ↪ array reports v == 0.
// EXACTNESS: exact on integers, the aggregate never leaves the value
    ↪ type. Counts of subarrays
// need ll ( $(n+1)/2$  overflows int at  $n = 65536$ ), the run endpoints
    ↪ stay int.
template <class T> struct Run { T v; int lo, hi; }; //
    ↪  $op(a[l..r]) = v$  for every l in [lo, hi]
template <class T, class Op, class F>
void subarrayAgg(const vector<T>& a, Op op, F cb) {
    vector<Run<T>> cur, nxt; // runs
    ↪ for r-1, then the ones for r
    for (int r = 0; r < sz(a); r++) {
        nxt.clear();
        nxt.pb({a[r], r, r}); // l ==
        ↪ r: the element on its own
        for (auto& e : cur) { //
            ↪ associativity:  $op(a[l..r-1], a[r])$ 
            T v = op(e.v, a[r]);
            if (v == nxt.back().v) nxt.back().lo = e.lo; //
            ↪ unchanged: swallow the whole old run
            else nxt.pb({v, e.lo, e.hi}); //
            ↪ changed: at most  $O(\log A)$  times
        }
        cur.swap(nxt);
        cb(r, cur);
    }
}
// --- CONSUMER 1: cnt[x] = number of subarrays whose gcd is exactly
    ↪ x.  $O(n \log A (\log A + \log n))$ .
// std::gcd (C++17) and not __gcd: __gcd is a libstdc++ internal and
    ↪ rejects signed types on clang.
map<ll, ll> gcdHistogram(const vector<ll>& a) {
    map<ll, ll> cnt;
    subarrayAgg(a, [](ll x, ll y) { return gcd(x, y); }, [&](int,
    ↪ auto& v) {
        for (auto& e : v) cnt[e.v] += e.hi - e.lo + 1;
    });
    return cnt;
}
// --- CONSUMER 2: number of subarrays with  $or(a[l..r]) \geq k$ , and the
    ↪ set of distinct subarray ORs
// (which has size  $O(n \log A)$ , not  $O(n^2)$  - that bound is the reason
    ↪ the question gets asked).
ll orAtLeast(const vector<ll>& a, ll k, set<ll>* distinct = nullptr) {
    ll c = 0;
    subarrayAgg(a, [](ll x, ll y) { return x | y; }, [&](int, auto&
    ↪ v) {
        for (auto& e : v) { // v is
            ↪ sorted by value, so you could
            if (e.v >= k) c += e.hi - e.lo + 1; // binary
            ↪ search - but it has  $O(\log A)$ 

```

```

    if (distinct) distinct->insert(e.v);          //
    ↪ entries, a scan is already free
    }
    });
    return c;
}
// VARIANTS
// - LONGEST / SHORTEST subarray with a target property: the runs
    ↪ are sorted by l, so per r the
// - best l is an endpoint of the first / last run that satisfies
    ↪ it.  $O(n \log A)$  overall.
// - "gcd of  $a[l..r] == 1$ " for all pairs: gcdHistogram(a)[1].
// - LCM needs a cap or it overflows instantly:  $g = \gcd(x, y)$ , then
    ↪  $x / g > LIM / y ? LIM :$ 
// -  $x / g * y$ . Capping keeps monotonicity, so the run structure
    ↪ survives.
// - Two-pointer twin: for or / and / gcd the aggregate is monotone
    ↪ in l, so "count subarrays with
// - op == k" also falls to two pointers plus a sliding structure,
    ↪ but that needs undo; this does
// - not and generalises to any associative monotone op in the same
    ↪ 10 lines.
// - The same list carried from the LEFT (fix l, grow r) works
    ↪ identically; pick the end your
// - query is indexed by.

```

1.29 MexOfAllSubarrays

MEX OF EVERY SUBARRAY - which values occur as $\text{mex}(a[l..r])$ over all $O(n^2)$ subarrays, $O(n \log n)$.

```

// Input a[0..n-1] with a[i] >= 0. Output ok[0..n], ok[m] == 1 iff
    ↪ some NON-EMPTY subarray has mex
// exactly m. Nothing above n is reachable: mex m needs m distinct
    ↪ values, so  $m \leq n$ .
// WHY IT IS ONLY  $O(n \log n)$ : m is reachable iff some MAXIMAL m-free
    ↪ block contains all of  $0..m-1$ .
// If a block avoids m and holds  $0..m-1$  its mex is exactly m; if the
    ↪ maximal block misses some
// value < m then so does every sub-block of it, so testing the
    ↪ maximal blocks alone is complete.
// The blocks for m are the gaps between consecutive occurrences of
    ↪ m, so over all m there are only
//  $\text{sum}(\text{cnt}[m] + 1) \leq 2n + 1$  blocks to test. Each test is one
    ↪ range-mex query, answered offline
// sorted by right end: keep last[v] = the last position of v seen so
    ↪ far in a MIN segment tree over
// v, and  $\text{mex}(a[L..R])$  is the leftmost v with  $\text{last}[v] < L$  - one
    ↪ descend, go left iff the left
// child's min is < L. That descend is the reusable half of this file.
// WHEN: Codeforces 1436E ("mex of the set of all subarray mexes"),
    ↪ "is m achievable", and any
// range-mex problem - the tree below answers those directly.
// COMPLEXITY:  $O(n \log n)$  time,  $O(n)$  memory (4n tree nodes plus  $\leq 2n$ 
    ↪ + 1 buffered queries).
// DEGENERATE: n == 0 returns {0} (no subarray, nothing reachable).
    ↪ All-zero returns only ok[1].
// All-distinct  $0..n-1$  returns every ok[0..n]. Duplicates are fine.
    ↪ Values > n are ignored - they
// can neither BE a mex (too big) nor block one - but they still
    ↪ occupy a position and split
// nothing, which is correct. Negative values are undefined input,
    ↪ mex assumes non-negative.
// EXACTNESS: pure integer index arithmetic, no overflow anywhere
    ↪ (positions fit int).
struct MexTree {                                // min, over a
    ↪ range of VALUES, of "last seen at"
    int V; vector<int> t;                        // values 0..V; -1
    ↪ = never seen
    MexTree(int V) : V(V), t(4 * (V + 1), -1) {}
    void see(int v, int p, int x, int l, int r) {
        if (l == r) { t[x] = p; return; }
        int m = (l + r) / 2;
        v <= m ? see(v, p, 2 * x, l, m) : see(v, p, 2 * x + 1, m + 1,
    ↪ r);
        t[x] = min(t[2 * x], t[2 * x + 1]);
    }
    void see(int v, int p) { if (v <= V) see(v, p, 1, 0, V); } //
    ↪ v > V cannot affect a mex <= V
    int mex(int L, int x, int l, int r) {                //
    ↪ leftmost v with last[v] < L
        if (l == r) return l;                          // a
    ↪ leaf always exists: [L..R] has
        int m = (l + r) / 2;                            //
    ↪ <= V+1 elements so some value
        return t[2 * x] < L ? mex(L, 2 * x, l, m)       //
    ↪ in 0..V must be missing
                                : mex(L, 2 * x + 1, m + 1, r);
    }
    int mex(int L) { return mex(L, 1, 0, V); }          // ==
    ↪ mex(a[L..R]), R = last position fed to see
};
vector<char> allSubarrayMex(const vector<int>& a) {
    int n = sz(a);
    vector<char> ok(n + 1, 0);
    if (!n) return ok;
    vector<vi> pos(n + 1);                            //
    ↪ pos[m] = positions holding m
    for (int i = 0; i < n; i++) if (a[i] <= n) pos[a[i]].pb(i);
    vector<vector<pii>> qs(n);                          //
    ↪ at R: (L, m), "is mex[L..R] m?"
    for (int m = 0; m <= n; m++) {                    //
    ↪ the maximal m-free blocks
        int lo = 0;
        for (int p : pos[m]) { if (lo < p) qs[p - 1].pb({lo, m}); lo
    ↪ = p + 1; }
        if (lo < n) qs[n - 1].pb({lo, m});
    }
    MexTree T(n);
    for (int R = 0; R < n; R++) {
        T.see(a[R], R);
        for (auto& [L, m] : qs[R]) if (T.mex(L) == m) ok[m] = 1; //
    ↪ block avoids m, so mex <= m
    }
    return ok;
}

```

```

}
// CF1436E wants the mex OF THAT SET: int r = 0; while (r <= n &&
    ↳ ok[r]) r++;
// VARIANTS
// - RANGE MEX, offline: drop the block machinery, bucket the real
    ↳ queries by R and run the same
// - sweep - T.mex(L) after feeding a[0..R] is mex(a[L..R]). O((n +
    ↳ q) log n).
// - RANGE MEX, online: make the same "last occurrence" tree
    ↳ PERSISTENT, one version per R, and
// - descend in version R. O(log n) per query, O(n log n) memory.
    ↳ Merge-sort tree also works.
// - "smallest missing value >= x in a range": identical descend
    ↳ restricted to values [x, V].
// - MEX OF A MULTISSET under insert/erase: segment tree over counts,
    ↳ descend to the leftmost zero.
// - MEX OF THE WHOLE ARRAY is O(n) with a bucket array; never build
    ↳ a tree for that.
// - mex(a[l..r]) is non-increasing in l for fixed r, and
    ↳ non-decreasing in r for fixed l - that
// - monotonicity is what lets a two-pointer answer "shortest
    ↳ subarray with mex >= k".

```

1.30 XorEquationRange

COUNT THE SOLUTIONS OF $x_0 \text{ xor } x_1 \text{ xor } \dots \text{ xor } x_{n-1} == X$ with $0 \leq x_i \leq a[i]$, modulo M.

```

// Returns the count mod M. a[i] >= 0, X >= 0, M >= 1.
// WHEN: "how many tuples bounded componentwise xor to X" -
    ↳ Nim-position counting, POJ 3986, the
// - stone-game family. The bounds are ARBITRARY; if every a[i] were
    ↳  $2^k - 1$  the answer would just be
// -  $2^k(k(n-1))$  [ $X < 2^k$ ], and this file is what you type when they are
    ↳ not.
// HOW: process bit k from the top. Split on whether every  $x_i$  copies
    ↳ a[i]'s bit k ("all tight") or
// - at least one  $x_i$  drops a bit that a[i] has ("relaxed"). Relaxed is
    ↳ countable in one pass: the
// - dropper is then free over ALL k lower bits, so designate the
    ↳ LOWEST-indexed dropper as the
// - absorber - its lower bits are forced to whatever makes the xor
    ↳ come out right, factor 1 - and
// - every other variable contributes its own independent count: a[i] -
    ↳  $2^k + 1$  if it keeps bit k,
// -  $2^k$  if it drops it too, a[i] + 1 if a[i] has no bit k at all. A
    ↳ 4-state DP over
// - (parity of chosen k-bits, has anyone dropped yet) does that in
    ↳ O(n) per bit. The all-tight case
// - forces the k-th bit of the xor to the parity of  $\# \{i : a[i] \text{ has bit}$ 
    ↳  $k\}$ ; if that misses X's bit k
// - nothing survives, otherwise strip bit k everywhere and recurse to
    ↳ bit k-1. Bottoming out leaves
// - exactly the single tuple  $x_i == a[i]$ , hence the final +1.
// - COMPLEXITY: O(n log(max a)) time, O(1) extra memory. No recursion,
    ↳ no globals.
// - DEGENERATE: n == 0 returns [X == 0]. All a[i] == 0 returns [X ==
    ↳ 0]. X with a bit above every
// - a[i] returns 0. X == 0 always returns >= 1 (the all-zero tuple). M
    ↳ == 1 returns 0.
// - EXACTNESS: exact modulo M for any M up to  $9.2e18$  - every product
    ↳ goes through __int128. Bounds
// - a[i] up to  $9.2e18$  are fine (a[i] + 1 must not overflow, so keep
    ↳ a[i] <  $2^{63} - 1$ ).
ll xorEqCount(vector<ll> a, ll X, ll M) {
    auto mul = [&](ll p, ll q) { return (ll)((__int128)p * q % M); };
    int K = 0;
    for (ll v : a) while (v >> K) K++;
    ↳ every a[i] <  $2^K$ 
    if (X >> K) return 0;
    ↳ of values <  $2^K$  stays <  $2^K$ 
    ll res = 0;
    for (int k = K - 1; k >= 0; k--) {
        ll two = (1LL << k) % M, dp[2][2] = {{1 % M, 0}, {0, 0}};
        ↳ // dp[parity of bit k][dropped?]
        for (ll v : a) {
            ll nd[2][2] = {}, cnt = ((v & ~(1LL << k)) + 1) % M;
            ↳ v -  $2^k + 1$  if bit k is set,
            for (int p = 0; p < 2; p++) for (int f = 0; f < 2; f++) {
                ↳ // v + 1 if it is not: the
                ll c = dp[p][f];
                ↳ // SAME expression
                if (!c) continue;
                if (v >> k & 1) {
                    nd[p ^ 1][f] = (nd[p ^ 1][f] + mul(c, cnt)) % M;
                    ↳ // keep it:  $x_i$  in  $[2^k, v]$ 
                    nd[p][1] = (nd[p][1] + (f ? mul(c, two) : c)) %
                    ↳ M; // drop it:  $x_i$  in  $[0, 2^k)$ ,
                    } else nd[p][f] = (nd[p][f] + mul(c, cnt)) % M;
                    ↳ // and the FIRST dropper is
                }
                ↳ // the absorber: factor 1
                memcpy(dp, nd, sizeof dp);
            }
            res = (res + dp[X >> k & 1][1]) % M;
            ↳ at least one variable dropped
            int par = 0;
            for (ll& v : a) if (v >> k & 1) par ^= 1, v ^= 1LL << k;
            ↳ // strip, stay all-tight
            if (par != (int)(X >> k & 1)) return res;
            ↳ // all-tight branch dies
        }
        return (res + 1) % M;
        ↳ surviving tuple is  $x_i == a[i]$ 
    }
    // VARIANTS
    // - LOWER BOUNDS TOO (lo_i <= x_i <= hi_i): you cannot shift, xor
    ↳ is not linear under +. Split
    // - each range into O(log) aligned blocks "fixed prefix + j free
    ↳ bits"; a block with j >= 1 free
    // - bits plays the role of a dropper (one absorber, the rest
    ↳ contribute  $2^j$ ), so the same count

```

```
// works over the  $O(n \log A)$  blocks - at the cost of a product
//   ↳ over per-variable block choices.
// - AND / OR instead of xor: per bit the constraint decouples
//   ↳ completely, so it is a plain
//   per-bit product, no absorber trick needed.
// - Sum over all  $X$  of the answer is  $\text{prod}(a[i] + 1)$  - a free sanity
//   ↳ check on the implementation.
// - CONSTRUCTING one solution: run the same DP keeping the counts,
//   ↳ then walk it back greedily.
// - If all  $a[i]$  are equal to  $2^k - 1$  the answer is  $2^k(k(n-1))$  for  $X$ 
//   ↳  $< 2^k$ , 0 otherwise.
```

1.31 SubsetUnionOfBitsets

SUBSET UNION SIZES - given $m \leq \sim 20$ sets over the ground set $\{0..n-1\}$, get $|\text{union of the chosen}$

```
// sets| for EVERY one of the  $2^m$  subsets, in  $O(n + 2^m * m)$ . Output
//   ↳  $u[S]$  for every mask  $S$  over the
//  $m$  sets,  $u[S]$  = number of ground elements covered by at least one
//   ↳ set of  $S$ ;  $u[0] == 0$ .
// THE TRICK IS THE TRANSPOSE, everything else is bookkeeping. You
//   ↳ are handed "set  $j$  contains these
// elements"; flip it to  $\text{own}[e]$  = the mask of sets that contain
//   ↳ element  $e$ . Then element  $e$  is MISSED
// by  $S$  iff  $S$  is disjoint from  $\text{own}[e]$  iff  $S$  is a SUBSET of  $\sim \text{own}[e]$ .
//   ↳ So drop one token at  $\sim \text{own}[e]$  for
// each  $e$  and run a SUPERSET-sum (SOS) transform:  $\text{cnt}[S] = \#\{e :$ 
//   ↳  $\sim \text{own}[e]$  contains  $S\}$  = number of
// elements  $S$  misses, and the union size is  $n - \text{cnt}[S]$ . Counting the
//   ↳ complements is what turns a
// union (no closed form under subsets) into a plain subset-lattice
//   ↳ sum.
// WHEN: "cheapest set of towers covering  $\geq k$  houses", "for each
//   ↳ mask, how much does it cover",
// any  $2^m$  enumeration whose inner cost is a union - it removes the
//   ↳  $O(n / 64)$  bitset OR per mask.
// COMPLEXITY:  $O(n + 2^m * m)$  time,  $2^m$  ints = 4 MB at  $m = 20$  (plus 8
//   ↳ MB if you also build costs).
// DEGENERATE:  $m == 0$  gives the single answer  $u[0] == 0$ .  $n == 0$  gives
//   ↳ all zeros. Duplicate or empty
// sets are fine. An element in no set has  $\text{own}[e] == 0$ , is missed by
//   ↳ every  $S$ , and is never counted.
// EXACTNESS: pure integer counting, no overflow ( $u$  fits int as long
//   ↳ as  $n$  does). Weighted version:
// replace the  $++$  by  $+= w[e]$  and use  $ll$  - then the budget is  $\text{sum}|w|$ 
//   ↳  $\leq 9e18$ .
vi transposeSets(int n, const vector<vi>& sets) { // sets[j]
//   ↳ = the elements of set  $j$ 
    vi own(n, 0); // own[e]
//   ↳ = mask of sets holding  $e$ 
    for (int j = 0; j < sz(sets); j++) for (int e : sets[j]) own[e]
//   ↳ |= 1 << j;
    return own;
}
vi subsetUnionSize(int n, int m, const vi& own) {
    int full = (1 << m) - 1;
    vi u(1 << m, 0);
    for (int e = 0; e < n; e++) u[own[e] ^ full]++; // bucket
//   ↳  $e$  at "the sets that MISS  $e$ "
    for (int b = 0; b < m; b++) // SOS
//   ↳ over SUPERSETS: pull from  $S | \text{bit}$ 
        for (int S = 0; S <= full; S++) // into  $S$ .
//   ↳ Entries with bit  $b$  set are not
            if (!(S >> b & 1)) u[S] += u[S | 1 << b]; // touched
//   ↳ in this pass, so order is free
        for (int S = 0; S <= full; S++) u[S] = n - u[S]; // missed
//   ↳ -> covered
    return u;
}
// --- THE DUAL CONSUMER: cheapest subset covering AT LEAST  $k$ 
//   ↳ elements, for every  $k$ .  $O(2^m * m)$ .
vector<ll> minCostToCover(int n, int m, const vi& own, const
//   ↳ vector<ll>& cost) {
    vi u = subsetUnionSize(n, m, own);
    int full = (1 << m) - 1;
    vector<ll> c(1 << m, best(n + 1, llinf));
    for (int S = 1; S <= full; S++) c[S] = c[S ^ (1 << lsb(S))] +
//   ↳ cost[lsb(S)]; // cost by lowest
    for (int S = 0; S <= full; S++) best[u[S]] = min(best[u[S]],
//   ↳ c[S]); // set bit
    for (int k = n - 1; k >= 0; k--) best[k] = min(best[k], best[k +
//   ↳ 1]); // exactly -> at least
    return best; // best[k]
//   ↳ == llinf:  $k$  is unreachable
}
// VARIANTS
// - INTERSECTION SIZES for every  $S$ : bucket at  $\text{own}[e]$  instead of
//   ↳  $\sim \text{own}[e]$  and run the SAME superset
// SOS -  $\#\{e : \text{own}[e] \text{ contains } S\}$  is exactly  $|\text{intersection of } S|$ .
//   ↳ One character of difference.
// - "covered by EXACTLY the sets in  $S$ ": the raw histogram of
//   ↳  $\text{own}[e]$ , no transform at all.
// - MAX / MIN over the covered set instead of a count: SOS with max
//   ↳ / min in place of  $+$ .
// - SUBSET sum (Zeta over subsets) is the mirror: if  $(S >> b \& 1)$ 
//   ↳  $f[S] += f[S \wedge 1 << b]$ . Mobius
// (the inverse) is the same loop with  $-=$ , which recovers
//   ↳ "exactly" counts from "at least".
// -  $m > 20$ : meet in the middle over two halves of the sets, or drop
//   ↳ to a greedy / randomised
```



```
// cover - 2^m unions is the whole budget and it is 1e6 masks at m
//   ↪ = 20.
// - The ground set may be huge: only n and 2^m matter, never n * m
//   ↪ or n / 64 words per mask.
```

1.32 NegativeBase

NEGATIVE-BASE ARITHMETIC. `toNegBase(n, b)` writes ANY integer `n`, either sign, in base `-b` with

```
// digits in [0, b): n == sum d[i] * (-b)^i, digits LEAST SIGNIFICANT
//   ↪ FIRST, no leading zero, and
// zero is the single digit {0}. Every integer has exactly one such
//   ↪ representation and there is no
// sign anywhere - that is the entire point of the base. fromNegBase
//   ↪ inverts it.
// WHEN: "print n in base -2", "add two negabinary strings", puzzle
//   ↪ problems where a value must be
// encoded without a sign, and the folklore fact that negabinary
//   ↪ addition needs a NEGATIVE carry.
// THE DIVISION RULE: d = n mod b forced into [0, b), then n = (n -
//   ↪ d) / (-b), an EXACT division
// since b divides n - d. Do not write n /= -b and patch d
//   ↪ afterwards: C++ truncates toward zero,
// so after d += b the quotient also needs n += 1, and that is the
//   ↪ step everyone forgets.
// WHY IT TERMINATES: |n - d| <= |n| + b - 1, so the new |n| is at
//   ↪ most |n|/b + (b-1)/b, which is
// strictly smaller than |n| whenever |n| >= 2; from |n| == 1 it
//   ↪ reaches 0 in at most two steps.
// The digit count is O(log_b |n|) + 1, and the sign of n alternates
//   ↪ as you descend.
// ADDITION digit by digit: s = x[i] + y[i] + carry lands in [-b, 2b
//   ↪ - 1]; force the digit into
// [0, b) by moving b, and since the place value of the next digit is
//   ↪ -b, moving +b DOWN here means
// carrying -1 UP. A leftover carry of +1 appends the digit 1; a
//   ↪ leftover of -1 appends TWO digits,
// b-1 then 1, because -1 == (b-1) + 1 * (-b). The same loop does
//   ↪ subtraction (pass sgn = -1): the
// range of s only widens to [-b, b], still inside what the one carry
//   ↪ line handles.
// COMPLEXITY: O(log_b |n|) per conversion, O(len) per add.
//   ↪ DEGENERATE: n == 0 gives {0}; adding to
// or from {0} works; results are stripped of leading zeros so
//   ↪ equality is plain vector ==.
// EXACTNESS: exact integers. Budget: |n| <= 9.2e18 minus b for
//   ↪ toNegBase (n - d must not overflow,
// so LLONG_MIN is out), and fromNegBase must be told digits whose
//   ↪ value fits in ll.
vi toNegBase(ll n, int b) {                                     // b >= 2; the
//   ↪ base used is -b
    vi d;
    while (n) {
        int r = (int)(n % b);
        if (r < 0) r += b;                                     // C++ % keeps the
//   ↪ sign of n; digits must be >= 0
        d.pb(r);
        n = (n - r) / -b;                                       // exact division,
//   ↪ never a truncation
    }
    return d.empty() ? vi{0} : d;
}
ll fromNegBase(const vi& d, int b) {
    ll v = 0;
    for (int i = sz(d) - 1; i >= 0; i--) v = v * -b + d[i];    //
//   ↪ Horner, top digit down
    return v;
}
vi negAdd(vi x, const vi& y, int b, int sgn = 1) {            // x
//   ↪ + sgn * y, sgn is 1 or -1
    x.resize(max(sz(x), sz(y)), 0);
    int c = 0;
    for (int i = 0; i < sz(x); i++) {
        int s = x[i] + (i < sz(y) ? sgn * y[i] : 0) + c;
        c = s < 0 ? 1 : (s >= b ? -1 : 0);                    // next place is
//   ↪ worth -b, so shifting the digit
        x[i] = s + b * c;                                       // by -b here is a
//   ↪ carry of +1, and vice versa
    }
    if (c == 1) x.pb(1);
    else if (c == -1) x.pb(b - 1), x.pb(1);                    // -1 == (b-1) * 1
//   ↪ + 1 * (-b)
    while (sz(x) > 1 && !x.back()) x.pop_back();              // normalise, so
//   ↪ == is equality of values
    return x;
}
// VARIANTS
// - NEGATE: negAdd({0}, x, b, -1). Do not hunt for a bit pattern,
//   ↪ subtraction already does it.
// - SIGN: a normalised nonzero number is NEGATIVE iff its digit
//   ↪ count is even, i.e. the leading
// digit sits on an odd power of -b. Compare x and y by the sign
//   ↪ of negAdd(x, y, b, -1).
```

```
// - MULTIPLY BY -b: prepend a zero digit (shift). Divide by -b:
    ↳ drop d[0].
// - MULTIPLY: schoolbook into a wide accumulator, then normalise
    ↳ from the bottom with the same
// rule - while (d[i] < 0) d[i] += b, d[i+1] += 1; while (d[i] >=
    ↳ b) d[i] -= b, d[i+1] -= 1.
// - NEGABINARY (b == 2) IN ONE LINE, no loop: the digits of n as a
    ↳ bit pattern are
// (n + 0xAAAAAAAAAAAAAAAAuLL) ^ 0xAAAAAAAAAAAAAAAAuLL, bit i
    ↳ being d[i] - the mask is the whole
// carry cascade done in parallel. All 64 digits must fit, which
    ↳ covers every negative ll but
// only n <= (4^32 - 1) / 3 = 6148914691236517205 upward; past
    ↳ that it silently truncates.
// - BASE -1+i represents every Gaussian integer with digits {0, 1}:
    ↳ take d = (re + im) & 1, then
// (re, im) -> ((im - re + d) / 2, -(re + im - d) / 2). Same loop,
    ↳ same termination argument.
```

1.33 TestGenerators

RANDOM TEST GENERATORS for stress testing. 12 - StressTest.cpp is the harness; this is the

```
// library it feeds on. Writing a generator from scratch under
    ↳ pressure is where stress testing
// usually dies, so keep these on the page.
// SEED FROM argv SO A FAILING CASE IS REPRODUCIBLE:
// int main(int argc, char** argv) { if (argc > 1)
    ↳ rng.seed(stoull(argv[1])); ... }
// then the driver loop passes the iteration number and you can
    ↳ replay any failure exactly.
ll rnd(ll l, ll r) { return uniform_int_distribution<ll>(l, r)(rng);
    ↳ } // inclusive
bool coin(int pct = 50) { return rnd(1, 100) <= pct; }
vector<ll> randArray(int n, ll lo, ll hi) {
    vector<ll> a(n);
    for (ll& x : a) x = rnd(lo, hi);
    return a;
}
vector<int> randPermutation(int n) { //
    ↳ 0-indexed
    vector<int> p(n);
    iota(all(p), 0);
    shuffle(all(p), rng);
    return p;
}
string randString(int n, char lo = 'a', char hi = 'z') {
    string s;
    while (n-->) s += (char)rnd(lo, hi);
    return s;
}
vector<string> randGrid(int n, int m, const string& alpha = ".#") {
    vector<string> g(n, string(m, '.'));
    for (auto& row : g) for (char& c : row) c = alpha[rnd(0,
    ↳ alpha.size() - 1)];
    return g;
}
// --- TREES. All 1-indexed, n-1 edges, always connected. ---
vector<pair<int, int>> randTree(int n) { //
    ↳ uniform-ish, shuffled labels
    vector<int> lab = randPermutation(n);
    vector<pair<int, int>> e;
    for (int i = 1; i < n; i++) e.push_back({lab[rnd(0, i - 1)] + 1,
    ↳ lab[i] + 1});
    return e;
}
vector<pair<int, int>> pathTree(int n) { // the
    ↳ DEEP case: recursion, HLD
    vector<pair<int, int>> e;
    for (int i = 2; i <= n; i++) e.push_back({i - 1, i});
    return e;
}
vector<pair<int, int>> starTree(int n) { // the
    ↳ WIDE case: heavy-child logic
    vector<pair<int, int>> e;
    int c = rnd(1, n);
    for (int i = 1; i <= n; i++) if (i != c) e.push_back({c, i});
    return e;
}
vector<pair<int, int>> caterpillar(int n) { // a
    ↳ spine with legs: breaks naive
    vector<pair<int, int>> e; //
    ↳ centroid and diameter code
    int spine = max(1LL, (ll)rnd(1, n));
    for (int i = 2; i <= spine; i++) e.push_back({i - 1, i});
    for (int i = spine + 1; i <= n; i++) e.push_back({(int)rnd(1,
    ↳ spine), i});
    return e;
}
// --- GRAPHS. m edges, no self loops, no duplicates unless you ask
    ↳ for them. ---
vector<pair<int, int>> randGraph(int n, int m) {
    vector<pair<int, int>> all;
    for (int i = 1; i <= n; i++) for (int j = i + 1; j <= n; j++)
    ↳ all.push_back({i, j});
    shuffle(all(all), rng);
    all.resize(min((size_t)m, all.size()));
    return all;
}
vector<pair<int, int>> connectedGraph(int n, int m) { // a
    ↳ random tree plus extra edges
    auto e = randTree(n);
    set<pair<int, int>> used;
    for (auto [u, v] : e) used.insert({min(u, v), max(u, v)});
    m = min<ll>(m, (ll)n * (n - 1) / 2);
    while ((int)e.size() < m) {
        int u = rnd(1, n), v = rnd(1, n);
```

```

    if (u == v) continue;
    if (u > v) swap(u, v);
    if (used.insert({u, v}).second) e.push_back({u, v});
}
shuffle(all(e), rng);
return e;
}
vector<pair<int, int>> randDAG(int n, int m) {
    //
    ⇨ acyclic by a random topo order
    vector<int> p = randPermutation(n);
    vector<pair<int, int>> all;
    for (int i = 0; i < n; i++) for (int j = i + 1; j < n; j++)
    ⇨ all.push_back({p[i] + 1, p[j] + 1});
    shuffle(all, rng);
    all.resize(min((size_t)m, all.size()));
    return all;
}
vector<pair<int, int>> randQueries(int q, int n) {
    //
    ⇨ random inclusive [l, r]
    vector<pair<int, int>> v;
    while (q--) {
        int l = rnd(1, n), r = rnd(1, n);
        if (l > r) swap(l, r);
        v.push_back({l, r});
    }
    return v;
}
/* THE CASES A UNIFORM GENERATOR NEVER PRODUCES - and which is
    ⇨ exactly where the bug is
    n = 1, n = 2, and an EMPTY structure
    ⇨ (off-by-one, .back() on empty)
    ALL VALUES EQUAL, all zero, all negative
    ⇨ (tie-breaks, strict vs non-strict)
    values at the EXTREMES of the stated range (overflow)
    a SORTED or REVERSE-SORTED array
    ⇨ (quicksort, two pointers, hulls)
    a PATH tree and a STAR tree
    ⇨ (recursion depth, heavy-child logic)
    a DISCONNECTED graph, an isolated vertex, a self loop, a DOUBLED
    ⇨ edge (bridges, DSU, flows)
    COLLINEAR / DUPLICATE points, a degenerate polygon (all of
    ⇨ geometry)
    Generate with a SMALL value range (say [1,3]) most of the time:
    ⇨ collisions and ties are what
    break code, and a uniform draw from [1, 1e9] never creates one.
    SHRINK a failure before reading it: re-run the generator with
    ⇨ progressively smaller bounds
    until the counterexample fits on one line, then stare at that. */

```

2 Data Structures (DS)

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2.1 Segment Tree

2.1.1 SegmentTree

SEGMENT TREE, point update + range query, $O(\log n)$ each. Swap the merge in SEG to change the

// operation (sum/min/max/gcd/matrix); any associative monoid works.

⇨ Size is rounded up to a power

// of two so the tree is perfect and indices are simple.

```

struct SEG {
    ll sum = 0;
    SEG() {}
    SEG(ll x){
        sum = x;
    }
};

struct segTree {
    vector<SEG> seg;
    int sz = 1, n;
    segTree(int nn){
        n = nn;
        while (sz < nn)
            sz *= 2;
        seg.assign(2 * sz, SEG());
    }

    SEG merge(SEG seg1, SEG seg2) {
        SEG ret;
        ret.sum = seg1.sum + seg2.sum;
        return ret;
    }

    void build(vector<int> &v, int x, int lx, int rx) {
        if (lx == rx) {
            seg[x] = SEG(v[lx]);
            return;
        }
        int mid = (lx + rx) / 2;
        int lf = 2 * x + 1, rt = 2 * x + 2;

        build(v, lf, lx, mid);
        build(v, rt, mid + 1, rx);
        seg[x] = merge(seg[lf], seg[rt]);
    }

    void build(vector<int> &v){
        build(v, 0, 0, n-1);
    }

    void update(int i, ll val, int x, int lx, int rx) {
        if (lx == rx) {
            seg[x] = SEG(val);
            return;
        }
        int mid = (lx + rx) / 2, lf = 2*x+1, rt = 2*x+2;
        if (i <= mid)
            update(i, val, lf, lx, mid);
        else
            update(i, val, rt, mid + 1, rx);
        seg[x] = merge(seg[lf], seg[rt]);
    }

    void update(int i, ll val) {
        update(i, val, 0, 0, n-1);
    }

    SEG query(int l, int r, int x, int lx, int rx) {
        if (l <= lx && r >= rx)
            return seg[x];
        if (l > rx || r < lx)
            return SEG();

        int mid = (lx + rx) / 2, lf = 2 * x + 1, rt = 2*x+2;

        return merge(query(l, r, lf, lx, mid), query(l, r, rt, mid +
        ⇨ 1, rx));
    }

    SEG query(int l, int r) {
        return query(l, r, 0, 0, n-1);
    }
};

```

2.1.2 LazySegmentTree

Lazy Segment Tree (range add, range sum) - build $O(n)$, update/query $O(\log n)$

```
// 1-based node indices, 0-based positions, ALL recursions on [0,
//  ⇨ n-1].
// To change the op: edit merge(), apply() and the identity in qry().
struct LazySeg {
    int n;
    vector<ll> t, lz;

    LazySeg(int n = 0) : n(n), t(4 * n, 0), lz(4 * n, 0) {}

    ll merge(ll a, ll b) { return a + b; }
    void apply(int v, int l, int r, ll x) { t[v] += x * (r - l + 1);
    ⇨ lz[v] += x; }

    void push(int v, int l, int r) {
        if (!lz[v]) return;
        int m = (l + r) / 2;
        apply(2 * v, l, m, lz[v]);
        apply(2 * v + 1, m + 1, r, lz[v]);
        lz[v] = 0;
    }

    void build(const vector<ll>& a, int v, int l, int r) {
        if (l == r) { t[v] = a[l]; return; }
        int m = (l + r) / 2;
        build(a, 2 * v, l, m), build(a, 2 * v + 1, m + 1, r);
        t[v] = merge(t[2 * v], t[2 * v + 1]);
    }

    void build(const vector<ll>& a) { if (n) build(a, 1, 0, n - 1); }

    void upd(int ql, int qr, ll x, int v, int l, int r) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr) return apply(v, l, r, x);
        push(v, l, r);
        int m = (l + r) / 2;
        upd(ql, qr, x, 2 * v, l, m), upd(ql, qr, x, 2 * v + 1, m + 1,
    ⇨ r);
        t[v] = merge(t[2 * v], t[2 * v + 1]);
    }

    void upd(int l, int r, ll x) { upd(l, r, x, 1, 0, n - 1); }

    ll qry(int ql, int qr, int v, int l, int r) {
        if (qr < l || r < ql) return 0; // identity
        if (ql <= l && r <= qr) return t[v];
        push(v, l, r);
        int m = (l + r) / 2;
        return merge(qry(ql, qr, 2 * v, l, m), qry(ql, qr, 2 * v + 1,
    ⇨ m + 1, r));
    }

    ll qry(int l, int r) { return qry(l, r, 1, 0, n - 1); }
};
```

2.1.3 PersistentSegmentTree

PERSISTENT SEGMENT TREE - every update returns a NEW root and old versions stay queryable.

// $O(\log n)$ time and $O(\log n)$ new nodes per update. The standard use
 ⇨ is offline k-th smallest in a
 // range (build a version per prefix, then query version[r] minus
 ⇨ version[l-1]).

```
struct Node {
    Node *l, *r;
    ll val = 0;

    Node(ll val) : l(NULL), r(NULL), val(val) {}

    Node() : l(NULL), r(NULL) {}

    Node(Node* l, Node* r) : l(l), r(r) {
        if (l != NULL) val += l->val;
        if (r != NULL) val += r->val;
    }

    void addChild() {
        l = new Node();
        r = new Node();
    }
};

struct PST {
    int n;

    PST(int n) : n(n + 1) {}

    Node merge(Node x, Node y) {
        Node ret;
        ret.val = x.val + y.val;
        return ret;
    }

    Node* update(Node* v, int i, ll val, int lx, int rx) {
        if (lx == rx) return new Node(v->val + val);
        int mid = (lx + rx) / 2;
        if (!v->l) v->addChild();
        if (i <= mid) return new Node(update(v->l, i, val, lx, mid),
    ⇨ v->r);
        return new Node(v->l, update(v->r, i, val, mid + 1, rx));
    }

    Node* update(Node* v, int i, ll val) { return update(v, i, val,
    ⇨ 0, n - 1); }

    Node query(Node* v, int l, int r, int lx, int rx) {
        if (l > rx || r < lx) return {};
        if (l <= lx && r >= rx) return *v;
        if (!v->l) v->addChild();
        int mid = (lx + rx) / 2;
        return merge(query(v->l, l, r, lx, mid), query(v->r, l, r,
    ⇨ mid + 1, rx));
    }

    Node query(Node* v, int l, int r) { return query(v, l, r, 0, n -
    ⇨ 1); }

    int getKth(Node* a, Node* b, int k, int lx, int rx) {
        if (lx == rx) return lx;
        if (!a->l) a->addChild();
        if (!b->l) b->addChild();
        int rem = b->l->val - a->l->val;
        int mid = (lx + rx) / 2;
        if (rem >= k) return getKth(a->l, b->l, k, lx, mid);
        return getKth(a->r, b->r, k - rem, mid + 1, rx);
    }

    int getKth(Node* a, Node* b, int k) { return getKth(a, b, k, 0, n
    ⇨ - 1); }
};
```

2.1.4 SegmentTreeBeats

Rename one if you ever paste two of them together.

// NOTE: several segment-tree files here declare their own 'struct
↪ Node'.

// Rename one if you ever paste two of them together.

```
struct Node {
    ll mn;
    int mnCnt;
    ll secondMn;

    ll mx;
    int mxCnt;
    ll secondMx;

    ll sum;

    ll lazyAdd;
    ll lazyEqual;
    bool isLazyEqual;

    Node() {
        mx = -inf;
        secondMx = -inf;
        mxCnt = 0;

        mn = inf;
        secondMn = inf;
        mnCnt = 0;

        sum = 0;

        lazyAdd = 0;
        lazyEqual = 0;
        isLazyEqual = 0;
    };

    Node(int x) {
        mx = x;
        secondMx = -inf;
        mxCnt = 1;

        mn = x;
        secondMn = inf;
        mnCnt = 1;

        sum = x;

        lazyAdd = 0;
        lazyEqual = 0;
        isLazyEqual = 0;
    }
};

struct SegTree {
    int tree_size;
    vector<Node> seg_data;

    SegTree(int n) {
        tree_size = 1;
        while (tree_size < n) tree_size *= 2;
        seg_data.resize(2 * tree_size, Node());
    }

    Node merge(Node& lf, Node& ri) {
        Node ans = Node();

        ans.sum = lf.sum + ri.sum;

        ans.mx = max(lf.mx, ri.mx);
        ans.secondMx = max(lf.secondMx, ri.secondMx);
        ans.mxCnt = 0;
        if (lf.mx == ans.mx) ans.mxCnt += lf.mxCnt;
        else ans.secondMx = max(ans.secondMx, lf.mx);
        if (ri.mx == ans.mx) ans.mxCnt += ri.mxCnt;
        else ans.secondMx = max(ans.secondMx, ri.mx);

        ans.mn = min(lf.mn, ri.mn);
        ans.secondMn = min(lf.secondMn, ri.secondMn);
        ans.mnCnt = 0;
        if (lf.mn == ans.mn) ans.mnCnt += lf.mnCnt;
        else ans.secondMn = min(ans.secondMn, lf.mn);
        if (ri.mn == ans.mn) ans.mnCnt += ri.mnCnt;
        else ans.secondMn = min(ans.secondMn, ri.mn);

        return ans;
    }
};
```

```
void init(vector<int>& nums, int ni, int lx, int rx) {
    if (rx - lx == 1) {
        if (lx < nums.size())
            seg_data[ni] = Node(nums[lx]);
        return;
    }

    int mid = lx + (rx - lx) / 2;
    init(nums, 2 * ni + 1, lx, mid);
    init(nums, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
↪ 2]);
}

void init(vector<int>& nums) {
    init(nums, 0, 0, tree_size);
}

void doPushEq(int ni, int lx, int rx, int x) {
    seg_data[ni].mn = seg_data[ni].mx = seg_data[ni].lazyEqual =
↪ x;
    seg_data[ni].isLazyEqual = 1;
    seg_data[ni].mnCnt = seg_data[ni].mxCnt = rx - lx;
    seg_data[ni].secondMx = -inf;
    seg_data[ni].secondMn = inf;

    seg_data[ni].sum = (ll)x * (rx - lx);
    seg_data[ni].lazyAdd = 0;
}

void doPushMinEq(int ni, int lx, int rx, int x) {
    if (seg_data[ni].mn >= x) doPushEq(ni, lx, rx, x);
    else if (seg_data[ni].mx > x) {
        if (seg_data[ni].secondMn == seg_data[ni].mx)
            seg_data[ni].secondMn = x;
        seg_data[ni].sum -= (ll)(seg_data[ni].mx - x) *
↪ seg_data[ni].mxCnt;
        seg_data[ni].mx = x;
    }
}

void doPushMaxEq(int ni, int lx, int rx, int x) {
    if (seg_data[ni].mx <= x) doPushEq(ni, lx, rx, x);
    else if (seg_data[ni].mn < x) {
        if (seg_data[ni].secondMx == seg_data[ni].mn)
            seg_data[ni].secondMx = x;

        seg_data[ni].sum += (ll)(x - seg_data[ni].mn) *
↪ seg_data[ni].mnCnt;
        seg_data[ni].mn = x;
    }
}

void doPushSum(int ni, int lx, int rx, int x) {
    if (seg_data[ni].mn == seg_data[ni].mx)
        doPushEq(ni, lx, rx, seg_data[ni].mn + x);
    else {
        seg_data[ni].mx += x;
        if (seg_data[ni].secondMx != -inf)
            seg_data[ni].secondMx += x;

        seg_data[ni].mn += x;
        if (seg_data[ni].secondMn != inf)
            seg_data[ni].secondMn += x;

        seg_data[ni].sum += (ll)(rx - lx) * x;
        seg_data[ni].lazyAdd += x;
    }
}

void propagete(int ni, int lx, int rx) {
    if (rx - lx == 1) return;

    int mid = lx + (rx - lx) / 2;
    if (seg_data[ni].isLazyEqual) {
        doPushEq(2 * ni + 1, lx, mid, seg_data[ni].lazyEqual);
        doPushEq(2 * ni + 2, mid, rx, seg_data[ni].lazyEqual);
        seg_data[ni].lazyEqual = seg_data[ni].isLazyEqual = 0;
    }
    else {
        doPushSum(2 * ni + 1, lx, mid, seg_data[ni].lazyAdd);
        doPushSum(2 * ni + 2, mid, rx, seg_data[ni].lazyAdd);
        seg_data[ni].lazyAdd = 0;

        doPushMinEq(2 * ni + 1, lx, mid, seg_data[ni].mx);
        doPushMinEq(2 * ni + 2, mid, rx, seg_data[ni].mx);
    }
}
```

```

        doPushMaxEq(2 * ni + 1, lx, mid, seg_data[ni].mn);
        doPushMaxEq(2 * ni + 2, mid, rx, seg_data[ni].mn);
    }

void minimize(int l, int r, int x, int ni, int lx, int rx) {
    if (rx <= l || lx >= r || seg_data[ni].mx <= x)
        return;
    if (lx >= l && rx <= r && seg_data[ni].secondMx < x) {
        doPushMinEq(ni, lx, rx, x);
        return;
    }

    propagete(ni, lx, rx);

    int mid = lx + (rx - lx) / 2;
    minimize(l, r, x, 2 * ni + 1, lx, mid);
    minimize(l, r, x, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
↪ 2]);
}

void minimize(int l, int r, int x) {
    minimize(l, r, x, 0, 0, tree_size);
}

void maximize(int l, int r, int x, int ni, int lx, int rx) {
    if (rx <= l || lx >= r || seg_data[ni].mn >= x)
        return;
    if (lx >= l && rx <= r && seg_data[ni].secondMn > x) {
        doPushMaxEq(ni, lx, rx, x);
        return;
    }

    propagete(ni, lx, rx);

    int mid = lx + (rx - lx) / 2;
    maximize(l, r, x, 2 * ni + 1, lx, mid);
    maximize(l, r, x, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
↪ 2]);
}

void maximize(int l, int r, int x) {
    maximize(l, r, x, 0, 0, tree_size);
}

void add(int l, int r, int x, int ni, int lx, int rx) {
    if (rx <= l || lx >= r)
        return;
    if (lx >= l && rx <= r) {
        doPushSum(ni, lx, rx, x);
        return;
    }

    propagete(ni, lx, rx);

    int mid = lx + (rx - lx) / 2;
    add(l, r, x, 2 * ni + 1, lx, mid);
    add(l, r, x, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
↪ 2]);
}

void add(int l, int r, int x) {
    add(l, r, x, 0, 0, tree_size);
}

void set(int l, int r, int x, int ni, int lx, int rx) {
    if (rx <= l || lx >= r)
        return;
    if (lx >= l && rx <= r) {
        doPushEq(ni, lx, rx, x);
        return;
    }

    propagete(ni, lx, rx);

    int mid = lx + (rx - lx) / 2;
    set(l, r, x, 2 * ni + 1, lx, mid);
    set(l, r, x, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +

```

```

↪ 2]);
}

void set(int l, int r, int x) {
    set(l, r, x, 0, 0, tree_size);
}

Node get_range(int l, int r, int ni, int lx, int rx) {
    propagete(ni, lx, rx);

    if (lx >= l && rx <= r)
        return seg_data[ni];
    if (rx <= l || lx >= r)
        return Node();

    int mid = (lx + rx) / 2;

    Node lf = get_range(l, r, 2 * ni + 1, lx, mid);
    Node ri = get_range(l, r, 2 * ni + 2, mid, rx);
    return merge(lf, ri);
}

ll get_sum(int l, int r) {
    return get_range(l, r, 0, 0, tree_size).sum;
}

int get_mx(int l, int r) {
    return get_range(l, r, 0, 0, tree_size).mx;
}

int get_mn(int l, int r) {
    return get_range(l, r, 0, 0, tree_size).mn;
}
};

```


2.1.5 DynamicSegTreeDynamic (Implicit) Segment Tree - Space: $O(Q \log N)$ per query// Allocates nodes on demand for ranges up to $1e9$ or $1e18$

```

template<typename T = ll>
struct DynamicSegTree {
    struct Node {
        T val = 0;
        int lc = -1, rc = -1;
    };

    vector<Node> tree;
    ll L, R;

    DynamicSegTree(ll L = 0, ll R = 1e9) : L(L), R(R) {
        add_node();
    }

    int add_node() {
        tree.pb(Node());
        return tree.size() - 1;
    }

    void update(int node, ll l, ll r, ll pos, T val) {
        tree[node].val += val;
        if (l == r) return;
        ll mid = l + (r - l) / 2;
        if (pos <= mid) {
            if (tree[node].lc == -1) {
                int nxt = add_node();
                tree[node].lc = nxt;
            }
            update(tree[node].lc, l, mid, pos, val);
        } else {
            if (tree[node].rc == -1) {
                int nxt = add_node();
                tree[node].rc = nxt;
            }
            update(tree[node].rc, mid + 1, r, pos, val);
        }
    }

    void update(ll pos, T val) {
        update(0, L, R, pos, val);
    }

    T query(int node, ll l, ll r, ll ql, ll qr) {
        if (node == -1 || ql > r || qr < l) return 0;
        if (ql <= l && r <= qr) return tree[node].val;
        ll mid = l + (r - l) / 2;
        return query(tree[node].lc, l, mid, ql, qr) +
            query(tree[node].rc, mid + 1, r, ql, qr);
    }

    T query(ll ql, ll qr) {
        return query(0, L, R, ql, qr);
    }
};

```

2.1.6 MergeSortTreeMerge Sort Tree - Space: $O(N \log N)$, Build: $O(N \log N)$, Query: $O(\log^2 N)$ // Stores sorted elements in each node to answer 2D range count
→ queries

```

template<typename T = int>
struct MergeSortTree {
    int n;
    vector<vector<T>> tree;

    MergeSortTree(const vector<T>& a) : n(a.size()), tree(4 * n) {
        build(1, 0, n - 1, a);
    }

    void build(int node, int l, int r, const vector<T>& a) {
        if (l == r) {
            tree[node] = {a[l]};
            return;
        }
        int mid = (l + r) / 2;
        build(2 * node, l, mid, a);
        build(2 * node + 1, mid + 1, r, a);
        merge(all(tree[2 * node]), all(tree[2 * node + 1]),
            → back_inserter(tree[node]));
    }

    // Returns number of elements <= k in index range [ql, qr]
    int query_count_le(int node, int l, int r, int ql, int qr, T k) {
        if (ql > r || qr < l) return 0;
        if (ql <= l && r <= qr) {
            return upper_bound(all(tree[node]), k) -
                → tree[node].begin();
        }
        int mid = (l + r) / 2;
        return query_count_le(2 * node, l, mid, ql, qr, k) +
            query_count_le(2 * node + 1, mid + 1, r, ql, qr, k);
    }

    int query_count_le(int ql, int qr, T k) {
        return query_count_le(1, 0, n - 1, ql, qr, k);
    }
};

```

2.1.7 IterativeSegTree

Iterative (Non-Recursive) Segment Tree - Space: $O(N)$, Update: $O(\log N)$, Query: $O(\log N)$

```
// Fast, cache-friendly implementation for point update and range
    ↪ query
template<typename T = ll>
struct IterativeSegTree {
    int n;
    vector<T> tree;
    T neutral;
    function<T(T, T)> combine;

    IterativeSegTree(int n = 0, T neutral = 0, function<T(T, T)>
    ↪ combine = [](T a, T b) { return a + b; })
        : n(n), tree(2 * n, neutral), neutral(neutral),
    ↪ combine(combine) {}

    IterativeSegTree(const vector<T>& a, T neutral = 0, function<T(T,
    ↪ T)> combine = [](T a, T b) { return a + b; })
        : n(a.size()), tree(2 * a.size(), neutral), neutral(neutral),
    ↪ combine(combine) {
        for (int i = 0; i < n; i++) tree[n + i] = a[i];
        for (int i = n - 1; i > 0; i--) tree[i] = combine(tree[i <<
    ↪ 1], tree[i << 1 | 1]);
    }

    void update(int p, T val) {
        // combine(LEFT, RIGHT) - do NOT write combine(tree[p],
    ↪ tree[p ^ 1]): when p is the right
        // child that reverses the arguments, which is wrong for any
    ↪ non-commutative combine
        // (matrix product, affine composition, "merge two structs").
        for (tree[p += n] = val; p > 1; p >>= 1)
            tree[p >> 1] = combine(tree[p & ~1], tree[p | 1]);
    }

    // Range query on [l, r] inclusive
    T query(int l, int r) {
        T resl = neutral, resr = neutral;
        for (l += n, r += n + 1; l < r; l >>= 1, r >>= 1) {
            if (l & 1) resl = combine(resl, tree[l++]);
            if (r & 1) resr = combine(tree[--r], resr);
        }
        return combine(resl, resr);
    }
};
```

2.1.8 SegTree2D

2D Segment Tree - Space: $O(N * M)$, Update: $O(\log N * \log M)$, Query: $O(\log N * \log M)$

```
// Supports point updates and 2D subgrid range sum queries
template<typename T = ll>
struct SegTree2D {
    int n, m;
    vector<vector<T>> tree;

    SegTree2D(int n = 0, int m = 0) : n(n), m(m), tree(4 * n,
    ↪ vector<T>(4 * m, 0)) {}

    void update_y(int vx, int lx, int rx, int vy, int ly, int ry, int
    ↪ x, int y, T val) {
        if (lx == rx) {
            if (lx == rx) tree[vx][vy] += val;
            else tree[vx][vy] = tree[2 * vx][vy] + tree[2 * vx +
    ↪ 1][vy];
            return;
        }
        int my = (ly + ry) / 2;
        if (y <= my) update_y(vx, lx, rx, 2 * vy, ly, my, x, y, val);
        else update_y(vx, lx, rx, 2 * vy + 1, my + 1, ry, x, y, val);
        tree[vx][vy] = tree[vx][2 * vy] + tree[vx][2 * vy + 1];
    }

    void update_x(int vx, int lx, int rx, int x, int y, T val) {
        if (lx != rx) {
            int mx = (lx + rx) / 2;
            if (x <= mx) update_x(2 * vx, lx, mx, x, y, val);
            else update_x(2 * vx + 1, mx + 1, rx, x, y, val);
        }
        update_y(vx, lx, rx, 1, 0, m - 1, x, y, val);
    }

    void add(int x, int y, T val) {
        update_x(1, 0, n - 1, x, y, val);
    }

    T query_y(int vx, int vy, int ly, int ry, int qy1, int qy2) {
        if (qy1 > ry || qy2 < ly) return 0;
        if (qy1 <= ly && ry <= qy2) return tree[vx][vy];
        int my = (ly + ry) / 2;
        return query_y(vx, 2 * vy, ly, my, qy1, qy2) + query_y(vx, 2
    ↪ * vy + 1, my + 1, ry, qy1, qy2);
    }

    T query_x(int vx, int lx, int rx, int qx1, int qx2, int qy1, int
    ↪ qy2) {
        if (qx1 > rx || qx2 < lx) return 0;
        if (qx1 <= lx && rx <= qx2) return query_y(vx, 1, 0, m - 1,
    ↪ qy1, qy2);
        int mx = (lx + rx) / 2;
        return query_x(2 * vx, lx, mx, qx1, qx2, qy1, qy2) +
    ↪ query_x(2 * vx + 1, mx + 1, rx, qx1, qx2, qy1, qy2);
    }

    T query(int x1, int y1, int x2, int y2) {
        return query_x(1, 0, n - 1, x1, x2, y1, y2);
    }
};
```

2.1.9 MergeSortTreeFractional

MERGE SORT TREE WITH FRACTIONAL CASCADING - static "how many values $\leq x$ in $[l, r]$ " in $O(\log n)$

// instead of $O(\log^2 n)$: each node stores its sorted values plus
 ↳ pointers into the children, so the
 // binary search happens ONCE at the root and is then followed down
 ↳ for free.

```
const int MAXN = 30005;
int a[MAXN];
```

```
struct Node {
    vector<int> vals;
    vector<int> to_left;
    vector<int> to_right;
} tree[4 * MAXN];
```

```
void build(int node, int start, int end) {
    if (start == end) {
        tree[node].vals.push_back(a[start]);
        tree[node].to_left.assign(2, 0);
        tree[node].to_right.assign(2, 0);
        return;
    }

    int mid = (start + end) / 2;
    int left_child = 2 * node;
    int right_child = 2 * node + 1;

    build(left_child, start, mid);
    build(right_child, mid + 1, end);

    int left_sz = tree[left_child].vals.size();
    int right_sz = tree[right_child].vals.size();
    int total_sz = left_sz + right_sz;

    tree[node].vals.resize(total_sz);
    tree[node].to_left.resize(total_sz + 1);
    tree[node].to_right.resize(total_sz + 1);
```

```
    int p1 = 0, p2 = 0;

    for (int i = 0; i < total_sz; i++) {
        tree[node].to_left[i] = p1;
        tree[node].to_right[i] = p2;

        if (p1 < left_sz && (p2 == right_sz ||
        ↳ tree[left_child].vals[p1] <= tree[right_child].vals[p2])) {
            tree[node].vals[i] = tree[left_child].vals[p1++];
        } else {
            tree[node].vals[i] = tree[right_child].vals[p2++];
        }
    }

    tree[node].to_left[total_sz] = p1;
    tree[node].to_right[total_sz] = p2;
}

int query(int node, int start, int end, int l, int r, int idx) {
    if (l > end || r < start) {
        return 0;
    }

    if (l <= start && end <= r) {
        return idx;
    }

    int mid = (start + end) / 2;
    int next_idx_left = tree[node].to_left[idx];
    int next_idx_right = tree[node].to_right[idx];

    return query(2 * node, start, mid, l, r, next_idx_left) +
        query(2 * node + 1, mid + 1, end, l, r, next_idx_right);
}
```

2.1.10 RangeMajorityCandidates

```
const int MAXN = 200005; // array bound
↳ (Template.cpp calls this N)
int a[MAXN];
vector<int> pos[MAXN];
int K_LIMIT = 2;

struct Node {
    vector<pair<int, int>> cand;
};

Node tree_seg[4 * MAXN];

Node merge_nodes(const Node& left, const Node& right) {
    vector<pair<int, int>> combined;
    combined.reserve(left.cand.size() + right.cand.size());

    int i = 0, j = 0;
    while (i < left.cand.size() && j < right.cand.size()) {
        if (left.cand[i].first == right.cand[j].first) {
            combined.push_back({left.cand[i].first,
        ↳ left.cand[i].second + right.cand[j].second});
            i++; j++;
        } else if (left.cand[i].first < right.cand[j].first) {
            combined.push_back(left.cand[i++]);
        } else {
            combined.push_back(right.cand[j++]);
        }
    }
    while (i < left.cand.size()) combined.push_back(left.cand[i++]);
    while (j < right.cand.size())
        ↳ combined.push_back(right.cand[j++]);

    if (combined.size() <= K_LIMIT - 1) {
        return {combined};
    }

    vector<int> counts;
    counts.reserve(combined.size());
    for (const auto& p : combined) {
        counts.push_back(p.second);
    }

    nth_element(counts.begin(), counts.begin() + K_LIMIT - 1,
    ↳ counts.end(), greater<int>());
    int penalty = counts[K_LIMIT - 1];

    vector<pair<int, int>> survived;
    survived.reserve(K_LIMIT - 1);
    for (const auto& p : combined) {
        if (p.second > penalty) {
            survived.push_back({p.first, p.second - penalty});
        }
    }

    return {survived};
}

void build(int node, int start, int end) {
    if (start == end) {
        tree_seg[node].cand.push_back({a[start], 1});
        return;
    }
    int mid = start + (end - start) / 2;
    build(2 * node, start, mid);
    build(2 * node + 1, mid + 1, end);
    tree_seg[node] = merge_nodes(tree_seg[2 * node], tree_seg[2 *
    ↳ node + 1]);
}

Node query_candidates(int node, int start, int end, int l, int r) {
    if (r < start || end < l) return {{{}}};
    if (l <= start && end <= r) return tree_seg[node];

    int mid = start + (end - start) / 2;
    Node left_res = query_candidates(2 * node, start, mid, l, r);
    Node right_res = query_candidates(2 * node + 1, mid + 1, end, l,
    ↳ r);

    if (left_res.cand.empty()) return right_res;
    if (right_res.cand.empty()) return left_res;

    return merge_nodes(left_res, right_res);
}

// O(log N) fast static frequency count
int get_frequency(int val, int l, int r) {
```

```

    auto& v = pos[val];
    auto right_it = upper_bound(v.begin(), v.end(), r);
    auto left_it = lower_bound(v.begin(), v.end(), l);
    return distance(left_it, right_it);
}

void solve() {
    int n, q;
    cin >> n >> q;

    for (int i = 1; i <= n; i++) {
        cin >> a[i];
        pos[a[i]].push_back(i);
    }

    build(1, 1, n);

    while (q--) {
        int l, r;
        cin >> l >> r;

        int len = r - l + 1;
        Node res = query_candidates(1, 1, n, l, r);

        int max_freq = 0;
        for (const auto& p : res.cands) {
            int cand = p.first;
            int freq = get_frequency(cand, l, r);
            max_freq = max(max_freq, freq);
        }

        if (max_freq > (len + 1) / 2) {
            cout << 2 * max_freq - len << "\n";
        } else {
            cout << 1 << "\n";
        }
    }
}

```

2.1.11 SegTreeDescent

Segment Tree DESCENT - answer "first / last index satisfying P" in $O(\log n)$, not $O(\log^2 n)$.

```

// Here: max-segtree. first_ge(l, x) = smallest i >= l with a[i] >=
// ↪ x, or n if none.
// Swap the stored aggregate + the two 'good()' tests to get: first
// ↪ prefix-sum >= S,
// first zero, k-th set bit, "how far right can I extend". This is
// ↪ the workhorse behind
// greedy "extend as far as possible" and interval-cover problems.
struct SegMax {
    int n;
    vector<ll> t;

    SegMax(int n = 0) : n(n), t(4 * n, LLONG_MIN) {}

    void build(const vector<ll>& a, int v, int l, int r) {
        if (l == r) { t[v] = a[l]; return; }
        int m = (l + r) / 2;
        build(a, 2 * v, l, m); build(a, 2 * v + 1, m + 1, r);
        t[v] = max(t[2 * v], t[2 * v + 1]);
    }
    void build(const vector<ll>& a) { if (n) build(a, 1, 0, n - 1); }

    void set(int p, ll x, int v, int l, int r) {
        if (l == r) { t[v] = x; return; }
        int m = (l + r) / 2;
        p <= m ? set(p, x, 2 * v, l, m) : set(p, x, 2 * v + 1, m + 1,
        ↪ r);
        t[v] = max(t[2 * v], t[2 * v + 1]);
    }
    void set(int p, ll x) { set(p, x, 1, 0, n - 1); }

    ll qry(int ql, int qr, int v, int l, int r) {
        if (qr < l || r < ql) return LLONG_MIN;
        if (ql <= l && r <= qr) return t[v];
        int m = (l + r) / 2;
        return max(qry(ql, qr, 2 * v, l, m), qry(ql, qr, 2 * v + 1, m
        ↪ + 1, r));
    }
    ll qry(int l, int r) { return qry(l, r, 1, 0, n - 1); }

    // smallest i in [ql, n) with a[i] >= x (returns n if none)
    int first_ge(int ql, ll x, int v, int l, int r) {
        if (r < ql || t[v] < x) return n;
        if (l == r) return l;
        int m = (l + r) / 2, res = first_ge(ql, x, 2 * v, l, m);
        return res < n ? res : first_ge(ql, x, 2 * v + 1, m + 1, r);
    }
    int first_ge(int ql, ll x) { return first_ge(ql, x, 1, 0, n - 1); }
    ↪ }

    // largest i in [0, qr] with a[i] >= x (returns -1 if none)
    int last_ge(int qr, ll x, int v, int l, int r) {
        if (l > qr || t[v] < x) return -1;
        if (l == r) return l;
        int m = (l + r) / 2, res = last_ge(qr, x, 2 * v + 1, m + 1,
        ↪ r);
        return res >= 0 ? res : last_ge(qr, x, 2 * v, l, m);
    }
    int last_ge(int qr, ll x) { return last_ge(qr, x, 1, 0, n - 1); }
}

```

2.1.12 PersistentLazySegTree

PERSISTENT LAZY SEGMENT TREE - range add and range sum on ANY past version, $O(\log n)$ time and

```
// O(log n) new nodes per update.
// WHEN: you need RANGE updates and old versions. 03 -
    ↳ PersistentSegmentTree only does point
// updates, which is all the "k-th smallest in a range" trick needs -
    ↳ prefer it when that suffices,
// it is half the code and half the memory. If you only ever query
    ↳ the LATEST version, use
// 02 - LazySegmentTree.
// KEY INVARIANT: a tag is NEVER pushed down - that would mutate a
    ↳ node an old version still points
// at. Instead t[u].sum is always the CORRECT sum of u's whole range
    ↳ (tag included) and t[u].add is
// the amount still owed to the two children. So a query carries the
    ↳ accumulated ancestor tags down
// with it, and an update fixes sum on the way back up as sum(l) +
    ↳ sum(r) + add * len.
struct PLazySeg {
    struct Node { int l = 0, r = 0; ll sum = 0, add = 0; };
    vector<Node> t;
    int n;
    PLazySeg(int n) : t(1), n(n) {}
    ↳ node 0 = the shared all-zero node
    int build(const vector<ll>& a, int lo, int hi) {
        int u = t.size();
        t.push_back({});
        if (lo == hi) { t[u].sum = a[lo]; return u; }
        int m = (lo + hi) / 2, L = build(a, lo, m), R = build(a, m +
        ↳ 1, hi);
        t[u].l = L, t[u].r = R, t[u].sum = t[L].sum + t[R].sum;
        return u;
    }
    int build(const vector<ll>& a) { return build(a, 0, n - 1); }
    int update(int u, int ql, int qr, ll v, int lo, int hi) { // add
    ↳ v to [ql, qr], returns new root
        int c = t.size();
        t.push_back(t[u]);
        if (ql <= lo && hi <= qr) { t[c].sum += v * (hi - lo + 1),
        ↳ t[c].add += v; return c; }
        int m = (lo + hi) / 2;
        if (ql <= m) t[c].l = update(t[c].l, ql, qr, v, lo, m);
        if (qr > m) t[c].r = update(t[c].r, ql, qr, v, m + 1, hi);
        t[c].sum = t[t[c].l].sum + t[t[c].r].sum + t[c].add * (hi -
        ↳ lo + 1);
        return c;
    }
    int update(int u, int l, int r, ll v) { return update(u, l, r, v,
    ↳ 0, n - 1); }
    ll query(int u, int ql, int qr, ll acc, int lo, int hi) { // acc
    ↳ = sum of the ancestors' tags
        if (ql <= lo && hi <= qr) return t[u].sum + acc * (hi - lo +
        ↳ 1);
        int m = (lo + hi) / 2;
        acc += t[u].add;
        ll res = 0;
        if (ql <= m) res += query(t[u].l, ql, qr, acc, lo, m);
        if (qr > m) res += query(t[u].r, ql, qr, acc, m + 1, hi);
        return res;
    }
    ll query(int u, int l, int r) { return query(u, l, r, 0, 0, n -
    ↳ 1); }
};
// MEMORY is the real constraint:  $O((n + q \log n))$  nodes, ~32 bytes
    ↳ each. reserve() up front.
// RANGE ASSIGN instead of range add: the same no-push-down trick
    ↳ works for any tag that composes,
// but the accumulated 'acc' must then be "the last assign seen on
    ↳ the way down", so pass the tag
// itself rather than a sum, and let a deeper tag override a
    ↳ shallower one.
// MIN/MAX aggregate: replace  $v * \text{len}$  by  $v$  and the pull by
    ↳  $\min(\text{sum}(l), \text{sum}(r)) + \text{add}$ .
// A COMMON ALTERNATIVE that avoids all of this: make the update
    ↳ offline, sort by version, and use
// an ordinary lazy segment tree with rollback (14 -
    ↳ OfflineDynamicConnectivity's timeline trick).
```

2.1.13 ArithmeticProgressionSegTree

ARITHMETIC-PROGRESSION SEGMENT TREE - "add a, a+d, a+2d, ... to a[l..r]", range sum, $O(\log n)$.

```
// WHEN: updates whose value DEPENDS ON THE POSITION inside the range
    ↳ - "add i-l+1 to a[l..r]",
// "add w to every node on a path, weighted by depth", any linear
    ↳ ramp. A plain lazy segment tree
// cannot do it because its tag must be one number.
// KEY INVARIANT: a progression restricted to a sub-range is still a
    ↳ progression, so the tag stays
// two numbers (a, d) and stays CLOSED under composition: two tags
    ↳ added componentwise. The tag is
// ANCHORED at the node's left endpoint, so pushing to the right
    ↳ child skips (m+1-b) terms.
struct APSeg {
    struct Tag { ll a = 0, d = 0; };
    ↳ a+d, a+2d, ... from node's left
    int n;
    vector<ll> s;
    vector<Tag> lz;
    APSeg(int n) : n(n), s(4 * n, 0), lz(4 * n) {}
    static ll tot(Tag t, ll len) { return t.a * len + t.d * (len *
    ↳ (len - 1) / 2); }
    static Tag skip(Tag t, ll k) { return {t.a + k * t.d, t.d}; }
    ↳ // drop the first k terms
    void apply(int x, Tag t, int len) { s[x] += tot(t, len), lz[x].a
    ↳ += t.a, lz[x].d += t.d; }
    void push(int x, int b, int e) {
        if (!lz[x].a && !lz[x].d) return;
        int m = (b + e) / 2;
        apply(2 * x, lz[x], m - b + 1), apply(2 * x + 1, skip(lz[x],
        ↳ m + 1 - b), e - m);
        lz[x] = {};
    }
    void build(const vector<ll>& a, int x, int b, int e) {
        lz[x] = {};
        if (b == e) { s[x] = a[b]; return; }
        int m = (b + e) / 2;
        build(a, 2 * x, b, m), build(a, 2 * x + 1, m + 1, e);
        s[x] = s[2 * x] + s[2 * x + 1];
    }
    void build(const vector<ll>& a) { build(a, 1, 0, n - 1); }
    void upd(int x, int b, int e, int l, int r, Tag t) { // t
    ↳ is anchored at index l
        if (r < b || e < l) return;
        if (l <= b && e <= r) { apply(x, skip(t, b - l), e - b + 1);
        ↳ return; }
        push(x, b, e);
        int m = (b + e) / 2;
        upd(2 * x, b, m, l, r, t), upd(2 * x + 1, m + 1, e, l, r, t);
        s[x] = s[2 * x] + s[2 * x + 1];
    }
    void upd(int l, int r, ll a, ll d) { upd(1, 0, n - 1, l, r, {a,
    ↳ d}); } // a[l]+=a, a[l+1]+=a+d..
    ll qry(int x, int b, int e, int l, int r) {
        if (r < b || e < l) return 0;
        if (l <= b && e <= r) return s[x];
        push(x, b, e);
        int m = (b + e) / 2;
        return qry(2 * x, b, m, l, r) + qry(2 * x + 1, m + 1, e, l,
        ↳ r);
    }
    ll qry(int l, int r) { return qry(1, 0, n - 1, l, r); }
};
// OVERFLOW: tot() is  $O(\text{len}^2 * d)$  - with  $n = 2e5$  and  $d = 1e9$  that is
    ↳  $2e19$ , past ll. Work mod p
// (replace /2 by * inv2) or negative d.
// DECREASING ramps: pass a bound d; "add a, a-d, a-2d" to [l,r]
    ↳ read right-to-left is the same
// tag anchored at r, so mirror the array or negate d and anchor at l
    ↳ with  $a' = a + (r-l)*d$ .
// LIGHTER ALTERNATIVES, use these when they fit:
// Range AP add + POINT query only: one BIT over the second
    ↳ difference  $b[i] = a[i] - 2a[i-1] + a[i-2]$ ;
// an AP add touches 4 cells, and  $a[i]$  is a double prefix sum. 5
    ↳ lines (02 - BIT).
// OFFLINE with all updates before all queries: apply the
    ↳ second-difference trick to a plain
// array and prefix-sum twice at the end.  $O(n)$  total.
// GENERAL RULE: any tag family closed under composition and
    ↳ restriction works. Polynomials of
// degree k need k+1 coefficients; affine " $a[i] = p*a[i] + q$ " needs
    ↳ (p, q) composed as function
// composition (that one is NOT commutative - compose in the right
    ↳ order).
```

2.1.14 XORSegTree

XOR SEGMENT TREE - sum of $a[p \oplus x]$ over $l \leq p \leq r$, for an x given per query, plus point

```
// updates. O(log n) both, and NO extra memory over an ordinary
//   ↳ segment tree.
// WHEN: "the array is permuted by xor-ing every index with x" -
//   ↳ subset/bit problems where a query
// asks about  $a[l..r]$  of a xor-shifted array, e.g. CF 1654F. Also the
//   ↳ fast way to combine two
// arrays under an xor convolution restricted to a range.
// KEY INVARIANT: an ALIGNED block  $[b, b + 2^k)$  has its low  $k$  bits
//   ↳ free, so xor-ing every element
// of it with  $x$  maps the block ONTO the aligned block starting at  $b \oplus$ 
//   ↳  $(x \gg k \ll k)$  - a block goes
// to a block, never to a straddling range. The canonical
//   ↳ decomposition of  $[l, r]$  is  $O(\log n)$ 
// aligned blocks, so the answer is  $O(\log n)$  node lookups; the only
//   ↳ work is finding each partner
// node, which at level  $k$  is just xor-ing the node's
//   ↳ position-within-level by  $x \gg k$ .
struct XorSeg {                                // n MUST be a
    int n;                                     // power of two; queries use  $x < n$ 
    vector<ll> t;
    XorSeg(const vector<ll>& a) : n(a.size()), t(2 * n) {
        copy(all(a), t.begin() + n);
        for (int i = n - 1; i; i--) t[i] = t[2 * i] + t[2 * i + 1];
    }
    void add(int i, ll v) { for (t[i += n] += v; i >>= 1; i) t[i] = t[2 * i] + t[2 * i + 1]; }
    ll query(int l, int r, int x) {                // sum of  $a[p \oplus x]$ 
    // over  $l \leq p \leq r$ 
        ll res = 0;
        for (int lo = l + n, hi = r + n + 1, b = n; lo < hi; lo >>=
        // 1, hi >>= 1, b >>= 1, x >>= 1) {
            if (lo & 1) res += t[(lo - b) ^ x] + b], lo++; // b =
            // number of nodes on this level
            if (hi & 1) res += t[(--hi - b) ^ x] + b]; //
        // and also the level's index base
        }
        return res;
    }
};
// THE AGGREGATE must be COMMUTATIVE (sum, xor, min, gcd): xor-ing
//   ↳ reorders the elements inside
// each block, so anything order-sensitive (max prefix sum, matrix
//   ↳ product) breaks.
// PAD n up to a power of two with identity elements - the alignment
//   ↳ argument needs it.
// THE SAME TRICK, no data structure: if you only need it once,
//   ↳ enumerate the  $O(\log n)$  blocks of
//  $[l, r]$  yourself and answer each with a prefix sum of the ORIGINAL
//   ↳ array. That is the version to
// write when there are no updates.
// RANGE UPDATES do not survive this: a lazy tag would have to travel
//   ↳ to the xor-partner subtree.
// If you need them, keep the tag OUTSIDE (a separate BIT of "add v
//   ↳ to  $a[l..r]$ " offsets) and add
// the contribution of the covered blocks analytically.
```

2.1.15 KineticSegTree

KINETIC SEGMENT TREE (kinetic tournament) - each index i holds a LINE $y = a[i]*T + b[i]$; the

```
// tree answers "max over  $i$  in  $[l, r]$  of  $a[i]*T + b[i]$ " at a global
//   ↳ time  $T$  that only MOVES FORWARD,
// and supports adding a constant to  $b$  over a range.
// WHEN: DP where the candidate values are linear in a parameter that
//   ↳ sweeps monotonically -
// "max over  $j$  of  $dp[j] + (i-j)*c$ ", scheduling with a moving
//   ↳ deadline, CHT where the queries are
// sorted but the LINES also change over time (a Li Chao tree cannot
//   ↳ delete or shift lines).
// If  $T$  is arbitrary (not monotone) use Li Chao / convex hull trick
//   ↳ instead - this degenerates.
// COMPLEXITY:  $O(\log n)$  per query and per range-add; advancing  $T$  is
//   ↳  $O(\log^2 n)$  AMORTISED, and the
// total over a whole run is  $O((n + q) \log^2 n)$ . A single advance can
//   ↳ cost  $O(n)$  in the worst case.
// KEY INVARIANT (the "certificate"): every node stores the winning
//   ↳ line of its subtree AND melt =
// the earliest future time at which the winner of some node below it
//   ↳ changes. Advancing to  $T$ 
// recurses ONLY into nodes with melt  $\leq T$ ; everything else is
//   ↳ provably still correct. Each
// recursion strictly changes a winner, and a winner can only change
//   ↳  $O(\log n)$  times amortised per
// element because two lines cross at most once.
struct KineticSeg {
    int n;
    ll T = 0;
    vector<ll> a, b, lz, melt;                                // a,
    // b = the node's WINNING line
    KineticSeg(const vector<ll>& A, const vector<ll>& B)
        : n(A.size()), a(4 * n), b(4 * n), lz(4 * n, 0), melt(4 * n,
    // llinf) {
        build(A, B, 1, 0, n - 1);
    }
    static ll fdiv(ll p, ll q) { return p / q - ((p % q != 0) && ((p
    // < 0) != (q < 0))); }
    ll cross(int w, int o) {                                //
    // first time o strictly beats w
        if (a[o] <= a[w]) return llinf;                    //
    // parallel or flatter: never
        return fdiv(b[w] - b[o], a[o] - a[w]) + 1;
    }
    void pull(int x) {
        int L = 2 * x, R = L + 1;
        int w = a[L] * T + b[L] >= a[R] * T + b[R] ? L : R, o = L + R
    // - w;
        a[x] = a[w], b[x] = b[w];
        melt[x] = min({melt[L], melt[R], cross(w, o)});
    }
    void apply(int x, ll v) { b[x] += v, lz[x] += v; }        //
    // uniform shift keeps every melt
    void push(int x) { if (lz[x]) apply(2 * x, lz[x]), apply(2 * x +
    // 1, lz[x]), lz[x] = 0; }
    void build(const vector<ll>& A, const vector<ll>& B, int x, int
    // lo, int hi) {
        lz[x] = 0, melt[x] = llinf;
        if (lo == hi) { a[x] = A[lo], b[x] = B[lo]; return; }
        int m = (lo + hi) / 2;
        build(A, B, 2 * x, lo, m), build(A, B, 2 * x + 1, m + 1, hi);
        pull(x);
    }
    void heaten(int x, int lo, int hi) {
        if (melt[x] > T || lo == hi) return;                //
    // certificate still valid
        int m = (lo + hi) / 2;
        push(x);
        heaten(2 * x, lo, m), heaten(2 * x + 1, m + 1, hi);
        pull(x);
    }
    void advance(ll t) { T = t; heaten(1, 0, n - 1); }        // t
    // must be >= the current T
    void add(int x, int lo, int hi, int l, int r, ll v) {        //
    // b[i] += v for  $i$  in  $[l, r]$ 
        if (r < lo || hi < l) return;
        if (l <= lo && hi <= r) { apply(x, v); return; }
        int m = (lo + hi) / 2;
        push(x);
        add(2 * x, lo, m, l, r, v), add(2 * x + 1, m + 1, hi, l, r,
    // v);
        pull(x);
    }
    void add(int l, int r, ll v) { add(1, 0, n - 1, l, r, v); }
    void set(int x, int lo, int hi, int i, ll na, ll nb) {        //
    // replace the line at index i
```



```

    if (lo == hi) { a[x] = na, b[x] = nb; return; }
    int m = (lo + hi) / 2;
    push(x);
    i <= m ? set(2 * x, lo, m, i, na, nb) : set(2 * x + 1, m + 1,
    ↪ hi, i, na, nb);
    pull(x);
}
void set(int i, ll na, ll nb) { set(1, 0, n - 1, i, na, nb); }
ll qry(int x, int lo, int hi, int l, int r) { // max
    ↪ at the CURRENT T
    if (r < lo || hi < l) return -llinf;
    if (l <= lo && hi <= r) return a[x] * T + b[x];
    int m = (lo + hi) / 2;
    push(x);
    return max(qry(2 * x, lo, m, l, r), qry(2 * x + 1, m + 1, hi,
    ↪ l, r));
}
ll qry(int l, int r) { return qry(1, 0, n - 1, l, r); }
};
// MIN instead of max: negate both a and b, or flip the comparison in
    ↪ pull() and cross().
// ADDING TO THE SLOPES over a range destroys every certificate below
    ↪ and is NOT amortised - do it
// only at a single index (set()) or rebuild.
// SEGMENT TREE BEATS (04) is the same idea with a different
    ↪ certificate: recurse only where the
// tag cannot be applied wholesale, and bound the recursions by a
    ↪ potential. If you can phrase your
// update as "the answer only changes when two stored candidates swap
    ↪ order", this is the shape.
// A CHEAPER SPECIAL CASE: if the lines are only ever ADDED (never
    ↪ removed) and queries are sorted,
// a monotone CHT is O(1) amortised per op - reach for that first.

```

2.1.16 HistoricSegTree

HISTORIC-MAXIMUM SEGMENT TREE - range add, range max of the CURRENT values, and range max of

```

// the HISTORIC value hmax[i] = max over every past moment of a[i].
    ↪ O(log n) per operation.
// WHEN: "what is the largest this cell ever reached", "peak stock
    ↪ price / peak queue length /
// high-water mark over a range", and as the engine behind offline
    ↪ problems that sweep an index
// while range-adding. If you only need the current max, use 02 -
    ↪ LazySegmentTree.
// KEY INVARIANT: the tag is a PAIR (add, mx) = "the total of this
    ↪ batch of adds" and "the largest
// PARTIAL SUM (prefix) any moment inside the batch reached". That
    ↪ pair is closed under
// composition: applying t and then u gives {t.add + u.add, max(t.mx,
    ↪ t.add + u.mx)}. So a whole
// run of pending updates can sit on a node and still be replayed
    ↪ exactly for the historic max.
// Careful: mx is a max over prefixes INCLUDING the empty one, so it
    ↪ is always >= 0.
struct HistoricSeg {
    struct Tag { ll add = 0, mx = 0; };
    int n;
    vector<ll> cur, his; //
    ↪ current max, historic max
    vector<Tag> lz;
    HistoricSeg(int n) : n(n), cur(4 * n, 0), his(4 * n, 0), lz(4 *
    ↪ n) {}
    void apply(int x, Tag t) {
        his[x] = max(his[x], cur[x] + t.mx); //
        ↪ peak inside this batch
        cur[x] += t.add;
        lz[x] = {lz[x].add + t.add, max(lz[x].mx, lz[x].add + t.mx)};
    }
    void push(int x) {
        if (!lz[x].add && !lz[x].mx) return;
        apply(2 * x, lz[x]), apply(2 * x + 1, lz[x]);
        lz[x] = {};
    }
    void pull(int x) {
        cur[x] = max(cur[2 * x], cur[2 * x + 1]);
        his[x] = max(his[2 * x], his[2 * x + 1]);
    }
    void build(const vector<ll>& a, int x, int lo, int hi) {
        lz[x] = {};
        if (lo == hi) { cur[x] = his[x] = a[lo]; return; }
        int m = (lo + hi) / 2;
        build(a, 2 * x, lo, m), build(a, 2 * x + 1, m + 1, hi);
        pull(x);
    }
    void build(const vector<ll>& a) { build(a, 1, 0, n - 1); }
    void upd(int x, int lo, int hi, int l, int r, ll v) {
        if (r < lo || hi < l) return;
        if (l <= lo && hi <= r) { apply(x, {v, max(v, 0LL)}); return;
        ↪ }
        int m = (lo + hi) / 2;
        push(x);
        upd(2 * x, lo, m, l, r, v), upd(2 * x + 1, m + 1, hi, l, r,
        ↪ v);
        pull(x);
    }
    void upd(int l, int r, ll v) { upd(1, 0, n - 1, l, r, v); }
    // hist = false -> max of current values; hist = true -> max of
    ↪ the all-time values
    ll qry(int x, int lo, int hi, int l, int r, bool hist) {
        if (r < lo || hi < l) return -llinf;
        if (l <= lo && hi <= r) return hist ? his[x] : cur[x];
        int m = (lo + hi) / 2;
        push(x);
        return max(qry(2 * x, lo, m, l, r, hist), qry(2 * x + 1, m +
        ↪ 1, hi, l, r, hist));
    }
    ll qry(int l, int r, bool hist) { return qry(1, 0, n - 1, l, r,
    ↪ hist); }
};
// HISTORIC MIN: mirror everything (track the smallest prefix of the
    ↪ batch).
// HISTORIC SUM (sum over time of every past value, i.e. integrate
    ↪ the array) needs a bigger tag:
// keep sum, historic sum, and a count of elapsed time steps, and let
    ↪ the tag carry (add, time)
// with hsum += sum * dt. Same idea, one more field - only write it
    ↪ if the problem forces it.
// RANGE ASSIGN + historic max is the same tag with "assign"
    ↪ replacing "add"; compose so the later
// assign wins and mx tracks the largest value ever assigned.

```

```
// THE OFFLINE ALTERNATIVE, almost always shorter: if the queries are
    ↳ known in advance, sweep time
// and answer with an ordinary max segment tree plus a BIT for "was
    ↳ this cell touched after t".
```

2.1.17 SegTree2DSparse

SPARSE 2D SEGMENT TREE (segment tree of TREAPS) - point assign $a[i][j] = v$ and rectangle sum on

```
// a grid whose columns are huge or unknown in advance.  $O(\log n * \log$ 
    ↳  $q)$  per operation and only
//  $O(q \log n)$  memory - one node per touched cell per level.
// WHEN: 08 - SegTree2D is a dense  $n \times m$  tree and needs  $O(nm)$  memory;
    ↳ a merge-sort tree (06) is
// static; an offline problem should use 02 - BIT/03 -
    ↳ Offline2DFenwickTree, which is smaller and
// faster because it can compress the coordinates first. Reach for
    ↳ THIS one only when the updates
// are ONLINE and the grid is too big to allocate.
// KEY INVARIANT: the outer tree is indexed by ROW and every outer
    ↳ node owns a treap keyed by
// COLUMN holding, for each column  $j$  that some row in its range
    ↳ touches, the aggregate of that
// column over the node's rows. So an outer node is only ever asked
    ↳ about columns that exist, and
// after a point update the parents repair a SINGLE column by
    ↳ recombining their two children's
// value at that one column.
struct Seg2D {
    struct T { int l = 0, r = 0, key = 0; unsigned pr = 0; ll val =
        ↳ 0, agg = 0; };
    vector<T> tr; //
    ↳ shared pool for every treap
    vector<int> rt; //
    ↳ treap root of each outer node
    int n;
    Seg2D(int n) : tr(1), rt(4 * n, 0), n(n) {}
    unsigned rnd() { //
        ↳ xorshift; any prng will do
        static unsigned s = 246353424u;
        return s ^= s << 13, s ^= s >> 17, s ^= s << 5;
    }
    void pull(int x) { tr[x].agg = tr[tr[x].l].agg + tr[x].val +
        ↳ tr[tr[x].r].agg; }
    void split(int x, int k, int& a, int& b) { // a
        ↳ keeps keys < k
        if (!x) { a = b = 0; return; }
        if (tr[x].key < k) a = x, split(tr[x].r, k, tr[a].r, b),
        ↳ pull(a);
        else b = x, split(tr[x].l, k, a, tr[b].l), pull(b);
    }
    int merge(int a, int b) {
        if (!a || !b) return a | b;
        if (tr[a].pr > tr[b].pr) { tr[a].r = merge(tr[a].r, b);
        ↳ pull(a); return a; }
        tr[b].l = merge(a, tr[b].l);
        pull(b);
        return b;
    }
    void set(int& root, int k, ll v) { //
        ↳ a[k] = v inside one treap
        int a, b, c;
        split(root, k, a, b), split(b, k + 1, b, c);
        if (b) tr[b].val = v, pull(b);
        else tr.push_back({0, 0, k, rnd(), v, v}), b = tr.size() - 1;
        root = merge(merge(a, b), c);
    }
    ll get(int& root, int lo, int hi) { // sum
        ↳ of keys in [lo, hi]
        int a, b, c;
        split(root, lo, a, b), split(b, hi + 1, b, c);
        ll res = tr[b].agg;
        root = merge(merge(a, b), c);
        return res;
    }
    void upd(int x, int lo, int hi, int i, int j, ll v) { //
        ↳ a[i][j] = v
        if (lo == hi) { set(tr[x], j, v); return; }
        int m = (lo + hi) / 2;
        i <= m ? upd(2 * x, lo, m, i, j, v) : upd(2 * x + 1, m + 1,
        ↳ hi, i, j, v);
        set(tr[x], j, get(tr[2 * x], j, j) + get(tr[2 * x + 1], j,
        ↳ j));
    }
    void upd(int i, int j, ll v) { upd(1, 0, n - 1, i, j, v); }
    ll qry(int x, int lo, int hi, int r1, int r2, int c1, int c2) {
        if (r2 < lo || hi < r1) return 0;
        if (r1 <= lo && hi <= r2) return get(tr[x], c1, c2);
        int m = (lo + hi) / 2;
        return qry(2 * x, lo, m, r1, r2, c1, c2) + qry(2 * x + 1, m +
        ↳ 1, hi, r1, r2, c1, c2);
    }
    ll qry(int r1, int r2, int c1, int c2) { return qry(1, 0, n - 1,
```

```

    ↪ r1, r2, c1, c2); }
};
// ANY IDEMPOTENT OR GROUP AGGREGATE works (sum, xor, gcd, min):
    ↪ upd() rebuilds a parent's single
// column from its two children, so the operation only has to be
    ↪ associative and commutative.
// ROWS TOO BIG TOO: make the outer tree dynamic as well (05 -
    ↪ DynamicSegTree) - allocate the outer
// node on first touch. Memory stays O(q log n).
// BIT OF TREAPS is the same shape with a shorter outer layer, but it
    ↪ cannot do "descend to find
// the k-th" and needs a group (no min/max).
// IF THE PROBLEM IS OFFLINE, do not write this. Sort the queries by
    ↪ row, sweep, and keep a single
// 1D BIT over columns: O((n + q) log n), five lines, and no memory
    ↪ pressure.

```

2.1.18 MexSegTree

MEX UNDER UPDATES - the smallest non-negative integer NOT present in the multiset, maintained

```

// under insert and erase in O(log V), read in O(log V) by ONE
    ↪ descent (not a scan).
// WHEN: "after each operation print the mex", Grundy/nim-value
    ↪ maintenance (10 - Game Theory),
// sliding-window mex. If the multiset only ever GROWS, a pointer
    ↪ that walks forward is O(1)
// amortised and this file is overkill. If you need the mex of a
    ↪ RANGE, see the recipe below.
// KEY INVARIANT: a segment tree over the VALUE axis [0, V] storing
    ↪ the MINIMUM count in each value
// range. The mex is the leftmost value with count 0, so the descent
    ↪ is "if the left child's
// minimum is 0 go left, else go right" - which is why the tree
    ↪ stores a MIN and not a sum.
// V only needs to be n + 1: with n elements the mex can never exceed
    ↪ n.
struct MexSet {
    int V;
    vector<int> cnt, mn;
    MexSet(int V) : V(V), cnt(V + 1, 0), mn(4 * (V + 1), 0) {}
    void upd(int x, int lo, int hi, int p) {
        if (lo == hi) { mn[x] = cnt[lo]; return; }
        int m = (lo + hi) / 2;
        p <= m ? upd(2 * x, lo, m, p) : upd(2 * x + 1, m + 1, hi, p);
        mn[x] = min(mn[2 * x], mn[2 * x + 1]);
    }
    void add(int v) { if (v <= V) cnt[v]++, upd(1, 0, V, v); }
    void del(int v) { if (v <= V) cnt[v]--, upd(1, 0, V, v); }
    int mex(int x, int lo, int hi) {
        if (lo == hi) return lo;
        int m = (lo + hi) / 2;
        return mn[2 * x] == 0 ? mex(2 * x, lo, m) : mex(2 * x + 1, m
    ↪ + 1, hi);
    }
    int mex() { return mex(1, 0, V); }
};
// MEX OF A RANGE [l, r], offline: sort the queries by r and keep a
    ↪ segment tree over VALUES whose
// leaf v holds the LAST POSITION where v occurs in the prefix ending
    ↪ at r (-1 if never). The mex
// of [l, r] is the leftmost v whose last position is < l - the same
    ↪ descent, comparing against l
// instead of against 0. Online: make that tree persistent (03 -
    ↪ PersistentSegmentTree), version r.
// MEX OF A RANGE with array updates: no clean polylog solution - use
    ↪ Mo's plus the O(1)-add/del
// counter in 06 - Square Root Decomposition/07, scanning for the
    ↪ first empty count bucket.
// MEX OF ALL SUBARRAYS / sum of mex over all subarrays: divide and
    ↪ conquer on the position of the
// value 0, or the "for each mex value, count the subarrays achieving
    ↪ it" sweep - not this file.
// FIRST MISSING POSITIVE (mex over 1..) is the same with the axis
    ↪ shifted by one.

```

2.2 Binary Indexed Tree (BIT)

2.2.1 FenwickTree

Fenwick Tree (Binary Indexed Tree) - 0-based indexing

```

// Space: O(N), Time: O(log N) per update/query
template<typename T = ll>
struct FenwickTree {
    int n;
    vector<T> tree;

    FenwickTree(int n = 0) : n(n), tree(n + 1, 0) {}

    void add(int i, T delta) {
        for (i++; i <= n; i += i & -i) tree[i] += delta;
    }

    void set(int i, T val) {
        add(i, val - query(i, i));
    }

    T query(int i) {
        T sum = 0;
        for (i++; i > 0; i -= i & -i) sum += tree[i];
        return sum;
    }

    T query(int l, int r) {
        if (l > r) return 0;
        return query(r) - query(l - 1);
    }

    // Returns first 0-based index where prefix sum >= val
    int lower_bound(T val) {
        int pos = 0;
        for (int sz = 1 << (n ? __lg(n) : 0); sz > 0; sz >= 1) {
            if (pos + sz <= n && tree[pos + sz] < val) {
                val -= tree[pos + sz];
                pos += sz;
            }
        }
        return pos;
    }
};

```

2.2.2 FenwickTree2D

2D Fenwick Tree - 0-based indexing

```

// Space: O(N * M), Time: O(log N * log M) per update/query
template<typename T = ll>
struct FenwickTree2D {
    int n, m;
    vector<vector<T>> tree;

    FenwickTree2D(int n = 0, int m = 0) : n(n), m(m), tree(n + 1,
    ↪ vector<T>(m + 1, 0)) {}

    void add(int x, int y, T val) {
        for (int i = x + 1; i <= n; i += i & -i) {
            for (int j = y + 1; j <= m; j += j & -j) {
                tree[i][j] += val;
            }
        }
    }

    void set(int x, int y, T val) {
        add(x, y, val - query(x, y, x, y));
    }

    T query(int x, int y) {
        T sum = 0;
        for (int i = min(x + 1, n); i > 0; i -= i & -i) {
            for (int j = min(y + 1, m); j > 0; j -= j & -j) {
                sum += tree[i][j];
            }
        }
        return sum;
    }

    // 2D Subgrid Range Sum from (x1, y1) to (x2, y2) inclusive
    T query(int x1, int y1, int x2, int y2) {
        if (x1 > x2 || y1 > y2) return 0;
        return query(x2, y2) - query(x1 - 1, y2) - query(x2, y1 - 1)
    ↪ + query(x1 - 1, y1 - 1);
    }
};

```

2.2.3 Offline2DFenwickTree

Offline 2D Fenwick Tree (Compressed 2D BIT)

```
// Space: O(N + Q log N), Time: O(log^2 N) per query
template<typename T = ll>
struct BIT2D {
    int n;
    vector<vector<int>> vals;
    vector<vector<T>> bit;

    int ind(const vector<int>& v, int x) {
        return upper_bound(all(v), x) - v.begin() - 1;
    }

    // mx: max row index, todo: all (r, c) update coordinates that
    // will be called
    BIT2D(int mx, vector<array<int, 2>> todo) : n(mx), vals(mx + 1),
    bit(mx + 1) {
        sort(all(todo), [](const auto& a, const auto& b) { return
        a[1] < b[1]; });

        for (int i = 0; i <= n; i++) vals[i].pb(INT_MIN);
        for (auto [r, c] : todo) {
            for (int x = r; x <= n; x |= x + 1) {
                if (vals[x].back() != c) vals[x].pb(c);
            }
        }
        for (int i = 0; i <= n; i++) bit[i].resize(vals[i].size(), 0);
    }

    void add(int r, int c, T val) {
        for (; r <= n; r |= r + 1) {
            for (int i = ind(vals[r], c); i < (int)bit[r].size(); i
            |= i + 1) {
                bit[r][i] += val;
            }
        }
    }

    T query(int r, int c) {
        T res = 0;
        for (; r >= 0; r = (r & (r + 1)) - 1) {
            for (int i = ind(vals[r], c); i >= 0; i = (i & (i + 1)) -
            1) {
                res += bit[r][i];
            }
        }
        return res;
    }

    T query(int r1, int c1, int r2, int c2) {
        return query(r2, c2) - query(r2, c1 - 1) - query(r1 - 1, c2)
        + query(r1 - 1, c1 - 1);
    }

    void reset() {
        for (auto& v : bit) fill(all(v), 0);
    }
};
```

2.2.4 BITRangeUpdateRangeQuery

BIT with RANGE UPDATE + RANGE QUERY (two BITs). 0-based, O(log n) per op.

```
// pre(i) = sum_{j<=i} a[j] = S1(i)*(i+1) - S2(i), where a range-add
// of v on [l, r] does
// S1: +v at l, -v at r+1          S2: +v*l at l, -v*(r+1) at r+1
template<typename T = ll>
struct BIT2 {
    int n;
    vector<T> b1, b2;

    BIT2(int n = 0) : n(n), b1(n + 1, 0), b2(n + 1, 0) {}
    void add(vector<T>& b, int i, T v) { for (i++; i <= n; i += i &
    -1) b[i] += v; }
    T sum(const vector<T>& b, int i) const { T s = 0; for (i++; i >
    0; i -= i & -1) s += b[i]; return s; }

    void upd(int l, int r, T v) { // add v to [l, r]
        add(b1, l, v), add(b1, r + 1, -v);
        add(b2, l, v * l), add(b2, r + 1, -v * (r + 1));
    }
    T pre(int i) const { return i < 0 ? 0 : sum(b1, i) * (i + 1) -
    sum(b2, i); }
    T qry(int l, int r) const { return pre(r) - pre(l - 1); }
};
```

2.2.5 BITOfSegTrees

BIT OF SEGMENT TREES ("tree in tree") - k-th SMALLEST in a[l..r] WITH point assignments, in

```
// O(log^2 n) per operation and O(n log^2 n) nodes.
// WHEN: the array changes. Without updates the answer is a
//   -> persistent segment tree over prefixes
// (01 - Segment Tree/03) at O(log n) per query and a third of the
//   -> memory, or a wavelet tree (13);
// offline, Mo's or parallel binary search is usually simpler. Reach
//   -> for this only for ONLINE
// "k-th smallest / count values <= x in a range" mixed with
//   -> assignments.
// KEY INVARIANT: the BIT is over POSITIONS and every BIT node owns a
//   -> dynamic segment tree over the
// VALUE axis counting the values of the positions that node covers.
//   -> A prefix [0, r] is a sum of
// O(log n) BIT nodes, so [l, r] is O(log n) value-trees to add and
//   -> O(log n) to subtract; the k-th
// smallest is then ONE simultaneous top-down descent through all of
//   -> them at once, adding up the
// left-child counts to decide which way to go.
struct BITSeg {
    int n, V; //
    vector<int> rt, lc, rc, cnt, a; //
    rt[i] = value tree of BIT node i
    BITSeg(int n, int V) : n(n), V(V), rt(n + 1, 0), lc(1, 0), rc(1,
    0), cnt(1, 0), a(n, -1) {}
    int upd(int x, int lo, int hi, int p, int d) { // +=
    d at value p, returns node id
        if ((1 < x) x = cnt.size(), lc.push_back(0), rc.push_back(0),
        cnt.push_back(0);
        cnt[x] += d;
        if (lo < hi) {
            int m = (lo + hi) / 2, c;
            if (p <= m) c = upd(lc[x], lo, m, p, d), lc[x] = c;
            else c = upd(rc[x], m + 1, hi, p, d), rc[x] = c;
        }
        return x;
    }
    void add(int i, int v, int d) {
        for (i++; i <= n; i += i & -1) rt[i] = upd(rt[i], 0, V - 1,
        v, d);
    }
    void set(int i, int v) { //
    a[i] = v (first call = insert)
        if (a[i] >= 0) add(i, a[i], -1);
        add(i, a[i] = v, 1);
    }
    int kth(int l, int r, int k) { //
    k-th smallest in [l, r], k >= 1
        vector<int> A, B;
        for (int i = r + 1; i > 0; i -= i & -1) A.push_back(rt[i]);
        for (int i = l; i > 0; i -= i & -1) B.push_back(rt[i]);
        int lo = 0, hi = V - 1;
        while (lo < hi) {
            int m = (lo + hi) / 2, c = 0;
            for (int x : A) c += cnt[lc[x]];
            for (int x : B) c -= cnt[lc[x]];
            bool left = k <= c;
            if (!left) k -= c, lo = m + 1;
            else hi = m;
            for (int& x : A) x = left ? lc[x] : rc[x];
            for (int& x : B) x = left ? lc[x] : rc[x];
        }
        return lo;
    }
    int less(int x, int lo, int hi, int ql, int qr) { //
    count of values in [ql, qr]
        if (!x || qr < lo || hi < ql) return 0;
        if (ql <= lo && hi <= qr) return cnt[x];
        int m = (lo + hi) / 2;
        return less(lc[x], lo, m, ql, qr) + less(rc[x], m + 1, hi,
        ql, qr);
    }
    int count(int l, int r, int vl, int vr) { //
    how many a[i] in [vl, vr]
        int res = 0;
        for (int i = r + 1; i > 0; i -= i & -1) res += less(rt[i], 0,
        V - 1, vl, vr);
        for (int i = l; i > 0; i -= i & -1) res -= less(rt[i], 0, V -
        1, vl, vr);
        return res;
    }
};
// COORDINATE COMPRESS first (01 - Miscellaneous/05) - and if updates
//   -> introduce NEW values, compress
```

```
// the update values too, offline, before starting.
// MEMORY is the trap:  $n \log^2 n$  nodes at 12 bytes is  $\sim 100$  MB at  $n = 2e5$ . If that is too much,
// replace the inner dynamic tree by a MERGE SORT TREE rebuilt in blocks, or go offline.
// THE SAME SHAPE solves: "sum of the k smallest in a range", "rank of x in a range", "predecessor"
// / successor of x in a range" - all are the same descent with a different accumulator.
```

2.2.6 OfflineDistinct

COUNT DISTINCT VALUES IN A RANGE, OFFLINE - the single most reused sweep in the whole notebook.

```
// Needs: 02 - Binary Indexed Tree (BIT)/01 - FenwickTree.cpp
//  $O((n + q) \log n)$ ,  $O(n)$  memory,  $\sim 20$  lines. (SPOJ DQUERY and a hundred clones.)
// WHEN: any "how many different / how many values appear exactly once / sum of distinct values"
// question with all queries known up front. Mo's (06 - Square Root Decomposition/01) also does it
// but is  $O(n \sqrt{n})$ ; a persistent segment tree does it ONLINE at  $O(\log n)$  per query with the very
// same invariant, so start here and upgrade only if the queries are forced online.
// KEY INVARIANT: sweep r left to right and keep the BIT holding a 1 at the LAST OCCURRENCE SO FAR
// of every distinct value and 0 everywhere else. Then the number of ones in  $[l, r]$  is exactly the
// number of distinct values in  $[l, r]$ : each value is counted at its rightmost occurrence, which
// lies in  $[l, r]$  if and only if the value occurs there at all.
// Moving r to a new position means
// clearing the previous occurrence of a[r] and setting the new one - two BIT updates.
struct OfflineDistinct {
    int n;
    vector<int> a; // values already compressed to [0,V)
    vector<array<int, 3>> qs; // {r, l, id}
    OfflineDistinct(const vector<int>& a) : n(a.size()), a(a) {}
    void ask(int l, int r, int id) { qs.push_back({r, l, id}); }
    vector<int> solve(int V) {
        sort(all(qs));
        FenwickTree<int> bit(n);
        vector<int> last(V, -1), res(qs.size());
        int at = 0;
        for (int r = 0; r < n; r++) {
            if (last[a[r]] >= 0) bit.add(last[a[r]], -1);
            bit.add(last[a[r]] = r, 1);
            while (at < (int)qs.size() && qs[at][0] == r)
                res[qs[at][2]] = bit.query(qs[at][1], r), at++;
        }
        return res;
    }
};
// VALUES APPEARING EXACTLY ONCE in  $[l, r]$ : keep TWO BITS, one with a 1 at the last occurrence and
// one with a 1 at the second-to-last; the answer is ones(last) - ones(secondLast) over  $[l, r]$ .
// SUM of the distinct values: put a[r] instead of 1 at the last occurrence. Any per-value weight
// works the same way, including "count values whose multiplicity is  $\geq k$ " (mark the k-th last).
// ONLINE version: build a persistent segment tree where version r is "the array of last
// occurrences after processing r", then the answer is a range count over version r. Same invariant,
//  $O(n \log n)$  memory (01 - Segment Tree/03).
// UPDATES to the array as well: this sweep dies; use Mo's with updates (06/05) or a BIT of segment
// trees (05) keyed by "previous occurrence" - both are much heavier, so check first whether the
// problem really needs them.
```

2.2.7 BIT2DRangeUpdate

2D BIT WITH RANGE UPDATE AND RANGE QUERY - add v to a RECTANGLE, ask the sum of a RECTANGLE, in

```
//  $O(\log n \log m)$  with four BITS and no lazy propagation anywhere.
// WHEN: grid problems with rectangle add / rectangle sum. 02 - FenwickTree2D only does point
// update; 04 - BITRangeUpdateRangeQuery is the 1D version of exactly this trick; a 2D segment tree
// with lazy propagation does NOT exist in any usable form (you cannot push a tag through two
// dimensions), which is why this identity is the standard answer.
// KEY INVARIANT: work on the 2D difference array d, where  $a[i][j] = \sum d$  over the rectangle
//  $[0..i] \times [0..j]$ . A rectangle add touches exactly FOUR cells of d.
// Expanding the double prefix
// sum gives
//  $pre(x, y) = (x+1)(y+1) * S(d) - (y+1) * S(d*i) - (x+1) * S(d*j) + S(d*i*j)$ 
// over  $i \leq x, j \leq y$  - four independent BIT-able quantities, hence four BITS.
template <typename T = ll>
struct BIT2DRange {
    int n, m;
    vector<vector<T>> b[4];
    BIT2DRange(int n, int m) : n(n), m(m) {
        for (int k = 0; k < 4; k++) b[k].assign(n + 2, vector<T>(m + 2, 0));
    }
    void add(int x, int y, T v) { //  $d[x][y] += v$  (internal)
        if (x > n || y > m) return;
        T c[4] = {v, v * x, v * y, v * x * y};
        for (int i = x + 1; i <= n + 1; i += i & -i)
            for (int j = y + 1; j <= m + 1; j += j & -j)
                for (int k = 0; k < 4; k++) b[k][i][j] += c[k];
    }
    void upd(int x1, int y1, int x2, int y2, T v) { // add v to the rectangle
        add(x1, y1, v), add(x1, y2 + 1, -v), add(x2 + 1, y1, -v),
        add(x2 + 1, y2 + 1, v);
    }
    T pre(int x, int y) { // sum over  $[0..x] \times [0..y]$ 
        if (x < 0 || y < 0) return 0;
        T s[4] = {0, 0, 0, 0};
        for (int i = x + 1; i > 0; i -= i & -i)
            for (int j = y + 1; j > 0; j -= j & -j)
                for (int k = 0; k < 4; k++) s[k] += b[k][i][j];
        return (T)(x + 1) * (y + 1) * s[0] - (T)(y + 1) * s[1] - (T)(x + 1) * s[2] + s[3];
    }
    T qry(int x1, int y1, int x2, int y2) {
        return pre(x2, y2) - pre(x1 - 1, y2) - pre(x2, y1 - 1) + pre(x1 - 1, y1 - 1);
    }
};
// MEMORY is  $4 * n * m$  of your value type - 12.8 GB at  $2000 \times 2000$  with ll. If that hurts, use int
// where safe, or compress one axis and go offline (03 - Offline2DFenwickTree).
// THE SAME EXPANSION generalises: range-add of a polynomial of degree k in each axis needs
//  $(k+1)^2$  BITS. In 1D the arithmetic-progression case is 01 - Segment Tree/13.
// XOR instead of sum: the  $(x+1)(y+1)$  coefficients become parities, so keep four BITS and multiply
// by the parity of  $(x+1)(y+1)$  etc. Usually simpler to just use a 2D segment tree of counts.
// SPARSE GRIDS: replace each inner vector by a hash map, or compress coordinates first - the outer
// structure is unchanged.
```

2.3 Sparse Table

2.3.1 SparseTable

SPARSE TABLE - O(1) range query after O(N log N) build, NO updates.

```
// The operation must be IDEMPOTENT (min, max, gcd, and, or) because
// → the two covering blocks
// OVERLAP. For sum/product/xor - anything not idempotent - use 12 -
// → DisjointSparseTable.cpp
// (still O(1) query) or a Fenwick tree. Guard l <= r: __lg(0) is
// → undefined.
template<typename T, typename F = function<T(T, T)>>
struct SparseTable {
    int n;
    vector<vector<T>> mat;
    F func;

    SparseTable() = default;
    SparseTable(const vector<T>& a, F f = [](T x, T y) { return
        → max(x, y); }) : n(a.size()), func(f) {
        int k = n ? __lg(n) + 1 : 0;
        mat.assign(k, a);
        for (int j = 1; j < k; j++) {
            for (int i = 0; i + (1 << j) <= n; i++) {
                mat[j][i] = func(mat[j - 1][i], mat[j - 1][i + (1 <<
        → (j - 1))]);
            }
        }

        T query(int l, int r) const {
            int j = __lg(r - l + 1);
            return func(mat[j][l], mat[j][r - (1 << j) + 1]);
        }
    };
};
```

2.3.2 SparseTable2D

2D Sparse Table - O(N * M * log N * log M) build, O(1) query

```
template<typename T = ll>
struct SparseTable2D {
    int n, m;
    // jmp[kr][kc][i][j] = max in 2^kr x 2^kc subgrid starting at (i,
    → j)
    vector<vector<vector<vector<T>>>> jmp;

    SparseTable2D(const vector<vector<T>>& grid) {
        n = grid.size();
        if (!n) return;
        m = grid[0].size();
        if (!m) return;

        int log_n = __lg(n) + 1, log_m = __lg(m) + 1;
        jmp.assign(log_n, vector<vector<vector<vector<T>>>>(log_m,
    → vector<vector<T>>(n, vector<T>(m))));

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) {
                jmp[0][0][i][j] = grid[i][j];
            }

            for (int kc = 1; kc < log_m; kc++) {
                for (int i = 0; i < n; i++) {
                    for (int j = 0; j + (1 << kc) <= m; j++) {
                        jmp[0][kc][i][j] = max(jmp[0][kc - 1][i][j],
    → jmp[0][kc - 1][i][j + (1 << (kc - 1))]);
                    }
                }

                for (int kr = 1; kr < log_n; kr++) {
                    for (int kc = 0; kc < log_m; kc++) {
                        for (int i = 0; i + (1 << kr) <= n; i++) {
                            for (int j = 0; j + (1 << kc) <= m; j++) {
                                jmp[kr][kc][i][j] = max(jmp[kr -
    → 1][kc][i][j], jmp[kr - 1][kc][i + (1 << (kr - 1))][j]);
                            }
                        }
                    }
                }
            }

            T query(int r1, int c1, int r2, int c2) const {
                int kr = __lg(r2 - r1 + 1), kc = __lg(c2 - c1 + 1);
                return max({jmp[kr][kc][r1][c1],
                    jmp[kr][kc][r1][c2 - (1 << kc) + 1],
                    jmp[kr][kc][r2 - (1 << kr) + 1][c1],
                    jmp[kr][kc][r2 - (1 << kr) + 1][c2 - (1 << kc) +
    → 1]});
            }
        };
    };
};
```


2.3.3 SquareSparseTable

SQUARE SPARSE TABLE - $O(1)$ query for any SQUARE sub-grid, $O(N*M*\log(\min(N,M)))$ build.

```
// The 2D sparse table (02 - SparseTable2D.cpp) stores one table per
//   (row-power, col-power) pair,
// so it costs  $\log(N)*\log(M)$  memory. If every query is a SQUARE -
//   "best k x k block", "does some
// k x k window contain only ones", template matching on a grid - you
//   only ever need the diagonal
// of that table, which is a factor of log cheaper in both time and
//   memory.
// Query a k x k square: cover it with four overlapping  $2^t \times 2^t$ 
//   squares,  $t = \text{floor}(\log_2 k)$ .
// The operation must be IDEMPOTENT (min/max/gcd/and/or) because
//   those four squares overlap.
template<typename T> ll>
struct SquareSparseTable {
    int n, m;
    // jmp[k][i][j] = max of  $2^k \times 2^k$  square starting at (i, j)
    vector<vector<vector<T>>> jmp;

    SquareSparseTable(const vector<vector<T>>& grid) {
        n = grid.size();
        if (!n) return;
        m = grid[0].size();
        if (!m) return;

        int log_max = __lg(min(n, m)) + 1;
        jmp.assign(log_max, vector<vector<T>>>(n, vector<T>(m)));

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) {
                jmp[0][i][j] = grid[i][j];
            }
        }

        for (int k = 1; k < log_max; k++) {
            int pw = 1 << (k - 1);
            for (int i = 0; i + (pw * 2) <= n; i++) {
                for (int j = 0; j + (pw * 2) <= m; j++) {
                    jmp[k][i][j] = max({jmp[k - 1][i][j],
                                         jmp[k - 1][i][j + pw],
                                         jmp[k - 1][i + pw][j],
                                         jmp[k - 1][i + pw][j + pw]});
                }
            }
        }

        // Queries max in square subgrid from (r1, c1) to (r2, c2)
        T query(int r1, int c1, int r2, int c2) const {
            int k = __lg(r2 - r1 + 1), pw = 1 << k;
            return max({jmp[k][r1][c1],
                       jmp[k][r1][c2 - pw + 1],
                       jmp[k][r2 - pw + 1][c1],
                       jmp[k][r2 - pw + 1][c2 - pw + 1]});
        }
    };
};
```

2.3.4 SqrtTree

SQRT TREE - $O(1)$ range query for ANY associative operation, WITH point updates in $O(\sqrt{n})$.

```
// Build  $O(n \log \log n)$ , memory  $O(n \log \log n)$ .
// WHEN: the operation is associative but NOT idempotent and NOT
//   invertible (matrix product, "max
// suffix sum", min-plus, string hash concat), so a sparse table (01)
//   is wrong and a BIT is
// impossible - and you need  $O(1)$  queries or updates on top. If the
//   op is idempotent use
// 01 - SparseTable; if there are no updates use 12 -
//   DisjointSparseTable ( $O(n \log n)$  build,  $O(1)$ 
// query, 20 lines); if  $O(\log n)$  queries are fine just use a segment
//   tree.
// KEY INVARIANT: cut the array into  $\sqrt{n}$  blocks of  $\sqrt{n}$ . Store
//   prefix and suffix aggregates
// inside every block and a table 'between[i][j]' for whole blocks
//   i..j. A query then costs one
// suffix + one between + one prefix. Blocks are handled RECURSIVELY
//   by the same structure, which
// is why the depth is  $\log \log n$ :  $n \rightarrow \sqrt{n} \rightarrow n^{1/4} \rightarrow \dots$  The
//   top-level 'between' table is
// itself a Sqrt Tree over the  $\sqrt{n}$  block aggregates, so an update
//   rebuilds only one block per
// layer plus one entry of the index structure.
struct SqrtTree {
    typedef ll T;
    static T op(T a, T b) { return a + b; } // ANY
    // associative op
    int n, lg, isz;
    vector<T> v;
    vector<int> clz, layers, onLayer;
    vector<vector<T>> pref, suf, betw;
    void buildBlock(int k, int l, int r) {
        pref[k][l] = v[l];
        for (int i = l + 1; i < r; i++) pref[k][i] = op(pref[k][i -
        1], v[i]);
        suf[k][r - 1] = v[r - 1];
        for (int i = r - 2; i >= l; i--) suf[k][i] = op(v[i],
        suf[k][i + 1]);
    }
    void buildBetween(int k, int lb, int rb, int off) {
        int sl = (layers[k] + 1) >> 1, cl = layers[k] >> 1, bs = 1 <<
        sl;
        int bc = (rb - lb + bs - 1) >> sl;
        for (int i = 0; i < bc; i++) {
            T cur = T();
            for (int j = i; j < bc; j++) {
                T add = suf[k][lb + (j << sl)];
                cur = i == j ? add : op(cur, add);
                betw[k - 1][off + lb + (i << cl) + j] = cur;
            }
        }
    }
    void build(int k, int lb, int rb, int off) {
        if (k >= (int)layers.size()) return;
        int bs = 1 << ((layers[k] + 1) >> 1);
        for (int l = lb; l < rb; l += bs) {
            int r = min(l + bs, rb);
            buildBlock(k, l, r), build(k + 1, l, r, off);
        }
        if (k) buildBetween(k, lb, rb, off);
        else { //
            // layer 0: rebuild the index array
            int sl = (lg + 1) >> 1;
            for (int i = 0; i < isz; i++) v[n + i] = suf[0][i << sl];
            build(1, n, n + isz, (1 << lg) - n);
        }
    }
    void update(int k, int lb, int rb, int off, int x) {
        if (k >= (int)layers.size()) return;
        int sl = (layers[k] + 1) >> 1, bs = 1 << sl, bi = (x - lb) >>
        sl;
        int l = lb + (bi << sl), r = min(l + bs, rb);
        buildBlock(k, l, r);
        if (k) buildBetween(k, lb, rb, off);
        else {
            v[n + bi] = suf[0][bi << ((lg + 1) >> 1)];
            update(1, n, n + isz, (1 << lg) - n, n + bi);
        }
        update(k + 1, l, r, off, x);
    }
    void update(int x, T val) { v[x] = val, update(0, 0, n, 0, x); }
    T query(int l, int r, int off, int base) { //
        // inclusive [l, r]
        if (l == r) return v[l];
        if (l + 1 == r) return op(v[l], v[r]);
    }
};
```

```

    int k = onLayer[clz[(l - base) ^ (r - base)]];
    int sl = (layers[k] + 1) >> 1, cl = layers[k] >> 1;
    int lb = (((l - base) >> layers[k]) << layers[k]) + base;
    int lblk = ((l - lb) >> sl) + 1, rblk = ((r - lb) >> sl) - 1;
    T res = suf[k][l];
    if (lblk <= rblk)
        res = op(res, k ? betw[k - 1][off + lb + (lblk << cl) +
    ↪ rblk]
                : query(n + lblk, n + rblk, (1 << lg) -
    ↪ n, n));
    return op(res, pref[k][r]);
}
T query(int l, int r) { return query(l, r, 0, 0); }
SqrtTree(const vector<T>& a) : n(a.size()), lg(0), v(a) {
    while ((1 << lg) < n) lg++;
    clz.assign(1 << lg, 0), onLayer.assign(lg + 1, 0);
    for (int i = 1; i < (1 << lg); i++) clz[i] = clz[i >> 1] + 1;
    for (int t = lg; t > 1; t = (t + 1) >> 1) onLayer[t] =
    ↪ layers.size(), layers.push_back(t);
    for (int i = lg - 1; i >= 0; i--) onLayer[i] =
    ↪ max(onLayer[i], onLayer[i + 1]);
    int sl = (lg + 1) >> 1, bs = 1 << sl;
    isz = (n + bs - 1) >> sl;
    v.resize(n + isz);
    pref.assign(layers.size(), vector<T>(n + isz));
    suf.assign(layers.size(), vector<T>(n + isz));
    betw.assign(max(0, (int)layers.size() - 1), vector<T>((1 <<
    ↪ lg) + bs));
    build(0, 0, n, 0);
}
};
// NON-COMMUTATIVE ops are fine and are the whole point - op(a, b) is
    ↪ always applied left to right.
// T MUST have a sensible default constructor; it is never used as an
    ↪ identity, only as a filler.
// THE O(1)-QUERY / O(sqrt n)-UPDATE trade is unusual. Its real niche
    ↪ is "answer 1e6 queries of a
// non-invertible op with occasional updates". If updates outnumber
    ↪ queries, a segment tree wins.
// THE TWO-LEVEL VERSION (one layer of blocks, no recursion) is 20
    ↪ lines and gives O(1) query with
// O(n) build only if you can afford a sqrt(n) x sqrt(n)
    ↪ between-table; recursion is what makes the
// update cheap. Write the two-level one first if the array is static
    ↪ and small.

```

2.4 Trie

2.4.1 StringTrie

PREFIX TRIE over an alphabet. O(L) per insert/query. Reach for it for prefix counting, autocomplete,

// "is any inserted word a prefix of this one", and as the skeleton

↪ Aho-Corasick is built on.

// Prefix Trie for Strings

// Time: O(L) per insert/query, Space: O(N * L * Alphabet)

```

struct Trie {
    vector<vector<int>> nxt;
    vector<int> cnt, end_cnt;
    int alpha;

    Trie(int alpha = 26) : alpha(alpha) {
        add_node();
    }

    int add_node() {
        nxt.emplace_back(alpha, -1);
        cnt.emplace_back(0);
        end_cnt.emplace_back(0);
        return nxt.size() - 1;
    }

    void insert(const string& s, int val = 1) {
        int u = 0;
        cnt[u] += val;
        for (char c : s) {
            int ch = c - 'a';
            if (nxt[u][ch] == -1) nxt[u][ch] = add_node();
            u = nxt[u][ch];
            cnt[u] += val;
        }
        end_cnt[u] += val;
    }

    int count_prefix(const string& s) {
        int u = 0;
        for (char c : s) {
            int ch = c - 'a';
            if (nxt[u][ch] == -1) return 0;
            u = nxt[u][ch];
        }
        return cnt[u];
    }
};

```

2.4.2 BinaryTrie

Binary trie over $[0, 2^{(B+1)})$. insert / ERASE / query all $O(B)$.

```
// erase() is what lets you keep the trie equal to the current
// ↳ root->node path during a DFS
// ("min XOR with any ancestor"): add on entry, add(-1) on exit.
struct BTrie {
    static const int B = 30;
    vector<array<int, 2>> ch = {{-1, -1}};
    vector<int> cnt = {0};

    int node() { ch.push_back({-1, -1}); cnt.push_back(0); return
    ↳ ch.size() - 1; }
    void add(ll x, int d = 1) { // d = +1
    ↳ insert, -1 erase
        int u = 0; cnt[u] += d;
        for (int i = B; i >= 0; i--) {
            int b = x >> i & 1;
            if (ch[u][b] < 0) ch[u][b] = node();
            u = ch[u][b], cnt[u] += d;
        }
    }
    int cn(int u, int b) const { return u < 0 || ch[u][b] < 0 ? 0 :
    ↳ cnt[ch[u][b]]; }
    int size() const { return cnt[0]; }

    ll bestXor(ll x, bool mx) const { // mx=1 ->
    ↳ max x^y, mx=0 -> min x^y
        if (!cnt[0]) return -1;
        int u = 0; ll r = 0;
        for (int i = B; i >= 0; i--) {
            int b = (x >> i & 1) ^ mx; // bit we'd
    ↳ like y to have
            if (!cn(u, b)) b ^= 1;
            r |= (ll)((x >> i & 1) ^ b) << i;
            u = ch[u][b];
        }
        return r;
    }
    ll maxXor(ll x) const { return bestXor(x, 1); }
    ll minXor(ll x) const { return bestXor(x, 0); }

    int cntLess(ll x, ll k) const { // #{y in
    ↳ trie : (x ^ y) < k}
        int u = 0, r = 0;
        for (int i = B; i >= 0 && u >= 0; i--) {
            int xb = x >> i & 1;
            if (k >> i & 1) r += cn(u, xb), u = ch[u][xb ^ 1]; //
    ↳ xor-bit 0 branch is all < k
            else u = ch[u][xb];
        }
        return r;
    }
};
```

2.4.3 PersistentTrie

Persistent Binary Trie - Space: $O(N * BITS)$, Time: $O(BITS)$ per insert/query

```
// Supports versioning and range max XOR queries between versions
// ↳ [l_ver, r_ver]
struct PersistentTrie {
    struct Node {
        int cnt = 0;
        array<int, 2> nxt = {-1, -1};
    };

    vector<Node> tree;
    vector<int> roots;
    int max_bit;

    PersistentTrie(int max_bit = 30) : max_bit(max_bit) {
        roots.pb(add_node());
    }

    int add_node(int old_node = -1) {
        if (old_node != -1) {
            tree.pb(tree[old_node]);
        } else {
            tree.pb(Node());
        }
        return tree.size() - 1;
    }

    // Inserts x starting from prev_root and returns the new version
    ↳ root
    int insert(int prev_root, ll x) {
        int new_root = add_node(prev_root);
        int u = new_root, prev_u = prev_root;
        tree[u].cnt++;

        for (int i = max_bit; i >= 0; i--) {
            int b = (x >> i) & 1;
            int prev_child = (prev_u != -1) ? tree[prev_u].nxt[b] :
    ↳ -1;
            tree[u].nxt[b] = add_node(prev_child);
            u = tree[u].nxt[b];
            if (prev_u != -1) prev_u = tree[prev_u].nxt[b];
            tree[u].cnt++;
        }
        roots.pb(new_root);
        return new_root;
    }

    // Returns max (x ^ val) over elements inserted in version range
    ↳ [l_ver, r_ver]
    ll max_xor(int l_ver, int r_ver, ll x) const {
        int u_r = roots[r_ver], u_l = (l_ver > 0) ? roots[l_ver - 1]
    ↳ : -1;
        ll res = 0;

        for (int i = max_bit; i >= 0; i--) {
            int b = (x >> i) & 1;
            int want = b ^ 1;

            int count_r = (u_r != -1 && tree[u_r].nxt[want] != -1) ?
    ↳ tree[tree[u_r].nxt[want]].cnt : 0;
            int count_l = (u_l != -1 && tree[u_l].nxt[want] != -1) ?
    ↳ tree[tree[u_l].nxt[want]].cnt : 0;

            if (count_r - count_l > 0) {
                res |= (1LL << i);
                u_r = tree[u_r].nxt[want];
                u_l = (u_l != -1 && tree[u_l].nxt[want] != -1) ?
    ↳ tree[u_l].nxt[want] : -1;
            } else {
                u_r = (u_r != -1 && tree[u_r].nxt[b] != -1) ?
    ↳ tree[u_r].nxt[b] : -1;
                u_l = (u_l != -1 && tree[u_l].nxt[b] != -1) ?
    ↳ tree[u_l].nxt[b] : -1;
            }
        }
        return res;
    }
};
```

2.5 Disjoint Set Union (DSU)

2.5.1 DSU

Disjoint Set Union (DSU) - Path Compression & Union by Size

```
// Time: O(alpha(N)) per operation
struct DSU {
    vector<int> par, sz;

    DSU(int n = 0) : par(n), sz(n, 1) {
        iota(all(par), 0);
    }

    int find(int x) {
        return par[x] == x ? x : par[x] = find(par[x]);
    }

    bool same(int x, int y) {
        return find(x) == find(y);
    }

    bool join(int x, int y) {
        x = find(x);
        y = find(y);
        if (x == y) return false;
        if (sz[x] < sz[y]) swap(x, y);
        sz[x] += sz[y];
        par[y] = x;
        return true;
    }

    int size(int x) {
        return sz[find(x)];
    }
};
```

2.5.2 DSURollback

DSU with Rollback - No path compression to allow backtracking

```
// Time: O(log N) per operation, O(1) rollback
struct DSURollback {
    vector<int> par, siz;
    vector<pair<int, int>> history; // {child_node, old_parent_size}

    DSURollback(int n = 0) : par(n), sz(n, 1) {
        iota(all(par), 0);
    }

    int find(int x) {
        return par[x] == x ? x : find(par[x]);
    }

    bool same(int x, int y) {
        return find(x) == find(y);
    }

    bool join(int x, int y) {
        x = find(x);
        y = find(y);
        if (x == y) {
            history.eb(-1, -1);
            return false;
        }
        if (siz[x] < siz[y]) swap(x, y);
        history.eb(y, siz[x]);
        siz[x] += siz[y];
        par[y] = x;
        return true;
    }

    int size(int x) {
        return siz[find(x)];
    }

    void rollback() {
        if (history.empty()) return;
        auto [y, old_sz_x] = history.back();
        history.pop_back();
        if (y != -1) {
            siz[par[y]] = old_sz_x;
            par[y] = y;
        }
    }

    int get_state() const {
        return history.size();
    }

    void rollback_to(int state) {
        while ((int)history.size() > state) rollback();
    }
};
```

2.5.3 WeightedDSU

DSU with POTENTIALS: maintains $\text{pot}[v] = \text{value}(v) - \text{value}(\text{root})$, so you can answer

```
// "what is value(u) - value(v)" and detect contradictions.
// Group is (Z, +) here; for parity/bipartiteness use XOR: change ALL
//   ↳ FOUR of  $\text{pot}[x] += p \rightarrow ^=$ ,
//   ↳  $\text{pu} - \text{pv} == w \rightarrow (\text{pu} \wedge \text{pv}) == w$ ,  $\text{pot}[\text{rv}] = \text{pu} - \text{pv} - w \rightarrow \text{pu} \wedge \text{pv} \wedge$ 
//   ↳  $w$ , and DELETE the  $w = -w$  line
// (negation is not the inverse in  $\text{GF}(2)$ ), or  $Z_k$  for mod- $k$  relations.
struct WDSU {
    vector<int> par;
    vector<ll> pot; // potential
    ↳ relative to the parent
    vector<int> sz;

    WDSU(int n = 0) : par(n), pot(n, 0), sz(n, 1) { iota(all(par),
    ↳ 0); }
    pair<int, ll> find(int x) { // {root, value(x) -
    ↳ value(root)}
        if (par[x] == x) return {x, 0};
        auto [r, p] = find(par[x]);
        par[x] = r, pot[x] += p; // path compression
    ↳ accumulates
        return {r, pot[x]};
    }
    // assert value(u) - value(v) == w. false = CONTRADICTION with
    ↳ what we already know.
    bool join(int u, int v, ll w) {
        auto [ru, pu] = find(u);
        auto [rv, pv] = find(v);
        if (ru == rv) return pu - pv == w;
        if (sz[ru] < sz[rv]) swap(ru, rv), swap(pu, pv), w = -w;
        par[rv] = ru, pot[rv] = pu - pv - w, sz[ru] += sz[rv];
        return true;
    }
    // value(u) - value(v), or nullopt if they are not related yet
    bool diff(int u, int v, ll& out) {
        auto [ru, pu] = find(u);
        auto [rv, pv] = find(v);
        if (ru != rv) return false;
        out = pu - pv;
        return true;
    }
};
// BIPARTITENESS under edge additions: use XOR potentials with  $w = 1$ 
//   ↳ per edge; join() returning
// false means an odd cycle appeared. Same structure answers "are u
//   ↳ and v the same colour".
```

2.5.4 RangeParallelDSU

RANGE-PARALLEL DSU ("DSU on a sparse table") - merge two whole RANGES elementwise in $O(\log n)$.

```
// The operation is: " $a[u..u+\text{len}-1]$  must equal  $a[v..v+\text{len}-1]$ "
//   ↳ elementwise", i.e. union  $u+i$  with  $v+i$ 
// for every  $i < \text{len}$ . Doing that naively is  $O(\text{len})$  per constraint;
//   ↳ this does it in  $O(\log n)$ .
// IDEA: keep LOGN separate DSUs.  $\text{par}[k]$  answers "these two blocks of
//   ↳ length  $2^k$  are identical".
// A range constraint of length  $\text{len}$  is covered by TWO blocks of
//   ↳ length  $2^k$  ( $k = \text{floor}(\log_2 \text{len})$ ),
// exactly like a sparse-table query - so one union at level  $k$ ,
//   ↳ twice. Afterwards, PUSH DOWN:
// if two blocks of length  $2^k$  are equal then their two halves of
//   ↳ length  $2^{k-1}$  are equal too.
// After the push-down, level 0 holds the true elementwise DSU.
// USES: "count the strings/arrays satisfying these equal-substring
//   ↳ constraints" (the answer is
// alphabet(#components at level 0)), grid/string pattern
//   ↳ unification, and any problem that says
// "these two intervals must match".
struct RangeDSU {
    int n, LOG;
    vector<vector<int>>> par;
    RangeDSU(int n) : n(n), LOG(__lg(max(n, 1)) + 1), par(LOG,
    ↳ vector<int>(n)) {
        for (int k = 0; k < LOG; k++) iota(all(par[k]), 0);
    }
    int find(int k, int u) { return par[k][u] == u ? u : par[k][u] =
    ↳ find(k, par[k][u]); }
    void join(int k, int u, int v) {
        u = find(k, u), v = find(k, v);
        if (u != v) par[k][u] = v;
    }
    //  $a[u..u+\text{len}-1]$  equals  $a[v..v+\text{len}-1]$ , elementwise.  $O(\log n)$ 
    ↳ amortised.
    void equalRanges(int u, int v, int len) {
        if (len <= 0) return;
        int k = __lg(len);
        join(k, u, v);
        join(k, u + len - (1 << k), v + len - (1 << k));
    }
    // Call ONCE after all the constraints. Then find(0, i) is the
    ↳ real elementwise component.
    void pushDown() {
        for (int k = LOG - 1; k > 0; k--) {
            int half = 1 << (k - 1);
            for (int i = 0; i + (1 << k) <= n; i++) {
                int j = find(k, i);
                if (j == i) continue;
                join(k - 1, i, j); //
            }
            join(k - 1, i + half, j + half);
        }
    }
    int components() { //
    ↳ at level 0, after pushDown
        int c = 0;
        for (int i = 0; i < n; i++) if (find(0, i) == i) c++;
        return c;
    }
};
// The same "cover a range with two power-of-two blocks, then push
//   ↳ down" trick works for any
// idempotent relation, not just equality - it is the DSU analogue of
//   ↳ a sparse table.
// If instead you need to union a range of CONSECUTIVE elements to
//   ↳ each other (" $a[l..r]$  all equal"),
// that is much simpler: a "next unvisited pointer" DSU where find(i)
//   ↳ jumps to the next index not
// yet merged, giving  $O(n \alpha)$  total over all range-unions.
```

2.5.5 PartiallyPersistentDSU

PARTIALLY PERSISTENT DSU - a DSU you may QUERY at any past moment, while only the present can be

```
// modified. O(log n) per query, O(log n) per union, and only O(n)
//   ↳ memory - far cheaper than the
// fully persistent version in 20 - PersistentAndUndo, which costs
//   ↳ O(log^2 n) and O(n log n).
// WHEN: "what is the earliest time at which u and v are connected",
//   ↳ "how big was u's component
// after the first t edges", offline MST / bottleneck-path questions.
//   ↳ Partial persistence is enough
// whenever the timeline is a LINE and not a tree of branching
//   ↳ versions - which is nearly always.
// If you also want to UNDO, use 02 - DSURollback; if you want to
//   ↳ delete arbitrary edges, use
// 14 - OfflineDynamicConnectivity.
// KEY INVARIANT: NO path compression and union by size, so once a
//   ↳ vertex becomes a child it keeps
// that parent FOREVER. Every vertex therefore has a monotone
//   ↳ timeline: a list of component sizes
// while it was a root, then (at most) one final "I became a child of
//   ↳ p at time t" entry. Walking
// up while the stamp is <= t reconstructs the forest at time t
//   ↳ exactly, and the height stays
// O(log n) because of the union by size.
struct PartialDSU {
    vector<vector<pair<int, int>>> par; //
    ↳ {-size or parent, timestamp}
    int T = 0; //
    ↳ number of unions performed
    PartialDSU(int n) : par(n, {{-1, 0}}) {}
    int root(int u, int t) { //
    ↳ representative of u at time t
        auto b = par[u].back();
        return b.fi >= 0 && b.se <= t ? root(b.fi, t) : u;
    }
    bool same(int u, int v, int t) { return root(u, t) == root(v, t);
    ↳ }
    int size(int u, int t) { //
    ↳ |component of u| at time t
        u = root(u, t);
        int lo = 0, hi = par[u].size() - 1, at = 0;
        while (lo <= hi) { //
        ↳ last size entry stamped <= t
            int m = (lo + hi) / 2;
            par[u][m].se <= t ? at = m, lo = m + 1 : hi = m - 1;
        }
        return -par[u][at].fi;
    }
    bool join(int u, int v) { //
    ↳ happens at time ++T
        ++T;
        if ((u = root(u, T)) == (v = root(v, T))) return false;
        if (par[u].back().fi > par[v].back().fi) swap(u, v); // u
        ↳ is the bigger (more negative)
        par[u].push_back({par[u].back().fi + par[v].back().fi, T});
        par[v].push_back({u, T});
        return true;
    }
    int firstConnected(int u, int v) { //
    ↳ earliest time, or -1 if never
        int lo = 1, hi = T, ans = -1;
        while (lo <= hi) {
            int m = (lo + hi) / 2;
            same(u, v, m) ? ans = m, hi = m - 1 : lo = m + 1;
        }
        return ans;
    }
};
// TIME IS THE UNION COUNTER, not the query index: a query "after the
//   ↳ first k edges" must first map
// k to the number of SUCCESSFUL unions among those k edges (keep an
//   ↳ array while you insert).
// BOTTLENECK PATH / minimax edge: sort the edges by weight, union
//   ↳ them in that order, then
// firstConnected(u, v) gives the index of the edge that answers
//   ↳ "minimise the maximum edge on a
// path". That is exactly the Kruskal reconstruction tree (03 -
//   ↳ Graph/08 - MST) in 25 lines less
// code, at O(log^2 n) per query instead of O(1) - use the KRT if you
//   ↳ have many queries.
// OFFLINE ALTERNATIVE: if every query is known up front, PARALLEL
//   ↳ BINARY SEARCH (01 - Misc/17)
// answers all of them with one plain DSU and no persistence at all.
```

2.6 Square Root Decomposition

2.6.1 MoAlgorithm

MO'S ALGORITHM - answers m OFFLINE range queries in $O((n + m) \sqrt{n})$ when you can move an endpoint

```
// by one in O(1) (add/remove an element and maintain the answer).
//   ↳ Sort queries by block of l, then by
// r (alternating direction per block). Use it for distinct counts,
//   ↳ frequency-of-frequency, XOR pairs.
class MoArray {
private:
    struct Query {
        int l, r, idx;
        int bnd;

        bool operator<(const Query& other) const {
            if (bnd != other.bnd)
                return bnd < other.bnd;
            return (bnd & 1) ? (r < other.r) : (r > other.r);
        }
    };

    int n;
    int block_size;
    vector<int> a;

    // --- PROBLEM SPECIFIC STATE ---
    long long ans;
    vector<int> freq;

    inline void add(int idx) {
        if (freq[a[idx]] == 0) ans++;
        freq[a[idx]]++;
    }

    inline void remove(int idx) {
        freq[a[idx]]--;
        if (freq[a[idx]] == 0) ans--;
    }

    inline long long get_answer() const {
        return ans;
    }
    // -----

public:
    MoArray(const vector<int>& arr) : n(arr.size()), a(arr) {
        int max_val = 0;
        for (int x : a) {
            max_val = max(max_val, x);
        }
        freq.assign(max_val + 1, 0);
        ans = 0;
    }

    vector<long long> solve(const vector<pair<int, int>>&
    ↳ raw_queries) {
        int q = raw_queries.size();
        if (q == 0) return {};

        block_size = max(1, (int)(n / sqrt(q)));
        vector<Query> queries(q);

        for (int i = 0; i < q; ++i) {
            queries[i].l = raw_queries[i].first;
            queries[i].r = raw_queries[i].second;
            queries[i].idx = i;
            queries[i].bnd = queries[i].l / block_size;
        }

        sort(queries.begin(), queries.end());

        vector<long long> answers(q);
        int cur_l = 0, cur_r = -1;

        for (const Query& qry : queries) {
            while (cur_l > qry.l) add(--cur_l);
            while (cur_r < qry.r) add(++cur_r);
            while (cur_l < qry.l) remove(cur_l--);
            while (cur_r > qry.r) remove(cur_r--);
            answers[qry.idx] = get_answer();
        }

        return answers;
    }
};
```


2.6.2 MoOnSubtrees

MO'S ON SUBTREES - flatten the tree with an Euler tour so every subtree is a contiguous range,

// then run plain Mo's on it. Same $O((n + q) \sqrt{n})$. Use for "query
 $\hookrightarrow$ over the subtree of v " when no
 // online structure fits (distinct colours in a subtree, k -th most
 $\hookrightarrow$ frequent, etc.).

```
class MoSubtree {
private:
    struct Query {
        int l, r, idx;
        int bnd;

        bool operator<(const Query& other) const {
            if (bnd != other.bnd) return bnd < other.bnd;
            return (bnd & 1) ? (r < other.r) : (r > other.r);
        }
    };

    int n;
    int block_size;
    vector<vector<int>> adj;
    vector<int> in, out, flat, clr;

    // --- PROBLEM SPECIFIC STATE ---
    int mx_freq;
    vector<int> freq;
    vector<long long> sum_freq;

    inline void add(int node_idx) {
        int c = clr[node_idx];
        sum_freq[freq[c]] -= c;
        freq[c]++;
        sum_freq[freq[c]] += c;
        if (freq[c] > mx_freq) mx_freq = freq[c];
    }

    inline void remove(int node_idx) {
        int c = clr[node_idx];
        sum_freq[freq[c]] -= c;
        freq[c]--;
        sum_freq[freq[c]] += c;
        while (mx_freq > 0 && sum_freq[mx_freq] == 0) mx_freq--;
    }

    inline long long get_answer() const {
        return sum_freq[mx_freq];
    }
    // -----

    void dfs(int u, int p) {
        in[u] = flat.size();
        flat.push_back(u);
        for (int v : adj[u]) {
            if (v == p) continue;
            dfs(v, u);
        }
        out[u] = flat.size() - 1;
    }

public:
    MoSubtree(int n, int max_val, const vector<int>& colors)
        : n(n), adj(n + 1), in(n + 1), out(n + 1), clr(colors) {
        mx_freq = 0;
        freq.assign(max_val + 1, 0);
        sum_freq.assign(n + 1, 0);
    }

    void add_edge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    vector<long long> solve(int root, const vector<int>& query_nodes)
 $\hookrightarrow$  {
        flat.reserve(n);
        dfs(root, 0);

        int q = query_nodes.size();
        if (q == 0) return {};

        block_size = max(1, (int)(n / sqrt(q)));
        vector<Query> queries(q);

        for (int i = 0; i < q; ++i) {
            int u = query_nodes[i];
            queries[i].l = in[u];
```

```
            queries[i].r = out[u];
            queries[i].idx = i;
            queries[i].bnd = queries[i].l / block_size;
        }

        sort(queries.begin(), queries.end());

        vector<long long> answers(q);
        int cur_l = 0, cur_r = -1;

        for (const Query& qry : queries) {
            while (cur_l > qry.l) add(flat[--cur_l]);
            while (cur_r < qry.r) add(flat[++cur_r]);
            while (cur_l < qry.l) remove(flat[cur_l++]);
            while (cur_r > qry.r) remove(flat[cur_r--]);
            answers[qry.idx] = get_answer();
        }

        return answers;
    }
};
```

2.6.3 MoOnPaths

MO'S ON TREE PATHS - the Euler tour trick where a vertex appearing TWICE in the range cancels out,

// plus the LCA added back by hand. Answers path queries offline in
 $\hookrightarrow O((n + q) \sqrt{n})$ with no HLD.

```
class MoTreePath {
private:
    int n, LOG, block_size;
    vector<vector<int>> adj, anc;
    vector<int> depth, in, out, flat;
    vector<long long> up, a;
    vector<bool> exist;

    // --- PROBLEM SPECIFIC STATE ---
    long long ans;
    vector<deque<int>> dep; // Tracks nodes by depth

    void add(int idx) {
        if (!dep[depth[idx]].empty()) {
            ans += 1ll * a[dep[depth[idx]].back()] * a[idx];
        }
        dep[depth[idx]].push_back(idx);
    }

    void remove(int idx) {
        if (dep[depth[idx]].size() == 2) {
            ans -= a[*dep[depth[idx]].begin()] *
 $\hookrightarrow$  a[*next(dep[depth[idx]].begin())];
        }
        if (dep[depth[idx]].front() == idx) {
            dep[depth[idx]].pop_front();
        } else {
            dep[depth[idx]].pop_back();
        }
    }

    void update(int idx) {
        int node = flat[idx];
        if (exist[node]) {
            remove(node);
        } else {
            add(node);
        }
        exist[node] = !exist[node];
    }

    inline long long get_answer() const {
        return ans;
    }
    // -----

    void dfs(int u, int p = -1) {
        if (p != -1) {
            anc[u][0] = p;
            up[u] = a[u] * a[u] + up[p];
        } else {
            depth[u] = 0;
            up[u] = a[u] * a[u];
        }
        in[u] = flat.size();
        flat.push_back(u);

        for (int v : adj[u]) {
            if (v == p) continue;
            depth[v] = depth[u] + 1;
            dfs(v, u);
        }

        out[u] = flat.size();
        flat.push_back(u); // Push again for Eulerian tour
    }

    void build_lca() {
        for (int k = 1; k < LOG; k++) {
            for (int i = 1; i <= n; i++) {
                anc[i][k] = anc[anc[i][k - 1]][k - 1];
            }
        }
    }

    int lift(int x, int k) {
        for (int bit = 0; bit < LOG; bit++) {
            if ((1 << bit) & k) x = anc[x][bit];
        }
        return x;
    }
}
```

```
int lca(int u, int v) {
    if (depth[u] < depth[v]) swap(u, v);
    u = lift(u, depth[u] - depth[v]);
    if (u == v) return u;
    for (int bit = LOG - 1; bit >= 0; bit--) {
        if (anc[u][bit] != anc[v][bit]) {
            u = anc[u][bit];
            v = anc[v][bit];
        }
    }
    return anc[u][0];
}

public:
    struct Query {
        int l, r, idx, x = -1;
        int bnd;

        bool operator<(const Query& other) const {
            if (bnd != other.bnd) return bnd < other.bnd;
            return (bnd & 1) ? (r < other.r) : (r > other.r);
        }
    };

    MoTreePath(int n_nodes, const vector<long long>& node_values)
        : n(n_nodes), a(node_values), LOG(log2(n_nodes) + 2) {

        adj.assign(n + 1, vector<int>());
        anc.assign(n + 1, vector<int>(LOG, 0));
        depth.assign(n + 1, 0);
        in.assign(n + 1, 0);
        out.assign(n + 1, 0);
        up.assign(n + 1, 0);
        exist.assign(n + 1, false);
        dep.assign(n + 1, deque<int>());
        ans = 0;
    }

    void add_edge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    // PRIMARY ENTRY POINT: give it the paths as vertex pairs and it
 $\hookrightarrow$  builds the queries itself.
    // (in[], out[] and lca() are only valid AFTER the dfs, so you
 $\hookrightarrow$  cannot build them yourself.)
    vector<long long> solve(int root, const vector<pair<int, int>>&
 $\hookrightarrow$  paths) {
        flat.clear();
        dfs(root);
        build_lca();
        vector<Query> queries;
        for (int i = 0; i < (int)paths.size(); i++) {
            int u = paths[i].first, v = paths[i].second;
            if (in[u] > in[v]) swap(u, v);
            int w = lca(u, v);
            // u inside v's subtree -> plain range; otherwise use
 $\hookrightarrow$  out[u]..in[v] and add back the LCA
            if (w == u) queries.push_back({in[u], in[v], i, -1, 0});
            else queries.push_back({out[u], in[v], i, w, 0});
        }
        return solveRanges(queries);
    }

    vector<long long> solveRanges(vector<Query>& queries) {
        int q = queries.size();
        if (q == 0) return {};

        block_size = max(1, (int)(flat.size() / sqrt(q)));

        for (auto& qry : queries) {
            qry.bnd = qry.l / block_size;
        }

        sort(queries.begin(), queries.end());

        vector<long long> answers(q);
        int cur_l = 0, cur_r = -1;

        for (const Query& qry : queries) {
            while (cur_l > qry.l) update(--cur_l); // GROW
 $\hookrightarrow$  first, then shrink - the other
            while (cur_r < qry.r) update(++cur_r); // order can
 $\hookrightarrow$  pass through an invalid window
            while (cur_l < qry.l) update(cur_l++);
            while (cur_r > qry.r) update(cur_r--);
        }
    }
}
```

```

        answers[qry.idx] = get_answer() + (qry.x != -1 ?
        ↪ up[qry.x] : 0);
    }
    return answers;
}
};

```

2.6.4 BlockDecomposition

SQRT / BLOCK DECOMPOSITION. Split $[0, n)$ into blocks of $B \sim \sqrt{n}$ and keep one aggregate per

```

// block. Point update  $O(1 \cdot B)$ , range query  $O(\sqrt{n})$ . A BIT/segtree
↪ beats it for sum, so reach for
// this when the aggregate is NOT a group / has no lazy tag:
↪ block-sorted arrays, block frequency
// tables, "count values > x in  $[l, r]$ ", range-assign + weird query,
↪ or a per-block rebuild that is
// cheap. General shape below with sum; swap 'agg' and the two
↪ combine lines for anything else.
struct Blocks {
    int n, B;
    vector<ll> a, agg;
    ↪ agg[b] = aggregate of block b
    Blocks(const vector<ll>& v) : n(v.size()), B(max(1, (int)sqrt(n)
    ↪ + 1)), a(v),
        agg((n + B - 1) / B, 0) {
        for (int i = 0; i < n; i++) agg[i / B] += a[i];
    }
    void set(int i, ll v) { agg[i / B] += v - a[i], a[i] = v; } //
    ↪  $O(1)$ ; for non-group aggregates
    ↪ rebuild the block instead,  $O(B)$ 
    ll query(int l, int r) {
    ↪ inclusive,  $O(n/B + B)$ 
        ll s = 0;
        int bl = l / B, br = r / B;
        if (bl == br) { for (int i = l; i <= r; i++) s += a[i];
    ↪ return s; }
        for (int i = l; i < (bl + 1) * B; i++) s += a[i];
        for (int b = bl + 1; b < br; b++) s += agg[b];
        for (int i = br * B; i <= r; i++) s += a[i];
        return s;
    }
};
// RANGE UPDATE, RANGE QUERY without lazy propagation: keep add[b]
↪ per block. A partial block is
// updated element-wise and its aggregate rebuilt; a full block only
↪ bumps add[b]. Query adds
// add[b] * (elements taken from b).
// BLOCK-SORTED ARRAY ("sqrt tree"): store each block also as a
↪ sorted copy. Then
// count of values  $\leq x$  in  $[l, r]$  = partial blocks scanned + binary
↪ search inside full blocks,
//  $O(\sqrt{n} \log n)$  per query,  $O(B \log B)$  per point update (re-sort
↪ one block). This is the
// offline-free alternative to a wavelet tree / merge-sort tree
↪ when updates are present.
// REBUILD TRICK: when an operation is expensive but rare (e.g.
↪ inserting into the middle), buffer
// changes and rebuild the whole decomposition every  $\sqrt{q}$ 
↪ operations  $\rightarrow O((n+q) \sqrt{q})$  total.
// CHOOSING B: query cost  $n/B + B \cdot c$  is minimised at  $B = \sqrt{n \cdot c}$ 
↪ where c is the per-element cost;
// if updates are much more frequent than queries, take bigger
↪ blocks, and vice versa.

```

2.6.5 MoWithUpdates

MO'S WITH UPDATES (3D Mo). Queries $\{l, r, t\}$ where $t = \# \text{updates applied before this query}$.

```

// Block size  $n^{2/3}$  gives  $O(n^{5/3})$  total. Use when the array
↪ changes between queries and no
// online structure fits.
struct MoUpd {
    struct Q { int l, r, t, idx; };
    struct U { int pos, oldV, newV; };

    int n, B;
    vector<int> a, rec;
    ↪ time 0; rec = while recording
    vector<Q> qs;
    vector<U> us;

    // ---- problem state: #distinct in the window ----
    vector<int> cnt;
    int cur = 0;
    void add(int v) { if (cnt[v]++ == 0) cur++; }
    void rem(int v) { if (--cnt[v] == 0) cur--; }
    int answer() const { return cur; }
    // -----

    MoUpd(vector<int> arr, int maxVal) : n(arr.size()), a(arr),
    ↪ rec(arr), cnt(maxVal + 1, 0) {
        B = max(1, (int)pow((double)max(n, 1), 2.0 / 3));
    }
    void addQuery(int l, int r) { qs.push_back({l, r, (int)us.size(),
    ↪ (int)qs.size()}); }
    void addUpdate(int pos, int newV) { us.push_back({pos, rec[pos],
    ↪ newV}); rec[pos] = newV; }

    void applyU(int i, int l, int r) {
    ↪ update i, forwards or backwards
        auto& u = us[i];
        if (l <= u.pos && u.pos <= r) rem(a[u.pos]), add(u.newV);
        a[u.pos] = u.newV;
        swap(u.oldV, u.newV);
    ↪ operation its own inverse
    }

    vector<int> solve() {
        sort(all(qs), [&](const Q& x, const Q& y) {
            int bx = x.l / B, by = y.l / B;
            if (bx != by) return bx < by;
            int cx = x.r / B, cy = y.r / B;
            if (cx != cy) return (bx & 1) ? cx > cy : cx < cy;
            return (cx & 1) ? x.t > y.t : x.t < y.t;
        });
        vector<int> res(qs.size());
        int L = 0, R = -1, T = 0;
        for (auto& q : qs) {
            while (T < q.t) applyU(T++, L, R);
            while (T > q.t) applyU(--T, L, R);
            while (R < q.r) add(a[++R]);
            while (L > q.l) add(a[--L]);
            while (R > q.r) rem(a[R--]);
            while (L < q.l) rem(a[L++]);
            res[q.idx] = answer();
        }
        return res;
    }
};
// NOTE: 'a' stays at time 0; 'rec' tracks values while you record
↪ updates, so applyU() can
// step forwards AND backwards through time using the same swap-based
↪ inverse.

```

2.6.6 MoWithRollback

MO'S WITH ROLLBACK - for structures where ADD is easy but REMOVE is hard or impossible

```
// (running maximum, DSU, "best pair so far", knapsack-ish state).
// Per block of left endpoints: sort by r, keep a persistent right
// ↳ part that only grows, and
// rebuild the short left part from scratch for every query, rolling
// ↳ back afterwards.
// O((n + q) * sqrt(n)) adds, ZERO removes.
struct MoRollback {
    struct Q { int l, r, idx; };
    int n, B;
    vector<ll> a;
    vector<Q> qs;

    // ---- problem state: max (value * count) seen; only add() and a
    // ↳ snapshot/rollback ----
    vector<int> cnt;
    ll best = 0;
    vector<pair<int, ll>> hist; // {value
    // ↳ index, previous best}
    void add(int v, bool rec) {
        if (rec) hist.push_back({v, best});
        cnt[v]++;
        best = max(best, (ll)cnt[v] * v);
    }
    void rollback(size_t to) {
        while (hist.size() > to) {
            auto [v, b] = hist.back(); hist.pop_back();
            cnt[v]--; best = b;
        }
    }
    void clearAll() { fill(all(cnt), 0), best = 0, hist.clear(); }
    ll answer() const { return best; }
    //
    // -----
    MoRollback(vector<ll> arr, int maxVal) : n(arr.size()), a(arr),
    // ↳ cnt(maxVal + 1, 0) {
        B = max(1, (int)sqrtl((ll)max(1, n)));
    }
    void addQuery(int l, int r) { qs.push_back({l, r,
    // ↳ (int)qs.size()}); }

    vector<ll> solve() {
        sort(all(qs), [&](const Q& x, const Q& y) {
            int bx = x.l / B, by = y.l / B;
            return bx != by ? bx < by : x.r < y.r;
        });
        vector<ll> res(qs.size());
        size_t i = 0;
        while (i < qs.size()) {
            int b = qs[i].l / B, rEnd = min(n - 1, (b + 1) * B - 1);
            size_t j = i;
            while (j < qs.size() && qs[j].l / B == b) j++;
            // pass 1: queries contained in this block - answer each
            // ↳ from a CLEAN state
            for (size_t k2 = i; k2 < j; k2++) if (qs[k2].r <= rEnd) {
                clearAll();
                for (int k = qs[k2].l; k <= qs[k2].r; k++) add(a[k],
            // ↳ false);
                res[qs[k2].idx] = answer();
            }
            // pass 2: long queries - right part grows monotonically,
            // ↳ left part is rolled back
            clearAll();
            int R = rEnd;
            for (size_t k2 = i; k2 < j; k2++) if (qs[k2].r > rEnd) {
                while (R < qs[k2].r) add(a[++R], false);
            }
            // ↳ grow-only, not recorded
            size_t snap = hist.size();
            for (int k = qs[k2].l; k <= rEnd; k++) add(a[k],
            // ↳ true); // short left part
            res[qs[k2].idx] = answer();
            rollback(snap);
        }
        i = j;
    }
    return res;
};
```

2.6.7 RangeMode

RANGE MODE - the most frequent value in a range, and its frequency. Both structures below have

```
// O(1) ADD and O(1) DELETE, so dropped into Mo's (01 - MoAlgorithm)
// ↳ they solve the problem in
// O(n * sqrt(n)) total.
// WHEN: mode / "maximum multiplicity" queries. There is no known
// ↳ linear-space polylog-per-query
// solution, so sqrt IS the intended answer here, not a fallback. The
// ↳ mode is NOT decomposable -
// the mode of a union is not a function of the two modes - which is
// ↳ why a segment tree cannot do
// it and why the counter has to be maintained incrementally.
// KEY INVARIANT (both): adding one element raises some count by
// ↳ exactly one, so the running
// maximum can only go UP by one; deleting lowers one count by one,
// ↳ so the maximum can only go DOWN
// by one, and only when the deleted value was the SOLE holder of it.
// ↳ A count-of-counts array turns
// "is it the sole holder" into an O(1) test.

// --- max FREQUENCY only. O(1) everything, O(V) memory. Use this
// ↳ unless you need the value. ---
struct FreqCounter {
    vector<int> cnt, f; //
    // ↳ f[c] = #values with count == c
    int best = 0;
    FreqCounter(int V, int n) : cnt(V, 0), f(n + 2, 0) {}
    void add(int v) { f[cnt[v]]--, f[++cnt[v]]++, best = max(best,
    // ↳ cnt[v]); }
    void del(int v) {
        f[cnt[v]]--;
        if (cnt[v] == best && !f[cnt[v]]) best--;
        f[--cnt[v]]++;
    }
    int query() { return best; } // the
    // ↳ frequency of the mode
};
// --- the VALUE too: bucket the value axis and keep a max per
// ↳ bucket. O(sqrt(V)) query. ---
struct ModeCounter {
    int V, B, W;
    vector<int> cnt, bmx, bc; //
    // ↳ bc[b * W + c] flattened
    ModeCounter(int V, int n) : V(V), B(max(1, (int)sqrtl((double)V) +
    // ↳ 1)), W(n + 2),
        cnt(V, 0), bmx(V / B + 2, 0), bc((V /
    // ↳ B + 2) * W, 0) {}
    void add(int v) {
        int b = v / B;
        bc[b * W + cnt[v]]++, bc[b * W + ++cnt[v]]++;
        bmx[b] = max(bmx[b], cnt[v]);
    }
    void del(int v) {
        int b = v / B;
        if (bmx[b] == cnt[v] && bc[b * W + cnt[v]] == 1) bmx[b]--;
        bc[b * W + cnt[v]]--, bc[b * W + --cnt[v]]++;
    }
    pii query() { //
    // ↳ {value, count}, O(sqrt(V))
        int best = 0;
        for (int x : bmx) best = max(best, x);
        for (int b = 0; b < (int)bmx.size(); b++)
            if (bmx[b] == best)
                for (int v = b * B; v < min(V, (b + 1) * B); v++)
                    if (cnt[v] == best) return {v, best};
        return {-1, 0};
    }
};
// MEMORY WARNING: bc is O(sqrt(V) * n) ints - about 160 MB at V = n
// ↳ = 2e5. Shrink it by passing
// the largest range length as n, by widening B (query cost O(V/B +
// ↳ B), memory O(V/B * n)), or by
// dropping to FreqCounter when the value itself is not needed.
// COORDINATE COMPRESS the values first so V is the number of
// ↳ DISTINCT values (01 - Misc/05).
// BECAUSE DELETE IS O(1) you do not need Mo's with rollback (06) -
// ↳ plain Mo's is enough.
// THE ONLINE VERSION, O(sqrt(n)) per query and no Mo's: cut the array
// ↳ into sqrt(n) blocks,
// precompute the mode of every block RANGE [i..j] and the prefix
// ↳ count of every value per block.
// A query starts from the precomputed mode of its maximal inner
// ↳ block range and folds in the
// O(sqrt(n)) leftover elements one at a time; each can only raise the
// ↳ answer, and the count of any
// value in the current range is (prefix count) + (leftovers seen so
```

```

    ↪ far), i.e.  $O(1)$ . The same
// scaffolding answers any statistic that is monotone under adding
    ↪ one element.
// MEX WITH UPDATES is the same shape: keep cnt[] plus f[], and scan
    ↪ f[1], f[2], ... for the first
// zero.  $O(\sqrt{n})$  per query inside Mo's,  $O(1)$  amortised if elements
    ↪ are only ever added.

```

2.6.8 MoOnTwoIntervals

MO'S ON TWO INTERVALS ("4D Mo") - each query is a PAIR of ranges $[l1, r1]$ and $[l2, r2]$ and the

```

// answer depends on the multiset union of both. Four pointers
    ↪ instead of two.
// COMPLEXITY: with block size B the cost is  $O(q * B)$  for the three
    ↪ bucketed pointers plus
//  $O(n^4 / B^3)$  for the monotone one;  $B \sim n^{(3/4)}$  balances them at
    ↪  $O(n^{(7/4)})$  when  $q \sim n$ .
// WHEN: "compare two subtrees / two subarrays" - e.g. the most
    ↪ common value in the union of two
// subtrees (CF 1767F). Both ranges must be flattened to intervals
    ↪ first: for subtrees use the
// Euler tour (09 - Trees/06), which is what makes a tree query a
    ↪ pair of intervals.
// Do NOT reach for this before checking whether the two ranges can
    ↪ be merged into one, or whether
// the answer is decomposable (then a segment tree merge, 16 -
    ↪ SegTreeMerging, is far faster).
// KEY INVARIANT: the same amortisation as ordinary Mo's, one level
    ↪ deeper. Sort by
//  $(l1/B, r1/B, l2/B, r2)$ : the first three keys change  $O(B)$  per query
    ↪ each because they are
// bucketed, and the last one is monotone inside every fixed triple
    ↪ of buckets, so it sweeps the
// array once per triple. The state must support ADD and DELETE in
    ↪  $O(1)$  - it is shared by both
// intervals, so an element can be inserted twice and that must be
    ↪ meaningful.
struct Mo2 {
    struct Q { int l1, r1, l2, r2, id; };
    int n, B;
    vector<Q> qs;
    Mo2(int n, int q) : n(n), B(max(1, (int)pow((double)n, 0.75))) {
        ↪ qs.reserve(q); }
    void addQuery(int l1, int r1, int l2, int r2, int id) {
        ↪ qs.push_back({l1, r1, l2, r2, id}); }
    // add(i) / del(i) update the shared state with position i; get()
        ↪ reads the answer.
    template <class A, class D, class G>
    void run(A add, D del, G get) {
        int Bl = B;
        sort(all(qs), [&](const Q& a, const Q& b) {
            array<int, 7> x{a.l1 / Bl, a.r1 / Bl, a.l2 / Bl, a.r2,
        ↪ a.l1, a.r1, a.l2};
            array<int, 7> y{b.l1 / Bl, b.r1 / Bl, b.l2 / Bl, b.r2,
        ↪ b.l1, b.r1, b.l2};
            return x < y;
        });
        int a = 0, b = -1, c = 0, d = -1; // [a,
        ↪ b] and [c, d], both empty
        for (auto& q : qs) {
            while (b < q.r1) add(++b);
            while (a > q.l1) add(--a);
            while (b > q.r1) del(b--);
            while (a < q.l1) del(a--);
            while (d < q.r2) add(++d);
            while (c > q.l2) add(--c);
            while (d > q.r2) del(d--);
            while (c < q.l2) del(c--);
            get(q.id);
        }
    }
};
// THE ORDER OF THE FOUR LOOPS MATTERS: always GROW before you
    ↪ SHRINK, exactly as in 1D Mo's,
// or the two ranges can transiently cross into an invalid state
    ↪ (negative counts).
// OVERLAPPING RANGES: if the two intervals may intersect, an element
    ↪ gets added twice. Decide up
// front whether the state should count it twice (multiset union) or
    ↪ once (set union) - for set
// union keep a per-position "times present" counter and only touch
    ↪ the state on  $0 \leftrightarrow 1$ .
// TREE PAIRS:  $l1 = tin[u], r1 = tout[u], l2 = tin[v], r2 = tout[v]$ .
    ↪ If u is an ancestor of v the
// intervals nest, so the "counted twice" question above is not
    ↪ hypothetical.
// 3+ INTERVALS: the same sort with more keys works but the exponent
    ↪ gets ugly fast - at that point
// look for a completely different decomposition.

```

2.6.9 BlockList

BLOCK LIST (chunked / unrolled list, a "sqrt rope") - an array that supports insert and erase in

```
// the middle, RANGE ASSIGN, RANGE REVERSE, and range statistics, all
// in O(sqrt n).
// WHEN: an implicit treap (07 - BBST/02) does the same in O(log n)
// and is usually the right answer.
// Reach for this instead when the per-block statistic is something a
// treap node cannot merge in
// O(1) - "how many elements >= x in this range" (each block keeps a
// SORTED copy), "the k-th
// smallest", "count of distinct" - or when you want a structure you
// can debug by printing.
// COMPLEXITY: O(sqrt n) per operation amortised, with a full rebuild
// every ~sqrt n operations to
// stop the blocks from fragmenting. Rebuild is O(n).
// KEY INVARIANT: the array is a LIST OF BLOCKS, each holding at most
// ~2B elements plus two lazy
// tags (assigned-value, reversed). Every range operation starts by
// SPLITTING at both endpoints, so
// afterwards the range is an exact run of whole blocks and each one
// can be tagged in O(1). A
// reverse then only has to flip each block's tag and reverse the
// ORDER OF THE BLOCK LIST itself.
struct BlockList {
    struct Block {
        vector<int> v, s; // s =
        // sorted copy of v
        int len = 0, asg = -1; // asg
        // >= 0: whole block is asg
        bool rev = false;
        void resort() { s = v, sort(all(s)); }
        void solid() { //
        // drop both tags
        if (asg >= 0) v.assign(len, asg), asg = -1, resort();
        if (rev) std::reverse(all(v)), rev = false;
        }
        int atLeast(int x) { // how
        // many elements >= x
        if (asg >= 0) return asg >= x ? len : 0;
        return s.end() - lower_bound(all(s), x);
        }
    };
    vector<Block> b;
    int n, B, ops = 0;
    BlockList(const vector<int>& a) : n(a.size()), B(max(1,
    // (int)sqrt((double)n + 1))) { chunk(a); }
    void chunk(const vector<int>& a) {
        b.clear();
        for (int i = 0; i < (int)a.size(); i += B) {
            Block x;
            x.v.assign(a.begin() + i, a.begin() + min((int)a.size(),
            // i + B));
            x.len = x.v.size(), x.resort();
            b.push_back(x);
        }
    }
    void rebuild() { //
    // O(n), every ~B operations
    vector<int> a;
    a.reserve(n);
    for (auto& x : b) {
        x.solid();
        for (int y : x.v) a.push_back(y);
    }
    chunk(a);
    }
    int split(int i) { //
    // index of the block starting at i
    if (i >= n) return b.size();
    int cur = 0;
    for (int k = 0; k < (int)b.size(); k++) {
        if (i < cur + b[k].len) {
            if (i == cur) return k;
            int off = i - cur;
            b[k].solid();
            Block nb;
            nb.v.assign(b[k].v.begin() + off, b[k].v.end());
            nb.len = nb.v.size(), nb.resort();
            b[k].v.resize(off), b[k].len = off, b[k].resort();
            b.insert(b.begin() + k + 1, nb);
            return k + 1;
        }
        cur += b[k].len;
    }
    return b.size();
    }
};
```

```
void tick() { if (++ops % B == 0) rebuild(); } //
// call after every modification
void assign(int l, int r, int x) {
    int p = split(l), q = split(r + 1);
    for (int k = p; k < q; k++) b[k].asg = x, b[k].rev = false;
    tick();
}
void reverse(int l, int r) {
    int p = split(l), q = split(r + 1);
    for (int k = p; k < q; k++) b[k].rev ^= 1;
    std::reverse(b.begin() + p, b.begin() + q);
    tick();
}
void insert(int i, int x) {
    int p = split(i);
    Block nb;
    nb.v = {x}, nb.len = 1, nb.resort();
    b.insert(b.begin() + p, nb), n++;
    tick();
}
void erase(int l, int r) {
    int p = split(l), q = split(r + 1);
    b.erase(b.begin() + p, b.begin() + q), n -= r - l + 1;
    tick();
}
int count(int l, int r, int x) { // how
// many a[i] >= x in [l, r]
    int p = split(l), q = split(r + 1), res = 0;
    for (int k = p; k < q; k++) res += b[k].atLeast(x);
    return res;
}
};
// THE TWO TAGS INTERACT: assigning kills a pending reverse (all
// values are equal), reversing an
// assigned block is a no-op. solid() is the only place that has to
// get the order right.
// OTHER PER-BLOCK STATISTICS drop straight in: a frequency map
// (count of a value), a prefix sum
// (range sum with range assign), a max, a gcd. Anything you can
// rebuild in O(B) works.
// WITHOUT insert / erase / reverse, do not write this: 04 -
// BlockDecomposition is half the code.
// WITHOUT the sorted-copy statistic, an implicit treap is faster and
// shorter.
```

2.7 Balanced Binary Search Tree (BBST)

2.7.1 Treap

TREAP (keyed) - a BST by key and a heap by random priority, so it is balanced in expectation.

```
// NOTE: 01 - Miscellaneous/04 - random.cpp declares the same 'rng'.
    ↳ Keep one copy.
// O(log n) insert/erase/find, plus split and merge, which is what
    ↳ makes it better than std::set:
// you can cut the tree at a key, operate on a whole range, and glue
    ↳ it back.
mt19937_64
    ↳ rng(chrono::steady_clock::now().time_since_epoch().count());
```

```
template <typename T>
struct Treap {
    struct Node {
        T key, val, sum;
        T lazy = 0; // 1. Added lazy tag
        int sz = 1;
        uint64_t p = rng();
        Node *l = nullptr, *r = nullptr;

        Node(T k, T v = T()) : key(k), val(v), sum(v) {}
    } * root = nullptr;

    ~Treap() { destroy(root); }

    void destroy(Node* t) {
        if (!t) return;
        destroy(t->l);
        destroy(t->r);
        delete t;
    }

    int sz(Node* t) { return t ? t->sz : 0; }
    T sub_val(Node* t) { return t ? t->sum : 0; }

    // 2. The Push Function: Propagates pending updates to children
    void push(Node* t) {
        if (!t || t->lazy == 0) return;

        if (t->l) {
            t->l->key += t->lazy; // Shift the key (critical for the
            ↳ DP solution)
            t->l->val += t->lazy; // Shift the value
            t->l->sum += t->lazy * sz(t->l); // Update subtree sum
            t->l->lazy += t->lazy; // Pass the tag down
        }
        if (t->r) {
            t->r->key += t->lazy;
            t->r->val += t->lazy;
            t->r->sum += t->lazy * sz(t->r);
            t->r->lazy += t->lazy;
        }
        t->lazy = 0; // Clear the tag on the current node
    }

    Node* update(Node* t) {
        if (t) {
            t->sz = 1 + sz(t->l) + sz(t->r);
            t->sum = t->val + sub_val(t->l) + sub_val(t->r);
        }
        return t;
    }

    Node* merge(Node* l, Node* r) {
        push(l); // 3. Push before making routing decisions
        push(r);
        if (!l || !r) return l ? l : r;
        if (l->p > r->p) {
            l->r = merge(l->r, r);
            return update(l);
        }
        r->l = merge(l, r->l);
        return update(r);
    }

    pair<Node*, Node*> split(Node* t, T k) {
        if (!t) return {nullptr, nullptr};
        push(t); // 3. Push before evaluating keys for splits
        if (t->key <= k) {
            auto [l, r] = split(t->r, k);
            t->r = l;
            return {update(t), r};
        }
        auto [l, r] = split(t->l, k);
        t->l = r;
        return {l, update(t)};
    }
};
```

```
}

// --- Core Operations ---

// Range Update Method
void add_range(T l_val, T r_val, T add_val) {
    if (l_val > r_val) return;

    auto [left_mid, right] = split(root, r_val);
    auto [left, mid] = split(left_mid, l_val - 1);

    // Apply the lazy tag to the entire isolated segment
    if (mid) {
        mid->key += add_val;
        mid->val += add_val;
        mid->sum += add_val * sz(mid);
        mid->lazy += add_val;
    }

    root = merge(merge(left, mid), right);
}

bool search(T x) {
    Node* curr = root;
    while (curr) {
        push(curr); // Push during traversal
        if (curr->key == x) return true;
        curr = (x < curr->key) ? curr->l : curr->r;
    }
    return false;
}

void insert(T x, T val = T()) {
    auto [l, r] = split(root, x);
    auto [l2, r2] = split(l, x - 1);
    if (!r2) {
        r2 = new Node(x, val);
    } else {
        push(r2); // Push before modifying
        r2->val += val;
        update(r2);
    }
    root = merge(merge(l2, r2), r);
}

void erase(T x) {
    auto [l, r] = split(root, x);
    auto [l2, r2] = split(l, x - 1);
    destroy(r2);
    root = merge(l2, r);
}

// --- Utility Operations ---

Node* find_by_order(Node* t, int k) {
    if (!t) return nullptr;
    push(t); // Push before accessing children sizes
    int left_sz = sz(t->l);
    if (k < left_sz) return find_by_order(t->l, k);
    if (k == left_sz) return t;
    return find_by_order(t->r, k - left_sz - 1);
}

Node* find_by_order(int k) { return find_by_order(root, k); }

int order_of_key(Node* t, T k) {
    if (!t) return 0;
    push(t); // Push before comparing keys
    if (t->key >= k) return order_of_key(t->l, k);
    return sz(t->l) + 1 + order_of_key(t->r, k);
}

int order_of_key(T k) { return order_of_key(root, k); }

void print(Node* t) {
    if (!t) return;
    push(t); // Push before printing to get true values
    print(t->l);
    cout << t->key << " ";
    print(t->r);
}

void print() {
    print(root);
    cout << '\n';
}

};
```

2.7.2 ImplicitTreap

IMPLICIT TREAP - a treap keyed by POSITION (subtree size) instead of value, i.e. an array that

```
// NOTE: 01 - Miscellaneous/04 - random.cpp declares the same 'rng'.
    ↳ Keep one copy.
// supports insert/erase in the middle, reverse a range, and move a
    ↳ block, all in O(log n).
// This is the structure for "cut [l,r] out and paste it elsewhere"
    ↳ problems.
mt19937_64
    ↳ rng(chrono::steady_clock::now().time_since_epoch().count());
```

```
template <typename T>
struct ImplicitTreap {
    struct Node {
        T val, sum;
        T lazy = 0; // Lazy tag for range additions
        bool rev = false; // Lazy tag for range reversals
        int sz = 1;
        uint64_t p = rng();
        Node *l = nullptr, *r = nullptr;

        Node(T v = T()) : val(v), sum(v) {}
    } * root = nullptr;

    // Default Constructor
    ImplicitTreap() = default;

    // Vector Constructor - Builds treap in O(N) time
    ImplicitTreap(const vector<T>& a) {
        build(a);
    }

    ~ImplicitTreap() { destroy(root); }

    void destroy(Node* t) {
        if (!t) return;
        destroy(t->l);
        destroy(t->r);
        delete t;
    }

    int sz(Node* t) { return t ? t->sz : 0; }
    T sub_val(Node* t) { return t ? t->sum : 0; }

    // Propagates pending lazy updates to children
    void push(Node* t) {
        if (!t) return;

        // 1. Process pending reversal
        if (t->rev) {
            swap(t->l, t->r);
            if (t->l) t->l->rev ^= true;
            if (t->r) t->r->rev ^= true;
            t->rev = false;
        }

        // 2. Process pending value addition
        if (t->lazy != 0) {
            if (t->l) {
                t->l->val += t->lazy;
                t->l->sum += t->lazy * sz(t->l);
                t->l->lazy += t->lazy;
            }
            if (t->r) {
                t->r->val += t->lazy;
                t->r->sum += t->lazy * sz(t->r);
                t->r->lazy += t->lazy;
            }
            t->lazy = 0; // Clear the tag
        }
    }

    Node* update(Node* t) {
        if (t) {
            t->sz = 1 + sz(t->l) + sz(t->r);
            t->sum = t->val + sub_val(t->l) + sub_val(t->r);
        }
        return t;
    }

    // --- O(N) Linear Time Build ---
    void build(const vector<T>& a) {
        destroy(root);
        root = nullptr;
        if (a.empty()) return;

```

```
vector<Node*> st;
for (const T& val : a) {
    Node* curr = new Node(val);
    Node* last = nullptr;

    // Maintain max-heap property on random priority 'p'
    while (!st.empty() && st.back()->p < curr->p) {
        last = st.back();
        st.pop_back();
    }

    curr->l = last;
    if (!st.empty()) {
        st.back()->r = curr;
    }
    st.push_back(curr);
}

// Post-order pass to compute subtree sizes and sums in O(N)
auto recompute = [&](auto& self, Node* t) -> void {
    if (!t) return;
    self(self, t->l);
    self(self, t->r);
    update(t);
};

root = st.front(); // Bottom of stack is the root
recompute(recompute, root);
}

// --- Dynamic Array Operations ---

Node* merge(Node* l, Node* r) {
    push(l);
    push(r);
    if (!l || !r) return l ? l : r;
    if (l->p > r->p) {
        l->r = merge(l->r, r);
        return update(l);
    }
    r->l = merge(l, r->l);
    return update(r);
}

pair<Node*, Node*> split(Node* t, int k) {
    if (!t) return {nullptr, nullptr};
    push(t);

    int left_sz = sz(t->l);
    if (left_sz < k) {
        auto [l, r] = split(t->r, k - left_sz - 1);
        t->r = l;
        return {update(t), r};
    } else {
        auto [l, r] = split(t->l, k);
        t->l = r;
        return {l, update(t)};
    }
}

int size() { return sz(root); }

void insert(int idx, T val) {
    auto [l, r] = split(root, idx);
    Node* node = new Node(val);
    root = merge(merge(l, node), r);
}

void erase(int idx) {
    if (idx < 0 || idx >= sz(root)) return;
    auto [left_mid, right] = split(root, idx + 1);
    auto [left, mid] = split(left_mid, idx);
    destroy(mid);
    root = merge(left, right);
}

void reverse_range(int l, int r) {
    if (l >= r || l < 0 || r >= sz(root)) return;

    auto [left_mid, right] = split(root, r + 1);
    auto [left, mid] = split(left_mid, l);

    if (mid) mid->rev ^= true;

    root = merge(merge(left, mid), right);
}
```

```

void add_range(int l, int r, T add_val) {
    if (l > r || l < 0 || r >= sz(root)) return;

    auto [left_mid, right] = split(root, r + 1);
    auto [left, mid] = split(left_mid, l);

    if (mid) {
        mid->val += add_val;
        mid->sum += add_val * sz(mid);
        mid->lazy += add_val;
    }

    root = merge(merge(left, mid), right);
}

T query_range(int l, int r) {
    if (l > r || l < 0 || r >= sz(root)) return T();

    auto [left_mid, right] = split(root, r + 1);
    auto [left, mid] = split(left_mid, l);

    T ans = sub_val(mid);

    root = merge(merge(left, mid), right);
    return ans;
}

T get(int idx) {
    return query_range(idx, idx);
}

void print(Node* t) {
    if (!t) return;
    push(t);
    print(t->l);
    cout << t->val << " ";
    print(t->r);
}

void print() {
    print(root);
    cout << '\n';
}
};

```

2.7.3 Persistent Treap

PERSISTENT IMPLICIT TREAP - a persistent ARRAY that supports insert / erase / concatenate / cut

```

// a block in the middle, with every version staying queryable. O(log
    ↳ n) time and O(log n) new
// nodes per operation.
// WHEN: "version v of the text, then insert this string at position
    ↳ k", copy-paste editors, and
// the killer application: COPY A RANGE OF AN OLD VERSION AND PASTE
    ↳ IT ANYWHERE - two splits of an
// old root share all their nodes with it, so the copy is free. A
    ↳ persistent segment tree
// (01 - Segment Tree/03) cannot do this because its index set is
    ↳ fixed; 02 - ImplicitTreap can,
// but destroys the old version.
// KEY INVARIANT: COPY ON WRITE. split and merge only ever touch one
    ↳ root-to-leaf path, so cloning
// each node before modifying it costs O(log n) extra nodes and
    ↳ leaves every earlier root pointing
// at a consistent tree. Priorities are copied too, so the shape (and
    ↳ hence the depth bound) is
// unchanged by cloning.
struct PTreap {
    struct Node { int l = 0, r = 0, sz = 0; unsigned pr = 0; ll val =
        ↳ 0, sum = 0; };
    vector<Node> t;
    PTreap() : t(1) {} //
    ↳ node 0 = the empty tree
    unsigned rnd() {
        static unsigned s = 88172645u;
        return s ^= s << 13, s ^= s >> 17, s ^= s << 5;
    }
    int cp(int x) { t.push_back(t[x]); return t.size() - 1; }
    void pull(int x) {
        t[x].sz = t[t[x].l].sz + 1 + t[t[x].r].sz;
        t[x].sum = t[t[x].l].sum + t[x].val + t[t[x].r].sum;
    }
    pair<int, int> split(int x, int k) { //
        ↳ first k elements go left
        if (!x) return {0, 0};
        int c = cp(x);
        if (t[t[c].l].sz < k) {
            auto [a, b] = split(t[c].r, k - t[t[c].l].sz - 1);
            t[c].r = a, pull(c);
            return {c, b};
        }
        auto [a, b] = split(t[c].l, k);
        t[c].l = b, pull(c);
        return {a, c};
    }
    int merge(int a, int b) {
        if (!a || !b) return a | b;
        if (t[a].pr > t[b].pr) {
            int c = cp(a), r = merge(t[c].r, b);
            t[c].r = r, pull(c);
            return c;
        }
        int c = cp(b), l = merge(a, t[c].l);
        t[c].l = l, pull(c);
        return c;
    }
    int mk(ll v) { t.push_back({0, 0, 1, rnd(), v, v}); return
        ↳ t.size() - 1; }
    int build(const vector<ll>& a) { // O(n
        ↳ log n), fine for a one-off
        int r = 0;
        for (ll v : a) r = merge(r, mk(v));
        return r;
    }
    int insert(int root, int pos, ll v) { //
        ↳ returns the new root
        auto [a, b] = split(root, pos);
        return merge(merge(a, mk(v)), b);
    }
    int erase(int root, int l, int r) { //
        ↳ drop positions [l, r]
        auto [a, b] = split(root, l);
        auto [m, c] = split(b, r - l + 1);
        return merge(a, c);
    }
    int cut(int root, int l, int r) { // the
        ↳ block [l, r] as its own tree
        auto [a, b] = split(root, l);
        return split(b, r - l + 1).first;
    }
    int paste(int root, int pos, int piece) { //
        ↳ splice another tree in

```

```

    auto [a, b] = split(root, pos);
    return merge(merge(a, piece), b);
}
ll sum(int x, int l, int r) {
    // read-only: allocates nothing
    if (!x || l > r || r < 0 || l >= t[x].sz) return 0;
    if (l <= 0 && r >= t[x].sz - 1) return t[x].sum;
    int L = t[t[x].l].sz;
    return sum(t[x].l, l, min(r, L - 1)) + (l <= L && L <= r ?
    t[x].val : 0) +
        sum(t[x].r, max(0, l - L - 1), r - L - 1);
}
ll get(int root, int i) { return sum(root, i, i); }
};

// LAZY TAGS (range add, range reverse) DO work, but push() must
//   ↳ CLONE both children before
// pushing into them - otherwise you mutate a node an old version
//   ↳ still owns. That is one extra
// line in push and doubles the node count; write it only if the
//   ↳ problem needs it.
// MEMORY is the binding constraint: O(q log n) nodes at ~32 bytes.
//   ↳ reserve() and, if a version is
// provably dead, do not bother freeing - contests end first.
// IF YOU ONLY NEED THE LATEST VERSION use 02 - ImplicitTreap (no
//   ↳ allocation per split).
// IF THE OPERATIONS ARE POSITION-STABLE (no insert / erase, only
//   ↳ updates) a persistent segment
// tree is 3x faster and half the memory.

```

2.8 Policy Based Data Structures

2.8.1 OrderedSet

Policy-Based Data Structure: ordered_set

```

// Time: O(log N) per operation
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>

using namespace __gnu_pbds;
template<typename T>
using ordered_set = tree<T, null_type, less<T>, rb_tree_tag,
    ↳ tree_order_statistics_node_update>;

// Usage:
// st.find_by_order(k): Iterator to k-th element (0-based)
// st.order_of_key(x): Number of elements strictly smaller than x

```

2.9 Monotonic Data Structures

2.9.1 MonotonicStack

Monotonic Stack - Time: O(N), Space: O(N)

```

// Computes Next/Previous Smaller and Greater element indices for all
//   ↳ positions
template<typename T = ll>
struct MonotonicStack {
    int n;
    vector<T> a;
    vector<int> prev_less, next_less;
    vector<int> prev_greater, next_greater;

    MonotonicStack(const vector<T>& arr) : n(arr.size()), a(arr),
        prev_less(n, -1), next_less(n, n),
        prev_greater(n, -1), next_greater(n, n) {

        vector<int> st;
        // Next Less & Previous Less
        for (int i = 0; i < n; i++) {
            while (!st.empty() && a[st.back()] >= a[i]) {
                next_less[st.back()] = i;
                st.pop_back();
            }
            if (!st.empty()) prev_less[i] = st.back();
            st.pb(i);
        }

        st.clear();
        // Next Greater & Previous Greater
        for (int i = 0; i < n; i++) {
            while (!st.empty() && a[st.back()] <= a[i]) {
                next_greater[st.back()] = i;
                st.pop_back();
            }
            if (!st.empty()) prev_greater[i] = st.back();
            st.pb(i);
        }
    }
};

```

2.9.2 TwoStackQueue

Monotonic Queue via Two Stacks (Sliding Window Min/Max)

// Space: O(N), Time: O(1) amortized per push/pop/get

```

struct TwoStackQ {
    struct Node {
        ll mx = -llinf, mn = llinf, val;

        Node() : val(0) {}
        Node(ll x) : mx(x), mn(x), val(x) {}
    };

    stack<Node> a, b;

    int size() {
        return a.size() + b.size();
    }

    void mrg(Node &a, Node &b) {
        a.mn = min(a.mn, b.mn);
        a.mx = max(a.mx, b.mx);
    }

    // Pushes value into the queue
    void push(ll val) {
        auto nd = Node(val);
        if (!a.empty()) mrg(nd, a.top());
        a.push(nd);
    }

    // Transfers elements from stack a to stack b when b is empty
    void move() {
        while (!a.empty()) {
            auto nd = Node(a.top().val);
            if (!b.empty()) mrg(nd, b.top());
            b.push(nd);
            a.pop();
        }
    }

    // Returns min/max of all current elements in the queue in O(1)
    Node get() {
        Node res;
        if (!b.empty()) mrg(res, b.top());
        if (!a.empty()) mrg(res, a.top());
        return res;
    }

    // Removes the oldest element from the queue
    void pop() {
        if (b.empty()) move();
        if (!b.empty()) b.pop();
    }
};

```

2.9.3 MonotonicQueue2D

2D Monotonic Queue (Sliding Window 2D Min/Max) - Time: $O(N * M)$, Space: $O(N * M)$

```
// Computes min in every K x L subgrid of an N x M matrix
template<typename T = ll>
struct MonotonicQueue2D {
    int n, m;

    // Returns a matrix res where res[i][j] is the min in subgrid
    // [i..i+K-1][j..j+L-1]
    vector<vector<T>> sliding_window_2d(const vector<vector<T>>&
    grid, int K, int L) {
        n = grid.size();
        m = grid[0].size();

        // 1. Sliding window min horizontally along each row
        vector<vector<T>> row_min(n, vector<T>(m - L + 1));
        for (int i = 0; i < n; i++) {
            deque<int> dq;
            for (int j = 0; j < m; j++) {
                while (!dq.empty() && grid[i][dq.back()] >=
                grid[i][j]) dq.pop_back();
                dq.pb(j);
                if (dq.front() <= j - L) dq.pop_front();
                if (j >= L - 1) row_min[i][j - L + 1] =
                grid[i][dq.front()];
            }

            // 2. Sliding window min vertically down each column of
            row_min
            vector<vector<T>> res(n - K + 1, vector<T>(m - L + 1));
            for (int j = 0; j < m - L + 1; j++) {
                deque<int> dq;
                for (int i = 0; i < n; i++) {
                    while (!dq.empty() && row_min[dq.back()][j] >=
                    row_min[i][j]) dq.pop_back();
                    dq.pb(i);
                    if (dq.front() <= i - K) dq.pop_front();
                    if (i >= K - 1) res[i - K + 1][j] =
                    row_min[dq.front()][j];
                }

                return res;
            }
        }
    };
};
```

2.10 DynamicMedian

RUNNING MEDIAN / k-th SMALLEST under insertions, via two multisets kept balanced around the split.

```
// O(log n) per insert, O(1) to read the median and either side's
// sum. The pattern generalises to
// "maintain the sum of the k smallest" - which is what
// cost-to-move-to-median problems need.
multiset<ll> L, R;
ll sum_L = 0, sum_R = 0;
int k;

// Rebalances the multisets so L always has the lower half/median of
// CURRENT elements
void balance() {
    int total_size = sz(L) + sz(R);
    int k_L = (total_size + 1) / 2; // Fixed: Target size based on
    // actual current elements

    // If L is too big, move its largest element to R
    while (sz(L) > k_L) {
        auto it = prev(L.end());
        ll val = *it;

        sum_L -= val;
        sum_R += val;

        R.insert(val);
        L.erase(it);
    }

    // If L is too small, move the smallest element of R to L
    while (sz(L) < k_L) {
        auto it = R.begin();
        ll val = *it;

        sum_R -= val;
        sum_L += val;

        L.insert(val);
        R.erase(it);
    }
}

void add_val(ll val) {
    if (L.empty() || val <= *L.rbegin()) {
        L.insert(val);
        sum_L += val;
    } else {
        R.insert(val);
        sum_R += val;
    }
    balance();
}

void remove_val(ll val) {
    auto it = L.find(val);
    if (it != L.end()) {
        sum_L -= val;
        L.erase(it);
    } else {
        it = R.find(val);
        if (it == R.end()) return; // not present
        sum_R -= val;
        R.erase(it);
    }
    balance();
}

ll get_cost() {
    // UB if empty -
    guard the caller
    ll median = *L.rbegin();
    return sum_R - sum_L + median * (sz(L) - sz(R));
}
```

2.11 IntervalSetODT

INTERVAL SET / "Chtholly tree" (ODT). Keeps the array as maximal runs of equal value.

```
// assign(l, r, v) is amortised near-O(log n) WHEN the data has many
// ↪ range-assigns (random or
// ↪ adversarial-but-assign-heavy); each assign destroys the intervals
// ↪ it covers, so the total
// ↪ number of intervals created is O(q).
// THIS is the standard way to support range bitwise-AND / OR / MOD /
// ↪ assign together with
// ↪ range sum, because those operations rapidly make long stretches
// ↪ constant.
struct ODT {
    map<int, pair<int, ll>> mp; // l -> {r,
    // ↪ value}, covering [l, r]

    ODT(int n, ll v = 0) { mp[0] = {n - 1, v}; }

    // split so that an interval starts exactly at p; returns the
    // ↪ iterator to it
    auto split(int p) {
        auto it = prev(mp.upper_bound(p));
        if (it->first == p) return it;
        auto [r, v] = it->second;
        it->second.first = p - 1;
        return mp.emplace(p, make_pair(r, v)).first;
    }
    void assign(int l, int r, ll v) {
        auto itr = split(r + 1), itl = split(l);
        mp.erase(itl, itr);
        mp[l] = {r, v};
    }
    // Generic scan over [l, r]: f(l, r, value) is called once per
    // ↪ run. O(#runs touched).
    template <class F> void forEach(int l, int r, F f) {
        auto itr = split(r + 1), itl = split(l);
        for (auto it = itl; it != itr; ++it) f(it->first,
        // ↪ it->second.first, it->second.second);
    }
    ll sum(int l, int r) {
        ll s = 0;
        for (auto it = split(l); it != split(r + 1); ++it)
        // ↪ s += it->second.second * (it->second.first - it->second.second + 1);
    }
    // RANGE ADD: touch each run once. Runs are not merged, so adds
    // ↪ alone do NOT keep the
    // ↪ amortised bound - this is only cheap when assigns keep
    // ↪ collapsing the structure.
    void add(int l, int r, ll x) {
        auto itr = split(r + 1), itl = split(l);
        for (auto it = itl; it != itr; ++it) it->second.second += x;
    }
    // RANGE AND / OR / MOD follow the same loop, applying the op to
    // ↪ each run's value.
    // ↪ They are fast because every value can only decrease O(log V)
    // ↪ times before it stops
    // ↪ changing, at which point neighbouring runs become equal and
    // ↪ can be merged.
    template <class Op> void apply(int l, int r, Op op) {
        auto itr = split(r + 1), itl = split(l);
        for (auto it = itl; it != itr; ++it) it->second.second =
        // ↪ op(it->second.second);
        merge(l, r);
    }
    void merge(int l, int r) { // glue
        // ↪ neighbouring runs of equal value
        auto it = split(l);
        while (it != mp.end() && it->first <= r) {
            auto nx = next(it);
            if (nx != mp.end() && nx->second.second ==
            // ↪ it->second.second &&
            it->second.first + 1 == nx->first) {
                it->second.first = nx->second.first;
                mp.erase(nx);
            } else it = nx;
        }
    }
};
// Careful: worst case WITHOUT assigns is O(n) per operation. If the
// ↪ problem has only adds and
// ↪ queries, use a segment tree instead.
```

2.12 DisjointSparseTable

DISJOINT SPARSE TABLE - O(1) query for ANY associative operation (not just idempotent ones).

```
// Build O(n log n) time and memory. Use when a plain sparse table
// ↪ would double-count:
// ↪ sum, product, matrix product, hashing, "merge two structs",
// ↪ min-count pairs, ...
// (For min/max/gcd/or/and a normal sparse table is smaller - use
// ↪ that instead.)
template <typename T, class F = function<T(T, T)>>
struct DisjointST {
    int n, LOG;
    vector<vector<T>>> t;
    F op;

    DisjointST(const vector<T>& a, F op) : n(a.size()), op(op) {
        LOG = 1;
        while ((1 << LOG) < n) LOG++;
        t.assign(LOG + 1, a);
        for (int lvl = 1; lvl <= LOG; lvl++) {
            int half = 1 << lvl;
            for (int mid = half; mid < n + half; mid += 2 * half) {
                if (mid - 1 < n) { //
                    // ↪ build leftwards from the split
                    t[lvl][mid - 1] = a[mid - 1];
                    for (int i = mid - 2; i >= mid - half && i >= 0;
                    // ↪ i--)
                        t[lvl][i] = op(a[i], t[lvl][i + 1]);
                }
                if (mid < n) { // and
                    // ↪ rightwards
                    t[lvl][mid] = a[mid];
                    for (int i = mid + 1; i < min(n, mid + half); i++)
                        t[lvl][i] = op(t[lvl][i - 1], a[i]);
                }
            }
        }
    }
    T query(int l, int r) const { //
        // ↪ inclusive [l, r]
        if (l == r) return t[0][l];
        int lvl = 63 - __builtin_clzll((unsigned long long)(l ^ r));
        // ↪ // highest differing bit
        return op(t[lvl][l], t[lvl][r]);
    }
};
// Every query splits at the unique power-of-two boundary between l
// ↪ and r, and both halves are
// ↪ already prefix/suffix-accumulated at that level - hence exactly
// ↪ one op() per query.
```


2.13 WaveletTree

WAVELET TREE over values in $[lo, hi]$. Build $O(n \log V)$.

```
// kth(l, r, k)      : k-th SMALLEST in a[l..r]
    ↪  $O(\log V)$ 
// countLeq(l,r,x)  : how many  $a[i] \leq x$  in  $a[l..r]$ 
    ↪  $O(\log V)$ 
// countEq(l,r,x)   : how many  $a[i] == x$  in  $a[l..r]$ 
    ↪  $O(\log V)$ 
// Online (no offline sorting, no persistence) - the practical
    ↪ alternative to a merge-sort tree.
struct Wavelet {
    int lo, hi;
    Wavelet *l = nullptr, *r = nullptr;
    vector<int> b;                                // b[i] =
    ↪ #elements going left among first i

    // build on a[from, to) - the array IS reordered in place (stable)
    ↪ partition by midpoint)
    Wavelet(vector<int>::iterator from, vector<int>::iterator to, int
    ↪ x, int y) : lo(x), hi(y) {
        if (from >= to) return;
        if (lo == hi) { b.assign(to - from + 1, 0); return; }
        int mid = lo + (hi - lo) / 2;
        auto f = [mid](int v) { return v <= mid; };
        b.reserve(to - from + 1);
        b.push_back(0);
        for (auto it = from; it != to; ++it) b.push_back(b.back() +
    ↪ f(*it));
        auto pivot = stable_partition(from, to, f);
        l = new Wavelet(from, pivot, lo, mid);
        r = new Wavelet(pivot, to, mid + 1, hi);
    }
    int kth(int L, int R, int k) {                // 1-indexed k,
    ↪ inclusive [L, R] (1-based)
        if (L > R) return 0;
        if (lo == hi) return lo;
        int inLeft = b[R] - b[L - 1], lb = b[L - 1], rb = b[R];
        if (k <= inLeft) return l->kth(lb + 1, rb, k);
        return r->kth(L - lb, R - rb, k - inLeft);
    }
    int countLeq(int L, int R, int k) {            // #{i in [L,R]
    ↪ : a[i] <= k}
        if (L > R || k < lo) return 0;
        if (hi <= k) return R - L + 1;
        int lb = b[L - 1], rb = b[R];
        return l->countLeq(lb + 1, rb, k) + r->countLeq(L - lb, R -
    ↪ rb, k);
    }
    int countEq(int L, int R, int k) {
        if (L > R || k < lo || k > hi) return 0;
        if (lo == hi) return R - L + 1;
        int lb = b[L - 1], rb = b[R], mid = lo + (hi - lo) / 2;
        if (k <= mid) return l->countEq(lb + 1, rb, k);
        return r->countEq(L - lb, R - rb, k);
    }
    ~Wavelet() { delete l; delete r; }
};
// Usage: compress values to  $[0, V)$ , copy the array, then Wavelet
    ↪ w(a.begin(), a.end(), 0, V-1);
// queries are 1-BASED on the original positions.
```

2.14 OfflineDynamicConnectivity

OFFLINE DYNAMIC CONNECTIVITY - edges appear and disappear over time; answer queries at each

```
// Needs: DSURollback (02 - DSU/02 - DSURollback.cpp).
// moment. Segment tree over the TIMELINE + DSU with rollback.  $O(q)$ 
    ↪  $\log q * \log n$  - DSURollback has no path compression, so find is
    ↪  $\log n$ , not alpha.
// The same skeleton answers ANY "structure supports add + rollback,
    ↪ but not delete" problem:
// put each element on the time interval where it is alive, DFS the
    ↪ segment tree, add on entry,
// roll back on exit. ("Deleting from a data structure in  $O(T \log$ 
    ↪  $n)$ ")
struct DynConn {
    int n, T;
    vector<vector<pair<int, int>>> seg;           // edges alive
    ↪ on each timeline node
    DSURollback d;
    vector<int> comps;                          // comps[t] =
    ↪ #components at time t

    DynConn(int n, int T) : n(n), T(T), seg(4 * max(1, T)), d(n),
    ↪ comps(T, 0) {}

    void addSeg(int v, int l, int r, int ql, int qr, pair<int, int>
    ↪ e) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr) { seg[v].push_back(e); return; }
        int m = (l + r) / 2;
        addSeg(2 * v, l, m, ql, qr, e), addSeg(2 * v + 1, m + 1, r,
    ↪ ql, qr, e);
    }
    // edge (u,v) exists during times [tl, tr] (inclusive)
    void addEdge(int u, int v, int tl, int tr) { if (tl <= tr)
    ↪ addSeg(1, 0, T - 1, tl, tr, {u, v}); }

    void dfs(int v, int l, int r, int cur) {
        int snap = d.get_state();
        for (auto [a, b] : seg[v]) cur -= d.join(a, b);
        if (l == r) comps[l] = cur;
        else {
            int m = (l + r) / 2;
            dfs(2 * v, l, m, cur), dfs(2 * v + 1, m + 1, r, cur);
        }
        d.rollback_to(snap);
    }
    void solve() { if (T) dfs(1, 0, T - 1, n); }
};
// USAGE: collect every edge's [birth, death] interval first (an edge
    ↪ that is never removed dies
// at T-1), then one solve() fills comps[]. To answer "are u and v
    ↪ connected at time t", instead
// record the queries per leaf and test d.same(u, v) inside the l ==
    ↪ r branch.
```

2.15 CartesianTree

CARTESIAN TREE in O(n): heap-ordered by VALUE, BST-ordered by INDEX.

```
// The root of any range is its minimum, so "range minimum" structure
    ↳ becomes a TREE -
// which turns range problems into subtree problems (and enables
    ↳ divide & conquer on the min).
// Built with a monotonic stack; ties go left so the tree is unique.
struct CartTree {
    int n, root = -1;
    vector<int> l, r, par;

    CartTree(const vector<int>& a) : n(a.size()), l(n, -1), r(n, -1),
    ↳ par(n, -1) {
        vector<int> st;
        for (int i = 0; i < n; i++) {
            int last = -1;
            while (!st.empty() && a[st.back()] > a[i]) last =
    ↳ st.back(), st.pop_back();
            l[i] = last;
            if (last != -1) par[last] = i;
            if (!st.empty()) r[st.back()] = i, par[i] = st.back();
            st.push_back(i);
        }
        root = st[0];
    }
};
// USES
// - "for every element, the maximal range where it is the minimum"
    ↳ = its subtree range,
// which is the contribution technique (sum over subarrays of the
    ↳ min) in tree form
// - largest rectangle in a histogram = best over nodes of value *
    ↳ subtreeSize
// - divide & conquer "solve(l, r) splits at the minimum" without
    ↳ recomputing RMQ
// - counting subarrays whose minimum is a given element
// - a Treap built on (index, random priority) is exactly a
    ↳ Cartesian tree - same code shape
```

2.16 SegTreeMerging

SEGMENT TREE MERGING - one dynamic segment tree per vertex over the VALUE domain, merged

```
// bottom-up along a tree. Total O(n log n) nodes touched over ALL
    ↳ merges (amortised), because
// each merge destroys as many nodes as it visits.
// Solves, per subtree and in one DFS: most frequent value, k-th
    ↳ value, count of a value,
// "sum of values in a range" - i.e. everything a merge-sort tree
    ↳ gives, but online per subtree.
struct SegMerge {
    struct Node { int l = 0, r = 0; ll sum = 0; int best = 0; ll
    ↳ bestCnt = 0; };
    vector<Node> t;
    int V; // value domain
    ↳ is [0, V)

    SegMerge(int V, int reserve = 0) : V(V) { t.reserve(reserve),
    ↳ t.push_back(Node()); } // 0 = null
    int newNode() { t.push_back(Node()); return t.size() - 1; }

    void pull(int v) {
        int L = t[v].l, R = t[v].r;
        t[v].sum = t[L].sum + t[R].sum;
        if (t[L].bestCnt >= t[R].bestCnt) t[v].best = t[L].best,
    ↳ t[v].bestCnt = t[L].bestCnt;
        else t[v].best = t[R].best, t[v].bestCnt = t[R].bestCnt;
    }
    // Returns the (possibly new) node index. NEVER take an 'int&'
    ↳ into t here: newNode() can
    // reallocate the vector and invalidate the reference
    ↳ mid-recursion.
    int update(int v, int lo, int hi, int p, ll d) {
        if (!v) v = newNode();
        if (hi - lo == 1) { t[v].sum += d, t[v].best = lo,
    ↳ t[v].bestCnt = t[v].sum; return v; }
        int m = (lo + hi) / 2;
        if (p < m) { int c = update(t[v].l, lo, m, p, d); t[v].l = c;
    ↳ }
        else { int c = update(t[v].r, m, hi, p, d); t[v].r = c; }
        pull(v);
        return v;
    }

    int merge(int a, int b, int lo, int hi) { // destroys b;
    ↳ returns the merged root
        if (!a || !b) return a | b;
        if (hi - lo == 1) { t[a].sum += t[b].sum, t[a].bestCnt =
    ↳ t[a].sum; return a; }
        int m = (lo + hi) / 2;
        int L = merge(t[a].l, t[b].l, lo, m), R = merge(t[a].r,
    ↳ t[b].r, m, hi);
        t[a].l = L, t[a].r = R;
        pull(a);
        return a;
    }

    ll query(int v, int lo, int hi, int ql, int qr) { // sum of
    ↳ counts in [ql, qr)
        if (!v || qr <= lo || hi <= ql) return 0;
        if (ql <= lo && hi <= qr) return t[v].sum;
        int m = (lo + hi) / 2;
        return query(t[v].l, lo, m, ql, qr) + query(t[v].r, m, hi,
    ↳ ql, qr);
    }

    int kth(int v, int lo, int hi, ll k) { // k-th
    ↳ smallest (1-indexed), -1 if k too big
        if (!v || k > t[v].sum) return -1;
        if (hi - lo == 1) return lo;
        int m = (lo + hi) / 2;
        return k <= t[t[v].l].sum ? kth(t[v].l, lo, m, k) :
    ↳ kth(t[v].r, m, hi, k - t[t[v].l].sum);
    }

    // wrappers
    void update(int& root, int p, ll d) { root = update(root, 0, V,
    ↳ p, d); }
    int merge(int a, int b) { return merge(a, b, 0, V); }
    ll query(int root, int ql, int qr) { return query(root, 0, V, ql,
    ↳ qr); }
    int kth(int root, ll k) { return kth(root, 0, V, k); }
};
// TREE USAGE: root[v] starts with just v's own value; after
    ↳ recursing, merge every child's root
// into v's. Answer v's query right there. Never reuse a root after
    ↳ merging it away.
```

2.17 BitsetTricks

BITSET SPEEDUPS - the /64 family. All of these are "obviously too slow" DPs that fit anyway.

```
// TRANSITIVE CLOSURE (reachability) in  $O(n^3 / 64)$ . n up to ~5000.
template<int N>
void transitiveClosure(vector<bitset<N>>& g, int n) {
    for (int k = 0; k < n; k++)
        for (int i = 0; i < n; i++) if (g[i][k]) g[i] |= g[k];
}
// LCS in  $O(n*m/64)$  - bit-parallel (Allison-Dix). 'a' is the string
// whose length bounds the
// bitset width; row is a bitmask over positions of 'a'.
// x = row | match[b_i]
// row' = x & ((x - (row << 1 | 1)) ^ x)
// The "| 1" is the carry-in that makes the shift borrow correctly.
typedef unsigned long long u64;
int bitLCS(const string& a, const string& b) {
    int n = a.size(), W = (n + 63) / 64;
    if (!n || b.empty()) return 0;
    vector<vector<u64>> match(256, vector<u64>(W, 0));
    for (int i = 0; i < n; i++) match[(unsigned char)a[i]][i >> 6] |=
        1ULL << (i & 63);
    vector<u64> row(W, 0), x(W), sh(W);
    for (char c : b) {
        const auto& M = match[(unsigned char)c];
        for (int i = 0; i < W; i++) x[i] = row[i] | M[i];
        u64 carry = 1; // sh = (row <<
        1) | 1
        for (int i = 0; i < W; i++) { u64 nc = row[i] >> 63; sh[i] =
        (row[i] << 1) | carry; carry = nc; }
        u64 borrow = 0; // row = x &
        ((x - sh) ^ x)
        for (int i = 0; i < W; i++) {
            u64 t = sh[i] + borrow;
            borrow = (t < sh[i]) || (x[i] < t); // t < sh[i]
            u64 d = x[i] - t;
            row[i] = x[i] & (d ^ x[i]);
        }
        // catches the carry overflow
        u64 d = x[i] - t;
        row[i] = x[i] & (d ^ x[i]);
    }
}
int r = 0;
for (u64 v : row) r += __builtin_popcountll(v);
return r;
}

// SUBSET SUM / KNAPSACK FEASIBILITY: dp |= dp << w (see
// KnapsackFamily.cpp)
// BIPARTITE MATCHING on dense graphs: keep each left vertex's
// neighbourhood as a bitset and
// pick the next free right vertex with (adj[u] &
// free)._Find_first().
// COUNTING PAIRS with a property: bitset per value class, then
// popcount of ANDs.
// GCC EXTRAS: b._Find_first(), b._Find_next(i) iterate set bits in
//  $O(N/64)$  total.
// RULE OF THUMB: bitset turns  $1e9$  "simple" operations into  $\sim 1.5e7$ 
// word operations - budget
// about  $n*m/64 \leq 1e8$  for a comfortable pass.
```

2.18 VeniceAndSWAG

VENICE TECHNIQUE - a multiset with an $O(1)$ "add x to EVERY element".

```
// Keep a global offset; store values shifted by it. 15 lines, and it
// turns a whole family of
// "everyone loses X health, remove the dead" problems into a
// one-liner.
struct Venice {
    multiset<ll> s;
    ll water = 0; // global
    // offset

    void add(ll v) { s.insert(v + water); } // insert the
    // true value v
    void erase(ll v) { auto it = s.find(v + water); if (it !=
    s.end()) s.erase(it); }
    void addAll(ll v) { water -= v; } //  $O(1)$ : add v
    // to every element
    ll getMin() const { return *s.begin() - water; }
    ll getMax() const { return *s.rbegin() - water; }
    int size() const { return s.size(); }
    // Pop everything whose TRUE value is  $\leq t$  (e.g. everyone who
    // died).
    template<class F> void popLE(ll t, F onPop) {
        while (!s.empty() && *s.begin() - water <= t) {
            onPop(*s.begin() - water); s.erase(s.begin()); }
    };
    // SAME IDEA elsewhere: a binary trie with a global XOR mask ("xor
    // every element by x"),
    // a BIT with a global additive offset, a lazy "add to all" on a heap.

    // SWAG (Sliding Window Aggregation) for ANY monoid - non-commutative
    // and non-invertible ops
    // included: matrix product, affine composition, gcd, "merge two
    // structs".
    // Amortised  $O(1)$  push/pop/fold. This generalises 09/02 -
    // TwoStackQueue.cpp, which is hardwired
    // to min/max. Prefix-product-and-divide does NOT work without an
    // inverse - this does.
    template<class T, class Op>
    struct SWAG {
        vector<pair<T, T>> front_, back_; // {value,
        // aggregate from here to the bottom}
        T id;
        Op op;
        SWAG(T id, Op op) : id(id), op(op) {}

        T foldFront() const { return front_.empty() ? id :
        front_.back().second; }
        T foldBack() const { return back_.empty() ? id :
        back_.back().second; }
        T fold() const { return op(foldFront(), foldBack()); } //
        // front first: order matters
        void push(T v) { back_.push_back({v, op(foldBack(), v)}); }
        void pop() {
            if (front_.empty())
                while (!back_.empty()) {
                    T v = back_.back().first; back_.pop_back();
                    front_.push_back({v, op(v, foldFront())}); //
                    // reversed accumulation
                }
            if (!front_.empty()) front_.pop_back();
        }
        int size() const { return front_.size() + back_.size(); }
    };
    // USAGE: SWAG<ll, function<ll(ll,ll)>> q(0, [](ll a, ll b){ return a
    // + b; });
    // For non-commutative ops keep the argument order exactly as written
    // above, or the window
    // product comes out reversed.
```

2.19 MergeableHeap

MERGEABLE (MELDABLE) HEAP - a priority queue that also supports MERGING two heaps in $O(\log n)$,

```
// which std::priority_queue cannot do. This is a skew heap: merge by
//   ↳ walking the two right spines
// and swapping children, which keeps the amortised depth logarithmic
//   ↳ with no stored rank.
// Supports a lazy "add v to every key in this heap" tag, because
//   ↳ that is what the two algorithms
// that need a mergeable heap actually want.
// USES: Chu-Liu/Edmonds directed MST in  $O(E \log V)$  (each contracted
//   ↳ cycle merges its in-edge
// heaps and subtracts the chosen edge's weight from the whole heap);
//   ↳ Dijkstra-style k-shortest
// paths (Eppstein); "merge the two children's candidate sets" DP on
//   ↳ trees; leftist-heap solutions
// to scheduling problems (repeatedly merge and pop the largest, e.g.
//   ↳ minimise total lateness).
```

```
struct MHeap {
    struct Node { ll key, lz = 0; Node *l = 0, *r = 0; };
    static void down(Node* t) { //
        ↳ apply the lazy tag downwards
        if (!t || !t->l) return;
        t->key += t->lz;
        if (t->l) t->l->lz += t->lz;
        if (t->r) t->r->lz += t->lz;
        t->lz = 0;
    }
    static Node* merge(Node* a, Node* b) { //
        ↳  $O(\log n)$  amortised, MIN-heap
        if (!a || !b) return a ? a : b;
        down(a), down(b);
        if (a->key > b->key) swap(a, b);
        a->r = merge(a->r, b);
        swap(a->l, a->r); // the
        ↳ "skew" step
        return a;
    }
    static Node* push(Node* t, ll key) { return merge(t, new
        ↳ Node{key}); }
    static ll top(Node* t) { return down(t), t->key; }
    static Node* pop(Node* t) { down(t); return merge(t->l, t->r); }
    static void addAll(Node* t, ll v) { if (t) t->lz += v; } //
        ↳  $O(1)$  add to every key
};

// TO GET A MAX-HEAP: negate the keys, or flip the comparison in
//   ↳ merge.
// LEFTIST HEAP is the same interface with a stored 'dist' (null-path
//   ↳ length) and the rule "keep the
// larger dist on the left"; it gives WORST-CASE  $O(\log n)$  instead of
//   ↳ amortised. Skew is shorter and
// fine unless you need the worst-case bound inside another amortised
//   ↳ argument. 24 - PersistentHeap
// is that leftist version, made persistent (meld without destroying
//   ↳ either input).
// PAIRING HEAP has  $O(1)$  merge and decrease-key in practice and is
//   ↳ what you want if you need
// decrease-key (Dijkstra on very dense graphs); in contests a binary
//   ↳ heap with lazy deletion wins.
// LAZY DELETION is usually enough: push (key, id), and pop until the
//   ↳ top is still valid. That
// removes the need for decrease-key entirely and is 5 lines.
// SMALL-TO-LARGE with std::priority_queue also "merges" heaps:
//   ↳ repeatedly pop from the smaller and
// push into the larger,  $O(n \log^2 n)$  total. Shorter than this file -
//   ↳ use it unless the  $\log^2$  hurts.
```

2.20 PersistentAndUndo

PERSISTENCE AND UNDO - three ways to travel in a structure's history, and when each applies.

```
// ROLLBACK (05 - DSU/02): undo the last op. Cheapest. Needs a
//   ↳ stack discipline (LIFO).
// OFFLINE DELETION (14 - OfflineDynamicConnectivity): segment tree
//   ↳ over time + rollback.
// Handles arbitrary insert/delete, but you must know every
//   ↳ operation in advance.
// PERSISTENCE (this file): branch the history. The only one that
//   ↳ works ONLINE with deletions,
// and the only one that lets you query a past version after
//   ↳ moving on.
```

```
// --- PERSISTENT ARRAY: a segment tree over indices. get/set in
//   ↳  $O(\log n)$ ,  $O(\log n)$  new nodes. ---
struct PArray {
    struct Node { int l = 0, r = 0; ll v = 0; };
    vector<Node> t;
    int n;
    PArray(int n) : t(1), n(n) {} //
        ↳ node 0 is the shared all-zero
    int build(const vector<ll>& a, int lo, int hi) {
        int id = t.size();
        t.push_back({});
        if (lo == hi) { t[id].v = a[lo]; return id; }
        int m = (lo + hi) / 2, L = build(a, lo, m), R = build(a, m +
        ↳ 1, hi);
        t[id].l = L, t[id].r = R;
        return id;
    }
    int build(const vector<ll>& a) { return build(a, 0, n - 1); }
    ll get(int u, int i, int lo, int hi) {
        if (lo == hi) return t[u].v;
        int m = (lo + hi) / 2;
        return i <= m ? get(t[u].l, i, lo, m) : get(t[u].r, i, m + 1,
        ↳ hi);
    }
    ll get(int u, int i) { return get(u, i, 0, n - 1); }
    int set(int u, int i, ll v, int lo, int hi) {
        int id = t.size();
        t.push_back(t[u]);
        if (lo == hi) { t[id].v = v; return id; }
        int m = (lo + hi) / 2;
        if (i <= m) t[id].l = set(t[u].l, i, v, lo, m);
        else t[id].r = set(t[u].r, i, v, m + 1, hi);
        return id;
    }
    int set(int u, int i, ll v) { return set(u, i, v, 0, n - 1); }
};

// --- PERSISTENT DSU on top of it.  $O(\log^2 n)$  per operation. ---
// The catch that makes it work: NO PATH COMPRESSION (that would
//   ↳ mutate an old version), so you
// MUST union by size/rank to keep the height  $O(\log n)$ .
```

```
struct PDSU {
    PArray par, siz;
    PDSU(int n) : par(n), siz(n) {}
    pair<int, int> init(int n) { //
        ↳ returns {parRoot, szRoot}
        vector<ll> p(n), s(n, 1);
        iota(all(p), 0);
        return {par.build(p), siz.build(s)};
    }
    int find(int pr, int x) { ll p = par.get(pr, x); return p == x ?
        ↳ x : find(pr, p); }
    pair<int, int> join(int pr, int sr, int a, int b) {
        a = find(pr, a), b = find(pr, b);
        if (a == b) return {pr, sr};
        if (siz.get(sr, a) < siz.get(sr, b)) swap(a, b);
        return {par.set(pr, b, a), siz.set(sr, a, siz.get(sr, a) +
        ↳ siz.get(sr, b))};
    }
};

/* THE OTHER TWO TIME-TRAVEL TRICKS, as recipes (code: 21 -
//   ↳ QueueUndoTrick, 25 - StaticToDynamic)
QUEUE UNDO. You have a structure that supports only "undo the last
//   ↳ op" (rollback DSU), but the
deletions arrive in FIFO order (a sliding window). Keep the undo
//   ↳ stack partitioned so the
OLDEST element is near the top; when it is not, pop a prefix and
//   ↳ re-push it in the other order,
always rebuilding the SMALLER half.  $O(\log n)$  amortised extra. This
//   ↳ is the online alternative to
the offline segment-tree-on-time trick, and it is what "sliding
//   ↳ window connectivity" needs.
STATIC TO DYNAMIC (the logarithmic method). Any STATIC structure
//   ↳ with  $O(B(n))$  build and  $O(Q(n))$ 
```

query becomes insert-only: keep $O(\log n)$ instances of sizes 2^0 ,
 $\hookrightarrow 2^1, 2^2, \dots$; inserting is
 binary-counter carrying, so you rebuild an instance of size 2^k
 $\hookrightarrow$ only every 2^k insertions.
 Insert becomes $O(B(n)/n * \log n)$ amortised and query $O(Q(n) \log n)$
 $\hookrightarrow$ (ask all instances).
 Use it to make a convex hull / a wavelet tree / an Aho-Corasick
 $\hookrightarrow$ automaton accept insertions.

PARTIALLY vs FULLY PERSISTENT: partially = you may query any past
 $\hookrightarrow$ version but only modify the
 latest - for a DSU that is 05 - DSU/05 - PartiallyPersistentDSU,
 $\hookrightarrow O(n)$ memory instead of this
 file's $O(n \log n)$; fully = you may modify any version, branching
 $\hookrightarrow$ the history into a tree.
 The code above is fully persistent.

WHEN NOT TO BOTHER: if the problem is offline and only needs
 $\hookrightarrow$ "delete", the segment tree over the
 timeline with a rollback DSU is shorter, faster, and has a smaller
 $\hookrightarrow$ constant than any of this. */

2.21 QueueUndoTrick

QUEUE UNDO TRICK - turns a structure that can only "undo the LAST operation" into one that can

```
// Needs: 05 - Disjoint Set Union (DSU)/02 - DSURollback.cpp
// also "undo the OLDEST operation", i.e. a QUEUE of operations
// instead of a stack.
// O(log n) amortised extra factor on top of the underlying structure.
// WHEN: a sliding window over operations - "for each l, the largest
// r such that edges l..r keep
// the graph bipartite / acyclic / with <= k components" (two
// pointers over edges), or any online
// problem where insertions and deletions are FIFO. Deletions in
// arbitrary order need
// 14 - OfflineDynamicConnectivity (offline) or a link-cut tree
// (forests only).
// KEY INVARIANT: the undo stack is kept partitioned into a "to be
// deleted soon" prefix and the
// rest. To pop the front, undo a suffix of the stack, then re-push
// it with the OLDEST elements
// nearest the top; the rebuild always processes the SMALLER of the
// two groups, which is what makes
// the potential argument work - each operation is re-pushed  $O(\log n)$ 
// times over its lifetime.
// The underlying structure must support: apply(op) and undo-last,
// both cheap; nothing else.
struct QueueUndo {
    struct Op { int u, v; bool old; }; // old
    // = "already re-pushed once"
    DSURollback d;
    vector<Op> st; //
    // mirrors the undo stack exactly
    int bot = 0; //
    // first index that is not yet 'old'
    QueueUndo(int n) : d(n) {}
    void push(Op o) { d.join(o.u, o.v), st.push_back(o); }
    void push(int u, int v) { push({u, v, false}); }
    void pop() { //
        // remove the OLDEST operation
        vector<Op> a[2];
        while (bot < (int)st.size() && st[bot].old) bot++;
        if (bot == (int)st.size()) { // no
            // old ops left: flip everything
            while (!st.empty()) a[0].push_back(st.back()),
            // st.pop_back(), d.rollback();
            for (auto& o : a[0]) o.old = true, push(o);
            a[0].clear(), bot = 0;
        }
        do { //
            // pull back until the front is up
            a[st.back().old].push_back(st.back());
            st.pop_back(), d.rollback();
        } while (a[1].size() < a[0].size() && (int)st.size() > bot);
        for (int i : {0, 1}) {
            reverse(all(a[i]));
            for (auto& o : a[i]) push(o);
        }
        st.pop_back(), d.rollback(); // the
    } // front is now on top
};

// THE STRUCTURE UNDERNEATH can be anything with rollback: a DSU
// (connectivity, bipartiteness by
// storing the parity to the parent, number of components), a XOR
// basis, a max-stack, a knapsack
// with rollback. Replace push()/pop() bodies and keep the stack
// discipline identical.
// SLIDING WINDOW RECIPE: for l = 1..m, push edges while the
// invariant holds, record the answer,
// then pop() the edge at l. Total  $O(m \log m * \alpha)$ .
// CHEAPER OPTIONS FIRST: if the problem is offline, the segment tree
// over the timeline
// (14 - OfflineDynamicConnectivity) is shorter and has a better
// constant. If the deletions are
// LIFO, plain rollback is enough. This file is only for FIFO +
// online.
```

2.22 PermutationTree

PERMUTATION TREE (PQ-tree / substitution decomposition) - represents ALL "common intervals" of a

```
// permutation: the ranges [l, r] whose VALUES also form a contiguous
//   ↳ range. There can be  $O(n^2)$  of
// them; this tree stores every one of them in  $O(n)$  space, in  $O(n \log n)$  time.
// EVERY node has a contiguous span of POSITIONS and a contiguous
//   ↳ range of VALUES, and is one of:
//   LEAF - a single position.
//   JOIN node - its children's value ranges are consecutive and
//   ↳ monotone (increasing = type 1,
//   decreasing = type 2). ANY consecutive run of  $\geq 2$ 
//   ↳ of its children is itself a
//   common interval. This is where the  $O(n^2)$  of them
//   ↳ hide.
// CUT (prime) - no proper consecutive run of its children is a
//   ↳ common interval; only the whole
//   node is. A cut node always has  $\geq 4$  children.
// WHEN: "count subarrays whose values are consecutive integers",
//   ↳ "the number of ways to split a
// permutation into monotone blocks", simple-permutation questions,
//   ↳ and the smallest common
// interval containing a given [l, r]. If you only need to COUNT
//   ↳ common intervals, the divide and
// conquer with min/max stacks is shorter - reach for the tree when
//   ↳ you need the STRUCTURE.
// KEY INVARIANT: sweep r and keep, for every l, the quantity (max -
//   ↳ min) - (r - l) over a[l..r],
// which is  $\geq 0$  and equals 0 exactly when [l, r] is a common
//   ↳ interval. Monotone stacks maintain it
// under range add in  $O(1)$  amortised each, and a min segment tree
//   ↳ finds the zeros. Whenever the new
// element extends an existing node's value range by one the node
//   ↳ absorbs it (join), and whenever a
// zero appears further left a cut node is created over everything
//   ↳ between.
struct PermTree {
    int n, id, root = 0;
    vector<int> p, ty, par, mn, lz; // ty:
    ↳ 0 cut, 1 up-join, 2 down-join
    vector<pii> val, span; //
    ↳ value range, position range
    vector<vector<int>> ch;
    PermTree(const vector<int>& perm) //
    ↳ perm[1..n], a permutation of 1..n
        : n(perm.size() - 1), id(0), p(perm), ty(2 * perm.size(), 0),
        par(2 * perm.size(), -1),
        mn(8 * perm.size(), 0), lz(8 * perm.size(), 0), val(2 *
        perm.size(),
        span(2 * perm.size(), ch(2 * perm.size()) { build(); }
    void apply(int x, int v) { mn[x] += v, lz[x] += v; }
    void push(int x) { if (lz[x]) apply(2 * x, lz[x]), apply(2 * x +
    ↳ 1, lz[x]), lz[x] = 0; }
    void upd(int x, int b, int e, int l, int r, int v) {
        if (r < b || e < l) return;
        if (l <= b && e <= r) { apply(x, v); return; }
        int m = (b + e) / 2;
        push(x);
        upd(2 * x, b, m, l, r, v), upd(2 * x + 1, m + 1, e, l, r, v);
        mn[x] = min(mn[2 * x], mn[2 * x + 1]);
    }
    int qry(int x, int b, int e, int l, int r) {
        if (r < b || e < l || l > r) return inf;
        if (l <= b && e <= r) return mn[x];
        int m = (b + e) / 2;
        push(x);
        return min(qry(2 * x, b, m, l, r), qry(2 * x + 1, m + 1, e,
        ↳ l, r));
    }
    void link(int u, int v) { par[v] = u, ch[u].push_back(v); }
    bool adj(int i, int j) { return val[i].se == val[j].fi - 1; }
    ↳ // j starts right after i
    int len(int i) { return val[i].se - val[i].fi + 1; }
    static pii un(pii a, pii b) { return {min(a.fi, b.fi), max(a.se,
    ↳ b.se)}; }
    void build() {
        id = n + 1;
        vector<int> mx{0}, mi{0}, st; //
        ↳ monotone stacks + node stack
        for (int i = 1; i <= n; i++) {
            while (mx.back() && p[mx.back()] < p[i]) {
                int r = mx.back();
                mx.pop_back();
                upd(1, 1, n, mx.back() + 1, r, p[i] - p[r]);
            }
            mx.push_back(i);

```

```
while (mi.back() && p[mi.back()] > p[i]) {
    int r = mi.back();
    mi.pop_back();
    upd(1, 1, n, mi.back() + 1, r, p[r] - p[i]);
}
mi.push_back(i);
val[i] = {p[i], p[i]}, span[i] = {i, i};
int cur = i;
while (true) {
    if (!st.empty() && (adj(st.back(), cur) || adj(cur,
    ↳ st.back()))) {
        int b = st.back(), want = adj(b, cur) ? 1 : 2;
        if (ty[b] == want) { //
            ↳ absorb into the existing join
            link(b, cur);
            val[b] = un(val[b], val[cur]), span[b] =
            ↳ un(span[b], span[cur]);
            cur = b;
        } else { //
            ↳ start a new join node
            ty[id] = want, link(id, b), link(id, cur);
            val[id] = un(val[b], val[cur]), span[id] =
            ↳ un(span[b], span[cur]);
            cur = id++;
        }
        st.pop_back();
    } else if (qry(1, 1, n, 1, i - len(cur)) == 0) {
        ↳ // a cut node closes here
        int L = len(cur);
        pii v = val[cur], s = span[cur];
        link(id, cur);
        do {
            int b = st.back();
            L += len(b), v = un(v, val[b]), s = un(s,
            ↳ span[b]);
            link(id, b), st.pop_back();
        } while (v.se - v.fi + 1 != L);
        reverse(all(ch[id]));
        val[id] = v, span[id] = s, cur = id++;
    } else break;
}
st.push_back(cur);
upd(1, 1, n, 1, i, -1);
}
root = st.back();
}
};
// COUNTING COMMON INTERVALS: sum over join nodes with k children of
//   ↳  $k*(k-1)/2$ , plus n (the leaves),
// plus the number of cut nodes. That is the classic "count subarrays
//   ↳ forming a value range".
// SMALLEST COMMON INTERVAL CONTAINING [l, r]: binary-lift from leaf
//   ↳ l to the highest ancestor whose
// span still ends before r, take its parent u. If u is a cut node
//   ↳ the answer is span[u]; if u is a
// join node the answer is the run of u's children from the one
//   ↳ containing l to the one containing
// r (binary search over the children, whose spans are sorted).
// NODE COUNT is  $< 2n$ , so size every array  $2n$ . Leaves are  $1..n$ ,
//   ↳ internal nodes  $n+1..id-1$ .
// SIMPLE PERMUTATIONS are exactly those whose tree is a single cut
//   ↳ node over n leaves.
// THIS IS MODULAR DECOMPOSITION specialised to permutation graphs -
//   ↳ the same "join / cut" shape
// appears for cographs and for general graphs, where it is much
//   ↳ harder to compute.

```


2.23 DynamicTreeDiameter

DYNAMIC TREE DIAMETER - the diameter of a weighted tree under ONLINE edge-weight changes,

```
// O(log n) per update, O(1) to read the answer. The tree SHAPE is
//   ↳ fixed; only the weights move.
// WHEN: "after each update print the diameter" (CF 1192B and every
//   ↳ problem shaped like it). If the
// weights are static, two DFS or the tree DP in Θ9 - Trees/05 is
//   ↳ O(n). If edges are ADDED or
// REMOVED you need a link-cut tree (Θ9 - Trees/11) or offline divide
//   ↳ and conquer instead.
// KEY INVARIANT: put the tree on its Euler tour and let f[p] = depth
//   ↳ of the vertex visited at
// position p. For any x ≤ y ≤ z in the tour, f[x] - 2f[y] + f[z]
//   ↳ ≤ dist between the vertices at
// x and z (with equality when y is their LCA, which is always
//   ↳ visited between them), so the
// diameter is exactly max over x ≤ y ≤ z of f[x] - 2f[y] + f[z].
//   ↳ Changing an edge's weight by
// delta adds delta to f over the CONTIGUOUS tour range of the
//   ↳ child's subtree - a range add. The
// segment tree node keeps the five best partial expressions so they
//   ↳ merge in O(1):
// a = max f[x], b = max -2f[y], c = max f[x]-2f[y], d = max
//   ↳ -2f[y]+f[z], e = the answer.
struct DynDiameter {
    struct Node { ll a = 0, b = 0, c = 0, d = 0, e = 0; };
    static Node mrg(const Node& L, const Node& R) {
        Node x;
        x.a = max(L.a, R.a), x.b = max(L.b, R.b);
        x.c = max({L.c, R.c, L.a + R.b});
        x.d = max({L.d, R.d, L.b + R.a});
        x.e = max({L.e, R.e, L.c + R.a, L.a + R.d});
        return x;
    }
    int n, T = 0;
    vector<Node> t;
    vector<ll> lz, w;
    vector<int> st, en, low;
    ↳ low[e] = deeper endpoint of edge e
    vector<vector<pii>> g;
    ↳ {to, edge id}
    DynDiameter(int n) : n(n), t(16 * n), lz(16 * n, 0), st(n),
    ↳ en(n), g(n) {}
    void addEdge(int u, int v, ll c) {
        int e = w.size();
        w.push_back(c), low.push_back(-1);
        g[u].push_back({v, e}), g[v].push_back({u, e});
    }
    void dfs(int u, int p) {
        st[u] = ++T;
        for (auto [v, e] : g[u])
            if (v != p) low[e] = v, dfs(v, u), ++T;
        en[u] = T;
    }
    void apply(int x, ll v) {
        t[x].a += v, t[x].b -= 2 * v, t[x].c -= v, t[x].d -= v, lz[x]
        ↳ += v;
    }
    void push(int x) { if (lz[x]) apply(2 * x, lz[x]), apply(2 * x +
    ↳ 1, lz[x]), lz[x] = 0; }
    void upd(int x, int lo, int hi, int l, int r, ll v) {
        if (r < lo || hi < l) return;
        if (l ≤ lo && hi ≤ r) { apply(x, v); return; }
        int m = (lo + hi) / 2;
        push(x);
        upd(2 * x, lo, m, l, r, v), upd(2 * x + 1, m + 1, hi, l, r,
        ↳ v);
        t[x] = mrg(t[2 * x], t[2 * x + 1]);
    }
    void build(int root = 0) {
        ↳ call after every addEdge
        dfs(root, -1);
        for (int e = 0; e < (int)w.size(); e++) {
            ll c = w[e];
            w[e] = 0;
            setW(e, c);
            ↳ all
        }
        ↳ f start at 0, so just add
    }
    void setW(int e, ll c) {
        ↳ set
        ↳ edge e's weight to c
        upd(1, 1, T, st[low[e]], en[low[e]], c - w[e]);
        w[e] = c;
    }
    ll diameter() { return t[1].e; }
};
```

```
// NEGATIVE WEIGHTS break it: the argument needs f[x] - 2f[LCA] +
//   ↳ f[z] to be maximised at the true
// deepest pair, which assumes non-negative edges. Shift all weights
//   ↳ up if you must.
// THE SAME TOUR TRICK gives dist(u, v) queries under updates: dist =
//   ↳ f[st[u]] + f[st[v]] -
// 2 * min f over the tour between them, so keep a min segment tree
//   ↳ alongside.
// DIAMETER ENDPOINTS: store argmaxes next to each of the five
//   ↳ maxima; the merge picks them up
// unchanged. Only do this if the problem asks - it triples the node.
// SIMPLER SPECIAL CASE: if updates only ever INCREASE weights, the
//   ↳ diameter is monotone and a
// small-to-large "merge the two deepest per subtree" DP is enough.
```

2.24 PersistentHeap

PERSISTENT MELDABLE HEAP - push / pop / meld that RETURN a new heap and leave the old ones

```
// intact. O(log n) time and O(log n) new nodes per operation.
// WHEN: an algorithm needs many heaps that share most of their
//   ↳ contents. The canonical one is
// EPPSTEIN's k shortest walks: every vertex's candidate heap is its
//   ↳ parent's heap plus its own
// out-edges, so you meld into a COPY and keep the parent's version
//   ↳ alive. Also "k best paths in a
// DAG", "k-th best subset", and persistent priority queues in
//   ↳ offline simulations.
// If nothing needs the old version, use 19 - MergeableHeap
//   ↳ (destructive, no allocation per op).
// KEY INVARIANT: leftist heap - every node stores d = its null-path
//   ↳ length and keeps the LONGER
// path on the left, so the right spine is O(log n) and meld only
//   ↳ ever walks right spines. Making
// it persistent then costs nothing extra: COPY each node you touch
//   ↳ on that spine instead of
// mutating it, which is exactly O(log n) copies.
struct PHeap {
    struct Node { int l = 0, r = 0, d = 0; ll key = 0; };
    vector<Node> t;
    PHeap() : t(1) {}
    ↳ node 0 is the empty heap, d = 0
    int meld(int a, int b) {
        ↳ returns a NEW root, min-heap
        if (!a || !b) return a | b;
        if (t[a].key > t[b].key) swap(a, b);
        int c = t.size();
        t.push_back(t[a]);
        ↳ copy, never mutate a
        int r = meld(t[c].r, b);
        t[c].r = r;
        if (t[t[c].l].d < t[r].d) swap(t[c].l, t[c].r);
        t[c].d = t[t[c].r].d + 1;
        return c;
    }
    int push(int a, ll key) {
        t.push_back({0, 0, 1, key});
        return meld(a, (int)t.size() - 1);
    }
    ll top(int a) { return t[a].key; }
    ↳ != 0
    int pop(int a) { return meld(t[a].l, t[a].r); }
    bool empty(int a) { return !a; }
};
// MEMORY: O((n + q) log n) nodes. If a heap is only ever pushed into
//   ↳ (never melded), a persistent
// BINARY heap over a persistent array (20 - PersistentAndUndo) is
//   ↳ smaller but has no meld.
// A LAZY ADD-TO-ALL tag does NOT combine with persistence for free:
//   ↳ a tag is a mutation, so store
// the offset OUTSIDE the heap (carry "this version's keys are all
//   ↳ shifted by s" in the caller) -
// that is what Eppstein and the directed-MST contraction actually do.
// PERSISTENT QUEUE, if that is what you came for, is much cheaper:
//   ↳ keep a persistent ARRAY plus
// two indices (head, tail) per version. push = write at tail and
//   ↳ store {head, tail+1}; pop = read
// at head and store {head+1, tail}. O(log n) per op, no heap needed.
// PERSISTENT STACK is a linked list: every version is just a node
//   ↳ pointer. O(1) per op.
```

2.25 StaticToDynamic

STATIC TO DYNAMIC (the LOGARITHMIC METHOD) - makes ANY static structure accept insertions.

```
// If the static one builds in  $O(B(n))$  and answers in  $O(Q(n))$ , you
//   ↳ get insert in  $O(B(n)/n * \log n)$ 
// AMORTISED and query in  $O(Q(n) * \log n)$  - you ask every bucket and
//   ↳ combine.
// WHEN: you have a build-once structure (convex hull, sorted array,
//   ↳ wavelet tree, suffix
// automaton, Aho-Corasick, KD-tree, sparse table) and the problem
//   ↳ inserts items online. Do NOT
// use it when a natively dynamic structure exists and is fast enough
//   ↳ a BIT beats "buckets of
// sorted arrays" for prefix counting, and a balanced BST beats
//   ↳ "buckets of sorted arrays" for
// order statistics.
// KEY INVARIANT: keep at most one instance of each size  $2^0, 2^1,$ 
//   ↳  $2^2, \dots$  - a BINARY COUNTER.
// Inserting sets the lowest empty bit and merges every full bucket
//   ↳ below it, so a bucket of size
//  $2^k$  is rebuilt only once per  $2^k$  insertions; summed over  $k$  that is
//   ↳  $O(\log n)$  rebuilds of
// amortised cost  $B(n)/n$  each. Because the buckets partition the
//   ↳ items, ANY decomposable query
// (one whose answer over a union is a cheap combination of the
//   ↳ answers over the parts: count, sum,
// min, max, "does there exist") can be answered by asking all  $O(\log$ 
//   ↳  $n)$  of them.
template <class Static, class Item>
struct StaticToDynamic {
    vector<vector<Item>> buf; //
    ↳ buf[k] is empty or has  $2^k$  items
    vector<Static> ds;
    void insert(Item x) {
        vector<Item> cur{x};
        int k = 0;
        while (k < (int)buf.size() && !buf[k].empty()) {
            for (auto& y : buf[k]) cur.push_back(y);
            buf[k].clear(), ds[k] = Static();
            k++;
        }
        while ((int)buf.size() <= k) buf.emplace_back(),
        ↳ ds.emplace_back();
        buf[k] = cur, ds[k] = Static(cur); // one
        ↳ rebuild, size  $2^k$ 
    }
    template <class F> void each(F f) { //
        ↳ f(instance) over every bucket
        for (int k = 0; k < (int)ds.size(); k++) if (!buf[k].empty())
        ↳ f(ds[k]);
    }
};
// EXAMPLE static structure - a sorted array answering "how many
//   ↳ items <= x".
struct SortedBucket {
    vector<ll> v;
    SortedBucket() {}
    SortedBucket(vector<ll> a) : v(a) { sort(all(v)); }
    int count(ll x) { return upper_bound(all(v), x) - v.begin(); }
};
// USAGE: StaticToDynamic<SortedBucket, ll> d; d.insert(5);
//   ↳ int c = 0; d.each([&](SortedBucket& b) { c += b.count(x);
//   ↳ });
// DELETIONS, if the aggregate is INVERTIBLE (counts, sums): keep a
//   ↳ SECOND instance holding the
// deleted items and answer query(all) - query(deleted). This is
//   ↳ "deletion by counting" and it is
// how you get a fully dynamic convex-hull-trick or a deletable
//   ↳ "count <= x".
// DELETIONS otherwise (min, max, "exists"): you cannot subtract -
//   ↳ use offline dynamic connectivity
// style divide and conquer over the timeline (14) instead.
// WORST CASE: a single insertion can cost  $O(B(n))$  (rebuilding the
//   ↳ whole thing). If the problem has
// a per-query time limit rather than a total one, spread the rebuild
//   ↳ over the next  $2^k$  insertions
// (the "global rebuilding" / de-amortisation trick) - rarely needed
//   ↳ in contests.
```

2.26 TreeBinarization

TREE BINARIZATION - rewrite a tree so that EVERY vertex has degree ≤ 3 , by inserting dummy

```
// vertices joined with ZERO-weight edges.  $O(n)$  time, at most  $n - 2$ 
//   ↳ new vertices, and every
// distance between original vertices is unchanged.
// WHEN: it is a PRECONDITION, not an algorithm. Reach for it
//   ↳ whenever a per-vertex cost is
// proportional to the DEGREE and you need it to be  $O(1)$ :
//   ↳ - PERSISTENT centroid decomposition (27): copying a centroid
//     ↳ node must be  $O(1)$ , so a centroid
//     ↳ may have at most 3 children in the centroid tree.
//   ↳ - "merge the children's DP tables" small-to-large /
//     ↳ knapsack-on-tree DPs: after binarization
//     ↳ each merge has exactly two operands, which is what makes the
//     ↳ classic  $O(n^2)$  tree knapsack
//     ↳ analysis (and the  $O(n \log n)$  small-to-large bound) actually
//     ↳ hold.
//   ↳ - dynamic-tree structures whose node stores a fixed number of
//     ↳ child pointers (top trees).
// KEY INVARIANT: replace the star  $u \rightarrow c_1, c_2, \dots, c_k$  by a
//   ↳ CATERPILLAR:  $u$  keeps  $c_1$  and a chain of
// dummies  $d_1 - d_2 - \dots$  carries the rest, each dummy holding one
//   ↳ real child. Zero-weight edges mean
// distances are preserved exactly; original vertex ids are
//   ↳ preserved, so any answer indexed by
// vertex still makes sense. Depth grows by at most  $O(\deg)$ , so a DFS
//   ↳ on the result is still  $O(n)$ .
struct Binarize {
    int n, cnt; //
    ↳ 1..n original,  $n+1..cnt$  dummies
    vector<vector<pii>> g; //
    ↳ result: {neighbour, weight}
    Binarize(const vector<vector<pii>>& G, int root = 1)
        : n(G.size() - 1), cnt(G.size() - 1), g(2 * G.size()) {
        ↳ rec(G, root, 0); }
    void link(int u, int v, int w) { g[u].push_back({v, w}),
    ↳ g[v].push_back({u, w}); }
    void rec(const vector<vector<pii>>& G, int u, int p) {
        int last = u, k = 0, deg = G[u].size() - (p != 0);
        for (auto [v, w] : G[u]) {
            if (v == p) continue;
            if (++k == 1 || k == deg) link(last, v, w); //
            ↳ first and last hang off 'last'
            else {
                int d = ++cnt; //
                ↳ dummy carrying one middle child
                link(last, d, 0), link(d, v, w), last = d;
            }
        }
        for (auto [v, w] : G[u]) if (v != p) rec(G, v, u);
    }
};
// SIZE EVERY ARRAY  $2n$  - the vertex count can nearly double.
// WEIGHTS: dummy edges are 0, so shortest paths, diameters and
//   ↳ centroid distances are identical.
// If your problem counts EDGES rather than weights, give the real
//   ↳ edges weight 1 and the dummies 0.
// VERTEX VALUES: dummies must be neutral for whatever you aggregate
//   ↳ ( $0$  for sums,  $+\infty$  for minima).
// BINARY (degree  $\leq 3$ ) IS THE RIGHT TARGET, not "binary rooted
//   ↳ tree": a rooted binarization that
// keeps exactly two children per node needs the same construction
//   ↳ plus a root of degree  $\leq 2$ .
```

2.27 PersistentCentroid

PERSISTENT CENTROID DECOMPOSITION - "sum of $\text{dist}(u, x)$ over all marked vertices x ", where the

```
// Needs: 26 - TreeBinarization.cpp
// set of marked vertices is versioned: version i = the first i
//   marks. Insert  $O(\log n)$ , query
//    $O(\log n)$ , memory  $O(n \log n)$ .
// WHEN: queries of the form "consider only the marks with index in
//    $[l, r]$ , answer for vertex  $u$ " -
// subtract two versions (CF 757G). Also "the  $k$ -th mark", online,
//   when marks arrive in a fixed
// order. If the mark set only ever grows and you never look back,
//   plain centroid decomposition
// ( $\Theta(9)$  - Trees/04) with the same accumulators is enough and far
//   cheaper.
// COMPLEXITY REQUIRES BINARIZATION FIRST: a centroid node must have
//    $O(1)$  children so that copying
// it is  $O(1)$ . Feed the tree through 26 - TreeBinarization.
// KEY INVARIANT: the centroid tree has depth  $O(\log n)$  and every
//   vertex  $u$  lies in exactly one
// component per level, so  $\text{dist}(u, x) = d[\text{level}][u] + d[\text{level}][x]$ 
//   where the level is that of the
// LOWEST common centroid of  $u$  and  $x$ . Each centroid node stores cnt
//   (marks in its component), sum
// (total distance from the centroid to them), and par (their total
//   distance to the centroid's
// PARENT centroid). Walking from the root of the centroid tree down
//   to  $u$  then adds, at each step,
// the contribution of exactly the marks that "branch off" there -
//   the standard inclusion-exclusion
// that stops double counting. Marking one vertex touches only the
//    $O(\log n)$  centroid nodes on that
// root-to- $u$  path, so persistence costs  $O(\log n)$  copies.
```

```
struct PCentroid {
    struct Node { array<int, 3> ct{{0, 0, 0}}; int id = 0, lvl = 0,
        cnt = 0; ll sum = 0, par = 0; };
    vector<Node> t;
    vector<vector<pii>> g; //
    BINARISED tree, 1-indexed
    vector<vector<ll>> d; //
    d[lvl][v]
    vector<int> st, en, sub, done;
    int n, DT = 0, tot = 0, root = 0;
    PCentroid(const vector<vector<pii>>& g, int nv, int LG = 20)
        : t(nv + 1), g(g), d(LG, vector<ll>(nv + 1, 0)), st(nv + 1),
    en(nv + 1), sub(nv + 1),
        done(nv + 1, 0), n(nv) { root = decompose(1, 0, 0); }
    void calcSz(int u, int p) {
        tot++, sub[u] = 1;
        for (auto [v, w] : g[u])
            if (v != p && !done[v]) calcSz(v, u), sub[u] += sub[v];
    }
    int findCen(int u, int p) {
        for (auto [v, w] : g[u])
            if (v != p && !done[v] && sub[v] > tot / 2) return
    findCen(v, u);
        return u;
    }
    void dist(int u, int p, ll cur, int l) {
        d[l][u] = cur;
        for (auto [v, w] : g[u])
            if (v != p && !done[v]) dist(v, u, cur + w, l);
    }
    int decompose(int u, int p, int l) {
        tot = 0, calcSz(u, p);
        int c = findCen(u, p);
        done[c] = 1, st[c] = ++DT, t[c].id = c, t[c].lvl = l;
        dist(c, 0, 0, l);
        int k = 0;
        for (auto [v, w] : g[c])
            if (!done[v]) t[c].ct[k++] = decompose(v, c, l + 1);
        en[c] = DT;
        return c;
    }
    bool insub(int x, int u) { return st[t[x].id] <= st[u] && en[u]
    <= en[t[x].id]; }
    int mark(int cur, int u) { //
    returns the NEW version root
        t.push_back(t[cur]);
        int c = t.size() - 1;
        ll dv = d[t[c].lvl][u];
        t[c].cnt++, t[c].sum += dv;
        for (int k = 0; k < 3; k++) {
            int v = t[c].ct[k];
            if (v && insub(v, u)) {
                int nv = mark(v, u);
                t[nv].par += dv, t[c].ct[k] = nv;
            }
        }
        return c;
    }
};
```

```
    }
    return c;
}
ll query(int cur, int u) { // sum
    of dist(u, x) over marks
    ll ans = 0;
    while (t[cur].id != u) {
        int v = 0;
        for (int k = 0; k < 3; k++)
            if (t[cur].ct[k] && insub(t[cur].ct[k], u)) v =
    t[cur].ct[k];
        ans += d[t[cur].lvl][u] * (t[cur].cnt - t[v].cnt) +
    t[cur].sum - t[v].par;
        cur = v;
    }
    return ans + t[cur].sum;
}
};
// USAGE: Binarize b(G); PCentroid pc(b.g, b.cnt); then ver[0] =
    pc.root and
// ver[i] = pc.mark(ver[i-1], x_i). The answer for marks l..r is
// pc.query(ver[r], u) - pc.query(ver[l-1], u).
// RESERVE t:  $n + (\text{marks}) * \log n$  nodes; each is ~40 bytes.
// OTHER ACCUMULATORS: replace cnt/sum by whatever is additive over
//   the marks - count within a
// distance bound (keep a BIT per centroid instead, which kills
//   persistence), sum of weights,
// number of pairs. Anything that needs a per-centroid ARRAY does not
//   persist cheaply.
// DUMMY VERTICES from binarization are never marked and never
//   queried, but they DO appear as
// centroids - which is fine, since query() walks the centroid tree
//   by vertex id and only stops at
// the id it was asked for.
```

3 Graph Theory

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3.1 Graph Traversal

3.1.1 TopoSort

Topological Sort (Kahn's Algorithm) - $O(V + E)$

```
struct TopoSort {
    int n;
    vector<vector<int>> adj;
    vector<int> deg, order;

    TopoSort(int n = 0) : n(n), adj(n + 1), deg(n + 1, 0) {}

    void add_edge(int u, int v) {
        adj[u].pb(v);
        deg[v]++;
    }

    bool solve() {
        queue<int> q;
        for (int i = 1; i <= n; i++) {
            if (!deg[i]) q.push(i);
        }
        while (!q.empty()) {
            int u = q.front();
            q.pop();
            order.pb(u);
            for (int v : adj[u]) {
                if (--deg[v]) q.push(v);
            }
        }
        return (int)order.size() == n; // false if graph has cycles
    }
};
```

3.1.2 BFS_DFS

BFS / DFS / 0-1 BFS / multi-source. 0-based. The bread and butter - keep them short.

```
// Unweighted shortest path + parent for reconstruction.
pair<vector<int>, vector<int>> bfs(int n, const vector<vector<int>>&
    ↪ adj, vector<int> src) {
    vector<int> d(n, -1), par(n, -1);
    queue<int> q;
    for (int s : src) d[s] = 0, q.push(s); // MULTI-SOURCE:
    ↪ pass every source at once
    while (!q.empty()) {
        int u = q.front(); q.pop();
        for (int v : adj[u]) if (d[v] < 0) d[v] = d[u] + 1, par[v] =
    ↪ u, q.push(v);
    }
    return {d, par};
}

vector<int> pathTo(int t, const vector<int>& par) {
    vector<int> p;
    for (int v = t; v >= 0; v = par[v]) p.push_back(v);
    reverse(all(p));
    return p;
}

// 0-1 BFS: edge weights are only 0 or 1 -> deque instead of a heap,
    ↪ O(V + E).
// Use for "turning costs 1", "breaking one wall is free", "flip at
    ↪ most k edges".
vector<ll> bfs01(int n, const vector<vector<pair<int, int>>& adj,
    ↪ int s) {
    vector<ll> d(n, llinf);
    deque<int> q;
    d[s] = 0, q.push_back(s);
    while (!q.empty()) {
        int u = q.front(); q.pop_front();
        for (auto [v, w] : adj[u]) if (d[u] + w < d[v]) {
            d[v] = d[u] + w;
            w ? q.push_back(v) : q.push_front(v);
        }
    }
    return d;
}

// Iterative DFS with entry/exit times (avoids stack overflow at n =
    ↪ 2e5).
void dfsIter(int n, const vector<vector<int>>& adj, int root,
    ↪ vector<int>& tin, vector<int>& tout) {
    tin.assign(n, -1), tout.assign(n, -1);
    vector<int> it(n, 0), st = {root};
    int timer = 0;
    tin[root] = timer++;
    while (!st.empty()) {
        int u = st.back();
        if (it[u] < (int)adj[u].size()) {
            int v = adj[u][it[u]++];
            if (tin[v] < 0) tin[v] = timer++, st.push_back(v);
        } else tout[u] = timer++, st.pop_back();
    }
}

// GRID: 4- and 8-neighbour offsets (see 03 - grid.cpp). Encode a
    ↪ cell as r*m+c and reuse the
// graph routines above; add extra state dimensions (mask, parity,
    ↪ keys) by widening the index.
// CYCLE DETECTION, directed: colour DFS (0 white, 1 grey, 2 black);
    ↪ a grey->grey edge closes a
// cycle - walk the recursion stack back to extract it. Undirected:
    ↪ any edge to an already-visited
// non-parent vertex closes one.
```

3.2 Shortest Paths

3.2.1 Dijkstra

Dijkstra's Shortest Path Algorithm - $O((V + E) \log V)$

```
struct Dijkstra {
    int n;
    vector<vector<pair<int, ll>>> adj;
    vector<ll> dist;
    vector<int> par;

    Dijkstra(int n = 0) : n(n), adj(n + 1), dist(n + 1, llinf), par(n
    ↪ + 1, -1) {}

    void add_edge(int u, int v, ll w, bool directed = false) {
        adj[u].eb(v, w);
        if (!directed) adj[v].eb(u, w);
    }

    void solve(int src) {
        fill(all(dist), llinf);
        fill(all(par), -1);
        priority_queue<pair<ll, int>, vector<pair<ll, int>>,
    ↪ greater<>> pq;

        dist[src] = 0;
        pq.emplace(0, src);

        while (!pq.empty()) {
            auto [d, u] = pq.top();
            pq.pop();
            if (d > dist[u]) continue;

            for (auto& [v, w] : adj[u]) {
                if (dist[u] + w < dist[v]) {
                    dist[v] = dist[u] + w;
                    par[v] = u;
                    pq.emplace(dist[v], v);
                }
            }
        }
    };
};
```

3.2.2 FloydWarshall

Floyd-Warshall All-Pairs Shortest Path - $O(V^3)$

```
struct FloydWarshall {
    int n;
    vector<vector<ll>> dist;

    FloydWarshall(int n = 0) : n(n), dist(n + 1, vector<ll>(n + 1,
    ↪ llinf)) {}

    for (int i = 0; i <= n; i++) dist[i][i] = 0;

    void add_edge(int u, int v, ll w, bool directed = false) {
        dist[u][v] = min(dist[u][v], w);
        if (!directed) dist[v][u] = min(dist[v][u], w);
    }

    void solve() {
        for (int k = 1; k <= n; k++) {
            for (int i = 1; i <= n; i++) {
                for (int j = 1; j <= n; j++) {
                    if (dist[i][k] < llinf && dist[k][j] < llinf) {
                        dist[i][j] = min(dist[i][j], dist[i][k] +
    ↪ dist[k][j]);
                    }
                }
            }
        }
    };
};
```

3.2.3 BellmanFord

If you paste both, keep only ONE copy of it.

```
// NOTE: 08 - Spanning Trees/01 - MST.cpp declares an identical
// ↳ 'struct Edge'.
// If you paste both, keep only ONE copy of it.
// Bellman-Ford Shortest Path with Negative Cycle Detection -  $O(V * E)$ 
struct Edge {
    int u, v;
    ll w;
};

struct BellmanFord {
    int n;
    vector<Edge> edges;
    vector<ll> dist;
    vector<int> par;

    BellmanFord(int n = 0) : n(n), dist(n + 1, llinf), par(n + 1, -1)
    ↳ {}

    void add_edge(int u, int v, ll w) {
        edges.pb({u, v, w});
    }

    // Returns false if a negative cycle is reachable from src
    bool solve(int src) {
        fill(all(dist), llinf);
        fill(all(par), -1);
        dist[src] = 0;

        for (int i = 0; i < n - 1; i++) {
            bool relaxed = false;
            for (const auto& e : edges) {
                if (dist[e.u] < llinf && dist[e.u] + e.w < dist[e.v])
                    ↳ {
                        dist[e.v] = dist[e.u] + e.w;
                        par[e.v] = e.u;
                        relaxed = true;
                    }
            }
            if (!relaxed) break;
        }

        for (const auto& e : edges) {
            if (dist[e.u] < llinf && dist[e.u] + e.w < dist[e.v]) {
                return false; // Negative cycle detected
            }
        }
        return true;
    }
};
```

3.2.4 DominatorTree

```
/*
=====
CATALOGUE: Lengauer-Tarjan Dominator Tree ( $O((N + M) \log N)$ )
=====
DESCRIPTION:
Finds the "bottlenecks" in any directed graph. A node 'u'
↳ dominates 'v' if
EVERY path from the root to 'v' is forced to go through 'u'.

USAGE:
1. Build graph: Create a 2D vector 'adj' (1-based) containing the
↳ directed edges.
2. Initialize: DominatorTree dt(n, root, adj);
(Automatically builds the tree on instantiation)

EXTRACTING ANSWERS:
- dt.dom[u]: The Immediate Dominator of 'u' (the closest
↳ bottleneck to 'u').
- dt.id[u]: If dt.id[u] == 0 after building, 'u' is
↳ UNREACHABLE from the root.
=====
*/
struct DominatorTree {
    int T = 0, n;
    vector<vector<int>> rg, bucket, adj;
    vector<int> dsu, par, sdom, idom, dom, label, id, rev;

    DominatorTree(int _n, int r, vector<vector<int>>& adj):
        n(_n + 1), rg(n), bucket(n), adj(adj), dsu(n),
        par(n), sdom(n), idom(n), dom(n), label(n), id(n), rev(n) {
        ↳ dfs(r); build(r); }

    int find(int u, int x = 0) {
        if (u == dsu[u]) return x ? -1 : u;
        int v = find(dsu[u], x + 1);
        if (v < 0) return u;
        if (sdom[label[dsu[u]]] < sdom[label[u]]) label[u] =
        ↳ label[dsu[u]];
        dsu[u] = v;
        return x ? v : label[u];
    }

    void dfs(int u) {
        id[u] = ++T, rev[T] = u;
        label[T] = sdom[T] = dsu[T] = T;
        for (int& w : adj[u]) {
            if (!id[w]) {
                dfs(w);
                par[id[w]] = id[u];
            }
            rg[id[w]].push_back(id[u]);
        }
    }

    void build(int r) {
        // FIXED: Loop MUST go down to i >= 1 to process the root's
        ↳ bucket!
        for (int i = T; i >= 1; i--) {
            for (int& u : rg[i]) sdom[i] = min(sdom[i],
            ↳ sdom[find(u)]);
            if (i > 1) bucket[sdom[i]].push_back(i);

            for (int& w : bucket[i]) {
                int v = find(w);
                idom[w] = sdom[v] == sdom[w] ? sdom[w] : v;
            }

            if (i > 1) dsu[i] = par[i];
        }

        for (int i = 2; i <= T; i++) {
            if (idom[i] != sdom[i]) idom[i] = idom[idom[i]];
        }

        for (int u = 1; u <= T; ++u) {
            dom[rev[u]] = rev[idom[u]];
        }
    }
};

void testCase() {
    int n, m, s;
    cin >> n >> m >> s;
    s++; // Convert to 1-based indexing

    vector<vector<int>> adj(n + 1);
```

```

for (int i = 0; i < m; i++) {
    int u, v;
    cin >> u >> v;
    adj[u + 1].push_back(v + 1);
}

// Automatically builds upon instantiation
DominatorTree dt(n, s, adj);

for (int i = 1; i <= n; i++) {
    if (i == s) {
        // pS is S
        cout << s - 1 << (i == n ? "" : " ");
    } else if (!dt.id[i]) {
        // If we can not reach i from S, print -1
        cout << -1 << (i == n ? "" : " ");
    } else {
        // Print the 0-based parent (which is now guaranteed to
        // be calculated)
        cout << dt.dom[i] - 1 << (i == n ? "" : " ");
    }
}
cout << "\n";
}

```

3.2.5 ShortestPathEdgeModify

SHORTEST PATH WITH ONE EDGE CHANGED, answered OFFLINE in $O(1)$ per query after $O((n+m) \log n)$.

```

// Query: "if edge id had weight x instead, what is dist(s,t)?" -
//   ↳ each query is independent.
// IDEA. Let ds/dt be the distance arrays from s and to t, P the
//   ↳ shortest path, base = ds[t].
// * Edge NOT on P: the new path either ignores it (base) or
//   ↳ crosses it once:
//   min(base, ds[u] + x + dt[v], ds[v] + x + dt[u]).
// * Edge ON P at position i: either still use it (base - w + x),
//   ↳ or avoid it entirely, and
//   "the best s->t path avoiding path-edge i" is precomputed for
//   ↳ every i at once:
//   for a non-path edge (u,v,w), the path ds[u] -> u -> v -> dt[v]
//   ↳ skips exactly the path edges
//   in the window [L[u], R[v]], where L[x] = last path vertex on
//   ↳ some shortest s->x path and
//   R[x] = first path vertex on some shortest x->t path. Sweep i
//   ↳ from 0 to k-1 with a heap.
struct SPEdgeModify {
    struct Edge { int u, v; ll w; };
    int n, k; // k =
    ↳ #edges on the shortest path
    ll base;
    vector<Edge> E;
    vector<ll> ds, dt, best; //
    ↳ best[i] = s->t avoiding path edge i
    vector<int> pos; //
    ↳ pos[e] = index on P, or -1
    SPEdgeModify(int n, vector<Edge> E, int s, int t) : n(n), E(E) {
        int m = E.size();
        vector<vector<pair<int, int>>> adj(n);
        for (int i = 0; i < m; i++)
            adj[E[i].u].push_back({E[i].v, i});
        ↳ adj[E[i].v].push_back({E[i].u, i});
        vector<int> par(n, -1), pare(n, -1), ordS, ordT;
        auto dij = [&](int src, vector<ll> &d, vector<int> &ord, bool
        ↳ tree) {
            d.assign(n, llinf);
            priority_queue<pair<ll, int>, vector<pair<ll, int>>,
            ↳ greater<>> pq;
            d[src] = 0, pq.push({0, src});
            while (!pq.empty()) {
                auto [dd, u] = pq.top(); pq.pop();
                if (dd > d[u]) continue;
                ord.push_back(u);
                for (auto [v, id] : adj[u]) if (d[u] + E[id].w <
                ↳ d[v]) {
                    d[v] = d[u] + E[id].w;
                    if (tree) par[v] = u, pare[v] = id;
                    pq.push({d[v], v});
                }
            }
            dij(s, ds, ordS, true), dij(t, dt, ordT, false);
            base = ds[t];
            vector<int> pv, pe; // the
            ↳ path, s ... t
            for (int c = t; c != s; c = par[c]) pv.push_back(c),
            ↳ pe.push_back(pare[c]);
            pv.push_back(s), reverse(all(pv)), reverse(all(pe));
            k = pe.size();
            pos.assign(m, -1);
            vector<char> onP(n, 0);
            for (int x : pv) onP[x] = 1;
            for (int i = 0; i < k; i++) pos[pe[i]] = i;
            vector<int> L(n, 0), R(n, k);
            for (int i = 0; i <= k; i++) L[pv[i]] = R[pv[i]] = i;
            for (int u : ordS) for (auto [v, id] : adj[u]) //
            ↳ relaxed in shortest-path order
                if (!onP[v] && ds[v] == ds[u] + E[id].w) L[v] = max(L[v],
            ↳ L[u]);
            for (int u : ordT) for (auto [v, id] : adj[u])
                if (!onP[v] && dt[v] == dt[u] + E[id].w) R[v] = min(R[v],
            ↳ R[u]);
            vector<vector<pair<ll, int>>> ev(max(k, 1)); //
            ↳ ev[start] = {cost, end}
            for (int i = 0; i < m; i++) {
                if (pos[i] >= 0) continue;
                auto [u, v, w] = E[i];
                if (L[u] < R[v]) ev[L[u]].push_back({ds[u] + w + dt[v],
            ↳ R[v]});
                if (L[v] < R[u]) ev[L[v]].push_back({ds[v] + w + dt[u],
            ↳ R[u]});
            }
            best.assign(max(k, 1), llinf);

```



```

    priority_queue<pair<ll, int>, vector<pair<ll, int>>,
    greater<>> sw;
    for (int i = 0; i < k; i++) {
        for (auto& e : ev[i]) sw.push(e);
        while (!sw.empty() && sw.top().second <= i) sw.pop();
        if (!sw.empty()) best[i] = sw.top().first;
    }
}
ll query(int id, ll x) const {
    edge id gets weight x
    auto [u, v, w] = E[id];
    if (pos[id] < 0) return min({base, ds[u] + x + dt[v], ds[v] +
    x + dt[u]});
    return min(base - w + x, best[pos[id]]);
}
};
// RELATED, same precomputation:
// * "Does edge e lie on SOME shortest path?" <=> ds[u] + w +
    dt[v] == base (either direction).
// * "Does e lie on EVERY shortest path?" <=> removing it
    increases the distance; the sweep
// above answers this for path edges (best[i] > base means path
    edge i is essential).
// * REPLACEMENT PATHS (delete one edge, ask the new distance) =
    query(id, +infinity).
// * SECOND SHORTEST WALK (may reuse edges): min over edges of
    ds[u] + w + dt[v] strictly > base.
// * If instead the whole graph changes per query, you need a fresh
    Dijkstra - this trick only
// works because s, t and every other weight stay fixed.

```

3.2.6 CyclesAndConstraints

MINIMUM MEAN CYCLE, and SYSTEMS OF DIFFERENCE CONSTRAINTS - the two shortest-path tools that

// are not about shortest paths.

```

// --- KARP'S MINIMUM MEAN CYCLE,  $O(V \cdot E)$ . Returns the smallest
    possible (cycle weight)/(length).
// dp[k][v] = min weight of a walk of EXACTLY k edges ending at v
    (from a virtual source that
// reaches everything). Karp: the answer is min over v of max over
    k < n of (dp[n][v] - dp[k][v]) / (n - k).
// Reach for it whenever the objective is an AVERAGE around a cycle:
    "minimum cost per unit time",
// "is there a cycle with negative average", profit-per-day loops.
ld minMeanCycle(int n, const vector<array<ll, 3>>& e) { // e =
    {u, v, w}, directed
    vector<vector<ll>> dp(n + 1, vector<ll>(n, llinf));
    fill(all(dp[0]), 0);
    // virtual source reaches every v
    for (int k = 1; k <= n; k++)
        for (auto [u, v, w] : e)
            if (dp[k - 1][u] < llinf) dp[k][v] = min(dp[k][v], dp[k -
            1][u] + w);
    ld best = 1e18;
    for (int v = 0; v < n; v++) {
        if (dp[n][v] == llinf) continue;
        ld cur = -1e18;
        for (int k = 0; k < n; k++)
            if (dp[k][v] < llinf) cur = max(cur, (ld)(dp[n][v] -
            dp[k][v]) / (n - k));
        best = min(best, cur);
    }
    return best;
    // 1e18 if the graph is acyclic
}
// MINIMUM COST-TO-TIME RATIO CYCLE (minimise sum(cost)/sum(time),
    time > 0): binary search x and
// test for a negative cycle with weights cost - x*time. Same
    skeleton, one Bellman-Ford per step.
// MAXIMUM mean cycle: negate the weights.

```

```

// --- SYSTEM OF DIFFERENCE CONSTRAINTS. Every constraint  $x_j - x_i$ 
    <= w becomes an edge  $i \rightarrow j$  of
// weight w; add a super-source with a 0-edge to every vertex and run
    Bellman-Ford.
// FEASIBLE iff there is no negative cycle. The distances are then a
    solution, and they are the
// POINTWISE MAXIMUM one (every other solution is <= it
    componentwise, after fixing  $x_{\text{source}} = 0$ ).
// TRANSLATIONS you will need:  $x_j - x_i \geq w$  is  $x_i - x_j \leq -w$ .
     $x_j - x_i = w$  is both.
//  $x_i \leq c$  is  $x_i - x_0 \leq c$  against the source. Strict "<" on
    integers is "<= w-1".
// USES: scheduling with precedence and deadlines, "arrange these
    values so all the gaps hold",
// consistency of a set of relative measurements, and the dual of a
    min-cost-flow.

```

```

bool differenceConstraints(int n, const vector<array<ll, 3>>& c,
    vector<ll>& x) {
    vector<array<ll, 3>> e = c; // c =
    {i, j, w} meaning  $x_j - x_i \leq w$ 
    for (int v = 0; v < n; v++) e.push_back({(ll)n, (ll)v, 0}); //
    super-source n
    x.assign(n + 1, 0);
    for (int it = 0; it <= n; it++) {
        bool any = false;
        for (auto [u, v, w] : e)
            if (x[u] + w < x[v]) x[v] = x[u] + w, any = true;
        if (!any) { x.resize(n); return true; }
    }
    return false; //
    negative cycle: infeasible
}
// --- BELLMAN-FORD EXTRAS worth remembering
// NEGATIVE CYCLE RECOVERY: after n rounds, take a vertex that still
    relaxed, walk n parent
// pointers back (this lands you ON the cycle - the relaxed vertex
    may only be reachable from it),
// then follow parents until a vertex repeats.
// SPFA is Bellman-Ford with a queue; it is  $O(VE)$  worst case and
    killable, but the SLF heuristic
// (push to the front if the new distance beats the front) makes it
    fast in practice.
// ONLY SHORTEST PATHS WITH AT MOST K EDGES: run exactly k rounds,
    updating from a SNAPSHOT of the
// previous round (otherwise one round can propagate along many
    edges).

```

```
// JOHNSON'S APSP for sparse graphs with negative edges: Bellman-Ford
    ↪ from a super-source gives
// potentials h[], reweight  $w'(u,v) = w + h[u] - h[v] \geq 0$ , then run
    ↪ n Dijkstras.  $O(VE + V E \log V)$ .
```

3.2.7 SPFA

SPFA (Bellman-Ford with a queue) - shortest paths with NEGATIVE edges, plus negative-cycle

```
// detection.  $O(V \cdot E)$  worst case and constructible-against, but
    ↪ usually far faster than plain
// Bellman-Ford because only vertices whose distance actually
    ↪ improved get re-examined.
// Returns false iff a negative cycle is reachable from s: a vertex
    ↪ relaxed n times must be on
// (or reachable from) one, since a shortest walk uses at most n-1
    ↪ edges.
// WHEN: negative weights and you also want the cycle check. With
    ↪ NON-negative weights always use
// Dijkstra - SPFA is asymptotically worse and hackable. For a
    ↪ guaranteed bound use Bellman-Ford.
bool spfa(int n, int s, const vector<vector<pair<int, ll>>>& g,
    ↪ vector<ll>& d) {
    d.assign(n, llinf);
    vector<int> cnt(n, 0);
    vector<char> inq(n, 0);
    deque<int> q;
    d[s] = 0, q.push_back(s), inq[s] = 1;
    while (!q.empty()) {
        int v = q.front(); q.pop_front();
        inq[v] = 0;
        for (auto [to, w] : g[v]) if (d[v] + w < d[to]) {
            d[to] = d[v] + w;
            if (!inq[to]) {
                if (++cnt[to] >= n) return false; //
            }
            inq[to] = 1;
            // SLF: a smaller tentative distance goes to the
            ↪ FRONT. Cheap, and it is what
            // makes SPFA fast in practice; without it the queue
            ↪ degenerates.
            if (!q.empty() && d[to] < d[q.front()])
                ↪ q.push_front(to);
            else q.push_back(to);
        }
    }
    return true;
}
// RECOVERING THE CYCLE: keep par[] and, on detection, walk n parent
    ↪ pointers back (that lands you
// ON the cycle - the detected vertex may only be reachable from it),
    ↪ then follow parents until a
// vertex repeats. See 06 - CyclesAndConstraints.cpp, which uses
    ↪ exactly this for difference
// constraints and for the minimum mean cycle.
// KILLER TESTS EXIST: grid-like graphs make SPFA quadratic. If the
    ↪ problem looks adversarial,
// use Bellman-Ford (predictable  $O(VE)$ ) or reweight with Johnson
    ↪ potentials and run Dijkstra.
```

3.2.8 SegmentTreeGraph

SEGMENT-TREE GRAPH - add "u-> every vertex in [l, r]" (or the reverse) with $O(\log n)$ edges

```
// instead of  $O(n)$ . Then run Dijkstra/BFS as usual. Total  $O((n + q \log n) \log n)$ .
// Two trees: OUT (parent -> children, cost 0) to broadcast INTO a
    ↪ range,
// IN (children -> parent, cost 0) to collect FROM a
    ↪ range.
// Leaves of both trees are the original vertices.
struct SegGraph {
    int n, cnt;
    vector<int> outId, inId; // node ids of
    ↪ the two trees
    vector<vector<pair<int, ll>>> g;

    SegGraph(int n) : n(n) {
        cnt = n;
        outId.assign(4 * n, -1), inId.assign(4 * n, -1);
        g.assign(n + 8 * n, {});
        buildOut(1, 0, n - 1), buildIn(1, 0, n - 1);
    }
    int node() { return cnt++; }
    void buildOut(int v, int l, int r) {
        if (l == r) { outId[v] = l; return; }
        outId[v] = node();
        int m = (l + r) / 2;
        buildOut(2 * v, l, m), buildOut(2 * v + 1, m + 1, r);
        g[outId[v]].push_back({outId[2 * v], 0});
        g[outId[v]].push_back({outId[2 * v + 1], 0});
    }
    void buildIn(int v, int l, int r) {
        if (l == r) { inId[v] = l; return; }
        inId[v] = node();
        int m = (l + r) / 2;
        buildIn(2 * v, l, m), buildIn(2 * v + 1, m + 1, r);
        g[inId[2 * v]].push_back({inId[v], 0});
        g[inId[2 * v + 1]].push_back({inId[v], 0});
    }
    void addRange(int v, int l, int r, int ql, int qr, int u, ll w,
    ↪ bool toRange) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr) {
            if (toRange) g[u].push_back({outId[v], w}); // u
            ↪ -> everything in [ql,qr]
            else g[inId[v]].push_back({u, w}); //
            ↪ everything in [ql,qr] -> u
            return;
        }
        int m = (l + r) / 2;
        addRange(2 * v, l, m, ql, qr, u, w, toRange);
        addRange(2 * v + 1, m + 1, r, ql, qr, u, w, toRange);
    }
    void edgeToRange(int u, int l, int r, ll w) { addRange(1, 0, n -
    ↪ 1, l, r, u, w, true); }
    void edgeFromRange(int l, int r, int u, ll w) { addRange(1, 0, n
    ↪ - 1, l, r, u, w, false); }
    void edge(int u, int v, ll w) { g[u].push_back({v, w}); }

    vector<ll> dijkstra(int s) {
        vector<ll> d(cnt, llinf);
        priority_queue<pair<ll, int>, vector<pair<ll, int>>,
        ↪ greater<>> pq;
        d[s] = 0, pq.push({0, s});
        while (!pq.empty()) {
            auto [dd, u] = pq.top(); pq.pop();
            if (dd > d[u]) continue;
            for (auto [v, w] : g[u]) if (dd + w < d[v]) d[v] = dd +
            ↪ w, pq.push({d[v], v});
        }
        d.resize(n); // only the
        ↪ original vertices matter
        return d;
    }
};
// The same construction gives 2-SAT range clauses and "range ->
    ↪ range" edges (route both
// through one fresh auxiliary node).
```

3.2.9 KShortestWalks

EPPSTEIN'S ALGORITHM - the k SHORTEST s-t WALKS (repeated vertices and edges allowed), in

```
// O(m log m + k log k). Directed, 0-indexed, non-negative weights.
// Returns the k lengths in
// non-decreasing order, padding with -1 when fewer than k walks
// exist.
// Reach for it for "the k cheapest routes", "the k-th smallest path
// length", "is there a path of
// length exactly L" (enumerate until you pass L), and for problems
// that sum over the k best plans.
// THE IDEA. Fix the shortest path TREE towards t, with d[v] =
// dist(v, t). Any walk is described by
// the SIDETRACK edges it takes - the edges it uses that are not tree
// edges - and its length is
// d[s] + (sum of the sidetracks' reduced costs w + d[to] - d[from]),
// each of which is >= 0. So the
// k shortest walks = the k cheapest ANTICHAINS in a heap-ordered
// structure, and a best-first search
// over that structure emits them one at a time.
// H[v] is a PERSISTENT leftist heap holding the sidetracks available
// at v and at all its ancestors
// in the tree; persistence is what makes "inherit the parent's heap"
// O(log n) instead of O(n).
struct KShortestWalks {
    struct Node { Node *l = 0, *r = 0; int sz = 1, to; ll val; };
    Node* meld(Node* a, Node* b) {
        if (!a) return b;
        if (!b) return a;
        if (a->val > b->val) swap(a, b);
        Node* c = new Node(*a);
        copy-on-write: keeps old versions
        c->r = meld(c->r, b);
        if (!c->l || c->l->sz < c->r->sz) swap(c->l, c->r);
        c->sz = (c->r ? c->r->sz : 0) + 1;
        return c;
    }
    Node* push(Node* h, ll val, int to) {
        Node* c = new Node();
        c->val = val, c->to = to;
        return meld(h, c);
    }
    // e = {u, v, w} directed arcs. Returns the k shortest s -> t
    // walk lengths.
    vector<ll> solve(int n, const vector<array<ll, 3>>& e, int s, int
    t, int k) {
        vector<vector<array<ll, 3>>> fwd(n), rev(n);
        {other endpoint, weight, edge id}
        for (int i = 0; i < (int)e.size(); i++) {
            fwd[e[i][0]].push_back({e[i][1], e[i][2], (ll)i});
            rev[e[i][1]].push_back({e[i][0], e[i][2], (ll)i});
        }
        vector<ll> d(n, llinf);
        vector<int> par(n, -1), tid(n, -1);
        tid[v] = v's tree edge id
        priority_queue<pair<ll, int>, vector<pair<ll, int>>,
        greater<>> pq;
        d[t] = 0, pq.push({0, t});
        while (!pq.empty()) {
            auto [dd, u] = pq.top(); pq.pop();
            if (dd > d[u]) continue;
            for (auto [v, w, id] : rev[u]) if (dd + w < d[v])
                d[v] = dd + w, par[v] = u, tid[v] = id,
            pq.push({d[v], (int)v});
        }
        vector<ll> ans;
        if (d[s] == llinf) return vector<ll>(k, -1);
        vector<vector<int>> kids(n);
        for (int v = 0; v < n; v++) if (par[v] >= 0)
            kids[par[v]].push_back(v);
        vector<Node*> H(n, nullptr);
        vector<int> bfs{t};
        build H down the tree from t
        for (int qi = 0; qi < (int)bfs.size(); qi++) {
            int u = bfs[qi];
            if (par[u] >= 0) H[u] = meld(H[u], H[par[u]]);
            for (auto [v, w, id] : fwd[u])
                if ((int)id != tid[u] && d[v] < llinf) H[u] =
                push(H[u], w + d[v] - d[u], (int)v);
            for (int v : kids[u]) bfs.push_back(v);
        }
        auto cmp = [](const pair<ll, Node*>& a, const pair<ll,
        Node*>& b) {
            return a.first > b.first;
        };
        priority_queue<pair<ll, Node*>, vector<pair<ll, Node*>>,
```

```
> decltype(cmp)> Q(cmp);
Node* root = new Node();
root->val = d[s], root->to = s;
Q.push({d[s], root});
while (!Q.empty() && (int)ans.size() < k) {
    auto [w, cur] = Q.top(); Q.pop();
    ans.push_back(w);
    if (cur->l) Q.push({w + cur->l->val - cur->val, cur->l});
    // swap the last sidetrack
    if (cur->r) Q.push({w + cur->r->val - cur->val, cur->r});
    if (H[cur->to]) Q.push({w + H[cur->to]->val,
    H[cur->to]}); // append a new sidetrack
}
while ((int)ans.size() < k) ans.push_back(-1);
return ans;
};
// WALKS, NOT PATHS. The second-shortest walk may repeat a vertex. If
// the problem insists on SIMPLE
// paths you need YEN'S ALGORITHM: keep the best path, then for each
// prefix of it (the "spur"), ban
// the edges that would recreate an already-found path, run one
// Dijkstra from the spur vertex, and
// keep the best candidate. O(k * n * (m + n log n)) - much slower,
// but it is the only correct
// answer for "k shortest SIMPLE paths". On a DAG walks and paths
// coincide, so use this file.
// K-TH SHORTEST PATH WITH A COUNT LIMIT PER VERTEX: a plain Dijkstra
// that pops each vertex up to k
// times is O(k m log) and far simpler - reach for that first if k is
// small (<= a few hundred).
// SECOND SHORTEST PATH ONLY: just take ans[1] here, or do the
// two-array trick (best and
// second-best distance per vertex) in one Dijkstra.
```

3.3 Graph Connectivity

3.3.1 LowLink

LowLink: BRIDGES + ARTICULATION POINTS + 2-EDGE-CONNECTED COMPONENTS - $O(V + E)$

```
// Skips the parent by EDGE ID, so MULTI-EDGES are handled correctly
// (a doubled edge is never a bridge). 0-based vertices.
// comp[v] = id of v's 2-edge-connected component; contracting those
//   ↳ gives the BRIDGE TREE,
// on which "number of bridges between u and v" is just a tree path
//   ↳ length.
struct LowLink {
    int n, timer = 0, nc = 0;
    vector<vector<pair<int, int>>> adj;           // {to, edge
    ↳ id}
    vector<int> tin, low, comp;
    vector<bool> isCut;
    vector<pair<int, int>> bridges;
    vector<int> st;

    LowLink(int n = 0) : n(n), adj(n), tin(n, -1), low(n), comp(n,
    ↳ -1), isCut(n, false) {}
    void add_edge(int u, int v, int id) { adj[u].push_back({v, id}),
    ↳ adj[v].push_back({u, id}); }

    void dfs(int u, int pe) {
        tin[u] = low[u] = timer++;
        st.push_back(u);
        int kids = 0;
        for (auto [v, id] : adj[u]) {
            if (id == pe) continue;           // skip the
            ↳ edge we came in on, not the vertex
            if (tin[v] != -1) low[u] = min(low[u], tin[v]);
            else {
                dfs(v, id), kids++;
                low[u] = min(low[u], low[v]);
                if (low[v] > tin[u]) bridges.push_back({u, v});
                if (low[v] >= tin[u] && pe != -1) isCut[u] = true;
            }
        }
        if (pe == -1 && kids > 1) isCut[u] = true;
        if (low[u] == tin[u]) {               // u is the
            ↳ top of a 2-edge-connected comp
            while (true) {
                int v = st.back(); st.pop_back();
                comp[v] = nc;
                if (v == u) break;
            }
            nc++;
        }
    }
    void solve() { for (int i = 0; i < n; i++) if (tin[i] == -1)
    ↳ dfs(i, -1); }

    // Bridge tree: nc nodes, one edge per bridge.
    vector<vector<int>> bridgeTree() {
        vector<vector<int>> t(nc);
        for (auto [u, v] : bridges) t[comp[u]].push_back(comp[v]),
        ↳ t[comp[v]].push_back(comp[u]);
        return t;
    }
};
```

3.3.2 SCC

Strongly Connected Components (Tarjan's Algorithm) - $O(V + E)$

```
// IMPORTANT: component ids come out in REVERSE topological order -
//   ↳ for every edge u->v with
// comp[u] != comp[v] we have comp[u] > comp[v]. Iterate ids DOWNWARD
//   ↳ to walk the condensation
// forwards. build_dag() may emit duplicate arcs (fine for
//   ↳ reachability, wrong for counting).
struct SCC {
    int n, timer = 0, num_comps = 0;
    vector<vector<int>> adj;
    vector<int> tin, low, comp;
    vector<bool> in_st;
    stack<int> st;
    vector<vector<int>> comps;

    SCC(int n = 0) : n(n), adj(n + 1), tin(n + 1, 0), low(n + 1, 0),
    ↳ comp(n + 1, -1), in_st(n + 1, false) {}

    void add_edge(int u, int v) {
        adj[u].pb(v);
    }

    void dfs(int u) {
        tin[u] = low[u] = ++timer;
        st.push(u);
        in_st[u] = true;

        for (int v : adj[u]) {
            if (!tin[v]) {
                dfs(v);
                low[u] = min(low[u], low[v]);
            } else if (in_st[v]) {
                low[u] = min(low[u], tin[v]);
            }
        }

        if (low[u] == tin[u]) {
            comps.pb({});
            while (true) {
                int v = st.top();
                st.pop();
                in_st[v] = false;
                comp[v] = num_comps;
                comps.back().pb(v);
                if (u == v) break;
            }
            num_comps++;
        }
    }

    void solve() {
        for (int i = 1; i <= n; i++) {
            if (!tin[i]) dfs(i);
        }
    }

    vector<vector<int>> build_dag() {
        vector<vector<int>> dag(num_comps);
        for (int u = 1; u <= n; u++) {
            for (int v : adj[u]) {
                if (comp[u] != comp[v]) dag[comp[u]].pb(comp[v]);
            }
        }
        return dag;
    }
};
```

3.3.3 BlockCutTree

BLOCK-CUT TREE (vertex biconnected components). $O(V + E)$.

```
// Builds a forest whose nodes are: the original CUT VERTICES, plus
//   ↳ one node per BICONNECTED
// COMPONENT ("block"). Every cut vertex is joined to each block
//   ↳ containing it.
// Answers "is there a path u -> v avoiding vertex w", "which blocks
//   ↳ does a path cross",
// "articulation structure of the graph" - all become tree queries.
struct BlockCut {
    int n, timer = 0;
    vector<vector<int>> adj, comps, tree;           // comps[i] =
    ↳ vertices of block i
    vector<int> tin, low, stk, id;                 // id[v] = tree
    ↳ node of original vertex v
    vector<bool> isCut;

    BlockCut(int n = 0) : n(n), adj(n), tin(n, -1), low(n), id(n,
    ↳ -1), isCut(n, false) {}
    void add_edge(int u, int v) { adj[u].push_back(v),
    ↳ adj[v].push_back(u); }

    void dfs(int u, int p) {
        tin[u] = low[u] = timer++;
        stk.push_back(u);
        int kids = 0;
        for (int v : adj[u]) {
            if (v == p) { p = -1; continue; }      // skip ONE
    ↳ parent edge (multi-edge safe)
            if (tin[v] != -1) low[u] = min(low[u], tin[v]);
            else {
                dfs(v, u), kids++;
                low[u] = min(low[u], low[v]);
                if (low[v] >= tin[u]) {              // u closes a
    ↳ block
                    isCut[u] = (tin[u] > 0 || kids > 1);
                    comps.push_back({u});
                    while (true) {
                        int w = stk.back(); stk.pop_back();
                        comps.back().push_back(w);
                        if (w == v) break;
                    }
                }
            }
        }
    }

    void build() {
        for (int v = 0; v < n; v++) if (adj[v].empty())
    ↳ comps.push_back({v}); // isolated vertex
        for (int i = 0; i < n; i++) if (tin[i] == -1) { timer = 0;
    ↳ dfs(i, -1); stk.clear(); }
        tree.assign(comps.size(), {});
        for (int v = 0; v < n; v++) if (isCut[v]) id[v] =
    ↳ tree.size(), tree.push_back({});
        for (int i = 0; i < (int)comps.size(); i++)
            for (int v : comps[i]) {
                if (!isCut[v]) id[v] = i;
                else tree[i].push_back(id[v]),
    ↳ tree[id[v]].push_back(i);
            }
    }
};

// A vertex that is NOT a cut vertex lies in exactly one block ->
//   ↳ id[v] is that block.
// An edge (u,v) always lies in exactly one block. Leaves of the tree
//   ↳ are blocks; a graph is
// biconnected iff the tree has a single node.
```

3.3.4 TriangleAndCycleCounting

COUNTING SMALL SUBGRAPHS in $O(m \sqrt{m})$ - the "degree ordering" trick. Orient every edge from

```
// the lower (degree, index) endpoint to the higher one. In that DAG
//   ↳ every out-degree is  $O(\sqrt{m})$ ,
// because a vertex with out-degree d has d neighbours of degree  $\geq$ 
//   ↳ d, so  $d^2 \leq 2m$ .
// Undirected simple graph, 0-indexed, edges given as pairs.

// --- TRIANGLES: enumerate each once.  $O(m \sqrt{m})$ . ---
ll countTriangles(int n, const vector<pair<int, int>>& es) {
    vector<int> deg(n, 0);
    for (auto [u, v] : es) deg[u]++, deg[v]++;
    auto better = [&](int a, int b) { return deg[a] != deg[b] ?
    ↳ deg[a] < deg[b] : a < b; };
    vector<vector<int>> out(n);
    for (auto [u, v] : es) better(u, v) ? out[u].push_back(v) :
    ↳ out[v].push_back(u);
    vector<char> mark(n, 0);
    ll cnt = 0;
    for (int u = 0; u < n; u++) {
        for (int v : out[u]) mark[v] = 1;
        for (int v : out[u]) for (int w : out[v]) cnt += mark[w];
        for (int v : out[u]) mark[v] = 0;
    }
    return cnt;
    ↳ each triangle counted exactly once
}

// LISTING them: inside the inner loop, (u,v,w) with mark[w] IS a
//   ↳ triangle.
// PER-VERTEX / PER-EDGE triangle counts: add 1 to each of u,v,w (or
//   ↳ to each of the three edges)
// at that moment - this is what "local clustering coefficient" and
//   ↳ many CF problems want.

// --- 4-CYCLES:  $O(m \sqrt{m})$  with the same orientation. ---
// For each u, walk two steps (u -> v -> w) in the oriented graph and
//   ↳ count how many times each w
// is reached; a pair of distinct paths u->w gives a 4-cycle. Sum
//   ↳  $C(\text{cnt}[w], 2)$ .
ll countFourCycles(int n, const vector<pair<int, int>>& es) {
    vector<int> deg(n, 0);
    for (auto [u, v] : es) deg[u]++, deg[v]++;
    auto better = [&](int a, int b) { return deg[a] != deg[b] ?
    ↳ deg[a] < deg[b] : a < b; };
    vector<vector<int>> adj(n), out(n);
    for (auto [u, v] : es) {
        adj[u].push_back(v), adj[v].push_back(u);
        better(u, v) ? out[u].push_back(v) : out[v].push_back(u);
    }
    vector<ll> cnt(n, 0);
    ll r = 0;
    for (int u = 0; u < n; u++) {
        for (int v : adj[u]) for (int w : out[v]) if (better(u, w)) r
    ↳ += cnt[w]++;
        for (int v : adj[u]) for (int w : out[v]) if (better(u, w))
    ↳ cnt[w] = 0;
    }
    return r;
}

// WHY IT IS  $O(m \sqrt{m})$ : the inner loop runs sum over v of  $\deg(v) \times$ 
//   ↳  $\text{outdeg}(v) \leq O(m \sqrt{m})$ .
//
// OTHER COUNTING FACTS WORTH HAVING:
// * #triangles =  $\text{trace}(A^3) / 6$  for the adjacency matrix A; with
//   ↳ bitsets that is  $O(n^3 / 64)$  and
//   ↳ is often the shorter answer when  $n \leq 2000$  (also gives
//   ↳ per-pair common-neighbour counts).
// * #paths of length 2 = sum over v of  $C(\deg v, 2)$ . Subtract 3 *
//   ↳ #triangles for "paths that are
//   ↳ not part of a triangle".
// * A graph is TRIANGLE-FREE  $\Rightarrow m \leq n^2/4$  (Mantel / Turan for
//   ↳  $K_3$ ). More generally Turan:
//   ↳ a  $K_{r+1}$ -free graph has at most  $(1 - 1/r) n^2 / 2$  edges.
// * #closed walks of length k =  $\text{trace}(A^k)$  - use matrix power for
//   ↳ k up to 1e18 on small n.
// * COUNTING CYCLES OF LENGTH  $\leq 5$  is polynomial by similar
//   ↳ tricks; length  $\geq 6$  is as hard as
//   ↳ matrix multiplication in general. For counting ALL simple
//   ↳ cycles use inclusion-exclusion
//   ↳ over subsets ( $2^n$ ) or DP over bitmask start-vertex.
// * GIRTH (shortest cycle) of an unweighted graph: BFS from every
//   ↳ vertex,  $O(nm)$ . For the
//   ↳ shortest cycle THROUGH a given edge, delete it and run BFS.
```

3.3.5 ThreeEdgeConnectivity

THREE-EDGE-CONNECTED COMPONENTS in ONE DFS, $O(n + m)$. Undirected, 0-indexed, may be

```
// disconnected; self loops and PARALLEL EDGES are handled (parallel
    ↳ edges matter here - two
// vertices joined by three parallel edges are 3-edge-connected).
// u and v land in the same component iff no set of at most TWO edges
    ↳ separates them, i.e. iff
// there are 3 edge-disjoint u-v paths (Menger). This is the next
    ↳ step after LowLink (01), which
// gives the 2-edge-connected components (= no single edge separates
    ↳ them).
// Reach for it on "the network survives any two cable failures",
    ↳ "contract every pair of parallel
// bridges", and as the preprocessing for 3-edge-connectivity
    ↳ augmentation problems.
// Tsin's algorithm: a DFS that maintains, for each vertex, a PATH of
    ↳ still-undecided descendants;
// whenever a back edge proves two of them 3-edge-connected the path
    ↳ is absorbed into one class.
// The classes live in a circular linked list, so "merge" is a single
    ↳ swap.
vector<vector<int>> threeEdgeCC(const vector<vector<int>>& g) {
    int n = g.size(), cnt = 0;
    vector<int> in(n), out(n), low(n, n), deg(n, 0), path(n, n),
    ↳ nxt(n), it(n, 0), stk;
    vector<char> vis(n, 0);
    iota(all(nxt), 0);
    ↳ circular list = the components
    auto absorb = [&](int v, int w) { swap(nxt[v], nxt[w]), deg[v] +=
    ↳ deg[w]; };
    for (int s = 0; s < n; s++) {
    ↳ iterative DFS over the forest
        if (vis[s]) continue;
        stk = {s}, vis[s] = 1, in[s] = cnt++;
        vector<int> par(1, n), pe(1, -1);
        ↳ = "parent edge already used"
        while (!stk.empty()) {
            int v = stk.back(), p = par.back();
            if (it[v] < (int)g[v].size()) {
                int w = g[v][it[v]++];
                if (w == v) continue;
                if (w == p && !pe.back()) { pe.back() = 1; continue;
            } // skip ONE parent edge
            if (vis[w]) {
                if (in[w] < in[v]) deg[v]++, low[v] = min(low[v],
            ↳ in[w]);
            else {
            ↳ back edge to a descendant
                deg[v]--;
                int u = path[v];
                while (u != n && in[u] <= in[w] && in[w] <
            ↳ out[u])
                    absorb(v, u), u = path[u];
                path[v] = u;
            }
            continue;
        }
        vis[w] = 1, in[w] = cnt++, stk.push_back(w),
    ↳ par.push_back(v), pe.push_back(0);
    } else {
    ↳ leaving v: fold it into its parent
        out[v] = cnt, stk.pop_back(), par.pop_back(),
    ↳ pe.pop_back();
        if (p == n) continue;
        int w = v;
        if (path[w] == n && deg[w] <= 1) { deg[p] += deg[w],
    ↳ low[p] = min(low[p], low[w]); }
        else {
            if (deg[w] == 0) w = path[w];
            if (low[w] < low[p]) low[p] = low[w], swap(w,
            ↳ path[p]);
            while (w != n) absorb(p, w), w = path[w];
        }
    }
    }
    vector<vector<int>> comp;
    ↳ walk each circular list once
    vector<char> seen(n, 0);
    for (int i = 0; i < n; i++) if (!seen[i]) {
        comp.push_back({});
        for (int j = i; !seen[j]; j = nxt[j]) seen[j] = 1,
    ↳ comp.back().push_back(j);
    }
    return comp;
}
```

```
// CONTRACTING each component gives a graph in which every 3-edge cut
    ↳ is visible; the 2-edge-cuts
// are exactly the bridges plus the "cactus" pairs from LowLink.
// 3-VERTEX-CONNECTIVITY (SPQR / triconnected components) is a
    ↳ genuinely harder, ~400-line
// algorithm. If you only need to test 3-vertex-connectivity for
    ↳ small n, delete every pair of
// vertices and check connectivity in  $O(n^2 (n + m))$ .
```


3.3.6 OnlineBridges

ONLINE (INCREMENTAL) BRIDGES - maintain the number of bridges while edges are ADDED, in

```
// O(alpha) amortised per edge. Undirected, 0-indexed, edges may
// ↳ repeat (a doubled edge kills the
// bridge, which is exactly right).
// LowLink (01) recomputes bridges from scratch in O(n + m); use this
// ↳ when the graph grows and you
// must answer after every insertion: "after each new road, how many
// ↳ roads are still critical",
// "the first moment u and v are 2-edge-connected".
// STRUCTURE. Contract every 2-edge-connected component to a node;
// ↳ what is left is the BRIDGE
// FOREST, kept as a rooted forest with parent pointers. Adding (u,
// ↳ v):
// * same 2ecc → nothing changes;
// * different trees → a new bridge; re-root the smaller
// ↳ tree and hang it under v;
// * same tree, different 2ecc → the tree path u..v becomes a
// ↳ cycle: contract it, losing
// (path length) bridges.
// Two DSUs: dsu2ecc = the contracted components, dsuCC = which tree
// ↳ of the forest you are in.
// DELETIONS are NOT supported - that needs offline divide and
// ↳ conquer or a link-cut tree.
struct OnlineBridges {
    vector<int> par, e2, cc, ccSize, stamp; // par
    ↳ is in the BRIDGE FOREST
    int bridges = 0, iter = 0;
    OnlineBridges(int n) : par(n, -1), e2(n), cc(n), ccSize(n, 1),
    ↳ stamp(n, 0) {
        iota(all(e2), 0), iota(all(cc), 0);
    }
    int find2(int u) { return u < 0 || e2[u] == u ? u : e2[u] =
    ↳ find2(e2[u]); }
    int findCC(int u) { u = find2(u); return cc[u] == u ? u : cc[u] =
    ↳ findCC(cc[u]); }
    void makeRoot(int u) { //
    ↳ reverse the parent chain above u
        u = find2(u);
        int root = u, child = -1;
        while (u != -1) { int p = find2(par[u]); par[u] = child,
    ↳ cc[u] = root, child = u, u = p; }
        ccSize[root] = ccSize[child];
    }
    void mergePath(int u, int v) { //
    ↳ contract the u..v tree path
        ++iter;
        vector<int> pu, pv;
        int lca = -1;
        while (lca < 0) {
            if (u != -1) {
                u = find2(u), pu.push_back(u);
                if (stamp[u] == iter) { lca = u; break; }
                stamp[u] = iter, u = par[u];
            }
            if (v != -1) {
                v = find2(v), pv.push_back(v);
                if (stamp[v] == iter) { lca = v; break; }
                stamp[v] = iter, v = par[v];
            }
        }
        for (auto& p : {pu, pv})
            for (int x : p) { if (x == lca) break; e2[x] = lca,
    ↳ bridges--; }
    }
    void addEdge(int u, int v) {
        u = find2(u), v = find2(v);
        if (u == v) return; //
    ↳ already 2-edge-connected
        int a = findCC(u), b = findCC(v);
        if (a != b) {
            bridges++;
            if (ccSize[a] > ccSize[b]) swap(u, v), swap(a, b); //
    ↳ re-root the SMALLER tree
            makeRoot(u);
            par[u] = cc[u] = v, ccSize[b] += ccSize[u];
        } else mergePath(u, v);
    }
    bool sameComponent(int u, int v) { return find2(u) == find2(v); }
    ↳ // 2-edge-connected?
};
// WHY IT IS AMORTISED O(alpha): makeRoot walks the smaller tree,
// ↳ which is the union-by-size
// argument (O(log n) total per vertex), and mergePath destroys one
// ↳ bridge per step.
// ONLINE ARTICULATION POINTS / biconnected components can be
```

```
↳ maintained the same way, but the
// bookkeeping is much heavier - use offline divide and conquer over
// ↳ time instead.
// OFFLINE DELETIONS: process the queries in reverse so deletions
// ↳ become insertions, or run
// divide-and-conquer on the time axis with a rollback DSU (see 02 -
// ↳ DS/05 - DSU).
```

3.3.7 STNumbering

ST-NUMBERING (bipolar orientation) - order the vertices $v_1 = s, \dots, v_n = t$ so that EVERY

```
// other vertex has at least one neighbour BEFORE it and one AFTER
    ↳ it.  $O(n + m)$ , 0-indexed,
// undirected. Returns the order, or an EMPTY vector when none exists.
// EXISTS iff  $G + \text{edge}(s, t)$  is 2-VERTEX-CONNECTED (that is exactly
    ↳ what the first DFS checks).
// Why you want it: orienting every edge from the smaller number to
    ↳ the larger gives a DAG with a
// UNIQUE source  $s$  and a UNIQUE sink  $t$ , and every vertex lies on some
    ↳  $s$ - $t$  path. That is the tool
// for "orient the edges so the graph stays connected / has one
    ↳ source and one sink", for planarity
// testing and st-planar embeddings, and for "peel the graph one
    ↳ vertex at a time keeping both
// halves connected" (any prefix AND its complement induce connected
    ↳ subgraphs).
// Even-Tarjan: DFS from  $t$ , then insert the vertices into a list in
    ↳ preorder, each one just before
// or just after its parent according to a sign flipped at every
    ↳ insertion.
vector<int> stNumbering(int n, vector<vector<int>> adj, int s, int t)
    ↳ {
    if (n == 1) return {0};
    bool has = false; // add
    ↳ the s-t edge if missing
    for (int v : adj[s]) has |= (v == t);
    if (!has) adj[s].push_back(t), adj[t].push_back(s);
    vector<int> dis(n, 0), low(n, 0), par(n, -1), sg(n + 2, 0), pre;
    int T = 0;
    bool bicon = true;
    function<void(int, int)> dfs1 = [&](int u, int p) { //
    ↳ articulation-point check
        low[u] = dis[u] = ++T;
        int kids = 0;
        for (int v : adj[u]) {
            if (v == p) { p = -1; continue; } //
    ↳ skip ONE parent edge
            if (!dis[v]) {
                kids++; dfs1(v, u), low[u] = min(low[u], low[v]);
                if (par[u] != -2 && low[v] >= dis[u]) bicon = false;
    ↳ // par[u] == -2 marks the root
            } else low[u] = min(low[u], dis[v]);
        }
        if (par[u] == -2 && kids > 1) bicon = false;
    };
    par[t] = -2, dfs1(t, -1);
    for (int i = 0; i < n; i++) if (!dis[i]) bicon = false;
    if (!bicon) return {};
    fill(all(dis), 0), fill(all(low), 0), fill(all(par), -1), T = 0;
    function<void(int, int)> dfs2 = [&](int u, int p) { //
    ↳ tree rooted at t; s is fixed
        low[u] = dis[u] = ++T;
        for (int v : adj[u]) {
            if (v == p) { p = -1; continue; }
            if (!dis[v]) pre.push_back(v), par[v] = u, dfs2(v, u),
    ↳ low[u] = min(low[u], low[v]);
            else low[u] = min(low[u], dis[v]);
        }
    };
    dis[s] = low[s] = ++T, sg[dis[s]] = -1; // s
    ↳ takes number 1, never entered
    dfs2(t, -1);
    list<int> L{s, t};
    vector<list<int>::iterator> it(n + 2);
    it[dis[s]] = L.begin(), it[dis[t]] = next(L.begin());
    for (int v : pre) { //
    ↳ insert before/after the parent
        if (sg[low[v]] == -1) it[dis[v]] = L.insert(it[dis[par[v]]],
    ↳ v);
        else it[dis[v]] = L.insert(next(it[dis[par[v]]]), v);
        sg[dis[par[v]]] = -sg[low[v]];
    }
    return vector<int>(all(L)); //
    ↳ order[0] = s, order[n-1] = t
}
// THE DAG. With pos[order[i]] = i, orient every edge from the
    ↳ smaller pos to the larger. Then
// every vertex is on an s-t path, so "assign directions keeping
    ↳ everything reachable from s and
// able to reach t" is solved. Reversing all edges swaps the roles of
    ↳ s and t.
// EAR DECOMPOSITION is the other standard certificate of
    ↳ 2-connectivity and is often what a
// problem really wants: an open ear decomposition exists iff the
    ↳ graph is 2-vertex-connected, and
```

```
// the ears are found from the same DFS (each back edge starts one
    ↳ ear).
// If the graph is only 2-EDGE-connected, use the bridge tree
    ↳ instead; st-numbering needs vertex
// connectivity.
```

3.3.8 CactusTree

CACTUS -> TREE. A CACTUS is a connected graph in which every EDGE lies on at most one simple

```
// cycle (equivalently  $m \leq 3(n-1)/2$  and the cycles are
//   edge-disjoint).  $O(n + m)$ , 0-indexed, no
//   self loops and no parallel edges.
// Build a tree on  $n + (\text{number of cycles})$  nodes: node  $n + i$ 
//   represents the  $i$ -th cycle and is joined
// to every vertex of that cycle; every BRIDGE stays a tree edge.
//   Then path/subtree queries on the
// cactus become ordinary tree queries (LCA, HLD, rerooting), which
//   is the whole point.
// Reach for it whenever the input says "each edge belongs to at most
//   one cycle" or "n vertices and
// n edges" (a single cycle with trees hanging off - a "functional
//   graph" shape) and the question
// is about paths, distances, or counting: "number of simple paths
//   between u and v" (2 per cycle
// the path enters and leaves through two different vertices),
//   "shortest path in a cactus", "how
// many vertices can a path from u to v visit".
// This is really the BLOCK-CUT TREE (03) specialised: on a cactus
//   every block is a single edge or
// a single cycle, so the block nodes ARE the cycle nodes and the
//   construction is much shorter.
struct CactusTree {
    int n, cycles = 0;
    vector<vector<int>>> tree; //
    size n + cycles after build()
    vector<char> inCycle; // per
    EDGE id
    vector<int> par, parE, state;
    vector<vector<pair<int, int>>> g; //
    g[u] = {neighbour, edge id}
    CactusTree(int n) : n(n), par(n, -1), parE(n, -1), state(n, 0),
    g(n) {}
    void addEdge(int u, int v) {
        int id = inCycle.size();
        inCycle.push_back(0), g[u].push_back({v, id}),
        g[v].push_back({u, id});
    }
    void dfs(int u, int p) {
        state[u] = 1, par[u] = p;
        for (auto [v, id] : g[u]) {
            if (v == p) continue;
            if (!state[v]) parE[v] = id, dfs(v, u);
            else if (state[v] == 1) { //
                back edge: close a cycle
                int c = n + cycles++;
                tree.push_back({});
                for (int x = u, e = parE[u]; x != v; x = par[x], e =
                parE[x]) tree[x].push_back(c), tree[c].push_back(x),
                inCycle[e] = 1;
                tree[v].push_back(c), tree[c].push_back(v),
                inCycle[id] = 1;
            }
            state[u] = 2;
        }
    }
    void build() {
        tree.assign(n, {});
        for (int i = 0; i < n; i++) if (!state[i]) dfs(i, -1);
        for (int u = 0; u < n; u++) for (auto [v, id] : g[u]) //
        the bridges stay as tree edges
            if (!inCycle[id]) tree[u].push_back(v);
    }
};
// The tree is BIPARTITE between real vertices and cycle nodes only
//   in the cycle parts; a bridge
// joins two real vertices directly, so do not assume alternation.
// NUMBER OF SIMPLE u-v PATHS =  $2^{(\text{number of CYCLE NODES on the u-v}$ 
//   tree path)}, because inside a
// cycle you may go either way round. That is the standard use of
//   this decomposition.
// SHORTEST PATH IN A CACTUS: on the tree path, a bridge contributes
//   its weight and a cycle node
// contributes min(clockwise, anticlockwise) between its two
//   attachment points - store each
// vertex's position and prefix sum along its cycle to get that in
//    $O(1)$ .
// VERTEX CACTUS (every VERTEX in at most one cycle) is a strictly
//   smaller class and the same code
// builds its tree.
// IF THE GRAPH IS NOT A CACTUS this silently produces nonsense -
//   verify m and the block sizes, or
// just use the block-cut tree, which is correct on any graph.
```

3.4 Satisfiability

3.4.1 TwoSat

2-SAT via Kosaraju - $O(n + m)$. Self-contained, 0-based variables.

```
// Literal encoding:  $2*i = (x_i \text{ false}), 2*i+1 = (x_i \text{ true})$ .
// Satisfiable iff x and !x are never in the same SCC; then take the
//   literal whose SCC is LATER
// in topological order.
struct TwoSat {
    int n;
    vector<vector<int>>> adj, radj;
    vector<int> ord, comp;
    vector<bool> used, val;

    TwoSat(int n = 0) : n(n), adj(2 * n), radj(2 * n) {}
    int lit(int i, bool t) { return 2 * i + t; }
    int add_var() { n++; adj.resize(2 * n), radj.resize(2 * n);
    return n - 1; }

    void imply(int a, int b) { // literal a
        // => literal b (and contrapositive)
        adj[a].push_back(b), radj[b].push_back(a);
        adj[b ^ 1].push_back(a ^ 1), radj[a ^ 1].push_back(b ^ 1);
    }
    void either(int a, int b) { imply(a ^ 1, b); } // (a OR b)
    void force(int a) { imply(a ^ 1, a); } // a must be
    true
    void nand_(int a, int b) { either(a ^ 1, b ^ 1); } // not both
    void eq(int a, int b) { imply(a, b), imply(b, a); } // a <=> b
    void xor_(int a, int b) { eq(a, b ^ 1); }

    // AT MOST ONE of 'lits' is true, with  $O(k)$  edges instead of
    //  $O(k^2)$  (prefix/"lightbulb" trick):
    // aux_i means "someone at index <= i was chosen".
    void at_most_one(const vector<int>& lits) {
        int prev = -1;
        for (int c : lits) {
            int cur = lit(add_var(), true);
            imply(c, cur);
            if (prev != -1) imply(prev, cur), imply(c, prev ^ 1);
            prev = cur;
        }
    }

    void dfs1(int u) { used[u] = true; for (int v : adj[u]) if
    (!used[v]) dfs1(v); ord.push_back(u); }
    void dfs2(int u, int c) { comp[u] = c; for (int v : radj[u]) if
    (comp[v] < 0) dfs2(v, c); }

    bool solve() {
        used.assign(2 * n, false), comp.assign(2 * n, -1),
        val.assign(n, false), ord.clear();
        for (int i = 0; i < 2 * n; i++) if (!used[i]) dfs1(i);
        int c = 0;
        for (int i = 2 * n - 1; i >= 0; i--) if (comp[ord[i]] < 0)
        dfs2(ord[i], c++);
        for (int i = 0; i < n; i++) {
            if (comp[2 * i] == comp[2 * i + 1]) return false;
            val[i] = comp[2 * i + 1] > comp[2 * i];
        }
        return true; // read
    }
};
// RANGE CLAUSE ("if x then ALL of [l,r]") in  $O(\log n)$  edges: build a
//   segment tree over the
// variables, give every internal node an aux variable, add parent =>
//   both children, then
// imply(x, node) for the  $O(\log n)$  covering nodes. Mirror the tree
//   (child => parent) for
// "if any of [l,r] then x".
```

3.4.2 DPLL

GENERAL SAT (3-SAT and beyond) by DPLL - Davis-Putnam-Logemann-Loveland.
0-indexed variables.

```
// Exponential in the worst case, but with unit propagation and
    ↳ pure-literal elimination it decides
// most contest-sized instances (n in the hundreds, even thousands)
    ↳ instantly.
// 2-SAT is polynomial (01 - TwoSat) - always check whether the
    ↳ clauses really need three literals.
// Reach for this when a problem reduces to "assign booleans so that
    ↳ all these constraints hold" and
// some clause has three or more literals: exact cover with few
    ↳ options, graph colouring encoded as
// clauses, puzzle solvers, "is this configuration reachable"
    ↳ searches with heavy pruning.
// LITERAL ENCODING: the positive literal of variable v is v, the
    ↳ negative one is ~v (= -v-1).
// A clause is a vector of literals meaning their OR. Internally
    ↳ literal u lives at index u + n.
// THE THREE MOVES.
// UNIT PROPAGATION: a clause with all but one literal already
    ↳ false forces that last literal.
// PURE LITERAL: a variable that only ever appears with one sign
    ↳ can be set that way for free.
// BRANCH: pick the unassigned variable with the MOST remaining
    ↳ occurrences, try it, backtrack.
struct SAT {
    int n;
    vector<int> occ, pos, neg, unitS, pureS;
    vector<vector<int>> g, cls, level; //
    ↳ g[lit] = clauses containing lit
    vector<char> val; //
    ↳ val[u + n] = "literal u is true"
    SAT(int n) : n(n), occ(2 * n, 0), g(2 * n) {}
    void addClause(const vector<int>& c) {
        for (int u : c) g[u + n].push_back(cls.size()), occ[u + n]++;
        cls.push_back(c);
    }
    void set(int u) { //
        ↳ assign literal u = true
        val[u + n] = 1, level.back().push_back(u);
        for (int i : g[u + n]) if (pos[i]++ == 0) for (int w :
    ↳ cls[i]) --occ[w + n];
        for (int i : g[~u + n]) { ++neg[i]; if (pos[i] == 0)
    ↳ unitS.push_back(i); }
    }
    void undo() {
        int u = level.back().back();
        level.back().pop_back(), val[u + n] = 0;
        for (int i : g[u + n]) if (--pos[i] == 0) for (int w :
    ↳ cls[i]) ++occ[w + n];
        for (int i : g[~u + n]) --neg[i];
    }
    bool reduce() { //
        ↳ propagate until nothing is forced
        while (!unitS.empty() || !pureS.empty()) {
            if (!pureS.empty()) {
                int u = pureS.back(); pureS.pop_back();
                if (occ[u + n] == 1 && occ[~u + n] == 0) set(u);
            } else {
                int i = unitS.back(); unitS.pop_back();
                if (pos[i] > 0) continue; //
            }
            ↳ clause already satisfied
            if (neg[i] == (int)cls[i].size()) return false; //
            ↳ clause already violated
            if (neg[i] + 1 == (int)cls[i].size()) {
                int w = 0;
                bool found = false;
                for (int u : cls[i]) if (!val[u + n] && !val[~u +
    ↳ n]) w = u, found = true;
                if (!found) return false;
                set(w);
            }
        }
        return true;
    }
    bool solve() { //
        ↳ val[v + n] = the model afterwards
        val.assign(2 * n, 0);
        pos.assign(cls.size(), 0), neg.assign(cls.size(), 0);
        level.assign(1, {}), unitS.clear(), pureS.clear();
        for (;;) {
            if (reduce()) {
                int s = -1, bc = 0; //
                ↳ branch on the most constrained
                for (int u = 0; u < n; u++) { //
```

```
↳ UNASSIGNED variable
        if (val[u + n] || val[~u + n]) continue;
        int c = occ[u + n] + occ[~u + n];
        if (c > bc) bc = c, s = u;
    }
    if (s < 0) return true; //
    ↳ every clause is satisfied
    level.push_back({}), set(s);
} else {
    if (level.back().empty()) return false;
    int s = level.back()[0];
    while (!level.back().empty()) undo();
    level.pop_back();
    if (level.empty()) return false;
    set(~s); // the
    ↳ other branch
}
}
}
bool value(int v) const { return val[v + n]; }
};
// EXAMPLE. (x0 OR NOT x1) AND (NOT x1) AND (NOT x1 OR x3 OR x5) is
// addClause({0, ~1}); addClause({~1}); addClause({~1, 3, 5});
// WATCHED LITERALS + CLAUSE LEARNING (CDCL) is what real solvers add
    ↳ on top; it is far more code
// and rarely needed in a contest - if DPLL is too slow, the intended
    ↳ solution is not SAT.
// EXACTLY-ONE OF k VARIABLES: one clause (v1 OR ... OR vk) plus
    ↳ C(k,2) clauses (NOT vi OR NOT vj);
// for large k use the commander/sequential encoding to keep the
    ↳ clause count linear.
// AT MOST ONE with 2-SAT structure? Only for k = 2. Beyond that you
    ↳ are in this file.
// MAX-SAT and #SAT are harder still; for #SAT on n ≤ 20 variables,
    ↳ use inclusion-exclusion or a
// subset DP over the clause set instead.
```

3.5 Eulerian Path

3.5.1 EulerianPath

Hierholzer's Algorithm for Eulerian Path / Circuit - $O(V + E)$

```
// Works for both Directed and Undirected graphs
struct EulerianPath {
    int n, m;
    vector<vector<pair<int, int>>> adj; // {neighbor, edge_id}
    vector<bool> used_edge;
    vector<int> in_deg, out_deg, path;

    EulerianPath(int n = 0, int m = 0) : n(n), m(m), adj(n + 1),
    ↪ used_edge(m, false), in_deg(n + 1, 0), out_deg(n + 1, 0) {}

    void add_edge(int u, int v, int edge_id, bool directed = true) {
        adj[u].pb({v, edge_id});
        out_deg[u]++;
        in_deg[v]++;
        if (!directed) {
            adj[v].pb({u, edge_id});
            out_deg[v]++;
            in_deg[u]++;
        }
    }

    // Finds Eulerian path starting at start_node. Returns false if
    ↪ graph is not Eulerian
    bool solve(int start_node) {
        vector<int> ptr(n + 1, 0);
        stack<int> st;
        st.push(start_node);

        while (!st.empty()) {
            int u = st.top();
            if (ptr[u] < adj[u].size()) {
                auto [v, id] = adj[u][ptr[u]++];
                if (!used_edge[id]) {
                    used_edge[id] = true;
                    st.push(v);
                }
            } else {
                path.pb(u);
                st.pop();
            }
        }
        reverse(all(path));

        // Verify that every edge was traversed
        for (bool used : used_edge) {
            if (!used) return false;
        }
        return true;
    }
};
```

3.6 Flow and Min Cut

3.6.1 Dinic

DINIC MAX FLOW. $O(V^2 E)$ in general, $O(E \sqrt{V})$ on unit-capacity graphs, $O(V^2 E)$ worst case but

```
// fast in practice. After maxflow, vertices reachable from s in the
    ↪ residual graph form the min cut.
// Scaling (start with big capacities) helps on dense graphs with
    ↪ large capacities.
// DINIC MAX FLOW.  $O(V^2 E)$  general,  $O(E \sqrt{E})$  with unit
    ↪ capacities,  $O(E \sqrt{V})$  for bipartite
// matching. This is the SCALING variant (big capacities first), so
    ↪ it stays fast on wide ranges.
// addEdge(a, b, cap, rcap): pass rcap = cap for an UNDIRECTED edge.
    ↪ Edge::flow() recovers the flow
// on each arc. After calc(s,t), leftOfMinCut(v) is true exactly for
    ↪ the SOURCE side of a minimum
// cut - the cut edges are those from a left vertex to a right
    ↪ vertex. 0-indexed.
struct Dinic {
    struct Edge {
        int to, rev;
        ll c, oc;
        ll flow() { return max(oc - c, 0LL); } // if you need flows
    };

    vector<int> lvl, ptr, q;
    vector<vector<Edge>> adj;

    Dinic(int n) : lvl(n), ptr(n), q(n), adj(n) {}

    void addEdge(int a, int b, ll c, ll rcap = 0) {
        adj[a].push_back({b, (int) adj[b].size(), c, c});
        adj[b].push_back({a, (int) adj[a].size() - 1, rcap, rcap});
    }

    ll dfs(int v, int t, ll f) {
        if (v == t || !f) return f;
        for (int &i = ptr[v]; i < (int) adj[v].size(); i++) {
            Edge &e = adj[v][i];
            if (lvl[e.to] == lvl[v] + 1)
                if (ll p = dfs(e.to, t, min(f, e.c))) {
                    e.c -= p, adj[e.to][e.rev].c += p;
                    return p;
                }
        }
        return 0;
    }

    ll calc(int s, int t) {
        ll flow = 0;
        q[0] = s;
        for (int L = 0; L < 31; L++)
            do {
                // 'int L=30' maybe faster for random data
                lvl = ptr = vector<int>((int) q.size());
                int qi = 0, qe = lvl[s] = 1;
                while (qi < qe && !lvl[t]) {
                    int v = q[qi++];
                    for (Edge e : adj[v])
                        if (!lvl[e.to] && e.c >> (30 - L))
                            q[qi++] = e.to, lvl[e.to] = lvl[v] + 1;
                }
                while (ll p = dfs(s, t, LLONG_MAX)) flow += p;
            } while (lvl[t]);
        return flow;
    }

    bool leftOfMinCut(int a) { return lvl[a] != 0; }
};
```

3.6.2 MCMF

MIN-COST MAX-FLOW with JOHNSON POTENTIALS (Dijkstra instead of SPFA on every augmentation).

```
// O(F * E log V) where F = number of augmentations. Much faster than
//   ↳ the SPFA version and the
//   ↳ one to use by default. Negative edge costs are allowed at
//   ↳ construction PROVIDED there is no negative-cost CYCLE:
// run one Bellman-Ford
// to initialise the potentials (setPotentialsBF), after that every
//   ↳ reduced cost is  $\geq 0$ .
struct MCMF {
    struct E { int to, rev; ll cap, cost, flow = 0; };
    int n;
    vector<vector<E>> g;
    vector<ll> pot, dist;
    vector<int> pv, pe;

    MCMF(int n) : n(n), g(n), pot(n, 0), dist(n), pv(n), pe(n) {}
    void addEdge(int u, int v, ll cap, ll cost) {
        g[u].push_back({v, (int)g[v].size(), cap, cost});
        g[v].push_back({u, (int)g[u].size() - 1, 0, -cost});
    }
    void setPotentialsBF(int s) { // only needed
        ↳ if some cost < 0
        fill(all(pot), llinf);
        pot[s] = 0;
        for (int it = 0; it < n; it++) {
            bool ch = false;
            for (int u = 0; u < n; u++) if (pot[u] < llinf)
                for (auto& e : g[u] if (e.cap - e.flow > 0 && pot[u]
                    ↳ + e.cost < pot[e.to])
                    pot[e.to] = pot[u] + e.cost, ch = true;
            if (!ch) break;
        }
        for (auto& x : pot) if (x == llinf) x = 0;
    }
    bool dijkstra(int s, int t) {
        fill(all(dist), llinf);
        priority_queue<pair<ll, int>, vector<pair<ll, int>>,
        ↳ greater<>> pq;
        dist[s] = 0, pq.push({0, s});
        while (!pq.empty()) {
            auto [d, u] = pq.top(); pq.pop();
            if (d > dist[u]) continue;
            for (int i = 0; i < (int)g[u].size(); i++) {
                auto& e = g[u][i];
                if (e.cap - e.flow <= 0) continue;
                ll nd = d + e.cost + pot[u] - pot[e.to]; //
                ↳ reduced cost, always  $\geq 0$ 
                if (nd < dist[e.to]) dist[e.to] = nd, pv[e.to] = u,
                ↳ pe[e.to] = i, pq.push({nd, e.to});
            }
        }
        return dist[t] < llinf;
    }
    // Returns {flow, cost}. Pass maxf to cap the flow (e.g. for
    ↳ min-cost k-flow).
    pair<ll, ll> run(int s, int t, ll maxf = llinf) {
        ll flow = 0, cost = 0;
        while (flow < maxf && dijkstra(s, t)) {
            for (int i = 0; i < n; i++) if (dist[i] < llinf) pot[i]
            ↳ += dist[i];
            ll aug = maxf - flow;
            for (int v = t; v != s; v = pv[v]) aug = min(aug,
            ↳ g[pv[v]][pe[v]].cap - g[pv[v]][pe[v]].flow);
            for (int v = t; v != s; v = pv[v]) {
                auto& e = g[pv[v]][pe[v]];
                e.flow += aug, g[v][e.rev].flow -= aug;
                cost += aug * e.cost;
            }
            flow += aug;
        }
        return {flow, cost};
    }
    // MIN-COST flow of exactly K units: run(s, t, K) and check flow
    ↳ == K.
    // MIN-COST (not max) flow: stop as soon as dist[t] + pot[t] -
    ↳ pot[s] > 0 (no profitable path).
};
// MODELLING: assignment on a SPARSE graph; for a dense/complete one
//   ↳ use 07 - Matching/03 - Hungarian * transportation * "k disjoint
//   ↳ paths of minimum total cost" *
// min-cost bipartite matching * scheduling with penalties * MCMF is
//   ↳ also how you solve
// "choose k items with a matroid-ish constraint minimising cost".
```

3.6.3 FlowWithLowerBounds

FLows WITH LOWER BOUNDS (circulation with demands). Every edge needs $d(e) \leq f(e) \leq c(e)$.

```
// Build an auxiliary network, run one max flow; feasible iff every
//   ↳ edge out of s' saturates.
// excess(v) = (sum of d over edges INTO v) - (sum of d over edges
//   ↳ OUT of v)
// excess(v) > 0 -> edge s' -> v of capacity excess(v); < 0 -> edge
//   ↳ v -> t' of capacity -excess
// every original edge (u,v) keeps capacity c - d
// For an s-t flow, add t->s with capacity INF first, which turns it
//   ↳ into a circulation.
// Needs Dinic (01 - Dinic.cpp) with addEdge(u, v, cap) and calc(s,
//   ↳ t).
struct LowerBoundFlow {
    int n, S, T; // S, T are the
    ↳ auxiliary super source/sink
    Dinic din;
    vector<ll> excess, low;
    vector<array<int, 2>> eid; // per original
    ↳ edge: {u, index in din.adj[u]}

    LowerBoundFlow(int n) : n(n), S(n), T(n + 1), din(n + 2),
    ↳ excess(n + 2, 0) {}
    void addEdge(int u, int v, ll lo, ll hi) {
        low.push_back(lo);
        eid.push_back({u, (int)din.adj[u].size()});
        din.addEdge(u, v, hi - lo);
        excess[v] += lo, excess[u] -= lo;
    }
    bool feasible() { // circulation
        ↳ only (no s, t)
        ll need = 0;
        for (int v = 0; v < n; v++) {
            if (excess[v] > 0) din.addEdge(S, v, excess[v]), need +=
            ↳ excess[v];
            else if (excess[v] < 0) din.addEdge(v, T, -excess[v]);
        }
        return din.calc(S, T) == need;
    }
    // Min / max s-t flow subject to the bounds - see the RECIPES
    ↳ block below for the ONLY
    // correct order of operations. You must DELETE the t->s edge
    ↳ (zero both residual directions)
    // before the second max flow, otherwise Dinic's first augmenting
    ↳ path runs s -> t backwards
    // along that arc and CANCELS the feasible flow instead of
    ↳ extending it.
    // feasible() appends the S/T edges every time it is called -
    ↳ call it exactly once.
    ll flowOn(int i) { return low[i] +
    ↳ din.adj[eid[i][0]][eid[i][1]].flow(); }
};
/* RECIPES
    "each edge must be used at least once" -> d(e) = 1
    "each vertex must be visited between L and R" -> split v into
    ↳ v_in -> v_out with [L, R]
    "exact degree subgraph" -> d = c =
    ↳ required degree on the split edge
    MAX s-t FLOW WITH BOUNDS: add t->s with capacity INF, run
    ↳ feasible(); then delete that edge
    and run one more max flow from s to t on the residual graph. The
    ↳ answer is the flow that
    was on t->s plus the extra augmentation.
    MIN s-t FLOW WITH BOUNDS: same, but augment from t to s to cancel
    ↳ as much as possible. */
```


3.6.4 StoerWagner

STOER-WAGNER GLOBAL MIN CUT - minimum cut over ALL vertex pairs, undirected weighted, $O(V^3)$.

```
// No source/sink. Returns {cut weight, one side of the cut}.
// Key lemma: in a maximum-adjacency order, if s is the
    ↳ second-to-last and t the last vertex
// added, then the minimum s-t cut is exactly {t}. Merge s and t and
    ↳ repeat n-1 times.
pair<ll, vector<int>> globalMinCut(vector<vector<ll>> w) {
    int n = w.size();
    if (n < 2) return {0, {}};
    vector<char> gone(n, 0);
    vector<vector<int>> grp(n);
    for (int i = 0; i < n; i++) grp[i] = {i};
    ll bestW = llinf;
    vector<int> best;
    for (int phase = 0; phase + 1 < n; phase++) {
        vector<ll> ww(n, 0);
        vector<char> added(n, 0);
        int prev = -1, last = -1;
        for (int it = 0; it < n - phase; it++) { //
            ↳ maximum-adjacency order
            int sel = -1;
            for (int j = 0; j < n; j++)
                if (!gone[j] && !added[j] && (sel < 0 || ww[j] >
                    ↳ ww[sel])) sel = j;
            added[sel] = 1, prev = last, last = sel;
            for (int j = 0; j < n; j++) if (!gone[j] && !added[j])
                ↳ ww[j] += w[sel][j];
            }
            ll cut = 0; //
            ↳ cut-of-the-phase separates {last}
            for (int j = 0; j < n; j++) if (!gone[j] && j != last) cut +=
                ↳ w[last][j];
            if (cut < bestW) bestW = cut, best = grp[last];
            gone[last] = 1; // merge last
            ↳ into prev
            grp[prev].insert(grp[prev].end(), all(grp[last]));
            for (int j = 0; j < n; j++) if (!gone[j] && j != prev)
                w[prev][j] += w[last][j], w[j][prev] = w[prev][j];
        }
        return {bestW, best};
    }
// USES: "split the network into two non-empty parts as cheaply as
    ↳ possible", edge connectivity
// of a weighted undirected graph, "minimum number of edges to delete
    ↳ to disconnect the graph"
// (unit weights). Doing this with n-1 max-flows is both slower and
    ↳ longer to write.
// KARGER (randomized contraction) finds a min cut with probability
    ↳ >= 1/C(n,2) per run; the
// useful corollary is that a graph has AT MOST C(n,2) distinct
    ↳ minimum cuts.
```

3.6.5 MinCutModelling

MIN-CUT MODELLING - the reason flow appears at ICPC. The code is never the hard part; choosing

```
// Needs: Dinic (01 - Dinic.cpp).
// what the vertices and the infinite capacities mean is. Rule of
    ↳ thumb: if the objective contains
// an "infinity", you are CUTTING, not flowing.

// --- MAXIMUM WEIGHT CLOSURE / PROJECT SELECTION ---
// Choose a set S closed under the dependency arcs (if v in S and v
    ↳ -> u is a dependency, u in S)
// maximising the total weight. Build: s -> v with capacity w(v) for
    ↳ every w(v) > 0;
// v -> t with capacity -w(v) for every w(v) < 0; u -> v with INF for
    ↳ every dependency u needs v.
// Answer = (sum of positive weights) - maxflow. S = the source side
    ↳ of the min cut.
// USES: "each project earns p but needs these machines that cost c";
    ↳ "pick cells maximising
// profit but a chosen cell forces its neighbours"; densest-subgraph
    ↳ style selections.
ll maxClosure(int n, const vector<ll>& w, const vector<pair<int,
    ↳ int>>& need, vector<int>* chosen = nullptr) {
    Dinic din(n + 2);
    int s = n, t = n + 1;
    ll pos = 0;
    for (int v = 0; v < n; v++)
        if (w[v] > 0) pos += w[v], din.addEdge(s, v, w[v]);
        else if (w[v] < 0) din.addEdge(v, t, -w[v]);
    for (auto [u, v] : need) din.addEdge(u, v, llinf); //
    ↳ choosing u forces choosing v
    ll cut = din.calc(s, t);
    if (chosen) for (int v = 0; v < n; v++) if (din.leftOfMinCut(v))
        ↳ chosen->push_back(v);
    return pos - cut;
}
/* THE OTHER STANDARD REDUCTIONS
TWO LABELS WITH PAIR PENALTIES ("cats or dogs", image segmentation)
Every vertex gets label A or B. Cost a_v for A, b_v for B, and
    ↳ c_uv extra if u and v differ.
Build s -> v with b_v, v -> t with a_v, and u <-> v with c_uv in
    ↳ both directions.
The min cut is the minimum total cost; the source side takes label
    ↳ A.
PRECONDITION: the pair penalty must be SUBMODULAR (cost(A,A) +
    ↳ cost(B,B) <= cost(A,B) +
    ↳ cost(B,A)). A penalty that rewards disagreement cannot be modelled
    ↳ by a cut - it is NP-hard.

VERTEX CAPACITIES / VERTEX-DISJOINT PATHS
Split v into v_in -> v_out with the vertex capacity.
    ↳ Vertex-disjoint s-t paths use capacity 1.

BIPARTITE FAMILY (see 07 - Matching and 10 - Theorems)
max matching = max flow with unit capacities
min vertex cover = max matching (Konig) -> recover via
    ↳ the alternating-BFS set
max independent set = n - max matching
min path cover of a DAG (vertex-disjoint) = n - max matching of
    ↳ the split graph

"MUST BE ON THE SOURCE SIDE" -> add s -> v with INF. "MUST BE ON
    ↳ THE SINK SIDE" -> v -> t INF.
That is how you force decisions inside an otherwise free min cut.

EDGE / VERTEX DISJOINT PATHS = Menger. Recover the paths by
    ↳ decomposing the integral flow.

LOWER BOUNDS ON EDGES -> 03 - FlowWithLowerBounds.cpp.

MAXIMUM DENSITY SUBGRAPH (maximise |E(S)| / |S|): binary search
    ↳ lambda, then min cut with
    s -> v capacity m, v -> t capacity m + 2*lambda - deg(v), u <-> v
    ↳ capacity 1. Feasible iff
    the min cut is < n*m.  $O(\log(n^2) * \text{maxflow})$  since the density is a
    ↳ ratio of integers <=  $n^2$ .

MINIMUM CUT WITHOUT s AND t (global) -> 04 - StoerWagner.cpp,
    ↳  $O(V^3)$ , or n-1 max flows.

ALL-PAIRS MIN CUT -> the Gomory-Hu tree: n-1 max flows build a
    ↳ weighted tree in which the min
    u-v cut equals the MINIMUM EDGE on the u-v tree path. Use it when
    ↳ many pairs are asked.

COMPLEXITY TABLE - read this at minute 3, not minute 90
Dinic  $O(V^2 E)$  general |  $O(E \sqrt{E})$  unit caps |
```

```

→ O(E sqrt V) bipartite
Edmonds-Karp          O(V E^2)          -- do not
Push-relabel / HLPP    O(V^2 sqrt E)      -- dense
→ graphs, n ~ 1000, m ~ n^2/2
MCMF (SSP + potentials) O(F * E log V)    -- F = total
→ flow, so bound F first
Hungarian              O(n^2 m)           -- dense
→ assignment, n <= ~1000
Hopcroft-Karp          O(E sqrt V)        -- bipartite
→ matching, n >= few thousand
Kuhn                   O(V E)             -- shorter,
→ fine to ~2000 vertices
Blossom (general)      O(V^3)
Stoer-Wagner           O(V^3)
Gomory-Hu              O(V * maxflow)
If the flow value itself can be huge (capacities up to 1e9), an O(F
→ ...) bound is not a bound -
scale the capacities or use a strongly polynomial algorithm. */

```

3.6.6 GomoryHu

GOMORY-HU TREE - ALL-PAIRS MINIMUM CUT from only n-1 max-flow runs.

```

// Needs: Dinic (01 - Dinic.cpp).
// The result is a weighted tree on the same vertex set with the
→ property that, for EVERY pair
// (u, v), the minimum u-v cut in the original graph equals the
→ MINIMUM EDGE WEIGHT on the u-v
// path in the tree. (There are at most n-1 distinct min-cut values
→ in any graph - that is the
// whole content of the theorem.)
// Reach for it when a problem asks about min cuts between many
→ pairs: "sum/max/k-th of all
// pairwise min cuts", "order the vertices to maximise the sum of
→ consecutive cuts", "how many
// pairs have min cut >= k". Undirected graphs only.
// Complexity: (n-1) x Dinic. The tree is returned as parent + weight
→ arrays (par[0] = -1).
struct GomoryHu {
    int n;
    vector<array<ll, 3>> edges;           // {u,
→ v, capacity}, undirected
    vector<int> par;
    vector<ll> w;
    GomoryHu(int n) : n(n), par(n, 0), w(n, 0) {}
    void addEdge(int u, int v, ll c) { edges.push_back({(ll)u, (ll)v,
→ c}); }
    void build() {
        par.assign(n, 0), w.assign(n, 0), par[0] = -1;
        for (int i = 1; i < n; i++) {
            Dinic din(n);
            for (auto [u, v, c] : edges) din.addEdge(u, v, c);
→ // undirected: both directions
            w[i] = din.calc(i, par[i]);
            for (int j = i + 1; j < n; j++)                //
→ re-parent the same-side vertices
                if (par[j] == par[i] && din.leftOfMinCut(j)) par[j] =
→ i;
        }
        ll minCut(int u, int v) const {                    // min
→ edge on the tree path
            ll best = llinf;
            vector<int> du(n, 0);
            for (int x = u; x != -1; x = par[x]) du[x] = 1;
            vector<int> pv;
            int m = v;
            while (!du[m]) pv.push_back(m), m = par[m];      // m
→ = the meeting vertex
            for (int x = u; x != m; x = par[x]) best = min(best, w[x]);
            for (int x : pv) best = min(best, w[x]);
            return best;
        }
    };
// EQUIVALENT-FLOW TREE (Gusfield's variant) is the same n-1 flows
→ without the re-parenting step;
// it gets the min-cut VALUES right but not the actual cuts. The
→ version above is the real
// Gomory-Hu, so the tree edges are genuine cuts.
// GLOBAL MIN CUT is the minimum edge of the whole tree - but
→ Stoer-Wagner (04) is O(V^3) and
// needs no flow at all, so prefer it when that is all you want.
// DIRECTED graphs have no Gomory-Hu tree; you genuinely need O(n^2)
→ flows there.

```

3.6.7 PushRelabel

HIGHEST-LABEL PUSH-RELABEL max flow, $O(V^2 \sqrt{E})$, 0-indexed. Same interface as Dinic (01).

```

// WHEN TO PREFER IT OVER DINIC: dense graphs (V ~ 1000-5000, E ~
→ V^2/2) and graphs with large
// capacities, where Dinic's O(V^2 E) bound bites - usually 3-10x
→ faster there. On unit-capacity
// or bipartite-matching graphs Dinic is O(E sqrt V) and wins, so
→ keep both.
// It does NOT maintain a flow at intermediate times - only at the
→ end - so it cannot be used for
// the incremental "add capacity and continue" tricks; Dinic can.
// IDEA: give every vertex a HEIGHT and let excess flow downhill
→ (push); a vertex with excess and
// nowhere to go is lifted (relabel). Source starts at height V, so
→ nothing can return to it.
// The GAP HEURISTIC (if no vertex has height h any more, lift
→ everything above h out of range) is
// what makes it fast in practice - it is the 'co[]' bookkeeping
→ below, do not drop it.
// addEdge(a, b, cap, rcap): rcap = cap gives an UNDIRECTED edge.
→ After calc, leftOfMinCut(v) marks
// the SOURCE side of a minimum cut, and e.f is the flow on that arc.
struct PushRelabel {
    struct Edge { int to, rev; ll f, c; };
    int n;
    vector<vector<Edge>> g;
    vector<ll> ec;
    vector<int> H, cur, co;
    vector<vector<int>> hs;
    PushRelabel(int n) : n(n), g(n), ec(n), H(n), cur(n), co(2 * n +
→ 1), hs(2 * n + 1) {}
    void addEdge(int a, int b, ll cap, ll rcap = 0) {
        if (a == b) return;
        g[a].push_back({b, (int)g[b].size(), 0, cap});
        g[b].push_back({a, (int)g[a].size() - 1, 0, rcap});
    }
    void addFlow(int u, Edge& e, ll f) {
        Edge& back = g[e.to][e.rev];
        if (!ec[e.to] && f) hs[H[e.to]].push_back(e.to);
        e.f += f, e.c -= f, ec[e.to] += f;
        back.f -= f, back.c += f, ec[u] -= f;
    }
    ll calc(int s, int t) {
        fill(all(H), 0), fill(all(ec), 0), fill(all(co), 0),
→ fill(all(cur), 0);
        for (auto& v : hs) v.clear();
        H[s] = n, ec[t] = 1, co[0] = n - 1;
        for (auto& e : g[s]) addFlow(s, e, e.c);
        for (int hi = 0;;) {
            while (hs[hi].empty()) if (!hi--) return -ec[s];
            int u = hs[hi].back(); hs[hi].pop_back();
            while (ec[u] > 0) {                //
→ discharge u
                if (cur[u] == (int)g[u].size()) {                //
→ relabel + gap heuristic
                    H[u] = 2 * n;
                    for (int i = 0; i < (int)g[u].size(); i++)
                        if (g[u][i].c && H[u] > H[g[u][i].to] + 1)
                            H[u] = H[g[u][i].to] + 1, cur[u] = i;
                    if (++co[H[u]], !--co[hi] && hi < n)
                        for (int i = 0; i < n; i++)
                            if (hi < H[i] && H[i] < n) --co[H[i]],
→ H[i] = n + 1;
                    hi = H[u];
                } else if (g[u][cur[u]].c && H[u] ==
→ H[g[u][cur[u]].to] + 1)
                    addFlow(u, g[u][cur[u]], min(ec[u],
→ g[u][cur[u]].c));
                else ++cur[u];
            }
        }
        bool leftOfMinCut(int a) { return H[a] >= n; }
    };
// COMPLEXITY NOTES. FIFO push-relabel is O(V^3); the highest-label
→ rule above is O(V^2 sqrt E).
// The gap heuristic and the global relabel (BFS from t every so
→ often) are the two practical
// speedups; the version here has the gap only, which is already
→ enough for contests.
// TIE-BREAKING WITH DINIC: if you also need the actual min cut
→ EDGES, both give them the same way
// (an edge from a leftOfMinCut vertex to a non-leftOfMinCut one). If
→ you need the flow DECOMPOSED
// into paths, repeatedly DFS from s along positive-flow arcs and
→ subtract the bottleneck.

```

3.6.8 CostScalingMCMF

MIN COST MAX FLOW BY COST SCALING (push-relabel), $O(V^2 E \log(V \cdot C))$ but very fast in practice -

```
// 3e4 edges are comfortable. 0-indexed, INTEGER costs and capacities.
// WHY THIS EXISTS ALONGSIDE 02 - MCMF. The successive-shortest-path
// ↪ version needs the residual
// graph to have NO NEGATIVE CYCLE, so it forbids negative-cost edges
// ↪ (Johnson potentials only fix
// negative EDGES on a DAG-like start, not negative CYCLES). This one
// ↪ accepts ARBITRARY costs,
// including negative cycles: it first pushes a max flow ignoring
// ↪ cost, then repeatedly "refines"
// the eps-optimality of the flow, halving eps each round. It also
// ↪ does not depend on the flow
// VALUE, so a huge max flow is fine, whereas SSP is  $O(F \cdot E \log V)$ .
// Reach for it when: costs can be negative, or the flow value is
// ↪ large, or the graph is dense.
// The source and sink are fixed at construction. Costs are
// ↪ internally multiplied by V, so keep
// |cost| * V * V inside the cost type - use long long unless you
// ↪ have checked.
struct CostScalingMCMF {
    struct Edge { ll c, f; int to, rev; };
    ll eps = 0;
    int n, S, T;
    vector<vector<Edge>> g;
    vector<vector<int>> hs;
    vector<int> cur, isq, co;
    vector<ll> ex, h;
    CostScalingMCMF(int n, int s, int t) : n(n), S(s), T(t), g(n) {}
    void addEdge(int a, int b, ll cost, ll cap) {
        ↪ cost per unit, cap >= 0
        if (a == b) return;
        cost *= n, eps = max(eps, abs(cost));
        g[a].push_back({cost, cap, b, (int)g[b].size()});
        g[b].push_back({-cost, 0, a, (int)g[a].size() - 1});
    }
    void addFlow(int u, Edge& e, ll f) {
        Edge& bk = g[e.to][e.rev];
        if (!ex[e.to] && f) hs[h[e.to]].push_back(e.to);
        e.f += f, ex[e.to] += f, bk.f += f, ex[u] -= f;
    }
    ll maxFlow() {
        ↪ plain highest-label push-relabel
        ex.assign(n, 0), h.assign(n, 0), hs.assign(2 * n + 1, {}),
        ↪ co.assign(2 * n + 1, 0);
        cur.assign(n, 0), h[S] = n, ex[T] = 1, co[0] = n - 1;
        for (auto& e : g[S]) addFlow(S, e, e.f);
        if (hs[0].size()) for (int hi = 0; hi >= 0; hi++) {
            int u = hs[hi].back(); hs[hi].pop_back();
            while (ex[u] > 0) {
                if (cur[u] == (int)g[u].size()) {
                    h[u] = 2 * n;
                    for (int i = 0; i < (int)g[u].size(); i++)
                        if (g[u][i].f && h[u] > h[g[u][i].to] + 1)
                            h[u] = h[g[u][i].to] + 1, cur[u] = i;
                    if (++co[h[u]], !--co[hi] && hi < n)
                        for (int i = 0; i < n; i++)
                            if (hi < h[i] && h[i] < n) --co[h[i]],
                                h[i] = n + 1;
                    hi = h[u];
                } else if (g[u][cur[u]].f && h[u] ==
                    ↪ h[g[u][cur[u]].to] + 1)
                    addFlow(u, g[u][cur[u]], min(ex[u],
                    ↪ g[u][cur[u]].f));
                else ++cur[u];
            }
            while (hi >= 0 && hs[hi].empty()) --hi;
        }
        return -ex[S];
    }
    void push(int u, Edge& e, ll amt) {
        amt = min(amt, e.f);
        e.f -= amt, ex[e.to] += amt, g[e.to][e.rev].f += amt, ex[u]
        ↪ -= amt;
    }
    void relabel(int v) {
        ll nh = -llinf;
        for (int i = 0; i < (int)g[v].size(); i++)
            if (g[v][i].f && nh < h[g[v][i].to] - g[v][i].c)
                nh = h[g[v][i].to] - g[v][i].c, cur[v] = i;
        h[v] = nh - eps;
    }
    pair<ll, ll> solve() {
        ↪ {max flow, its minimum cost}
        ll cost = 0;
        for (int i = 0; i < n; i++) for (Edge& e : g[i]) cost += e.c
```

```
↪ * e.f;
    ll flow = maxFlow();
    h.assign(n, 0), ex.assign(n, 0), isq.assign(n, 0),
    ↪ cur.assign(n, 0);
    queue<int> q;
    for (; eps; eps >>= 2) {
        ↪ refine: halve eps twice per round
        fill(all(cur), 0);
        for (int i = 0; i < n; i++) for (auto& e : g[i])
            if (h[i] + e.c - h[e.to] < 0 && e.f) push(i, e, e.f);
        for (int i = 0; i < n; i++) if (ex[i] > 0) q.push(i),
        ↪ isq[i] = 1;
        while (!q.empty()) {
            ↪ restore feasibility
            int u = q.front(); q.pop(), isq[u] = 0;
            while (ex[u] > 0) {
                if (cur[u] == (int)g[u].size()) relabel(u);
                for (int& i = cur[u]; i < (int)g[u].size(); i++) {
                    Edge& e = g[u][i];
                    if (h[u] + e.c - h[e.to] < 0) {
                        push(u, e, ex[u]);
                        if (ex[e.to] > 0 && !isq[e.to])
                            ↪ q.push(e.to), isq[e.to] = 1;
                        if (!ex[u]) break;
                    }
                }
            }
            if (eps > 1 && (eps >> 2) == 0) eps = 4;
        }
        for (int i = 0; i < n; i++) for (Edge& e : g[i]) cost -= e.c
        ↪ * e.f;
        return {flow, cost / 2 / n};
    }
    ll flowOn(const Edge& e) const { return g[e.to][e.rev].f; }
};
// MIN COST FLOW OF A GIVEN VALUE k (not necessarily maximum): cap
// ↪ the source arc at k, or add an
// arc T → S of capacity INF and cost -BIG and take a min cost
// ↪ CIRCULATION.
// NEGATIVE COST CYCLES in the input are handled: the answer is the
// ↪ true minimum, which may be
// negative. With 02 - MCMF you would have to cancel the cycles by
// ↪ hand first (saturate every
// negative edge, then fix the resulting excesses with a circulation)
// ↪ - that is what this replaces.
// ASSIGNMENT PROBLEMS with n <= 500 are still better served by
// ↪ Hungarian (07 - Matching/03).
```

3.6.9 MinCostFlowLowerBounds

MIN COST FLOW WITH LOWER BOUNDS (min cost circulation with demands). 0-indexed, costs ≥ 0 .

```
// Needs: MCMF (02 - MCMF.cpp).
// This is 03 - FlowWithLowerBounds with a price tag on every unit:
//   ↳ each edge needs
// lo <= f(e) <= hi and costs c per unit, and among all FEASIBLE
//   ↳ flows you want the cheapest.
// Reach for it for "every task must be done at least once, minimise
//   ↳ total cost", "each shift needs
// between L and R workers", "cover every row of the matrix at least
//   ↳ k times as cheaply as
// possible", and generally any min-cut/flow model where a constraint
//   ↳ reads "at least".
// CONSTRUCTION. Send the mandatory lo units for free by hand, then
//   ↳ repair the resulting imbalance:
// base cost += lo * c ; the arc keeps capacity hi - lo at cost c
// excess(v) = (sum of lo entering v) - (sum of lo leaving v)
// excess(v) > 0 -> SS -> v with capacity excess(v), cost 0; < 0
//   ↳ -> v -> TT, cost -excess(v)
// One min cost max flow from SS to TT. FEASIBLE iff it saturates
//   ↳ every SS arc; the answer is
// base + that cost. For an s-t flow, add t -> s with capacity INF
//   ↳ and cost 0 FIRST.
struct MinCostLowerBounds {
    int n, S, T;
    MCMF f;
    ll base = 0, need = 0;
    vector<ll> low, exc;
    vector<array<int, 2>> eid;
    ↳ index in f.g[u] per edge
    MinCostLowerBounds(int n) : n(n), S(n), T(n + 1), f(n + 2), exc(n
    ↳ + 2, 0) {}
    void addEdge(int u, int v, ll lo, ll hi, ll cost) {
        low.push_back(lo), eid.push_back({u, (int)f.g[u].size()});
        f.addEdge(u, v, hi - lo, cost);
        base += lo * cost, exc[v] += lo, exc[u] -= lo;
    }
    pair<bool, ll> solve() {
        ↳ {feasible, minimum total cost}
        for (int v = 0; v < n; v++) {
            if (exc[v] > 0) f.addEdge(S, v, exc[v], 0), need +=
            ↳ exc[v];
            else if (exc[v] < 0) f.addEdge(v, T, -exc[v], 0);
        }
        auto [fl, c] = f.run(S, T);
        return {fl == need, base + c};
    }
    ll flowOn(int i) { return low[i] +
    ↳ f.g[eid[i][0]][eid[i][1]].flow; }
};
// MIN COST *MAX* s-t FLOW WITH BOUNDS: add t -> s with capacity INF
//   ↳ and cost 0, run solve(); the
// flow that ended up on that arc is the feasible s-t flow value.
//   ↳ Then delete the arc and run one
// more MCMF from s to t on the residual to push the flow up to
//   ↳ maximum.
// MIN COST *MIN* s-t FLOW: same, but afterwards cancel flow by
//   ↳ running MCMF from t to s.
// NEGATIVE COSTS: this construction is still correct, but MCMF's
//   ↳ Dijkstra is not - either call
// f.setPotentialsBF(S) first (no negative cycle allowed) or switch
//   ↳ to 08 - CostScalingMCMF.
// b-FLOW (each vertex has a supply/demand b(v) instead of a single
//   ↳ source): put b(v) directly into
// exc[v] before solving - the SS/TT construction is exactly the same
//   ↳ machinery.
// SANITY CHECK: if lo > hi on any edge the instance is infeasible;
//   ↳ the code will not notice.
```

3.6.10 MaxDensitySubgraph

MAXIMUM DENSITY SUBGRAPH - choose a NON-EMPTY vertex set S maximising $|E(S)| / |S|$, the number

```
// Needs: Dinic (01 - Dinic.cpp).
// of induced edges per vertex. Undirected, 0-indexed,  $O(\log(n^2) *
//   ↳ maxflow)$  via parametric min cut.
// (Note this is NOT the maximum CLIQUE relaxation - it is
//   ↳ polynomial, unlike everything nearby.)
// Reach for it for "find the most tightly knit group", "the subgraph
//   ↳ with the best edges-to-people
// ratio", spam/fraud ring detection, and as the LP-relaxation
//   ↳ certificate for densest-subgraph
// style scoring problems.
// FRACTIONAL PROGRAMMING. Ask "is there S with  $|E(S)| - g|S| > 0$ ?"
//   ↳ for a guess g. Build
// s -> v (capacity m), v -> t (capacity m + 2g - deg(v)), u <-> v
//   ↳ (capacity 1 each way)
// then min cut =  $n*m - 2 * \max \text{ over } S \text{ of } (|E(S)| - g|S|)$ , so the
//   ↳ answer is yes iff the min cut is
// strictly below  $n*m$ . Binary search g. Two distinct densities differ
//   ↳ by at least  $1/(n(n-1))$ , so
//  $O(\log(n^2))$  iterations of the binary search are enough for an
//   ↳ EXACT answer with rationals.
// Below every capacity is scaled by  $K = n^2$  so they stay INTEGRAL
//   ↳ and the guess is  $g = g_2 / K$ .
pair<ld, vector<int>> maxDensitySubgraph(int n, const
    ↳ vector<pair<int, int>>& e) {
    int m = e.size();
    if (n == 1 || m == 0) return {0, {0}};
    ll K = (ll)n * n;
    vector<int> deg(n, 0), best{0};
    for (auto [u, v] : e) deg[u]++, deg[v]++;
    ll lo = 0, hi = (ll)m * K;
    ↳ density * K, density <= m
    while (lo < hi) {
        ll g2 = (lo + hi + 1) / 2;
        ↳ test the density  $g = g_2 / K$ 
        Dinic din(n + 2);
        int s = n, t = n + 1;
        for (auto [u, v] : e) din.addEdge(u, v, K, K);
        ↳ undirected, capacity 1 scaled
        for (int v = 0; v < n; v++) {
            din.addEdge(s, v, (ll)m * K);
            din.addEdge(v, t, (ll)m * K + 2 * g2 - (ll)deg[v] * K);
        }
        if (din.calc(s, t) < (ll)n * m * K) {
            ↳ some S beats density g
            lo = g2, best.clear();
            for (int v = 0; v < n; v++) if (din.leftOfMinCut(v))
            ↳ best.push_back(v);
            else hi = g2 - 1;
        }
        return {(ld)lo / K, best};
    }
    ↳ {density, the vertex set}
}
// The source side of the min cut IS the densest set - no separate
//   ↳ extraction pass is needed.
// MAXIMUM DENSITY with VERTEX WEIGHTS or edge weights: replace
//   ↳ deg(v) by the weighted degree and
// the capacity-1 edges by the edge weights; the same cut works.
// AT LEAST k VERTICES / AT MOST k VERTICES turns the problem NP-hard
//   ↳ - do not try to patch this.
// MINIMUM density is trivial (a single vertex, density 0), so the
//   ↳ question is always maximum.
// RELATED: "maximum (sum of vertex profits) - (sum of edge
//   ↳ penalties)" is plain project selection
// (05 - MinCutModelling); the RATIO is what forces the binary search.
// GOLDBERG'S ORIGINAL formulation uses capacities  $m + 2g - \deg(v)$ 
//   ↳ exactly as above; if you prefer
// floating point, binary search g as a double for ~40 iterations and
//   ↳ skip the scaling.
```

3.6.11 UniqueMinCut

IS THE MINIMUM s-t CUT UNIQUE? One extra pair of DFS over the residual graph, $O(V + E)$ after the

```
// Needs: Dinic (01 - Dinic.cpp).
// max flow. 0-indexed, directed or undirected.
// After a max flow, let A = the vertices REACHABLE FROM s in the
//   ↳ residual graph (the smallest
// minimum cut) and B = the vertices that can REACH t in the residual
//   ↳ graph (the largest minimum
// cut's source side is  $V \setminus B$ ). Every minimum cut lies "between"
//   ↳ these two, so:
// the minimum cut is UNIQUE  $\Leftrightarrow$  A and B together cover every
//   ↳ vertex.
// A vertex in neither set can be put on either side without changing
//   ↳ the cut value, and moving it
// gives a genuinely different cut of the same weight.
// Reach for it for "is the answer unique / is the bottleneck
//   ↳ forced", "which edges are in EVERY
// minimum cut", and for problems that ask you to output AMBIGUOUS
//   ↳ when a tie exists.
bool uniqueMinCut(Dinic& din, int s, int t) {
    ↳ call AFTER din.calc(s, t)
    int n = din.adj.size();
    vector<char> a(n, 0), b(n, 0);
    vector<int> st[s];
    a[s] = 1;
    while (!st.empty()) {
        ↳ forward residual reachability
        int u = st.back(); st.pop_back();
        for (auto& e : din.adj[u]) if (e.c > 0 && !a[e.to]) a[e.to] =
        ↳ 1, st.push_back(e.to);
    }
    st = {t}, b[t] = 1;
    while (!st.empty()) {
        ↳ backward: residual arcs INTO u
        int u = st.back(); st.pop_back();
        for (auto& e : din.adj[u]) if (din.adj[e.to][e.rev].c > 0 &&
        ↳ !b[e.to])
            b[e.to] = 1, st.push_back(e.to);
    }
    for (int v = 0; v < n; v++) if (!a[v] && !b[v]) return false;
    return true;
}
// THE TWO EXTREME CUTS come out of the same two searches: A is the
//   ↳ source side of the minimum cut
// with the FEWEST source-side vertices, and  $V \setminus B$  is the one with
//   ↳ the MOST.
// AN EDGE IS IN SOME MINIMUM CUT iff it is saturated and its
//   ↳ endpoints are in different strongly
// connected components of the residual graph (contract the residual
//   ↳ SCCs and the min cuts are
// exactly the closed sets of the resulting DAG).
// AN EDGE IS IN EVERY MINIMUM CUT iff it is saturated and removing
//   ↳ its capacity lowers the max
// flow, i.e. iff u is in A and v is in B - checkable directly from
//   ↳ these two arrays.
// COUNTING MINIMUM CUTS = counting antichains/closed sets in that
//   ↳ residual-SCC DAG, which is
// #P-hard in general but easy when the DAG is a path or has few
//   ↳ vertices.
```

3.7 Matching

3.7.1 HopcroftKarp

HOPCROFT-KARP - maximum bipartite matching in $O(E \sqrt{V})$, the fast alternative to Kuhn's $O(VE)$.

```
// Use it when n is above a few thousand; below that Kuhn is shorter
//   ↳ and fast enough.
// HOPCROFT-KARP - maximum bipartite matching in  $O(E \sqrt{V})$ , the
//   ↳ fast alternative to Kuhn's
//  $O(VE)$ . Use above a few thousand vertices; below that Kuhn (02) is
//   ↳ shorter and fast enough.
// INDEXING TRAP: leftMatch has size m and is indexed by a RIGHT
//   ↳ vertex; rightMatch has size n and
// is indexed by a LEFT vertex. Reading the names the obvious way is
//   ↳ the default way to get this
// wrong. For the minimum vertex cover / maximum independent set
//   ↳ (Konig), see 02 - KuhnAndKonig.
```

```
struct HopcroftKarp {
    vector<int> leftMatch, rightMatch, dist, cur;
    vector<vector<int>> > a;
    int n, m;

    HopcroftKarp() {}

    HopcroftKarp(int n, int m) {
        this->n = n;
        this->m = m;
        a = vector<vector<int>>(n);
        leftMatch = vector<int>(m, -1);
        rightMatch = vector<int>(n, -1);
        dist = vector<int>(n, -1);
        cur = vector<int>(n, -1);
    }

    void addEdge(int x, int y) {
        a[x].push_back(y);
    }

    int bfs() {
        int found = 0;
        queue<int> q;
        for (int i = 0; i < n; i++)
            if (rightMatch[i] < 0) dist[i] = 0, q.push(i);
            else dist[i] = -1;

        while (!q.empty()) {
            int x = q.front();
            q.pop();
            for (int i = 0; i < int(a[x].size()); i++) {
                int y = a[x][i];
                if (leftMatch[y] < 0) found = 1;
                else if (dist[leftMatch[y]] < 0)
                    dist[leftMatch[y]] = dist[x] + 1,
                    ↳ q.push(leftMatch[y]);
            }
        }

        return found;
    }

    int dfs(int x) {
        for (; cur[x] < int(a[x].size()); cur[x]++) {
            int y = a[x][cur[x]];
            if (leftMatch[y] < 0 || (dist[leftMatch[y]] == dist[x] +
            ↳ 1 && dfs(leftMatch[y]))) {
                leftMatch[y] = x;
                rightMatch[x] = y;
                return 1;
            }
        }
        return 0;
    }

    int maxMatching() {
        int match = 0;
        while (bfs()) {
            for (int i = 0; i < n; i++) cur[i] = 0;
            for (int i = 0; i < n; i++)
                if (rightMatch[i] < 0) match += dfs(i);
        }
        return match;
    }
};
```


3.7.2 KuhnAndKonig

Kuhn's bipartite matching, $O(V \cdot E)$ but tiny and easy to modify. Left size n , right size m .

```
// In practice fast; randomise the adjacency order + greedy pre-match
// ↳ if you need more speed.
struct Kuhn {
    int n, m;
    vector<vector<int>> adj;           // left -> right
    vector<int> mt;                  // mt[right] =
    ↳ matched left, or -1
    vector<bool> used;

    Kuhn(int n = 0, int m = 0) : n(n), m(m), adj(n), mt(m, -1) {}
    void add_edge(int u, int v) { adj[u].push_back(v); }

    bool try_(int u) {
        for (int v : adj[u]) if (!used[v]) {
            used[v] = true;
            if (mt[v] < 0 || try_(mt[v])) { mt[v] = u; return true; }
        }
        return false;
    }
    int maxMatching() {
        int r = 0;
        vector<int> ord(n);
        iota(all(ord), 0);
        shuffle(all(ord), rng);
        for (int u : ord) { used.assign(m, false); r += try_(u); }
        return r;
    }
    // KONIG: minimum VERTEX COVER = maximum matching. Recover it:
    // ↳ alternate from every UNMATCHED left vertex; let Z = reached
    ↳ set.
    // cover = (left \ Z) union (right and Z); max independent set
    ↳ = complement of the cover.
    pair<vector<int>, vector<int>> minVertexCover() {
        vector<int> matchL(n, -1);
        for (int v = 0; v < m; v++) if (mt[v] >= 0) matchL[mt[v]] = v;
        vector<bool> zL(n, false), zR(m, false);
        vector<int> st;
        for (int u = 0; u < n; u++) if (matchL[u] < 0) zL[u] = true,
        ↳ st.push_back(u);
        while (!st.empty()) {
            int u = st.back(); st.pop_back();
            for (int v : adj[u]) if (!zR[v]) {
                zR[v] = true;
                if (mt[v] >= 0 && !zL[mt[v]]) zL[mt[v]] = true,
                ↳ st.push_back(mt[v]);
            }
        }
        vector<int> L, R;
        for (int u = 0; u < n; u++) if (!zL[u]) L.push_back(u);
        for (int v = 0; v < m; v++) if (zR[v]) R.push_back(v);
        return {L, R};
        ↳ |L| + |R| ==
    }
    ↳ maxMatching()
};

/* THEOREMS you actually quote (bipartite graphs):
Konig:    max matching = min vertex cover
max independent set = V - min vertex cover
MIN PATH COVER of a DAG (VERTEX-DISJOINT paths covering all
↳ vertices)
    = n - max matching of the split graph (u_out -> v_in for
↳ every edge)
    CAREFUL: this is the vertex-disjoint cover. To get
↳ Dilworth (FINITE poset)'s width (minimum
    number of CHAINS, i.e. paths allowed to share vertices)
↳ run it on the
    TRANSITIVE CLOSURE of the DAG first.
Hall:    a perfect matching on the left exists iff |N(S)| >= |S|
↳ for every subset S of the LEFT part X
Dilworth: min chain cover of a poset = longest antichain (build
↳ the DAG, TRANSITIVELY CLOSE
    it, then use the path cover above)
Mirsky:  the dual - longest chain = minimum number of antichains
↳ partitioning the poset
Gallai:  alpha + tau = n always; nu + rho = n only if there are
↳ NO ISOLATED VERTICES
    (alpha = max independent set, tau = min vertex cover, nu
↳ = max matching,
    rho = min edge cover)
Hall defect: nu(G) = |X| - max over W subset of X of ( |W| -
↳ |N(W)| )
Tutte:    G has a perfect matching iff odd(G - U) <= |U| for every
↳ U
Tutte-Berge: nu(G) = (1/2) * min over U of ( |V| + |U| - odd(G -
↳ U) )
```

Menger: min edge cut(s,t) = max edge-disjoint s-t paths;
 min vertex cut(s,t) = max internally-vertex-disjoint s-t
 ↳ paths (s,t non-adjacent) */

3.7.3 Hungarian

HUNGARIAN ALGORITHM - minimum-cost perfect matching on a DENSE bipartite graph, $O(n^2 m)$.

```
// a is 1-indexed: a[1..n][1..m] with n <= m. Returns the cost;
↳ ans[j] = row matched to column j.
// For MAXIMUM cost, negate the matrix.
pair<ll, vector<int>> hungarian(const vector<vector<ll>>& a) {
    int n = a.size() - 1, m = a[0].size() - 1;
    vector<ll> u(n + 1, 0), v(m + 1, 0);
    vector<int> p(m + 1, 0), way(m + 1, 0);
    for (int i = 1; i <= n; i++) {
        p[0] = i;
        int j0 = 0;
        vector<ll> minv(m + 1, llinf);
        vector<char> used(m + 1, false);
        do {
            used[j0] = true;
            int i0 = p[j0], j1 = 0;
            ll delta = llinf;
            for (int j = 1; j <= m; j++) if (!used[j]) {
                ll cur = a[i0][j] - u[i0] - v[j];
                if (cur < minv[j]) minv[j] = cur, way[j] = j0;
                if (minv[j] < delta) delta = minv[j], j1 = j;
            }
            for (int j = 0; j <= m; j++)
                if (used[j]) u[p[j]] += delta, v[j] -= delta;
            else minv[j] -= delta;
            j0 = j1;
        } while (p[j0] != 0);
        do { int j1 = way[j0]; p[j0] = p[j1], j0 = j1; } while (j0);
    }
    vector<int> ans(m + 1);
    for (int j = 1; j <= m; j++) ans[j] = p[j];
    return {-v[0], ans};
    ↳ optimal total cost
}

// Use it for: assignment problems, "match every task to a worker
↳ minimising total time",
// minimum-cost perfect matching in a complete bipartite graph.
// FORBIDDEN PAIRS: use a large finite cost (1e15), NOT llinf - the
↳ reduced cost a[i][j]-u[i]-v[j]
// would overflow. ans[j] == 0 means column j is unmatched (possible
↳ only when n < m).
// WHICH ONE: on a DENSE / complete graph this beats MCMF - n^2*m
↳ here versus n augmentations of
// Dijkstra over n^2 arcs (n^3 log n) there. Use MCMF only when the
↳ graph is genuinely SPARSE.
```


3.7.4 Blossom

MAXIMUM MATCHING IN A GENERAL (NON-BIPARTITE) GRAPH - Edmonds' blossom algorithm, $O(V^3)$.

```
// Everything else in this folder assumes bipartite. Reach for this
// ↳ the moment the two sides are
// not fixed: "pair up people who are compatible", "tile a board
// ↳ where the colouring argument
// fails", "maximum set of disjoint edges" on an arbitrary graph.
// The trick it exists for: an ODD cycle can be traversed in two
// ↳ directions, so an alternating
// search can enter and leave it on the same side. Edmonds CONTRACTS
// ↳ such a cycle (a "blossom")
// into a single vertex, searches the contracted graph, and expands
// ↳ afterwards.
// Certificate: the answer equals the Tutte-Berge bound (see 10 -
// ↳ Theorems/01 - GraphTheorems).
struct Blossom {
    int n;
    vector<vector<char>> > g; //
    ↳ adjacency MATRIX
    vector<int> match, p, base;
    vector<char> used, inBlossom;
    Blossom(int n) : n(n), g(n, vector<char>(n, 0)), match(n, -1),
    ↳ p(n), base(n),
        used(n), inBlossom(n) {}
    void addEdge(int u, int v) { g[u][v] = g[v][u] = 1; }
    int lca(int a, int b) {
        vector<char> seen(n, 0);
        for (;;) { a = base[a], seen[a] = 1; if (match[a] < 0) break;
    ↳ a = p[match[a]]; }
        for (;;) { b = base[b]; if (seen[b]) return b; b =
    ↳ p[match[b]]; }
    }
    void mark(int v, int b, int child) { //
    ↳ walk v up to the blossom base
        while (base[v] != b) {
            inBlossom[base[v]] = inBlossom[base[match[v]]] = 1;
            p[v] = child, child = match[v], v = p[match[v]];
        }
    }
    bool augment(int root) { // one
    ↳ BFS for an augmenting path
        fill(all(used), 0), fill(all(p), -1);
        iota(all(base), 0);
        used[root] = 1;
        queue<int> q({root});
        while (!q.empty()) {
            int v = q.front(); q.pop();
            for (int to = 0; to < n; to++) {
                if (!g[v][to] || base[v] == base[to] || match[v] ==
    ↳ to) continue;
                if (to == root || (match[to] >= 0 && p[match[to]] >=
    ↳ 0)) { // odd cycle -> contract
                    int b = lca(v, to);
                    fill(all(inBlossom), 0);
                    mark(v, b, to), mark(to, b, v);
                    for (int i = 0; i < n; i++) if
    ↳ (inBlossom[base[i]]) {
                        base[i] = b;
                        if (!used[i]) used[i] = 1, q.push(i);
                    }
                } else if (p[to] < 0) {
                    p[to] = v;
                    if (match[to] < 0) { //
    ↳ free vertex: flip the path
                        for (int u = to, pv, nv; u >= 0; u = nv)
                            pv = p[u], nv = match[pv], match[u] = pv,
    ↳ match[pv] = u;
                        return true;
                    }
                    used[match[to]] = 1, q.push(match[to]);
                }
            }
        }
        return false;
    }
    int solve() {
        int r = 0;
        for (int v = 0; v < n; v++) //
    ↳ greedy start: fewer BFS rounds
            if (match[v] < 0)
                for (int u = 0; u < n; u++)
                    if (g[v][u] && match[u] < 0) { match[v] = u,
    ↳ match[u] = v, r++; break; }
                for (int v = 0; v < n; v++) if (match[v] < 0) r += augment(v);
        return r; //
    ↳ match[v] = partner, or -1
```

```
}
};
// PERFECT MATCHING EXISTS iff solve() * 2 == n. Tutte's theorem
// ↳ gives the certificate when it
// does not: some U with odd(G - U) > |U|.
// MINIMUM EDGE COVER (no isolated vertices) = n - maximum matching
// ↳ (Gallai).
// MAXIMUM INDEPENDENT SET is still NP-hard here - Konig's alpha = n
// ↳ - nu is BIPARTITE ONLY.
// WEIGHTED general matching needs the dual-variable version (~150
// ↳ lines); if the graph is small
// (n <= 20) a bitmask DP over matched vertices is far shorter and
// ↳ usually enough.
// CHINESE POSTMAN (undirected): make every degree even by
// ↳ duplicating a minimum-weight perfect
// matching on the ODD-degree vertices, using all-pairs shortest
// ↳ paths as the edge weights, then
// take an Eulerian circuit. The MIXED version (some edges directed)
// ↳ is NP-hard.
// THE MATCHING GAME (a token moves along edges to unvisited
// ↳ vertices; stuck loses): the first
// player wins iff the start vertex is covered by EVERY maximum
// ↳ matching, i.e. iff deleting it
// decreases the matching size. With this file that works on a
// ↳ general graph, not just bipartite.
```

3.7.5 BipartiteEdgeColouring

BIPARTITE EDGE COLOURING in exactly D colours (D = maximum degree), $O(m * D)$. 0-indexed.

```
// KONIG'S EDGE COLOURING THEOREM: a bipartite (multi)graph is always
//    ↳ D-edge-colourable, and  $D$  is
// obviously necessary. Equivalently: the edges split into  $D$  perfect
//    ↳ matchings of the  $D$ -regular
// completion, which is the "round robin timetable" primitive.
// Reach for it for "schedule all the games/lectures in the fewest
//    ↳ rounds so nobody plays twice in
// one round", "decompose a bipartite multigraph into matchings",
//    ↳ "Latin square completion",
// and for the routing step of a Clos/Benes network.
// The insertion argument: give edge (u, v) the smallest colour cu
//    ↳ free at u and look at the
// smallest colour cv free at v. If they differ, the cu/cv
//    ↳ alternating path starting at v cannot
// reach u (it would have to ENTER u by cu, which is free there), so
//    ↳ flipping the two colours
// along it frees cu at v as well - then colour the edge cu.
vector<int> bipartiteEdgeColouring(int nl, int nr, const
    ↳ vector<pair<int, int>>& e) {
    int m = e.size(), D = 0;
    vector<int> dl(nl, 0), dr(nr, 0);
    for (auto [u, v] : e) D = max({D, ++dl[u], ++dr[v]});
    vector<vector<int>> L(nl, vector<int>(D, -1)), R(nr,
    ↳ vector<int>(D, -1)); // colour -> edge id
    vector<int> col(m, -1);
    for (int i = 0; i < m; i++) {
        auto [u, v] = e[i];
        int cu = 0, cv = 0;
        while (L[u][cu] >= 0) cu++;
        while (R[v][cv] >= 0) cv++;
        if (cu != cv) { //
            ↳ walk the cu/cv alternating path
            vector<int> path;
            for (int x = v, side = 1, c = cu;;) {
                int id = side ? R[x][c] : L[x][c];
                if (id < 0) break;
                path.push_back(id);
                x = side ? e[id].first : e[id].second, side ^= 1, c =
            ↳ (c == cu ? cv : cu);
            }
            for (int id : path) L[e[id].first][col[id]] =
            ↳ R[e[id].second][col[id]] = -1;
            for (int id : path) {
                col[id] = (col[id] == cu ? cv : cu);
                L[e[id].first][col[id]] = R[e[id].second][col[id]] =
            ↳ id;
            }
            col[i] = cu, L[u][cu] = R[v][cu] = i;
        }
        return col; //
    ↳ col[i] in [0, D)
}
// The number of colours actually used IS the maximum degree - no
//    ↳ search needed.
// EQUAL-SIZE ROUNDS: to split the edges into k groups whose sizes
//    ↳ differ by at most one and where
// each vertex has at most ceil(deg/k) edges per group, split every
//    ↳ vertex into ceil(deg/k) copies
// of degree <= k and edge-colour that graph.
// GENERAL (non-bipartite) graphs need  $D$  or  $D+1$  colours - Vizing
//    ↳ (06), and deciding which is
// NP-hard. On a bipartite graph the answer is always  $D$ , which is why
//    ↳ this file is separate.
// A REGULAR bipartite graph decomposes into perfect matchings;
//    ↳ taking the colour classes one at a
// time is much faster than running a matching algorithm  $D$  times.
```

3.7.6 VizingEdgeColouring

VIZING EDGE COLOURING - colour the edges of a SIMPLE undirected graph with at most $D+1$ colours

```
// ( $D$  = maximum degree), constructively, in  $O(n * m)$ . 0-indexed in,
//    ↳ colours returned 0-indexed.
// VIZING'S THEOREM says  $D$  or  $D+1$  always suffices; deciding WHICH is
//    ↳ NP-hard, so  $D+1$  is the answer
// you ship. (Bipartite graphs always need only  $D$  - use 05 instead,
//    ↳ it is much shorter.)
// Reach for it for "split the matches/tasks into as few rounds as
//    ↳ possible so no player/machine is
// used twice per round" on a general graph, and for
//    ↳ "orient/partition the edges into  $D+1$  matchings"
// arguments. A  $D+1$  colouring also proves the  $D+1$  bound
//    ↳ constructively, which some problems ask for.
// MULTIGRAPHS are NOT covered (Shannon's bound is  $3D/2$  there);
//    ↳ remove parallel edges first.
// Method: insert the edges one at a time. For a new edge (u, v)
//    ↳ build a MAXIMAL FAN around u -
// a sequence of neighbours where each one's free colour is used by
//    ↳ the next - and either rotate the
// fan (when some colour is free at both ends) or flip an alternating
//    ↳ two-colour path and rotate.
// C[u][c] = the neighbour joined to u by colour c (0 = none); X[u] =
//    ↳ smallest free colour at u.
vector<int> vizing(int n, const vector<pair<int, int>>& e) {
    int D = 0;
    vector<int> deg(n, 0);
    for (auto [u, v] : e) D = max({D, ++deg[u], ++deg[v]});
    int K = D + 2; //
    ↳ colours 1..D+1, 0 = uncoloured
    vector<vector<int>> C(n, vector<int>(K + 1, 0)), col(n,
    ↳ vector<int>(n, 0));
    vector<int> X(n, 1);
    auto update = [&](int u) { for (X[u] = 1; C[u][X[u]]; X[u]++); };
    auto color = [&](int u, int v, int c) { //
        ↳ give edge (u,v) colour c
        int p = col[u][v];
        col[u][v] = col[v][u] = c, C[u][c] = v, C[v][c] = u, C[u][p]
        ↳ = C[v][p] = 0;
        if (p) X[u] = X[v] = p; else update(u), update(v);
        return p;
    };
    auto flip = [&](int u, int c1, int c2) { //
        ↳ swap colours c1/c2 at u
        int p = C[u][c1];
        swap(C[u][c1], C[u][c2]);
        if (p) col[u][p] = col[p][u] = c2;
        if (!C[u][c1]) X[u] = c1;
        if (!C[u][c2]) X[u] = c2;
        return p;
    };
    for (int t = 0; t < (int)e.size(); t++) {
        int u = e[t].first, v0 = e[t].second, v = v0, c0 = X[u], c =
        ↳ c0, d = 0, a;
        vector<pair<int, int>> L; // the
        ↳ fan: (neighbour, its free col)
        vector<char> vis(K + 1, 0);
        while (!col[u][v0]) {
            L.push_back({v, d = X[v]});
            if (!C[v][c]) for (a = (int)L.size() - 1; a >= 0; a--) c
            ↳ = color(u, L[a].first, c);
            else if (!C[u][d])
                for (a = (int)L.size() - 1; a >= 0; a--) color(u,
            ↳ L[a].first, L[a].second);
            else if (vis[d]) break; // fan
            ↳ is maximal -> flip a path
            else vis[d] = 1, v = C[u][d];
        }
        if (!col[u][v0]) {
            for (; v; v = flip(v, c, d), swap(c, d)); //
            ↳ alternating c/d path from v
            if (C[u][c0]) {
                for (a = (int)L.size() - 2; a >= 0 && L[a].second !=
            ↳ c; a--);
                for (; a >= 0; a--) color(u, L[a].first, L[a].second);
            } else t--; //
            ↳ retry this edge, now it fits
        }
    }
    vector<int> res(e.size());
    for (int i = 0; i < (int)e.size(); i++) res[i] =
    ↳ col[e[i].first][e[i].second] - 1;
    return res; //
    ↳ res[i] in [0, D]
}
// D IS ENOUGH (class 1) for: bipartite graphs, every graph with a
```

```

    ↪ vertex of degree D that is not
// "overfull", and every planar graph with  $D \geq 7$ .  $D+1$  is forced
    ↪ (class 2) for odd cycles, the
// Petersen graph, and any OVERFULL graph ( $m > D * \text{floor}(n/2)$ ) - a
    ↪ cheap sufficient test.
// EDGE COLOURING = VERTEX COLOURING OF THE LINE GRAPH, so  $\chi'(G) =$ 
    ↪  $\chi(L(G))$ ; that is usually the
// wrong direction to compute, but it is the right way to see a
    ↪ problem stated on the line graph.
// MEMORY is  $O(n^2)$  for the col matrix; at  $n \sim 2000$  that is 16 MB of
    ↪ int - use short if it is tight.

```

3.7.7 WeightedBlossom

MAXIMUM WEIGHT MATCHING IN A GENERAL GRAPH - blossom with DUAL VARIABLES, $O(n^3)$, $O(n^2)$ memory.

```

// 1-INDEXED (vertices 1..n; index 0 is the "none" sentinel and 0
    ↪ weight means "no edge").
// This is the weighted counterpart of 04 - Blossom. It returns the
    ↪ maximum TOTAL WEIGHT over all
// matchings (NOT necessarily a maximum-cardinality one) together
    ↪ with how many edges it used.
// Reach for it when the pairs are not bipartite and the pairing has
    ↪ a value: "pair up the people
// to maximise total compatibility", Chinese postman (08), "minimum
    ↪ cost to make every degree even",
// and any problem where Hungarian would work if only the graph had
    ↪ two sides.
// RECIPES
// MINIMUM weight PERFECT matching: add a big constant B to every
    ↪ edge ( $B \geq \text{sum of } |w|$ ) and
// negate, i.e. addEdge(u, v, B - w); a perfect matching then
    ↪ dominates any smaller one, and the
// answer is  $B * n/2 - \text{result}$ . Verify  $\text{solve().second} * 2 == n$ 
    ↪ before trusting it.
// MAXIMUM CARDINALITY first, weight second: use  $w * (n + 1) + 1$ 
    ↪ per edge.
// The duals lab[] are Edmonds' y-values; a blossom b keeps lab[b] =
    ↪  $2 * (\text{its dual})$ . Every step
// either tightens an edge (slack 0), contracts a blossom, or expands
    ↪ one with zero dual.
struct WeightedBlossom {
    struct E { int u, v; ll w; };
    int n, nx, tk = 0;
    vector<vector<E>> g;
    vector<vector<int>> ff, flower; // ff
    ↪ = which sub-blossom x came from
    vector<int> match, slack, st, pa, S, vis;
    vector<ll> lab;
    deque<int> q;
    WeightedBlossom(int n) : n(n), nx(n), g(2 * n + 1, vector<E>(2 *
    ↪ n + 1)),
        ff(2 * n + 1, vector<int>(2 * n + 1, 0)), flower(2 * n + 1,
    ↪ match(2 * n + 1, 0),
        slack(2 * n + 1, 0), st(2 * n + 1, 0), pa(2 * n + 1, 0), S(2
    ↪ * n + 1, 0),
        vis(2 * n + 1, 0), lab(2 * n + 1, 0) {
        for (int u = 1; u <= 2 * n; u++) for (int v = 1; v <= 2 * n;
    ↪ v++) g[u][v] = {u, v, 0};
    }
    void addEdge(int u, int v, ll w) { g[u][v].w = g[v][u].w =
    ↪ max(g[u][v].w, w); }
    ll d(const E& e) { return lab[e.u] + lab[e.v] - g[e.u][e.v].w *
    ↪ 2; } // slack
    void upSlack(int u, int x) { if (!slack[x] || d(g[u][x]) <
    ↪ d(g[slack[x]][x])) slack[x] = u; }
    void setSlack(int x) {
        slack[x] = 0;
        for (int u = 1; u <= n; u++)
            if (g[u][x].w > 0 && st[u] != x && S[st[u]] == 0)
    ↪ upSlack(u, x);
    }
    // res[i] in [0, D]
}
// D IS ENOUGH (class 1) for: bipartite graphs, every graph with a
    ↪ vertex of degree D that is not
// "overfull", and every planar graph with  $D \geq 7$ .  $D+1$  is forced
    ↪ (class 2) for odd cycles, the
// Petersen graph, and any OVERFULL graph ( $m > D * \text{floor}(n/2)$ ) - a
    ↪ cheap sufficient test.
// EDGE COLOURING = VERTEX COLOURING OF THE LINE GRAPH, so  $\chi'(G) =$ 
    ↪  $\chi(L(G))$ ; that is usually the
// wrong direction to compute, but it is the right way to see a
    ↪ problem stated on the line graph.
// MEMORY is  $O(n^2)$  for the col matrix; at  $n \sim 2000$  that is 16 MB of
    ↪ int - use short if it is tight.

```

3.7.8 WeightedBlossom

MAXIMUM WEIGHT MATCHING IN A GENERAL GRAPH - blossom with DUAL VARIABLES, $O(n^3)$, $O(n^2)$ memory.

```

// 1-INDEXED (vertices 1..n; index 0 is the "none" sentinel and 0
    ↪ weight means "no edge").
// This is the weighted counterpart of 04 - Blossom. It returns the
    ↪ maximum TOTAL WEIGHT over all
// matchings (NOT necessarily a maximum-cardinality one) together
    ↪ with how many edges it used.
// Reach for it when the pairs are not bipartite and the pairing has
    ↪ a value: "pair up the people
// to maximise total compatibility", Chinese postman (08), "minimum
    ↪ cost to make every degree even",
// and any problem where Hungarian would work if only the graph had
    ↪ two sides.
// RECIPES
// MINIMUM weight PERFECT matching: add a big constant B to every
    ↪ edge ( $B \geq \text{sum of } |w|$ ) and
// negate, i.e. addEdge(u, v, B - w); a perfect matching then
    ↪ dominates any smaller one, and the
// answer is  $B * n/2 - \text{result}$ . Verify  $\text{solve().second} * 2 == n$ 
    ↪ before trusting it.
// MAXIMUM CARDINALITY first, weight second: use  $w * (n + 1) + 1$ 
    ↪ per edge.
// The duals lab[] are Edmonds' y-values; a blossom b keeps lab[b] =
    ↪  $2 * (\text{its dual})$ . Every step
// either tightens an edge (slack 0), contracts a blossom, or expands
    ↪ one with zero dual.
struct WeightedBlossom {
    struct E { int u, v; ll w; };
    int n, nx, tk = 0;
    vector<vector<E>> g;
    vector<vector<int>> ff, flower; // ff
    ↪ = which sub-blossom x came from
    vector<int> match, slack, st, pa, S, vis;
    vector<ll> lab;
    deque<int> q;
    WeightedBlossom(int n) : n(n), nx(n), g(2 * n + 1, vector<E>(2 *
    ↪ n + 1)),
        ff(2 * n + 1, vector<int>(2 * n + 1, 0)), flower(2 * n + 1,
    ↪ match(2 * n + 1, 0),
        slack(2 * n + 1, 0), st(2 * n + 1, 0), pa(2 * n + 1, 0), S(2
    ↪ * n + 1, 0),
        vis(2 * n + 1, 0), lab(2 * n + 1, 0) {
        for (int u = 1; u <= 2 * n; u++) for (int v = 1; v <= 2 * n;
    ↪ v++) g[u][v] = {u, v, 0};
    }
    void addEdge(int u, int v, ll w) { g[u][v].w = g[v][u].w =
    ↪ max(g[u][v].w, w); }
    ll d(const E& e) { return lab[e.u] + lab[e.v] - g[e.u][e.v].w *
    ↪ 2; } // slack
    void upSlack(int u, int x) { if (!slack[x] || d(g[u][x]) <
    ↪ d(g[slack[x]][x])) slack[x] = u; }
    void setSlack(int x) {
        slack[x] = 0;
        for (int u = 1; u <= n; u++)
            if (g[u][x].w > 0 && st[u] != x && S[st[u]] == 0)
    ↪ upSlack(u, x);
    }
    void qPush(int x) { if (x <= n) q.push_back(x); else for (int y :
    ↪ flower[x]) qPush(y); }
    void setSt(int x, int b) { st[x] = b; if (x > n) for (int y :
    ↪ flower[x]) setSt(y, b); }
    int getPr(int b, int xr) {
        int pr = find(all(flower[b]), xr) - flower[b].begin();
        if (pr & 1) {
            reverse(flower[b].begin() + 1, flower[b].end());
            return (int)flower[b].size() - pr;
        }
        return pr;
    }
    void setMatch(int u, int v) {
        match[u] = g[u][v].v;
        if (u <= n) return;
        E e = g[u][v];
        int xr = ff[u][e.u], pr = getPr(u, xr);
        for (int i = 0; i < pr; i++) setMatch(flower[u][i],
    ↪ flower[u][i ^ 1]);
        setMatch(xr, v);
        rotate(flower[u].begin(), flower[u].begin() + pr,
    ↪ flower[u].end());
    }
    void augment(int u, int v) {
        int nv = st[match[u]];
        setMatch(u, v);
        if (!nv) return;
        setMatch(nv, st[pa[nv]]), augment(st[pa[nv]], nv);
    }
}

```

```

}
int getLca(int u, int v) {
    for (++tk; u || v; swap(u, v)) {
        if (!u) continue;
        if (vis[u] == tk) return u;
        vis[u] = tk, u = st[match[u]];
        if (u) u = st[pa[u]];
    }
    return 0;
}

void addBlossom(int u, int lca, int v) { //
    ↪ contract the odd cycle u..lca..v
    int b = n + 1;
    while (b <= nx && st[b]) ++b;
    if (b > nx) ++nx;
    lab[b] = 0, S[b] = 0, match[b] = match[lca];
    flower[b].assign(1, lca);
    for (int x = u, y; x != lca; x = st[pa[y]])
        flower[b].push_back(x), flower[b].push_back(y =
    ↪ st[match[x]]), qPush(y);
    reverse(flower[b].begin() + 1, flower[b].end());
    for (int x = v, y; x != lca; x = st[pa[y]])
        flower[b].push_back(x), flower[b].push_back(y =
    ↪ st[match[x]]), qPush(y);
    setSt(b, b);
    for (int x = 1; x <= nx; x++) g[b][x].w = g[x][b].w = 0;
    for (int x = 1; x <= n; x++) ff[b][x] = 0;
    for (int xs : flower[b]) {
        for (int x = 1; x <= nx; x++)
            if (g[b][x].w == 0 || d(g[xs][x]) < d(g[b][x]))
                g[b][x] = g[xs][x], g[x][b] = g[x][xs];
        for (int x = 1; x <= n; x++) if (ff[xs][x]) ff[b][x] = xs;
    }
    setSlack(b);
}

void expandBlossom(int b) { //
    ↪ only if S[b] == 1, lab[b] == 0
    for (int x : flower[b]) setSt(x, x);
    int xr = ff[b][g[b][pa[b]].u], pr = getPr(b, xr);
    for (int i = 0; i < pr; i += 2) {
        int xs = flower[b][i], xns = flower[b][i + 1];
        pa[xs] = g[xns][xs].u, S[xs] = 1, S[xns] = 0;
        slack[xs] = 0, setSlack(xns), qPush(xns);
    }
    S[xr] = 1, pa[xr] = pa[b];
    for (int i = pr + 1; i < (int)flower[b].size(); i++)
    ↪ S[flower[b][i]] = -1,
        setSlack(flower[b][i]);
    st[b] = 0;
}

bool onFound(const E& e) {
    int u = st[e.u], v = st[e.v];
    if (S[v] == -1) {
        pa[v] = e.u, S[v] = 1;
        int nu = st[match[v]];
        slack[v] = slack[nu] = 0, S[nu] = 0, qPush(nu);
    } else if (S[v] == 0) {
        int lca = getLca(u, v);
        if (!lca) return augment(u, v), augment(v, u), true;
        addBlossom(u, lca, v);
    }
    return false;
}

bool matching() { // one
    ↪ augmentation, or false
    fill(S.begin(), S.begin() + nx + 1, -1);
    fill(slack.begin(), slack.begin() + nx + 1, 0);
    q.clear();
    for (int x = 1; x <= nx; x++) if (st[x] == x && !match[x])
    ↪ pa[x] = 0, S[x] = 0, qPush(x);
    if (q.empty()) return false;
    for (;;) {
        while (!q.empty()) {
            int u = q.front(); q.pop_front();
            if (S[st[u]] == 1) continue;
            for (int v = 1; v <= n; v++) if (g[u][v].w > 0 &&
    ↪ st[u] != st[v]) {
                if (d(g[u][v]) == 0) { if (onFound(g[u][v]))
    ↪ return true; }
                else upSlack(u, st[v]);
            }
        }
        ll dt = llinf; //
    ↪ dual update step
        for (int b = n + 1; b <= nx; b++) if (st[b] == b && S[b]
    ↪ == 1) dt = min(dt, lab[b] / 2);
        for (int x = 1; x <= nx; x++) if (st[x] == x && slack[x])

```

```

    ↪ {
        if (S[x] == -1) dt = min(dt, d(g[slack[x]][x]));
        else if (S[x] == 0) dt = min(dt, d(g[slack[x]][x]) /
    ↪ 2);
    }
    for (int u = 1; u <= n; u++) {
        if (S[st[u]] == 0) { if (lab[u] <= dt) return false;
    ↪ lab[u] -= dt; }
        else if (S[st[u]] == 1) lab[u] += dt;
    }
    for (int b = n + 1; b <= nx; b++) if (st[b] == b) {
        if (S[st[b]] == 0) lab[b] += dt * 2;
        else if (S[st[b]] == 1) lab[b] -= dt * 2;
    }
    q.clear();
    for (int x = 1; x <= nx; x++)
        if (st[x] == x && slack[x] && st[slack[x]] != x &&
    ↪ d(g[slack[x]][x]) == 0)
            if (onFound(g[slack[x]][x])) return true;
    for (int b = n + 1; b <= nx; b++)
        if (st[b] == b && S[b] == 1 && lab[b] == 0)
    ↪ expandBlossom(b);
    }
}

pair<ll, int> solve() { //
    ↪ {total weight, #matched edges}
    fill(match.begin(), match.begin() + n + 1, 0);
    nx = n;
    ll ans = 0, wmax = 0;
    int cnt = 0;
    for (int u = 0; u <= n; u++) st[u] = u, flower[u].clear();
    for (int u = 1; u <= n; u++) for (int v = 1; v <= n; v++)
        ff[u][v] = (u == v ? 0 : 0), wmax = max(wmax, g[u][v].w);
    for (int u = 1; u <= n; u++) lab[u] = wmax;
    while (matching()) ++cnt;
    for (int u = 1; u <= n; u++) if (match[u] && match[u] < u)
    ↪ ans += g[u][match[u]].w;
    return {ans, cnt}; //
    ↪ match[u] = partner of u, or 0
}

};
// WEIGHTS MUST BE NON-NEGATIVE here (an edge of weight 0 is
    ↪ indistinguishable from a non-edge);
// shift them if needed. Everything is integral - with real weights
    ↪ use long double and an eps.
// FASTER ALTERNATIVES: if n <= 20, a bitmask DP over matched
    ↪ vertices is 5 lines and beats this.
// If the graph is bipartite, use Hungarian (O3) - O(n^3) too but
    ↪ with a far smaller constant.
// TUTTE MATRIX: for the UNWEIGHTED decision "is there a perfect
    ↪ matching", a random skew-symmetric
// matrix over a large field has full rank with high probability iff
    ↪ one exists - O(n^3) and 10
// lines, and rank/2 is the maximum matching size.

```

3.7.9 ChinesePostman

CHINESE POSTMAN (route inspection) - the cheapest CLOSED WALK that uses EVERY edge at least

```
// Needs: WeightedBlossom (07 - WeightedBlossom.cpp).
// once, on an undirected connected weighted graph. 0-indexed in,
//    ↳  $O(n^3)$  via general matching.
// THEORY. A closed walk using every edge exists with no repeats iff
//    ↳ every degree is even (Euler).
// Otherwise you must DUPLICATE some edges, and the duplicated set
//    ↳ must fix exactly the odd-degree
// vertices - i.e. it is a union of paths pairing them up. So: take
//    ↳ all-pairs shortest paths, build
// the complete graph on the ODD-degree vertices (there is an even
//    ↳ number of them, by handshake),
// and find a MINIMUM WEIGHT PERFECT MATCHING there. Answer = total
//    ↳ edge weight + matching weight.
// Reach for it for "walk every street and come back, minimum
//    ↳ distance", "traverse every corridor",
// and "make every degree even at minimum cost" (that last one IS the
//    ↳ matching part alone).
ll chinesePostman(int n, const vector<array<ll, 3>>& e) {    // e =
//    ↳ {u, v, w}, undirected
    vector<vector<ll>> d(n, vector<ll>(n, llinf));
    vector<int> deg(n, 0);
    ll total = 0;
    for (int i = 0; i < n; i++) d[i][i] = 0;
    for (auto [u, v, w] : e) {
        d[u][v] = d[v][u] = min(d[u][v], w), total += w, deg[u]++,
    ↳ deg[v]++;
    }
    for (int k = 0; k < n; k++) for (int i = 0; i < n; i++) for (int
    ↳ j = 0; j < n; j++)
        if (d[i][k] < llinf && d[k][j] < llinf) d[i][j] =
    ↳ min(d[i][j], d[i][k] + d[k][j]);
    vector<int> odd;
    for (int i = 0; i < n; i++) if (deg[i] & 1) odd.push_back(i);
    int k = odd.size();
    if (!k) return total;    //
    ↳ already Eulerian
    ll mx = 0;    //
    ↳ shift so that MIN becomes MAX and
    for (int i = 0; i < k; i++) for (int j = 0; j < k; j++)    // a
    ↳ PERFECT matching always wins
        if (d[odd[i]][odd[j]] < llinf) mx = max(mx,
    ↳ d[odd[i]][odd[j]]);
    ll B = mx * k + 1;
    WeightedBlossom M(k);
    for (int i = 0; i < k; i++) for (int j = i + 1; j < k; j++)
        if (d[odd[i]][odd[j]] < llinf) M.addEdge(i + 1, j + 1, B -
    ↳ d[odd[i]][odd[j]]);
    auto [val, cnt] = M.solve();
    if (cnt * 2 != k) return -1;    // odd
    ↳ vertices cannot be paired
    return total + (ll)cnt * B - val;
}
// RECOVERING THE ROUTE: duplicate every edge on the shortest path
//    ↳ between each matched pair, then
// run an Eulerian circuit (05 - Eulerian Path) on the resulting
//    ↳ even-degree multigraph.
// DIRECTED CHINESE POSTMAN is a MIN COST FLOW, not a matching: give
//    ↳ each vertex a supply of
// (outdeg - indeg), send flow from the surplus vertices to the
//    ↳ deficit ones along the original
// arcs with cost = weight and infinite capacity, and add that cost
//    ↳ to the total weight.
// MIXED (some edges directed, some not) is NP-hard.
// RURAL POSTMAN (only a SUBSET of edges must be covered) is NP-hard
//    ↳ as well - it contains TSP.
// OPEN version ("start anywhere, end anywhere"): leave two odd
//    ↳ vertices unmatched, i.e. run a
// maximum-weight (not perfect) matching, which is what solve() gives
//    ↳ you without the count check.
```

3.8 Spanning Trees

3.8.1 MST

MINIMUM SPANNING TREE. Kruskal $O(m \log m)$ - use on sparse graphs / when you also want the

```
// Needs: DSU (02 - Data Structures/05 - DSU/01).
// KRUSKAL RECONSTRUCTION TREE. Prim  $O(n^2)$  - use on dense or
//    ↳ IMPLICIT complete graphs.
struct Edge { int u, v; ll w; bool operator<(const Edge& o) const {
//    ↳ return w < o.w; } };

ll kruskal(int n, vector<Edge> e, vector<Edge>* used = nullptr) {
    sort(all(e));
    DSU d(n);
    ll tot = 0;
    for (auto& x : e) if (d.join(x.u, x.v)) { tot += x.w; if (used)
    ↳ used->push_back(x); }
    return tot;    // check d.size(0)
//    ↳ == n for connectivity
}
// Dense / implicit graphs: cost(i,j) given by a function.  $O(n^2)$ , no
//    ↳ edge list needed.
template <class F>
ll prim(int n, F cost) {
    vector<ll> best(n, llinf);
    vector<bool> in(n, false);
    best[0] = 0;
    ll tot = 0;
    for (int it = 0; it < n; it++) {
        int u = -1;
        for (int i = 0; i < n; i++) if (!in[i] && (u < 0 || best[i] <
    ↳ best[u])) u = i;
        in[u] = true, tot += best[u];
        for (int v = 0; v < n; v++) if (!in[v]) best[v] =
    ↳ min(best[v], cost(u, v));
    }
    return tot;
}
// KRUSKAL RECONSTRUCTION TREE: binary tree with  $2n-1$  nodes; internal
//    ↳ node = merging edge weight.
// Then min possible MAX edge on a path  $u \rightarrow v = \text{val}[\text{lca}(u,v)]$ , and
//    ↳ "all vertices reachable using
// edges of weight  $\leq w$ " is exactly a subtree. Turns "binary search
//    ↳ the threshold + DSU" into  $O(1)$ .
struct KRT {
    int n, cnt;
    vector<ll> val;
    vector<array<int, 2>>& ch;
    KRT(int n, vector<Edge> e) : n(n), cnt(n), val(2 * n, 0), ch(2 *
    ↳ n, {-1, -1}) {
        sort(all(e));
        DSU d(2 * n);
        vector<int> top(2 * n);
        iota(all(top), 0);
        for (auto& x : e) {
            int a = top[d.find(x.u)], b = top[d.find(x.v)];
            if (a == b) continue;
            val[cnt] = x.w, ch[cnt] = {a, b};
            d.join(x.u, x.v);
            top[d.find(x.u)] = cnt++;
        }
    }
//    ↳ connected; then build LCA on it
};
// SECOND-BEST MST: build the MST, then for each non-tree edge
//    ↳ (u,v,w) the answer candidate is
// mst - maxEdgeOnPath(u,v) + w. Get maxEdgeOnPath with binary
//    ↳ lifting or the KRT.
```


3.8.2 DirectedMSTandExtras

DIRECTED MST (Chu-Liu / Edmonds) - minimum spanning ARBORESCENCE rooted at r , $O(V^2E)$.

```
// Needs: struct Edge {int u,v; ll w;} and DSU (both from 01 -
    ↳ MST.cpp).
// Every non-root vertex takes its cheapest incoming edge; if that
    ↳ creates a cycle, contract it,
// subtract the chosen weights, and repeat. Returns -1 if some vertex
    ↳ is unreachable from  $r$ .
ll directedMST(int n, int r, vector<Edge> e) {      // Edge = {u,
    ↳ v, w} meaning u -> v
    ll total = 0;
    vector<int> id(n), par(n);
    while (true) {
        vector<ll> best(n, llinf);
        vector<int> from(n, -1);
        for (auto& x : e) if (x.u != x.v && x.w < best[x.v])
            ↳ best[x.v] = x.w, from[x.v] = x.u;
        best[r] = 0;
        for (int i = 0; i < n; i++) if (i != r && from[i] < 0) return
            ↳ -1; // unreachable
        int cnt = 0;
        fill(all(id), -1), fill(all(par), -1);
        for (int i = 0; i < n; i++) {
            if (i != r) total += best[i];
            int v = i;
            while (v != r && par[v] == -1 && id[v] == -1) par[v] = i,
                ↳ v = from[v];
            if (v != r && id[v] == -1 && par[v] == i) { // found a
                ↳ new cycle
                for (int u = from[v]; u != v; u = from[u]) id[u] =
                    ↳ cnt;
                id[v] = cnt++;
            }
            if (!cnt) break; // no cycle
            ↳ -> done
            for (int i = 0; i < n; i++) if (id[i] == -1) id[i] = cnt++;
            for (auto& x : e) {
                int u = x.u, v = x.v;
                x.u = id[u], x.v = id[v];
                if (x.u != x.v) x.w -= best[v]; // reduced
            }
            ↳ cost inside the contraction
            n = cnt, r = id[r];
        }
        return total;
    }
}

// SECOND-BEST MST,  $O(E \log V + E * \log V)$ . The second-best tree
    ↳ differs from the MST by exactly
// ONE edge swap. For each non-tree edge (u,v,w) the candidate is mst
    ↳ - maxEdgeOnPath(u,v) + w.
// You must track BOTH the maximum and the strict second maximum on
    ↳ the path, because if the
// maximum equals w the swap is not a real change - use the second
    ↳ maximum instead.
// With binary lifting: up[k][v], mx1[k][v], mx2[k][v] merged
    ↳ pairwise while lifting.
// (The Kruskal Reconstruction Tree in 01 - MST.cpp gives
    ↳ maxEdgeOnPath directly as val[lca].)

// BORUVKA - MST in  $O(E \log V)$  by phases; each phase every component
    ↳ picks its cheapest outgoing
// edge and they are all merged at once. Use it when the edge set is
    ↳ IMPLICIT:
// XOR-MST : cheapest cross edge per component via a binary
    ↳ trie
// Manhattan MST: cheapest cross edge per octant (see
    ↳ Transforms.cpp)
ll boruvka(int n, const vector<Edge>& e) {
    DSU d(n);
    ll total = 0;
    int comps = n;
    while (comps > 1) {
        vector<int> best(n, -1);
        for (int i = 0; i < (int)e.size(); i++) {
            int a = d.find(e[i].u), b = d.find(e[i].v);
            if (a == b) continue;
            if (best[a] < 0 || e[i].w < e[best[a]].w) best[a] = i;
            if (best[b] < 0 || e[i].w < e[best[b]].w) best[b] = i;
        }
        bool any = false;
        for (int i = 0; i < n; i++) if (best[i] >= 0) {
            auto& x = e[best[i]];
            if (d.join(x.u, x.v)) total += x.w, comps--, any = true;
        }
    }
}
```

```
        if (!any) break;
        ↳ disconnected
    }
    return comps == 1 ? total : -1;
}
```


3.8.3 SecondBestMST

SECOND-BEST SPANNING TREE - cheapest spanning tree whose weight is STRICTLY greater than the MST.

```
// Needs: Edge, kruskal (01 - MST.cpp).
// It differs from the MST in exactly one edge: add a non-tree edge
//   (u,v,w), drop the heaviest edge
// on the tree path u->v whose weight is < w. By the cycle property
//   every edge on that path has
// weight <= w, so "heaviest edge < w" is either the path maximum (if
//   it is < w) or the second
// LARGEST DISTINCT value on the path. Binary lifting carries both.
//   O(m log n).
ll secondBestMST(int n, const vector<Edge>& e) {
    vector<Edge> used;
    ll mst = kruskal(n, e, &used);
    if ((int)used.size() != n - 1) return -1; // disconnected
    vector<vector<pair<int, ll>>> g(n);
    for (auto& x : used) g[x.u].push_back({x.v, x.w});
    for (g[x.v].push_back({x.u, x.w});
    typedef array<ll, 2> T; // {largest, 2nd largest DISTINCT}
    const T NONE = {LLONG_MIN, LLONG_MIN};
    auto join = [&](T a, T b) {
        array<ll, 4> v = {a[0], a[1], b[0], b[1]};
        sort(v.rbegin(), v.rend());
        T r = {v[0], LLONG_MIN};
        for (ll x : v) if (x < v[0]) { r[1] = x; break; }
        return r;
    };
    int L = 1;
    while ((1 << L) < n) L++;
    vector<vector<int>> up(L + 1, vector<int>(n, 0));
    vector<vector<T>> mx(L + 1, vector<T>(n, NONE));
    vector<int> dep(n, 0), st{0};
    vector<char> vis(n, 0);
    vis[0] = 1;
    while (!st.empty()) { // iterative DFS from 0
        int u = st.back(); st.pop_back();
        for (auto [v, w] : g[u]) if (!vis[v])
            vis[v] = 1, dep[v] = dep[u] + 1, up[0][v] = u, mx[0][v] =
            {w, LLONG_MIN}, st.push_back(v);
        for (int k = 1; k <= L; k++)
            for (int v = 0; v < n; v++) {
                int m = up[k - 1][v];
                up[k][v] = up[k - 1][m], mx[k][v] = join(mx[k - 1][v],
                {mx[k - 1][m]});
            }
        auto path = [&](int u, int v) { // top-2 distinct weights on u->v
            T r = NONE;
            if (dep[u] < dep[v]) swap(u, v);
            for (int k = L; k >= 0; k--) if (dep[up[k][u]] >= dep[v]) r =
            join(r, mx[k][u]), u = up[k][u];
            if (u == v) return r;
            for (int k = L; k >= 0; k--) if (up[k][u] != up[k][v])
                r = join(join(r, mx[k][u]), mx[k][v]), u = up[k][u], v =
            up[k][v];
            return join(join(r, mx[0][u]), mx[0][v]);
        };
        ll best = LLONG_MAX;
        for (auto& x : e) { // tree edges give p = {w, MIN} and
            T p = path(x.u, x.v); // fall through to the MIN branch
            ll rem = p[0] < x.w ? p[0] : p[1];
            if (rem != LLONG_MIN) best = min(best, mst - rem + x.w);
        }
        return best == LLONG_MAX ? -1 : best; // -1: no second tree exists
    }
}
// K-TH BEST SPANNING TREE: repeat the swap argument with a priority
//   queue over (tree, forced-in /
// forced-out edge sets) - Gabow's algorithm, O(k m log n). Rare; the
//   second-best case is the one
// that actually shows up.
// MINIMUM BOTTLENECK SPANNING TREE = any MST (minimising the maximum
//   edge is implied by Kruskal).
// MST OF A GRAPH PLUS ONE NEW EDGE: add it, find the cycle, drop the
//   cycle maximum. O(n).
// MINIMUM SPANNING TREE WITH DEGREE CONSTRAINT at one vertex r:
//   build the MST of G - r, then add
// the cheapest edge from r into each component, then improve by
//   swaps while degree < k.
```

3.8.4 MinDiameterSpanningTree

MINIMUM DIAMETER SPANNING TREE - the spanning tree whose longest path is as SHORT as possible.

```
// O(V^3) (Floyd-Warshall plus one sweep per edge). Undirected,
//   non-negative weights, 0-indexed.
// This is NOT the MST: the MST minimises total weight and can have a
//   terrible diameter.
// KEY FACT. An MDST is always a SHORTEST PATH TREE from the ABSOLUTE
//   1-CENTRE - a point that may
// sit in the MIDDLE OF AN EDGE, not only at a vertex. So: for every
//   edge (u, v) of weight w, slide
// a point c at distance x along it and minimise its eccentricity
//   ecc(x) = max over i of
// min(x + d[u][i], w - x + d[v][i]). Each i contributes a "tent"
//   function of x, the upper envelope
// is piecewise linear, and it is enough to test the breakpoints -
//   which is the sweep below, after
// sorting the vertices by d[u][.] descending. The answer is the
//   minimum eccentricity, doubled.
// Reach for it for "build a network minimising the worst-case delay
//   between any two nodes",
// "connect all cities so the farthest pair is as close as possible",
//   and broadcast-tree problems.
// ALL WEIGHTS ARE DOUBLED INTERNALLY so the centre's position stays
//   an integer; the returned
// diameter is already in the ORIGINAL units (it is twice the
//   eccentricity, which cancels out).
struct MDST {
    ll diameter = llinf;
    int edge = -1; // the
    // edge holding the centre
    ll offset = 0; // 2 *
    // distance from e[edge][0]
    vector<ll> distFromCentre; // 2 *
    // dist(centre, i), after solve
    ll solve(int n, const vector<array<ll, 3>>& e) { // e =
    // {u, v, w}
        if (n == 1) return diameter = 0;
        vector<vector<ll>> d(n, vector<ll>(n, llinf));
        for (int i = 0; i < n; i++) d[i][i] = 0;
        for (auto [u, v, w] : e) d[u][v] = min(d[u][v], 2 * w),
        d[v][u] = min(d[v][u], 2 * w);
        for (int k = 0; k < n; k++) for (int i = 0; i < n; i++) for
        (int j = 0; j < n; j++)
            if (d[i][k] < llinf && d[k][j] < llinf) d[i][j] =
            min(d[i][j], d[i][k] + d[k][j]);
        vector<int> id(n);
        iota(all(id), 0);
        for (int u = 0; u < n; u++) {
            sort(all(id), [&](int a, int b) { return d[u][a] >
            d[u][b]; });
            for (int k = 0; k < (int)e.size(); k++) {
                int a = e[k][0], b = e[k][1];
                if (a != u && b != u) continue;
                int v = a ^ b ^ u;
                ll W = 2 * e[k][2], opt = 0;
                int last = id[0];
                ll val = d[u][last]; //
                // centre at u itself
                for (int t = 1; t < n; t++) { //
                // walk the envelope breakpoints
                    int i = id[t];
                    if (d[v][i] <= d[v][last]) continue;
                    ll cx = (W - d[u][i] + d[v][last]) / 2, cy = (W +
                    d[u][i] + d[v][last]) / 2;
                    if (val > cy) val = cy, opt = cx;
                    last = i;
                }
                if (val > d[v][last]) val = d[v][last], opt = W; //
                // centre at v itself
                if (val < diameter) diameter = val, edge = k, offset
                = opt;
            }
        }
        distFromCentre.assign(n, llinf); //
        // Dijkstra from the virtual centre
        int a = e[edge][0], b = e[edge][1];
        priority_queue<pair<ll, int>, vector<pair<ll, int>>,
        greater<>> pq;
        distFromCentre[a] = offset, distFromCentre[b] = 2 *
        e[edge][2] - offset;
        pq.push({distFromCentre[a], a}), pq.push({distFromCentre[b],
        b});
        vector<vector<pair<int, ll>>> g(n);
        for (auto [x, y, w] : e) g[x].push_back({(int)y, 2 * w});
        g[y].push_back({(int)x, 2 * w});
        while (!pq.empty()) {
```

```

    auto [dd, x] = pq.top(); pq.pop();
    if (dd > distFromCentre[x]) continue;
    for (auto [y, w] : g[x]) if (dd + w < distFromCentre[y])
        distFromCentre[y] = dd + w,
        pq.push({distFromCentre[y], y});
    }
    return diameter;
}
};
// RECOVERING THE TREE: take the shortest path tree of distFromCentre
//   (each vertex keeps one
//   incoming edge that is tight), plus the two halves of the centre
//   edge. If offset is 0 or 2*w the
//   centre is a vertex and the tree is just that vertex's shortest
//   path tree.
// UNWEIGHTED graphs: the answer is the shortest path tree from the
//   middle of the graph's own
//   diameter path, and BFS from every vertex (O(VE)) is enough - do
//   not bother with this file.
// MINIMUM RADIUS spanning tree is the same computation without the
//   final doubling; the absolute
//   1-centre is the object both problems share.
// MINIMUM DIAMETER with a bound on the total weight, or "diameter <=
//   k with fewest edges", is
// NP-hard - the polynomial case is exactly the unconstrained one
//   above.

```

3.9 Exact Exponential

3.9.1 CliqueColouringMIS

EXACT ALGORITHMS FOR NP-HARD GRAPH PROBLEMS ON SMALL n . The constraint IS the hint:

```

// n <= 20-24 -> plain subset DP,  $O(2^n * n)$ 
// n <= 40-50 -> meet in the middle, or Bron-Kerbosch (fast in
//   practice on sparse graphs)
// n <= 100+ -> only if the graph is bipartite/perfect/a tree -
//   then it is polynomial (Konig)
// adj[] is an ADJACENCY BITMASK: bit j of adj[i] is set iff i-j is
//   an edge. No self loops.

// --- MAXIMUM CLIQUE, Bron-Kerbosch with PIVOTING. Worst case
//    $O(3^{n/3})$ , very fast in practice. ---
// R = clique built so far, P = candidates, X = already-explored
//   vertices (avoids duplicates).
// The pivot u in P|X is chosen to maximise |P & adj[u]|; only P \
//   adj[u] needs branching, because
// any maximal clique either omits u or contains a non-neighbour of u.
int bestClique;
unsigned bestSet;
void bk(unsigned R, unsigned P, unsigned X, const vector<unsigned>&
    adj) {
    if (!P && !X) {
        if (__builtin_popcount(R) > bestClique) bestClique =
        __builtin_popcount(R), bestSet = R;
        return;
    }
    int u = __builtin_ctz(P | X), best = -1;
    for (unsigned t = P | X; t; t &= t - 1) {
        int v = __builtin_ctz(t), c = __builtin_popcount(P & adj[v]);
        if (c > best) best = c, u = v;
    }
    for (unsigned cand = P & ~adj[u]; cand; cand &= cand - 1) {
        int v = __builtin_ctz(cand);
        bk(R | 1u << v, P & adj[v], X & adj[v], adj);
        P &= ~(1u << v), X |= 1u << v;
    }
}
int maxClique(const vector<unsigned>& adj) { // n
    // <= 32
    bestClique = 0, bestSet = 0;
    bk(0, (1u << adj.size()) - 1, 0, adj);
    return bestClique; // the
    // clique itself is in bestSet
}
// MAXIMUM INDEPENDENT SET = maximum clique of the COMPLEMENT graph.
//   Just flip adj and mask off i.
// MINIMUM VERTEX COVER = n - MIS. On BIPARTITE graphs both are
//   polynomial: Konig says
// min vertex cover = maximum matching, and MIS = n - matching (see
//   07 - Matching).

// --- CHROMATIC NUMBER,  $O(2^n * n)$  by inclusion-exclusion
//   (Bjorklund-Husfeldt-Koivisto). ---
// Let ind[S] = 1 if S is independent. The number of ways to cover V
//   with k ORDERED independent
//   sets is  $\sum_S (-1)^{(n-|S|)} * I(S)^k$ , where  $I(S)$  = #independent
//   subsets of S. The chromatic
//   number is the smallest k making that non-zero. Compute I over
//   subsets with one SOS pass.
int chromatic(const vector<unsigned>& adj) { // n
    // <= 20 with this modulus
    int n = adj.size(), N = 1 << n;
    if (!n) return 0;
    const ll M = (1LL << 61) - 1; // any
    // large prime works
    vector<ll> I(N), pw(N, 1);
    I[0] = 1;
    for (int S = 1; S < N; S++) { //
        // I[S] = #independent subsets of S
        int v = __builtin_ctz(S), rest = S ^ 1 << v; // (v
        // in it, or not)
        I[S] = (I[rest] + I[rest & ~adj[v]]) % M;
    }
    for (int k = 1; k <= n; k++) {
        ll tot = 0;
        for (int S = 0; S < N; S++) {
            pw[S] = (__int128)pw[S] * I[S] % M; //
        }
        // I(S)^k
        ll t = (n - __builtin_popcount(S)) & 1 ? M - pw[S] :
        pw[S];
        tot = (tot + t) % M;
    }
    if (tot % M) return k;
    return n;
}

```

```
// The count is done MODULO a prime, so a zero could in principle be
↳ a false negative; with a
// 61-bit modulus that never happens on contest inputs. Check
↳ bipartiteness first -  $k \leq 2$  is  $O(n+m)$ .

// --- MINIMUM STEINER TREE (connect a set of  $k$  terminals in a
↳ weighted graph),  $O(3^k n + 2^k n^2)$ .
//  $dp[S][v]$  = cheapest tree containing terminal set  $S$  and touching
↳ vertex  $v$ .
// merge:  $dp[S][v] = \min$  over submasks  $T$  of  $dp[T][v] + dp[S \setminus T][v]$ 
// grow:  $dp[S][v] = \min$  over edges  $(u,v)$  of  $dp[S][u] + w$ 
↳ (run Dijkstra per  $S$ )
ll steinerTree(int n, const vector<vector<pair<int, ll>>> &g, const
↳ vector<int>& term) {
    int k = term.size();
    if (!k) return 0;
    vector<vector<ll>> dp(1 << k, vector<ll>(n, llinf));
    for (int i = 0; i < k; i++) dp[1 << i][term[i]] = 0;
    for (int S = 1; S < 1 << k; S++) {
        for (int T = (S - 1) & S; T; T = (T - 1) & S)
            for (int v = 0; v < n; v++)
                if (dp[T][v] < llinf && dp[S ^ T][v] < llinf)
                    dp[S][v] = min(dp[S][v], dp[T][v] + dp[S ^ T][v]);
        priority_queue<pair<ll, int>, vector<pair<ll, int>>,
↳ greater<>> pq;
        for (int v = 0; v < n; v++) if (dp[S][v] < llinf)
↳ pq.push({dp[S][v], v});
        while (!pq.empty()) {
            //
            // Dijkstra to relax the tree outward
            auto [d, u] = pq.top(); pq.pop();
            if (d > dp[S][u]) continue;
            for (auto [v, w] : g[u]) if (dp[S][u] + w < dp[S][v])
                dp[S][v] = dp[S][u] + w, pq.push({dp[S][v], v});
        }
    }
    return *min_element(all(dp[(1 << k) - 1]));
}

// STEINER TREE IN A GRID / with  $k$  up to  $\sim 10$  is the usual contest
↳ size. If  $k = 2$  it is a shortest
// path; if  $k = n$  it is the MST; the DP interpolates between them.
// RELATED EXACT-EXPONENTIAL TOOLS:
// * HAMILTONIAN PATH / TSP:  $dp[mask][v]$ ,  $O(2^n n^2)$ . See 08 -
↳ DP/02 - BitmaskDP.
// * SET COVER:  $dp$  over covered elements,  $O(2^n * \#sets)$ , or IE
↳ like the chromatic number.
// * GRAPH COLOURING with  $k$  fixed: try to 3-colour by 2-SAT? No -
↳ 3-colouring is NP-hard; use
// the IE formula above, or branch on the highest-degree vertex
↳ with memoisation.
// * MAXIMUM INDEPENDENT SET for  $n \leq 40-50$ : meet in the middle -
↳ split the vertices in half,
// enumerate independent subsets of the left half, and for each
↳ store the best independent
// subset of the right half compatible with its neighbourhood
↳ (SOS over the right mask).
// * COUNTING INDEPENDENT SETS / colourings on a graph of small
↳ TREewidth: DP over a tree
// decomposition,  $O(2^{tw} * n)$  - usually a grid problem in
↳ disguise (see broken-profile DP).
```

3.9.2 MaxCliqueBitset

MAXIMUM CLIQUE FOR n UP TO ~150 (about 1 s at n = 155, much faster on sparse graphs). 0-indexed.

```
// 01 - CliqueColouringMIS is capped at 32 vertices by its unsigned
    ↪ mask; this is the same problem
// with a bitset adjacency matrix and a far stronger pruning rule, so
    ↪ it is the one to paste when
// n is in the 40..200 range. MAXIMUM INDEPENDENT SET = run it on the
    ↪ COMPLEMENT graph.
// The bound: greedily colour the candidate set (vertices of the same
    ↪ colour are pairwise
// non-adjacent, so at most one of them can join the clique). If
    ↪ |current| + #colours <= |best|,
// prune. The extra trick is the ADAPTIVE re-sorting - it recomputes
    ↪ the degree order in a
// sub-branch only when that branch has been visited often enough to
    ↪ pay for it (the 'limit').
// PRECONDITION: symmetric matrix, NO self edges. Set MV to the
    ↪ smallest power-of-two-ish bound
// above n; a bitset<MV> is scanned in MV/64 words, so an over-large
    ↪ MV costs real time.
const int MV = 160; // <--
    ↪ adjust to your n
struct MaxClique {
    typedef vector<bitset<MV>> Graph;
    struct Vertex { int i, d = 0; };
    double limit = 0.025, pk = 0;
    Graph e;
    vector<Vertex> V;
    vector<vector<int>> C;
    vector<int> qmax, q, S, old;
    MaxClique(Graph g) : e(g), C(g.size() + 1), S(C.size()), old(S) {
        for (int i = 0; i < (int)e.size(); i++) V.push_back({i});
    }
    void init(vector<Vertex>& r) { //
        ↪ order by degree inside r
        for (auto& v : r) v.d = 0;
        for (auto& v : r) for (auto& j : r) v.d += e[v.i][j.i];
        sort(all(r), [](Vertex a, Vertex b) { return a.d > b.d; });
        int mxD = r[0].d;
        for (int i = 0; i < (int)r.size(); i++) r[i].d = min(i, mxD)
    ↪ + 1;
    }
    void expand(vector<Vertex>& R, int lev = 1) {
        S[lev] += S[lev - 1] - old[lev], old[lev] = S[lev - 1];
        while (R.size()) {
            if ((int)q.size() + R.back().d <= (int)qmax.size())
        ↪ return; // colour bound
            q.push_back(R.back().i);
            vector<Vertex> T;
            for (auto v : R) if (e[R.back().i][v.i])
        ↪ T.push_back({v.i});
            if (T.size()) {
                if (S[lev]++ / ++pk < limit) init(T);
                int j = 0, mxk = 1, mnk = max((int)qmax.size() -
        ↪ (int)q.size() + 1, 1);
                C[1].clear(), C[2].clear();
                for (auto v : T) { //
        ↪ greedy colouring
                    int k = 1;
                    auto f = [&](int i) { return e[v.i][i]; };
                    while (any_of(all(C[k]), f)) k++;
                    if (k > mxk) mxk = k, C[mxk + 1].clear();
                    if (k < mnk) T[j++].i = v.i;
                    C[k].push_back(v.i);
                }
                if (j > 0) T[j - 1].d = 0;
                for (int k = mnk; k <= mxk; k++) for (int i : C[k])
        ↪ T[j].i = i, T[j++].d = k;
                expand(T, lev + 1);
            } else if (q.size() > qmax.size()) qmax = q;
            q.pop_back(), R.pop_back();
        }
    }
    vector<int> solve() { init(V), expand(V); return qmax; } // the
    ↪ clique itself
};
// MAXIMUM INDEPENDENT SET: build h with h[i][j] = (i != j &&
    ↪ !g[i][j]) and run this on h.
// MINIMUM VERTEX COVER = n - MIS. MINIMUM CLIQUE COVER of G =
    ↪ chromatic number of the complement.
// WEIGHTED maximum clique needs a different bound (sum of the
    ↪ per-colour maxima); the unweighted
// colour count above is not valid there.
// MAXIMUM CLIQUE ON A PERFECT GRAPH (bipartite, chordal, interval,
    ↪ comparability) is POLYNOMIAL -
// check the structure before reaching for exponential search.
```

```
// COUNTING all maximal cliques instead: Bron-Kerbosch with pivoting
// → (01), 0(3^(n/3)) of them.
```

3.9.3 ChromaticPolynomial

CHROMATIC POLYNOMIAL $P(x)$ = the number of proper colourings with at most x colours, for $n \leq 20$.

```
// Needs: Mint (06 - Math/01 - Mint.cpp).
// 0(2^n * n^2) to get all of P(0..n), then 0(n^2) per evaluation
// → elsewhere. 0-indexed; adj[] is an
// adjacency BITMASK (bit j of adj[i] set iff i-j is an edge), no
// → self loops.
// 01 - CliqueColouringMIS finds the chromatic NUMBER with the same
// → inclusion-exclusion; this gets
// the whole polynomial, which is what you need for "count the proper
// → k-colourings" (k possibly
// huge), for "count acyclic orientations", and for
// → deletion-contraction arguments.
// THE FORMULA. With I(S) = the number of independent subsets of S
// → (including the empty one),
// P(k) = sum over S of (-1)^(n-|S|) * I(S)^k.
// P has degree n, so evaluating it at 0..n pins it down; Lagrange
// → interpolation gives any other
// point. All arithmetic is mod MOD, so a zero result is "almost
// → surely" a genuine zero.
struct ChromaticPolynomial {
    int n;
    vector<Mint> p;
    // p[k] = P(k) for k = 0..n
    ChromaticPolynomial(const vector<int>& adj) : n(adj.size()), p(n)
    {
        int F = 1 << n;
        vector<Mint> I(F), pw(F, 1);
        I[0] = 1;
        for (int S = 1; S < F; S++) {
            // v
            // is in the subset, or it is not
            int v = __builtin_ctz(S), rest = S ^ 1 << v;
            I[S] = I[rest] + I[rest & ~adj[v]];
        }
        for (int k = 1; k <= n; k++) {
            Mint tot = 0;
            for (int S = 0; S < F; S++) {
                pw[S] *= I[S];
            }
            pw[S] = I(S)^k;
            tot += ((n - __builtin_popcount(S)) & 1) ? Mint(0) -
            pw[S] : pw[S];
        }
        p[k] = tot;
    }
    Mint at(ll x) const {
        // P(x) for any x, by interpolation
        if (x >= 0 && x <= n) return p[x];
        Mint ans = 0;
        for (int i = 0; i <= n; i++) {
            Mint num = p[i];
            for (int j = 0; j <= n; j++) if (j != i) num *= (Mint(x)
            - j) / (Mint(i) - j);
            ans += num;
        }
        return ans;
    }
    int chromaticNumber() const {
        for (int k = 1; k <= n; k++) if (!(p[k] == Mint(0))) return k;
        return n;
    }
};
// KNOWN VALUES, useful as checks: a tree on n vertices has P(x) = x
// → (x-1)^(n-1); the cycle C_n has
// (x-1)^n + (-1)^n (x-1); K_n has x(x-1)...(x-n+1); a forest of k
// → components on n vertices in total
// has P(x) = x^k (x-1)^(n-k) only when every component is a star...
// → just use the tree formula per
// component and multiply, since P is MULTIPLICATIVE over connected
// → components.
// ACYCLIC ORIENTATIONS of G = |P(-1)| = (-1)^n P(-1) (Stanley's
// theorem). That also counts the
// DAGs you can make by directing the edges of G, which is how "count
// → the DAGs whose underlying
// graph is G" problems are solved.
// DELETION-CONTRACTION: P_G(x) = P_{G-e}(x) - P_{G/e}(x).
// → Exponential in general, but it is the
// right tool when the graph is nearly a tree or you can memoise on
// → small components.
// LABELLED DAGs ON n VERTICES (a different question - no fixed
// → graph): with a_n the count,
// a_n = sum_{k=1..n} (-1)^(k+1) C(n,k) 2^k (k-n-k)) a_{n-k}, by
// → inclusion-exclusion on the sources.
```

3.9.4 Hafnian

HAFNIAN of a symmetric matrix = the NUMBER OF PERFECT MATCHINGS of a general weighted graph

```
// Needs: Mint (06 - Math/01 - Mint.cpp).
// (the sum over perfect matchings of the product of the chosen
// → entries). 0(2^(n/2) * n^3), which
// is about 2 s for a 38x38 matrix. 0-indexed, n must be EVEN, the
// → diagonal is ignored.
// The hafnian is to the permanent what the general graph is to the
// → bipartite one:
// permanent(A) = #perfect matchings of a BIPARTITE graph →
// → Ryser, 0(2^n n)
// hafnian(A) = #perfect matchings of a GENERAL graph → this
// → file
// determinant = the SIGNED version, polynomial, and counts
// → nothing on its own
// Reach for it for "count the perfect matchings / domino tilings /
// → pairings" when the graph is not
// bipartite and n is around 30-40. If the graph IS bipartite use the
// → permanent (2x more vertices
// here would be hopeless); if you only need EXISTENCE, use Blossom
// → or a Tutte matrix.
// METHOD (Bjorklund): keep, for every pair (i, j) of the first m
// → vertices, a POLYNOMIAL in a
// formal variable that counts partial matchings by size. Removing
// → the last two vertices splits
// into "they are matched to each other" and "they are not", and the
// → recursion has depth n/2 with
// two branches that share all the work below them.
struct Hafnian {
    int h;
    void add(vector<Mint>& ans, const vector<Mint>& a, const
    vector<Mint>& b) {
        for (int i = 0; i < h; i++)
            for (int j = 0; j < h - 1 - i; j++) ans[i + j + 1] +=
            a[i] * b[j];
    }
    vector<Mint> rec(const vector<vector<vector<Mint>>>& v) {
        vector<Mint> ans(h, 0);
        if (v.empty()) { ans[0] = 1; return ans; }
        int m = (int)v.size() - 2;
        auto V = v;
        V.resize(m);
        vector<Mint> zero = rec(V);
        // last two vertices dropped
        for (int i = 0; i < m; i++) for (int j = 0; j < i; j++) {
            add(V[i][j], V[m][i], V[m + 1][j]);
            // ...
            add(V[i][j], V[m + 1][i], V[m][j]);
        }
        vector<Mint> one = rec(V);
        for (int i = 0; i < h; i++) ans[i] += one[i] - zero[i];
        add(ans, one, V[m + 1][m]);
        return ans;
    }
    Mint solve(const vector<vector<Mint>>& a) {
        // a
        // symmetric, a[i][i] unused
        int n = a.size();
        h = n / 2 + 1;
        vector<vector<vector<Mint>>> v(n);
        for (int i = 0; i < n; i++) {
            v[i].resize(i);
            for (int j = 0; j < i; j++) v[i][j].assign(h, 0);
        }
        v[i][j][0] = a[i][j];
        return rec(v).back();
    }
};
// 0/1 MATRIX → plain count of perfect matchings. General weights →
// → weighted count, so this also
// answers "sum over perfect matchings of the product of edge
// → weights" (partition functions).
// PERMANENT by Ryser, for comparison, is 6 lines and 0(2^n n):
// perm(A) = (-1)^n * sum over subsets S of columns of (-1)^|S| *
// → prod_i (sum_{j in S} A[i][j]).
// COUNTING MATCHINGS OF ALL SIZES (not just perfect): pad the matrix
// → with a zero-weight vertex per
// vertex - or read off rec()'s whole coefficient vector, which is
// → exactly that generating function.
// PLANAR GRAPHS have a POLYNOMIAL count via the FKT algorithm
// → (Pfaffian orientation + determinant),
// 0(n^3) - that is the one to use for domino tilings of a grid.
```

3.9.5 MeetInTheMiddle

MEET IN THE MIDDLE ON GRAPHS - the $n \leq 40..50$ band, where 2^n is hopeless but $2^{n/2}$ is not.

```
// 0-indexed, adj[] is an adjacency BITMASK (ll, so n <= 63), no self
// ↳ loops unless stated.
// THE PATTERN. Split the vertices into halves L (first k) and R (the
// ↳ rest). Enumerate every valid
// configuration of L; each one imposes a REQUIREMENT on R that is a
// ↳ single bitmask; precompute for
// every R-mask the best/total over all valid R-configurations
// ↳ COMPATIBLE with it, using a SOS
// (subset-sum / superset-sum) transform. Then one pass over the  $2^k$ 
// ↳ left masks answers everything.

// --- COUNT ALL CLIQUES (including the empty one),  $O(2^{n/2} * n)$ .
// ↳ ---
// A set is a clique iff every chosen vertex is adjacent to all the
// ↳ others, i.e. the intersection of
// (adj[v] | bit v) over v in the set contains the set.
ll countCliques(int n, const vector<ll>& adj) {
    int k = n / 2, r = n - k;
    ll full = n == 64 ? -1LL : (1LL << n) - 1;
    vector<ll> g(n);
    for (int i = 0; i < n; i++) g[i] = adj[i] | 1LL << i;
    vector<ll> dp(1 << k, 0);
    dp[0] = 1;
    for (int m = 1; m < (1 << k); m++) { // is
        ↳ the left mask m a clique?
        ll nw = full;
        for (int j = 0; j < k; j++) if (m >> j & 1) nw &= g[j];
        if ((nw & m) == m) dp[m] = 1;
    }
    for (int i = 0; i < k; i++) for (int m = 0; m < (1 << k); m++)
        ↳ SOS: dp[m] = #subsets
        if (m >> i & 1) dp[m] += dp[m ^ 1 << i];
    ll ans = dp[(1 << k) - 1]; //
    ↳ right half empty
    for (int m = 1; m < (1 << r); m++) {
        ll nw = full, p = (ll)m << k;
        for (int j = 0; j < r; j++) if (m >> j & 1) nw &= g[k + j];
        if ((nw & p) == p) ans += dp[nw & ((1LL << k) - 1)]; //
        ↳ left must sit inside nw
    }
    return ans;
}

// --- MINIMUM COST VERTEX COVER OF A GENERAL GRAPH,  $O(2^{n/2} * n)$ .
// ↳ ---
// Every edge needs at least one endpoint chosen; minimise the total
// ↳ cost of the chosen vertices.
// A left mask fixes the left choices; the UNCHOSEN left vertices
// ↳ force their right neighbours in.
// SELF LOOP at v (bit v of adj[v]) = "v must be taken", which is how
// ↳ "forbidden" vertices are said.
// (On a BIPARTITE graph this is polynomial - Konig / min cut. This
// ↳ is the general case.)
ll minCostVertexCover(int n, const vector<ll>& adj, const vector<ll>&
    ↳ cost) {
    int k = n / 2, r = n - k;
    vector<ll> cl(1 << k, 0), need(1 << k, 0), dp(1 << r, 0);
    for (int m = 0; m < (1 << k); m++) {
        for (int i = 0; i < k; i++) {
            if (m >> i & 1) { cl[m] += cost[i]; continue; }
            if (adj[i] >> i & 1) { cl[m] = llinf; break; } //
        }
        ↳ self loop: i had to be taken
        for (int j = 0; j < n; j++) if (adj[i] >> j & 1) {
            if (j < k) { if (!(m >> j & 1)) cl[m] = llinf; } //
        }
        ↳ uncovered left edge
        else need[m] |= 1 << (j - k);
    }
}
for (int m = 0; m < (1 << r); m++) {
    for (int i = 0; i < r; i++) {
        if (m >> i & 1) { dp[m] += cost[k + i]; continue; }
        for (int j = k; j < n; j++)
            if ((adj[k + i] >> j & 1) && !(m >> (j - k) & 1))
                ↳ dp[m] = llinf;
    }
}
for (int i = 0; i < r; i++) for (int m = (1 << r) - 1; m >= 0;
    ↳ m--) // SOS over SUPERSETS
    if (!(m >> i & 1)) dp[m] = min(dp[m], dp[m ^ 1 << i]);
ll ans = llinf;
for (int m = 0; m < (1 << k); m++)
    if (cl[m] < llinf && dp[need[m]] < llinf) ans = min(ans,
        ↳ cl[m] + dp[need[m]]);
```

```
return ans;
}
// MAXIMUM INDEPENDENT SET = (total cost) - (minimum cost vertex
// ↳ cover) with unit costs, so the
// second routine solves MIS for  $n \leq 50$  too. MAXIMUM CLIQUE for that
// ↳ range: 02 - MaxCliqueBitset.
// OTHER PROBLEMS THAT SPLIT THE SAME WAY: subset sum / knapsack on n
// ↳ <= 40, "count the subsets
// with property X" when X only couples the halves through one mask,
// ↳ and 3-colouring by enumerating
// the colour classes of one half.
// THE SOS DIRECTION MATTERS: use the subset transform when the right
// ↳ half must fit INSIDE a mask
// (counting), and the superset transform when the right half must
// ↳ CONTAIN a mask (covering).
```

3.10 Theorems

3.10.1 GraphTheorems

GRAPH THEOREM SHEET =====

```
/* ===== GRAPH THEOREM SHEET =====
Every hypothesis here matters. A theorem quoted without its
  ↳ precondition is a wrong answer.
```

MATCHING AND COVERING

KONIG (BIPARTITE only): max matching = min vertex cover.

Constructive: Z = vertices reachable by alternating paths from
 ↳ the UNMATCHED left vertices;
 the cover is $(X \setminus Z) \cup (Y \cap Z)$.

GALLAI: for EVERY graph on n vertices, $\alpha + \tau = n$
 ↳ (independent set + vertex cover).

for every graph WITH NO ISOLATED VERTEX, $\nu + \rho = n$
 ↳ (matching + edge cover).

Bipartite + Konig gives $\alpha = n - \nu$.

HALL (bipartite): a matching saturating every vertex of the LEFT
 ↳ part X exists iff

$|N(S)| \geq |S|$ for every S contained in X . (Not every S of $V -$
 ↳ the side matters.)

ORE'S DEFECT FORM: $\nu(G) = |X| - \max \text{ over } S \text{ contained in } X \text{ of } ($
 ↳ $|S| - |N(S)|$).

HALL FOR REGULAR BIPARTITE: every k -regular bipartite graph ($k \geq$
 ↳ 1) has a perfect matching,
 so its edges split into exactly k perfect matchings.

↳ Equivalently KONIG'S EDGE COLOURING

THEOREM: $\chi'(G) = \Delta(G)$ for every bipartite MULTIGRAPH.

↳ (Timetabling, "split into
 Delta rounds", Latin-square completion.)

BERGE: a matching is maximum iff there is no augmenting path. (Why
 ↳ Kuhn / HK / Blossom stop.)

TUTTE (general graphs): G has a PERFECT matching iff $\text{odd}(G - U) \leq$
 ↳ $|U|$ for every U contained

in V , where $\text{odd}(H)$ = the number of components of H with an ODD
 ↳ number of vertices.

U = empty forces n even.

TUTTE-BERGE: $\nu(G) = (1/2) * \min \text{ over } U \text{ of } (|V| + |U| - \text{odd}(G -$
 ↳ $U))$.

DILWORTH (FINITE poset): min number of chains covering P = size of
 ↳ the maximum antichain.

MIRSKY (dual): min number of antichains partitioning P = length of
 ↳ the longest chain.

SPERNER: the largest antichain in the subset lattice of $[n]$ has
 ↳ size $C(n, \lfloor n/2 \rfloor)$.

PATH COVERS - the distinction that most notebooks get wrong:
 min VERTEX-DISJOINT path cover of a DAG = $n - (\text{max matching of}$
 ↳ the split graph, no closure).

min chain cover for Dilworth = $n - (\text{max matching of the}$
 ↳ TRANSITIVE CLOSURE's split graph).

CONNECTIVITY AND FLOW

MENGER: for NON-ADJACENT s, t : min s - t VERTEX cut = max
 ↳ internally-vertex-disjoint s - t paths.

For any $s \neq t$: min s - t EDGE cut = max edge-disjoint s - t paths.
 WHITNEY: $\kappa(G) \leq \lambda(G) \leq \delta(G)$ (vertex conn $\leq$ edge
 ↳ conn $\leq$ min degree).

A graph on ≥ 3 vertices is 2-connected iff every two vertices
 ↳ lie on a common cycle.

MAX-FLOW MIN-CUT, with INTEGRALITY: integer capacities $\Rightarrow$ some
 ↳ maximum flow is integral (and
 Dinic produces one). Corollary used constantly: a $0/1$ -capacity

↳ max flow decomposes into that
 many edge-disjoint s - t paths.

ROBBINS: an undirected graph has a STRONG orientation iff it is
 ↳ connected and BRIDGELESS.

Constructive in $O(V+E)$: DFS, orient tree edges away from the
 ↳ root and back edges toward it.

ESWARAN-TARJAN (augmentation):

directed - min arcs to make a digraph strongly connected =
 ↳ $\max(\#\text{sources}, \#\text{sinks})$ of the

condensation ($\emptyset$ if the condensation is a single node).
 undirected - for a CONNECTED graph, min edges to make it

↳ 2-edge-connected = $\lceil L/2 \rceil$ where

L is the number of leaves of the bridge tree ($\emptyset$ if the bridge
 ↳ tree is one node).

DEGREE SEQUENCES

ERDOS-GALLAI: $d_1 \geq \dots \geq d_n \geq 0$ is the degree sequence of a
 ↳ SIMPLE graph iff $\sum d_i$ is

even and for every k in $[1, n]$:

$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k)$.

Only k up to the largest index with $d_k \geq k$ needs checking (the
 ↳ Durfee square), so it is

$O(n)$ after sorting with prefix sums.

HAVEL-HAKIMI: graphical iff $d_1 \leq n-1$ and, after deleting d_1 and
 ↳ subtracting 1 from the next

d_1 entries (re-sorted), the rest is graphical. Gives a

↳ CONSTRUCTION, which Erdos-Gallai
 does not. $O(n^2)$ naive.

LANDAU (tournaments): $s_1 \leq \dots \leq s_n$ is a score sequence iff

↳ $\sum_{i=1}^k s_i \geq C(k, 2)$ for
 every k , WITH EQUALITY at $k = n$.

TOURNAMENTS

REDEI: every finite tournament has a Hamiltonian PATH (insertion
 ↳ sort, $O(n \log n)$).

CAMION: every STRONGLY CONNECTED tournament has a Hamiltonian
 ↳ CYCLE.

MOON: every strongly connected tournament is vertex-pancyclic -
 ↳ for every vertex v and every
 k in $[3, n]$ there is a k -cycle through v .

COUNTING

MATRIX-TREE (Kirchhoff), undirected: #spanning trees (or the sum
 ↳ over trees of the product of

weights) = ANY cofactor of $L = D - A$, loops excluded.

MATRIX-TREE, DIRECTED (Tutte), with $A_{ij} = \# \text{arcs } i \rightarrow j$:
 $L_{\text{out}} = D_{\text{out}} - A$, delete row AND column $r \rightarrow$ arborescences

↳ CONVERGING TO r (in-trees).

$L_{\text{in}} = D_{\text{in}} - A$, delete row AND column $r \rightarrow$ arborescences
 ↳ DIVERGING FROM r (out-trees).

These two are the pair everyone swaps. Verify on the 2-vertex
 ↳ graph $1 \rightarrow 2$.

BEST THEOREM: for a CONNECTED digraph with $\text{indeg}(v) = \text{outdeg}(v)$
 ↳ everywhere, the number of

Eulerian circuits up to starting point is $\text{ec}(G) = t_w(G) *$
 ↳ $\prod_v (\text{outdeg}(v) - 1)!$,

where t_w is the number of arborescences CONVERGING TO any fixed
 ↳ w (the L_{out} form above);

t_w is the same for every w in such a graph. Multiply by

↳ $\text{outdeg}(v)$ to fix a start vertex.

CAYLEY: $n^{(n-2)}$ labelled trees; with prescribed degrees, $(n-2)! /$
 ↳ $\prod (d_i - 1)!$;

labelled forests with k trees whose roots are k specified
 ↳ vertices: $k * n^{(n-k-1)}$.

EULERIAN CONDITIONS (the check EulerianPath.cpp needs before it
 ↳ starts)

UNDIRECTED: a circuit exists iff every degree is even and all
 ↳ EDGES lie in one component.

A trail exists iff exactly 0 or 2 vertices have odd degree
 ↳ (start at an odd one), same

connectivity condition.

DIRECTED: a circuit exists iff $\text{indeg} = \text{outdeg}$ at every vertex and
 ↳ all vertices of nonzero

degree lie in one component of the underlying undirected graph.
 ↳ A trail exists iff exactly

one vertex has $\text{out} - \text{in} = 1$ (the start), exactly one has $\text{in} -$
 ↳ $\text{out} = 1$ (the end), the rest

are balanced, same connectivity condition.

COLOURING

BROOKS: for connected G , $\chi(G) \leq \Delta(G)$ UNLESS G is complete
 ↳ or an odd cycle, where

$\chi = \Delta + 1$. Greedy on any order gives $\Delta + 1$; greedy on a
 ↳ degeneracy order gives $d + 1$.

VIZING (SIMPLE graphs): $\Delta \leq \chi' \leq \Delta + 1$. Multigraphs:
 ↳ $\chi' \leq \Delta + \mu$, and

Shannon's $\chi' \leq \lfloor 3\Delta / 2 \rfloor$. Deciding class 1 vs class
 ↳ 2 is NP-complete - do not

look for a polynomial algorithm mid-contest. Bipartite is the
 ↳ exception (Konig above).

GALLAI-HASSE-ROY-VITAVER: $\chi(G) = 1 + \min \text{ over orientations of}$
 ↳ (longest directed path length).

Equivalently every orientation contains a directed path on
 ↳ $\chi(G)$ vertices.

EXTREMAL

TURAN: a K_{r+1} -free graph has $e \leq (1 - 1/r) n^2 / 2$, with
 ↳ equality only for the balanced

complete r -partite Turan graph. $r = 2$ is MANTEL: triangle-free
 ↳ $\Rightarrow e \leq \lfloor n^2/4 \rfloor$.

Every graph has a bipartite subgraph with $\geq m/2$ edges (greedy
 ↳ max-cut).

A graph is bipartite iff it has no odd cycle.

RAMSEY, the known small values: $R(3,3)=6$, $R(3,4)=9$, $R(3,5)=14$,
 ↳ $R(3,6)=18$, $R(3,7)=23$,

$R(3,8)=28$, $R(3,9)=36$, $R(4,4)=18$, $R(4,5)=25$. $R(4,6)$ in $[36,40]$
 ↳ and $R(5,5)$ in $[43,46]$ are

OPEN. Bound: $R(s,t) \leq C(s+t-2, s-1)$.

NASH-WILLIAMS ARBORICITY: $a(G) = \max \text{ over subgraphs } S \text{ with } \geq 2$
 ↳ vertices of

$\lceil m_S / (n_S - 1) \rceil$. Consequences: degeneracy d satisfies a
 ↳ $\leq d \leq 2a - 1$; $m \leq a(n-1)$;

triangle enumeration is $O(m \cdot a)$.
 TUTTE / NASH-WILLIAMS TREE PACKING: G has t edge-disjoint spanning
 $\hookrightarrow$ trees iff every partition
 of V into k non-empty parts has at least $t(k-1)$ crossing edges.
 PETERSEN: every BRIDGELESS CUBIC graph has a perfect matching.
 $\hookrightarrow$ Every $2k$ -regular graph splits
 into k edge-disjoint 2-factors, so it has an Eulerian circuit
 $\hookrightarrow$ and an orientation with
 $\text{indeg} = \text{outdeg} = k$.

PLANARITY

EULER: connected plane graph $V - E + F = 2$; with c components, $V -$
 $\hookrightarrow E + F = 1 + c$.
 Simple planar with $V \geq 3$: $E \leq 3V - 6$. Triangle-free (girth $\geq$
 $\hookrightarrow 4$): $E \leq 2V - 4$.
 Every planar graph has a vertex of degree ≤ 5 , i.e. it is
 $\hookrightarrow$ 5-degenerate.
 KURATOWSKI / WAGNER: planar iff no K_5 and no $K_{3,3}$ subdivision
 $\hookrightarrow$ (minor).
 FOUR COLOUR: every planar graph is 4-colourable - but there is no
 $\hookrightarrow$ usable algorithm, so never
 plan around it. FIVE COLOUR is constructive in $O(n)$ from the
 $\hookrightarrow$ degree ≤ 5 vertex.
 K_5 , $K_{3,3}$ and the Petersen graph are non-planar. In a planar
 $\hookrightarrow$ graph min cut = shortest path
 in the dual.

HAMILTONICITY (sufficient conditions only - the general problem is

$\hookrightarrow$ NP-complete)
 DIRAC: $n \geq 3$ and min degree $\geq n/2 \Rightarrow$ Hamiltonian.
 ORE: $\deg(u) + \deg(v) \geq n$ for every non-adjacent pair $\Rightarrow$
 $\hookrightarrow$ Hamiltonian.
 Both are constructive by rotation-extension.

TREE FACTS

JORDAN: every tree has 1 or 2 CENTRES; if 2 they are adjacent;
 $\hookrightarrow$ they are the middle of any
 diameter path.
 CENTROID: every tree has 1 or 2 centroids; c is a centroid iff
 $\hookrightarrow$ every component of $T - c$ has
 $\leq n/2$ vertices iff c minimises the sum of distances. If 2 they
 $\hookrightarrow$ are adjacent and n is even.
 That $\leq n/2$ bound is exactly what makes centroid decomposition
 $\hookrightarrow$ $O(\log n)$ deep.
 DIAMETER: the greedy two-BFS is valid on trees and on positively
 $\hookrightarrow$ weighted trees, NOT on
 general graphs. Every longest path passes through a centre.
 $\hookrightarrow$ Joining two trees by an edge
 gives diameter $\max(d_1, d_2, r_1 + r_2 + 1)$, minimised by joining
 $\hookrightarrow$ centre to centre.
 A tree is bipartite, has ≥ 2 leaves, and sum of degrees = $2(n-1)$.

WHAT IS NP-HARD, AND WHAT ONLY LOOKS IT

NP-hard: max clique, chromatic number (≥ 3 colours), Hamiltonian
 $\hookrightarrow$ path/cycle, TSP, MIXED
 Chinese postman, vertex cover / independent set on general
 $\hookrightarrow$ graphs, max cut (general),
 Steiner tree with unbounded terminals, longest path, dominating
 $\hookrightarrow$ set, feedback vertex/arc
 set, edge-colouring class decision, bandwidth.
 Polynomial despite appearances: vertex cover and independent set
 $\hookrightarrow$ on BIPARTITE graphs (Konig)
 and on trees / interval / chordal graphs; 2-colouring; 2-SAT;
 $\hookrightarrow$ max cut on PLANAR graphs;
 min mean cycle; min-cost flow; directed MST; maximum matching in
 $\hookrightarrow$ a GENERAL graph (Blossom);
 min path cover of a DAG; Steiner tree with $k \leq \sim 15$ terminals;
 $\hookrightarrow$ TSP with $n \leq \sim 20$.

===== */

3.11 Special Structures

3.11.1 StableChordalComplement

THREE GRAPH TOOLS THAT ARE NOT ABOUT PATHS OR FLOW.

```
// --- GALE-SHAPLEY STABLE MARRIAGE,  $O(n^2)$ . ---
// n proposers, n receivers, each with a full preference ranking. A
//   matching is STABLE if no pair
//   mutually prefer each other to their assigned partners. One always
//   exists, and this finds the
// PROPOSER-OPTIMAL one: every proposer gets the best partner they
//   could have in ANY stable
//   matching, and simultaneously every receiver gets their worst.
//   (Swap the roles to flip that.)
// pref[i] = the proposers' preference lists; rank[j][i] = receiver
//   j's ranking of proposer i.
vector<int> stableMarriage(const vector<vector<int>>& pref, const
//   vector<vector<int>>& rank) {
//   int n = pref.size();
//   vector<int> match(n, -1), nxt(n, 0);
//   match[receiver] = proposer
//   vector<int> free_(n);
//   iota(all(free_), 0);
//   while (!free_.empty()) {
//     int m = free_.back(); free_.pop_back();
//     int w = pref[m][nxt[m]++];
//     propose down the list
//     if (match[w] < 0) match[w] = m;
//     else if (rank[w][m] < rank[w][match[w]])
//       free_.push_back(match[w]), match[w] = m;
//     else free_.push_back(m);
//   }
//   return match;
}

// HOSPITALS/RESIDENTS (one side has capacity c) is the same loop
//   with a heap of accepted
// proposers per hospital. STABLE ROOMMATES (one pool, no sides) may
//   have NO stable matching -
// Irving's algorithm decides it, and it is a different, longer
//   algorithm.

// --- TRAVERSAL OF THE COMPLEMENT GRAPH in  $O(n + m)$ , never
//   materialising the  $O(n^2)$  edges. ---
// Keep the not-yet-visited vertices in a linked structure; from u,
//   walk that set and step to
//   every w that is NOT a neighbour of u. Each vertex leaves the set
//   once, and each REAL edge is
//   inspected once, so the total is  $O(n + m)$ .
// Reach for it whenever the input describes the NON-edges:
//   "connected components of the graph of
//   pairs that do NOT conflict", complement colouring, complement BFS
//   distances.
vector<int> complementComponents(int n, const vector<vector<int>>&
//   adj) {
//   vector<char> isAdj(n, 0);
//   set<int> unvis;
//   for (int i = 0; i < n; i++) unvis.insert(i);
//   vector<int> comp(n, -1);
//   int c = 0;
//   while (!unvis.empty()) {
//     int s = *unvis.begin();
//     unvis.erase(unvis.begin());
//     vector<int> st{s};
//     comp[s] = c;
//     while (!st.empty()) {
//       int u = st.back(); st.pop_back();
//       for (int v : adj[u]) isAdj[v] = 1;
//       vector<int> take;
//       for (int w : unvis) if (!isAdj[w]) take.push_back(w);
//       for (int w : take) unvis.erase(w), comp[w] = c,
//         st.push_back(w);
//       for (int v : adj[u]) isAdj[v] = 0;
//     }
//     c++;
//   }
//   return comp;
//   comp[v] = component id
}

// The complement of a DISCONNECTED graph is connected, so a
//   complement graph almost always has
//   very few components - that is why this is fast in practice as well
//   as in theory.

// --- CHORDAL GRAPHS: maximum cardinality search gives a PERFECT
//   ELIMINATION ORDERING. ---
// A graph is CHORDAL if every cycle of length  $\geq 4$  has a chord.
//   Equivalently it has a PEO: an
//   order in which every vertex's LATER neighbours form a clique.
// Why you care: on a chordal graph, MAXIMUM CLIQUE, CHROMATIC
```

```

    ⇨ NUMBER, MAXIMUM INDEPENDENT SET
// and MINIMUM CLIQUE COVER are all POLYNOMIAL - the four problems
    ⇨ that are NP-hard in general.
// Interval graphs and trees are chordal, so "these intervals
    ⇨ conflict" problems land here.
vector<int> mcsOrder(int n, const vector<vector<int>>& adj) {
    ⇨ reverse of a PEO if chordal
    vector<int> w(n, 0), ord;
    vector<char> used(n, 0);
    for (int it = 0; it < n; it++) {
        int b = -1;
        for (int v = 0; v < n; v++) if (!used[v] && (b < 0 || w[v] >
    ⇨ w[b])) b = v;
        used[b] = 1, ord.push_back(b);
        for (int u : adj[b]) if (!used[u]) w[u]++;
    }
    reverse(all(ord));
    return ord;
    ⇨ this IS a PEO iff chordal
}

bool isChordal(int n, const vector<vector<int>>& adj) {
    auto ord = mcsOrder(n, adj);
    vector<int> pos(n);
    for (int i = 0; i < n; i++) pos[ord[i]] = i;
    vector<char> mark(n, 0);
    for (int i = 0; i < n; i++) {
        ⇨ later neighbours must be
        int v = ord[i], par = -1;
        ⇨ clique; it is enough to check
        for (int u : adj[v]) if (pos[u] > i && (par < 0 || pos[u] <
    ⇨ pos[par])) par = u;
        if (par < 0) continue;
        for (int u : adj[par]) mark[u] = 1;
        for (int u : adj[v]) if (pos[u] > i && u != par && !mark[u]) {
            for (int x : adj[par]) mark[x] = 0;
            return false;
        }
        for (int u : adj[par]) mark[u] = 0;
    }
    return true;
}

// GIVEN A PEO: greedy colouring in PEO order uses exactly omega
    ⇨ colours (so chi = omega); the
// maximum clique is the largest "v plus its later neighbours"; the
    ⇨ maximum independent set is the
// greedy "take v, delete its later neighbours" over the PEO.

```

3.11.2 MinimumWeightCycle

MINIMUM WEIGHT CYCLE THROUGH EVERY VERTEX (and hence the global minimum weight cycle), by one

```

// Dijkstra per source,  $O(V^3)$  with the dense version below or  $O(V(E + V \log V))$  with a heap.
// UNDIRECTED, non-negative weights, 0-indexed, g is an adjacency
    ⇨ MATRIX with llinf for "no edge".
// Returns best[u] = the weight of the lightest simple cycle
    ⇨ CONTAINING u (llinf if there is none);
// the global answer is the minimum over u, and the GIRTH is the
    ⇨ unweighted case (all weights 1).
// WHY IT WORKS. A lightest cycle through u splits into two shortest
    ⇨ paths from u that leave along
// DIFFERENT first edges and are joined by one more edge (cur, i). So
    ⇨ run Dijkstra from u, tag every
// vertex with the first edge (the "branch") its shortest path used,
    ⇨ and combine any two settled
// vertices in different branches. The 'par[cur] != cur' guard
    ⇨ rejects the degenerate 2-cycle that
// walks one edge out and back.
vector<ll> minWeightCycles(int n, const vector<vector<ll>>& g) {
    vector<ll> best(n, llinf), d(n);
    vector<int> par(n), vis(n);
    for (int u = 0; u < n; u++) {
        fill(all(vis), 0), fill(all(d), llinf);
        iota(all(par), 0);
        d[u] = 0;
        for (;) {
            int cur = -1;
            ⇨ dense Dijkstra:  $O(n^2)$  per source
            for (int i = 0; i < n; i++) if (!vis[i] && d[i] < llinf
    ⇨ && (cur < 0 || d[i] < d[cur]))
                cur = i;
            if (cur < 0) break;
            vis[cur] = 1;
            if (par[cur] != cur && g[cur][u] < llinf)
                ⇨ close the cycle straight to u
                best[u] = min(best[u], d[cur] + g[cur][u]);
            for (int i = 0; i < n; i++) {
                if (i == u || g[cur][i] == llinf) continue;
                if (vis[i]) {
                    if (par[i] != par[cur])
                        ⇨ different branches meet
                        best[u] = min(best[u], d[cur] + d[i] +
    ⇨ g[cur][i]);
                    } else if (d[cur] + g[cur][i] < d[i])
                        d[i] = d[cur] + g[cur][i], par[i] = (cur == u ? i
    ⇨ : par[cur]);
                }
            }
            ⇨ return best;
        }
    }

// GLOBAL MINIMUM WEIGHT CYCLE: min over u of best[u]. If that is all
    ⇨ you want and the graph is
// DIRECTED, Floyd-Warshall is shorter: run it, then take min over v
    ⇨ of d[v][v] (initialise the
// diagonal to llinf, not 0).
// GIRTH of an unweighted graph: replace Dijkstra by BFS -  $O(V * E)$ ,
    ⇨ and for a cycle through a
// FIXED vertex the two-branch rule is the same.
// MINIMUM CYCLE THROUGH A GIVEN EDGE (u, v): delete the edge and
    ⇨ take dist(u, v) + w.
// MINIMUM MEAN cycle (weight divided by length) is a different
    ⇨ problem - Karp, see 02 - Shortest
// Paths/06 - CyclesAndConstraints.
// NEGATIVE WEIGHTS break this completely (Dijkstra); a minimum
    ⇨ weight cycle with negative edges is
// NP-hard in general, since a Hamiltonian cycle can be encoded in it.
// LONGEST cycle is NP-hard. Shortest cycle THROUGH k GIVEN VERTICES
    ⇨ is FPT via colour coding.

```

Number Theory

3.11.3 IncrementalDAGReachability

INCREMENTAL DAG REACHABILITY (Italiano) - answer "is there a path $u \rightarrow v$?" while ARCS ARE BEING

```
// ADDED, in  $O(1)$  per query and  $O(n)$  amortised per insertion.
    ↳ Directed, 0-indexed,  $O(n^2)$  memory.
// addEdge(s, t) refuses (and returns false) any arc that would
    ↳ create a CYCLE, so the structure
// doubles as an online "would this arc break acyclicity?" oracle -
    ↳ which is the usual reason to
// reach for it: "process the constraints in order and report the
    ↳ first one that is contradictory",
// online topological ordering, incremental dependency graphs.
// STRUCTURE. For every vertex  $s$  it keeps a REACHABILITY TREE of
    ↳ everything  $s$  can reach, stored as
// par[s][v] = v's parent in that tree. Adding  $s \rightarrow t$  only ever ADDS
    ↳ vertices to those trees, and a
// vertex is added to a tree at most once, so the total work over all
    ↳ insertions is  $O(n^2)$  per
// source - i.e.  $O(n)$  amortised per update. Nothing is ever removed,
    ↳ hence "incremental only".
struct IncrementalDAG {
    int n;
    vector<vector<int>> par; //
    ↳ par[root][v], -1 = not reachable
    vector<vector<vector<int>>> ch; //
    ↳ ch[root][v] = children in the tree
    IncrementalDAG(int n) : n(n), par(n, vector<int>(n, -1)), ch(n,
    ↳ vector<vector<int>>(n)) {}
    bool reachable(int s, int t) const { return s == t || par[s][t]
    ↳ >= 0; }
    void meld(int root, int sub, int u, int v) { //
    ↳ graft v (and its subtree) under u
        par[root][v] = u, ch[root][u].push_back(v);
        for (int c : ch[sub][v]) if (!reachable(root, c)) meld(root,
    ↳ sub, v, c);
    }
    bool addEdge(int s, int t) { //
    ↳ false = it would close a cycle
        if (reachable(t, s)) return false;
        if (reachable(s, t)) return true; //
    ↳ nothing changes
        for (int p = 0; p < n; p++)
            if (reachable(p, s) && !reachable(p, t)) meld(p, t, s, t);
        return true;
    }
};
// FULLY DYNAMIC reachability (with deletions) has no simple
    ↳ solution; if the updates are known in
// advance, run OFFLINE divide and conquer over the time axis instead.
// DENSE ALTERNATIVE: bitset transitive closure,  $n/64$  words per
    ↳ vertex, rebuilt from scratch in
//  $O(n * m / 64)$  - simpler and often faster for  $n \leq 2000$  when the
    ↳ number of queries is small.
// TRANSITIVE CLOSURE of a static DAG: process in reverse topological
    ↳ order and OR the successors'
// bitsets -  $O(n * m / 64)$ .
// COUNTING PAIRS (u, v) with u reaching v is the same bitset sweep;
    ↳ there is no better general
// algorithm (it is equivalent to boolean matrix multiplication).
```

4 Number Theory

Euclidean Algorithm 101 · Primes and Divisors 106 · Modular Arithmetic 113 · Theorems 119 · Rational 124 · Sequences 125

4.1 Euclidean Algorithm

4.1.1 ExtendedEuclid

EXTENDED EUCLID - returns $g = \gcd(a, b)$ and x, y with $a*x + b*y = g$. This is the one routine that

```
// gives you modular inverses for a NON-prime modulus, solves linear
    ↳ Diophantine equations, and
// underpins CRT.  $|x| \leq b/(2g)$  and  $|y| \leq a/(2g)$ , so no overflow for
    ↳ 64-bit inputs.
long long extGCD(long long a, long long b, long long &x, long long
    ↳ &y) {
    if (b == 0) {
        x = 1; y = 0;
        return a;
    }
    long long x1, y1;
    long long d = extGCD(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return d;
}

long long modInverse(long long a, long long m) {
    long long x, y;
    long long g = extGCD(a, m, x, y);
    if (g != 1) return -1; // Inverse doesn't exist
    return (x % m + m) % m;
}
```

4.1.2 LinearDiophantine

Linear Diophantine Equations (2 Variables & N Variables)

```
// 1. Solves  $a * x + b * y = c$  (Particular solution & GCD check)
// 2. Solves  $a_1*x_1 + a_2*x_2 + \dots + a_n*x_n = c$  for N variables
struct LinearDiophantine {
    static ll ext_gcd(ll a, ll b, ll &x, ll &y) {
        if (b == 0) {
            x = 1; y = 0;
            return a;
        }
        ll x1, y1;
        ll d = ext_gcd(b, a % b, x1, y1);
        x = y1;
        y = x1 - y1 * (a / b);
        return d;
    }

    // Solves  $a * x + b * y = c$ . Returns false if no solution exists
    static bool solve_2var(ll a, ll b, ll c, ll &x, ll &y, ll &g) {
        if (a == 0 && b == 0) {
            if (c == 0) { x = y = g = 0; return true; }
            return false;
        }
        g = ext_gcd(abs(a), abs(b), x, y);
        if (c % g != 0) return false;
        x *= c / g;
        y *= c / g;
        if (a < 0) x = -x;
        if (b < 0) y = -y;
        return true;
    }

    // Solves  $a_1*x_1 + a_2*x_2 + \dots + a_n*x_n = c$  for N variables.
    // Returns true if a solution exists and populates x vector
    static bool solve_nvar(const vector<ll> &a, ll c, vector<ll> &x) {
        int n = a.size();
        if (n == 0) return c == 0;
        x.assign(n, 0);

        vector<ll> prefix_g(n);
        prefix_g[0] = abs(a[0]);
        for (int i = 1; i < n; i++) prefix_g[i] = std::gcd(prefix_g[i-1], a[i]);
        ↳ - 1];

        if (prefix_g.back() == 0) return c == 0;
        if (c % prefix_g.back() != 0) return false;

        ll cur_c = c;
        for (int i = n - 1; i > 0; i--) {
            ll x_prev, y_curr, g;
            solve_2var(prefix_g[i - 1], a[i], cur_c, x_prev, y_curr,
        ↳ g);
            x[i] = y_curr;
            cur_c = prefix_g[i - 1] * x_prev;
        }
        x[0] = cur_c / a[0];
        return true;
    }
};
```

4.1.3 FloorSum

EUCLIDEAN-LIKE ALGORITHM (AtCoder Library 'floor_sum').

```
// floorSum(n, m, a, b) = sum_{i=0}^{n-1} floor((a*i + b) / m)
//   ↳ in O(log m).
// This is THE tool for "sum over a lattice line": counting lattice
//   ↳ points under a line, sums of
// floor/mod expressions, and anything of the shape sum floor((a i +
//   ↳ b) / m) that a loop cannot do
// because n is up to 1e9 or more. Works like the Euclidean
//   ↳ algorithm: reduce a mod m, then SWAP
// the roles of the axes (count by rows instead of by columns) and
//   ↳ recurse.
// Requires n >= 0, m >= 1, a >= 0, b >= 0. Overflow: a*n + b must
//   ↳ fit in ll (use __int128 if not).
ll floorSum(ll n, ll m, ll a, ll b) {
    ll ans = 0;
    if (a >= m) ans += (n - 1) * n / 2 * (a / m), a %= m;
    if (b >= m) ans += n * (b / m), b %= m;
    ll ymax = (a * n + b) / m;
    if (ymax == 0) return ans;
    ll xmax = ymax * m - b;
    ans += ymax * (n - (xmax + a - 1) / a);
    return ans + floorSum(ymax, a, m, (a - xmax % a) % a);
}
// NEGATIVE a or b: shift them into range first.
// a < 0: let a2 = a mod m in [0,m); floorSum(n,m,a,b) =
//   ↳ floorSum(n,m,a2,b) - (a2-a)/m * (n-1)n/2
// b < 0: let b2 = b mod m in [0,m); subtract n * (b2-b)/m. (Same
//   ↳ trick, easier: use __int128.)
//
// WHAT IT SOLVES, directly:
// * #lattice points (x,y) with 1 <= x <= n and 1 <= y <= (a x +
//   ↳ b)/m   ↳ floorSum itself
// * sum_{i=0}^{n-1} ((a i + b) mod m) = (a * n(n-1)/2 + b*n) - m *
//   ↳ floorSum(n, m, a, b)
// * sum_{i=1}^{n} floor(n / i) in O(sqrt n) instead - use DIVISOR
//   ↳ BLOCKS (below), not this.
// * "how many multiples of q lie in [l, r]" = floor(r/q) -
//   ↳ floor((l-1)/q); the basic case.
//
// THE OTHER TWO SUMS of the same family (needed together often
//   ↳ enough to be worth knowing they
// exist): S1 = sum i * floor((a i + b)/m) and S2 = sum floor((a i +
//   ↳ b)/m)^2 obey the same
// recursion with a 3-tuple state - the "universal Euclidean
//   ↳ algorithm" generalises
// this to any monoid: you supply the two operators U ("move right")
//   ↳ and R ("move up") and it
// composes the word for the line in O(log) monoid multiplications.
//
// DIVISOR BLOCKS - the other floor trick, O(sqrt n):
// floor(n / i) is constant on maximal blocks; for a block starting
//   ↳ at l the block ends at
// r = n / (n / l). Use it for sum_{i=1}^{n} floor(n/i), sum of
//   ↳ tau/sigma, Du sieve, Mobius sums.
// for (ll l = 1, r; l <= n; l = r + 1) { r = n / (n / l); /*
//   ↳ value n/l on [l, r] */ }
// Two-argument version (blocks constant for BOTH n/i and m/i): r =
//   ↳ min(n/(n/l), m/(m/l)).
```

4.1.4 FloorSumExtended

EUCLIDEAN-LIKE ALGORITHM, THE FULL TRIPLE. With q_i = floor((a*i + b) / c) it returns

```
// f = sum_{i=0..n} q_i      g = sum_{i=0..n} i * q_i      h
//   ↳ = sum_{i=0..n} q_i^2
// all three at once in O(log c). NOTE i runs to n INCLUSIVE, unlike
//   ↳ floorSum in 03, which stops
// at n-1: floorSum(n, m, a, b) = fgh(n - 1, a, b, m).f.
// Reach for it when a lattice-line sum has an i or a square in it:
//   ↳ sum of i*floor, sum of
// (a i + b mod c)^2, the second moment of a modular AP, "sum over
//   ↳ (x + y)" and friends. The recursion is the same as floorSum: strip
//   ↳ a/c and b/c, then swap axes.
// Requires n >= -1, a >= 0, b >= 0, c >= 1. Exact (no modulus) in
//   ↳ __int128, which is good for
// n and m = (a n + b)/c up to ~1e11 (h is of order n*m^2). Take
//   ↳ everything mod p if they are
// bigger - the recursion only ever adds and multiplies, so it is
//   ↳ modulus-friendly.
struct FGH { __int128 f, g, h; };
FGH fgh(ll n, ll a, ll b, ll c) {
    if (n < 0) return {0, 0, 0};
    __int128 M = n, s1 = M * (M + 1) / 2, s2 = M * (M + 1) * (2 * M +
    ↳ 1) / 6;
    if (a >= c || b >= c) { // q_i = A*i
        ↳ + B + (reduced q_i)
        __int128 A = a / c, B = b / c;
        FGH t = fgh(n, a % c, b % c, c);
        return {t.f + A * s1 + B * (M + 1), t.g + A * s2 + B * s1,
        ↳ t.h + 2 * B * t.f + 2 * A * t.g + A * A * s2 + B * B
        ↳ * (M + 1) + 2 * A * B * s1};
    }
    __int128 m = ((__int128)a * n + b) / c; // the
    ↳ largest value any q_i takes
    if (m == 0) return {0, 0, 0};
    FGH t = fgh((ll)m - 1, c, c - b - 1, a); // count by
    ↳ rows instead of by columns
    __int128 f = M * m - t.f;
    return {f, (m * M * (M + 1) - t.f - t.h) / 2, M * m * (m + 1) - 2
    ↳ * t.g - 2 * t.f - f};
}
// DERIVED SUMS (r_i = (a*i + b) mod c = a*i + b - c*q_i):
// sum r_i = a*s1 + b*(n+1) - c*f
// sum r_i^2 = sum (a i + b)^2 - 2c * (a*g + b*f) + c^2 * h
// sum i*r_i = a*s2 + b*s1 - c*g
// The same recursion with a bigger state gives sum i^k1 * q_i^k2 for
//   ↳ any k1 + k2 <= K in
// O(K^3 log c) - rarely needed. The clean way to state the whole
//   ↳ family is the UNIVERSAL
// EUCLIDEAN ALGORITHM: walk the line as a word in {U = "step up", R
//   ↳ = "step right"} and
// multiply the letters in ANY monoid; supply U and R and you get
//   ↳ whatever statistic you like.
```

4.1.5 LatticeUnderLine

COUNT THE NONNEGATIVE LATTICE POINTS UNDER A LINE: $\#\{(x, y) : x, y \geq 0, a*x + b*y \leq c\}$

```
// in O(log min(a, b)) - a Euclidean-style descent that peels off a
//   ↳ full rectangle of solutions
// each round, never touching a single point individually.
// This is the "how many ways to pay at most c with coins a and b"
//   ↳ primitive, and the engine
// behind 06 - ModCountInRange.cpp. Requires a >= 1, b >= 1; returns
//   ↳ 0 for c < 0 (a = 0 or b = 0
// would be infinite - handle that case yourself).
// OVERFLOW: the answer is about c^2 / (2ab) and the intermediate
//   ↳ (f+1)*f/2*k is of the same
// order, so it must fit in ll; c up to ~4e18 is fine as long as the
//   ↳ count itself does.
ll latticeUnderLine(ll a, ll b, ll c) {
    if (c < 0) return 0;
    if (a > b) swap(a, b); // descend on
    ↳ the smaller coefficient
    ll ans = 0;
    while (c >= 0) { // b = k*a +
    ↳ l, then swap roles
        ll k = b / a, l = b % a, f = c / b, e = c % b / a, g = c % b
    ↳ % a;
        ans += (f + 1) * (e + 1) + (f + 1) * f / 2 * k; // the
    ↳ axis-aligned part
        c = f * l - a + g; // l = 0
    ↳ makes c < 0: the loop ends
        b = a, a = l;
    }
    return ans;
}
// VARIANTS, all from this one call:
// a*x + b*y < c -> latticeUnderLine(a, b, c - 1)
// a*x + b*y <= c, x >= 1, y >= 1 -> latticeUnderLine(a, b, c - a
//   ↳ - b)
// a*x + b*y == c exactly -> latticeUnderLine(a,b,c) -
//   ↳ latticeUnderLine(a,b,c-1), or solve the
// Diophantine equation and count the admissible k directly (02 -
//   ↳ LinearDiophantine.cpp).
// x in [0, X] as well -> subtract the shifted count
//   ↳ latticeUnderLine(a, b, c - a*(X+1)).
// Related: sum_{x} floor((c - a*x)/b) is the same quantity as a
//   ↳ floor sum, so 03 - FloorSum.cpp
// also answers it in O(log) - use whichever shape the problem hands
//   ↳ you.
```

4.1.6 ModCountInRange

HOW MANY x IN $[xl, xr]$ HAVE $(a*x \bmod m)$ INSIDE $[cl, cr]$? $O(\log m)$, any $a, m \geq 1$.

```
// Needs: latticeUnderLine (05 - LatticeUnderLine.cpp).
// The classic use: "how many of the first n multiples of a land in a
//   ↳ window mod m" - collision
// counting, clock/lap problems, counting i with (a*i + b) mod m <= c
//   ↳ (shift by b: replace the
// window [cl, cr] by [cl - b, cr - b] reduced mod m, splitting it if
//   ↳ it wraps past 0).
// modCountLE counts 0 <= x <= n with (a*x mod m) <= c; everything
//   ↳ else is inclusion-exclusion
// on that. OVERFLOW: latticeUnderLine is called with c ~ 2*m*n, so
//   ↳ m*n must stay below ~4e18
// (use __int128 inside latticeUnderLine if it does not).
ll modCountLE(ll a, ll m, ll c, ll n) { // count 0 <=
    ↳ x <= n with a*x mod m <= c
    if (n < 0) return 0;
    ++c, ++n; // work with
    ↳ half-open [0, c) and [0, n)
    a %= m; if (a < 0) a += m;
    ll ec = c / m; c %= m; if (c < 0) ec--, c += m; // c outside
    ↳ [0, m) means whole periods
    ll ans = ec * n, en = n / m; n %= m;
    if (n < 0) en--, n += m;
    auto blk = [&](ll len) { // x in [0,
    ↳ len): len < m, uses a + m > 0
        return latticeUnderLine(m, a + m, (a + m) * (len - 1))
            - latticeUnderLine(m, a + m, (a + m) * (len - 1) - c);
    };
    if (en) ans += en * blk(m); // en full
    ↳ periods of length m
    if (n) ans += blk(n);
    return ans;
}
ll modCountRange(ll a, ll m, ll cl, ll cr, ll xl, ll xr) {
    return modCountLE(a, m, cr, xr) - modCountLE(a, m, cr, xl - 1)
        - modCountLE(a, m, cl - 1, xr) + modCountLE(a, m, cl - 1, xl
    ↳ - 1);
}
// SANITY CHECK: with g = gcd(a, m) the values a*x mod m are exactly
//   ↳ the multiples of g, and over
// one full period x in [0, m) each of them is hit exactly g times.
//   ↳ So modCountRange over a full
// period must equal g * (number of multiples of g inside [cl, cr]).
// COMPANION ROUTINES: the smallest such x is 07 -
//   ↳ SmallestModInRange.cpp; if you need the whole
// set of x, note that it is a union of O(sqrt m) arithmetic
//   ↳ progressions
// (03 - Modular Arithmetic/09 - ModularAPBlocks.cpp).
```

4.1.7 SmallestModInRange

SMALLEST $x \geq 0$ WITH $l \leq (a \cdot x \bmod p) \leq r$, or -1 if no such x exists. $O(\log p)$.

```
// The "when does the ticking counter first land in the window"
//   ↳ problem: jumps of size a on a
// circle of circumference p, first time inside the arc [l, r]. Also
//   ↳ solves "smallest x with
// a*x mod p <= r" (l = 0) and, by symmetry, the first time OUTSIDE a
//   ↳ window.
// PRECONDITIONS: 0 <= a < p, 0 <= l <= r < p, p >= 1. p need NOT be
//   ↳ prime.
// Idea: if ceil(l/a)*a <= r we are done in one step. Otherwise write
//   ↳ p = k*a + b; asking for
// l <= a*x - p*y <= r turns into the same question one Euclidean
//   ↳ step down, with the roles of
// the modulus and the multiplier swapped - so the recursion depth is
//   ↳ O(log p), like gcd.
// OVERFLOW: the internal products are up to p*p, hence the __int128.
ll smallestModIn(ll p, ll a, ll l, ll r) {
    if (a == 0) return l == 0 ? 0 : -1;
    auto cdiv = [](__int128 x, __int128 y) { return (ll)((x + y - 1)
        ↳ / y); };
    ll c = cdiv(l, a); // first x
    ↳ with a*x >= l, ignoring wraps
    if ((__int128)a * c <= r) return c;
    ll b = p % a; // p =
    ↳ (p/a)*a + b
    ll y = smallestModIn(a, b, a - r % a, a - l % a); // -r <= b*y
    ↳ (mod a) <= -l
    return y == -1 ? -1 : cdiv((__int128)l + (__int128)b * y, a) + p
        ↳ / a * y;
}
// COMPANIONS
// count of such x in a range -> 06 - ModCountInRange.cpp
// the whole solution set -> it is a union of O(sqrt
//   ↳ p) APs, see
// 03 - Modular
//   ↳ Arithmetic/09 - ModularAPBlocks.cpp
// smallest x with a*x mod p == target -> solve the linear
//   ↳ congruence instead
// 03 - Modular
//   ↳ Arithmetic/04 - CongruencesAndTowers.cpp)
// TYPICAL USE: interactive/recovery problems where you know x mod M
//   ↳ for a huge M and want the
// smallest representative that also satisfies 1 <= q <= N; and
//   ↳ "minimum of (k*a mod p) over
// k <= n", which the three-distance theorem says is achieved at a
//   ↳ continued-fraction denominator.
```

4.1.8 CoinReachability

WHICH TOTALS ARE PAYABLE WITH COINS $a[0..n-1]$? (all $x_i \geq 0$, unbounded supply)

```
// sum a_i x_i = k solvable? how many solvable k in [l, r]?
//   ↳ largest unsolvable k?
// THE TRICK ("congruence shortest path" / Dijkstra on residues): fix
//   ↳ A = min a_i and build a
// graph on the residues mod A with an edge u -> (u + a_i) mod A of
//   ↳ weight a_i. Then
// d[u] = the SMALLEST payable value congruent to u (mod A)
// and k is payable iff d[k mod A] <= k. So every residue class
//   ↳ contributes the arithmetic
// progression d[u], d[u] + A, d[u] + 2A, ... and everything below
//   ↳ d[u] in that class is not.
// Build O(A * n * log A) time, O(A) memory. Take A = min a_i (that
//   ↳ is what bounds the table).
// PRECONDITIONS: all a_i >= 1. Values may be up to 1e18; A up to a
//   ↳ few million is the real cap.
struct CoinReach {
    ll A;
    vector<ll> d; // d[u] = min
    ↳ payable value = u (mod A)
    CoinReach(vector<ll> a) {
        A = *min_element(all(a));
        d.assign(A, llinf);
        d[0] = 0;
        priority_queue<pair<ll, ll>, vector<pair<ll, ll>>, greater<>>
        ↳ pq;
        pq.push({0, 0});
        while (!pq.empty()) {
            auto [val, u] = pq.top(); pq.pop();
            if (val != d[u]) continue;
            for (ll c : a) {
                ll nv = val + c, v = nv % A;
                if (nv < d[v]) d[v] = nv, pq.push({nv, v});
            }
        }
        bool payable(ll k) { return k >= 0 && d[k % A] <= k; }
        ll count(ll l, ll r) { // #payable k
            in [l, r], l >= 0
            ll res = 0;
            for (ll u = 0; u < A; u++) {
                if (d[u] <= r) res += (r - d[u]) / A + 1;
                if (d[u] <= l - 1) res -= (l - 1 - d[u]) / A + 1;
            }
            return res;
        }
        ll frobenius() { // largest
            ↳ UNpayable value, or -1
            ll mx = 0; // requires
            ↳ gcd(a_i) = 1, else infinite
            for (ll u = 1; u < A; u++) mx = max(mx, d[u]);
            return mx == 0 ? -1 : mx - A;
        }
    };
// FROBENIUS / CHICKEN McNUGGET closed forms exist only for two
//   ↳ coins: with gcd(a,b) = 1 the
// largest unpayable value is a*b - a - b and exactly (a-1)(b-1)/2
//   ↳ values are unpayable.
// For three or more coins there is no formula - this table IS the
//   ↳ algorithm.
// SANITY: if gcd of all a_i is g > 1, only multiples of g are ever
//   ↳ payable; divide everything
// by g first, otherwise frobenius() is meaningless (infinitely many
//   ↳ unpayable values).
// SAME TABLE ANSWERS: smallest payable value >= k (that is d[u] +
//   ↳ ceil((k - d[u])/A)*A minimised
// over u), and "can I pay exactly k with at most/at least c coins"
//   ↳ if you make d 2-dimensional.
```


4.1.9 ThreeVarDiophantine

NUMBER OF NONNEGATIVE SOLUTIONS OF $a*x + b*y + c*z = n$, in $O(\log \max(a,b,c))$.

```
// (Popoviciu's formula for two coins, extended to three by Binner -
// three-coin system.) Reach for it when a, b, c and n are all up to
// 1e18 and the DP over n or
// the  $O(n / \max)$  loop over one variable are both hopeless. For two
// coins use the much simpler
// count of lattice points under a line, 05 - LatticeUnderLine.cpp.
// PRECONDITIONS: a, b, c >= 1, n >= 0. Handles arbitrary gcds (the
// wrapper strips them).
// All arithmetic is __int128 because the numerator N1 is of order
//  $n^2$  and the products  $a*b*c$  and
//  $c*b*b1p$  reach  $\max(a,b,c)^3$ ; keep a, b, c below  $\sim 1e12$  and n below
//  $\sim 1e18$ .
__int128 gcd128(__int128 a, __int128 b) {
    while (b) { __int128 t = a % b; a = b, b = t; }
    return a;
}
__int128 xgcd128(__int128 a, __int128 b, __int128 &x, __int128 &y) {
    if (!b) { x = 1, y = 0; return a; }
    __int128 x1, y1, d = xgcd128(b, a % b, x1, y1);
    x = y1, y = x1 - y1 * (a / b);
    return d;
}
__int128 inv128(__int128 a, __int128 m) { // m >= 1,
    // gcd(a, m) = 1
    __int128 x, y; xgcd128((a % m) + m, m, x, y);
    return ((x % m) + m) % m;
}
__int128 fsum128(__int128 n, __int128 m, __int128 a, __int128 b) {
    // = floorSum, in __int128
    __int128 ans = 0;
    if (a >= m) ans += (n - 1) * n / 2 * (a / m), a %= m;
    if (b >= m) ans += n * (b / m), b %= m;
    __int128 ymax = (a * n + b) / m;
    if (ymax == 0) return ans;
    __int128 xmax = ymax * m - b;
    ans += ymax * (n - (xmax + a - 1) / a);
    return ans + fsum128(ymax, a, m, (a - xmax % a) % a);
}
__int128 denum3coprime(__int128 a, __int128 b, __int128 c, __int128
    // n) { // a,b,c PAIRWISE coprime
    auto fix = [ ](__int128 v, __int128 m) { return v == 0 ? m : v; };
    __int128 b1 = fix((-n % a * inv128(b, a) % a + a) % a, a);
    __int128 c1 = fix(b % a * inv128(c, a) % a, a);
    __int128 c2 = fix((-n % b * inv128(c, b) % b + b) % b, b);
    __int128 a2 = fix(c % b * inv128(a, b) % b, b);
    __int128 a3 = fix((-n % c * inv128(a, c) % c + c) % c, c);
    __int128 b3 = fix(a % c * inv128(b, c) % c, c);
    __int128 N1 = n * (n + a + b + c);
    N1 += c * b * b1 * (a + 1 - c1 * (b1 - 1));
    N1 += a * c * c2 * (b + 1 - a2 * (c2 - 1));
    N1 += b * a * a3 * (c + 1 - b3 * (a3 - 1));
    return N1 / (2 * a * b * c) + fsum128(b1, a, c1, 0) + fsum128(c2,
    // b, a2, 0)
    + fsum128(a3, c, b3, 0) - 2;
}
__int128 denum3(__int128 a, __int128 b, __int128 c, __int128 n) {
    __int128 g = gcd128(gcd128(a, b), c);
    if (n % g) return 0;
    a /= g, b /= g, c /= g, n /= g;
    __int128 g1 = gcd128(b, c), g2 = gcd128(c, a), g3 = gcd128(a, b);
    __int128 n1 = n * inv128(a, g1) % g1, n2 = n * inv128(b, g2) %
    // g2, n3 = n * inv128(c, g3) % g3;
    return denum3coprime(a / g2 / g3, b / g3 / g1, c / g1 / g2,
    // (n - a * n1 - b * n2 - c * n3) / g1 / g2 /
    // g3);
}
// TWO COINS, for reference: #solutions of  $a*x + b*y = n$  with
// gcd(a,b) = 1 and x, y >= 0 is
//  $n/(a*b) - \{n * \text{inv}(a) \bmod b\} - \{n * \text{inv}(b) \bmod a\} + 1$ 
// (Popoviciu),
// i.e. floor( $n/(ab)$ ) or that plus one; the largest n with NO
// solution is  $a*b - a - b$ .
// FOUR OR MORE coins: no such formula. Use 08 - CoinReachability.cpp
// (needs min  $a_i$  small), or
// a generating-function / polynomial approach when n is small.
```

4.1.10 IntegerLattice2D

A SYSTEM OF TWO LINEAR DIOPHANTINE EQUATIONS IN ANY NUMBER OF UNKNOWNNS:

```
// Needs: extGCD (01 - ExtendedEuclid.cpp).
// sum  $x_i * a_i = p$  and  $\sum x_i * b_i = q$ ,  $x_i$  ANY integers
// - is it solvable?
// Equivalently: is (p, q) in the subgroup of  $\mathbb{Z}^2$  generated by the
// vectors (a_i, b_i)? Keep that
// lattice in Hermite normal form as two vectors (g, s) and (0, t):
// every lattice point is
//  $m*(g, s) + k*(0, t)$ , with  $g \geq 0$ ,  $t \geq 0$  and  $0 \leq s < t$  when  $t > 0$ .
// Adding a generator is one
// extended-gcd step, so this is ONLINE - generators may arrive
// interleaved with queries, which
// is what makes it more than "solve a  $2 \times N$  system once".
// add() is  $O(\log)$ , contains() is  $O(1)$ . No preconditions; zero
// vectors and rank 0/1 are handled.
// OVERFLOW: add() forms  $s*(c/g)$  and  $d*(g/g')$ , so keep coordinates
// below  $\sim 2e9$  or switch to __int128.
struct Lattice2D {
    ll g = 0, s = 0, t = 0; // t = 0
    // means "no free second direction"
    void add(ll c, ll d) { // insert the
        // generator (c, d)
        if (c < 0) c = -c, d = -d; // (c, d) and
        // its negative are equivalent
        if (c == 0) { // pure
            // vertical generator
            t = gcd(t, d < 0 ? -d : d);
            if (t) s = ((s % t) + t) % t;
            return;
        }
        if (g == 0) { g = c, s = t ? ((d % t) + t) % t : d; return; }
        ll x, y, ng = extGCD(g, c, x, y); // ng =
        // gcd(g, c) =  $g*x + c*y$ 
        ll cross = s * (c / ng) - d * (g / ng); // the
        // leftover pure-vertical direction
        t = gcd(t, cross < 0 ? -cross : cross);
        ll ns = s * x + d * y;
        g = ng, s = t ? ((ns % t) + t) % t : ns;
    }
    bool contains(ll p, ll q) { // is (p, q)
        // representable?
        if (g == 0) return p == 0 && (t ? q % t == 0 : q == 0);
        if (p % g) return false;
        ll rem = q - (p / g) * s;
        return t ? rem % t == 0 : rem == 0;
    }
};
// WHY IT WORKS: extGCD writes  $(ng, ns) = x*(g, s) + y*(c, d)$  with ng
// = gcd(g, c), the smallest
// positive first coordinate achievable. What remains is the
// sublattice with first coordinate 0,
// generated by  $s*(c/ng) - d*(g/ng)$  - the determinant of the two old
// generators divided by ng.
// THE SAME OBJECT in higher dimension is the Hermite normal form of
// the generator matrix; with
// one row it degenerates to gcd, and 02 - LinearDiophantine.cpp is
// the single-equation version.
// TYPICAL USES: "can I reach (p, q) using these moves any integer
// number of times", reachability
// with two conserved quantities, and answering many (p, q) queries
// against a growing move set.
// NOTE this allows NEGATIVE  $x_i$ . Requiring  $x_i \geq 0$  is a completely
// different (much harder)
// problem - see 08 - CoinReachability.cpp for the one-equation
// nonnegative version.
```

4.2 Primes and Divisors

4.2.1 Sieve

Linear Sieve & Smallest Prime Factor (SPF) - Time: $O(N)$, Space: $O(N)$

```
// Computes primes list, SPF array, primality test, and  $O(\log N)$ 
// prime factorization
struct Sieve {
    int n;
    vector<int> primes;
    vector<int> spf;

    Sieve(int n = 1e7) : n(n), spf(n + 1, 0) {
        for (int i = 2; i <= n; i++) {
            if (spf[i] == 0) {
                spf[i] = i;
                primes.pb(i);
            }
            for (int p : primes) {
                if (p > spf[i] || (ll)i * p > n) break;
                spf[i * p] = p;
            }
        }

        bool is_prime(int x) const {
            return x >= 2 && x <= n && spf[x] == x;
        }

        //  $O(\log N)$  prime factorization using precomputed SPF array
        vector<int> get_prime_factors(int x) const {
            vector<int> factors;
            while (x > 1) {
                factors.pb(spf[x]);
                x /= spf[x];
            }
            return factors;
        }
    };
};
```

4.2.2 Factorization

Trial Division Prime Factorization - Time: $O(\sqrt{N})$

```
template <typename T = ll>
struct Factorization {
    static vector<T> get_prime_factors(T n) {
        vector<T> factors;
        for (T i = 2; i * i <= n; i++) {
            while (n % i == 0) {
                factors.pb(i);
                n /= i;
            }
        }
        if (n > 1) factors.pb(n);
        return factors;
    }
};
```

4.2.3 Divisors

Trial Division Divisor Generation - Time: $O(\sqrt{N})$

```
template <typename T = ll>
struct Divisors {
    static vector<T> get_divisors(T n) {
        vector<T> divs;
        for (T i = 1; i * i <= n; i++) {
            if (n % i == 0) {
                divs.pb(i);
                if (n / i != i) divs.pb(n / i);
            }
        }
        return divs;
    }
};
```

4.2.4 Sieve1e9

Fast Bitset Sieve up to $N = 10^9$ - Time: $\sim 0.3s$ for 10^9 , Space: $N / 16$ bytes ($\sim 60MB$)

```
// Stores odd numbers only (index i represents number  $2*i + 1$ )
// WARNING: the bitset is  $\sim 60$  MB - the instance MUST be
// global/static, never a local.
template <int MAXPR = (int)1e9>
struct Sieve1e9 {
    bitset<MAXPR / 2> is_prime_odd;

    Sieve1e9(int n = MAXPR) {
        is_prime_odd.set();
        is_prime_odd[0] = 0; // 1 is not prime
        for (int i = 1; (2 * i + 1) * (2 * i + 1) <= n; i++) {
            if (is_prime_odd[i]) {
                int p = 2 * i + 1;
                // bound is  $(n-1)/2$ , NOT  $n/2$ :  $n/2$  writes one index
                // past the end of the bitset
                for (int j = 2 * i * (i + 1); j <= (n - 1) / 2; j +=
                    p) {
                        is_prime_odd[j] = 0;
                    }
            }
        }

        bool is_prime(int x) const {
            if (x == 2) return true;
            if (x < 2 || x % 2 == 0) return false;
            return is_prime_odd[x / 2];
        }
    };
};
```

4.2.5 SegmentedSieve

Segmented Sieve for Range $[L, R]$ - Time: $O(\sqrt{R}) + (R - L) \log \log R$, Space: $O(\sqrt{R}) + R - L$

```
// Computes all prime numbers in range  $[L, R]$  where  $R \leq 1e12$  and  $(R - L) \leq 1e7$ 
struct SegmentedSieve {
    static vector<ll> get_primes_in_range(ll L, ll R) {
        ll limit = sqrtl((ld)R);
        while (limit * limit > R) limit--;
        while ((limit + 1) * (limit + 1) <= R) limit++;
        vector<bool> is_prime_small(limit + 1, true);
        vector<ll> primes;
        for (ll i = 2; i <= limit; i++) {
            if (is_prime_small[i]) {
                primes.pb(i);
                for (ll j = i * i; j <= limit; j += i)
                    is_prime_small[j] = false;
            }
        }

        vector<bool> is_prime_range(R - L + 1, true);
        for (ll p : primes) {
            ll start = max(p * p, (L + p - 1) / p * p);
            for (ll j = start; j <= R; j += p) {
                is_prime_range[j - L] = false;
            }
        }

        for (ll x = L; x <= min(R, 1LL); x++) is_prime_range[x - L] =
            false; // 0 and 1

        vector<ll> result;
        for (ll i = 0; i <= R - L; i++) {
            if (is_prime_range[i]) result.pb(L + i);
        }
        return result;
    }
};
```

4.2.6 MillerRabinPollard

Primality + factorization for n up to $1e18$. $\text{isPrime } O(\log^3 n)$, $\text{factor } O(n^{1/4})$.

```
// THE gate for any problem with a_i up to 1e18 - divisors, phi, tau,
// → sigma all follow from factor().
typedef unsigned long long u64;
u64 mulmod(u64 a, u64 b, u64 m) { return (__int128)a * b % m; }
u64 powmod(u64 a, u64 e, u64 m) {
    u64 r = 1; a %= m;
    for (; e >= 1, a = mulmod(a, a, m)) if (e & 1) r = mulmod(r,
    → a, m);
    return r;
}
// Deterministic for all n < 3.3e24 with these 7 bases (Miller-Rabin).
bool isPrime(u64 n) {
    if (n < 2 || n % 6 % 4 != 1) return (n | 1) == 3;
    u64 d = n - 1; int s = __builtin_ctzll(d); d >>= s;
    for (u64 a : {2, 325, 9375, 28178, 450775, 9780504, 1795265022}) {
        u64 p = powmod(a % n, d, n); int i = s;
        while (p != 1 && p != n - 1 && a % n && i--) p = mulmod(p, p,
    → n);
        if (p != n - 1 && i != s) return false;
    }
    return true;
}
u64 pollard(u64 n) { // returns a
    → non-trivial divisor (Brent)
    auto f = [n](u64 x) { return mulmod(x, x, n) + 1; };
    u64 x = 0, y = 0, t = 30, prd = 2, i = 1, q;
    while (t++ % 40 || gcd(prd, n) == 1) {
        if (x == y) x = ++i, y = f(x);
        if ((q = mulmod(prd, x > y ? x - y : y - x, n))) prd = q;
        x = f(x), y = f(f(y));
    }
    return gcd(prd, n);
}
vector<u64> factor(u64 n) { // unsorted
    → multiset of primes
    if (n <= 1) return {};
    → factor(0) would loop forever
    if (isPrime(n)) return {n};
    u64 x = pollard(n);
    auto l = factor(x), r = factor(n / x);
    l.insert(l.end(), all(r));
    return l;
}
// {prime, exponent} pairs, sorted
vector<pair<u64, int>> factorize(u64 n) {
    auto f = factor(n); sort(all(f));
    vector<pair<u64, int>> r;
    for (u64 p : f) (!r.empty() && r.back().first == p) ?
    → r.back().second++ : (r.push_back({p, 1}), 0);
    return r;
}
vector<u64> divisors(u64 n) { // all
    → divisors, unsorted
    vector<u64> d = {1};
    for (auto [p, e] : factorize(n)) {
        int sz = d.size();
        for (u64 pe = 1, k = 0; k < (u64)e; k++) {
            pe *= p;
            for (int i = 0; i < sz; i++) d.push_back(d[i] * pe);
        }
    }
    return d;
}
u64 phi(u64 n) { u64 r = n; for (auto [p, e] : factorize(n)) r -= r /
    → p; return r; } // Euler totient
u64 tau(u64 n) { u64 r = 1; for (auto [p, e] : factorize(n)) r *= e +
    → 1; return r; } // #divisors
u64 sigma(u64 n) {
    → // sum of divisors
    u64 r = 1;
    for (auto [p, e] : factorize(n)) { u64 t = 1, pe = 1; for (int i
    → = 0; i < e; i++) t += (pe *= p); r *= t; }
    return r;
}
```

4.2.7 MultiplicativeSieve

Linear sieve computing ANY multiplicative function for all $i \leq n$ in $O(N)$.

```
// Shipped: spf, phi, mu, tau (#divisors), sigma (sum of divisors).
// To add your own f: you need f(p), f(p^k) and f(a*b)=f(a)f(b) for
    → coprime a,b.
struct MultSieve {
    int n;
    vector<int> spf, primes, phi, mu, tau, cnt; // cnt[i] =
    → exponent of spf[i] in i
    vector<ll> sigma, pw; // pw[i] =
    → spf[i]^cnt[i]

    MultSieve(int n) : n(n), spf(n + 1), phi(n + 1), mu(n + 1), tau(n
    → + 1), cnt(n + 1),
        sigma(n + 1), pw(n + 1) {
        phi[1] = mu[1] = tau[1] = sigma[1] = pw[1] = 1;
        for (int i = 2; i <= n; i++) {
            if (!spf[i]) {
                spf[i] = i, primes.push_back(i);
                phi[i] = i - 1, mu[i] = -1, tau[i] = 2, cnt[i] = 1,
    → pw[i] = i, sigma[i] = i + 1LL;
            }
            for (int p : primes) {
                if (p > spf[i] || (ll)i * p > n) break;
                int j = i * p;
                spf[j] = p;
                if (p == spf[i]) { // p divides
    → i: same prime, exponent grows
                    cnt[j] = cnt[i] + 1, pw[j] = pw[i] * p;
                    phi[j] = phi[i] * p;
                    mu[j] = 0;
                    tau[j] = tau[i] / (cnt[i] + 1) * (cnt[j] + 1);
                    sigma[j] = sigma[i] / ((pw[i] * p - 1) / (p - 1))
    → * ((pw[j] * p - 1) / (p - 1));
                } else { // coprime:
    → multiply
                    cnt[j] = 1, pw[j] = p;
                    phi[j] = phi[i] * (p - 1);
                    mu[j] = -mu[i];
                    tau[j] = tau[i] * 2;
                    sigma[j] = sigma[i] * (p + 1);
                }
            }
        }
    }
    vector<int> factors(int x) const { // O(log x)
    → via spf
        vector<int> f;
        while (x > 1) f.push_back(spf[x]), x /= spf[x];
        return f;
    }
};
```

4.2.8 PrimeEnumeration1e9

ENUMERATE EVERY PRIME UP TO $1e9$ IN $\sim 0.5s$, and in a 30-smooth wheel that costs 1 BIT per 30

```
// integers (so  $\sim 33$  MB at  $n = 1e9$  versus  $\sim 60$  MB for the odds-only
//  $\hookrightarrow$  bitset in 04 - Sieve1e9.cpp).
// Credit: min_25. Use THIS when you need the primes themselves at  $n$ 
//  $\hookrightarrow \geq 1e8$ ; use 04 when you only
// need the primality bitmap and want six lines instead of sixty.
// HOW IT IS FAST: (1) a mod-30 wheel - of the 30 residues only
//  $\hookrightarrow \{1, 7, 11, 13, 17, 19, 23, 29\}$  can be
// prime, so one byte covers 30 integers; (2) the pattern of the
//  $\hookrightarrow$  small primes ( $\leq Q$ ) is computed
// ONCE into 'pre' and memcpy'd over each block; (3) the sieve runs
//  $\hookrightarrow$  block by block so everything
// stays in L1/L2 cache, with each prime remembering where it left
//  $\hookrightarrow$  off in each of the 8 residues.
vector<int> sievePrimes(const int N, const int Q = 17, const int L =
//  $\hookrightarrow 1 \ll 15$ ) {
    static const int rs[] = {1, 7, 11, 13, 17, 19, 23, 29};
    struct P { P(int p) : p(p) {} int p; int pos[8]; };
    auto approxPrimeCount = [](const int N) -> int {
        return N > 60184 ? N / (log(N) - 1.1) : max(1., N / (log(N) -
//  $\hookrightarrow 1.11$ )) + 1;
    };
    const int v = sqrt(N), vv = sqrt(v);
    vector<bool> isp(v + 1, true);
    for (int i = 2; i <= vv; ++i) if (isp[i])
        for (int j = i * i; j <= v; j += i) isp[j] = false;
    const int rsize = approxPrimeCount(N + 30);
    vector<int> primes = {2, 3, 5};
    int psize = 3;
    primes.resize(rsize);
    vector<P> sp;
    size_t pbeg = 0;
    int prod = 1;
    for (int p = 7; p <= v; ++p) {
        if (!isp[p]) continue;
        if (p <= Q) prod *= p, ++pbeg, primes[psize++] = p;
        auto pp = P(p);
        for (int t = 0; t < 8; ++t) {
            int j = (p <= Q) ? p : p * p;
            while (j % 30 != rs[t]) j += p << 1;
            pp.pos[t] = j / 30;
        }
        sp.push_back(pp);
    }
    vector<unsigned char> pre(prod, 0xFF); // the
//  $\hookrightarrow$  small primes' pattern, once
    for (size_t pi = 0; pi < pbeg; ++pi) {
        auto pp = sp[pi];
        const int p = pp.p;
        for (int t = 0; t < 8; ++t) {
            const unsigned char m = ~(1 << t);
            for (int i = pp.pos[t]; i < prod; i += p) pre[i] &= m;
        }
    }
    const int bs = (L + prod - 1) / prod * prod;
    vector<unsigned char> block(bs);
    unsigned char* pb = block.data();
    const int M = (N + 29) / 30;
    for (int beg = 0; beg < M; beg += bs, pb += bs) {
        int end = min(M, beg + bs);
        for (int i = beg; i < end; i += prod) copy(pre.begin(),
//  $\hookrightarrow$  pre.end(), pb + i);
        if (beg == 0) pb[0] &= 0xFE; // 1
//  $\hookrightarrow$  is not prime
        for (size_t pi = pbeg; pi < sp.size(); ++pi) {
            auto& pp = sp[pi];
            const int p = pp.p;
            for (int t = 0; t < 8; ++t) {
                int i = pp.pos[t];
                const unsigned char m = ~(1 << t);
                for (; i < end; i += p) pb[i] &= m;
                pp.pos[t] = i; //
//  $\hookrightarrow$  resume here in the next block
            }
        }
        for (int i = beg; i < end; ++i)
            for (int m = pb[i]; m > 0; m &= m - 1)
                primes[psize++] = i * 30 + rs[__builtin_ctz(m)];
    }
    while (psize > 0 && primes[psize - 1] > N) --psize;
    primes.resize(psize);
    return primes;
}
// pi(1e9) = 50847534, so the output vector alone is  $\sim 200$  MB of int -
//  $\hookrightarrow$  if you only need to ITERATE
```

```
// the primes, sieve block by block and consume them inside the loop
//  $\hookrightarrow$  instead of storing them.
// Q = 17 means the wheel pre-pattern covers  $7 \cdot 11 \cdot 13 \cdot 17 = 17017$ 
// bytes; raising Q makes 'pre'
// bigger but the inner loop shorter. L =  $1 \ll 15$  keeps a block inside
//  $\hookrightarrow$  L1.
// FOR PRIMALITY OF ONE NUMBER up to  $1e18$  use Miller-Rabin (06); for
//  $\hookrightarrow$  the FACTORISATION of one
// number use Pollard rho (06); for pi(n) without the primes use
//  $\hookrightarrow$  Lucy_Hedgehog (04 - Theorems/03).
```

4.2.9 Min25Sieve

MIN_25 SIEVE - prefix sum of ANY multiplicative f in $O(n^{3/4} / \log n)$, n up to $\sim 1e10$ - $1e11$.

```
// Needs: Mint (06 - Math/01 - Modular Arithmetic/01 - Mint.cpp).
// You must supply two things:
// c[] : f(p) written as a POLYNOMIAL in p, f(p) = c[0] + c[1]*p +
//       c[2]*p^2 (degree <= 2 here,
//       extend pref() with more power sums for higher degree),
// fpe : f(p^e) for a prime power, given (p, e, p^e).
// PHASE 1 sieves sum of p^k over primes <= v for every v =
//       floor(n/i) (this is exactly Lucy's
// method, 04 - Theorems/03 - PrimeCounting.cpp). PHASE 2 (S) walks
//       every "smallest prime factor"
// class and re-adds the prime-power contributions. Memory O(sqrt n *
//       deg).
// EXAMPLES: phi -> c = {-1, 1}, fpe = p^e - p^(e-1); mu -> c =
//       {-1}, fpe = (e == 1 ? -1 : 0);
// tau -> c = {2}, fpe = e + 1; Id -> c = {0,
//       1}, fpe = p^e;
// f(p^e) = p^e * (p^e - 1) -> c = {0, -1, 1} (that is f(p)
//       = p^2 - p).
// If f is COMPLETELY multiplicative, phase 1 alone is the answer
//       (use Lucy directly).
struct Min25 {
    ll n, rt;
    int np;
    vector<ll> pr, w; // primes <=
    // sqrt(n); values floor(n/i)
    vector<int> id1, id2; // position
    // of v in w, for v <= rt / v > rt
    vector<vector<Mint>> g, sp; // g[k][j]
    // over w[j]; sp[k][i] = prefix p^k
    vector<Mint> c;
    function<Mint(ll, int, ll)> fpe;

    int id(ll x) { return x <= rt ? id1[x] : id2[n / x]; }
    Mint pref(ll x, int k) { //
    // sum_{i=2..x} i^k (mod MOD)
    Mint z = x % MOD;
    if (k == 0) return z - 1;
    if (k == 1) return z * (z + 1) / 2 - 1;
    return z * (z + 1) * (2 * z + 1) / 6 - 1;
    }

    Min25(ll n, vector<Mint> c, function<Mint(ll, int, ll)> fpe) :
    // n(n), c(c), fpe(fpe) {
    rt = (ll)sqrtl((ld)n);
    while (rt * rt > n) rt--;
    while ((rt + 1) * (rt + 1) <= n) rt++;
    int d = c.size();
    vector<char> comp(rt + 2, 0);
    for (ll i = 2; i <= rt; i++)
        if (!comp[i]) { pr.push_back(i); for (ll j = i * i; j <=
    // rt; j += i) comp[j] = 1; }
    np = pr.size();
    sp.assign(d, vector<Mint>(np + 1, 0));
    for (int i = 0; i < np; i++) {
        Mint t = 1;
        for (int k = 0; k < d; k++) sp[k][i + 1] = sp[k][i] + t,
    // t *= pr[i] % MOD;
    }
    id1.assign(rt + 2, 0), id2.assign(rt + 2, 0);
    for (ll i = 1, j; i <= n; i = j + 1) { // w is
    // DECREASING: w[0] = n, back = 1
    ll x = n / i; j = n / x;
    w.push_back(x);
    (x <= rt ? id1[x] : id2[n / x]) = w.size() - 1;
    }
    g.assign(d, vector<Mint>(w.size()));
    for (int k = 0; k < d; k++)
        for (int j = 0; j < (int)w.size(); j++) g[k][j] =
    // pref(w[j], k);
    for (int i = 0; i < np; i++) // sieve out
    // multiples of pr[i]
    for (int j = 0; j < (int)w.size() && pr[i] <= w[j] /
    // pr[i]; j++) {
    int u = id(w[j] / pr[i]);
    Mint pk = 1;
    for (int k = 0; k < d; k++) g[k][j] -= pk * (g[k][u]
    // - sp[k][i]), pk *= pr[i] % MOD;
    }
    }
    Mint S(ll x, int j) { // sum of
    // f(i), i <= x, spf(i) > pr[j-1]
    if (j >= 1 && pr[j - 1] >= x) return 0;
    Mint res = 0;
    for (int k = 0; k < (int)c.size(); k++) res += c[k] *
    // (g[k][id(x)] - sp[k][j]);
    }
```

```
for (int i = j; i < np && pr[i] <= x / pr[i]; i++) {
    ll pw = pr[i];
    for (int e = 1; pw <= x; e++) {
        res += fpe(pr[i], e, pw) * (S(x / pw, i + 1) + Mint(e
    // != 1));
        if (pw > x / pr[i]) break;
        pw *= pr[i];
    }
    return res;
}

Mint solve() { return n < 1 ? Mint(0) : S(n, 0) + 1; } // the
// +1 is f(1)
};
// The "+ Mint(e != 1)" term is f(p^e) counted for the number p^e
// itself (only for e >= 2, since
// e = 1 is already inside the prime sum). No memoisation is needed
// anywhere - that is the point.
// SIBLINGS: 10 - PowerfulNumberSieve.cpp is usually FASTER (O(sqrt
// n)) when you can find a g with
// g(p) = f(p) and an easy prefix sum; 04 - Theorems/05 - DuSieve.cpp
// is the O(n^(2/3)) tool when
// a Dirichlet partner makes the convolution telescope.
```

4.2.10 PowerfulNumberSieve

POWERFUL NUMBER SIEVE - prefix sum of a multiplicative f in $O(\sqrt{n})$ calls to an easy prefix

```
// Needs: Mint (06 - Math/01 - Modular Arithmetic/01 - Mint.cpp).
// sum. Faster and shorter than Min_25 whenever you can find a helper
//   ↪ g with
//   (1)  $g(p) = f(p)$  for every prime  $p$ , and (2) the prefix sum  $G(x) =$ 
//       ↪  $\sum_{i \leq x} g(i)$  is cheap.
// Then  $f = g * h$  (Dirichlet), and  $h(p) = f(p) - g(p) = 0$ , so  $h$  is
//   ↪ supported ONLY on POWERFUL
// numbers (every exponent  $\geq 2$ ) - and there are only  $O(\sqrt{n})$  of
//   ↪ those below  $n$ . Hence
//  $F(n) = \sum_{d \text{ powerful}, d \leq n} h(d) * G(n / d)$ ,
// which this DFS evaluates.  $h(p^e)$  is derived on the fly from  $f$  and
//   ↪  $g$  by the convolution
// recurrence  $h(p^e) = f(p^e) - \sum_{i < 0..e-1} h(p^i) * g(p^{e-i})$ ,
//   ↪ so you never derive it by hand.
// PRECONDITION:  $g(p) = f(p)$  for ALL primes - check it, the algorithm
//   ↪ is silently wrong otherwise.
// COST:  $O(\sqrt{n})$  nodes, each one call to  $G$ . With  $G$  in  $O(1)$  that is
//   ↪  $O(\sqrt{n})$  total; with  $G$  in
//  $O(\sqrt{x})$  it is still about  $O(n^{2/3})$ . Needs primes up to  $\sqrt{n}$ 
//   ↪ (a plain sieve).
// EXAMPLE:  $f(p^e) = p^e + [e \geq 2]$  pairs with  $g = \text{Id}$  ( $G(x) =$ 
//   ↪  $x(x+1)/2$ ).  $f = \text{"phi times something"}$ 
// style problems usually pair with  $g = \text{Id}$  or  $g = 1$  or  $g = \text{sigma}$ 
//   ↪ ( $G(x) = \text{sum } d * \text{floor}(x/d)$ ).
struct PowerfulSieve {
    ll n;
    vector<ll> pr; // primes up
    ↪ to  $\sqrt{n}$ 
    function<Mint(ll, int)> f, g; //  $f(p^e)$ ,
    ↪  $g(p^e)$ ,  $e \geq 1$ 
    function<Mint(ll)> G; // prefix sum
    ↪ of  $g$ 
    PowerfulSieve(ll n, vector<ll> pr, function<Mint(ll, int)> f,
    ↪ function<Mint(ll, int)> g,
        function<Mint(ll)> G) : n(n), pr(pr), f(f), g(g),
    ↪ G(G) {}
    Mint dfs(int i, ll m, Mint h) { // m = a
    ↪ powerful number,  $h = h(m)$ 
        Mint res = h * G(n / m);
        for (ll lim = n / m; i < (int)pr.size() && pr[i] <= lim /
    ↪ pr[i]; i++) {
            ll p = pr[i], pw = p * p;
            vector<Mint> hp = {1, 0}; //  $h(p^e) =$ 
    ↪ 1,  $h(p^1) = 0$ 
            for (int e = 2; pw <= lim; e++) {
                Mint v = f(p, e);
                for (int j = 0; j < e; j++) v -= hp[j] * g(p, e - j);
                hp.push_back(v);
                res += dfs(i + 1, m * pw, h * v);
                if (pw > lim / p) break;
                pw *= p;
            }
        }
        return res;
    }
    Mint solve() { return dfs(0, 1, 1); }
};

// COUNTING POWERFUL NUMBERS is itself a useful fact: every powerful
//   ↪ number is uniquely  $a^2 b^3$ 
// with  $b$  squarefree, so there are about  $2.173 * \sqrt{n}$  of them below
//   ↪  $n$ .
// WHEN  $g$  IS HARD TO GUESS: try  $g = \text{Id}^k$  for the  $k$  with  $f(p) \sim p^k$ ,
//   ↪ then fix the difference; or
// fall back to 09 - Min25Sieve.cpp, which needs no partner function
//   ↪ at all.
```

4.2.11 DivisorSummatory

DIVISOR SUMMATORY FUNCTION $D(n) = \sum_{i=1..n} \tau(i) = \sum_{i=1..n} \text{floor}(n/i)$

```
// in  $O(n^{1/3})$  instead of the usual  $O(\sqrt{n})$  divisor blocks.
//   ↪ About 100 ms at  $n = 1e18$ .
// Reach for it when  $n$  is  $1e14$  or more (blocks are fine up to  $\sim 1e12$ ),
//   ↪ or when many queries make
// the sqrt too slow. The result exceeds 64 bits around  $n \sim 4e17$ ,
//   ↪ hence __int128.
// HOW: the hyperbola  $x*y = n$  is walked with a STERN-BROCOT search
//   ↪ over slopes. Starting from the
// point nearest the diagonal, we repeatedly take the longest lattice
//   ↪ segment of the current slope
// that stays under the hyperbola (counting the trapezoid under it in
//   ↪  $O(1)$ ), and refine the slope
// by mediant when the curvature no longer allows it. Below  $y \sim$ 
//   ↪  $n^{1/3}$  the slopes get shorter
// than one step, so the tail is finished with a plain loop.
__int128 divisorSummatory(unsigned long long n) {
    if (n == 0) return 0;
    auto out = [n](unsigned long long x, unsigned y) { return x * y >
    ↪ n; }; // above the curve?
    auto cut = [n](unsigned long long x, unsigned dx, unsigned dy) {
        return (__uint128_t)x * x * dy >= (__uint128_t)n * dx;
    ↪ // slope too steep
    };
    unsigned long long sn = (unsigned long long)sqrtl((ld)n);
    while (sn * sn > n) sn--;
    while ((sn + 1) * (sn + 1) <= n) sn++;
    unsigned long long cn = (unsigned long long)powl((ld)n, 0.34L);
    ↪ //  $\sim n^{1/3}$ 
    unsigned long long x = n / sn;
    unsigned y = (unsigned)(n / x + 1);
    __int128 ret = 0;
    stack<pair<unsigned, unsigned>> st;
    st.emplace(1, 0), st.emplace(1, 1); // the
    ↪ Stern-Brocot stack of slopes
    while (true) {
        unsigned lx, ly;
        tie(lx, ly) = st.top(); st.pop();
        while (out(x + lx, y - ly)) { // walk as
        ↪ long as the segment fits
            ret += (__int128)x * ly + (__int128)(ly + 1) * (lx - 1) /
        ↪ 2;
            x += lx, y -= ly;
        }
        if (y <= cn) break;
        unsigned rx = lx, ry = ly;
        while (true) { // pop until
        ↪ a usable slope is on top
            tie(lx, ly) = st.top();
            if (out(x + lx, y - ly)) break;
            rx = lx, ry = ly, st.pop();
        }
        while (true) { // mediant
        ↪ refinement
            unsigned mx = lx + rx, my = ly + ry;
            if (out(x + mx, y - my)) st.emplace(lx = mx, ly = my);
            else { if (cut(x + mx, lx, ly)) break; rx = mx, ry = my; }
        }
    }
    for (--y; y > 0; --y) ret += n / y; // the
    ↪ small-y tail, plain loop
    return ret * 2 - (__int128)sn * sn; // Dirichlet
    ↪ hyperbola symmetry
}

// EXACT VALUES for testing:  $D(10) = 27$ ,  $D(100) = 482$ ,  $D(1e6) =$ 
//   ↪ 13970034,  $D(1e9) = 20951330018$ ,
//  $D(1e18) = 31264402477813376929$ . Asymptotically  $D(n) = n \ln n +$ 
//   ↪  $(2\gamma - 1)n + O(n^{1/3})$ .
// RELATED SUMS with the same blocks:  $\sum \sigma(i) = \sum$ 
//   ↪  $i * \text{floor}(n/i)$ , and any  $\sum_{i=1}^n f(n/i)$ ;
// for those, plain  $O(\sqrt{n})$  blocks (03 - FloorSum.cpp) are the
//   ↪ right tool.
```


4.2.12 MeisselLehmer

MEISSEL-LEHMER PRIME COUNTING - $\pi(n)$ in about $O(n^{2/3} / \log n)$, well under a second at

```
// n = 1e12 and a few seconds at 1e14. Use it when you need  $\pi(n)$ 
// MANY times, or for n past the
// comfort zone of Lucy's  $O(n^{3/4})$  method (04 - Theorems/03 -
// PrimeCounting.cpp) - but Lucy
// stays the better choice when you also want the SUM of the primes,
// which this cannot give.
//  $\phi(n, k)$  = count of  $i \leq n$  free of the first  $k$  primes; the
// recursion  $\phi(n, k) =$ 
//  $\phi(n, k-1) - \phi(n/p_k, k-1)$  is memoised in a table for small
// arguments, and Lehmer's
// identity peels off the numbers with exactly two or three prime
// factors.
// CALL init() ONCE. Memory: LIM ints for the prefix table plus PN*PK
// ints (about 30 MB with the
// constants below) - shrink PN/PK if the limit is tight, raise LIM
// if n is huge.
namespace PCF {
    const int LIM = 5000000, PN = 40000, PK = 60; // sieve
    // bound; memo table dimensions
    vector<int> primes, pref; // pref[i] =
    //  $\pi(i)$  for  $i < LIM$ 
    vector<vector<int>> memo;
    ll isqrt(ll x) { ll r = (ll)sqrtl((ld)x); while (r * r > x) r--;
    // while ((r+1)*(r+1) <= x) r++;
    // return r; }
    ll icbrt(ll x) { ll r = (ll)cbrtl((ld)x); while (r * r * r > x)
    // r--;
    // while ((r+1)*(r+1)*(r+1) <= x) r++; return r; }

    void init() {
        vector<bool> comp(LIM, false);
        pref.assign(LIM, 0);
        for (int i = 2; i < LIM; i++) {
            if (!comp[i]) { primes.push_back(i);
            // for (ll j = (ll)i * i; j < LIM; j += i)
            // comp[j] = 1; }
            pref[i] = primes.size();
        }
        memo.assign(PN, vector<int>(PK));
        for (int i = 0; i < PN; i++) memo[i][0] = i;
        for (int k = 1; k < PK; k++)
            for (int i = 0; i < PN; i++)
                memo[i][k] = memo[i][k-1] - memo[i / primes[k-1]][k-1];
        // }
        ll phi(ll n, int k) { // count  $i \leq n$ 
        // free of the first  $k$  primes
        if (n < PN && k < PK) return memo[n][k];
        if (k == 1) return (n + 1) >> 1;
        if (primes[k-1] >= n) return 1;
        return phi(n, k-1) - phi(n / primes[k-1], k-1);
        // }
        ll legendre(ll n) { // simplest
        // version,  $O(n^{2/3} \log n)$ -ish
        if (n < LIM) return pref[n];
        int k = pref[isqrt(n)];
        return phi(n, k) + k - 1;
        // }
        ll lehmer(ll n) { // the fast
        // one
        if (n < LIM) return pref[n];
        ll b = isqrt(n), c = lehmer(icbrt(n)), a = lehmer(isqrt(b));
        b = lehmer(b);
        ll res = phi(n, a) + (b + a - 2) * (b - a + 1) / 2;
        for (ll i = a; i < b; i++) {
            ll w = n / primes[i];
            res -= lehmer(w);
            if (i <= c) {
                ll lim = lehmer(isqrt(w));
                for (ll j = i; j < lim; j++) res += j - lehmer(w /
                // primes[j]);
            }
        }
        return res;
        // }
    }
    // SANITY VALUES:  $\pi(1e6) = 78498$ ,  $\pi(1e9) = 50847534$ ,  $\pi(1e12) =$ 
    // 37607912018,
    //  $\pi(1e13) = 346065536839$ ,  $\pi(1e15) = 29844570422669$ .
    //  $\pi(n) \sim n / \ln n$ , and  $li(n)$  is a much better estimate; the  $k$ -th
    // prime is  $\sim k(\ln k + \ln \ln k)$ .
    // NEXT STEPS: the Lagarias-Miller-Odlyzko and Deleglise-Rivat
    // refinements reach  $n^{2/3} / \log^2$  and
    // are what a real prime-counting program uses - not worth writing
    // during a contest.
```

4.2.13 CountKDivisors

HOW MANY $x \leq n$ HAVE EXACTLY k DIVISORS? n up to $\sim 1e12$, k up to a few thousand.

```
// Needs: PCF::Lehmer, PCF::primes (12 - MeisselLehmer.cpp).
// A number with exactly  $k$  divisors has the shape  $p_1^{e_1-1} \dots$ 
//  $p_2^{e_2-1} \dots$  where  $e_1 e_2 \dots = k$ 
// and (choosing the exponents in decreasing order) the primes can be
// taken increasing. So we
// backtrack over the FACTORISATIONS of  $k$  into factors  $> 1$ , assigning
// a prime to each factor.
// Two things make it fast: (a) the case  $k = 2$  is just "count the
// primes  $\leq n$ ", answered by
// Meissel-Lehmer, and (b) a single remaining factor  $x$  means "count
// primes  $p$  with  $p^{x-1} \leq n$ ",
// answered by a binary search over the prime list.
// PRECONDITIONS: call PCF::init() first, and prep() once for the  $k$ 
// range you need. Sieve limit
// LIM in PCF must exceed  $n^{1/2}$  for the binary searches to be safe
// (5e6 covers  $n \leq 2.5e13$ ).
// COMPLEXITY: hard to state - it is a search whose branching is
// bounded by the divisors of  $k$ ;
// in practice a handful of milliseconds per query for  $n = 1e12$ .
namespace KDiv {
    const int MK = 2100; // max  $k + 1$ 
    vector<vector<int>> dv; //  $dv[k] =$ 
    // divisors of  $k$  that are  $> 1$ , desc
    vector<int> minexp; // minexp[k]
    // = sum  $(p-1)$  over primes  $p \mid k$ 
    void prep() {
        dv.assign(MK, {}), minexp.assign(MK, 0);
        for (int i = 2; i < MK; i++) {
            for (int j = i; j < MK; j += i) dv[j].push_back(i);
            for (int p : PCF::primes) {
                if (p > i) break;
                int cur = i;
                while (cur % p == 0) cur /= p, minexp[i] += p - 1;
            }
            for (int i = 2; i < MK; i++) reverse(all(dv[i])); // try
            // the big exponents first
        }
        ll powCap(ll b, int e, ll cap) { //  $b^e$ ,
        // saturating at  $cap + 1$ 
        ll r = 1;
        for (int i = 0; i < e; i++) { if (r > cap / b) return cap +
        // 1; r *= b; }
        // return r;
        // }
        // count of  $x \leq n$  with exactly  $k$  divisors, using only primes
        // from index 'first' on
        ll go(ll n, int k, int first = 0) {
            if (n <= 0 || k >= MK) return 0;
            if (k == 1) return 1;
            if (n == 1) return 0;
            if (k == 2) return max(0LL, PCF::lehmer(n) - first);
            if (dv[k].size() == 1) { //  $k$  is
            // prime: one prime power  $p^{k-1}$ 
            int e = dv[k][0] - 1;
            if (powCap(PCF::primes[first], e, n) > n) return 0;
            int lo = first, hi = (int)PCF::primes.size() - 1, ans =
            // first - 1;
            while (lo <= hi) { // last prime
            // with  $p^e \leq n$ 
            int mid = (lo + hi) / 2;
            if (powCap(PCF::primes[mid], e, n) > n) hi = mid - 1;
            else lo = mid + 1, ans = mid;
            }
            return ans - first + 1;
            // }
            ll res = 0;
            for (int x : dv[k]) // this prime
            // carries exponent  $x - 1$ 
            for (int i = first; i < (int)PCF::primes.size(); i++) {
                ll p = powCap(PCF::primes[i], x - 1, n);
                if (p > n) break;
                if (k / x == 1) { res++; continue; }
                if (powCap(PCF::primes[i+1], minexp[k / x], n) > n
                // / p) break;
                res += go(n / p, k / x, i + 1);
            }
            return res;
        }
    }
    // RANGE QUERY:  $answer(l, r) = go(r, k) - go(l - 1, k)$ .
    // SMALL  $n$ : just sieve tau with a linear sieve (07 -
    // MultiplicativeSieve.cpp) and count.
    // SMALLEST number with exactly  $k$  divisors is a different search - 15
```

```

    ↪ - SmallestWithKDivisors.cpp.
// FACTS: tau(x) is odd iff x is a perfect square; the count of x <=
    ↪ n with tau(x) = 2 is pi(n),
// with tau(x) = 3 it is pi(sqrt n), and with tau(x) = 4 it is
    ↪ pi(cbrt n) plus the number of
// products p*q of two distinct primes below n.

```

4.2.14 PrimeBasis

COPRIME (PRIME) BASIS - factor a set of numbers RELATIVE TO EACH OTHER, without ever factoring

```

// them. Given n values up to 1e18 (far past Pollard-rho comfort when
    ↪ n is large), it builds a set
// of PAIRWISE COPRIME basis elements such that every input is a
    ↪ product of basis powers. Then
// each input becomes an exponent vector, and gcd / lcm /
    ↪ divisibility / "product is a square" all
// become coordinatewise min / max / <= on those vectors.
// WHEN: "is a_i | prod a_j", "make all numbers equal by moving prime
    ↪ factors", "count subsets
// whose product is a perfect k-th power", any problem where only the
    ↪ RELATIVE factorisations of
// the given numbers matter and the numbers are too big to factor.
// SIZE: at most n * log log(max a) basis elements; build is O(n^2
    ↪ log(max a)) worst case, and
// far less in practice. factorize(x) is O(|basis| * log x).
// NOTE the basis elements are not necessarily prime - they are only
    ↪ pairwise coprime, which is
// all that matters. They ARE prime whenever the inputs force it.
template <typename T>
struct PrimeBasis {
    vector<T> basis;
    void reducePair(T& x, T& y) { // strip the
        ↪ common part greedily
        bool sw = 0;
        if (x > y) sw ^= 1, swap(x, y);
        while (x > 1 && y % x == 0) { y /= x; if (x > y) sw ^= 1,
        ↪ swap(x, y); }
        if (sw) swap(x, y);
    }
    void split(int pos, T& val) { // basis[pos]
        ↪ and val share a factor
        while (basis[pos] % val == 0) basis[pos] /= val;
        vector<T> cur = {basis[pos], val};
        for (int ptr = 1; ptr < (int)cur.size(); ptr++)
            for (int i = 0; i < ptr; i++) {
                reducePair(cur[i], cur[ptr]);
                if (cur[ptr] == 1 || cur[i] == 1) break;
                T g = gcd(cur[ptr], cur[i]);
                if (g > 1) cur[ptr] /= g, cur[i] /= g,
                ↪ cur.push_back(g);
            }
        basis[pos] = cur[0], val = cur[1];
        for (int i = 2; i < (int)cur.size(); i++) if (cur[i] > 1)
        ↪ basis.push_back(cur[i]);
        if (basis[pos] == 1) swap(basis[pos], basis.back());
        ↪ basis.pop_back();
    }
    void add(T val) {
        for (int i = 0; i < (int)basis.size(); i++) {
            reducePair(val, basis[i]);
            if (basis[i] == 1) { swap(basis[i], basis.back());
            ↪ basis.pop_back(); break; }
            if (val == 1) return;
            if (gcd(basis[i], val) > 1) split(i, val);
        }
        if (val > 1) basis.push_back(val);
    }
    vector<int> factorize(T x) { // exponent
        ↪ vector of x over the basis
        vector<int> f(basis.size());
        for (int i = 0; i < (int)basis.size(); i++) while (x %
        ↪ basis[i] == 0) f[i]++, x /= basis[i];
        return f;
    }
};
// AFTER BUILDING: add() EVERY input first, then factorize() them -
    ↪ the basis must be final, or
// old vectors have the wrong length. Verify with: every
    ↪ factorize(a_i) must multiply back to a_i
// (i.e. dividing a_i by all basis powers leaves 1), and all pairs of
    ↪ basis elements are coprime.
// The vectors support: gcd = coordinatewise min, lcm = max, a | b
    ↪ iff vec(a) <= vec(b)
// coordinatewise, a*b = sum, a/b = difference, x is a perfect square
    ↪ iff all coordinates are even.

```

4.2.15 SmallestWithKDivisors

THE SMALLEST NUMBER WITH EXACTLY k DIVISORS. The answer explodes (k = 1e9 already needs ~100

```

// Needs: powmod (06 - MillerRabinPollard.cpp), divisors of k (03 -
    ↪ Divisors.cpp or factorize).
// digits), so it is compared by the SUM OF LOGARITHMS and reported
    ↪ modulo MOD.
// Structure: x = p_0^{e_0} * p_1^{e_1} * ... over the FIRST primes
    ↪ in order, with
// (e_0+1)(e_1+1)... = k and e_0 >= e_1 >= ... (swapping a larger
    ↪ exponent onto a smaller prime
// never hurts). So: walk the primes in order, and at prime i choose
    ↪ a divisor x | k for the
// factor (e_i + 1) = x. Memoise on (prime index, remaining k).
// PRECONDITIONS: pr must hold the first ~64 primes (2, 3, 5, ...);
    ↪ dvs = all divisors of k that
// are > 1, ASCENDING (that order is what makes the break-early
    ↪ pruning valid). k >= 1.
// COMPLEXITY: O(#states * tau(k)) with tiny constants - milliseconds
    ↪ for k up to 1e12.
struct SmallestKDiv {
    vector<ll> pr, dvs;
    vector<ld> lg;
    vector<map<ll, pair<ld, ll>>> memo; //
    ↪ memo[i][rem] = {log of x, x mod MOD}
    SmallestKDiv(vector<ll> pr, vector<ll> dvs) : pr(pr), dvs(dvs) {
        for (ll p : pr) lg.push_back(logl((ld)p));
        memo.assign(pr.size() + 1, {});
    }
    pair<ld, ll> go(int i, ll rem) { // use primes
        ↪ pr[i], pr[i+1], ... only
        if (rem == 1) return {0.0L, 1};
        auto it = memo[i].find(rem);
        if (it != memo[i].end()) return it->second;
        pair<ld, ll> best = {1e30L, 0};
        for (ll x : dvs) {
            if (rem % x) continue;
            ld z = lg[i] * (ld)(x - 1); // cost of
            ↪ putting exponent x-1 on pr[i]
            if (z > best.first) break; // dvs
            ↪ ascending => all later are worse
            auto cur = go(i + 1, rem / x);
            cur.first += z;
            cur.second = cur.second * powmod(pr[i], x - 1, MOD) % MOD;
            best = min(best, cur);
        }
        return memo[i][rem] = best;
    }
    ll solve(ll k) { return k == 1 ? 1 : go(0, k).second; }
};
// EXACT SMALL ANSWERS (for a sanity check): k = 1,2,3,4,5,6,7,8,9,10
    ↪ -> 1, 2, 4, 6, 16, 12, 64,
// 24, 36, 48. k = 100 -> 45360. The sequence is OEIS A005179.
// LONG DOUBLE is enough here: two candidates tie only when the
    ↪ numbers are equal, and the log
// sums differ by more than 1e-9 otherwise for any k that fits in 64
    ↪ bits. If paranoid, compare
// the exponent vectors exactly with big integers, or compare logs
    ↪ and break ties by the vectors.
// RELATED: the MAXIMUM number of divisors below n (highly composite
    ↪ numbers) is found by the
// same DFS with the objective flipped - keep the primes in order and
    ↪ try e_i = 0..60 with
// e_0 >= e_1 >= ...; tau maxes at 1344 for n <= 1e9, 6720 for 1e12,
    ↪ 103680 for 1e18.

```

4.3 Modular Arithmetic

4.3.1 CRT

```
// NOTE: extGCD below is the same one as in 01 - Euclidean Algorithm.
↳ Paste only one copy.
long long extGCD(long long a, long long b, long long &x, long long
↳ &y) {
    if (b == 0) {
        x = 1; y = 0;
        return a;
    }
    long long x1, y1;
    long long d = extGCD(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return d;
}

// Solves x = a_i (mod m_i). Returns {x, lcm(m_1, ..., m_k)}.
pair<long long, long long> CRT(const vector<long long>& a, const
↳ vector<long long>& m) {
    long long res = a[0], lcm = m[0];
    for (int i = 1; i < a.size(); i++) {
        long long x, y;
        long long g = extGCD(lcm, m[i], x, y);
        if ((a[i] - res) % g != 0) return {-1, -1}; // No solution

        long long step = m[i] / g;
        long long k = ((a[i] - res) / g) % step * (x % step) % step;
        res = (long long)((__int128)k * lcm + res) % ((__int128)lcm
↳ * step);
        lcm *= step;
        res = (res % lcm + lcm) % lcm;
    }
    return {res, lcm};
}
```

4.3.2 DiscreteLogAndRoots

Discrete log, modular square root, primitive root, discrete k-th root, multiplicative order.

```
// Needs: powmod (02 - Primes and Divisors/06 -
↳ MillerRabinPollard.cpp).
// All need mulmod/powmod from MillerRabinPollard.

// BSGS: smallest x >= 0 with a^x = b (mod m). Works for ANY m
↳ (handles gcd(a,m) != 1). O(sqrt m).
ll discreteLog(ll a, ll b, ll m) {
    a %= m, b %= m;
    ll k = 1, add = 0, g;
    while ((g = gcd(a, m)) > 1) { // strip common
↳ factors
        if (b == k) return add;
        if (b % g) return -1;
        b /= g, m /= g, add += k * (a / g) % m;
    }
    ll n = (ll)sqrtl((ld)m) + 1, an = 1;
    for (ll i = 0; i < n; i++) an = an * a % m;
    unordered_map<ll, ll> vals;
    for (ll q = 0, cur = b; q <= n; q++) vals[cur] = q, cur = cur * a
↳ % m;
    for (ll p = 1, cur = k; p <= n; p++) {
        cur = cur * an % m;
        if (vals.count(cur)) { ll ans = n * p - vals[cur] + add; if
↳ (ans >= 0) return ans; }
    }
    return -1;
}

// Tonelli-Shanks: x with x^2 = a (mod p), p odd prime. Returns -1 if
↳ a is not a QR.
ll sqrtMod(ll a, ll p) {
    a %= p; if (a < 0) a += p;
    if (a == 0) return 0;
    if (powmod(a, (p - 1) / 2, p) != 1) return -1; // Euler's
↳ criterion
    if (p % 4 == 3) return powmod(a, (p + 1) / 4, p);
    ll s = p - 1, n = 2; int r = 0;
    while (s % 2 == 0) s /= 2, r++;
    while (powmod(n, (p - 1) / 2, p) != p - 1) n++; // find a
↳ non-residue
    ll x = powmod(a, (s + 1) / 2, p), b = powmod(a, s, p), g =
↳ powmod(n, s, p);
    for (int m = r; m = r) {
        ll t = b;
        for (r = 0; r < m && t != 1; r++) t = t * t % p;
        if (r == 0) return x;
        ll gs = powmod(g, 1LL << (m - r - 1), p);
        g = gs * gs % p, x = x * gs % p, b = b * g % p;
    }
}

// Smallest primitive root of a PRIME p (generator of Z_p^*).
ll primitiveRoot(ll p) {
    if (p == 2) return 1;
    auto f = factorize(p - 1);
    for (ll g = 2; g < p; g++) {
        bool ok = true;
        for (auto [q, e] : f) if (powmod(g, (p - 1) / q, p) == 1) {
↳ ok = false; break; }
        if (ok) return g;
    }
    return -1;
}

// Multiplicative order of a mod m (smallest e > 0 with a^e = 1).
↳ Requires gcd(a, m) = 1.
ll multOrder(ll a, ll m) {
    ll e = phi(m), r = e;
    for (auto [q, k] : factorize(e))
        while (r % q == 0 && powmod(a, r / q, m) == 1) r /= q;
    return r;
}

// All x with x^k = a (mod p), p prime. Reduce to a discrete log in
↳ base g.
vector<ll> discreteRoot(ll k, ll a, ll p) {
    if (a == 0) return {0};
    ll g = primitiveRoot(p), y = discreteLog(powmod(g, k, p), a, p);
    if (y < 0) return {};
    ll x0 = powmod(g, y, p), d = gcd(k, p - 1), step = powmod(g, (p -
↳ 1) / d, p);
    vector<ll> r;
    for (ll i = 0, cur = x0; i < d; i++, cur = cur * step % p)
↳ r.push_back(cur);
    sort(all(r)), r.erase(unique(all(r)), r.end());
}
```

```

    return r;
}

```

4.3.3 LucasAndBinomialMod

$C(n, r) \bmod m$ for HUGE n . Pick by what m is:

```

// Needs: Mint (06 - Math/01), and factorize (02 - Primes and
    ↳ Divisors/06) for the last section.
// m prime and small -> lucas
// m = p^e -> binomPrimePower (Andrew Granville
    ↳ / Wilson generalisation)
// m arbitrary (squarefree-ish)-> binomAnyMod (CRT over prime
    ↳ powers)

// LUCAS. p prime, needs fact/invFact built up to p. C(n,r) = prod
    ↳ C(n_i, r_i) over base-p digits.
Mint lucas(ll n, ll r) {
    if (r < 0 || r > n) return 0;
    Mint res = 1;
    while (n || r) {
        ll a = n % MOD, b = r % MOD;
        if (b > a) return 0;
        res *= fact[a] * invFact[b] * invFact[a - b];
        n /= MOD, r /= MOD;
    }
    return res;
}

// --- arbitrary modulus ---
ll pw(ll a, ll e, ll m) { ll r = 1; a %= m; for (; e >= 1, a = a
    ↳ * a % m) if (e & 1) r = r * a % m; return r; }
ll inv_mod(ll a, ll m) { // requires
    ↳ gcd(a, m) = 1
    ll g = m, x = 0, x1 = 1, a1 = a % m;
    while (a1) { ll q = g / a1; g -= q * a1, swap(g, a1); x -= q *
    ↳ x1, swap(x, x1); }
    return (x % m + m) % m;
}

// n! with all factors of p removed, mod p^e. O(p^e + log n).
ll factNoP(ll n, ll p, ll pe) {
    if (n == 0) return 1;
    vector<ll> f(pe);
    f[0] = 1;
    for (ll i = 1; i < pe; i++) f[i] = (i % p ? f[i - 1] * i : f[i -
    ↳ 1]) % pe;
    ll r = 1;
    for (ll m = n; m; m /= p) {
        r = r * pw(f[pe - 1], m / pe, pe) % pe;
        r = r * f[m % pe] % pe;
    }
    return r;
}

// C(n, r) mod p^e
ll binomPrimePower(ll n, ll r, ll p, ll pe) {
    if (r < 0 || r > n) return 0;
    ll k = 0; // exponent
    ↳ of p in C(n,r) (Kummer's theorem:
    for (ll m = n / p, a = r / p, b = (n - r) / p; m /= p, a /= p,
    ↳ b /= p) k += m - a - b;
    if (k >= 60 || pw(p, k, pe * p) % pe == 0) { ll t = pw(p, k, pe);
    ↳ if (t == 0) return 0; }
    ll num = factNoP(n, p, pe), d1 = factNoP(r, p, pe), d2 =
    ↳ factNoP(n - r, p, pe);
    ll res = num * inv_mod(d1, pe) % pe * inv_mod(d2, pe) % pe;
    return res * pw(p, k, pe) % pe;
}

// C(n, r) mod m for arbitrary m: factor m, solve mod each p^e, CRT.
ll binomAnyMod(ll n, ll r, ll m) {
    ll res = 0, M = m;
    for (auto [p, e] : factorize(m)) {
        ll pe = 1; for (int i = 0; i < e; i++) pe *= p;
        ll c = binomPrimePower(n, r, p, pe), t = M / pe;
        res = (res + c % pe * t % M * inv_mod(t % pe, pe)) % M;
    }
    return res;
}

// KUMMER'S THEOREM: the exponent of prime p in C(n,r) = number of
    ↳ CARRIES when adding
// r and n-r in base p. (Used above to pull out
    ↳ the p-part.)
// WILSON: (p-1)! = -1 (mod p) for prime p.

```

4.3.4 CongruencesAndTowers

Linear congruences and power towers. Needs powmod / phi / gcd.

```

// Needs: powmod, phi (04 - NT/02 - Primes and Divisors/06 -
    ↳ MillerRabinPollard.cpp).

// Solve a*x = b (mod m). Returns all solutions in [0, m) (there are
    ↳ gcd(a,m) of them, or none).
vector<ll> linCongruence(ll a, ll b, ll m) {
    a = ((a % m) + m) % m, b = ((b % m) + m) % m;
    ll g = gcd(a, m);
    if (b % g) return {};
    ll m2 = m / g, a2 = a / g, b2 = b / g;
    // a2 is invertible mod m2
    ll x0 = 0, x1 = 1, g2 = m2, a3 = a2 % m2;
    while (a3) { ll q = g2 / a3; g2 -= q * a3, swap(g2, a3); x0 -= q
    ↳ * x1, swap(x0, x1); }
    ll inv = ((x0 % m2) + m2) % m2, x = b2 * m2 * inv % m2;
    vector<ll> r;
    for (ll i = 0; i < g; i++) r.push_back((x + i * m2) % m);
    return r;
}

// GENERALISED EULER: a^k = a^(k mod phi(m) + phi(m)) (mod m)
    ↳ whenever k >= log2(m).
// This is what makes power towers computable even when gcd(a, m) !=
    ↳ 1.
ll capTower(const vector<ll>& a, int i, ll CAP = 64) { // min(true
    ↳ tower value, CAP)
    if (i == (int)a.size()) return 1;
    if (a[i] == 0) return capTower(a, i + 1, CAP) == 0 ? 1 : 0; //
    ↳ 0^0 = 1, 0^(k>0) = 0
    if (a[i] == 1) return 1;
    ll e = capTower(a, i + 1, CAP), r = 1;
    for (ll k = 0; k < e; k++) { r *= a[i]; if (r >= CAP) return CAP;
    ↳ }
    return r;
}

// a[i] ^ a[i+1] ^ ... ^ a[last] (mod m), right-associative.
ll powTower(const vector<ll>& a, int i, ll m) {
    if (m == 1) return 0;
    if (i == (int)a.size()) return 1 % m;
    ll ph = phi(m), t = capTower(a, i + 1);
    ll e = t < 64 ? t : powTower(a, i + 1, ph) + ph; // exact
    ↳ when tiny, +phi when large
    return powmod(a[i] % m, e, m);
}

// Extended CRT for NON-coprime moduli: x = a1 (mod m1), x = a2 (mod
    ↳ m2).
// Returns {x, lcm} or {-1, -1}. Chain it to merge many congruences.
pair<ll, ll> crt2(ll a1, ll m1, ll a2, ll m2) {
    ll g = gcd(m1, m2), l = m1 / g * m2;
    if ((a2 - a1) % g) return {-1, -1};
    ll m2g = m2 / g;
    ll x0 = 0, x1 = 1, gg = m2g, aa = (m1 / g) % m2g;
    while (aa) { ll q = gg / aa; gg -= q * aa, swap(gg, aa); x0 -= q
    ↳ * x1, swap(x0, x1); }
    ll inv = ((x0 % m2g) + m2g) % m2g;
    __int128 k = (__int128)((a2 - a1) / g % m2g) * inv % m2g;
    ll x = (ll)((__int128)k * m1 + a1) % l;
    return {(x % l + l) % l, l};
}

```

4.3.5 Garner

GARNER'S ALGORITHM - reconstruct x from residues modulo PAIRWISE COPRIME m_i , in mixed-radix

```
// form, so you can emit x modulo a target MOD without ever building
// prod(m_i).
// This is what CRT cannot do once the product overflows 64 bits.
// x = x1 + x2*m1 + x3*m1*m2 + ... , 0 <= xi < mi
ll garner(vector<ll> a, vector<ll> m, ll MODOUT) {
    int k = a.size();
    vector<ll> x(k);
    auto inv = [](ll a, ll M) { // requires
        gcd(a, M) = 1;
        ll g = M, X = 0, x1 = 1, A = ((a % M) + M) % M;
        while (A) { ll q = g / A; g -= q * A, swap(g, A); X -= q *
        x1, swap(X, x1); }
        return (X % M + M) % M;
    };
    for (int i = 0; i < k; i++) {
        x[i] = ((a[i] % m[i]) + m[i]) % m[i];
        ll mul = 1;
        for (int j = 0; j < i; j++) {
            x[i] = (x[i] - x[j] % m[j] * mul) % m[i];
            mul = mul * (m[j] % m[i]) % m[i];
        }
        x[i] = (x[i] % m[i] + m[i]) % m[i] * inv(mul, m[i]) % m[i];
    }
    ll res = 0, mul = 1;
    for (int i = 0; i < k; i++) {
        res = (res + x[i] % MODOUT * mul) % MODOUT;
        mul = mul * (m[i] % MODOUT) % MODOUT;
    }
    return (res % MODOUT + MODOUT) % MODOUT;
}
// MAIN USE: arbitrary-modulus convolution. Run NTT under three NTT
// primes, then Garner the
// three residues down to the target modulus.
// NTT PRIMES: 998244353 = 119*2^23 + 1 (g=3) 167772161 =
// 5*2^25 + 1 (g=3)
// 469762049 = 7*2^26 + 1 (g=3) 1004535809 =
// 479*2^21 + 1 (g=3)
// 754974721 = 45*2^24 + 1 (g=11) <- note: NOT
// 9*2^23+1, a common misprint
// The 3-prime product 5.95e25 covers n * (MOD-1)^2 for any MOD <
// 2^30 and n up to ~5e7.
```

4.3.6 Montgomery

MONTGOMERY MULTIPLICATION - modular multiplication for a FIXED ODD 64-bit modulus without any

```
// hardware division: 2-3x faster than ((__int128)a*b % m. Use it in
// the hot loop of Miller-Rabin,
// Pollard rho, BSGS, matrix power or any modpow-heavy routine with a
// 64-bit modulus. For a
// COMPILE-TIME modulus the compiler already turns % into a
// multiply-shift, so this buys nothing;
// the win is for a modulus known only at run time.
// TRICK: work with the representative a*R mod m where R = 2^64.
// reduce(x) computes x*R^-1 mod m
// with two multiplications and no division, so (aR)(bR)*R^-1 = (ab)R
// stays in the same form.
// PRECONDITIONS: m must be ODD (the Newton iteration inverts m
// modulo 2^64) and m < 2^63.
// Values returned by get() are in [0, m); mul/add work on the
// Montgomery form.
typedef unsigned long long u64; // same
// typedef as 02/06 - keep one copy
struct Mont {
    u64 mod, minv, r2; // minv =
    m^-1 mod 2^64
    void set(u64 m) {
        mod = minv = m;
        for (int i = 0; i < 5; i++) minv *= 2 - minv * m; //
        // Newton: doubles the bits each time
        r2 = (u64)(-((__uint128_t)m % m)); // R^2
    }
    u64 reduce((__uint128_t x) const {
        u64 y = (u64)(x >> 64) - (u64)((__uint128_t)((u64)x * minv)
        * mod) >> 64);
        return (long long)y < 0 ? y + mod : y;
    }
    u64 in(u64 a) const { return reduce((__uint128_t)(a % mod) * r2);
    } // to Montgomery form
    u64 get(u64 a) const { u64 y = reduce(a); return y >= mod ? y -
    mod : y; } // back to normal
    u64 mul(u64 a, u64 b) const { return reduce((__uint128_t)a * b); }
    u64 add(u64 a, u64 b) const { a += b - mod; return (long long)a <
    0 ? a + mod : a; }
    u64 sub(u64 a, u64 b) const { a -= b; return (long long)a < 0 ? a
    + mod : a; }
    u64 pow(u64 a, u64 e) const { // a already
    in Montgomery form
        u64 r = in(1);
        for (; e >= 1, a = mul(a, a)) if (e & 1) r = mul(r, a);
        return r;
    }
};
// USAGE: Mont M; M.set(n); u64 x = M.in(3); x = M.pow(x, d);
// cout << M.get(x);
// Do NOT mix forms: everything inside the loop stays in Montgomery
// form, converting only at the
// boundaries. Comparisons against 1 and m-1 (Miller-Rabin) must
// compare against M.in(1) and
// M.in(m-1), which you compute once.
// EVEN MODULUS: split m = 2^k * odd, run Montgomery on the odd part
// and the trivial bit-mask on
// the power of two, then CRT. Usually not worth it - just use
// __int128.
// ALTERNATIVE: BARRETT reduction works for even moduli too
// (precompute floor(2^64/m) and replace
// the division by a multiply-shift); it is a little slower but has
// no parity restriction.
```


4.3.7 QuadraticExtension

```

ARITHMETIC IN  $F_p[\sqrt{D}] = \{a + b\sqrt{D} : a, b \text{ in } F_p\}$  - the quadratic extension field

// Needs: powmod (02 - Primes and Divisors/06 - MillerRabinPollard.cpp).
// * CIPOLLA's square root mod p: always  $O(\log p)$  multiplications,
//   and unlike Tonelli-Shanks it
//   does not degrade when  $p - 1$  has a huge power of two. (Tonelli
//   is in 02 - DiscreteLog...)
// * Closed forms that need an irrational: Fibonacci via
//    $((1+\sqrt{5})/2)^n \bmod p$ , and ANY order-2
//   linear recurrence  $x_{n+1} = a x_n + b x_{n-1}$ , even when the
//   discriminant is a non-residue
//   mod p (no matrix needed - one field power).
// * Lucas sequences / Frobenius primality tests, and n-th terms of
//    $(a + b\sqrt{D})^n$  identities.
// PRECONDITIONS: QP an odd prime, all inputs reduced into  $[0, QP)$ .
//   QP up to  $\sim 4.6e18$  (the __int128
//   products need  $QP^2 < 2^{127}$ ).
ll QP, QD; // the
// modulus and the radicand
ll qmul(ll x, ll y) { return (ll)((__int128)x * y % QP); }
struct Quad {
    ll a, b; // a +
    // b*sqrt(QD)
    Quad(ll a = 0, ll b = 0) : a(a), b(b) {}
    Quad operator*(const Quad& o) const {
        return Quad((qmul(a, o.a) + qmul(QD, qmul(b, o.b))) % QP,
                    (qmul(a, o.b) + qmul(b, o.a)) % QP);
    }
    Quad operator+(const Quad& o) const { return Quad((a + o.a) % QP,
    // (b + o.b) % QP); }
    Quad operator-(const Quad& o) const { return Quad((a - o.a + QP)
    // % QP, (b - o.b + QP) % QP); }
    Quad conj() const { return Quad(a, (QP - b) % QP); } //
    // the Frobenius  $x \rightarrow x^p$ 
    ll norm() const { return (qmul(a, a) - qmul(QD, qmul(b, b)) % QP
    // + QP) % QP; }
};
Quad qpow(Quad x, ll e) { //  $O(\log e)$ 
    // with 3 multiplications a step
    Quad r(1, 0);
    for (; e >= 1, x = x * x) if (e & 1) r = r * x;
    return r;
}
// CIPOLLA: x with  $x^2 = a \pmod p$ , p an odd prime. Returns -1 if a
// is not a quadratic residue.
ll cipolla(ll a, ll p) {
    a %= p; if (a < 0) a += p;
    if (a <= 1) return a;
    if (powmod(a, (p - 1) / 2, p) != 1) return -1; // Euler's
    // criterion
    QP = p;
    ll t = 1;
    while (true) { // find t
        // with  $t^2 - a$  a NON-residue
        QD = ((__int128)t * t - a) % p; QD = (QD + p) % p;
        if (QD && powmod(QD, (p - 1) / 2, p) == p - 1) break;
        t++;
    }
    return qpow(Quad(t, 1), (p + 1) / 2).a; // the
    // sqrt(QD) part is provably 0
}
// The other root is  $p - x$ . Expected  $O(\log p)$  tries for t (half the
// residues work).
// FIBONACCI mod p: set QD = 5, phi =  $(1 + \sqrt{5})/2 = \text{Quad}((p+1)/2,$ 
//    $(p+1)/2)$ ; then
//    $F(n) = (\text{phi}^n - (1 - \text{phi})^n) / \sqrt{5}$ , i.e. qpow(phi, n).b * 2
//    $\pmod p$  since the two
//   conjugate parts cancel. Works for every odd p, whether or not 5
//   is a residue.
// GENERAL ORDER-2 RECURRENCE  $x_{n+1} = A x_n + B x_{n-1}$ : the roots
//   of  $z^2 = A z + B$  live in
//    $F_p[\sqrt{A^2 + 4B}]$ , so  $x_n$  is a linear combination of their n-th
//   powers - one qpow call.

```

4.3.8 SqrtModPrimePower

factorize (02 - Primes and Divisors/06 - MillerRabinPollard.cpp).

```

// Needs: sqrtMod (02 - DiscreteLogAndRoots.cpp), extGCD (01 -
//   Euclidean Algorithm), CRT (01),
//   factorize (02 - Primes and Divisors/06 - MillerRabinPollard.cpp).
// SQUARE ROOTS MODULO A PRIME POWER, AND MODULO A COMPOSITE - what
//   Tonelli-Shanks alone cannot
// do. Everything here is HENSEL LIFTING: a root mod  $p^j$  lifts to a
//   root mod  $p^{j+1}$  by one
//   Newton step  $x \leftarrow x - f(x)/f'(x)$ , valid whenever  $f'(x) = 2x$  is
//   invertible mod p, i.e. p odd and
//   p !| x. p = 2 is the exception (the derivative is even) and is
//   handled separately below.
// PRECONDITION for both routines: gcd(a, p) = 1. If p | a, first
//   pull out the largest  $p^t$ 
// dividing a (an ODD exponent of p in a means NO solution unless it
//   is  $\geq k$ ), solve for  $a/p^t$ 
// modulo  $p^{k-2t}$ , multiply the roots by  $p^t$ , and add every multiple
//   of  $p^{k-t}$ .
// COMPLEXITY:  $O(k \log p)$  per lift;  $O(\log^2)$  for the composite
//   version plus the factorisation.
ll invMod(ll a, ll m) { ll x, y; extGCD(((a % m) + m) % m, m, x, y);
    // return ((x % m) + m) % m; }
// one root of  $x^2 = a \pmod{p^k}$ , p an ODD prime, gcd(a, p) = 1. -1
//   if a is a non-residue mod p.
ll sqrtPrimePower(ll a, ll p, int k) {
    ll x = sqrtMod(((a % p) + p) % p, p);
    if (x < 0) return -1;
    ll pk = p;
    for (int j = 1; j < k; j++) {
        ll np = pk * p;
        ll fx = (((__int128)x * x - a) % np + np) % np;
        x = (ll)(((__int128)x - (__int128)fx * invMod(2 * x % np,
    // np)) % np + np) % np;
        pk = np;
    }
    return x; // the other
    // root is  $p^k - x$ 
}
// one root of  $x^2 = a \pmod{2^k}$ , a ODD. -1 if there is none.
ll sqrtPow2(ll a, int k) {
    a &= (1LL << k) - 1;
    if (k == 1) return 1; // a odd =>
    // x = 1
    if (k == 2) return (a & 3) == 1 ? 1 : -1;
    if ((a & 7) != 1) return -1; // a = 1
    // (mod 8) is necessary for k >= 3
    ll x = 1;
    for (int j = 3; j < k; j++) { // lift  $2^j$ 
        //  $\rightarrow 2^{j+1}$ ; pick the right one
        ll nm = 1LL << (j + 1);
        if (((__int128)x * x - a) % nm + nm) % nm) x += 1LL << (j -
    // 1);
    }
    return x;
}
// NUMBER OF SOLUTIONS of  $x^2 = a \pmod{p^k}$  with gcd(a, p) = 1:
//   p odd: 2 if a is a residue mod p, else 0. The roots are  $\pm x$ .
//   p = 2: k = 1  $\rightarrow$  1 root; k = 2  $\rightarrow$  2 roots (a = 1 mod 4); k >= 3
//    $\rightarrow$  4 roots (a = 1 mod 8),
//   namely  $\pm x$  and  $\pm x + 2^{k-1}$ .
// COMPOSITE m: factor m = prod  $p_i^{k_i}$ , solve each, and CRT every
//   solution count is the product of the per-prime-power counts, so it
//   can be  $2^{\omega(m)}$  many.
// THE SPECIAL CASE a = 1 is worth memorising: the number of x with
//    $x^2 = 1 \pmod m$  is
//   prod over odd  $p^k$  || m of 2, times (1 if  $2 \nmid m$  is  $2^1$ , 2 if  $4 \mid m$ ,
//   4 if  $8 \mid m$ ).
ll numSqrt1(ll m) { // count of
    // x in  $[0, m)$  with  $x^2 = 1$ 
    ll res = 1;
    for (auto [p, e] : factorize(m))
        res *= (p != 2 ? 2 : (e == 1 ? 1 : e == 2 ? 2 : 4));
    return res;
}
// USES: counting self-inverse elements, "how many x with  $x^2 = 1$ " in
//   group-theory counting,
// solving quadratic congruences  $ax^2 + bx + c = 0$  (complete the
//   square:  $(2ax + b)^2 = b^2 - 4ac$ ,
// needs gcd(2a, m) = 1), and lifting solutions of ANY polynomial f -
//   the same Newton step works
// whenever gcd(f'(x0), p) = 1.

```


4.3.9 ModularAPBlocks

THE SEQUENCE $(a \cdot k + b) \bmod m$, FOR $k = 0..n-1$, AS $O(\sqrt{m})$ ARITHMETIC PROGRESSIONS.

```
// Each returned block {start, diff, len} lists consecutive terms
// → start, start+diff, ... that do
// NOT wrap past m - so a value-sorted / value-structured query over
// → the whole sequence becomes
//  $O(\sqrt{m})$  range operations instead of  $n$  individual ones. This is
// → the constructive form of the
// THREE-DISTANCE THEOREM: the points  $k \cdot a \bmod m$  take at most three
// → distinct gaps.
// USE IT FOR: sum of  $x^{((a \cdot k + b) \bmod m)}$ , sum over  $k$  of  $f((a \cdot k + b) \bmod m)$  for an  $f$  with prefix
// sums, min/max of  $(a \cdot k + b) \bmod m$  over a range of  $k$ , counting terms
// → in a value window when the
// generic  $O(\log)$  count (01 - Euclidean Algorithm/06) does not fit
// → the shape of the problem.
// PRECONDITIONS for decomposeModAP:  $\gcd(a, m) = 1$  and  $n \leq m$ . The
// → wrapper solveModAP below
// removes both restrictions:  $\gcd$  is peeled off by rewriting the
// → sequence, and  $n > m$  is split
// into  $\text{floor}(n/m)$  identical full cycles plus a remainder.
// COMPLEXITY:  $O(\sqrt{m} \log m)$  for the decomposition (the sort
// → dominates).
struct APBlock { ll start, diff, len; }; // start +
// → i*diff for  $0 \leq i < \text{len}$ 
vector<APBlock> decomposeModAP(ll a, ll b, ll m, ll n) {
    vector<APBlock> res;
    ll w = (ll)sqrtl((ld)m) + 1;
    if (n <= w) { // too short
        // → to be worth the machinery
        for (ll i = 0; i < n; i++) res.push_back({(a * i + b) % m, 0,
        // → 1});
        return res;
    }
    vector<pair<ll, ll>> v(w + 1); // (residue,
    // → index) for the first w+1 terms
    for (ll i = 0; i <= w; i++) v[i] = {a * i % m, i};
    sort(all(v));
    ll d = 1, h = m; // a*d = +-h
    // → (mod m) with  $h \leq m/w$ ,  $|d| \leq w$ 
    for (ll i = 0; i < w; i++) { //
        // → pigeonhole: two residues within m/w
        ll diff = v[i + 1].first - v[i].first;
        if (diff < h) h = diff, d = v[i + 1].second - v[i].second;
    }
    if (d > 0) { // stepping k
        // → by d increases the value by h
        for (ll r = 0; r < d; r++)
            for (ll j = r; j < n; j++) {
                ll val = (a * j + b) % m;
                ll step = min((m - val - 1) / h + 1, (n - j - 1) / d
                // → + 1);
                res.push_back({val, h, step});
                j += step * d;
            }
    } else { // stepping k
        // → by -d decreases it by h
        for (ll r = n - 1; r >= n + d; r--)
            for (ll j = r; j >= 0; j--) {
                ll val = (a * j + b) % m;
                ll step = min((m - val - 1) / h + 1, j / -d + 1);
                res.push_back({val, h, step});
                j += step * d;
            }
    }
    return res;
}
// FULL WRAPPER for arbitrary a, b, m, n: reduce gcd, then full
// → cycles, then one decomposition.
// g = gcd(a, m) > 1: write  $b = b_1 \cdot g + b_2$  with  $b_2 < g$ ; then  $(a \cdot k + b) \bmod m$ 
// → =  $b_2 + g \cdot ((a/g \cdot k + b_1) \bmod (m/g))$  -
// → recurse on the smaller modulus.
// n >= m (after gcd = 1): each residue in  $[0, m)$  appears exactly
// → once per cycle, so
// → floor(n/m) full cycles contribute a constant,
// → and  $n \bmod m$  is left over.
// EXAMPLE (sum of  $x^{((a \cdot k + b) \bmod m)}$  mod p): per block the answer
// → is the geometric-like sum
//  $x^{\text{start}} \cdot (1 + x^{\text{diff}} + \dots + x^{(\text{diff} \cdot (\text{len} - 1))})$ , computable in
// →  $O(\log \text{len})$  by the standard
// "return  $(x^n, 1 + x + \dots + x^{(n-1)})$  together" doubling recursion.
```

4.3.10 RootsOfUnityFilter

SUM OF BINOMIALS OVER A CONGRUENCE CLASS: $r[j] = \sum_{k \equiv j \pmod m} \binom{n}{k}$, for every

```
// Needs: multiply (06 - Math/05 - Convolution/02 - NTT.cpp), modulus
// → NMOD = 998244353.
// j at once, with n up to  $1e18$  and m up to  $\sim 1e5$ .  $O(m \log m \log n)$ .
// THE IDEA:  $(1 + x)^n$  has  $\binom{n}{k}$  as its coefficients, so the answer
// → is that polynomial reduced
// MODULO  $x^m - 1$  (i.e. exponents folded by  $k \bmod m$ ).
// → Binary-exponentiate  $(1 + x)$  in the ring
//  $\mathbb{Z}_p[x] / (x^m - 1)$ : one NTT product plus a fold per bit.
// PRECONDITIONS: works modulo NMOD (998244353); no condition on m at
// → all - unlike the classical
// roots-of-unity filter, this needs no primitive m-th root to exist.
vector<ll> binomByResidue(ll n, int m) { // r[j] =
    // →  $\sum_{k \equiv j \pmod m} \binom{n}{k}$ 
    auto fold = [&](vector<ll> v) { // reduce
        // modulo  $x^m - 1$ 
        for (int i = (int)v.size() - 1; i >= m; i--) v[i - m] = (v[i]
        // → - m] + v[i]) % NMOD;
        v.resize(min<int>(v.size(), m));
        return v;
    };
    vector<ll> res = {1}, base = fold({1, 1});
    for (; n; n >>= 1) {
        if (n & 1) res = fold(multiply(res, base));
        base = fold(multiply(base, base));
    }
    res.resize(m);
    return res;
}
// THE CLASSICAL ROOTS-OF-UNITY FILTER, which is what you use when m
// → is TINY (2, 3, 4, ...) and
// you want a closed form instead of a convolution. With w a
// → primitive m-th root of unity:
//  $\sum_{k \equiv r \pmod m} \binom{n}{k} a_k = (1/m) \cdot \sum_{j=0}^{m-1} w^{(-j)r} \cdot A(w^j)$ 
// →  $A(w^j) = \sum a_k x^k$ 
// so for  $a_k = \binom{n}{k}$ ,  $A(x) = (1+x)^n$  and
//  $\sum_{k \equiv r \pmod m} \binom{n}{k} = (1/m) \cdot \sum_{j=0}^{m-1} w^{(-j)r} \cdot (1+w^j)^n$ 
// →  $\cdot (1+w^j)^n$ .
// Examples: m = 2 gives  $2^{(n-1)}$  for both classes ( $n \geq 1$ ); m = 4
// → gives the classic
//  $(2^n + 2^{((n+2)/2)} \cdot \cos(n \pi / 4)) / 4$  style answers.
// WHERE TO GET w: if  $m \mid p - 1$  use  $g^{((p-1)/m)}$  for a primitive root
// → g mod p; otherwise work in
// the extension  $\mathbb{F}_p[x]/(\text{cyclotomic})$ , or just use this polynomial
// → version, which sidesteps it.
// SAME TECHNIQUE, other sequences: replace  $(1 + x)$  by ANY generating
// → function and you filter its
// coefficients by residue - e.g. "number of subsets with sum
// → divisible by m", "number of paths of
// length n whose displacement is  $0 \bmod m$ ". Reduce modulo  $x^m - 1$  and
// → exponentiate.
```

4.3.11 BinomialPrefixQueries

PARTIAL SUMS OF A BINOMIAL ROW: $f(n, k) = C(n, 0) + C(n, 1) + \dots + C(n, k)$, for q queries.

```
// Needs: Mint, buildFact, C (06 - Math/01 - Modular Arithmetic/01 -
  ↳ Mint.cpp).
// There is NO closed form for  $f(n, k)$ , but both coordinates move in
  ↳  $O(1)$ :
//  $f(n+1, k) = 2*f(n, k) - C(n, k)$   $f(n, k+1) = f(n, k) +$ 
  ↳  $C(n, k+1)$ 
//  $f(n-1, k) = (f(n, k) + C(n-1, k)) / 2$   $f(n, k-1) = f(n, k) -$ 
  ↳  $C(n, k)$ 
// so treat the queries as points  $(n, k)$  and run MO'S ALGORITHM on
  ↳ them:  $O((n + q) \sqrt{n})$  with a
// tiny constant, versus  $O(n)$  per query naively.
// REACH FOR IT when a problem asks "probability that at most  $k$  of  $n$ 
  ↳ coins came up heads" or
// "number of subsets of size  $\leq k$ " for many  $(n, k)$  pairs offline.
// PRECONDITIONS: MOD prime, buildFact() called, all  $n < N$ . Queries
  ↳ must be answered OFFLINE.
vector<Mint> binomPrefixQueries(vector<pair<int, int>> qs) {
    int q = qs.size(), B = max(1, (int)(N / max(1.0,
  ↳ sqrt((double)q))));
    vector<int> ord(q);
    iota(all(ord), 0);
    sort(all(ord), [&](int x, int y) {
        int bx = qs[x].first / B, by = qs[y].first / B;
        if (bx != by) return bx < by;
        return (bx & 1) ? qs[x].second > qs[y].second : qs[x].second
  ↳ < qs[y].second;
    });
    vector<Mint> ans(q);
    Mint cur = 1, inv2 = Mint((MOD + 1) / 2); //  $f(1, 0) =$ 
  ↳  $C(1, 0) = 1$ 
    int n = 1, k = 0;
    for (int i : ord) {
        auto [L, R] = qs[i];
        while (n < L) cur = cur * 2 - C(n, k), n++;
        while (n > L) --n, cur = (cur + C(n, k)) * inv2;
        while (k < R) ++k, cur += C(n, k);
        while (k > R) cur -= C(n, k), k--;
        ans[i] = cur;
    }
    return ans;
}
// EXACT VALUES you can use as identities instead, when they apply:
//  $f(n, n) = 2^n$ ;  $f(n, k) = 2^n - f(n, n-k-1)$ ;  $f(2m, m) = (2^{2m})$ 
  ↳  $+ C(2m, m) / 2$ .
// A SINGLE query with  $n$  up to  $1e18$  and  $k$  small is just Cbig summed  $k$ 
  ↳ times.
// The same Mo trick works for any quantity with  $O(1)$  moves in both
  ↳ indices - partial sums of
//  $C(n, i) \cdot x^i$ , of Catalan-like rows, or of a fixed row of any
  ↳ Pascal-style triangle.
```

4.3.12 IntersectAP

INTERSECTION OF TWO ARITHMETIC PROGRESSIONS $\{a_1 + d_1 \cdot i : i \geq 0\}$ and $\{a_2 + d_2 \cdot j : j \geq 0\}$.

```
// Needs: crt2 (04 - CongruencesAndTowers.cpp) or CRT (01 - CRT.cpp).
// The intersection is itself an AP with difference  $\text{lcm}(d_1, d_2)$ ,
  ↳ whose first term is the smallest
// common value that is  $\geq \max(a_1, a_2)$ . Empty exactly when the CRT
  ↳ system is unsolvable, i.e.
// when  $(a_1 - a_2)$  is not divisible by  $\text{gcd}(d_1, d_2)$ .
// Returns {first, step}; step = 0 means EMPTY.  $O(\log)$  via one CRT.
// WHEN: merging periodic events ("both traffic lights green"),
  ↳ intersecting the answer sets of
// several modular constraints, and range-counting over a set defined
  ↳ by many congruences.
// OVERFLOW:  $\text{lcm}(d_1, d_2)$  can reach  $d_1 \cdot d_2$  - use __int128 inside CRT
  ↳ (the notebook version does).
pair<ll, ll> intersectAP(ll a1, ll d1, ll a2, ll d2) {
    auto [r, m] = crt2(((a1 % d1) + d1) % d1, d1, ((a2 % d2) + d2) %
  ↳ d2, d2);
    if (m == -1) return {0, 0}; // no common
  ↳ term at all
    ll start = max(a1, a2);
    if (r < start) r += (start - r + m - 1) / m * m; // advance to
  ↳ the first term  $\geq$  start
    return {r, m};
}
// COUNTING the common terms in  $[lo, hi]$  is then  $(hi - \text{first})/\text{step} +$ 
  ↳ 1 when first  $\leq hi$ .
// MANY progressions: fold them left to right with this function; the
  ↳ difference multiplies up to
// the lcm of all of them, so it overflows fast - keep an eye on it,
  ↳ or cap at "beyond the range".
// THE INFINITE-IN-BOTH-DIRECTIONS version ( $i$  and  $j$  any integer) is
  ↳ just the CRT answer with no
//  $\max(a_1, a_2)$  clamp - that is the subgroup  $a_1 + \text{gcd}$ -shifted lattice,
  ↳ see also
// 01 - Euclidean Algorithm/10 - IntegerLattice2D.cpp for the
  ↳ two-coordinate analogue.
```

4.3.13 PohligHellman

discreteLog (02 - DiscreteLogAndRoots.cpp), CRT (01 - CRT.cpp).

```
// Needs: powmod, factorize (02 - Primes and Divisors/06 -
//   ↳ MillerRabinPollard.cpp),
//   discreteLog (02 - DiscreteLogAndRoots.cpp), CRT (01 -
//   ↳ CRT.cpp).
// POHLIG-HELLMAN DISCRETE LOG - solve  $g^x = h \pmod p$  in  $O(\sum \text{over } q^e \mid n \text{ of } e \cdot (\log n + \sqrt{q}))$ 
// instead of the  $O(\sqrt{p})$  of plain baby-step giant-step, where  $n =$ 
//   ↳  $\text{ord}(g)$ . It is a massive win
// exactly when  $n$  is SMOOTH (all prime factors small), which is
//   ↳ common in constructed problems:
//  $p - 1 = 2^k \cdot \text{small stuff}$ , or a subgroup of small order.
// THE IDEA: the discrete log is determined modulo each prime power
//   ↳  $q^e$  dividing  $n$ . Inside the
// order- $q$  subgroup a log costs only  $O(\sqrt{q})$ , and the  $q$ -adic digits
//   ↳ of  $x$  are recovered one at a
// time; CRT glues the residues back together.
// PRECONDITIONS:  $p$  PRIME (the inverse below uses Fermat),  $g$  of
//   ↳ multiplicative order exactly  $n$ 
// (pass  $n = p - 1$  when  $g$  is a primitive root),  $0 < h < p$ . Returns  $x$ 
//   ↳ in  $[0, n)$ , or  $-1$  if  $h$  is not
// in the subgroup generated by  $g$ .
ll pohligHellman(ll g, ll h, ll p, ll n) {
    vector<ll> rs, ms;
    for (auto [q, e] : factorize(n)) {
        ll qe = 1;
        for (int i = 0; i < e; i++) qe *= q;
        ll gq = powmod(g, n / q, p), x = 0, cur = 1; // gq
        ↳ generates the subgroup of order q
        for (int k = 0; k < e; k++) { // recover
            ↳ the  $q$ -adic digits of  $x$ 
            ll gx = powmod(powmod(g, x, p), p - 2, p); //  $g^{-x}$ 
            ll hk = powmod((ll)((__int128)h * gx % p), n / (cur * q),
            ↳ p);
            ll d = discreteLog(gq, hk, p); // a log
            ↳ inside a group of size q
            if (d < 0) return -1;
            x += d * cur, cur *= q;
        }
        rs.push_back(x % qe), ms.push_back(qe);
    }
    return CRT(rs, ms).first;
}
// WHEN TO USE WHICH
// n smooth -> this file (can be near-instant even for
//   ↳  $p \sim 1e18$ )
// n has a large prime -> plain BSGS,  $O(\sqrt{p})$  time AND memory
//   ↳ (02 - DiscreteLogAndRoots.cpp)
// composite modulus -> the BSGS in 02 already strips gcd(a,
//   ↳ m); Pohlig-Hellman needs a
//   ↳ group, so factor m and CRT the
//   ↳ per-prime-power answers yourself.
// SANITY: check powmod(g, n, p) == 1 and that n really is the ORDER
//   ↳ of g (multOrder), otherwise
// the CRT reassembles a residue modulo the wrong number and the
//   ↳ answer is silently wrong.
// SECURITY TRIVIA THAT IS ACTUALLY A HINT: this is why cryptographic
//   ↳ primes are chosen so that
//  $p - 1$  has a large prime factor. If a problem hands you a "random"
//   ↳ prime whose  $p - 1$  factors
// into small pieces, that is the intended solution.
```

4.4 Theorems

4.4.1 NumberTheoryFormulas

NUMBER THEORY - FORMULA SHEET =====

```
/* ===== NUMBER THEORY - FORMULA SHEET =====
MULTIPLICATIVE FUNCTIONS (f(ab)=f(a)f(b) for gcd(a,b)=1)
tau(n)=#divisors=prod(e_i+1) sigma(n)=sum of divisors=prod
↳ ( $p^e+1$ )/( $p-1$ )
phi(n)=n*prod(1-1/p) mu(n)= 0 if NOT SQUAREFREE (some
↳  $p^2 \mid n$ ), else  $(-1)^{\#\text{primes}}$ 
sum_{d|n} phi(d) = n sum_{d|n} mu(d) = [n==1]
↳ (this is the key identity)
sum_{d|n} mu(d)/d = phi(n)/n sum_{d|n} mu(d)*tau(n/d) = 1

DIRICHLET CONVOLUTION (f*g)(n) = sum_{d|n} f(d) g(n/d)
1*1 = tau Id*1 = sigma phi*1 = Id mu*1 =
↳ eps([n==1])
mu is the inverse of 1, so  $f = g*1 \Leftrightarrow g = f*mu$ .

MOBIUS INVERSION
g(n) = sum_{d|n} f(d) <=> f(n) = sum_{d|n} mu(d) g(n/d)
g(n) = sum_{n|d, d<=N} f(d) <=> f(n) = sum_{n|d, d<=N} mu(d/n)
↳ g(d) (the "upward" form)
Standard use: [gcd(i,j)==1] = sum_{d | gcd(i,j)} mu(d).

GCD / LCM SUMS (all  $O(N \log N)$  or better with a mu/phi sieve)
#coprime pairs (i,j)<=n = sum_{d=1..n} mu(d) * floor(n/d)^2
sum_{i,j<=n} gcd(i,j) = sum_{d=1..n} phi(d) * floor(n/d)^2
sum_{d|n} d*phi(n/d) = sum_{i=1..n} gcd(i,n)
sum_{i=1..n} lcm(i,n) = n/2 * (1 + sum_{d|n} d*phi(d))
sum_{i,j<=n} lcm(i,j) = sum_g g * sum_{a,b<=n/g,
↳ gcd(a,b)=1} a*b
general: sum_{i,j} f(gcd(i,j)) = sum_d (f*mu)(d) * floor(n/d)^2

DIVISOR BLOCKING - evaluates sum_{i=1..n} F(floor(n/i)) in  $O(\sqrt{n})$ :
↳ n):
for (ll l = 1, r; l <= n; l = r + 1) { r = n / (n / l); } //
↳ floor(n/i) == n/l on [l, r]
Gives sum tau(i), sum sigma(i), sum floor(n/i), and Min_25 /
↳ Lucy-style prefix sums.

MODULAR
Fermat:  $a^{p-1} = 1 \pmod p$ ,  $p$  prime,  $p \nmid a \Rightarrow a^{p-1} =$ 
↳  $a^{p-2}$ 
Euler:  $a^{\phi(m)} = 1 \pmod m$  if gcd(a,m)=1
GENERALISED EULER (a and m NOT coprime), for  $n \geq \log^2(m)$ :
 $a^n = a^{(n \bmod \phi(m) + \phi(m))} \pmod m \quad \leftarrow$ 
↳ power towers
Wilson:  $(p-1)! = -1 \pmod p$  iff  $p$  prime
CRT:  $x = a_i \pmod{m_i}$  has a unique solution mod lcm(m_i) iff
↳  $a_i = a_j \pmod{\gcd(m_i, m_j)}$ 
Legendre: exponent of p in  $n!$  = sum_{i>=1} floor(n / p^i) = (n -
↳ s_p(n)) / (p - 1)
Kummer: exponent of p in  $C(n, r)$  = #carries when adding r and
↳ (n-r) in base p
LTE (p odd, p | a-b, p !| a, b):  $v_p(a^n - b^n) = v_p(a-b) + v_p(n)$ 
p=2 (a, b both ODD):
n odd :  $v_2(a^n - b^n) = v_2(a-b)$  <- the case
↳ everyone forgets
n even :  $v_2(a^n - b^n) = v_2(a-b) + v_2(a+b) + v_2(n) - 1$ 
n odd :  $v_2(a^n + b^n) = v_2(a+b)$ 
n even :  $v_2(a^n + b^n) = 1$ 
LTE for SUMS (p odd, p | a+b, p !| a, b):
n odd ->  $v_p(a^n + b^n) = v_p(a+b) + v_p(n)$ 
n even -> p does not divide  $a^n + b^n$  at all

QUADRATIC RESIDUES (p odd prime)
Euler's criterion:  $a^{(p-1)/2} = +1$  if a is a QR,  $-1$  if not
#QRs = (p-1)/2.  $x^2=a$  has 0 or 2 roots. Use Tonelli-Shanks.
-1 is a QR iff  $p \equiv 1 \pmod 4$ ; 2 is a QR iff  $p \equiv \pm 1 \pmod 8$ 

PRIMITIVE ROOTS exist exactly for  $m = 1, 2, 4, p^k, 2p^k$  (p odd
↳ prime). Count = phi(phi(m)).

REPRESENTATIONS
n is a sum of two squares iff every prime  $p \equiv 3 \pmod 4$  has an
↳ EVEN exponent
n is a sum of three squares iff  $n \neq 4^a(8b+7)$  every n is a
↳ sum of four squares
Primitive Pythagorean triples:  $(m^2-k^2, 2mk, m^2+k^2)$ ,  $m>k>0$ ,
↳ gcd(m,k)=1, m-k odd

SIZE FACTS (for choosing an algorithm)
max tau(n): 1344 for  $n \leq 1e9$ , 6720 for  $n \leq 1e12$ , 103680 for  $n \leq 1e18$ 
sum_{i<=n} tau(i) ~  $n \ln n$ ; sum_{i<=n} phi(i) ~  $3n^2/\pi^2$ 
pi(n) ~  $n/\ln n$ ; n-th prime ~  $n \ln n$ ; Bertrand: a prime always
↳ lies in (n, 2n)
gaps: for  $n \leq 1e18$  the max prime gap is < 1500
Pisano period pi(m) | ... : F(n) mod m repeats with period <= 6m
```

```

===== */

4.4.2 MobiusInversion
Working code for the identities on the formula sheet. Needs MultSieve (mu, phi).
// Needs: MultSieve (04 - NT/02 - Primes and Divisors/07 -
  ↳ MultiplicativeSieve.cpp) for mu/phi.

// #pairs (i,j) with 1<=i,j<=n and gcd(i,j)=1          0(n)
ll coprimePairs(int n, const MultSieve& S) {
    ll r = 0;
    for (int d = 1; d <= n; d++) r += (ll)S.mu[d] * (n / d) * (n / d);
    return r;
}

// sum over all pairs of gcd(i,j)                      0(n)
ll gcdSum(int n, const MultSieve& S) {
    ll r = 0;
    for (int d = 1; d <= n; d++) r += (ll)S.phi[d] * (n / d) * (n /
↳ d);
    return r;
}

// General: sum_{i,j<=n} f(gcd(i,j)) = sum_d (f*mu)(d) *
↳ floor(n/d)^2
// Build g = f*mu once by sieving over multiples: for d: for
↳ m=d,2d,...: g[m] += mu[m/d]*f[d]
vector<ll> dirichletMu(const vector<ll>& f, const MultSieve& S) {
    ↳ // g = f * mu    0(n log n)
    int n = f.size() - 1;
    vector<ll> g(n + 1, 0);
    for (int d = 1; d <= n; d++) if (f[d])
        for (int m = d; m <= n; m += d) g[m] += (ll)S.mu[m / d] *
↳ f[d];
    return g;
}

// DIVISOR BLOCKING: sum_{i=1..n} floor(n/i) in O(sqrt n). Same
↳ skeleton for sum tau / sum sigma.
ll sumFloorDiv(ll n) {
    ll r = 0;
    for (ll l = 1, rr; l <= n; l = rr + 1) { rr = n / (n / l); r +=
↳ (n / l) * (rr - l + 1); }
    return r;
}

// sum_{i=1..n} tau(i) = sum_{i=1..n} floor(n/i)          (count
↳ multiples)
// sum_{i=1..n} sigma(i) = sum_{i=1..n} i * floor(n/i)

// COUNT MULTIPLES / INCLUSION-EXCLUSION OVER DIVISORS:
// "how many i<=n are divisible by at least one of p1..pk" -> IE over
↳ the 2^k squarefree products,
// sign = mu of the product. For k up to ~20 enumerate submasks;
↳ beyond that, sieve mu.

// sum_{i=1..n} gcd(i, n) = sum_{d|n} d * phi(n/d)
ll gcdSumWithN(ll n) {
    ll r = 0;
    for (auto d : divisors(n)) r += (ll)d * phi(n / d);
    return r;
}

// sum_{i=1..n} lcm(i, n) = n/2 * (1 + sum_{d|n} d*phi(d))
ll lcmSumWithN(ll n) {
    ll s = 1;
    for (auto d : divisors(n)) s += (ll)d * phi(d);
    return n / 2 * s + (n % 2 ? s / 2 : 0);
}

```

4.4.3 PrimeCounting

LUCY_HEDGEHOG: $\pi(n) = \#\text{primes} \leq n$, and the SUM of primes $\leq n$, in $O(n^{3/4})$ time and

```

// O(sqrt n) memory. Handles n up to ~1e12 in well under a second;
↳ 1e13 in a few seconds.
// Idea: S(v, p) = (count or sum) of the integers in [2, v] left
↳ after sieving by all primes < p.
// Sieving by p only touches v >= p^2, so the whole table lives on
↳ the O(sqrt n) distinct
// values of floor(n / i).
ll primePi(ll n) {
    if (n < 2) return 0;
    ll r = (ll)sqrtl((ld)n);
    while (r * r > n) r--;
    while ((r + 1) * (r + 1) <= n) r++;
    vector<ll> S(r + 1), L(r + 1); // S[v] for
↳ v <= r, L[i] for v = n/i
    for (ll i = 1; i <= r; i++) S[i] = i - 1, L[i] = n / i - 1;
    for (ll p = 2; p <= r; p++) {
        if (S[p] == S[p - 1]) continue; // p is not
↳ prime
        ll sp = S[p - 1], p2 = p * p;
        for (ll i = 1; i <= min(r, n / p2); i++)
            L[i] -= (i * p <= r ? L[i * p] : S[n / (i * p)] - sp);
        for (ll v = r; v >= p2; v--) S[v] -= S[v / p] - sp;
    }
    return L[1];
}

// Sum of all primes <= n. Same recurrence with f(x) = x(x+1)/2 - 1
↳ and a factor of p.
ll primeSum(ll n) {
    if (n < 2) return 0;
    ll r = (ll)sqrtl((ld)n);
    while (r * r > n) r--;
    while ((r + 1) * (r + 1) <= n) r++;
    // __int128: n(n+1)/2 wraps for n > 4.29e9, and the ANSWER
↳ (~n^2/(2 ln n)) wraps at n ~ 3e10.
    auto f = [](ll x) { return x % 2 ? (lll)(x + 1) / 2 * x - 1 :
↳ (lll)x / 2 * (x + 1) - 1; };
    vector<ll> S(r + 1), L(r + 1);
    for (ll i = 1; i <= r; i++) S[i] = f(i), L[i] = f(n / i);
    for (ll p = 2; p <= r; p++) {
        if (S[p] == S[p - 1]) continue;
        ll sp = S[p - 1], p2 = p * p;
        for (ll i = 1; i <= min(r, n / p2); i++)
            L[i] -= p * ((i * p <= r ? L[i * p] : S[n / (i * p)] -
↳ sp);
        for (ll v = r; v >= p2; v--) S[v] -= p * (S[v / p] - sp);
    }
    return L[1];
}

// SAME SKELETON computes the prefix sum of ANY
↳ completely-multiplicative f (change the two
// initialisations and the multiplier). For general multiplicative f
↳ use the Min_25 sieve, which
// runs this as its first phase and then re-adds the prime-power
↳ contributions.
// APPROXIMATIONS for sanity checks: pi(n) ~ n/ln n, the k-th prime ~
↳ k ln k.

```

4.4.4 Representations

Representing integers: sums of two squares, Pythagorean triples, Legendre symbol.

```
// Needs: powmod, factorize (02 - Primes and Divisors/06 -
  ↳ MillerRabinPollard.cpp).
// Needs factorize / sqrtMod / isPrime.

int legendre(ll a, ll p) { // 1 QR, -1
  ↳ non-residue, 0 if p | a
  a %= p; if (a < 0) a += p;
  if (!a) return 0;
  return powmod(a, (p - 1) / 2, p) == 1 ? 1 : -1;
}
// n is a sum of two squares <=> every prime p = 3 (mod 4) appears
  ↳ to an EVEN power.
bool isSumOfTwoSquares(ll n) {
  if (n <= 0) return n == 0;
  for (auto [p, e] : factorize(n)) if (p % 4 == 3 && (e & 1))
    ↳ return false;
  return true;
}
// One representation n = x^2 + y^2 (Cornacchia). Returns {-1,-1} if
  ↳ none exists.
pair<ll, ll> sumOfTwoSquares(ll n) {
  if (n == 0) return {0, 0};
  ll t = n, pw2 = 1;
  while (t % 4 == 0) t /= 4, pw2 *= 2;
  ↳ squares of 2
  if (!isSumOfTwoSquares(t)) return {-1, -1};
  // Gaussian-integer product over the prime factorisation
  ll xr = 1, yr = 0; // current
  ↳ value as xr + yr*i
  for (auto [p, e] : factorize(t)) {
    if (p == 2) { for (int i = 0; i < e; i++) { ll nx = xr - yr,
    ↳ ny = xr + yr; xr = nx, yr = ny; } continue; }
    if (p % 4 == 3) { ll k = 1; for (int i = 0; i < e / 2; i++) k
    ↳ *= p; xr *= k, yr *= k; continue; }
    ll r = sqrtMod(p - 1, p); // r^2 = -1
    ↳ (mod p)
    ll a = p, b = r;
    while (b * b > p) { ll c = a % b; a = b, b = c; } //
    ↳ Cornacchia descent: p = b^2 + c^2
    ll c = (ll)sqrtl((ld)(p - b * b));
    while (c * c < p - b * b) c++;
    while (c * c > p - b * b) c--;
    for (int i = 0; i < e; i++) { ll nx = xr * b - yr * c, ny =
    ↳ xr * c + yr * b; xr = nx, yr = ny; }
  }
  return {llabs(xr) * pw2, llabs(yr) * pw2};
}
// #representations of n as an ORDERED pair of squares (with signs) =
  ↳ 4 * (d_1(n) - d_3(n)),
// where d_i counts divisors = i (mod 4). [Jacobi's two-square
  ↳ theorem]

// PRIMITIVE Pythagorean triples with hypotenuse <= L: (m^2-k^2,
  ↳ 2mk, m^2+k^2),
// m > k > 0, gcd(m,k) = 1, m-k odd. Every triple is a multiple of
  ↳ exactly one primitive triple.
vector<array<ll, 3>> pythagorean(ll L) {
  vector<array<ll, 3>> r;
  for (ll m = 2; m * m <= L; m++)
    for (ll k = 1 + (m % 2); k < m; k += 2) // m-k must
    ↳ be ODD => opposite parity
      if (gcd(m, k) == 1 && m * m + k * k <= L)
        r.push_back({m * m - k * k, 2 * m * k, m * m + k *
        ↳ k});
  return r;
}
// n is a sum of THREE squares <=> n != 4^a * (8b + 7). Every n
  ↳ is a sum of FOUR squares.
```

4.4.5 DuSieve

DU SIEVE (Dirichlet prefix sums) - prefix sums of a multiplicative f in SUBLINEAR time.

```
// Pick g with an easy prefix sum G and an easy (f*g) prefix sum; then
// g(1)*S(n) = sum_{i=1..n} (f*g)(i) - sum_{i=2..n} g(i) *
  ↳ S(floor(n/i))
// Sieve S(x) directly for x <= LIM (take LIM ~ n^(2/3)); recurse
  ↳ with memoisation above that.
// Total O(n^(2/3)) with the sieve, O(n^(3/4)) without.
// mu * 1 = eps -> M(n) = 1 - sum_{i=2..n} M(n/i)
// phi * 1 = Id -> Phi(n) = n(n+1)/2 - sum_{i=2..n} Phi(n/i)
struct DuSieve {
  ll LIM;
  vector<ll> mu, phi; // prefix sums
  ↳ for x <= LIM
  unordered_map<ll, ll> memoM, memoP;

  DuSieve(ll n) {
    LIM = max((ll)1000, (ll)powl((ld)n, 2.0L / 3));
    vector<int> spf(LIM + 1, 0), primes, m(LIM + 1, 0), p(LIM +
    ↳ 1, 0);
    m[1] = p[1] = 1;
    for (ll i = 2; i <= LIM; i++) {
      if (!spf[i]) spf[i] = i, primes.push_back(i), m[i] = -1,
    ↳ p[i] = i - 1;
      for (int q : primes) {
        if (q > spf[i] || i * q > LIM) break;
        spf[i * q] = q;
        if (q == spf[i]) m[i * q] = 0, p[i * q] = p[i] * q;
        else m[i * q] = -m[i], p[i * q] = p[i] * (q - 1);
      }
    }
    mu.assign(LIM + 1, 0), phi.assign(LIM + 1, 0);
    for (ll i = 1; i <= LIM; i++) mu[i] = mu[i - 1] + m[i],
    ↳ phi[i] = phi[i - 1] + p[i];
  }
  ll M(ll n) { // sum_{i=1..n}
    ↳ mu(i) (Mertens)
    if (n <= LIM) return mu[n];
    auto it = memoM.find(n);
    if (it != memoM.end()) return it->second;
    ll r = 1;
    for (ll l = 2, rr; l <= n; l = rr + 1) { rr = n / (n / l); r
    ↳ -= (rr - l + 1) * M(n / l); }
    return memoM[n] = r;
  }
  ll P(ll n) { // sum_{i=1..n}
    ↳ phi(i)
    if (n <= LIM) return phi[n];
    auto it = memoP.find(n);
    if (it != memoP.end()) return it->second;
    ll r = n % 2 ? (n + 1) / 2 * n : n / 2 * (n + 1);
    for (ll l = 2, rr; l <= n; l = rr + 1) { rr = n / (n / l); r
    ↳ -= (rr - l + 1) * P(n / l); }
    return memoP[n] = r;
  }
};
// USES: squarefree count up to n = sum_{d<=sqrt n} mu(d) *
  ↳ floor(n/d^2);
// #coprime pairs up to n = 2*Phi(n) - 1; sum of gcd/lcm over
  ↳ pairs at n ~ 1e10;
// any "sum over 1..n of a multiplicative function" that a
  ↳ linear sieve cannot reach.
```


4.4.6 MoreFormulaCards

NUMBER THEORY, SECOND FORMULA SHEET =====

```

/* ===== NUMBER THEORY, SECOND FORMULA SHEET =====
Everything here was machine-verified against brute force / sympy.

QUADRATIC RECIPROCITY. p, q distinct ODD primes.
(p/q)(q/p) = (-1)^((p-1)/2 * (q-1)/2)
=> (p/q) = (q/p), UNLESS p = q = 3 (mod 4), where (p/q) = -(q/p)
SUPPLEMENTS: (-1/p) = 1 iff p = 1 (mod 4); (2/p) = 1 iff p = +-1
-> (mod 8);
(3/p) = 1 iff p = +-1 (mod 12)
EULER'S CRITERION (p odd prime AND p !| a): a^((p-1)/2) = (a/p)
-> (mod p).
If p | a the value is 0, not +-1 - state the hypothesis.
#k-th ROOTS of a != 0 mod p: d = gcd(k, p-1); the count is d if
-> a^((p-1)/d) = 1, else 0.
JACOBI SYMBOL (a/n) for ODD n > 0, n NOT necessarily prime - O(log
-> n), no factoring: */
int jacobi(ll a, ll n) { // n
-> odd, n > 0
int r = 1;
a = ((a % n) + n) % n; //
-> (a/n) is periodic in a mod n
while (a) {
while (a % 2 == 0) { a /= 2; if (n % 8 == 3 || n % 8 == 5) r
-> = -r; }
swap(a, n);
if (a % 4 == 3 && n % 4 == 3) r = -r;
a %= n;
}
return n == 1 ? r : 0;
}
/* CAUTION: (a/n) = 1 does NOT prove a is a QR mod a composite n.
-> (a/n) = -1 DOES prove it is
not. Use it as a cheap filter, and to find a non-residue fast for
-> Tonelli-Shanks.

FIBONACCI (F_0 = 0, F_1 = 1)
FAST DOUBLING, O(log n), no matrices:
F(2k) = F(k) * (2 F(k+1) - F(k)), F(2k+1) = F(k)^2 + F(k+1)^2
gcd(F_m, F_n) = F_gcd(m,n) => F_m | F_n iff m | n (m >= 3)
Cassini: F_{n-1} F_{n+1} - F_n^2 = (-1)^n
d'Ocagne: F_m F_{n+1} - F_{m+1} F_n = (-1)^n F_{m-n}
sum_{i<n} F_i = F_{n+2} - 1 sum_{i<n} F_i^2 = F_n F_{n+1}
F_92 is the last Fibonacci number fitting in a signed 64-bit
-> integer.
PISANO PERIOD pi(m) = the period of F(n) mod m
pi(1)=1, pi(2)=3, and pi(m) is EVEN for every m > 2
pi(lcm(a,b)) = lcm(pi(a), pi(b)) pi(p^k) = p^(k-1) * pi(p)
pi(m) <= 6m, with EQUALITY exactly at m = 2 * 5^k (verified: m =
-> 10, 50, 250)
p = +-1 (mod 5) => pi(p) | p-1; p = +-2 (mod 5) => pi(p) | 2(p+1)
ZECKENDORF: every n >= 1 is a UNIQUE sum of NON-CONSECUTIVE
-> Fibonacci numbers, found
greedily from the top. The basis of "Fibonacci base" digit DP and
-> of Wythoff's game.

PELL: x^2 - D y^2 = 1, D > 0 non-square. Always infinitely many
-> solutions.
sqrt(D) = [a0; a1, ..., a_L] is periodic with a_L = 2 a0 and a
-> palindromic middle.
FUNDAMENTAL SOLUTION = the convergent h_k / k_k with
k = L - 1 if the period L is EVEN, k = 2L - 1 if L is
-> ODD.
ALL solutions: x_m + y_m sqrt(D) = (x1 + y1 sqrt(D))^m, i.e.
x_{m+1} = x1 x_m + D y1 y_m, y_{m+1} = x1 y_m + y1 x_m
NEGATIVE PELL x^2 - D y^2 = -1 is solvable IFF the period L is
-> ODD; then the fundamental
solution is the convergent at k = L - 1.
SIZE WARNING: solutions explode. D = 61 gives x = 1766319049, but
-> D = 661 gives a
39-digit x. 'll' is useless past D ~ 150 - use bignum, or work
-> modulo the answer.
CF MACHINERY (a0 = floor(sqrt(D)); m = 0, d = 1, a = a0), all exact
-> integers:
m' = d*a - m, d' = (D - m'^2) / d, a' = floor((a0 + m')
-> / d')
BEST RATIONAL APPROXIMATION: the closest p/q to x with q <= N is a
-> convergent OR a
semiconvergent (t h1 + h2)/(t k1 + k2) with 1 <= t <= a_next.
-> Check both.
FAREY MEDIANT: for neighbours a/b < c/d (bc - ad = 1), (a+c)/(b+d)
-> is the unique fraction
with the SMALLEST denominator strictly between them.

SUMS OF SQUARES
TWO: n = x^2 + y^2 possible IFF every prime p = 3 (mod 4) has an

```

-> EVEN exponent.

$r_2(n) = 4 (d_1(n) - d_3(n))$, $d_i = \# \text{divisors congruent to } i \pmod 4$.

-> i mod 4.

THREE: $n = x^2 + y^2 + z^2$ possible IFF n is NOT of the form 4^a

-> $(8b + 7)$.

FOUR: ALWAYS possible (Lagrange). $r_4(n) = 8 * \sum_{d|n} 1$

-> with 4 !| d of d.

PRIMITIVE PYTHAGOREAN TRIPLES: $(m^2 - k^2, 2mk, m^2 + k^2)$, $m > k$

-> > 0 , $\gcd(m,k) = 1$,

$m - k$ odd.

GAUSSIAN INTEGERS $Z[i]$: the norm $N(a+bi) = a^2 + b^2$ is

-> multiplicative. Primes of $Z[i]$:

$1+i$ (over 2); $p = 1 \pmod 4$ SPLITS as $\pi_i * \text{conj}(\pi_i)$; $p = 3 \pmod$

-> 4) stays prime.

$\gcd$ by rounded division: $q = \text{round}(a \text{ conj}(b) / N(b))$, $r = a -$

-> qb , then Euclid on norms.

FROBENIUS / CHICKEN McNUGGET. $a, b \geq 2$, $\gcd(a,b) = 1$:

largest non-representable $g(a,b) = ab - a - b$;

-> #non-representable = $(a-1)(b-1)/2$.

For $k \geq 3$ coins there is NO closed form: run Dijkstra on residues

-> mod a_1 ($d[r] = \text{the}$

smallest representable value congruent to r), then $g = \max_r d[r]$

-> $-a_1$. $0(a_1 k \log a_1)$.

BEATTY / RAYLEIGH. $r, s > 1$ IRRATIONAL with $1/r + 1/s = 1$ =>

-> floor(n r) and floor(n s),

$n \geq 1$, PARTITION the positive integers exactly. (Wythoff's losing

-> positions are exactly

(floor(n phi), floor(n phi^2)).)

THREE-DISTANCE (Steinhaus). The points $\{a\}, \{2a\}, \dots, \{na\}$ cut

-> $[0,1)$ into intervals of AT

MOST 3 distinct lengths, and when there are 3 the largest is the

-> sum of the other two.

Use: min over $k \leq n$ of $(k*a \bmod m)$ in $O(\log)$ via the continued

-> fraction of a/m .

DETERMINISTIC MILLER-RABIN BASE SETS (all machine-verified)

$n < 2^{32}$: $\{2, 7, 61\}$ (valid

-> to 4759123141)

$n < 3.2e18$: the first 12 primes $\{2..37\}$

$n < 3.317e24 > 2^{64}$: $\{2, 325, 9375, 28178, 450775, 9780504,$

-> 1795265022}

$\{2,3,5,7\}$ is NOT enough even for 32 bits: $3215031751 =$

-> $151*751*28351$ passes all four.

Skip a base a when n divides a, or small primes are reported

-> composite.

NTT PRIMES (form, primality and primitive root all machine-verified)

p	= c * 2^k + 1	g	max transform length
998244353	= 119 * 2^23 + 1	3	2^23 <- the

-> default

167772161	= 5 * 2^25 + 1	3	2^25
-----------	----------------	---	------

469762049	= 7 * 2^26 + 1	3	2^26
-----------	----------------	---	------

754974721	= 45 * 2^24 + 1	11	2^24 <- NOT
-----------	-----------------	----	-------------

-> $9*2^{23}+1$ (common misprint)

1004535809	= 479 * 2^21 + 1	3	2^21
------------	------------------	---	------

2013265921	= 15 * 2^27 + 1	31	2^27
------------	-----------------	----	------

1224736769	= 73 * 2^24 + 1	3	2^24
------------	-----------------	---	------

3221225473	= 3 * 2^30 + 1	5	2^30
------------	----------------	---	------

922337203673735297	= 549755813881 * 2^24 + 1, g = 3		(64-bit,
--------------------	----------------------------------	--	----------

-> needs __int128 mulmod)

THE 3-PRIME TRIPLE for an arbitrary modulus: $167772161 * 469762049$

-> $* 754974721 = 5.95e25$.

THE FINDING CONSTRAINT IS LENGTH, NOT MAGNITUDE: 754974721 caps

-> the transform at 2^{24} , so

the input length is capped at 2^{23} . At that length the magnitude

-> headroom is still 6x.

64-BIT CEILINGS - memorise these

$a*b$ fits in signed ll iff $a, b < 3037000499$ (=

-> floor(sqrt($2^{63} - 1$)))

$n(n+1)/2$ fits in ll iff $n \leq 4294967295$

sum of primes $\leq n$ fits iff $n < 3e10$ (the sum is ~

-> $n^2 / (2 \ln n)$)

sum of $\phi(i)$, $i \leq n$, fits iff $n \leq 5508509576$ (the sum is ~

-> $0.3040 n^2$)

max tau(n): 1344 for $n \leq 1e9$, 6720 for $1e12$, 103680 for $1e18$

NEED __int128 FOR: any modmul with modulus $> 3.04e9$ (Pollard,

-> BSGS, Tonelli, CRT, Garner);

floorSum with $n > 1e9$; Cornacchia's $b*b > p$ test; prime/totient

-> SUMS at $n \geq 1e10$.

FFT PRECISION, the real criterion (not $|\text{result}| < 1e15$):

safe iff $(\sum a_i^2 + \sum b_i^2) * \log_2(N) < 9e14$.

Splitting at sqrt(M) additionally needs $M^{1.5} < 9.2e18$, i.e. M

-> $< 4.4e12$, with the

inputs already reduced into $[0, M)$.

FWHT: the forward transform inflates magnitudes by up to N, so the

↪ pointwise product is
 bounded by $(N * \max|a|)^2$. Under a modulus the XOR inverse must
 ↪ MULTIPLY BY $\text{inv}(N) \bmod p$ -
 an integer ' $x \neq n$ ' is exact over Z and WRONG mod p .
 ===== */

4.4.7 MultiplicativeGroupStructure

THE STRUCTURE OF Z_n^* (the units mod n) - the facts that decide whether a modular equation

```

// Needs: factorize (02 - Primes and Divisors/06 -
    ↪ MillerRabinPollard.cpp).
// has solutions, how many, and what the true exponent of the group
    ↪ is.
// Z_n^* = product over p^e || n of Z_(p^e)^*, and
// p odd : Z_(p^e)^* is CYCLIC of order  $\phi(p^e) = p^{e-1}(p-1)$ 
    ↪ -> primitive root exists
// p = 2 : Z_2^* trivial, Z_4^* cyclic of order 2, and for e >= 3
// Z_(2^e)^* = Z_2 x Z_(2^{e-2}) -> NOT cyclic, no
    ↪ primitive root
// A PRIMITIVE ROOT mod n exists iff n = 1, 2, 4, p^e or 2*p^e for
    ↪ an odd prime p.
// CARMICHAEL LAMBDA = the true exponent of Z_n^*: the smallest L
    ↪ with a^L = 1 for every unit a.
// It divides  $\phi(n)$  and is often much smaller - use it (not  $\phi$ ) to
    ↪ reduce exponents, and as the
// modulus bound in power towers.
ll carmichael(ll n) {
    ll res = 1;
    for (auto [p, e] : factorize(n)) {
        ll pe = 1;
        for (int i = 0; i < e; i++) pe *= p;
        ll l = pe / p * (p - 1); //  $\phi(p^e)$ 
        if (p == 2 && e >= 3) l /= 2; // the one
        ↪ exception
        res = res / gcd(res, l) * l; // lcm
    }
    return res;
}
// HOW MANY DISTINCT VALUES DOES a^k mod n TAKE, over ALL a in [0,
    ↪ n)? (Not just the units - the
// non-units contribute the recursive term, which is what makes this
    ↪ more than  $|Z_n^*| / \gcd$ .)
// Multiplicative in n, so handle each prime power and multiply.
ll ipow(ll b, ll e) { ll r = 1; while (e-- > 0) r *= b; return r; }
ll kthPowersPrimePower(ll p, ll e, ll k) { // count of
    ↪ distinct a^k mod p^e
    if (e <= 0 || k == 0) return 1;
    if (p == 2) {
        if (e == 1) return 2;
        ll u = ipow(2, e - 2) / gcd(k, ipow(2, e - 2));
        if (k % 2) u *= 2;
        return u + kthPowersPrimePower(2, e - k, k);
    }
    ll ph = ipow(p, e) - ipow(p, e - 1);
    return ph / gcd(k, ph) + kthPowersPrimePower(p, e - k, k);
}
ll kthPowersMod(ll k, ll n) { // #distinct
    ↪ a^k mod n over all a
    ll res = 1;
    for (auto [p, e] : factorize(n)) res *= kthPowersPrimePower(p, e,
        ↪ k);
    return res;
}
// COMPANION FACTS (all about  $x^k = a \bmod p$ , p prime, a != 0):
// the k-th powers form the subgroup of index  $d = \gcd(k, p-1)$ , so
    ↪ there are  $(p-1)/d$  of them;
// a is a k-th power iff  $a^{((p-1)/d)} = 1$ , and then it has exactly d
    ↪ roots.
// For a MODULUS n,  $x^k = a$  is solved prime-power by prime-power
    ↪ and glued with CRT.
// ORDER OF AN ELEMENT: ord(a) | lambda(n); compute it by starting
    ↪ from lambda(n) and dividing
// out prime factors while  $a^{(L/q)} = 1$  (that is multOrder in 03 -
    ↪ Modular Arithmetic/02).
// EXPONENT REDUCTION:  $a^e = a^{(e \bmod \lambda(n) + \lambda(n))} \bmod n$ 
    ↪ for e >= log2(n), which is
// the correct form even when gcd(a, n) > 1 - the same guard the
    ↪ power tower uses.
  
```

4.4.8 FourSquares

sumOfTwoSquares (04 - Representations.cpp).

```

// Needs: isPrime (02 - Primes and Divisors/06 -
    ↪ MillerRabinPollard.cpp),
// sumOfTwoSquares (04 - Representations.cpp).
// LAGRANGE'S FOUR-SQUARE DECOMPOSITION - actually CONSTRUCT x, y, z,
    ↪ w with
// n = x^2 + y^2 + z^2 + w^2
// for n up to 1e18, in expected  $O(\log^2 n)$  primality tests. The
    ↪ existence theorem is famous; the
// construction is the part people cannot reproduce under pressure,
    ↪ and brute force over two of
// the four variables is  $O(n)$  - hopeless past 1e12.
// THE ALGORITHM (Rabin-Shallit, simplified):
// 1. strip factors of 4:  $n = 4^a * m$ , solve for m and multiply
    ↪ every answer by  $2^a$ ;
// 2. if  $m \equiv 7 \pmod{8}$  then m is NOT a sum of three squares, so
    ↪ spend one square:  $m -= 1$ ;
// 3. now m IS a sum of three squares. Pick x at random and test
    ↪ whether  $m - x^2$  is a PRIME
// congruent to 1 mod 4 (or a trivial 0/1/2). Such x are dense,
    ↪ so a handful of tries suffice;
// 4. split that prime into two squares with Cornacchia (04 -
    ↪ Representations.cpp).
// PRECONDITION: n >= 0. Uses randomness, so it is expected-time, not
    ↪ worst-case.
array<ll, 4> fourSquares(ll n) {
    array<ll, 4> r{0, 0, 0, 0};
    if (n <= 0) return r;
    ll mul = 1;
    while (n % 4 == 0) n /= 4, mul *= 2; // 4^a
    ↪ factors out of every square
    int idx = 0;
    if (n % 8 == 7) r[idx++] = 1, n -= 1; // 7 mod 8
    ↪ needs a fourth nonzero square
    if (n < 2000) { // small:
        ↪ brute force, and always succeeds
        for (ll x = 0; x * x <= n && idx < 4; x++)
            for (ll y = x; x * x + y * y <= n; y++) {
                ll t = n - x * x - y * y, z = (ll)sqrtl((ld)t);
                while (z * z > t) z--;
                while ((z + 1) * (z + 1) <= t) z++;
                if (z * z == t) { r[idx++] = x, r[idx++] = y,
                    ↪ r[idx++] = z; break; }
            }
    } else {
        mt19937_64
        ↪ rng(chrono::steady_clock::now().time_since_epoch().count());
        ll lim = (ll)sqrtl((ld)n);
        while (lim * lim > n) lim--;
        while (true) {
            ll x = rng() % (lim + 1), m = n - x * x;
            if (m % 4 != 1 && m != 2) continue; // want a
            ↪ prime = 1 (mod 4), or 2
            if (m != 2 && !isPrime(m)) continue;
            auto [u, v] = sumOfTwoSquares(m);
            if (u < 0) continue;
            r[idx++] = x, r[idx++] = u, r[idx++] = v;
            break;
        }
    }
    for (ll& v : r) v *= mul;
    return r;
}
// FACTS WORTH KNOWING
// THREE squares suffice unless  $n = 4^a * (8b + 7)$  (Legendre); TWO
    ↪ suffice iff every prime
// 3 mod 4 has an even exponent; ONE iff n is a perfect square.
//  $r_4(n) = 8 * \text{sum of the divisors of n that are not multiples of}$ 
    ↪ 4 (Jacobi) - so the number
// of ORDERED signed representations is easy even though finding
    ↪ one is not.
// Every n is a sum of at most 9 cubes and at most 19 fourth powers
    ↪ (Waring); the "at most 4
// squares" bound is tight exactly on the numbers  $4^a(8b+7)$ .
// IF n IS ALREADY FACTORED, a deterministic route exists: multiply
    ↪ Hurwitz quaternions the way
// sumOfTwoSquares multiplies Gaussian integers. The randomized
    ↪ version above is far shorter.
  
```

4.5 Rational

4.5.1 ContinuedFractions

Continued fractions, convergents, best rational approximation, Stern-Brocot search.

```
// x = [a0; a1, a2, ...]. For a rational p/q this is just the
// ↪ Euclidean algorithm's quotients.
vector<ll> cfrac(ll p, ll q) {
    vector<ll> a;
    // FLOOR division: C++ truncates toward zero, so cfrac(-7,2)
    // ↪ would give [-3,-2] instead of
    // the correct [-4,2] and every convergent would be wrong.
    while (q) { ll d = p / q - ((p % q) && ((p < 0) ^ (q < 0)));
        ↪ a.push_back(d); ll r = p - d * q; p = q, q = r; }
    return a;
}
// Convergents h[i]/k[i] of a continued fraction: h = a*h1 + h2, k =
// ↪ a*k1 + k2.
// Every convergent is a BEST approximation: no fraction with a
// ↪ smaller denominator is closer.
vector<pair<ll, ll>> convergents(const vector<ll>& a) {
    vector<pair<ll, ll>> c;
    ll h1 = 1, h2 = 0, k1 = 0, k2 = 1;
    for (ll x : a) {
        ll h = x * h1 + h2, k = x * k1 + k2;
        c.push_back({h, k});
        h2 = h1, h1 = h, k2 = k1, k1 = k;
    }
    return c;
}
// Best rational approximation to p/q with denominator <= N.
// The answer is a convergent OR a SEMICONVERGENT (t*h1+h2)/(t*k1+k2)
// ↪ - you must compare both.
pair<ll, ll> bestApprox(ll p, ll q, ll N) {
    ll h1 = 1, h2 = 0, k1 = 0, k2 = 1, P = p, Q = q;
    pair<ll, ll> best = {p >= 0 ? 0 : -((-p + q - 1) / q), 1};
    ↪ // 0/1 (or floor) is legal
    auto better = [&](pair<ll, ll> a, pair<ll, ll> b) {
        ↪ // is a closer to p/q than b?
        __int128 da = (__int128)llabs(p * a.second - a.first * q) *
        ↪ b.second;
        __int128 db = (__int128)llabs(p * b.second - b.first * q) *
        ↪ a.second;
        return da != db ? da < db : a.second < b.second;
    };
    while (Q) {
        ll a = P / Q, r = P % Q;
        if (k1 && a * k1 + k2 > N) {
            ↪ // cannot take the full step
            ll t = (N - k2) / k1;
            if (t > 0) { pair<ll, ll> c = {t * h1 + h2, t * k1 + k2};
                ↪ if (better(c, best)) best = c; }
            break;
        }
        ll h = a * h1 + h2, k = a * k1 + k2;
        if (k > N) break;
        h2 = h1, h1 = h, k2 = k1, k1 = k;
        if (better({h1, k1}, best)) best = {h1, k1};
        P = Q, Q = r;
    }
    return best;
}
// STERN-BROCOT SEARCH: find the simplest fraction p/q with f(p/q)
// ↪ true, when f is monotone.
// Walk the mediant tree; take exponential jumps in each direction so
// ↪ it is O(log^2).
// Classic uses: "smallest denominator between two reals", recovering
// ↪ a fraction from a decimal,
// and answering "is the answer rational with small denominator"
// ↪ style problems.
// FAREY / mediant fact: if a/b < c/d are neighbours (bc - ad = 1)
// ↪ then their mediant
// (a+c)/(b+d) is the unique simplest fraction strictly between them.
```

4.5.2 PellEquation

PELL'S EQUATION $x^2 - D*y^2 = 1$ ($D > 0$ and NOT a perfect square) - the fundamental solution,

```
// then all of them. Every solution is a CONVERGENT of the continued
// ↪ fraction of sqrt(D), so walk
// the convergents and stop at the first one that satisfies the
// ↪ equation: 0(sqrt D) steps, since
// the period of sqrt(D) is 0(sqrt D log D).
// ALL SOLUTIONS from the fundamental (x1, y1): x_{m+1} = x1*x_m +
// ↪ D*y1*y_m,
// y_{m+1} = x1*y_m +
// ↪ y1*x_m,
// i.e. x_m + y_m sqrt(D) = (x1 + y1 sqrt(D))^m. So there are
// ↪ infinitely many, growing
// EXPONENTIALLY - D = 61 already needs x = 1766319049, and D = 661
// ↪ needs 39 digits. __int128
// carries you to about D = 200; past that use bignum, or work modulo
// ↪ the required output.
// Returns {0, 0} when D is a square or <= 0 (only the trivial (1, 0)
// ↪ exists).
pair<__int128, __int128> pell(ll D) {
    ll a0 = (ll)sqrtl((ll)D);
    while (a0 * a0 > D) a0--;
    while ((a0 + 1) * (a0 + 1) <= D) a0++;
    if (D <= 0 || a0 * a0 == D) return {0, 0};
    ll m = 0, d = 1, a = a0; // the exact
    ↪ CF recurrence for sqrt(D)
    __int128 h1 = 1, h = a0, k1 = 0, k = 1; //
    ↪ convergents h/k
    while (h * h - (__int128)D * k * k != 1) {
        m = d * a - m;
        d = (D - m * m) / d;
        a = (a0 + m) / d;
        __int128 h2 = h1; h1 = h; h = a * h1 + h2;
        __int128 k2 = k1; k1 = k; k = a * k1 + k2;
    }
    return {h, k};
}
// NEGATIVE PELL x^2 - D*y^2 = -1 is solvable IFF the period of the
// ↪ CF of sqrt(D) is ODD; the
// same loop finds it if you test for -1 instead (and it always
// ↪ appears before the +1 solution).
// Necessary condition: D has no prime factor congruent to 3 mod 4
// ↪ (not sufficient: D = 34).
// GENERAL x^2 - D*y^2 = k: find one solution by brute force /
// ↪ Lagrange-Matthews-Mollin, then
// multiply by powers of the fundamental unit to walk the whole
// ↪ class; there can be several
// inequivalent classes, so collect every solution with y below the
// ↪ standard bound.
// CF MACHINERY, exact in integers (a0 = floor(sqrt D), start m = 0,
// ↪ d = 1, a = a0):
// m' = d*a - m, d' = (D - m'^2)/d, a' = floor((a0 + m')/d')
// ↪ - the period ends when
// a' = 2*a0, and the a-sequence between is a palindrome.
// USES: minimal solutions of "x^2 - D y^2 = 1" style Diophantine
// ↪ puzzles, the fundamental unit of
// the ring Z[sqrt D], square-triangular numbers (D = 8), and any
// ↪ recurrence whose growth factor
// is a quadratic irrational.
```

4.5.3 RationalReconstruction

RATIONAL RECONSTRUCTION - recover the fraction p/q from its value $r = p * q^{-1} \pmod{m}$.

```
// The answer is UNIQUE as long as |p| and q are both below
//   ↳ sqrt(m/2), and it is found by running
// the extended Euclidean algorithm on (m, r) and stopping halfway:
//   ↳ the remainder is p and the
// cofactor is q. O(log m).
// WHY YOU WANT IT: do a whole computation in modular arithmetic
//   ↳ (Gaussian elimination, a DP over
// fractions, an interpolation, a determinant) and then read off the
//   ↳ EXACT rational answer -
// no bignum fractions, no gcd-ing at every step. Also the standard
//   ↳ way to guess a closed form
// from modular data, and to answer "print the answer as an
//   ↳ irreducible fraction" when the
// intermediate values would explode.
// PRECONDITIONS: m >= 2, gcd(q, m) = 1 (implicit in r existing).
//   ↳ Returns {0, 0} on failure -
// which means no such small fraction exists, so take a bigger m (or
//   ↳ several and CRT them).
pair<ll, ll> ratRecon(ll r, ll m) {
    ll a = m, b = ((r % m) + m) % m, x = 0, y = 1;    // invariant:
    ↳ b = y * r (mod m)
    ll lim = (ll)sqrtl((ld)m / 2);
    while (b > lim) {
        ll q = a / b;
        ll t = a - q * b; a = b, b = t;
        t = x - q * y; x = y, y = t;
    }
    if (y < 0) b = -b, y = -y;
    if (y == 0 || gcd(b < 0 ? -b : b, y) != 1) return {0, 0};
    return {b, y};    // b / y = r
    ↳ (mod m), in lowest terms
}
// CHECK YOUR RESULT: verify b * inverse(y) mod m == r before
//   ↳ trusting it - the reconstruction can
// return a wrong small fraction if the true one is not small enough.
//   ↳ Verifying is O(log m).
// LARGER BOUNDS: if you know |p| <= P and q <= Q with 2*P*Q < m,
//   ↳ stop the loop at b <= P instead
// and the same argument applies. Choosing m as a product of several
//   ↳ NTT primes (Garner) is the
// usual way to make m big enough.
// RELATED: the same halfway-Euclid stopping rule is the
//   ↳ Berlekamp-Massey / Pade approximant of a
// power series, and 01 - ContinuedFractions.cpp is the
//   ↳ exact-arithmetic version of this walk.
```

4.6 Sequences

4.6.1 FibonacciAndPisano

FIBONACCI, FAST. The notebook had the identities on a formula sheet and no routine; this is it.

```
// F(0)=0, F(1)=1. Fast doubling is O(log n) with two multiplications
//   ↳ per bit and NO matrices -
// shorter and ~2x faster than 2x2 matrix exponentiation.
//   F(2k)   = F(k) * (2*F(k+1) - F(k))
//   F(2k+1) = F(k)^2 + F(k+1)^2
pair<ll, ll> fibPair(ll n, ll mod) {
    ↳ returns {F(n), F(n+1)} mod
    if (!n) return {0, 1 % mod};
    auto [a, b] = fibPair(n >> 1, mod);
    ll c = a * ((2 * b % mod - a % mod + mod) % mod) % mod;    //
    ↳ F(2k)
    ll d = (a * a + b * b) % mod;    //
    ↳ F(2k+1)
    return (n & 1) ? make_pair(d, (c + d) % mod) : make_pair(c, d);
}
ll fib(ll n, ll mod) { return fibPair(n, mod).first; }
// LUCAS NUMBERS L(n) = F(n-1) + F(n+1) = 2F(n+1) - F(n).
// NEGATIVE index: F(-n) = (-1)^(n+1) F(n).
// The mod must be < ~3e9 for the ll products above, or use __int128
//   ↳ / Mint.

// --- PISANO PERIOD pi(m): the period of F(n) mod m. Needed to
//   ↳ reduce a huge index. ---
// pi(1) = 1, pi(2) = 3, and pi(m) is EVEN for every m > 2.
// pi(lcm(a,b)) = lcm(pi(a), pi(b))    pi(p^k) = p^(k-1) * pi(p)
// pi(m) <= 6m, with equality exactly at m = 2 * 5^k.
// p = +-1 (mod 5) => pi(p) divides p-1;    p = +-2 (mod 5) => pi(p)
//   ↳ divides 2(p+1).
// So: factor m, get pi for each prime power by testing the divisors
//   ↳ of that bound, then lcm.
ll pisanoPrime(ll p) {
    ↳ prime
    if (p == 2) return 3;
    if (p == 5) return 20;
    ll cand = (p % 5 == 1 || p % 5 == 4) ? p - 1 : 2 * (p + 1);
    vector<ll> divs;
    for (ll d = 1; d * d <= cand; d++)
        if (cand % d == 0) divs.push_back(d), divs.push_back(cand /
    ↳ d);
    sort(all(divs));
    for (ll d : divs) {
        auto [a, b] = fibPair(d, p);
        if (a == 0 && b == 1 % p) return d;
    }
    return cand;
}
// Full pi(m): factorise m (Pollard rho), take pisanoPrime(p) *
//   ↳ p^(e-1) per prime power, lcm them.
/* MORE FIBONACCI FACTS worth the page
gcd(F_m, F_n) = F_gcd(m,n)    =>    F_m | F_n iff m | n    (m >=
↳ 3)
Cassini:    F_{n-1} F_{n+1} - F_n^2 = (-1)^n
Catalan:    F_n^2 - F_{n-r} F_{n+r} = (-1)^{n-r} F_r^2
d'Ocagne:    F_m F_{n+1} - F_{m+1} F_n = (-1)^n F_{m-n}
sum_{i<=n} F_i = F_{n+2} - 1    sum_{i<=n} F_i^2 = F_n F_{n+1}
sum_{i<=n} F_{2i-1} = F_{2n}    sum_{i<=n} F_{2i} = F_{2n+1} -
↳ 1
F_n = (phi^n - psi^n) / sqrt(5); F_92 is the last one fitting in a
↳ signed 64-bit integer.
ZECKENDORF: every n >= 1 is a UNIQUE sum of NON-CONSECUTIVE
↳ Fibonacci numbers, found greedily
from the largest. This is the "Fibonacci base", and it is what
↳ makes Fibonacci-nim work.
LCM of F_1..F_n and "which F are divisible by k" both follow from
↳ the gcd identity.
Fibonacci mod p when 5 is a QR mod p: sqrt(5) exists, so the
↳ closed form works in F_p directly.
When it is not, work in F_p[x]/(x^2-5) (a "Phi field") - two
↳ components, same O(log n). */

// --- TRIBONACCI / any linear recurrence: matrix exponentiation (06
↳ - Math/03), or Kitamasa /
// Bostan-Mori for a long recurrence. Berlekamp-Massey RECOVERS the
↳ recurrence from ~2k terms.
// --- LUCAS SEQUENCES U_n, V_n with U_{n+1} = P U_n - Q U_{n-1}
↳ generalise all of this and have
// the same doubling identities; they are what Baillie-PSW primality
↳ is built on.
```

4.6.2 FibonacciLCM

LCM OF FIBONACCI NUMBERS $\text{lcm}(F(a_1), F(a_2), \dots, F(a_n)) \bmod p$ - the numbers themselves have

```
// Needs: fib (01 - FibonacciAndPisano.cpp), powmod (02 - Primes and
// Divisors/06).
// billions of digits, so everything is done modulo p and division is
// a modular inverse.
// THE KEY IDENTITY is  $\text{gcd}(F(i), F(j)) = F(\text{gcd}(i, j))$ , which turns
// the usual
//  $\text{lcm}(S) = \text{lcm}(S') * x / \text{gcd}(\text{lcm}(S'), x)$  ( $x$  = the last
// element)
// into --
// all indices, no big numbers --
//  $\text{lcm}(F(A)) = \text{lcm}(F(A')) * F(x) / \text{lcm}(F(\text{gcd}(a_i, x))$  for  $a_i$  in
//  $A'$ 
// i.e. the denominator is again an lcm of Fibonacci numbers, on the
// SMALLER index set of gcfs.
// So recurse on sets of indices; memoise on the sorted, deduplicated
// index vector.
// COMPLEXITY:  $O(\# \text{distinct gcd-closed subsets} * n * \log)$  - bounded by
// the number of divisors of
// the  $a_i$  in practice, fast for  $n$  up to a few hundred with  $a_i$  up to
//  $1e9$ .
// CAUTION: this DIVIDES modulo  $p$ , so  $p$  must be prime AND no  $F(k)$ 
// involved may be  $0 \bmod p$ . With
//  $p = 1e9+7$  that is safe for the usual constraints; if you hit a
// zero, use a different prime.
map<vector<int>, ll> fibLcmMemo;
ll fibLcm(vector<int> a) {
    sort(a.rbegin(), a.rend());
    while (!a.empty() && a.back() <= 2) a.pop_back(); // F(1) =
    // F(2) = 1 contribute nothing
    a.erase(unique(all(a)), a.end());
    if (a.empty()) return 1;
    if (a.size() == 1) return fib(a[0], MOD);
    auto it = fibLcmMemo.find(a);
    if (it != fibLcmMemo.end()) return it->second;
    vector<int> b(a.begin(), a.end() - 1);
    ll res = fibLcm(b);
    for (int& x : b) x = gcd(x, a.back());
    res = res * fib(a.back(), MOD) % MOD * powmod(fibLcm(b), MOD - 2,
    // MOD) % MOD;
    return fibLcmMemo[a] = res;
}
// SAME PATTERN, other sequences: any sequence with  $\text{gcd}(u_i, u_j) =$ 
//  $u_{\text{gcd}(i,j)}$  (a "strong
// divisibility sequence") works verbatim -  $F(n)$ ,  $2^n - 1$  (lcm of
// Mersenne numbers), and the
// Lucas sequences  $U_n(P, Q)$  with  $\text{gcd}(P, Q) = 1$  in general.
// USEFUL COMPANIONS:  $F(m) \mid F(n)$  iff  $m \mid n$  (for  $m \geq 3$ );  $\text{gcd}(F(m),$ 
//  $F(n)) = F(\text{gcd}(m,n))$ ;
// lcm over ALL  $i \leq n$  of  $F(i)$  has a product formula via the
// primitive parts (Zsigmondy), but the
// recursion above is what you want in a contest.
```

5 Combinatorics

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5.1 StarsAndBars

Stars and Bars (Number of Solutions to $x_1 + x_2 + \dots + x_k = n$) - $O(1)$

```
// 1. Non-negative integer solutions ( $x_i \geq 0$ ):  $C(n + k - 1, k - 1)$ 
// 2. Positive integer solutions ( $x_i \geq 1$ ):  $C(n - 1, k - 1)$ 
struct StarsAndBars {
    // Requires precomputed Combination  $C(n, k)$ 
    static ll count_non_negative(ll n, ll k, function<ll(ll, ll)> nCr) {
        // nCr {
        if (n < 0 || k <= 0) return 0;
        return nCr(n + k - 1, k - 1);
        }

    static ll count_positive(ll n, ll k, function<ll(ll, ll)> nCr) {
        if (n < k || k <= 0) return 0;
        return nCr(n - 1, k - 1);
        }
};
```

5.2 BurnsidePolya

Burnside / Polya - count colourings up to symmetry. $\# \text{orbits} = (1/|G|) \sum_g |\text{fix}(g)|$

```
// Needs: Mint (06 - Math/01); phi (02 - Primes and Divisors/06) for
// the necklace formulas.
// Needs Mint and phi() (single-value Euler totient).

// Necklaces: n beads in a circle, k colours, ROTATIONS only.
// A rotation by r has  $\text{gcd}(r,n)$  cycles  $\rightarrow k^{\text{gcd}(r,n)}$  fixed
// colourings; group by  $d = \text{gcd}$ .
Mint necklaces(ll n, ll k) {
    Mint s = 0;
    for (ll d = 1; d * d <= n; d++) if (n % d == 0) {
        s += Mint(phi(n / d)) * Mint(k).pow(d);
        if (d != n / d) s += Mint(phi(d)) * Mint(k).pow(n / d);
    }
    return s * Mint(n).inv();
}
// Bracelets: rotations AND reflections (dihedral group,  $|G| = 2n$ ).
Mint bracelets(ll n, ll k) {
    Mint s = necklaces(n, k) * Mint(n); // = sum over
    // the n rotations
    if (n & 1) s += Mint(n) * Mint(k).pow((n + 1) / 2);
    else s += Mint(n / 2) * (Mint(k).pow(n / 2) + Mint(k).pow(n / 2 +
    // 1));
    return s * Mint(2 * n).inv();
}
// GENERAL RECIPE: for each group element g, count its CYCLES c(g) on
// the objects being
// coloured; it fixes  $k^{c(g)}$  colourings. Answer =  $(1/|G|) * \sum_g$ 
//  $k^{c(g)}$ .
// square grid n x n under the 4 rotations: cycles =  $n^2$ ,
//  $\text{ceil}(n^2/4)$ ... (work them out per problem)
// cube faces under 24 rotations: cycle counts 6,3,3,4,2,2,...  $\rightarrow$ 
//  $(k^6 + 3k^4 + 12k^3 + 8k^2)/24$ 
// POLYA (colours have weights): replace  $k^c$  by prod over cycles of
// (sum of  $x_i^{\text{len}}$ ).
Mint burnside(const vector<ll>& cyclesPerElement, ll k) {
    Mint s = 0;
    for (ll c : cyclesPerElement) s += Mint(k).pow(c);
    return s * Mint((ll)cyclesPerElement.size()).inv();
}
```

5.3 CombinatoricsFormulas

COMBINATORICS - FORMULA SHEET =====

```
/* ===== COMBINATORICS - FORMULA SHEET =====
BASICS  C(n,k)=n!/(k!(n-k)!)  P(n,k)=n!/(n-k)!  multinomial
  ⇨ n!/(k1!..km!)
STARS AND BARS:  x1+...+xk = n, xi>=0 -> C(n+k-1, k-1);  xi>=1 ->
  ⇨ C(n-1, k-1)
              with upper bounds xi<=c: IE over which variables
  ⇨ overflow
Compositions of n into k parts C(n-1,k-1); into any parts 2^(n-1)
```

```
BINOMIAL IDENTITIES  (memorise these - recognising one is worth 30
  ⇨ minutes)
Pascal      C(n,k) = C(n-1,k-1) + C(n-1,k)
Symmetry    C(n,k) = C(n,n-k)
Absorption  k*C(n,k) = n*C(n-1,k-1)
Hockey stick  sum_{i=r..n} C(i,r) = C(n+1, r+1)
Vandermonde  sum_k C(m,k) C(n,p-k) = C(m+n, p)
sum_k C(n,k) = 2^n          sum_k (-1)^k C(n,k) = [n==0]
sum_k k*C(n,k) = n*2^(n-1)  sum_k C(n,k)^2 = C(2n,n)
sum_k C(k,r) x^k ... upper negation: C(-n,k) = (-1)^k C(n+k-1, k)
sum_{k} C(n,k) C(k,m) = C(n,m) 2^(n-m)
Binomial inversion: f(n)=sum_k C(n,k) g(k)  <=>  g(n)=sum_k
  ⇨ (-1)^(n-k) C(n,k) f(k)
```

INCLUSION-EXCLUSION

```
|A1 ∪ ... ∪ An| = sum |Ai| - sum |Ai ∩ Aj| + ...
#surjections [n]->[k] = sum_{i=0..k} (-1)^i C(k,i) (k-i)^n = k! *
  ⇨ S2(n,k)
"exactly m" from "at least m":  E(m) = sum_{j>=m} (-1)^(j-m)
  ⇨ C(j,m) A(j)
derangements D(n) = n! sum_{k=0..n} (-1)^k/k! =
  ⇨ (n-1)(D(n-1)+D(n-2)), D0=1,D1=0
permutations with exactly k fixed points = C(n,k) * D(n-k)
```

```
CATALAN  Cat(n) = C(2n,n)/(n+1) = C(2n,n) - C(2n,n+1),  Cat(n+1) =
  ⇨ sum Cat(i)Cat(n-i)
1,1,2,5,14,42,132,429,1430,4862  - counts: balanced bracket
  ⇨ strings of length 2n;
  binary trees with n nodes; triangulations of an (n+2)-gon;
  ⇨ monotone lattice paths under
  the diagonal; stack-sortable permutations; non-crossing
  ⇨ partitions/chords.
BALLOT / CYCLE LEMMA: paths from 0 to k in n steps staying > 0 =
  ⇨ (k/n) C(n, (n+k)/2)
REFLECTION: #paths (0,0)->(a,b) touching the line y=x+c = #paths
  ⇨ to the reflected point
```

STIRLING

```
2nd kind S2(n,k) = #partitions of n LABELLED items into k
  ⇨ non-empty UNLABELLED blocks
S2(n,k) = k*S2(n-1,k) + S2(n-1,k-1);  S2(n,k) = (1/k!) sum_i
  ⇨ (-1)^i C(k,i) (k-i)^n
x^n = sum_k S2(n,k) * x_falling(k)          surjections = k!
  ⇨ S2(n,k)
1st kind (unsigned) c(n,k) = #permutations of n with exactly k
  ⇨ cycles
c(n,k) = c(n-1,k-1) + (n-1) c(n-1,k);  sum_k c(n,k) x^k =
  ⇨ x(x+1)...(x+n-1)
BELL B(n) = sum_k S2(n,k) = #set partitions.  B(n+1) = sum_k
  ⇨ C(n,k) B(k).  Bell triangle.
```

PARTITIONS (unordered, into positive parts)

```
p(n) via pentagonal number theorem, O(n sqrt n):
p(n) = sum_{k>=1} (-1)^(k+1) [ p(n - k(3k-1)/2) + p(n -
  ⇨ k(3k+1)/2) ]
partitions into <= k parts = partitions into parts <= k (conjugate)
```

TWELVEFOLD WAY (n balls into k boxes)

balls	boxes	any	injective	surjective
dist.	dist.	k^n	k!/(k-n)!	k! S2(n,k)
ident.	dist.	C(n+k-1, n)	C(k,n)	C(n-1, n-k)
dist.	ident.	sum_{i<=k} S2(n,i)	[n<=k]	S2(n,k)
ident.	ident.	p_k(n)	[n<=k]	p(n into

⇨ exactly k parts)

```
BURNSIDE / POLYA  #orbits = (1/|G|) sum_{g in G} |fix(g)|
necklaces (rotations), n beads, k colours:  (1/n) sum_{d|n}
  ⇨ phi(n/d) k^d
bracelets (+ reflections): (necklace sum + reflection sum) / (2n),
  reflection sum = n*k^((n+1)/2) if n odd, (n/2)(k^(n/2) +
  ⇨ k^(n/2+1)) if n even
```

TREES / GRAPHS

```
Cayley: n^(n-2) labelled trees on n vertices; with given degrees
  ⇨ d_i: (n-2)! / prod (d_i-1)!
Prufer code <-> labelled tree, bijective
```

```
Matrix-Tree: #spanning trees = any cofactor of the Laplacian (D -
  ⇨ A)
LGV: #non-intersecting path systems = det( M ) with M_ij = #paths
  ⇨ a_i -> b_j
```

GENERATING FUNCTIONS

```
OGF  1/(1-x) = sum x^n          1/(1-x)^k = sum C(n+k-1, k-1)
  ⇨ x^n
      x/(1-x-x^2) = Fibonacci    (1-sqrt(1-4x))/(2x) = Catalan
      prod 1/(1-x^i) = partitions  prod (1+x^i) = distinct-part
  ⇨ partitions
EGF  e^x = sum x^n/n!          EGF of set partitions = e^(e^x
  ⇨ - 1)
EXPONENTIAL FORMULA: if C(x) is the EGF of connected structures,
  ⇨ e^C(x) is the EGF of all
Coefficient extraction: [x^n] A(x)B(x) = sum_k a_k b_{n-k}  (a
  ⇨ convolution -> NTT)
Lagrange inversion, for F(0)=0, F'(0)!=0:  [x^n] G(F^-1)(x) =
  ⇨ (1/n)[x^(n-1)] G'(x)(x/F(x))^n
```

MISC

```
hook length: #standard Young tableaux of shape lambda = n! /
  ⇨ prod(hook lengths)
Erdos-Ko-Rado, Dilworth (min chain cover = max antichain), Sperner
  ⇨ (max antichain C(n,n/2))
===== */
```


5.4 CatalanStirlingBell

Catalan, Stirling (both kinds), Bell, derangements, partitions. Needs Mint + buildFact().

```
Mint catalan(int n) { return C(2 * n, n) * inv_[n + 1]; }
// paths from 0 to k in n steps (+-1) that stay strictly positive -
// ↪ ballot / cycle lemma
Mint ballot(int n, int k) {
    if (n == 0) return k == 0; // the empty
    // ↪ path
    if (k <= 0 || k > n || (n + k) % 2) return 0; // k <= 0 was
    // ↪ unguarded: Mint(-2) wraps
    return Mint(k) * inv_[n] * C(n, (n + k) / 2);
}

Mint derange(int n) { // D(n) =
    // ↪ (n-1)(D(n-1)+D(n-2))
    Mint a = 1, b = 0; // D0=1, D1=0
    for (int i = 2; i <= n; i++) { Mint c = Mint(i - 1) * (a + b); a
    // ↪ = b, b = c; }
    return n ? b : a;
}

// permutations of n with exactly k fixed points
Mint fixedPoints(int n, int k) { return C(n, k) * derange(n - k); }

// Stirling 2nd kind, single value: S2(n,k) = (1/k!) sum_i (-1)^i
// ↪ C(k,i) (k-i)^n O(k log n)
Mint stirling2(ll n, int k) {
    Mint r = 0;
    for (int i = 0; i <= k; i++) {
        Mint t = C(k, i) * Mint(k - i).pow(n);
        (i & 1) ? r -= t : r += t;
    }
    return r * invFact[k];
}

Mint surjections(ll n, int k) { return stirling2(n, k) * fact[k]; }

// Whole ROW of Stirling 2nd (all k for fixed n) in O(n^2), or O(n
// ↪ log n) as a convolution of
// a_i = (-1)^i/i! and b_i = i^n/i! (NTT).
vector<Mint> stirling2Row(int n) {
    vector<Mint> a(n + 1), b(n + 1), r(n + 1, 0);
    for (int i = 0; i <= n; i++) a[i] = (i & 1 ? Mint(0) - invFact[i]
    // ↪ : invFact[i]), b[i] = Mint(i).pow(n) * invFact[i];
    for (int k = 0; k <= n; k++) for (int i = 0; i <= k; i++) r[k] +=
    // ↪ a[i] * b[k - i];
    return r;
}

// Stirling 1st kind (unsigned), whole row: prod_{i=0..n-1} (x + i)
// ↪ O(n^2)
vector<Mint> stirling1Row(int n) {
    vector<Mint> p = {1};
    for (int i = 0; i < n; i++) {
        vector<Mint> q(p.size() + 1, 0);
        for (size_t j = 0; j < p.size(); j++) q[j + 1] += p[j], q[j]
    // ↪ += p[j] * Mint(i);
        p = q;
    }
    return p; // p[k] =
    // ↪ c(n, k)
}

// Bell numbers B(0..n) via the Bell triangle, O(n^2)
vector<Mint> bell(int n) {
    vector<Mint> B(n + 1), row = {1};
    B[0] = 1;
    for (int i = 1; i <= n; i++) {
        vector<Mint> nr = {row.back()};
        for (Mint x : row) nr.push_back(nr.back() + x);
        row = nr, B[i] = row[0];
    }
    return B;
}

// Integer partitions p(0..n) via the pentagonal number theorem, O(n
// ↪ sqrt n)
vector<Mint> partitions(int n) {
    vector<Mint> p(n + 1, 0);
    p[0] = 1;
    for (int i = 1; i <= n; i++)
        for (int k = 1;; k++) {
            ll g1 = (ll)k * (3 * k - 1) / 2, g2 = (ll)k * (3 * k + 1)
    // ↪ / 2;
            if (g1 > i) break;
            Mint t = p[i - g1] + (g2 <= i ? p[i - g2] : Mint(0));
            (k & 1) ? p[i] += t : p[i] -= t;
        }
    return p;
}
```

5.5 InclusionExclusion

INCLUSION-EXCLUSION recipes. Needs Mint + buildFact().

```
// x1+...+xk = n with 0 <= xi <= ub[i]. IE over which variables
// ↪ overflow their bound.
// O(2^k) - fine for k <= ~20. With equal bounds use the closed form
// ↪ below instead.
Mint boundedStarsBars(ll n, const vector<ll>& ub) {
    int k = ub.size();
    Mint r = 0;
    for (int m = 0; m < (1 << k); m++) {
        ll rem = n;
        for (int i = 0; i < k; i++) if (m >> i & 1) rem -= ub[i] + 1;
        if (rem < 0) continue;
        Mint t = C(rem + k - 1, k - 1);
        (__builtin_popcount(m) & 1) ? r -= t : r += t;
    }
    return r;
}

// Same with a COMMON bound c: sum_j (-1)^j C(k,j) C(n - j(c+1) +
// ↪ k-1, k-1). O(k)
Mint boundedStarsBarsEqual(ll n, ll k, ll c) {
    Mint r = 0;
    for (ll j = 0; j <= k && j * (c + 1) <= n; j++) {
        Mint t = C(k, j) * C(n - j * (c + 1) + k - 1, k - 1);
        (j & 1) ? r -= t : r += t;
    }
    return r;
}

// #surjections [n] -> [k] (equivalently k! * S2(n,k))
Mint surj(ll n, ll k) {
    Mint r = 0;
    for (ll i = 0; i <= k; i++) {
        Mint t = C(k, i) * Mint(k - i).pow(n);
        (i & 1) ? r -= t : r += t;
    }
    return r;
}

// "AT LEAST" -> "EXACTLY" (binomial / Mobius inversion over the
// ↪ subset lattice):
// if A(j) = #configurations having at least a chosen set of j
// ↪ properties (summed over sets),
// then E(m) = sum_{j>=m} (-1)^(j-m) C(j,m) A(j) counts those with
// ↪ EXACTLY m.
Mint exactlyFromAtLeast(const vector<Mint>& A, int m) {
    Mint r = 0;
    for (int j = m; j < (int)A.size(); j++) {
        Mint t = C(j, m) * A[j];
        ((j - m) & 1) ? r -= t : r += t;
    }
    return r;
}

// BINOMIAL INVERSION: f(n) = sum_k C(n,k) g(k) <=> g(n) = sum_k
// ↪ (-1)^(n-k) C(n,k) f(k)
vector<Mint> binomialInvert(const vector<Mint>& f) {
    int n = f.size();
    vector<Mint> g(n, 0);
    for (int i = 0; i < n; i++)
        for (int k = 0; k <= i; k++) {
            Mint t = C(i, k) * f[k];
            ((i - k) & 1) ? g[i] -= t : g[i] += t;
        }
    return g;
}

// COUNTING COPRIME PAIRS / "gcd is exactly g": count multiples then
// ↪ Mobius-invert downwards:
// exact[g] = cntMultiples[g] - sum_{g | d, d > g} exact[d]
// ↪ (iterate g downwards)
vector<ll> exactGcdCounts(const vector<ll>& cntMultiples) {
    int n = cntMultiples.size() - 1;
    vector<ll> e(n + 1);
    for (int g = n; g >= 1; g--) {
        e[g] = cntMultiples[g];
        for (int d = 2 * g; d <= n; d += g) e[g] -= e[d];
    }
    return e;
}
```


5.6 TreeCounting

COUNTING TREES AND GRAPHS.

```
// Needs: Mint (06 - Math/01) and determinant (06 - Math/04 - Linear
  ↳ Algebra/02).

// PRUFER CODE: bijection between labelled trees on n>=2 vertices and
  ↳ sequences in [0,n)^(n-2).
// Immediate corollaries: Cayley n^(n-2) labelled trees; with
  ↳ prescribed degrees d_i the count is
// (n-2)! / prod (d_i - 1)! (because vertex i appears exactly d_i -
  ↳ 1 times in the code).
vector<int> toPrufer(const vector<vector<int>>& adj) {    //
  ↳ repeatedly strip the SMALLEST leaf
  int n = adj.size();
  if (n <= 2) return {};
  vector<int> deg(n), code;
  vector<bool> gone(n, false);
  priority_queue<int, vector<int>, greater<int>> q;
  for (int i = 0; i < n; i++) if ((deg[i] = adj[i].size()) == 1)
  ↳ q.push(i);
  for (int it = 0; it < n - 2; it++) {
    int leaf = q.top(); q.pop();
    gone[leaf] = true;
    int nb = -1;
    for (int v : adj[leaf]) if (!gone[v]) nb = v;
    code.push_back(nb);
    if (--deg[nb] == 1) q.push(nb);
  }
  return code;
}
vector<pair<int, int>> fromPrufer(vector<int> code) {
  int n = code.size() + 2;
  vector<int> deg(n, 1);
  for (int x : code) deg[x]++;
  priority_queue<int, vector<int>, greater<int>> q;
  for (int i = 0; i < n; i++) if (deg[i] == 1) q.push(i);
  vector<pair<int, int>> e;
  for (int x : code) {
    int leaf = q.top(); q.pop();
    e.push_back({leaf, x});
    if (--deg[x] == 1) q.push(x);
  }
  int a = q.top(); q.pop();
  e.push_back({a, q.top()});
  return e;
}

// MATRIX-TREE THEOREM: #spanning trees of a graph = any cofactor of
  ↳ the Laplacian L = D - A.
// Multigraphs work (put the multiplicity in A). Needs determinant()
  ↳ from GaussianElimination.
Mint spanningTrees(int n, const vector<pair<int, int>>& edges) {
  if (n == 1) return 1;
  vector<vector<Mint>> L(n, vector<Mint>(n, 0));
  for (auto [u, v] : edges) {
    L[u][u] += 1, L[v][v] += 1;
    L[u][v] -= 1, L[v][u] -= 1;
  }
  vector<vector<Mint>> M(n - 1, vector<Mint>(n - 1));    // delete
  ↳ row/col 0
  for (int i = 1; i < n; i++) for (int j = 1; j < n; j++) M[i -
  ↳ 1][j - 1] = L[i][j];
  return determinant(M);
}

// DIRECTED version (Tutte): #arborescences rooted at r = cofactor of
  ↳ (Dout - A) deleting row/col r
// for in-trees, or (Din - A) for out-trees.
// WEIGHTED version: put edge weights in A and the weighted degree in
  ↳ D -> sum over spanning
// trees of the product of weights (useful mod p for "sum over all
  ↳ spanning trees" problems).

// EXPONENTIAL FORMULA: if C(x) is the EGF of CONNECTED structures
  ↳ then e^(C(x)) is the EGF of
// all structures. Concretely, with g_n = #labelled graphs on n
  ↳ nodes = 2^C(n,2):
// c_n = g_n - sum_{k=1}^{n-1} C(n-1, k-1) * c_k * g_{n-k}
  ↳ (connected labelled graphs)
vector<Mint> connectedLabelledGraphs(int n) {
  vector<Mint> g(n + 1), c(n + 1, 0);
  for (int i = 0; i <= n; i++) g[i] = Mint(2).pow((ll)i * (i - 1) /
  ↳ 2);
  for (int i = 1; i <= n; i++) {
    c[i] = g[i];
    for (int k = 1; k < i; k++) c[i] -= C(i - 1, k - 1) * c[k] *
    ↳ g[i - k];
  }
}
```

```
return c;
}
```

5.7 SubsetConvolution

TRANSFORMS OVER THE SUBSET LATTICE. n bits, arrays of size 2^n.

```
// Needs: Mint (06 - Math/01 - Modular Arithmetic/01 - Mint.cpp).

// ZETA (sum over subsets) and its inverse MOBIUS. O(n 2^n). This is
  ↳ SOS DP.
void zetaSub(vector<Mint>& f, int n) {    // f[m] :=
  ↳ sum_{s subset of m} f[s]
  for (int i = 0; i < n; i++)
    for (int m = 0; m < (1 << n); m++) if (m >> i & 1) f[m] +=
    ↳ f[m ^ (1 << i)];
}
void mobiusSub(vector<Mint>& f, int n) {    // exact
  ↳ inverse of zetaSub
  for (int i = 0; i < n; i++)
    for (int m = 0; m < (1 << n); m++) if (m >> i & 1) f[m] -=
    ↳ f[m ^ (1 << i)];
}
void zetaSuper(vector<Mint>& f, int n) {    // f[m] :=
  ↳ sum_{m subset of s} f[s]
  for (int i = 0; i < n; i++)
    for (int m = 0; m < (1 << n); m++) if (!(m >> i & 1)) f[m] +=
    ↳ f[m | (1 << i)];
}
void mobiusSuper(vector<Mint>& f, int n) {
  for (int i = 0; i < n; i++)
    for (int m = 0; m < (1 << n); m++) if (!(m >> i & 1)) f[m] -=
    ↳ f[m | (1 << i)];
}
// OR-convolution: h[m] = sum_{a|b = m} f[a] g[b]    -> zetaSub,
  ↳ multiply, mobiusSub
// AND-convolution: h[m] = sum_{a&b = m} f[a] g[b]    -> zetaSuper,
  ↳ multiply, mobiusSuper
vector<Mint> orConv(vector<Mint> f, vector<Mint> g, int n) {
  zetaSub(f, n), zetaSub(g, n);
  for (int m = 0; m < (1 << n); m++) f[m] *= g[m];
  return mobiusSub(f, n), f;
}
vector<Mint> andConv(vector<Mint> f, vector<Mint> g, int n) {
  zetaSuper(f, n), zetaSuper(g, n);
  for (int m = 0; m < (1 << n); m++) f[m] *= g[m];
  return mobiusSuper(f, n), f;
}

// SUBSET (disjoint) CONVOLUTION: h[m] = sum_{a|b = m, a&b = 0} f[a]
  ↳ g[b].    O(2^n n^2).
// The ranked-zeta trick: index by popcount, convolve the ranks like
  ↳ polynomials.
vector<Mint> subsetConv(const vector<Mint>& f, const vector<Mint>& g,
  ↳ int n) {
  int N = 1 << n;
  vector<vector<Mint>> F(n + 1, vector<Mint>(N, 0)), G = F, H = F;
  for (int m = 0; m < N; m++) F[__builtin_popcount(m)][m] = f[m],
  ↳ G[__builtin_popcount(m)][m] = g[m];
  for (int i = 0; i <= n; i++) zetaSub(F[i], n), zetaSub(G[i], n);
  for (int i = 0; i <= n; i++)
    for (int j = 0; i + j <= n; j++)
      for (int m = 0; m < N; m++) H[i + j][m] += F[i][m] *
      ↳ G[j][m];
  for (int i = 0; i <= n; i++) mobiusSub(H[i], n);
  vector<Mint> h(N);
  for (int m = 0; m < N; m++) h[m] = H[__builtin_popcount(m)][m];
  return h;
}

// USES: exact set cover / "partition the ground set into k valid
  ↳ groups" (repeated subset
// convolution, or exponentiate the EGF-like series); counting
  ↳ perfect matchings by subsets;
// "for each mask, best way to split it into two disjoint valid
  ↳ halves".
// SUBMASK ENUMERATION over all masks is O(3^n): for (s = m; s; s =
  ↳ (s-1) & m)
```

5.8 SpecialSequences

Named sequences that keep showing up, with what each one counts. Needs Mint + buildFact().

```
// MOTZKIN M(n): lattice paths (0,0)->(n,0) with steps U/D/FLAT
  ↳ staying >= 0; also the number of
// ways to draw non-crossing chords on n points on a circle. M:
  ↳ 1,1,2,4,9,21,51,127
vector<Mint> motzkin(int n) {
    vector<Mint> m(n + 1, 0);
    m[0] = 1;
    if (n >= 1) m[1] = 1;
    for (int i = 2; i <= n; i++) { // M(i) =
        ↳ M(i-1) + sum M(k)M(i-2-k)
        m[i] = m[i - 1];
        for (int k = 0; k <= i - 2; k++) m[i] += m[k] * m[i - 2 - k];
    }
    return m;
}
// NARAYANA N(n,k): Dyck paths of length 2n with exactly k peaks.
  ↳ sum_k N(n,k) = Catalan(n).
Mint narayana(int n, int k) { return k < 1 || k > n ? Mint(0) : C(n,
  ↳ k) * C(n, k - 1) * inv_n[n]; }
// SCHRODER: large R(n) counts lattice paths (0,0)->(n,n) with steps
  ↳ E/N/NE staying below y=x.
// R: 1,2,6,22,90,394. R(n) = R(n-1) + sum_{k} R(k) R(n-1-k)
  ↳ (little schroeder s(n) = R(n)/2 for n >= 1 only (s(0) = 1, not
  ↳ 1/2))
vector<Mint> schroder(int n) {
    vector<Mint> r(n + 1, 0);
    r[0] = 1;
    for (int i = 1; i <= n; i++) { r[i] = r[i - 1]; for (int k = 0; k
    ↳ < i; k++) r[i] += r[k] * r[i - 1 - k]; }
    return r;
}
// EULERIAN A(n,k): permutations of n with exactly k ascents.
// A(n,k) = (k+1) A(n-1,k) + (n-k) A(n-1,k-1). sum_k A(n,k) = n!
vector<vector<Mint>> eulerian(int n) {
    vector<vector<Mint>> a(n + 1, vector<Mint>(n + 1, 0));
    a[0][0] = 1;
    for (int i = 1; i <= n; i++)
        for (int k = 0; k < i; k++)
            a[i][k] = Mint(k + 1) * a[i - 1][k] + Mint(i - k) * (k ?
    ↳ a[i - 1][k - 1] : Mint(0));
    return a;
}
// q-BINOMIAL (Gaussian) coefficient: #k-dim subspaces of an n-dim
  ↳ space over F_q; also
// #partitions fitting in a k x (n-k) box, weighted by q^area.
// PRECONDITION: ord_p(q) > k. The denominator is prod (q^i - 1),
  ↳ which vanishes in F_p
// whenever ord_p(q) <= k - NOT only at q == 1. (q = -1 mod 998244353
  ↳ breaks every k >= 2.)
// Root-of-unity-safe fallback, 0(nk): Pascal's rule [n,k]_q =
  ↳ [n-1,k-1]_q + q^k [n-1,k]_q.
Mint qBinom(int n, int k, Mint q) { //
    ↳ prod_{i=1..k} (q^(n-k+i)-1)/(q^i-1)
    if (k < 0 || k > n) return 0;
    if (q == Mint(1)) return C(n, k); // q=1 is
    ↳ the 0/0 limit
    Mint num = 1, den = 1;
    for (int i = 1; i <= k; i++) num *= q.pow(n - k + i) - 1, den *=
    ↳ q.pow(i) - 1;
    return num / den;
}
// HOOK LENGTH FORMULA: #standard Young tableaux of shape lambda = n!
  ↳ / prod(hook lengths).
// hook(i,j) = (arm to the right) + (leg below) + 1.
Mint youngTableaux(const vector<int>& lam) {
    int n = 0;
    for (int x : lam) n += x;
    vector<int> conj(lam.empty() ? 0 : lam[0], 0);
    for (int i = 0; i < (int)lam.size(); i++) for (int j = 0; j <
    ↳ lam[i]; j++) conj[j]++;
    Mint den = 1;
    for (int i = 0; i < (int)lam.size(); i++)
        for (int j = 0; j < lam[i]; j++) den *= Mint(lam[i] - j +
    ↳ conj[j] - i - 1);
    return fact[n] / den;
}
// LGV LEMMA: for sources a_1..a_k and sinks b_1..b_k in a DAG, the
  ↳ number of NON-INTERSECTING
// path systems equals det(M) with M[i][j] = #paths a_i -> b_j (when
  ↳ only the identity
// permutation can be non-intersecting). Counts plane partitions,
  ↳ rhombus tilings, etc.
```

5.9 EulerTransformAndSymbolic

```
SYMBOLIC METHOD: spec -> generating function =====
// Needs: Mint (06 - Math/01 - Modular Arithmetic/01 - Mint.cpp).
/* ===== SYMBOLIC METHOD: spec -> generating function
  ↳ =====
Decide labelled (EGF) or unlabelled (OGF), then translate the
  ↳ construction:
    UNLABELLED (OGF), A built from atoms with GF B(z):
      SEQ (ordered list) A = 1 / (1 - B)
      MSET (unordered, rep) A = exp( sum_{k>=1} B(z^k) / k )
    ↳ <- Euler transform
      PSET (unordered, no rep) A = exp( sum_{k>=1} (-1)^(k-1) B(z^k)/k )
      CYC (up to rotation) A = sum_{k>=1} (phi(k)/k) * log(1/(1 -
    ↳ B(z^k)))
    LABELLED (EGF):
      SEQ A = 1/(1-B) SET A = exp(B) CYC A =
    ↳ log(1/(1-B))
    "SET of connected pieces = exp(connected)" is the exponential
    ↳ formula.
=====
  ↳ */
// EULER TRANSFORM: a[n] = #atoms of size n -> b[n] = #MULTISSETS of
  ↳ atoms of total size n.
// prod_{n>=1} (1 - x^n)^(-a_n) = 1 + sum b_n x^n. 0(n log n) via
  ↳ the harmonic sum.
vector<Mint> eulerTransform(const vector<Mint>& a) { // a[0]
    ↳ unused; returns b with b[0] = 1
    int n = a.size() - 1;
    vector<Mint> c(n + 1, 0), b(n + 1, 0);
    for (int d = 1; d <= n; d++) // c_n =
    ↳ sum_{d | n} d * a_d
        for (int m = d; m <= n; m += d) c[m] += Mint(d) * a[d];
    b[0] = 1;
    for (int i = 1; i <= n; i++) { // b_n = (c_n
    ↳ + sum_{k<n} c_k b_{n-k}) / n
        Mint s = c[i];
        for (int k = 1; k < i; k++) s += c[k] * b[i - k];
        b[i] = s * inv_i[i];
    }
    return b;
}
int muSmall(int n) { // Mobius of
    ↳ one small n, by trial division
    int r = 1;
    for (int p = 2; (ll)p * p <= n; p++) if (n % p == 0) {
        n /= p;
        if (n % p == 0) return 0;
        r = -r;
    }
    return n > 1 ? -r : r;
}
// INVERSE Euler transform: b -> a (given multiset counts, recover
  ↳ the atom counts).
vector<Mint> eulerInverse(const vector<Mint>& b) {
    int n = b.size() - 1;
    vector<Mint> c(n + 1, 0), a(n + 1, 0);
    for (int i = 1; i <= n; i++) { // c_n = n*b_n
    ↳ - sum_{d<n} c_d b_{n-d}
        Mint s = Mint(i) * b[i];
        for (int d = 1; d < i; d++) s -= c[d] * b[i - d];
        c[i] = s;
    }
    for (int i = 1; i <= n; i++) { // a_n = (1/n)
    ↳ sum_{d|n} mu(n/d) c_d
        Mint s = 0;
        for (int d = 1; d <= i; d++) if (i % d == 0) s +=
    ↳ Mint(muSmall(i / d)) * c[d];
        a[i] = s * inv_i[i];
    }
    return a;
}
// UNLABELLED ROOTED TREES: r_1 = 1, and r_{n+1} = (Euler transform
  ↳ of r_1..r_n)[n]
// (a rooted tree = a root plus a MULTISSET of rooted subtrees).
vector<Mint> rootedTrees(int n) {
    vector<Mint> r(n + 1, 0);
    if (n >= 1) r[1] = 1;
    for (int i = 1; i < n; i++) {
        vector<Mint> a(i + 1, 0);
        for (int j = 1; j <= i; j++) a[j] = r[j];
        auto b = eulerTransform(a);
        r[i + 1] = b[i];
    }
    return r; //
    ↳ 0,1,1,2,4,9,20,48,115,286,719,...
}
```

```
// PARTITIONS INTO PARTS FROM A SET S = eulerTransform with a[s] = 1
  ↪ for s in S.
// COMPOSITIONS (ordered) use SEQ instead: 1/(1 - sum x^s).
```

5.10 PermanentAndSymmetric

RYSER'S FORMULA - permanent of an $n \times n$ matrix in $O(2^n \cdot n)$.

```
// perm(A) counts PERFECT MATCHINGS of a bipartite graph (weighted:
  ↪ sum over matchings of the
// product of weights). Use for  $n \leq \sim 20$ : "count ways to assign n
  ↪ tasks to n people",
// "count systems of distinct representatives", "count permutations
  ↪ avoiding forbidden cells".
Mint permanent(const vector<vector<Mint>>& a) {
    int n = a.size();
    vector<Mint> row(n, 0);
    ↪ sum_{j in S} a[i][j]
    Mint total = 0;
    int prev = 0;
    for (int g = 1; g < (1 << n); g++) {
        ↪ one column toggles per step
        int cur = g ^ (g >> 1), diff = cur ^ prev, j =
        ↪ __builtin_ctz(diff);
        if (cur & diff) for (int i = 0; i < n; i++) row[i] +=
        ↪ a[i][j]; // column j entered S
        else for (int i = 0; i < n; i++) row[i] -= a[i][j];
        ↪ // column j left S
        prev = cur;
        Mint p = 1;
        for (int i = 0; i < n; i++) p *= row[i];
        (__builtin_popcount(cur) & 1) ? total -= p : total += p;
    }
    return (n & 1) ? Mint(0) - total : total; // perm =
    ↪ (-1)^n * sum_S (-1)^|S| prod_i rowSum
}
// PERMANENT vs DETERMINANT: identical formula without the signs. det
  ↪ is polynomial-time,
// perm is #P-hard - never expect an  $O(n^3)$  permanent.

// NEWTON'S IDENTITIES - convert between power sums  $p_k = \sum x_i^k$ 
  ↪ and elementary symmetric  $e_k$ .
//  $e_k = \sum$  over  $k$ -subsets of the product; this is exactly "for
  ↪ every k, the sum over all
//  $k$ -subsets of the product of the chosen values".
//  $e_0 = 1$ ;  $k * e_k = \sum_{i=1..k} (-1)^{i-1} * e_{k-i} * p_i$ 
vector<Mint> powerToElementary(const vector<Mint>& p, int n) { //
    ↪ p[1..n] -> e[0..n]
    vector<Mint> e(n + 1, 0);
    e[0] = 1;
    for (int k = 1; k <= n; k++) {
        Mint s = 0;
        for (int i = 1; i <= k; i++) {
            Mint t = e[k - i] * p[i];
            (i & 1) ? s += t : s -= t;
        }
        e[k] = s * inv_[k];
    }
    return e;
}
//  $p_k = (-1)^{k-1} * k * e_k + \sum_{i=1..k-1} (-1)^{k-1+i} * e_{k-i} * p_i$ 
vector<Mint> elementaryToPower(const vector<Mint>& e, int n) { //
    ↪ e[0..n] -> p[1..n]
    vector<Mint> p(n + 1, 0);
    for (int k = 1; k <= n; k++) {
        Mint s = Mint(k) * e[k];
        if ((k - 1) & 1) s = Mint(0) - s; // (-1)^{k-1}
        ↪ * k * e_k
        for (int i = 1; i < k; i++) {
            Mint t = e[k - i] * p[i];
            ((k - 1 + i) & 1) ? s -= t : s += t;
        }
        p[k] = s;
    }
    return p;
}
// e_k are also the coefficients of prod (x + x_i): prod_{i} (x +
  ↪ x_i) = sum_k e_k * x^{n-k}.
// So "sum over k-subsets of the product" = coefficient extraction
  ↪ from that product, which you
// can also get in  $O(n \log^2 n)$  by divide-and-conquer multiplication
  ↪ of the linear factors.
```

5.11 BernoulliFaulhaber

BERNOULLI NUMBERS and the CLOSED-FORM FAULHABER polynomial.

```
// Use this (not Lagrange interpolation) when you need the POLYNOMIAL
  ↪ COEFFICIENTS of
//  $\sum_{k=1..n} k^p$ , or when p is large and many n are queried.
// Convention here:  $B_{-1} = -1/2$  (the "B^-" convention that the
  ↪ formula below expects).
//  $B_0 = 1$ ;  $\sum_{k=0}^m C(m+1, k) * B_k = [m == 0]$ 
//  $\Rightarrow B_m = -(1/(m+1)) * \sum_{k=0}^{m-1} C(m+1, k) * B_k$ 
vector<Mint> bernoulli(int m) { // B[0..m],
    ↪  $O(m^2)$ 
    vector<Mint> B(m + 1, 0);
    B[0] = 1;
    for (int i = 1; i <= m; i++) {
        Mint s = 0;
        for (int k = 0; k < i; k++) s += C(i + 1, k) * B[k];
        B[i] = Mint(0) - s * inv_[i + 1];
    }
    return B; // 1, -1/2,
    ↪ 1/6, 0, -1/30, 0, 1/42, ...
}
//  $O(m \log m)$  alternative:  $B_m / m! = [t^m] \text{inv}(\sum_{i \geq 0} t^i / (i+1)!) \text{ -- one Poly::inv.}$ 

// FAULHABER:  $\sum_{k=1}^n k^p = (1/(p+1)) * \sum_{r=0}^p C(p+1, r) * B_r * n^{p+1-r}$ 
  ↪  $C(p+1, r) * B_r * n^{p+1-r}$ 
Mint powerSumClosed(ll n, int p, const vector<Mint>& B) {
    Mint s = 0, np = Mint(n).pow(p + 1); //  $n^{p+1-r}$ ,
    ↪ decreasing r
    vector<Mint> pw(p + 2);
    pw[0] = 1;
    for (int i = 1; i <= p + 1; i++) pw[i] = pw[i - 1] * Mint(n);
    for (int r = 0; r <= p; r++) {
        Mint t = C(p + 1, r) * B[r] * pw[p + 1 - r];
        (r & 1) ? s -= t : s += t;
    }
    return s * inv_[p + 1];
}
// The COEFFICIENTS of the Faulhaber polynomial in n (degree p+1):
//  $[n^{p+1-r}] = (-1)^r * C(p+1, r) * B_r / (p+1)$ 
vector<Mint> faulhaberPoly(int p, const vector<Mint>& B) {
    vector<Mint> c(p + 2, 0);
    for (int r = 0; r <= p; r++) {
        Mint t = C(p + 1, r) * B[r] * inv_[p + 1];
        c[p + 1 - r] = (r & 1) ? Mint(0) - t : t;
    }
    return c; // c[i] =
    ↪ coefficient of  $n^i$ 
}
// RELATED SUMS
//  $\sum_{k=0}^n k^p = C(n+1, p+1)$ 
  ↪ (hockey stick)
//  $\sum_{k=1}^n k^p * r^k$  : degree-p polynomial times  $r^n$ ;
  ↪ solve by ansatz or by
// differentiating the geometric
  ↪ series p times.
// Bernoulli also gives Euler-Maclaurin and the von Staudt-Clausen
  ↪ denominator facts.
```

5.12 CycleLemmaFussCatalan

CYCLE LEMMA (Dvoretzky-Motzkin) and the FUSS-CATALAN / rational-Catalan family.

```
// The general-k version of the ballot problem: it handles every "+1
//   -> / -k step" path problem.
//
// CYCLE LEMMA: a word with m a's and n b's, m >= k*n, has EXACTLY m
//   -> - k*n cyclic shifts that
// are k-dominating (every non-empty prefix has #a > k * #b).
// k = 1 ballot: p A's and q B's with p > q -> exactly p - q
//   -> dominating shifts,
// so the probability that A leads the whole count is
//   -> (p - q) / (p + q).

// #paths with p up-steps and q down-steps that stay STRICTLY
//   -> positive after the first step.
Mint ballotStrict(ll p, ll q) { // p > q >= 0
    if (p <= q) return 0;
    return C(p + q, q) * Mint(p - q) * inv_[(int)(p + q)];
}

// FUSS-CATALAN: A_m(p, r) = (r / (m*p + r)) * C(m*p + r, m)
Mint fussCatalan(ll m, ll p, ll r) {
    if (m < 0) return 0;
    return C(m * p + r, m) * Mint(r) * Mint(m * p + r).inv();
}

// #p-ary trees with m internal nodes = A_m(p, 1) = C(m*p + 1, m) /
//   -> (m*p + 1)
Mint pAryTrees(ll m, ll p) { return fussCatalan(m, p, 1); }
// Catalan is the case p = 2, r = 1.
// #sequences of m opens and (p-1)*m closes where every prefix keeps
//   -> the stack non-negative
// is also A_m(p, 1).

// RATIONAL CATALAN: for gcd(a, b) = 1, the number of monotone
//   -> lattice paths from (0,0) to
// (b, a) that never rise above the line a*x = b*y is C(a + b, a) /
//   -> (a + b).
Mint rationalCatalan(ll a, ll b) { return C(a + b, a) * Mint(a +
//   -> b).inv(); }

// CHUNG-FELLER: among the C(2n,n) balanced +-1 paths of length 2n,
//   -> the number having exactly
// 2r steps above the axis is Catalan(n) for EVERY r in 0..n. Hence
//   -> C(2n,n) = (n+1)*Cat(n).
// Use it for "count sequences with exactly k flaws" - the answer
//   -> does not depend on k.

// REFLECTION PRINCIPLE (the tool behind all of the above):
// #paths (0,0)->(x,y) touching the line y = m = #paths
//   -> (0,0)->(x, 2m - y)
// so "#paths avoiding the line" = C(total, up) - C(total, up
//   -> shifted by the reflection).
// PRECONDITION: the barrier must SEPARATE the start (height 0) from
//   -> the end y, i.e. m != 0
// and 0, y strictly on the same side of m. Without it the reflection
//   -> subtracts too much:
// (x,y,m) = (2,2,1) returns -1 where the true count is 0.
Mint pathsAvoidingLine(ll x, ll y, ll m) {
    if (m == 0 || (m > 0 && y >= m) || (m < 0 && y <= m)) return 0;
    // steps +1/-1, must never reach y = m
    if (((x + y) & 1) || y > x) return 0;
    ll up = (x + y) / 2, up2 = (x + 2 * m - y) / 2;
    return C(x, up) - ((x + 2 * m - y) % 2 || up2 < 0 || up2 > x ?
//   -> Mint(0) : C(x, up2));
}
```

5.13 LGVandPolya

LGV LEMMA and the POLYA CYCLE INDEX - the two theorems that were stated but never coded.

```
// Needs: Mint, determinant (06 - Math/04/02), phi (04 - NT/02/06),
//   -> muSmall (09 - EulerTransform...).

// LINDSTROM-GESSEL-VIENNOT: with sources a_1..a_k and sinks b_1..b_k
//   -> in a DAG,
// det(M), M[i][j] = #paths a_i -> b_j
// counts NON-INTERSECTING path systems, PROVIDED the only
//   -> permutation that admits a
// non-intersecting system is the identity (true for "sorted"
//   -> sources/sinks in a grid).
// Grid instance: monotone lattice paths, so M[i][j] = C(dx + dy, dx).
Mint lgvGrid(const vector<pair<ll, ll>>& src, const vector<pair<ll,
//   -> ll>>& dst) {
    int k = src.size();
    vector<vector<Mint>>> M(k, vector<Mint>(k));
    for (int i = 0; i < k; i++)
        for (int j = 0; j < k; j++) {
            ll dx = dst[j].first - src[i].first, dy = dst[j].second -
//   -> src[i].second;
            M[i][j] = (dx < 0 || dy < 0) ? Mint(0) : C(dx + dy, dx);
        }
    return determinant(M); // needs
//   -> GaussianElimination.cpp
}

// USES: k non-crossing monotone paths, rhombus/domino tilings of a
//   -> region, plane partitions,
// "k tokens walk on a grid and must never share a cell".

// POLYA CYCLE INDEX. Z(G)(a_1..a_n) = (1/|G|) sum_{g in G} prod_k
//   -> a_k^{#k-cycles of g}.
// The UNWEIGHTED orbit count is Z(G)(m, m, ..., m) for m colours -
//   -> that is what necklaces()
// and bracelets() already give. The reason to carry the cycle index
//   -> is the WEIGHTED case:
// "necklaces with exactly c_1 of colour 1, c_2 of colour 2, ..."
//   -> which the numeric form cannot do.
// Z(C_n) = (1/n) * sum_{d | n} phi(d) * a_d^{(n/d)}
// Z(D_n) = (1/2) Z(C_n) + (1/2) a_1 a_2^{((n-1)/2)}
//   -> [n odd]
// Z(D_n) = (1/2) Z(C_n) + (1/4) (a_1^2 a_2^{((n-2)/2)} + a_2^{(n/2)})
//   -> [n even]
// Z(S_n) = (1/n) * sum_{l=1..n} a_l * Z(S_{(n-l)}), Z(S_0) = 1
// Weighted PET: substitute a_k -> (t_1^k + t_2^k + ... + t_m^k) and
//   -> read off the coefficient
// of t_1^{c_1} * ... * t_m^{c_m}.
// Necklaces with a PRESCRIBED colour multiset, n beads, counts c[]
//   -> summing to n (rotations only):
Mint necklacesWithCounts(int n, const vector<int>& c) {
    Mint total = 0;
    for (int d = 1; d <= n; d++) {
        if (n % d) continue; // rotations
// of order n/d have d cycles
        bool ok = true;
        for (int x : c) if (x % (n / d)) ok = false; // each
// colour must split evenly
        if (!ok) continue;
        Mint ways = fact[d]; // multinomial
// over the d cycle slots
        for (int x : c) ways *= invFact[x / (n / d)];
        total += Mint(phi(n / d)) * ways;
    }
    return total * Mint(n).inv();
}

// APERIODIC necklaces (Lyndon words) with k colours: M(n,k) = (1/n)
//   -> sum_{d|n} mu(d) k^{(n/d)}.
Mint lyndonCount(ll n, ll k) {
    Mint s = 0;
    for (ll d = 1; d <= n; d++) if (n % d == 0) s += Mint(muSmall(d))
//   -> * Mint(k).pow(n / d);
    return s * Mint(n).inv();
}

// TRANSFER MATRIX: count length-L walks in a finite automaton. Build
//   -> A[s][t] = #transitions,
// then (A^L)[s][t]. The generating function sum_L (A^L)[i][j] x^L =
//   -> ((I - xA)^{-1})[i][j] is
// RATIONAL with denominator det(I - xA) of degree <= #states, so the
//   -> sequence satisfies a linear
// recurrence of that order - feed a prefix to Berlekamp-Massey and
//   -> finish in O(k^2 log L)
// instead of O(states^3 log L).
```

5.14 Hafnian

HAFNIAN - the number of PERFECT MATCHINGS of a GENERAL (non-bipartite) graph, weighted:

```
// Needs: Mint (06 - Math/01 - Modular Arithmetic/01 - Mint.cpp).
// haf(A) = sum over perfect matchings M of prod over {i,j} in M of
//   ↪ A[i][j],
// for a symmetric A with a zero diagonal and n EVEN.  $O(2^{n/2} * n^2)$ , n up to ~38 in a couple of
// seconds. This is the general-graph analogue of the permanent
//   ↪ (which counts perfect matchings of
// a BIPARTITE graph, 10 - PermanentAndSymmetric.cpp); haf(A) is the
//   ↪ "square root" of the permanent
// of the corresponding symmetric matrix.
// REACH FOR IT for "count the ways to pair up 2m items with
//   ↪ forbidden or weighted pairs", domino
// tilings of an irregular board, "partition into pairs" counting,
//   ↪ and any #matching problem where
// Blossom only gives you the MAXIMUM matching and you need the COUNT
//   ↪ of perfect ones.
// COUNTING is much harder than finding: max matching is polynomial
//   ↪ (Blossom), counting is #P-hard.
// For PLANAR graphs use the FKT algorithm (a Pfaffian orientation,
//   ↪ then one determinant) - that is
// polynomial, and is how you count domino tilings of a grid.
// The recursion removes the last two rows/columns and tracks a
//   ↪ truncated polynomial in a marker
// variable whose exponent counts how many pairs have been committed;
//   ↪ only degree <= n/2 survives.
namespace Hafnian {
int H; // = n/2 +
//   ↪ 1, the truncation length
void addTo(vector<Mint>& r, const vector<Mint>& a, const
//   ↪ vector<Mint>& b) {
    for (int i = 0; i < H; i++) if (a[i].v)
        for (int j = 0; j + i + 1 < H; j++) r[i + j + 1] += a[i] *
//   ↪ b[j];
}
vector<Mint> rec(const vector<vector<vector<Mint>>>& v) {
    vector<Mint> r(H, 0);
    if (v.empty()) { r[0] = 1; return r; }
    int m = (int)v.size() - 2;
    auto V = v;
    V.resize(m);
    vector<Mint> zero = rec(V); // vertex
//   ↪ pair (m, m+1) left unmatched here
    for (int i = 0; i < m; i++)
        for (int j = 0; j < i; j++) { // ... or
//   ↪ routed through i and j
            addTo(V[i][j], v[m][i], v[m + 1][j]);
            addTo(V[i][j], v[m + 1][i], v[m][j]);
        }
    vector<Mint> one = rec(V);
    for (int i = 0; i < H; i++) r[i] += one[i] - zero[i];
    addTo(r, one, v[m + 1][m]); // ... or
//   ↪ matched to each other
    return r;
}
Mint solve(const vector<vector<Mint>>& a) { // a
//   ↪ symmetric, zero diagonal, n even
    int n = a.size();
    H = n / 2 + 1;
    vector<vector<vector<Mint>>> v(n);
    for (int i = 0; i < n; i++) {
        v[i].resize(i);
        for (int j = 0; j < i; j++) v[i][j].assign(H, 0), v[i][j][0]
//   ↪ = a[i][j];
    }
    return rec(v).back();
}
}
// SMALL n: a plain bitmask DP over "set of matched vertices", always
//   ↪ matching the lowest unmatched
// vertex, is  $O(2^n * n)$  - simpler and faster up to n = 22. Use the
//   ↪ Hafnian only past that.
// RELATED
// PFAFFIAN: for a SKEW-symmetric matrix,  $\text{pf}(A)^2 = \det(A)$  and pf is
//   ↪ computable in  $O(n^3)$ ; the
// Hafnian is its "no signs" version, which is exactly why one is
//   ↪ easy and the other is not.
// NUMBER OF MATCHINGS OF EVERY SIZE (not just perfect): that is the
//   ↪ matching polynomial; the same
// recursion with a longer truncation, or  $O(2^n)$  subset DP.
// MODULUS: everything here is Mint arithmetic, so the count comes
//   ↪ out mod MOD; for exact small
// counts use unsigned long long and drop Mint.
```

5.15 SubsetSumCounts

NUMBER OF SUBSETS WITH EACH SUM, for every sum 0..m, in $O(m \log m)$ - not $O(n * m)$.

```
// Needs: multiply / npow (02 - NTT.cpp), Poly / exp_ / log_ / cut
//   ↪ (05 - Convolution/04).
// Given a multiset of n positive values, the generating function is
//   ↪  $\text{prod}_i (1 + x^{a_i})$ , which
// naively costs n multiplications. Take logarithms: with  $\text{cnt}[v]$  =
//   ↪ how many times v occurs,
// log  $\text{prod}_v (1 + x^v)^{\text{cnt}[v]} = \sum_v \text{cnt}[v] * \sum_{j \geq 1} \log(1 + x^{jv})$ 
//   ↪  $\log(1 + x^{jv}) = \sum_{k \geq 1} (-1)^{k+1} x^{kjv} / k$ ,
// and the double sum is a HARMONIC sum,  $O(m \log m)$  terms in total.
//   ↪ Then one exp brings it back.
// REACH FOR IT for Yosupo "#P subset sum", "for every k, how many
//   ↪ subsets sum to k", partition
// counting with a bounded multiset of part sizes, and knapsack
//   ↪ COUNTING when m is 1e5 and n is
// large. A plain  $O(n*m)$  knapsack (or the bitset one) is better when
//   ↪ n is small or you only need
// feasibility rather than counts.
// PRECONDITION: all values >= 1 (handle  $a_i = 0$  separately - each
//   ↪ zero doubles every count).
Poly subsetSumCounts(const vector<int>& a, int m) { // result[k]
//   ↪ = #subsets summing to k
    vector<int> cnt(m + 1, 0);
    for (int v : a) if (v >= 1 && v <= m) cnt[v]++;
    Poly p(m + 1, 0);
    for (int v = 1; v <= m; v++) if (cnt[v])
        for (int j = 1; (ll)v * j <= m; j++) {
            ll t = (ll)cnt[v] % NMOD * npow(j, NMOD - 2) % NMOD;
            if (j & 1) p[v * j] = (p[v * j] + t) % NMOD;
            else p[v * j] = (p[v * j] - t + NMOD) % NMOD;
        }
    return exp_(p, m + 1);
}
// THE SAME MACHINE, DIFFERENT SIGN - this is the symbolic method (09
//   ↪ - EulerTransformAndSymbolic):
// MULTISSET (each value usable any number of times) =  $\exp(\sum_j B(x^j) / j)$ 
//   ↪  $B(x^j) = \sum_v \text{cnt}[v] x^{jv}$ 
// POWERSET (each value usable at most once) =  $\exp(\sum_j (-1)^{j+1} B(x^j) / j)$ 
//   ↪ with  $B(x) = \sum_v \text{cnt}[v] x^v$ . Swapping the sign is the only
//   ↪ difference between "partitions into
// parts from a set" and "subsets of a multiset".
// GOING BACKWARDS: given the counts, log_ recovers B and hence the
//   ↪ multiset - that is the inverse
// Euler transform, the standard "recover the item sizes from the
//   ↪ subset-sum counts" problem.
// ADD OR REMOVE ONE ITEM in  $O(m)$ : multiply or divide the answer by
//   ↪  $(1 + x^v)$  directly (division by
//  $1 + x^v$  is the recurrence  $r[k] = r[k - v]$ ) - much cheaper than
//   ↪ rebuilding.
// K ITEMS ONLY: to also track the SIZE of the subset you need a
//   ↪ second variable; do the same
// construction with  $x^v(v)$  y and truncate in both, or run a 2-D
//   ↪ convolution (05 - Convolution/08).
```

6 Math

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6.1 Modular Arithmetic

6.1.1 Mint

Modular integer (MOD must be PRIME for inv/division) + O(N) factorial tables.

```
struct Mint {
    int v;
    Mint(ll x = 0) { v = int(x % MOD); if (v < 0) v += MOD; }
    Mint& operator+=(Mint o) { if ((v += o.v) >= MOD) v -= MOD;
    ↪ return *this; }
    Mint& operator-=(Mint o) { if ((v -= o.v) < 0) v += MOD; return
    ↪ *this; }
    Mint& operator*=(Mint o) { v = int(1LL * v * o.v % MOD); return
    ↪ *this; }
    Mint& operator/=(Mint o) { return *this *= o.inv(); }
    friend Mint operator+(Mint a, Mint b) { return a += b; }
    friend Mint operator-(Mint a, Mint b) { return a -= b; }
    friend Mint operator*(Mint a, Mint b) { return a *= b; }
    friend Mint operator/(Mint a, Mint b) { return a /= b; }
    friend bool operator==(Mint a, Mint b) { return a.v == b.v; }
    Mint pow(ll e) const { Mint r = 1, b = *this; for (; e > 0; e >>=
    ↪ 1, b *= b) if (e & 1) r *= b; return r; }
    Mint inv() const { return pow(MOD - 2); } // Fermat:
    ↪ a^(p-2) = a^-1 (mod p)
    friend ostream& operator<<(ostream& o, Mint m) { return o << m.v;
    ↪ }
    friend istream& operator>>(istream& i, Mint& m) { ll x; i >> x; m
    ↪ = Mint(x); return i; }
};

const int N = 200005; // ALSO in 01
    ↪ - Template.cpp: keep one
Mint fact[N], invFact[N], inv_[N];
void buildFact() { // O(N)
    ↪ total, not O(N log MOD)
    fact[0] = invFact[0] = inv_[1] = 1;
    for (int i = 1; i < N; i++) fact[i] = fact[i - 1] * i;
    invFact[N - 1] = fact[N - 1].inv();
    for (int i = N - 1; i > 0; i--) invFact[i - 1] = invFact[i] * i;
    for (int i = 2; i < N; i++) inv_[i] = Mint(MOD - MOD / i) *
    ↪ inv_[MOD % i];
}
Mint C(ll n, ll r) { if (r < 0 || r > n) return 0; return fact[n] *
    ↪ invFact[r] * invFact[n - r]; }
Mint Perm(ll n, ll r) { if (r < 0 || r > n) return 0; return fact[n]
    ↪ * invFact[n - r]; }
// C(n, r) for HUGE n and small r (no tables): prod_{i<r} (n-i) / r!
Mint Cbig(ll n, ll r) {
    if (r < 0 || n < 0 || r > n) return 0;
    Mint num = 1;
    for (ll i = 0; i < r; i++) num *= Mint(n - i);
    return num * invFact[r];
}
```

6.1.2 BinaryFieldGF2k

ARITHMETIC IN THE FINITE FIELD $GF(2^K)$: elements are K-bit masks = polynomials over $GF(2)$

```
// reduced mod a fixed IRREDUCIBLE polynomial of degree K. Add = xor,
    ↪ multiply = carry-less
// multiply then reduce. mul is O(K), pow / inverse O(K log) - or
    ↪ O(1) with the CLMUL intrinsic.
// REACH FOR IT when a problem is "linear algebra over a field of
    ↪ characteristic 2 with more than
// two elements": random algebraic identity testing over  $GF(2^K)$ 
    ↪ (Freivalds / Schwartz-Zippel with
// xor-shaped constraints), Reed-Solomon-flavoured coding, "assign
    ↪ field values to symbols and test
// whether a determinant vanishes", and Wiedemann over  $GF(2)$ .
// PRECONDITION: POLY below must be irreducible over  $GF(2)$  and of
    ↪ degree exactly K.
// K=8: 0x11B K=16: 0x1002B K=30: 0x40000003 K=31: 0x80000009
    ↪ K=32: 0x1000000AF
// K=64: x^64 + x^4 + x^3 + x + 1 (constant 0x1B), the usual choice
    ↪ for unsigned long long.
namespace GF2K {
const int K = 30;
const ll POLY = (1LL << K) | 3; // x^30 + x + 1,
    ↪ irreducible

ll add(ll a, ll b) { return a ^ b; } // subtraction is
    ↪ the same operation
ll mul(ll a, ll b) {
    ll r = 0;
    while (b) {
        if (b & 1) r ^= a;
        b >>= 1, a <<= 1;
        if (a >> K & 1) a ^= POLY; // reduce as soon as
    ↪ the degree reaches K
    }
    return r;
}
ll pw(ll a, ll e) {
    ll r = 1;
    for (; e; e >>= 1, a = mul(a, a)) if (e & 1) r = mul(r, a);
    return r;
}
ll inv(ll a) { return pw(a, (1LL << K) - 2); } // a^(2^K - 1) = 1,
    ↪ so a^-1 = a^(2^K - 2)
}
// FACTS YOU NEED WHEN USING IT
// |GF(2^K)| = 2^K, the multiplicative group is CYCLIC of order 2^K
    ↪ - 1, and every element
// satisfies a^(2^K) = a. Squaring is a field automorphism
    ↪ (FROBENIUS): (a + b)^2 = a^2 + b^2,
// so squaring is LINEAR over GF(2) and can be tabulated as a matrix.
// GF(2^K) contains GF(2^d) exactly when d | K.
// x^(2^K) - x factors as the product of all irreducible polynomials
    ↪ whose degree divides K.
// A random degree-K polynomial is irreducible with probability
    ↪ about 1/K; test with
// gcd(x^(2^d) - x, f) == 1 for all d < K dividing K.
// NOT THE SAME AS NIMBERS. Nim-multiplication (10 - Game Theory/05 -
    ↪ NimbersAndSums.cpp) is a
// DIFFERENT field structure on the same set: it needs no irreducible
    ↪ polynomial, it is defined by
// the mex rules, and 2^(2^k) values form subfields. Use nimbers for
    ↪ game values, this for coding
// and linear-algebra style problems - they are isomorphic but NOT
    ↪ equal element by element.
// SPEED: on x86 with -mpclmul, _mm_clmulepi64_si128 does the
    ↪ carry-less product in one
// instruction; you then only need the Barrett-style reduction. Worth
    ↪ it only in a hot loop.
```

6.2 Multiplicative Functions

6.2.1 EulerTotientPhi

EULER TOTIENT SIEVE - $\text{phiArr}[i] = \#\{1 \leq k \leq i : \gcd(k,i)=1\}$ for all $i < N$ in $O(N \log \log N)$.

```
// Needed for: a^b mod m with gcd(a,m)=1 (Euler's theorem), counting
//   ↳ coprime pairs, Mobius identities
// (sum_{d|n} phi(d) = n), and the order of the multiplicative group
//   ↳ mod n.
// NAME: phiArr, not phi - a global 'phi' collides with the phi()
//   ↳ FUNCTION in
// 04 - Number Theory/02 - Primes and Divisors/06 -
//   ↳ MillerRabinPollard.cpp.
int phiArr[N];

void calculatePhi() {
    for (int i = 0; i < N; ++i) phiArr[i] = i & 1 ? i : i / 2;
    for (int i = 3; i < N; i += 2)
        if (phiArr[i] == i)
            for (int j = i; j < N; j += i) phiArr[j] -= phiArr[j] / i;
}
```

6.2.2 MobiusFunction

Mobius Function $\mu(n)$ Sieve - Time: $O(N)$, Space: $O(N)$

```
// mu(n) = 1 if n is square-free with an even number of prime factors
// mu(n) = -1 if n is square-free with an odd number of prime factors
// mu(n) = 0 if n has a squared prime factor
struct Mobius {
    int n;
    vector<int> mu, primes;
    vector<bool> is_prime;

    Mobius(int n = 1e7) : n(n), mu(n + 1, 0), is_prime(n + 1, true) {
        is_prime[0] = is_prime[1] = false;
        mu[1] = 1;

        for (int i = 2; i <= n; i++) {
            if (is_prime[i]) {
                primes.pb(i);
                mu[i] = -1;
            }
            for (int p : primes) {
                if ((ll)i * p > n) break;
                is_prime[i * p] = false;
                if (i % p == 0) {
                    mu[i * p] = 0;
                    break;
                } else {
                    mu[i * p] = -mu[i];
                }
            }
        }
    }
};
```

6.3 Matrices

6.3.1 MatrixExponentiation

Matrix over Mint. mul $O(n^3)$ (cache-friendly i-k-j order), pow $O(n^3 \log e)$.

```
// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp).
// Fix the size at compile time (array<array<Mint,K>,K>) if you need
//   ↳ the last 2x speedup.
typedef vector<vector<Mint>> Mat;

Mat operator*(const Mat& a, const Mat& b) {
    int n = a.size(), k = b.size(), m = b[0].size();
    Mat c(n, vector<Mint>(m));
    for (int i = 0; i < n; i++)
        for (int t = 0; t < k; t++) if (a[i][t].v)
            for (int j = 0; j < m; j++) c[i][j] += a[i][t] * b[t][j];
    return c;
}

Mat eye(int n) { Mat r(n, vector<Mint>(n)); for (int i = 0; i < n; i++)
    ↳ i++) r[i][i] = 1; return r; }

Mat mpow(Mat a, ll e) {
    Mat r = eye(a.size());
    for (; e > 0; e >>= 1, a = a * a) if (e & 1) r = r * a;
    return r;
}

// RECIPES
// Linear recurrence f(n) = c1*f(n-1) + ... + ck*f(n-k):
//   companion matrix, first row = (c1..ck), subdiagonal = 1. f(n)
//   ↳ = (M^(n-k) * [f(k-1)..f(0)]^T)[0]
// Add a "+d" constant term: enlarge by one row/col with a 1 on the
//   ↳ diagonal, put d in the first row.
// Count walks of length L between u and v: (A^L)[u][v] where A is
//   ↳ the adjacency matrix.
// Shortest walk with exactly L edges: same, but replace (+,*) with
//   ↳ (min,+) - "min-plus product".
```

6.3.2 CharacteristicPolynomial

CHARACTERISTIC POLYNOMIAL $p(x) = \det(xI - A)$ of an $n \times n$ matrix in $O(n^3)$. MOD must be PRIME.

```
// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp), Mat (03 -
//   ↳ Matrices/01).
// REACH FOR IT when you need  $\det(A - tI)$  for MANY  $t$  ( $n = 500$ ,  $q =$ 
//   ↳  $1e5$  queries: build  $p$  once,
// then each query is one Horner evaluation), or  $A^k$  for a huge  $k$ : by
//   ↳ Cayley-Hamilton  $p(A) = 0$ , so
//  $A^k = (x^k \bmod p(x))(A)$ , which costs  $O(n^2 \log k)$  polynomial work
//   ↳ plus  $O(n^3)$  to expand - much
// better than  $O(n^3 \log k)$  when  $n$  is large. Also:  $\det(A) = (-1)^n \times$ 
//   ↳  $p[0]$ ,  $\text{trace}(A) = -p[n-1]$ .
// Gaussian elimination CANNOT do this directly because the entries
//   ↳ are polynomials in  $x$ .

// STEP 1: reduce to upper HESSENBERG form ( $a[i][j] = 0$  for  $i > j +$ 
//   ↳  $1$ ) by SIMILARITY transforms.
// The row op "row  $i \leftarrow f \times \text{row}(c+1)$ " is undone by "col  $(c+1) \leftarrow f \times$ 
//   ↳ col  $i$ ", so  $A \rightarrow E A E^{-1}$  and
// the whole spectrum (hence the characteristic polynomial) is
//   ↳ unchanged.  $O(n^3)$ .
Mat hessenberg(Mat a) {
    int n = a.size();
    for (int c = 0; c + 2 < n; c++) {
        int p = -1;
        for (int i = c + 1; i < n; i++) if (a[i][c].v) { p = i;
        ↳ break; }
        if (p < 0) continue; // column already
        ↳ clear below the subdiagonal
        if (p != c + 1) { // swapping rows AND
        ↳ cols keeps it a similarity;
        swap(a[c + 1], a[p]); // column c is
        ↳ untouched because  $c < c+1 \leq p$ 
        for (int i = 0; i < n; i++) swap(a[i][c + 1], a[i][p]);
        }
        for (int i = c + 2; i < n; i++) if (a[i][c].v) {
            Mint f = a[i][c] / a[c + 1][c];
            for (int j = 0; j < n; j++) a[i][j] -= a[c + 1][j] * f;
            for (int j = 0; j < n; j++) a[j][c + 1] += a[j][i] * f;
        ↳ // uses the UPDATED col i
        }
    }
    return a;
}

// STEP 2: expand  $\det(xI - H)$  along the last row. With  $f[m] = \text{char}$ 
//   ↳ poly of the leading  $m \times m$  block,
//  $f[m] = (x - h[m-1][m-1]) f[m-1] - \sum_{i=1..m-1} h[m-1-i][m-1] \times$ 
//   ↳  $(\prod_{j=m-i..m-1} h[j][j-1])$ 
//
//   ↳  $f[m-1-i]$ 
// Only Hessenberg matrices satisfy such a short recurrence; that is
//   ↳ the whole point of step 1.
vector<Mint> charPoly(const Mat& a) { // returns  $p[0..n]$ ,
    ↳  $p[i] = \text{coeff of } x^i$ ,  $p[n] = 1$ 
    int n = a.size();
    Mat h = hessenberg(a);
    vector<vector<Mint>> f(n + 1);
    f[0] = {1};
    for (int m = 1; m <= n; m++) {
        f[m].assign(m + 1, 0);
        for (int i = 0; i < m; i++) {
            f[m][i + 1] += f[m - 1][i];
            f[m][i] -= h[m - 1][m - 1] * f[m - 1][i];
        }
        Mint prod = 1;
        for (int i = 1; i < m; i++) {
            prod *= h[m - i][m - i - 1];
            Mint c = prod * h[m - 1 - i][m - 1];
            if (!c.v) continue;
            for (int j = 0; j <= m - 1 - i; j++) f[m][j] -= c * f[m -
        ↳ 1 - i][j];
        }
    }
    return f[n];
}

//  $A^k$  VIA CAYLEY-HAMILTON:  $r(x) = x^k \bmod p(x)$  ( $O(n^2 \log k)$  with
//   ↳ schoolbook,  $O(n \log n \log k)$ 
// with NTT), then  $A^k = \sum r[i] A^i$ . If you only need  $u^T A^k v$ ,
//   ↳ that is  $\sum r[i] \times (u^T A^i v)$ 
// and the  $A^i v$  are  $n$  matrix-vector products -  $O(n \times \text{nnz} \times n)$  for a
//   ↳ sparse  $A$ , no  $n^3$  anywhere.
// EIGENVALUES over  $F_p$  are the roots of  $p$ ; over  $R$  use the QR
//   ↳ algorithm (not a contest routine).
//  $\det(A + tB)$  as a polynomial in  $t$ : reduce  $(A, B)$  to
//   ↳ Hessenberg/triangular pencil form, or just
// evaluate at  $n+1$  values of  $t$  with plain Gaussian determinants and
```

```
↳ interpolate -  $O(n^4)$ , fine to
//  $n = 100$  and much easier to get right.
// SIMILAR-LOOKING BUT DIFFERENT: the MINIMAL polynomial divides  $p$ ;
//   ↳ get it from the Krylov
// sequence  $u^T A^k v$  with Berlekamp-Massey (see 03 -
//   ↳ SparseDeterminant.cpp).
```

6.3.3 SparseDeterminant

rnd (01 - Miscellaneous/04 - random.cpp).

```
// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp),
//   ↳ berlekampMassey (06 - Interpolation/02),
//   ↳ rnd (01 - Miscellaneous/04 - random.cpp).
// WIEDEMANN: determinant and minimal polynomial of a SPARSE  $n \times n$ 
//   ↳ matrix in  $O(n \times \text{nnz} + n^2)$ ,
// using  $O(n)$  memory. MOD must be PRIME. Gaussian elimination is
//   ↳  $O(n^3)$  and fills in; this is the
// routine for  $n = 5e4$  with  $1e5$  non-zeros (Yosupo "sparse matrix
//   ↳ det"), for adjacency-like
// matrices, and for the Matrix-Tree determinant of a sparse graph.
// IDEA: for random vectors  $u, v$  the scalar sequence  $s_k = u^T A^k v$ 
//   ↳ obeys the same linear
// recurrence as  $A$  itself (Cayley-Hamilton), so Berlekamp-Massey on
//   ↳  $2n+2$  samples recovers the
// minimal polynomial. Each sample costs one sparse mat-vec, hence  $n$ 
//   ↳  $\times$  nnz total.
// PRECONDITIONER: right-multiplying by a random diagonal  $D$  makes
//   ↳  $\text{minpoly}(AD) = \text{charpoly}(AD)$  with
// probability  $\geq 1 - O(n^2 / \text{MOD})$ , which is what lets us read the
//   ↳ determinant off the last
// coefficient.  $\det(AD) = \det(A) \times \prod d_i$ , so divide the  $d_i$  back
//   ↳ out.
struct Elem { int r, c; Mint v; }; //  $A[r][c] = v$ ; all
//   ↳  $(r, c)$  distinct, rest zero

vector<Mint> sparseMinPoly(int n, const vector<Elem>& a) {
    vector<Mint> u(n), x(n), s;
    for (int i = 0; i < n; i++) u[i] = rnd(1, MOD - 1), x[i] = rnd(1,
    ↳ MOD - 1);
    for (int k = 0; k < 2 * n + 2; k++) {
        Mint t = 0;
        for (int i = 0; i < n; i++) t += u[i] * x[i];
        s.push_back(t);
        vector<Mint> y(n, 0); //  $y = A \times x$ , one
        ↳ pass over the non-zeros
        for (const Elem& e : a) y[e.r] += e.v * x[e.c];
        x = y;
    }
    return berlekampMassey(s); //  $s[i] = \sum_j C[j]$ 
    ↳  $\times s[i-1-j]$ 
}

Mint sparseDet(int n, vector<Elem> a) {
    vector<Mint> d(n);
    for (int i = 0; i < n; i++) d[i] = rnd(1, MOD - 1);
    for (Elem& e : a) e.v *= d[e.c]; //  $A \rightarrow A \times \text{diag}(d)$ 
    vector<Mint> c = sparseMinPoly(n, a);
    if ((int)c.size() < n) return 0; // minpoly shorter
    ↳ than  $n \Rightarrow$  singular (whp)
    Mint r = c.back(); // charpoly =  $x^n -$ 
    ↳  $c_0 x^{(n-1)} - \dots - c_{(n-1)}$ 
    if (n % 2 == 0) r = Mint(0) - r; //  $\det = (-1)^{(n+1)}$ 
    ↳  $\times c[n-1]$ 
    for (int i = 0; i < n; i++) r /= d[i];
    return r;
}

// RANK of a sparse matrix: same trick on a random projection;
//   ↳ simpler in practice is to run
// blocked elimination only if the fill-in stays small.
// SOLVING  $Ax = b$  sparsely: with the minimal polynomial  $m(x) = \sum$ 
//   ↳  $m_i x^i$  and  $m_0 \neq 0$ ,
//  $x = -(1/m_0) \times \sum_{i \geq 1} m_i A^{(i-1)} b$  -- again only sparse
//   ↳ mat-vecs.
// FAILURE MODE: if the returned recurrence is shorter than  $n$ , either
//   ↳ the matrix is singular or
// the random draw was unlucky; rerun once before believing the 0.
// INTEGER determinant with no modulus: run this under several primes
//   ↳ and CRT, sizing with
// Hadamard's bound  $|\det| \leq \prod \text{of the row 2-norms}$ .
```

6.3.4 CirculantDeterminant

DETERMINANT OF A CIRCULANT in $O(n \log^2 n)$ instead of $O(n^3)$.

```
// Needs: resultant (05 - Convolution/11 - PolyGcdResultant.cpp),
//   ↳ which needs NTT + Polynomial.
// C[i][j] = c[(j - i) mod n] (every row is the previous one rotated
//   ↳ right by 1). Then
// det C = prod_{k=0}^{n-1} f(w^k) = Res(x^n - 1, f), f(x) = c0 +
//   ↳ c1 x + ... + c(n-1) x^(n-1),
// where w is a primitive n-th root of unity. Eigenvalue f(w^k) has
//   ↳ eigenvector (1, w^k, w^2k, ...)
// - the Fourier modes diagonalise EVERY circulant simultaneously,
//   ↳ which is why circulants commute
// and why multiplying two of them is just a cyclic convolution of
//   ↳ their first rows.
// The resultant form is what you use when n does NOT divide MOD - 1,
//   ↳ so the roots of unity do not
// exist in the field. Everything below is mod NMOD = 998244353 (the
//   ↳ Poly modulus).
ll circulantDet(const vector<ll>& c) {
    int n = c.size();
    Poly xn(n + 1, 0);
    xn[0] = NMOD - 1, xn[n] = 1; // x^n - 1
    return resultant(xn, c);
}
// If n | (NMOD - 1) you do not need the resultant at all: evaluate f
//   ↳ at the n-th roots of unity
// with one length-n NTT and multiply the results,  $O(n \log n)$ .
// PERMUTANT MATRIX (a circulant along a permutation): M[i][j] = a[k]
//   ↳ where  $\sigma^k(i) = j$ ,
// for a permutation sigma that is a SINGLE n-cycle. Relabel the
//   ↳ indices along the cycle
// (q[0] = 0, q[t] = sigma^t(0)) and M becomes an honest circulant;
//   ↳ the determinant picks up the
// SIGN of that relabelling permutation. If sigma has more than one
//   ↳ cycle the matrix is not a
// permutant and the trick does not apply.
// BLOCK-CIRCULANT (blocks of size m, n blocks): det = prod_k
//   ↳ det(F(w^k)) with F the matrix-valued
// symbol - same argument, m x m determinants instead of scalars.
// OTHER DETERMINANTS YOU CAN DO WITHOUT ELIMINATION
// MATRIX DETERMINANT LEMMA:  $\det(A + u v^T) = \det(A) * (1 + v^T A^{-1} u)$ . Rank-k update:
//   ↳  $\det(A + U V^T) = \det(A) * \det(I_k + V^T A^{-1} U)$ . So  $\det(xI + a$ 
//   ↳  $b^T) = x^{n-1} * (x + a.b)$ ,
// which closes "M[i][j] = a_i b_j off the diagonal, x + a_i b_i
//   ↳ on it" in  $O(n)$ .
// CAUCHY-BINET: for A (n x m) and B (m x n),  $\det(AB) = \sum \text{over}$ 
//   ↳ n-subsets S of [m] of
//  $\det(A[:,S]) * \det(B[S,:])$ . This is the LGV lemma's algebraic
//   ↳ twin.
// VANDERMONDE:  $\det = \prod_{i < j} (x_j - x_i)$ . CAUCHY:
//   ↳  $\det(1/(x_i + y_j)) =$ 
//  $\prod_{i < j} (x_i - x_j)(y_i - y_j) / \prod_{i,j} (x_i + y_j)$ .
// TRIDIAGONAL:  $d_k = b_k d_{k-1} - a_k c_{k-1} d_{k-2}$ ,  $O(n)$  (this
//   ↳ is the Hessenberg recurrence).
```

6.3.5 PolyMatrixPower

MATRIX EXPONENTIATION WHERE THE ENTRIES ARE POWER SERIES, truncated mod x^K .

```
// Needs: multiply (05 - Convolution/02 - NTT.cpp), Poly / cut (05 -
//   ↳ Convolution/04).
// M^e in  $O(n^3 * K \log K * \log e)$ . Everything is mod NMOD =
//   ↳ 998244353.
// REACH FOR IT for a linear recurrence in n that also tracks a
//   ↳ SECOND statistic you want the full
// distribution of: put  $x^{(\text{contribution})}$  in the transition instead of
//   ↳ 1, and the (i, j) entry of
// M^e becomes the generating function over that statistic. Classic:
//   ↳ "count length-n walks by
// number of times a marked edge is used", "f(n) for every k = number
//   ↳ of ..." (CF 755G), any DP
// whose answer is asked for ALL values of a parameter at once rather
//   ↳ than one value.
// A plain Mat over Mint gives you one number; this gives you K
//   ↳ numbers for a log K factor.
typedef vector<vector<Poly>> PMat;

PMat pmul(const PMat& a, const PMat& b, int K) {
    int n = a.size(), m = b[0].size(), k = b.size();
    PMat c(n, vector<Poly>(m, Poly(1, 0)));
    for (int i = 0; i < n; i++)
        for (int t = 0; t < k; t++) {
            if (a[i][t].size() == 1 && !a[i][t][0]) continue; //
            ↳ skip the zero series
            for (int j = 0; j < m; j++) c[i][j] = c[i][j] +
            ↳ cut(multiply(a[i][t], b[t][j]), K);
        }
    return c;
}

PMat peye(int n) {
    PMat r(n, vector<Poly>(n, Poly(1, 0)));
    for (int i = 0; i < n; i++) r[i][i] = Poly(1, 1);
    return r;
}

PMat ppow(PMat a, ll e, int K) { // a^e mod x^K
    PMat r = peye(a.size());
    for (; e > 0; e >>= 1, a = pmul(a, a, K)) if (e & 1) r = pmul(r,
    ↳ a, K);
    return r;
}

// EXAMPLE (Fibonacci refined by the number of "1" steps, CF 755G
//   ↳ shape):
// M = {{1 + x, x}, {1, 0}} and  $(M^{n-1} * v)[0]$  is the generating
//   ↳ function you want.
// CHEAPER ALTERNATIVES, try these first:
// - if K is tiny ( $\leq 3$ ), just hand-roll the truncated series as k
//   ↳ separate Mint matrices.
// - if the statistic is "how many times did we take transition t",
//   ↳ one variable suffices; if it
// is a pair of statistics you need a bivariate series - flatten
//   ↳ it to  $x^{(i*W + j)}$ .
// - if n (the matrix size) is 1, this is just power-series
//   ↳ exponentiation: use pow_ directly.
// - if you want ONE coefficient rather than all K, Bostan-Mori on
//   ↳ the rational function is
//  $O(d \log d \log e)$  and much faster (06 - Interpolation/03 -
//   ↳ BostanMori.cpp).
// COST WARNING:  $n^3$  poly multiplications per squaring. n = 8, K =
//   ↳ 1e5, e = 1e18 is already at the
// edge; keep the matrix small, and truncate (cut to K) after EVERY
//   ↳ product, never at the end.
```

6.3.6 Freivalds

FREIVALDS: verify $A * B == C$ in $O(t * n^2)$ instead of computing the product in $O(n^3)$.

```
// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp), rnd (01 -
// ↳ Miscellaneous/04 - random.cpp).
// Error probability <= 2^-t (one-sided: it never rejects a correct
// ↳ product). t = 30..40 is plenty.
// REACH FOR IT when the problem is "is this claimed product right",
// ↳ when A or B changes between
// queries so recomputing is too slow, or as a subroutine inside a
// ↳ bigger randomised check.
bool freivalds(const vector<vector<Mint>>& a, const
// ↳ vector<vector<Mint>>& b,
// ↳ const vector<vector<Mint>>& c, int t = 40) {
    int n = a.size();
    vector<Mint> r(n), x(n), y(n);
    while (t--> 0) {
        for (int i = 0; i < n; i++) r[i] = rnd(0, 1); //
        ↳ random 0/1 vector is enough
        for (int i = 0; i < n; i++) { // x
            ↳ = B r, y = C r
            x[i] = y[i] = 0;
            for (int j = 0; j < n; j++) x[i] += b[i][j] * r[j], y[i]
            ↳ += c[i][j] * r[j];
        }
        for (int i = 0; i < n; i++) { //
            ↳ check A x == y
            Mint s = 0;
            for (int j = 0; j < n; j++) s += a[i][j] * x[j];
            if (!(s == y[i])) return false;
        }
    }
    return true;
}

// WHY IT WORKS: D = AB - C is non-zero, so some row of D is
// ↳ non-zero; for a uniform 0/1 vector r,
// Pr[D r == 0] <= 1/2 (fix all coordinates but one - at most one of
// ↳ the two choices can cancel).
// Independent rounds multiply, hence 2^-t. Do the products RIGHT to
// ↳ LEFT: A(Br), never (AB)r.

// THE GENERAL PATTERN - RANDOMISED ALGEBRAIC IDENTITY TESTING
// SCHWARTZ-ZIPPEL: a non-zero polynomial of total degree d over a
// ↳ field, evaluated at a uniform
// point of S^n, is zero with probability <= d / |S|. Every check
// ↳ below is one application.
// - "are these two expressions equal?" - evaluate both at a random
// ↳ point mod a large prime.
// - PERFECT MATCHING in a general graph: build the TUTTE matrix T
// ↳ with T[i][j] = +x_ij and
// T[j][i] = -x_ij for each edge i < j, substitute random values,
// ↳ and G has a perfect matching
// iff det T != 0 (whp). O(n^3) existence check with no Blossom.
// ↳ rank(T) = 2 * (max matching).
// - BIPARTITE matching: same with the Edmonds matrix (random value
// ↳ on each edge, 0 elsewhere).
// - "do these two multisets/strings/trees agree?" - hash them as
// ↳ polynomial evaluations.
// - CHECKING A PRODUCT OF SPARSE OBJECTS: a random-vector probe
// ↳ generalises far past matrices
// (it is exactly the idea behind Wiedemann in 03 -
// ↳ SparseDeterminant.cpp).
// Prefer a random FIELD element over 0/1 when the identity has
// ↳ degree > 1; the failure bound is
// then d / MOD rather than 1/2 per round.
```

6.3.7 MinPlusMatrixPower

TROPICAL (MIN-PLUS) MATRIX POWER: replace (+, *) by (min, +) and matrix exponentiation answers

```
// "cheapest walk using EXACTLY k edges" in  $O(n^3 \log k)$ . The
// ↳ min-plus semiring has additive
// identity llinf and multiplicative identity 0, and it is
// ↳ associative - which is all matrix
// exponentiation ever needed. Everything below works verbatim for
// ↳ MAX-PLUS by flipping min to max
// and llinf to -llinf.
// REACH FOR IT when the edge count is a hard constraint and k is
// ↳ large: "shortest path with
// exactly k edges", "cheapest k-step schedule", periodic/cyclic
// ↳ transport problems, and DPs whose
// transition is the same min-plus matrix every layer (n <= 100-200,
// ↳ k up to 1e18).
// If k is small, just do k layers of DP -  $O(k * n^2)$  beats  $O(n^3 \log k)$ 
// ↳ whenever  $k < n \log k$ .
typedef vector<vector<ll>> TMat;

TMat tmul(const TMat& a, const TMat& b) {
    int n = a.size(), m = b[0].size(), p = b.size();
    TMat c(n, vector<ll>(m, llinf));
    for (int i = 0; i < n; i++)
        for (int t = 0; t < p; t++) {
            if (a[i][t] >= llinf) continue; //
            ↳ skip, and avoid overflow
            for (int j = 0; j < m; j++)
                if (b[t][j] < llinf) c[i][j] = min(c[i][j], a[i][t] +
                ↳ b[t][j]);
        }
    return c;
}

TMat teye(int n) { //
    ↳ min-plus identity
    TMat r(n, vector<ll>(n, llinf));
    for (int i = 0; i < n; i++) r[i][i] = 0;
    return r;
}

TMat tpow(TMat a, ll k) { //
    ↳ exactly k edges
    TMat r = teye(a.size());
    for (; k > 0; k >>= 1, a = tmul(a, a)) if (k & 1) r = tmul(r, a);
    return r;
}

// SET-UP: a[i][j] = weight of edge i->j, llinf when there is no
// ↳ edge. Then tpow(a, k)[u][v] is the
// minimum total weight over walks u->v with exactly k edges (llinf
// ↳ if none exists).
// AT MOST k EDGES: add a zero-weight self-loop at every vertex
// ↳ (a[i][i] = min(a[i][i], 0)).
// COUNTING instead of minimising: the same code over (+, *) is 03 -
// ↳ Matrices/01, and over
// (or, and) with bitsets it answers reachability in exactly k steps
// ↳ in  $O(n^3 / 64)$ .
// LONGEST path with exactly k edges: max-plus, but ONLY if the graph
// ↳ is a DAG or k is bounded;
// otherwise a positive cycle makes it unbounded.
// OVERFLOW is the standard bug: guard both operands against llinf
// ↳ before adding, as above, or
// llinf + llinf silently wraps negative and poisons the whole matrix.
// FASTER SPECIAL CASES
// - if the matrix is CONVEX in the Monge sense, min-plus products
// ↳ drop to  $O(n^2)$  via SMAWK, so
// the whole power costs  $O(n^2 \log k)$  (see 08 - DP/01/11 -
// ↳ MinPlusConvolution.cpp).
// - if you only need one row (a fixed source), keep a VECTOR and
// ↳ multiply it by the precomputed
// powers  $a^{(2^i)}$ :  $O(n^2 \log k)$  total instead of  $n^3 \log k$ .
// - for eventually-periodic behaviour, the tropical eigenvalue (the
// ↳ minimum mean cycle, Karp's
// algorithm) gives the per-step growth rate and lets you skip the
// ↳ huge exponent entirely.
```

6.4 Linear Algebra

6.4.1 LinearXorBasis

XOR basis (vector space over GF(2)). insert / query O(B).

```
// span size = 2^sz. Use MERGE to keep a basis in each segment-tree
// node -> max-xor over a RANGE.
struct Basis {
    static const int B = 62; // 60 silently drops bits 61-62 of
    // an ll up to 9e18
    ll b[B + 1] = {}; // b[i] has
    // leading bit i
    int sz = 0;

    bool insert(ll x) { // false if x
        // already in the span
        for (int i = B; i >= 0; i--) if (x >> i & 1) {
            if (!b[i]) { b[i] = x, sz++; return true; }
            x ^= b[i];
        }
        return false;
    }
    bool contains(ll x) const {
        for (int i = B; i >= 0; i--) if (x >> i & 1) { if (!b[i])
        // return false; x ^= b[i]; }
        return true;
    }
    ll maxXor(ll r = 0) const { for (int i = B; i >= 0; i--) r =
    // max(r, r ^ b[i]); return r; }
    ll minXor(ll r = 0) const { for (int i = B; i >= 0; i--) if (r >>
    // i & 1) r ^= b[i]; return r; }
    void reduce() { // reduced
        // row echelon form
        for (int i = B; i >= 0; i--) if (b[i])
            for (int j = i - 1; j >= 0; j--) if (b[i] >> j & 1) b[i]
        // ^= b[j];
    }
    ll kth(ll k) { // k-th
        // SMALLEST value in the span, 0-indexed
        reduce(); // kth(0) ==
        // 0; -1 if k >= 2^sz
        ll r = 0;
        for (int i = 0; i <= B; i++) if (b[i]) { if (k & 1) r ^=
        // b[i]; k >>= 1; }
        return k ? -1 : r;
    }
    ll spanSize() const { return sz >= 63 ? -1 : 1LL << sz; }
    friend Basis merge(Basis a, const Basis& c) {
        for (int i = 0; i <= B; i++) if (c.b[i]) a.insert(c.b[i]);
        return a;
    }
};
```

6.4.2 GaussianElimination

Gaussian elimination mod a PRIME. Returns rank; solves $Ax = b$; also det and inverse.

```
// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp) for the
// modular routines.
// 0(n^2 m). Pass the augmented matrix (n rows, m+1 cols) to solve().
int gaussMod(vector<vector<Mint>>& a, int m) { // reduce
    // first m columns to RREF
    int n = a.size(), rank = 0;
    for (int col = 0; col < m && rank < n; col++) {
        int piv = -1;
        for (int i = rank; i < n; i++) if (a[i][col].v) { piv = i;
        // break; }
        if (piv < 0) continue;
        swap(a[rank], a[piv]);
        Mint iv = a[rank][col].inv();
        for (auto& x : a[rank]) x *= iv;
        for (int i = 0; i < n; i++) if (i != rank && a[i][col].v) {
            Mint f = a[i][col];
            for (size_t j = 0; j < a[i].size(); j++) a[i][j] -= f *
        // a[rank][j];
        }
        rank++;
    }
    return rank;
}
// Solve Ax = b. Returns {} if inconsistent, else one solution (free
// vars = 0).
// 'free' receives the indices of the free variables (solution space
// has MOD^|free| points).
vector<Mint> solveLinear(vector<vector<Mint>> a, vector<Mint> b,
    // vector<int>* freeIdx = nullptr) {
    int n = a.size(), m = a[0].size();
    for (int i = 0; i < n; i++) a[i].push_back(b[i]);
    int rank = gaussMod(a, m);
    for (int i = rank; i < n; i++) if (a[i][m].v) return {}; //
    // 0 = nonzero
    vector<Mint> x(m, 0);
    int r = 0;
    for (int c = 0; c < m; c++) { // must run over
        // EVERY column: stopping at
        if (r < rank && a[r][c].v) x[c] = a[r][m], r++; // r ==
        // rank drops the free columns
        else if (freeIdx) freeIdx->push_back(c); // that
        // come after the last pivot
    }
    return x;
}
Mint determinant(vector<vector<Mint>> a) { // 0(n^3),
    // destroys a
    int n = a.size();
    Mint det = 1;
    for (int col = 0; col < n; col++) {
        int piv = -1;
        for (int i = col; i < n; i++) if (a[i][col].v) { piv = i;
        // break; }
        if (piv < 0) return 0;
        if (piv != col) swap(a[col], a[piv]), det = Mint(0) - det;
        det *= a[col][col];
        Mint iv = a[col][col].inv();
        for (int i = col + 1; i < n; i++) if (a[i][col].v) {
            Mint f = a[i][col] * iv;
            for (int j = col; j < n; j++) a[i][j] -= f * a[col][j];
        }
    }
    return det;
}
vector<vector<Mint>> inverse(vector<vector<Mint>> a) { // {} if
    // singular
    int n = a.size();
    for (int i = 0; i < n; i++) { a[i].resize(2 * n, 0); a[i][n + i]
    // = 1; }
    if (gaussMod(a, n) < n) return {};
    vector<vector<Mint>> r(n, vector<Mint>(n));
    for (int i = 0; i < n; i++) for (int j = 0; j < n; j++) r[i][j] =
    // a[i][n + j];
    return r;
}
// MATRIX-TREE: #spanning trees = determinant of the Laplacian (deg
// on diagonal, -1 per edge)
// with any one row AND the matching column deleted.
```

6.4.3 GaussXorAndReal

XOR-equation systems over GF(2) with bitset - $O(n \cdot m / 64)$. n equations, m variables,

```
// row[i][m] holds the RHS. Returns rank, or -1 if inconsistent.
template<int M>
int gaussXor(vector<bitset<M + 1>>& a, int m, bitset<M>* sol =
    nullptr) {
    int n = a.size(), rank = 0;
    vector<int> where(m, -1);
    for (int col = 0; col < m && rank < n; col++) {
        int piv = -1;
        for (int i = rank; i < n; i++) if (a[i][col]) { piv = i;
        break; }
        if (piv < 0) continue;
        swap(a[rank], a[piv]);
        for (int i = 0; i < n; i++) if (i != rank && a[i][col]) a[i]
        ^= a[rank];
        where[col] = rank++;
    }
    for (int i = rank; i < n; i++) if (a[i][m]) return -1;
    if (sol) { sol->reset(); for (int c = 0; c < m; c++) if (where[c]
    >= 0) (*sol)[c] = a[where[c]][m]; }
    return rank; // #solutions
    // = 2^(m - rank)
}
// USES: "light switch" puzzles, xor-equation counting, rank of a set
// of vectors over GF(2),
// linear independence of xor values (dense sibling of the XOR
// basis).

// ---- Gaussian elimination over the reals, with partial pivoting
// ----
// Returns rank; a is n x (m+1) augmented. Solution in x (free vars =
// 0). eps-based.
int gaussReal(vector<vector<ld>>& a, int m, vector<ld>& x) {
    int n = a.size(), rank = 0;
    vector<int> where(m, -1);
    for (int col = 0; col < m && rank < n; col++) {
        int piv = rank;
        for (int i = rank; i < n; i++) if (fabs(a[i][col]) >
        fabs(a[piv][col])) piv = i;
        if (fabs(a[piv][col]) < 1e-12) continue;
        swap(a[rank], a[piv]);
        for (int i = 0; i < n; i++) if (i != rank) {
            ld f = a[i][col] / a[rank][col];
            for (int j = col; j <= m; j++) a[i][j] -= f * a[rank][j];
        }
        where[col] = rank++;
    }
    x.assign(m, 0);
    for (int c = 0; c < m; c++) if (where[c] >= 0) x[c] =
    a[where[c]][m] / a[where[c]][c];
    return rank;
}
// The classic use is EXPECTED-VALUE DP with cyclic state
// dependencies: write one equation per
// state,  $E[s] = 1 + \sum p(s \rightarrow t) E[t]$ , and solve.  $n \leq \sim 500$  fits
//  $O(n^3)$ .
```

6.4.4 WeightedXorBasis

WEIGHTED XOR BASIS - keep the MAXIMUM-WEIGHT independent set of vectors over GF(2). $O(B)$ per

```
// insert, online, no sorting and no rollback needed.
// The vectors of a matroid's ground set with a "keep the heaviest"
// greedy: linear independence is
// a MATROID, so the greedy is optimal. Standard basis insertion
// keeps whichever vector arrives
// first; this one keeps the heavier and pushes the lighter down to
// be re-inserted, which is the
// same as running Kruskal's exchange argument implicitly.
// REACH FOR IT: "pick a maximum-weight subset of vectors that stays
// linearly independent",
// "maximum spanning forest over GF(2) constraints", cycle-space
// problems where each edge carries
// both an xor value and a cost, and any "add items online, report
// the best independent set now".
struct WBasis {
    static const int B = 62;
    ll b[B + 1] = {}; // b[i] has
    // leading bit i
    ll w[B + 1] = {}; // weight of
    // the vector stored in b[i]
    int sz = 0;

    void insert(ll x, ll wt) {
        for (int i = B; i >= 0; i--) if (x >> i & 1) {
            if (!b[i]) { b[i] = x, w[i] = wt, sz++; return; }
            if (w[i] < wt) swap(b[i], x), swap(w[i], wt); // keep
            // the heavier one at this pivot
            x ^= b[i];
        }
        ll totalWeight() const { ll s = 0; for (int i = 0; i <= B; i++) s
        += w[i]; return s; }
        ll maxXor(ll r = 0) const { for (int i = B; i >= 0; i--) r =
        max(r, r ^ b[i]); return r; }
};
// MINIMUM-weight independent set: negate the weights (or compare
// with > instead of <).
// THE TRAP: you may NOT just sort by weight and insert greedily into
// a plain basis if the vectors
// arrive online - but if they are all known up front, sorting
// descending and using the ordinary
// Basis::insert gives exactly the same answer and is simpler. Use
// this version when insertions are
// interleaved with queries.
// MATROID INTERSECTION generalises this to "independent in TWO
// matroids at once" (e.g. linearly
// independent AND a forest); that needs augmenting paths in an
// exchange graph, not a greedy.
// LINEAR MATROID over a general field: same greedy with Gaussian
// row-reduction instead of xor.
// TYPICAL ENCODING TRICK: to force "at most one vector per group",
// append group-identifier bits
// above the value bits (a 128-bit word via __int128 gives you 62
// value bits plus 66 tag bits).
// That turns "spanning-forest + xor" constraints into one basis
// insert per edge.
```


6.4.5 XorSystemExtremes

LEXICOGRAPHICALLY EXTREME SOLUTION of a GF(2) system, on top of 03 - GaussXorAndReal.cpp.

```
// gaussXor returns SOME solution (free variables = 0), which is the
// ↳ lexicographically SMALLEST one
// when variables are compared in index order x[0], x[1], ... .
// ↳ Problems that ask for the
// lexicographically LARGEST assignment, or that weight the
// ↳ variables, need this wrapper.
// THE TRICK: reverse the variable order before eliminating and
// ↳ reverse the answer back. Reduced
// row echelon form always puts a pivot as early as possible, so
// ↳ "free variables = 0" minimises the
// reversed vector, i.e. maximises the original.  $O(n * m / 64)$ , same
// ↳ as one elimination.
// Careful: "lexicographically largest" here means as a SEQUENCE x[0]
// ↳ > x[1] > ...; if the problem
// means "largest as a binary NUMBER with x[0] the most significant
// ↳ bit", that is the same thing.
template<int M>
int gaussXorLexMax(vector<bitset<M + 1>> a, int m, bitset<M>& sol) {
    int n = a.size();
    for (int i = 0; i < n; i++) { // reverse
        ↳ the m variable columns
        bitset<M + 1> t;
        for (int j = 0; j < m; j++) t[j] = a[i][m - 1 - j];
        t[m] = a[i][m];
        a[i] = t;
    }
    bitset<M> s;
    int rank = gaussXor<M>(a, m, &s);
    if (rank < 0) return -1; //
    ↳ inconsistent
    for (int j = 0; j < m; j++) sol[j] = s[m - 1 - j]; // and
    ↳ reverse the answer back
    return rank; //
    ↳ #solutions =  $2^{\text{m} - \text{rank}}$ 
}
// PRIORITY ORDER other than "by index": permute the columns by the
// ↳ priority you want (most
// important variable first), solve, permute back. Only the ORDER of
// ↳ the columns matters.
// MAXIMISE A LINEAR OBJECTIVE sum  $w_i x_i$  over the solution set:
// ↳ that is NOT lexicographic and is
// NP-hard in general (max-XOR-SAT). It is easy only when all  $w_i \geq$ 
// ↳ 0 and you may take each pivot
// greedily - i.e. when the objective is itself lexicographic.
// MAXIMISE THE XOR of the solution vector viewed as a number: build
// ↳ a basis of the NULL SPACE
// (one vector per free variable: set that free variable to 1,
// ↳ back-substitute the pivots), then
// run Basis::maxXor starting from the particular solution (04 -
// ↳ Linear Algebra/01).
// COUNTING: the solution set is a coset, so every count is  $2^{\text{m} -$ 
// ↳ rank) or 0; "number of solutions
// with  $x[i] = 1$ " is either 0, the full count, or half of it,
// ↳ depending on whether  $e_i$  is forced.
```

6.4.6 GF2MatrixInverse

INVERSE OF A MATRIX OVER GF(2) with bitsets - $O(n^3 / 64)$. $n = 2000$ runs in well under a second,

```
// where a Mint-based inverse at the same size is 8e9 operations and
// ↳ hopeless.
// Returns {} if the matrix is singular. Rows are bitset<M> with bit
// ↳ j = entry (i, j).
// REACH FOR IT for xor-system problems that need the FULL inverse
// ↳ rather than one solution:
// - "for each edge, is it in EVERY / SOME spanning structure" style
// ↳ queries, which reduce to a
// single entry of the inverse (CF 736D: after inverting the
// ↳ incidence-style matrix, entry
// inv[b][a] decides whether edge (a, b) is forced);
// - answering many right-hand sides against one fixed matrix ( $x =$ 
// ↳  $A^{-1}b$  in  $O(n^2 / 64)$  each);
// - determinant over GF(2) (= 1 iff invertible), rank, and the null
// ↳ space in one pass.
template<int M>
vector<bitset<M>> inverseGF2(vector<bitset<M>>> a) {
    int n = a.size();
    vector<bitset<M>> inv(n);
    for (int i = 0; i < n; i++) inv[i][i] = 1;
    for (int i = 0; i < n; i++) { // forward
        ↳ elimination
        int piv = -1;
        for (int j = i; j < n; j++) if (a[j][i]) { piv = j; break; }
        if (piv < 0) return {}; // singular
        swap(a[i], a[piv]), swap(inv[i], inv[piv]);
        for (int j = i + 1; j < n; j++) if (a[j][i]) a[j] ^= a[i],
        ↳ inv[j] ^= inv[i];
    }
    for (int i = n - 1; i >= 0; i--) // back
        ↳ substitution
        for (int j = i + 1; j < n; j++) if (a[i][j]) inv[i] ^= inv[j];
    return inv;
}
// NO PIVOT VALUES to divide by: over GF(2) every non-zero entry is
// ↳ 1, so elimination is pure xor
// and there is no numerical anything. That is why this is the
// ↳ cleanest Gaussian elimination there
// is - and why you should reduce to GF(2) whenever a problem lets
// ↳ you.
// DETERMINANT over GF(2) is 1 iff the forward pass finds a pivot in
// ↳ every column; equivalently
// rank == n. For the rank alone, or a solution of  $Ax = b$ , use
// ↳ gaussXor (04 - Linear Algebra/03)
// which is the same loop with an augmented column.
// NULL SPACE: run gaussXor to RREF; for each free column c, the
// ↳ vector with 1 at c and the pivot
// entries copied from column c is a basis vector of the kernel. Its
// ↳ size is  $n - \text{rank}$ , and the
// solution set of  $Ax = b$  is (any particular solution) + (that span).
// PRODUCT over GF(2) in  $O(n^3 / 64)$ : transpose b, then  $c[i][j] =$ 
// ↳  $((a[i] \& bt[j]).count() \& 1)$ .
// Or, faster,  $c[i] = \text{xor of } a\text{'s rows selected by } b - \text{that is } O(n^2 /$ 
// ↳  $64)$  per row with no popcounts.
// M IS A COMPILE-TIME BOUND: bitset<M> must be at least n wide; keep
// ↳ M a multiple of 64.
```

6.5 Convolution

6.5.1 FFT

FFT over complex doubles. `conv()` is exact while $|\text{result coefficient}| < \sim 1e15$.

```
// Prefer NTT when the answer is taken modulo an NTT prime; use
↳ convMod for arbitrary moduli.
typedef complex<double> C;
void fft(vector<C>& a) {
    int n = a.size(), L = 31 - __builtin_clz(n);
    static vector<complex<long double>> R(2, 1);
    static vector<C> rt(2, 1);
    for (static int k = 2; k < n; k *= 2) {
        R.resize(n), rt.resize(n);
        auto x = polar(1.0/L, acos(-1.0/L) / k);
        for (int i = k; i < 2 * k; i++) rt[i] = R[i] = i & 1 ? R[i / 2] * x : R[i / 2];
    }
    vector<int> rev(n);
    for (int i = 0; i < n; i++) rev[i] = (rev[i / 2] | (i & 1) << L) <> / 2;
    for (int i = 0; i < n; i++) if (i < rev[i]) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 * k)
            for (int j = 0; j < k; j++) {
                C z = rt[j + k] * a[i + j + k];
                a[i + j + k] = a[i + j] - z;
                a[i + j] += z;
            }
}

// Real convolution. Result is rounded to the nearest integer.
vector<ll> conv(const vector<ll>& a, const vector<ll>& b) {
    if (a.empty() || b.empty()) return {};
    int rs = a.size() + b.size() - 1, n = 1;
    while (n < rs) n <<= 1;
    vector<C> in(n), out(n);
    for (size_t i = 0; i < a.size(); i++) in[i].real((double)a[i]);
    for (size_t i = 0; i < b.size(); i++) in[i].imag((double)b[i]);
    fft(in);
    for (C& x : in) x *= x; // (a+bi)^2
    ↳ = a^2 - b^2 + 2abi
    for (int i = 0; i < n; i++) out[i] = in[-i & (n - 1)] -
    ↳ conj(in[i]);
    fft(out);
    vector<ll> res(rs);
    for (int i = 0; i < rs; i++) res[i] = llround(imag(out[i]) / (4 *
    ↳ n));
    return res;
}

// Convolution modulo an ARBITRARY M, splitting coefficients at
↳ sqrt(M) to keep precision.
vector<ll> convMod(const vector<ll>& a, const vector<ll>& b, ll M) {
    if (a.empty() || b.empty()) return {};
    int rs = a.size() + b.size() - 1, B = 32 - __builtin_clz(rs), n =
    ↳ 1 << B;
    ll cut = (ll)sqrtl((ld)M);
    vector<C> L(n), R(n), os(n), ol(n);
    for (size_t i = 0; i < a.size(); i++) L[i] = C((double)a[i] /
    ↳ cut), (double)a[i] % cut);
    for (size_t i = 0; i < b.size(); i++) R[i] = C((double)b[i] /
    ↳ cut), (double)b[i] % cut);
    fft(L), fft(R);
    for (int i = 0; i < n; i++) {
        int j = -i & (n - 1);
        ol[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
        os[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / C(0, 1);
    }
    fft(ol), fft(os);
    vector<ll> res(rs);
    for (int i = 0; i < rs; i++) {
        ll av = llround(real(ol[i])), cv = llround(imag(os[i]));
        ll bv = llround(imag(ol[i])) + llround(real(os[i]));
        res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
    }
    return res;
}
```

6.5.2 NTT

NTT modulo 998244353 ($= 119 \cdot 2^{23} + 1$, primitive root 3). $O(n \log n)$.

```
// multiply() is the entry point. MAX LENGTH  $2^{23}$  for 998244353 (see
↳ the assert in ntt()).
// For an ARBITRARY modulus: run this under three of the primes below
↳ and Garner them together
// (06 - Garner.cpp is written for exactly that), or split the
↳ coefficients at sqrt(M) and use
// convMod in 01 - FFT.cpp.
const ll NMOD = 998244353, NROOT = 3;
ll npow(ll a, ll e, ll m = NMOD) { ll r = 1; a %= m; for (; e; e >>=
    ↳ 1, a = a * a % m) if (e & 1) r = r * a % m; return r; }

void ntt(vector<ll>& a, bool inv) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int len = 2; len <= n; len <<= 1) {
        assert((NMOD - 1) % len == 0); // 998244352 =  $119 \cdot 2^{23}$ : len
        ↳  $> 2^{23}$  truncates the division
        ll w = npow(NROOT, (NMOD - 1) / len); // and returns
        ↳ GARBAGE with no other symptom
        if (inv) w = npow(w, NMOD - 2);
        for (int i = 0; i < n; i += len) {
            ll cur = 1;
            for (int j = 0; j < len / 2; j++) {
                ll u = a[i + j], v = a[i + j + len / 2] * cur % NMOD;
                a[i + j] = (u + v) % NMOD;
                a[i + j + len / 2] = (u - v + NMOD) % NMOD;
                cur = cur * w % NMOD;
            }
        }
    }
    if (inv) { ll ninv = npow(n, NMOD - 2); for (ll& x : a) x = x *
    ↳ ninv % NMOD; }
}

vector<ll> multiply(vector<ll> a, vector<ll> b) {
    if (a.empty() || b.empty()) return {};
    int rs = a.size() + b.size() - 1, n = 1;
    while (n < rs) n <<= 1;
    a.resize(n), b.resize(n);
    ntt(a, false), ntt(b, false);
    for (int i = 0; i < n; i++) a[i] = a[i] * b[i] % NMOD;
    ntt(a, true);
    a.resize(rs);
    return a;
}

// ARBITRARY MODULUS: split each coefficient as  $a = hi \cdot 2^{15} + lo$  and
↳ combine three products
// (or run NTT under three NTT-friendly primes and CRT). Values must
↳ stay below  $\sim 1e18$ .
// COMMON NTT PRIMES: 998244353 (g=3), 167772161 (g=3), 469762049
↳ (g=3), 1004535809 (g=3).
// USES: polynomial multiplication; "count pairs/triples with a given
↳ sum"; convolving
// distributions; string matching with wildcards; sum of  $C(a_i, k)$ 
↳ over all i for every k.
```

6.5.3 FWHT

Fast Walsh-Hadamard Transform (FWHT) - Time: $O(N \log N)$

```
// Computes Bitwise Convolutions: XOR, AND, OR for array of size N =
  ↪  $2^k$ 
struct FWHT {
    enum Type { XOR, AND, OR };

    static void transform(vector<ll>& a, bool invert, Type type =
  ↪ XOR) {
        int n = a.size();
        for (int len = 1; 2 * len <= n; len <= 1) {
            for (int i = 0; i < n; i += 2 * len) {
                for (int j = 0; j < len; j++) {
                    ll u = a[i + j], v = a[i + len + j];
                    if (type == XOR) {
                        a[i + j] = u + v;
                        a[i + len + j] = u - v;
                    } else if (type == AND) {
                        a[i + j] = invert ? u - v : u + v;
                    } else if (type == OR) {
                        a[i + len + j] = invert ? v - u : u + v;
                    }
                }
            }
        }
        if (type == XOR && invert) {
            for (ll &x : a) x /= n;
        }
    }

    // Computes  $C[k] = \sum_{i \text{ op } j = k} (A[i] * B[j])$  where op in
  ↪ {XOR, AND, OR}
    static vector<ll> multiply(vector<ll> a, vector<ll> b, Type type
  ↪ = XOR) {
        int n = 1;
        while (n < max(a.size(), b.size())) n <= 1;
        a.resize(n); b.resize(n);

        transform(a, false, type);
        transform(b, false, type);
        for (int i = 0; i < n; i++) a[i] *= b[i];
        transform(a, true, type);
        return a;
    }
};
```

6.5.4 Polynomial

FORMAL POWER SERIES over $NMOD = 998244353$, on top of multiply() from 02 - NTT.cpp.

```
// Every routine works "mod  $x^k$ ". inv/log/exp/sqrt/pow are all Newton
  ↪ iteration: each step
// doubles the number of correct coefficients, so the cost is the
  ↪ cost of the last multiply.
typedef vector<ll> Poly;

Poly cut(Poly a, int k) { a.resize(k, 0); return a; }
Poly operator+(Poly a, const Poly& b) {
    a.resize(max(a.size(), b.size()), 0);
    for (size_t i = 0; i < b.size(); i++) a[i] = (a[i] + b[i]) % NMOD;
    return a;
}
Poly operator-(Poly a, const Poly& b) {
    a.resize(max(a.size(), b.size()), 0);
    for (size_t i = 0; i < b.size(); i++) a[i] = (a[i] - b[i] + NMOD)
  ↪ % NMOD;
    return a;
}
Poly operator*(const Poly& a, ll c) { Poly r = a; for (ll& x : r) x =
  ↪ x * (c % NMOD) % NMOD; return r; }

Poly deriv(const Poly& a) {
    if (a.size() <= 1) return {0};
    Poly r(a.size() - 1);
    for (size_t i = 1; i < a.size(); i++) r[i - 1] = a[i] * i % NMOD;
    return r;
}
Poly integ(const Poly& a) {
    Poly r(a.size() + 1, 0);
    for (size_t i = 0; i < a.size(); i++) r[i + 1] = a[i] * npow(i +
  ↪ 1, NMOD - 2) % NMOD;
    return r;
}
//  $1/a \bmod x^k$ . Requires  $a[0] \neq 0$ .  $g_{2m} = g_m * (2 - a * g_m)$ 
Poly inv(const Poly& a, int k) {
    Poly g = {npow(a[0], NMOD - 2)};
    for (int m = 1; m < k; m <= 1) {
        Poly t = multiply(cut(a, 2 * m), g);
        t = cut(t, 2 * m);
        for (ll& x : t) x = (NMOD - x) % NMOD;
        t[0] = (t[0] + 2) % NMOD;
        g = cut(multiply(g, t), 2 * m);
    }
    return cut(g, k);
}
//  $\log(a) \bmod x^k$ . Requires  $a[0] = 1$ .  $\log a = \text{integral}(a' / a)$ 
Poly log_(const Poly& a, int k) { return
  ↪ cut(integ(cut(multiply(deriv(a), inv(a, k)), k - 1)), k); }
//  $\exp(a) \bmod x^k$ . Requires  $a[0] = 0$ .  $g_{2m} = g_m * (1 + a - \log$ 
  ↪  $g_m)$ 
Poly exp_(const Poly& a, int k) {
    Poly g = {1};
    for (int m = 1; m < k; m <= 1) {
        Poly t = cut(a, 2 * m) - log_(cut(g, 2 * m), 2 * m);
        t[0] = (t[0] + 1) % NMOD;
        g = cut(multiply(g, t), 2 * m);
    }
    return cut(g, k);
}
//  $\sqrt{a} \bmod x^k$ , for  $a[0] = 1$  (general  $a[0]$  needs a modular
  ↪ square root of  $a[0]$ ).
Poly sqrt_(const Poly& a, int k) {
    Poly g = {1};
    ll i2 = npow(2, NMOD - 2);
    for (int m = 1; m < k; m <= 1) {
        Poly t = cut(a, 2 * m) + multiply(g, g);
        t = cut(t, 2 * m);
        g = cut(multiply(t, inv(cut(g, 2 * m) * 2, 2 * m)), 2 * m);
    }
    return cut(g, k);
}
//  $a^e \bmod x^k$ , for  $a[0] = 1$ :  $\exp(e * \log a)$ . For  $a[0] \neq 1$  factor
  ↪ out  $c * x^d$  first.
Poly pow_(const Poly& a, ll e, int k) { return exp_(log_(a, k) * (e %
  ↪ NMOD), k); }

// Polynomial DIVISION with remainder:  $a = q * b + r$ ,  $\deg r < \deg b$ .
  ↪  $O(n \log n)$ 
pair<Poly, Poly> divmod(Poly a, Poly b) {
    while (a.size() > 1 && !a.back()) a.pop_back(); // trim BOTH,
  ↪ or the reversal is wrong
    while (b.size() > 1 && !b.back()) b.pop_back();
    int n = a.size(), m = b.size();
    if (n < m) return {{0}, a};
```

```

    Poly ra(a.rbegin(), a.rend()), rb(b.rbegin(), b.rend());
    Poly q = cut(multiply(cut(ra, n - m + 1), inv(cut(rb, n - m + 1),
    ↪ n - m + 1)), n - m + 1);
    reverse(all(q));
    Poly r = cut(a - multiply(q, b), max(1, m - 1)); // cut AFTER
    ↪ subtracting, not before
    return {q, r};
}
// USES: counting with generating functions (exp of a "connected"
    ↪ series = all structures),
// partition numbers via prod 1/(1-x^i) = exp(sum ...), Catalan-style
    ↪ algebraic equations via
// sqrt, k-th power of a GF, linear recurrence coefficients via inv,
    ↪ and Lagrange inversion.

```

6.5.5 CyclicConvolution

CYCLIC CONVOLUTION and CYCLIC CORRELATION of two length- n arrays, $O(n \log n)$.

```

// Needs: multiply (02 - NTT.cpp) [or swap in conv() from 01 -
    ↪ FFT.cpp].
// Cyclic = multiplication in the ring modulo  $x^n - 1$ : indices wrap
    ↪ around instead of growing.
// REACH FOR IT whenever the index set is a CIRCLE rather than a
    ↪ line: necklaces and rotations,
// "for every rotation  $r$ , the score of aligning  $a$  against  $b$  rotated
    ↪ by  $r$ ", counting pairs  $(i, j)$ 
// with  $(i + j) \bmod n = k$ , and any DP on residues modulo  $n$ .
typedef vector<ll> Poly;
Poly cyclicConv(const Poly& a, const Poly& b, int n) { // c[k] =
    ↪ sum_{(i+j) mod n == k} a[i]b[j]
    Poly p = multiply(a, b), c(n, 0);
    for (size_t i = 0; i < p.size(); i++) c[i % n] = (c[i % n] +
    ↪ p[i]) % NMOD;
    return c;
}
// c[k] = sum_i a[i] * b[(i + k) mod n] -- "slide b by k over a and
    ↪ take the dot product".
// This is the one you actually want for rotations. Reverse a
    ↪ cyclically, then convolve.
Poly cyclicCorr(const Poly& a, const Poly& b, int n) {
    Poly ar(n);
    for (int i = 0; i < n; i++) ar[i] = a[(n - i) % n];
    return cyclicConv(ar, b, n);
}
// NEGACYCLIC convolution (modulo  $x^n + 1$ , "wrap with a sign flip"):
    ↪ same fold but subtract the
// wrapped part,  $c[i \% n] += (i / n \text{ odd} ? -p[i] : p[i])$ . Shows up in
    ↪ ring-LWE style problems and in
// the Schoenhage-Strassen recursion.
// FASTER when  $n \mid (NMOD - 1)$ : a length- $n$  NTT IS the cyclic
    ↪ convolution - transform, multiply
// pointwise, transform back, with no padding to a power of two and
    ↪ no fold. When  $n$  is arbitrary,
// Bluestein (06 - ChirpZ.cpp) gives you the length- $n$  DFT anyway.
// USES
// ROTATION SCORING: given two binary necklaces, the number of
    ↪ positions where both are 1 under
// rotation  $r$  is exactly cyclicCorr(a, b)[ $r$ ] - one call answers all
    ↪  $n$  rotations.
// RESIDUE DP: "number of pairs whose sum is congruent to  $k \bmod n$ "
    ↪ for all  $k$  at once.
// CIRCULAR STRING MATCHING with wildcards: encode as sums of
    ↪ products (see the wildcard trick in
// 07 - Strings/01 - String Matching/04) and fold cyclically.
// A circulant matrix times a vector IS a cyclic convolution, so
    ↪ this is also how you apply a
// circulant in  $O(n \log n)$  instead of  $O(n^2)$ .

```

6.5.6 ChirpZ

CHIRP-Z / BLUESTEIN: evaluate a polynomial at a GEOMETRIC progression of points,

```

// Needs: multiply, npow (02 - NTT.cpp); Poly / cut (04 -
    ↪ Polynomial.cpp).
// P(1), P(z), P(z^2), ..., P(z^(m-1)) in  $O((n + m) \log(n + m))$ ,
// where  $n = \deg P$ . Multipoint evaluation at arbitrary points is  $O(n$ 
    ↪  $\log^2 n$ ); this is a full log
// factor cheaper and much shorter, and it covers most of the cases
    ↪ that actually come up.
// PRECONDITION:  $z$  must be INVERTIBLE mod  $NMOD$  ( $z \neq 0$ ). Any  $m$  and  $n$ 
    ↪ are fine - no power of two.
// REACH FOR IT for: a DFT of ARBITRARY length (set  $z =$  a primitive
    ↪  $m$ -th root of unity - this is
// how you do a length-7 or length-1000000 transform with an NTT that
    ↪ only likes powers of two);
// "evaluate  $f$  at 1,  $r$ ,  $r^2$ , ..." which is what  $q$ -analogue and
    ↪ geometric-weight problems ask for;
// and cyclic convolution of a non-power-of-two length.
// IDENTITY:  $i * j = C(i + j, 2) - C(i, 2) - C(j, 2)$ , which turns the
    ↪ exponent  $z^{i * j}$  into a product
// of three one-index factors, so the sum becomes a convolution
    ↪ ("chirp" = the  $z^C(k, 2)$  sequence).
Poly chirpZ(const Poly& a, ll z, int m) {
    int n = a.size() - 1; // deg P
    if (m <= 0) return {};
    ll iz = npow(z, NMOD - 2);
    auto tri = [&](ll k) { return k * (k - 1) / 2 % (NMOD - 1); };
    ↪ // C(k, 2) mod (p-1)
    Poly f(n + 1), g(n + m);
    for (int j = 0; j <= n; j++) f[n - j] = a[j] * npow(iz, tri(j)) %
    ↪ NMOD; // reversed
    for (int t = 0; t < n + m; t++) g[t] = npow(z, tri(t));
    Poly h = multiply(f, g);
    Poly r(m);
    for (int k = 0; k < m; k++) r[k] = h[n + k] * npow(iz, tri(k)) %
    ↪ NMOD;
    return r;
}
// ARBITRARY-LENGTH DFT: with  $w$  a primitive  $m$ -th root of unity mod
    ↪  $NMOD$  (needs  $m \mid NMOD - 1$ ),
// chirpZ(a, w, m) is the length- $m$  DFT of  $a$ ; the inverse is chirpZ(.,
    ↪  $w^{-1}$ , m) scaled by  $1/m$ .
// That gives cyclic convolution of ANY length  $m$  in  $O(m \log m)$ 
    ↪ without a fold.
// THE OTHER DIRECTION: given the  $m$  values  $P(z^k)$ , recovering  $P$  is a
    ↪ chirp-z with  $z^{-1}$  followed by
// a rescale - the transform is its own inverse up to the chirp
    ↪ factors, exactly like a DFT.
// EVALUATING AT AN ARITHMETIC progression instead ( $P(a)$ ,  $P(a+d)$ ,
    ↪ ...): that is a Taylor SHIFT plus
// a scale, see 10 - PolyShiftAndComposition.cpp, also  $O(n \log n)$ .
// OVERFLOW: tri(k) is taken mod  $(NMOD - 1)$  by Fermat, so  $k$  may be as
    ↪ large as you like; without
// that reduction  $k * (k - 1) / 2$  overflows around  $k = 4e9$ .
// RELATED: the same "split the exponent" trick underlies baby-step
    ↪ giant-step for discrete log and
// the  $O(\sqrt{n})$  evaluation of  $n! \bmod p$ .

```

6.5.7 RelaxedConvolution

ONLINE (RELAXED) CONVOLUTION. You feed $a[i]$ and $b[i]$ one index at a time and immediately get

```
// Needs: multiply (02 - NTT.cpp).
// c[i] = sum_{j+k=i} a[j] b[k], BEFORE you have to supply a[i+1] -
//   ↳ which is exactly what a
// self-referential DP needs. Total  $O(n \log^2 n)$  for all  $n$  steps;  $\sim 1s$ 
//   ↳ at  $n = 2e5$ .
// REACH FOR IT whenever f[i] depends on a convolution of f with
//   ↳ itself (or with something f
// determines) up to index  $i - 1$ :  $f[n] = \sum f[k] f[n-1-k]$ 
//   ↳ (Catalan-shaped DPs),  $f[n] = \sum$ 
//  $g[k] f[n-k]$  where you only learn  $g[n]$  after computing  $f[n-1]$ ,
//   ↳ probability DPs with a feedback
// loop, ABC315-H style. Divide and conquer over the recursion is the
//   ↳ alternative and is usually
// harder to write; a plain  $O(n^2)$  double loop is the thing you are
//   ↳ replacing.
// The blocks are scheduled so that the product of two length- $w$ 
//   ↳ blocks is done exactly when both
// halves are known, and every coefficient is covered exactly once.
struct Relaxed {
    vector<ll> a, b, c;
    int k = 0;
    Relaxed(int n) : a(n), b(n), c(2 * n, 0) {}
    // supply a[i] = x and b[i] = y (i must equal the current index);
    // returns c[i].
    ll extend(ll x, ll y) {
        a[k] = x % NMOD, b[k] = y % NMOD;
        int s = k + 2;
        for (int w = 1; s % w == 0 && w < s; w <= 1) {
            for (int t = 0; t < 2; t++) {
                if (t == 0 || w * 2 != s) { // the
                    vector<ll> f(a.begin() + w - 1, a.begin() + 2 * w
                    ↳ symmetric block is done once only
                    ↳ - 1);
                    vector<ll> g(b.begin() + k - w + 1, b.begin() + k
                    ↳ + 1);
                    f = multiply(f, g);
                    for (size_t i = 0; i < f.size() && k + i <
                    ↳ c.size(); i++)
                        c[k + i] = (c[k + i] + f[i]) % NMOD;
                    }
                    swap(a, b); // and
                    ↳ again, so a/b are restored
                }
            }
            return c[k++];
        }
    };
};
// SELF-CONVOLUTION shorthand: pass the same value twice,
//   ↳ extend(f[i], f[i]).
// TYPICAL SHAPE
//   Relaxed R(n); f[0] = 1;
//   for (i = 1; i <= n; i++) { ll t = R.extend(f[i-1], f[i-1]); f[i]
//   ↳ = /* uses t */; }
// Note the OFF-BY-ONE: after the  $i$ -th extend you know coefficient
//   ↳  $i-1$  of the product, so the
// value you feed in must be the one that is already final.
// WHEN NOT T0: if the recurrence is  $f[n] = \sum_{k < n} g[k] f[n-k]$ 
//   ↳ with  $g$  KNOWN in advance, the
// cheaper standard answer is divide and conquer over  $[l, r)$ 
//   ↳ (contribute  $[l, mid)$  to  $[mid, r)$  with
// one multiply, same  $O(n \log^2 n)$  but a smaller constant and no
//   ↳ bookkeeping.
// If instead the whole system is polynomial ( $f = F(f)$  as power
//   ↳ series), Newton iteration on the
// series (04 - Polynomial.cpp: inv / log / exp / sqrt) is  $O(n \log n)$ 
//   ↳ and beats both.
```

6.5.8 NTT2D

2-D CONVOLUTION of two bivariate polynomials mod $NMOD = 998244353$:

```
// Needs: ntt, npow (02 - NTT.cpp).
// c[i][j] = sum_{p+r=i, q+s=j} a[p][q] * b[r][s], in
//   ↳  $O(R * C * \log(R * C))$  with  $R, C$  the padded
// dimensions. The transform SEPARATES: NTT every row, then every
//   ↳ column, multiply pointwise,
// invert columns then rows. Same idea works for any number of
//   ↳ dimensions.
// REACH FOR IT when the state has TWO additive coordinates: "count
//   ↳ paths with a given (x, y)
// displacement", "polynomials in two variables" (a variable per
//   ↳ dimension; you must track jointly),
// 2-D pattern matching by sums of products, and grid DPs whose
//   ↳ transition is a translation.
typedef vector<vector<ll>> Poly2;
void ntt2d(Poly2& a, bool inv) {
    int R = a.size(), C = a[0].size();
    for (int i = 0; i < R; i++) ntt(a[i], inv);
    vector<ll> col(R);
    for (int j = 0; j < C; j++) {
        for (int i = 0; i < R; i++) col[i] = a[i][j];
        ntt(col, inv);
        for (int i = 0; i < R; i++) a[i][j] = col[i];
    }
}
Poly2 multiply2D(const Poly2& a, const Poly2& b) {
    int rr = a.size() + b.size() - 1, cc = a[0].size() + b[0].size()
    ↳ - 1, R = 1, C = 1;
    while (R < rr) R <= 1;
    while (C < cc) C <= 1;
    Poly2 x(R, vector<ll>(C, 0)), y(R, vector<ll>(C, 0));
    for (size_t i = 0; i < a.size(); i++)
        for (size_t j = 0; j < a[0].size(); j++) x[i][j] = a[i][j];
    for (size_t i = 0; i < b.size(); i++)
        for (size_t j = 0; j < b[0].size(); j++) y[i][j] = b[i][j];
    ntt2d(x, false), ntt2d(y, false);
    for (int i = 0; i < R; i++)
        for (int j = 0; j < C; j++) x[i][j] = x[i][j] * y[i][j] %
    ↳ NMOD;
    ntt2d(x, true);
    x.resize(rr);
    for (auto& row : x) row.resize(cc);
    return x;
}
// CHEAPER ALTERNATIVE - FLATTENING. If one dimension is small ( $C0$ 
//   ↳ columns in a and  $C1$  in b), pack
// the 2-D array into a 1-D one with stride  $W \geq C0 + C1 - 1$  (index
//   ↳  $i * W + j$ ) and do a SINGLE 1-D
// convolution: the strides are wide enough that the coordinates
//   ↳ never interfere. One NTT of length
//  $R * W$  instead of  $R + C$  transforms - usually faster, and it is three
//   ↳ lines. Use the 2-D routine
// when both dimensions are large and comparable.
// The same separability argument gives 2-D FWHT (xor in both
//   ↳ coordinates) and 2-D subset-sum
// transforms; nothing about it is special to the NTT.
// MEMORY:  $R * C$  longs.  $4096 \times 4096$  is already 128 MB - flatten instead
//   ↳ when the product is large.
```


6.5.9 FWHTAnyBase

FWHT IN AN ARBITRARY BASE k : convolution where indices combine by DIGIT-WISE ADDITION MOD k in

```
// Needs: npow (02 - NTT.cpp).
// base k. For k = 2 this is exactly xor and the routine collapses to
//   ↳ the usual FWHT; for k = 10 it
// is "add the decimal digits without carrying"; for k = 3 it is the
//   ↳ ternary "no-carry" addition
// that Nim-like and ternary-state problems produce.
// Length must be  $n = k^d$ . Cost  $O(n * k * \log_k n)$  - the transform
//   ↳ along each digit is a length- $k$ 
// DFT, done with the  $k \times k$  matrix  $M[p][q] = w^{pq}$ .
// PRECONDITION: a primitive  $k$ -th root of unity must exist mod  $NMOD$ ,
//   ↳ i.e.  $k \mid NMOD - 1$ .
// 998244352 =  $2^{23} * 7 * 17$ , so  $k$  in  $\{2, 4, 8, \dots, 2^{23}, 7, 14, 17, 34,$ 
//   ↳ 119, ... $\}$  all work;  $k = 3$ 
// does NOT - see the note at the bottom. Use a different NTT prime
//   ↳ if you need another  $k$ .
void fwhtBase(vector<ll>& a, int k, bool inv) {
    int n = a.size();
    ll w = npow(NROOT, (NMOD - 1) / k); // primitive
    ↳  $k$ -th root of unity
    if (inv) w = npow(w, NMOD - 2);
    vector<ll> wp(k);
    wp[0] = 1;
    for (int i = 1; i < k; i++) wp[i] = wp[i - 1] * w % NMOD;
    vector<ll> v(k);
    for (int len = 1; len < n; len *= k)
        for (int i = 0; i < n; i += len * k)
            for (int j = 0; j < len; j++) {
                for (int p = 0; p < k; p++) v[p] = a[i + j + p * len];
                for (int p = 0; p < k; p++) {
                    ll s = 0;
                    for (int q = 0; q < k; q++) s = (s + v[q] * wp[p
↳ * q % k]) % NMOD;
                    a[i + j + p * len] = s;
                }
            }
    if (inv) { ll ni = npow(n, NMOD - 2); for (ll& x : a) x = x * ni
↳ % NMOD; }
}
vector<ll> multiplyBase(vector<ll> a, vector<ll> b, int k) {
    int n = 1;
    while (n < (int)max(a.size(), b.size())) n *= k; // n must be
    ↳ a power of k
    a.resize(n, 0); b.resize(n, 0);
    fwhtBase(a, k, false); fwhtBase(b, k, false);
    for (int i = 0; i < n; i++) a[i] = a[i] * b[i] % NMOD;
    fwhtBase(a, k, true);
    return a;
}
// k-TH POWER OF THE OPERATOR: to convolve one array with itself e
//   ↳ times, transform once, raise
// every entry to the  $e$ -th power, transform back -  $O(n \log n + n \log$ 
//   ↳  $e)$ , not  $e$  convolutions.
// TERNARY when 3 does not divide  $MOD - 1$ : work in  $F_p[w] / (w^2 + w$ 
//   ↳  $+ 1)$ , i.e. represent each
// value as  $a + b*w$  with  $w^2 = -1 - w$ . That is a genuine cube root of
//   ↳ unity as long as  $3 \mid p + 1$ .
// Multiplication is  $(a_1 + b_1 w)(a_2 + b_2 w) = (a_1 a_2 - b_1 b_2) + (a_1 b_2 +$ 
//   ↳  $a_2 b_1 - b_1 b_2) w$ . Everything else
// in the routine above is unchanged.
// PACKING TRICK: if your indices are BITMASKS but the operation is
//   ↳ digit-wise addition mod 3
// (e.g. "each element used 0, 1 or 2 times"), rewrite each mask in
//   ↳ base 3 first: the value with
// binary digits  $d_i$  becomes  $\sum d_i 3^i$ .
// WHY NOT SUBSET-SUM/OR: OR-convolution is the zeta/Mobius transform
//   ↳ ( $08 - DP/02 - SOS_DP.cpp$ );
// this file is only for the group  $\mathbb{Z}_k^d$ , where the transform is
//   ↳ invertible pointwise.
```

6.5.10 PolyShiftAndComposition

TAYLOR SHIFT $p(x) \rightarrow p(x + c)$, and COMPOSITION $p(g(x))$. Both mod $NMOD = 998244353$.

```
// Needs: multiply / npow (02 - NTT.cpp), Poly / cut (04 -
//   ↳ Polynomial.cpp).
// TAYLOR SHIFT in  $O(n \log n)$ .  $[x^k] p(x+c) = \sum_{j \geq k} a_j * C(j,$ 
//   ↳  $k) * c^{(j-k)}$ , which is a
// correlation once you divide by the factorials, hence ONE
//   ↳ multiplication.
// REACH FOR IT to re-centre an interpolation ("I have  $f$  sampled at
//   ↳  $0..n$ , I want it at  $m..m+n$ "),
// to turn "evaluate at an ARITHMETIC progression  $a, a+d, a+2d, \dots$ "
//   ↳ into a shift plus a scale, and
// inside falling-factorial / Stirling conversions.
Poly taylorShift(Poly a, ll c) {
    int n = a.size();
    vector<ll> f(n), fi(n);
    f[0] = 1;
    for (int i = 1; i < n; i++) f[i] = f[i - 1] * i % NMOD;
    fi[n - 1] = npow(f[n - 1], NMOD - 2);
    for (int i = n - 1; i > 0; i--) fi[i - 1] = fi[i] * i % NMOD;
    Poly A(n), B(n);
    for (int i = 0; i < n; i++) A[n - 1 - i] = a[i] * fi[i] % NMOD;
    ↳ // reversed  $a_j * j!$ 
    ll p = 1;
    for (int i = 0; i < n; i++) B[i] = p * fi[i] % NMOD, p = p * (c %
↳ NMOD + NMOD) % NMOD;
    Poly h = multiply(A, B);
    Poly r(n);
    for (int k = 0; k < n; k++) r[k] = h[n - 1 - k] * fi[k] % NMOD;
    return r;
}
// COMPOSITION  $p(g(x)) \bmod x^m$  in  $O(\sqrt{n} * m \log m)$  by the BLOCK
//   ↳ (baby-step giant-step) method:
// write  $p(y) = \sum_{i < K} y^{(i * D)} * q_i(y)$  with  $\deg q_i < D$ ,
//   ↳ precompute  $g^{0..g^D}$  once, build each
//  $q_i(g)$  by linear combinations, then Horner over the  $K$  blocks with
//   ↳ the single power  $g^D$ .
// REACH FOR IT for "substitute one generating function into
//   ↳ another": exp/log of a series with a
// non-constant argument, EGF of a labelled construction, "apply a
//   ↳ transform given as a series".
Poly composition(const Poly& p, const Poly& g, int m) {
    int n = p.size(), K = 1;
    while (K * K < n) K++; // K blocks
    ↳ of  $D = \text{ceil}(n / K)$  terms
    int D = (n + K - 1) / K;
    vector<Poly> pw(D + 1, Poly{1});
    for (int i = 1; i <= D; i++) pw[i] = cut(multiply(pw[i - 1], g),
↳ m);
    Poly res(m, 0), gd{1};
    for (int i = 0; i < K; i++) {
        Poly blk(m, 0);
        for (int j = 0; j < D && i * D + j < n; j++) {
            ll c = p[i * D + j];
            if (!c) continue;
            for (int t = 0; t < (int)pw[j].size() && t < m; t++)
                blk[t] = (blk[t] + c * pw[j][t]) % NMOD;
        }
        blk = cut(multiply(blk, gd), m);
        for (int t = 0; t < m && t < (int)blk.size(); t++) res[t] =
↳ (res[t] + blk[t]) % NMOD;
        gd = cut(multiply(gd, pw[D]), m);
    }
    return res;
}
// COMPOSITIONAL INVERSE  $f^{-1}$  (the series  $h$  with  $f(h(x)) = x$ ), for
//   ↳  $f[0] = 0, f[1] \neq 0$ :
// LAGRANGE INVERSION gives a single coefficient directly,
//  $[x^n] h = (1/n) * [x^{(n-1)}] (x / f(x))^n$ , and more generally
//  $[x^n] G(h(x)) = (1/n) * [x^{(n-1)}] G'(x) * (x / f(x))^n$ .
// One coefficient costs  $O(n \log n)$ ; all  $n$  of them by this route
//   ↳ costs  $O(n^2 \log n)$ , so use it when
// you need ONE answer (which is the usual case: Catalan,
//   ↳ Fuss-Catalan, tree counts).
// K-TH ROOT of a series with  $a[0] = 1$ :  $\text{pow}_k(a, \text{inv}(k) \bmod (NMOD -$ 
//   ↳  $1))$  - i.e.  $\exp(\log(a)/k)$ . For
//  $a[0] \neq 1$  factor out  $c * x^d$  first; a  $k$ -th root exists only if  $k \mid$ 
//   ↳  $d$  and  $c$  is a  $k$ -th power
// residue, and then it is unique up to a  $k$ -th root of unity.
// COST WARNING: composition is the expensive one. If  $g$  is a MONOMIAL
//   ↳  $c * x^t$ , substitute by hand; if
//  $p$  is short, plain Horner with truncated products is  $O(n * m \log m)$ 
//   ↳ and simpler.
```


6.5.11 PolyGcdResultant

POLYNOMIAL GCD and RESULTANT over F_p ($p = \text{NMOD} = 998244353$, prime).

```
// Needs: multiply / npow (02 - NTT.cpp), Poly / divmod / cut (04 -
  ↳ Polynomial.cpp).
// deg gcd(a, b) = the number of COMMON ROOTS of a and b counted with
  ↳ multiplicity (over the
// algebraic closure), so gcd answers "do these two polynomials share
  ↳ a root" and "how many".
// Euclid with the fast divmod is  $O(n \log n)$  per step and  $O(n)$  steps;
  ↳ the half-gcd algorithm brings
// it to  $O(n \log^2 n)$  but is long - this is the version you write in
  ↳ a contest.
Poly trim(Poly a) { while (a.size() > 1 && !a.back()) a.pop_back();
  ↳ return a; }

Poly polyGcd(Poly a, Poly b) { // monic gcd;
  ↳ returns {1} for coprime inputs
  a = trim(a), b = trim(b);
  while (b.size() > 1 || b[0]) { Poly r = trim(divmod(a,
    ↳ b).second); a = b, b = r; }
  if (a.size() == 1 && !a[0]) return a; // gcd(0, 0)
  ll ic = npow(a.back(), NMOD - 2); // normalise
  ↳ to monic
  for (ll& x : a) x = x * ic % NMOD;
  return a;
}

// RESULTANT Res(a, b) =  $lc(b)^{\deg(a)} \cdot \prod \text{over roots } \beta \text{ of } b \text{ of } a(\beta)$ 
  ↳  $a(\beta) = (-1)^{(\deg a \cdot \deg b)} \cdot lc(a)^{\deg(b)} \cdot \prod \text{over}$ 
  ↳ roots  $\alpha$  of  $a$  of  $b(\alpha)$ 
// It is 0 EXACTLY when  $a$  and  $b$  share a root, and it is the
  ↳ determinant of the Sylvester matrix -
// so it is the polynomial-time way to eliminate a variable from two
  ↳ equations.
// Euclidean recursion: with  $r = a \bmod b$ ,
//  $\text{Res}(a, b) = lc(b)^{(\deg a - \deg r)} \cdot (-1)^{(\deg b \cdot \deg r)} \cdot$ 
  ↳  $\text{Res}(b, r)$ .
//  $O(n \log n)$  per step,  $O(n)$  steps.
ll resultant(Poly a, Poly b) {
  a = trim(a), b = trim(b);
  if ((a.size() == 1 && !a[0]) || (b.size() == 1 && !b[0])) return
    ↳ 0;
  ll res = 1;
  while (b.size() > 1) {
    int da = a.size() - 1, db = b.size() - 1;
    Poly r = trim(divmod(a, b).second);
    int dr = r.size() - 1;
    if (r.size() == 1 && !r[0]) return 0; // common
    ↳ root
    res = res * npow(b.back(), da - dr) % NMOD;
    if ((ll)db * dr % 2) res = (NMOD - res) % NMOD;
    a = b, b = r;
  }
  return res * npow(b[0], a.size() - 1) % NMOD; // Res(a,
  ↳ const c) =  $c^{\deg(a)}$ 
}

// USES
// DISCRIMINANT of  $f$ :  $\text{Res}(f, f') / lc(f)$  - zero iff  $f$  has a repeated
  ↳ root. gcd( $f, f'$ ) isolates the
// squarefree part, which is the first step of any polynomial
  ↳ factorisation.
// ELIMINATION: given  $P(x, y) = 0$  and  $Q(x, y) = 0$ ,  $\text{Res}_y(P, Q)$  is a
  ↳ polynomial in  $x$  alone whose
// roots are exactly the  $x$ -coordinates of the common solutions.
//  $\det(A + x \cdot B)$  style problems and "determinant of a circulant" (03
  ↳ - Matrices/04) are resultants.
// COMMON ROOT COUNTING mod  $p$  for two polynomials given as products,
  ↳ without factoring either.
// PRECONDITION on the Euclidean chain:  $p$  must be PRIME (we invert
  ↳ leading coefficients). Over  $a$ 
// composite modulus use the Sylvester determinant with  $\detMod$  (07 -
  ↳ Numerical/02) instead.
// DEGREE GOTCHA: divmod in 04 - Polynomial.cpp keeps a padded
  ↳ remainder, so trim() before reading
// any degree - the sign in the resultant recursion depends on the
  ↳ TRUE degree.
```

6.5.12 CyclotomicPolynomial

CYCLOTOMIC POLYNOMIALS $\Phi_n(x) = \prod \text{over primitive } n\text{-th roots of unity of } (x - w)$.

```
// They are the IRREDUCIBLE factors of  $x^n - 1$  over the integers:
  ↳  $x^n - 1 = \prod_{d|n} \Phi_d(x)$ ,
// deg  $\Phi_n = \phi(n)$ , and every  $\Phi_n$  has integer coefficients (tiny
  ↳ ones - the first  $n$  with  $a$ 
// coefficient outside  $\{-1, 0, 1\}$  is  $n = 105$ ).
// By Mobius inversion  $\Phi_n(x) = \prod_{d|n} (x^d - 1)^{\mu(n/d)}$ ,
  ↳ and both multiplying and
// dividing by  $(x^d - 1)$  are  $O(\deg)$  in place, so one  $\Phi_n$  costs
  ↳  $O(\sigma(n))$  - linear, no FFT.
// REACH FOR IT when a problem factors  $x^n - 1$  (roots-of-unity
  ↳ filters, "count sequences whose sum
// is  $0 \bmod n$ ", cyclic codes), when you need the minimal polynomial
  ↳ of a root of unity, or for
// determinants of circulants factored per divisor.
void mulXd(vector<ll>& a, int d) { // a *=  $(x^d - 1)$ 
  int n = a.size();
  a.insert(a.begin(), d, 0);
  for (int i = 0; i < n; i++) a[i] -= a[i + d];
}

void divXd(vector<ll>& a, int d) { // a /=  $(x^d - 1)$ ,
  ↳ exact division assumed
  int n = a.size();
  for (int i = n - 1 - d; i >= 0; i--) a[i] += a[i + d];
  a.erase(a.begin(), a.begin() + d);
}

int muOne(int n) { // Mobius of a
  ↳ single n, by trial division
  int r = 1;
  for (int p = 2; (ll)p * p <= n; p++) if (n % p == 0) {
    n /= p;
    if (n % p == 0) return 0;
    r = -r;
  }
  return n > 1 ? -r : r;
}

vector<ll> cyclotomic(int n) { // coefficients, low
  ↳ degree first; size  $\phi(n)+1$ 
  vector<ll> a{1};
  for (int d = 1; d <= n; d++) if (n % d == 0 && muOne(n / d) == 1)
    ↳ mulXd(a, d);
  for (int d = 1; d <= n; d++) if (n % d == 0 && muOne(n / d) ==
    ↳ -1) divXd(a, d);
  return a; // do the
  ↳ multiplications FIRST, then the divisions
}

// FACTS
//  $\Phi_p(x) = 1 + x + \dots + x^{p-1}$  for prime  $p$ ;  $\Phi_{2n}(x) =$ 
  ↳  $\Phi_n(-x)$  for odd  $n > 1$ ;
//  $\Phi_{\{n p\}}(x) = \Phi_n(x^p)$  if  $p \mid n$ , and  $\Phi_n(x^p) / \Phi_n(x)$ 
  ↳ otherwise.
//  $\Phi_n(1) = p$  if  $n$  is a prime power  $p^k$ , and 1 for any other  $n > 1$ .
// ORDER OF 2 MOD  $n$  and friends: the irreducible factors of  $x^n - 1$ 
  ↳ over  $F_q$  are the factors of
// the  $\Phi_d$ , each splitting into  $\phi(d) / \text{ord}_d(q)$  factors of
  ↳ degree  $\text{ord}_d(q)$ .
//  $x^n - 1$  over  $F_p$  ( $p$  not dividing  $n$ ) is squarefree; if  $p \mid n$ ,  $x^n$ 
  ↳  $- 1 = (x^{(n/p)} - 1)^p$ .
// ROOTS-OF-UNITY FILTER (the usual reason you are here):
//  $\sum_{k=r \bmod m} [x^k] F(x) = (1/m) \cdot \sum_{j=0}^{m-1} w^{(-jr)} \cdot$ 
  ↳  $F(w^j)$ ,  $w$  a primitive  $m$ -th
// root of unity. Mod  $p$  you need  $m \mid p - 1$  for  $w$  to exist;
  ↳ otherwise work in  $F_p[x] / \Phi_m(x)$ ,
// where  $x$  IS a primitive  $m$ -th root by construction - which is
  ↳ exactly what this file gives you.
// OVERFLOW: coefficients stay small (bounded by a few hundred for  $n$ 
  ↳  $< 1e5$ ) so ll is safe; if you
// want them mod  $p$ , reduce afterwards.
```

6.5.13 ConvAnyMod

NTT, because 02 - NTT.cpp hardcodes NMOD.

```
// Needs: npow (02 - NTT.cpp) -- self-contained otherwise: it
// ↳ carries its own modulus-parametric
// CONVOLUTION MODULO AN ARBITRARY M (composite, not NTT-friendly):
// ↳ run the NTT under THREE
// NTT-friendly primes and recombine with Garner. O(n log n) with a
// ↳ constant of three transforms.
// The true coefficients are bounded by len * (M-1)^2; the three
// ↳ primes below multiply to ~9.6e25,
// so for M <= 1e9 and len <= 1e7 the CRT reconstruction is EXACT,
// ↳ and only then reduced mod M.
// PREFER THIS over convMod in 01 - FFT.cpp when M is large (> ~1e9),
// ↳ when the arrays are long
// (>= 1e6, where the double-based split loses precision), or when
// ↳ you do not want to reason about
// floating point at all. It is exact by construction; the FFT
// ↳ version is not.
const ll CP[3] = {167772161, 469762049, 1224736769}; // all
// ↳ c*2^k + 1, primitive root 3
void nttP(vector<ll>& a, bool inv, ll p) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int len = 2; len <= n; len <= 1) {
        ll w = npow(3, (p - 1) / len, p);
        if (inv) w = npow(w, p - 2, p);
        for (int i = 0; i < n; i += len) {
            ll cur = 1;
            for (int j = 0; j < len / 2; j++) {
                ll u = a[i + j], v = a[i + j + len / 2] * cur % p;
                a[i + j] = (u + v) % p, a[i + j + len / 2] = (u - v +
                // ↳ p) % p;
                cur = cur * w % p;
            }
        }
        if (inv) { ll ni = npow(n, p - 2, p); for (ll& x : a) x = x * ni
        // ↳ % p; } }
    }
}
vector<ll> convP(const vector<ll>& a, const vector<ll>& b, ll p) {
    int rs = a.size() + b.size() - 1, n = 1;
    while (n < rs) n <= 1;
    vector<ll> x(n, 0), y(n, 0);
    for (size_t i = 0; i < a.size(); i++) x[i] = (a[i] % p + p) % p;
    for (size_t i = 0; i < b.size(); i++) y[i] = (b[i] % p + p) % p;
    nttP(x, false, p), nttP(y, false, p);
    for (int i = 0; i < n; i++) x[i] = x[i] * y[i] % p;
    nttP(x, true, p);
    x.resize(rs);
    return x;
}
vector<ll> convAnyMod(const vector<ll>& a, const vector<ll>& b, ll M)
// ↳ {
// if (a.empty() || b.empty()) return {};
// vector<ll> r0 = convP(a, b, CP[0]), r1 = convP(a, b, CP[1]), r2 =
// ↳ convP(a, b, CP[2]);
// ll i01 = npow(CP[0] % CP[1], CP[1] - 2, CP[1]);
// ll i02 = npow(CP[0] % CP[2], CP[2] - 2, CP[2]);
// ll i12 = npow(CP[1] % CP[2], CP[2] - 2, CP[2]);
// vector<ll> res(r0.size());
// for (size_t i = 0; i < r0.size(); i++) { //
// ↳ Garner: mixed-radix digits x0,x1,x2
// ll x0 = r0[i];
// ll x1 = (r1[i] - x0 % CP[1] + CP[1]) % CP[1] * i01 % CP[1];
// ll t = (r2[i] - x0 % CP[2] + CP[2]) % CP[2] * i02 % CP[2];
// ll x2 = (t - x1 % CP[2] + CP[2]) % CP[2] * i12 % CP[2];
// __int128 v = (__int128)x2 * CP[1] % M * CP[0] % M +
// ↳ (__int128)x1 * CP[0] % M + x0;
// res[i] = (ll)(v % M);
// }
// return res;
// }
// PRIME MENU (all with primitive root 3; the 2-power in p-1 caps the
// ↳ transform length):
// 167772161 = 5*2^25 + 1 469762049 = 7*2^26 + 1 754974721 =
// ↳ 45*2^24 + 1
// 998244353 = 119*2^23 + 1 1004535809 = 479*2^21 + 1 1224736769
// ↳ = 73*2^24 + 1
// TWO PRIMES are enough only when len * (M-1)^2 < P0 * P1 (~7.9e16)
// ↳ - i.e. tiny M. Use three.
// SPLIT-COEFFICIENT ALTERNATIVE: convMod in 01 - FFT.cpp writes each
// ↳ coefficient as hi*2^15 + lo
```

```
// and does 3 complex FFTs - shorter, but its error grows with len
// ↳ and with M.
// EXACT INTEGER convolution with no modulus: pass an M larger than
// ↳ any possible coefficient, or
// drop the final reduction and reconstruct into __int128.
```

6.6 Interpolation

6.6.1 LagrangeAndFaulhaber

Lagrange interpolation + power sums. Needs Mint + buildFact().

// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp).

```
// Given y[0..n] = f(0), f(1), ..., f(n) for a polynomial f of degree
// ↳ <= n, evaluate f(x). O(n).
// The consecutive-points case is the one you actually want; it kills
// ↳ a whole class of problems
// whose answer is "a polynomial in n of degree d" (guess d, sample
// ↳ d+1 values, interpolate).
Mint lagrangeConsecutive(const vector<Mint>& y, ll x) {
    int n = y.size() - 1;
    if (0 <= x && x <= n) return y[x]; // x < 0 would index
    // ↳ y[negative]
    vector<Mint> pre(n + 2, 1), suf(n + 2, 1);
    for (int i = 0; i <= n; i++) pre[i + 1] = pre[i] * Mint(x - i);
    for (int i = n; i >= 0; i--) suf[i + 1] = suf[i] * Mint(x - i);
    Mint r = 0;
    for (int i = 0; i <= n; i++) {
        Mint t = y[i] * pre[i] * suf[i + 1] * invFact[i] * invFact[n
        // ↳ - i];
        ((n - i) & 1) ? r -= t : r += t;
    }
    return r;
}
// Arbitrary sample points, O(n^2).
Mint lagrange(const vector<Mint>& xs, const vector<Mint>& ys, Mint x)
// ↳ {
// int n = xs.size();
// Mint r = 0;
// for (int i = 0; i < n; i++) {
// Mint num = ys[i], den = 1;
// for (int j = 0; j < n; j++) if (j != i) num *= x - xs[j], den
// ↳ *= xs[i] - xs[j];
// r += num / den;
// }
// return r;
// }
// FAULHABER: sum_{i=1..n} i^k is a polynomial in n of degree k+1 ->
// ↳ sample k+2 points. O(k log k).
Mint powerSum(ll n, int k) {
    vector<Mint> y(k + 2, 0);
    for (int i = 1; i <= k + 1; i++) y[i] = y[i - 1] + Mint(i).pow(k);
    return lagrangeConsecutive(y, n);
}
// Same trick works for: sum of C(i,k), sum of i^k * r^i (degree k in
// ↳ n after dividing by r^n),
// and any DP whose answer is eventually polynomial.
```

6.6.2 BerlekampMassey

Berlekamp-Massey: find the shortest linear recurrence fitting a sequence, $O(n^2)$.

```
// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp).
// Then jump to the k-th term in  $O(\text{len}^2 \log k)$ . THE weapon for
  ↳ "brute force small n, find the
// pattern, extrapolate" - works on any sequence that satisfies a
  ↳ constant-coefficient recurrence.
// Feed it ~2*len terms (60-100 is usually plenty).
vector<Mint> berlekampMassey(vector<Mint> s) {
    int n = s.size(), L = 0, m = 0;
    vector<Mint> C(n), B(n), T;
    C[0] = B[0] = 1;
    Mint b = 1;
    for (int i = 0; i < n; i++) {
        m++;
        Mint d = s[i];
        for (int j = 1; j <= L; j++) d += C[j] * s[i - j];
        if (d.v == 0) continue;
        T = C;
        Mint coef = d / b;
        for (int j = m; j < n; j++) C[j] -= coef * B[j - m];
        if (2 * L > i) continue;
        L = i + 1 - L, B = T, b = d, m = 0;
    }
    C.resize(L + 1), C.erase(C.begin());
    for (auto& x : C) x = Mint(0) - x;
    return C;
}
  ↳ sum_j C[j] * s[i-1-j]
// k-th term (0-indexed) of the recurrence tr with initial values S.
  ↳  $O(|\text{tr}|^2 \log k)$ .
Mint linearRec(const vector<Mint>& S, const vector<Mint>& tr, ll k) {
    if (tr.empty()) return 0;
     ↳ of bounds
    int n = tr.size();
    auto comb = [&](vector<Mint> a, vector<Mint> b) {
        vector<Mint> r(2 * n + 1, 0);
        for (int i = 0; i <= n; i++) if (a[i].v)
            for (int j = 0; j <= n; j++) r[i + j] += a[i] * b[j];
        for (int i = 2 * n; i > n; i--)
            for (int j = 0; j < n; j++) r[i - 1 - j] += r[i] * tr[j];
        r.resize(n + 1);
        return r;
    };
    vector<Mint> pol(n + 1, 0), e(pol);
    pol[0] = e[1] = 1;
    for (++k; k; k /= 2) {
        if (k % 2) pol = comb(pol, e);
        e = comb(e, e);
    }
    Mint r = 0;
    for (int i = 0; i < n; i++) r += pol[i + 1] * S[i];
    return r;
}
```

6.6.3 BostanMori

BOSTAN-MORI: the k-th term of a linear recurrence of order d in $O(d \log d \log k)$, mod NMOD.

```
// Needs: multiply / npow (02 - NTT.cpp).
// This is the fast replacement for Kitamasa ( $O(d^2 \log k)$ ), 06 -
  ↳ Interpolation/02 - linearRec) and
// for companion-matrix exponentiation ( $O(d^3 \log k)$ ). At d = 500, k
  ↳ = 1e18 it is the only one that
// runs. Pair it with Berlekamp-Massey: guess the recurrence from
  ↳ brute-forced terms, then jump.
// INPUT: a[0..d-1] the first d terms, c[0..d-1] with a[n] = sum_i
  ↳ c[i] * a[n-1-i] for n >= d.
// IDEA: the generating function is RATIONAL,  $\sum a_n x^n = P(x) /$ 
  ↳  $Q(x)$  with
//  $Q = 1 - c_0 x - \dots - c_{d-1} x^d$ ,  $P = (A * Q) \bmod x^d$ .
// Then  $[x^k] P/Q$ : multiply top and bottom by  $Q(-x)$ . The new
  ↳ denominator  $V(x) = Q(x)Q(-x)$  is EVEN,
// so  $V(x) = W(x^2)$ ; keeping the even or odd part of the numerator
  ↳ (by the parity of k) halves k.
ll bostanMori(vector<ll> a, const vector<ll>& c, ll k) {
    int d = c.size();
    if (k < (ll)a.size() && k < d) return a[k] % NMOD;
    vector<ll> Q(d + 1, 0);
    Q[0] = 1;
    for (int i = 0; i < d; i++) Q[i + 1] = (NMOD - c[i] % NMOD) %
     ↳ NMOD;
    a.resize(d, 0);
    vector<ll> P = multiply(a, Q);
    P.resize(d);
     ↳  $P = (A * Q) \bmod x^d$ 
    while (k) {
        vector<ll> Qm = Q;
        for (size_t i = 1; i < Qm.size(); i += 2) Qm[i] = (NMOD -
     ↳ Qm[i]) % NMOD;
        vector<ll> U = multiply(P, Qm), V = multiply(Q, Qm);
        P.assign((U.size() + 1) / 2, 0);
        for (size_t i = k & 1; i < U.size(); i += 2) P[i / 2] = U[i];
        Q.assign((V.size() + 1) / 2, 0);
        for (size_t i = 0; i < V.size(); i += 2) Q[i / 2] = V[i];
        P.resize(d), Q.resize(d + 1);
        k >>= 1;
    }
    return P[0] * npow(Q[0], NMOD - 2) % NMOD;
}
// THE RATIONAL GENERATING FUNCTION IS THE POINT, not just the k-th
  ↳ term:
//  $\sum_n a_n = P(1) / Q(1)$  (formally; use it when the series
  ↳ really does terminate/converge,
// e.g. absorbing-state probabilities where  $a_n$  is the chance of
  ↳ stopping at step n).
//  $\sum_n n * a_n = (P'(1) Q(1) - P(1) Q'(1)) / Q(1)^2$  - this is the
  ↳ EXPECTED STOPPING TIME, the
// reason this trick shows up in probability problems: brute-force
  ↳ 2d terms, run Berlekamp-
// Massey, build P/Q, differentiate at  $x = 1$ . Beware  $Q(1) = 0$ ,
  ↳ which means the series does not
// sum and you must handle the pole separately.
// PARTIAL FRACTIONS of P/Q give the closed form  $a_n = \sum c_i r_i^n$ 
  ↳ over the roots of Q.
// PREFIX SUMS:  $\sum_{n \leq k} a_n$  is the k-th coefficient of  $P / ((1-x)$ 
  ↳  $Q)$  - just multiply Q by
//  $(1 - x)$  and run the same routine.
// PRECONDITION: NMOD prime,  $Q[0] = 1$  (automatic here). The
  ↳ recurrence must be exact; if you got it
// from Berlekamp-Massey, feed BM at least  $2*d$  terms or the answer is
  ↳ silently wrong.
```

6.6.4 MultipointEval

MULTIPOINT EVALUATION and FAST INTERPOLATION at ARBITRARY points, both $O(n \log^2 n)$, mod NMOD.

```
// Needs: multiply (02 - NTT.cpp), Poly / divmod / deriv / cut /
//   ↳ operator+ (04 - Polynomial.cpp).
// Evaluating a degree-n polynomial at n points naively is  $O(n^2)$ ;
//   ↳ interpolating from n arbitrary
// pairs naively is  $O(n^2)$ . These are the log-squared versions, built
//   ↳ on the same PRODUCT TREE:
// node [l, r] stores  $\prod_{i \in [l, r]} (x - x_i)$ .
// REACH FOR IT when n is  $1e5$  and the points are NOT consecutive
//   ↳ integers (if they are, use
// lagrangeConsecutive,  $O(n)$ ) and NOT a geometric progression (then
//   ↳ use chirp-z,  $O(n \log n)$ ).
// Typical: "evaluate this generating function at n given points",
//   ↳ "recover the polynomial behind
// n scattered samples", shifting a sampled polynomial to a new set
//   ↳ of arguments.
vector<Poly> ptree; // 4n nodes;
//   ↳ call buildTree first
void buildTree(int v, int l, int r, const vector<ll>& x) {
    if (l == r) { ptree[v] = Poly{(NMOD - x[l] % NMOD) % NMOD, 1};
    return; }
    int m = (l + r) / 2;
    buildTree(2 * v, l, m, x), buildTree(2 * v + 1, m + 1, r, x);
    ptree[v] = multiply(ptree[2 * v], ptree[2 * v + 1]);
}
void evalRec(int v, int l, int r, Poly f, vector<ll>& out) {
    f = divmod(f, ptree[v]).second; // f mod
    //   ↳ prod (x - x_i) keeps the values
    if (l == r) { out[l] = f.empty() ? 0 : f[0]; return; }
    int m = (l + r) / 2;
    evalRec(2 * v, l, m, f, out), evalRec(2 * v + 1, m + 1, r, f,
    //   ↳ out);
}
vector<ll> evalAll(const Poly& f, const vector<ll>& x) {
    int n = x.size();
    if (!n) return {};
    ptree.assign(4 * n, {});
    buildTree(1, 0, n - 1, x);
    vector<ll> out(n);
    evalRec(1, 0, n - 1, f, out);
    return out;
}
// INTERPOLATION: with  $M(x) = \prod (x - x_i)$ , the Lagrange basis
//   ↳ denominator at  $x_i$  is  $M'(x_i)$ , so
// one multipoint evaluation of  $M'$  gives every denominator; then
//   ↳ merge the leaves bottom-up with
// result = A * (right subtree product) + B * (left subtree product).
Poly interpRec(int v, int l, int r, const vector<ll>& d) {
    if (l == r) return Poly{d[l]};
    int m = (l + r) / 2;
    Poly A = interpRec(2 * v, l, m, d), B = interpRec(2 * v + 1, m +
    //   ↳ 1, r, d);
    return multiply(A, ptree[2 * v + 1]) + multiply(B, ptree[2 * v]);
}
Poly interpolate(const vector<ll>& x, const vector<ll>& y) { //
    //   ↳ all x_i must be DISTINCT
    int n = x.size();
    if (!n) return {};
    ptree.assign(4 * n, {});
    buildTree(1, 0, n - 1, x);
    Poly root = ptree[1];
    vector<ll> out(n);
    evalRec(1, 0, n - 1, deriv(root), out); // out[i] =
    //   ↳  $M'(x_i) \neq 0$  since x_i distinct
    vector<ll> d(n);
    for (int i = 0; i < n; i++) d[i] = y[i] % NMOD * npow[out[i],
    //   ↳ NMOD - 2] % NMOD;
    return cut(interpRec(1, 0, n - 1, d), n);
}
// COMPLEXITY: each level of the tree does  $O(n)$  total polynomial work
//   ↳ with logarithmic factors, so
//  $n \log^2 n$ ; the constant is big - expect ~1s at  $n = 1e5$ , and prefer
//   ↳ the specialised routines.
// PRECONDITIONS: NMOD prime; the x_i must be pairwise distinct for
//   ↳ interpolate (duplicates make
//  $M'(x_i) = 0$  and the inverse blows up). deriv/divmod come from 04 -
//   ↳ Polynomial.cpp.
// APPLICATIONS BEYOND THE OBVIOUS
// Composing "evaluate then re-interpolate" implements ANY linear
//   ↳ map given by its action on
// points, e.g. shifting a sampled polynomial by a constant in  $O(n)$ 
//   ↳  $\log^2 n$  (or  $O(n \log n)$  with
// the Taylor shift in 05 - Convolution/10).
// The product tree alone is worth having: prod (x - x_i) in  $O(n)$ 
//   ↳  $\log^2 n$  is how you build the
```

```
// characteristic polynomial of a set of roots, the elementary
//   ↳ symmetric polynomials of n values,
// and "the number of ways to pick k of these items" for every k at
//   ↳ once.
```

6.6.5 InterpolationCoefficients

GETTING THE COEFFICIENTS, not just one value. 01 - LagrangeAndFaulhaber.cpp evaluates an

```
// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp); binomialBasis
//   ↳ needs 02 - NTT.cpp instead.
// interpolating polynomial at a point; sometimes the problem wants
//   ↳ the polynomial itself - to
// integrate it, to compose it, to read off a single coefficient, or
//   ↳ to answer queries at points
// that are not known in advance.

// ARBITRARY sample points, explicit coefficients, O(n^2). x_i must
//   ↳ be pairwise DISTINCT.
// Build M(x) = prod (x - x_i) once, then divide out each (x - x_i)
//   ↳ by synthetic division.
vector<Mint> interpCoeffs(const vector<Mint>& x, const vector<Mint>&
    ↳ y) {
    int n = x.size();
    vector<Mint> M(n + 1, 0), r(n, 0);
    M[0] = 1;
    for (int i = 0; i < n; i++) // M *= (x -
    ↳ x_i)
        for (int j = n; j >= 0; j--) { M[j] *= Mint(0) - x[i]; if (j)
            ↳ M[j] += M[j - 1]; }
    for (int i = 0; i < n; i++) {
        vector<Mint> q(n); // q = M /
        ↳ (x - x_i), synthetic division
        Mint rem = M[n];
        for (int j = n - 1; j >= 0; j--) q[j] = rem, rem = M[j] + rem
        ↳ * x[i];
        Mint den = 1;
        for (int j = 0; j < n; j++) if (j != i) den *= x[i] - x[j];
        Mint sc = y[i] / den;
        for (int j = 0; j < n; j++) r[j] += q[j] * sc;
    }
    return r; // r[j] =
    ↳ coefficient of x^j
}

// THE INVERSE VANDERMONDE MATRIX is exactly this: V[i][j] = x_i^j,
//   ↳ and (V^-1)[j][i] is the
// coefficient of x^j in the i-th Lagrange basis polynomial - the
//   ↳ vector interpCoeffs produces when
// y = e_i. So "solve V a = y" IS interpolation, in O(n^2) instead of
//   ↳ O(n^3), and
// det V = prod_{i < j} (x_j - x_i).
// If the first column starts at x_i instead of 1 (i.e. V[i][j] =
//   ↳ x_i^(j+1)), divide row i by x_i.

// BINOMIAL (Newton forward-difference) BASIS, from CONSECUTIVE
//   ↳ samples p(0..n), in O(n log n):
// p(x) = sum_i a_i C(x, i), a_i = i-th finite difference at 0 =
//   ↳ sum_j (-1)^(i-j) C(i, j) p(j)
// which is one convolution of p[j]/j! with (-1)^j/j!. The a_i are
//   ↳ integers whenever p maps
// integers to integers, so this is THE basis for "p has integer
//   ↳ values" arguments, and evaluating
// p at a HUGE x is then sum a_i * C(x, i) with C(x, i) built by a
//   ↳ running product - O(n) and no
// division by anything that could vanish.
// Written mod NMOD on top of multiply() from 02 - NTT.cpp (Poly =
//   ↳ vector<ll>); if your modulus is
// not NTT-friendly, run the same two lines with an O(n^2)
//   ↳ convolution or with convMod.
vector<ll> binomialBasis(const vector<ll>& p) { // p[j] =
    ↳ value at j, j = 0..n
    int n = p.size();
    vector<ll> f(n), fi(n), u(n), v(n);
    f[0] = 1;
    for (int i = 1; i < n; i++) f[i] = f[i - 1] * i % NMOD;
    fi[n - 1] = npow(f[n - 1], NMOD - 2);
    for (int i = n - 1; i > 0; i--) fi[i - 1] = fi[i] * i % NMOD;
    for (int i = 0; i < n; i++) {
        u[i] = p[i] % NMOD * fi[i] % NMOD;
        v[i] = (i & 1) ? (NMOD - fi[i]) % NMOD : fi[i];
    }
    vector<ll> w = multiply(u, v);
    vector<ll> a(n);
    for (int i = 0; i < n; i++) a[i] = w[i] * f[i] % NMOD;
    return a;
}

// FALLING FACTORIAL basis x^(i) = x(x-1)...(x-i+1) is the same thing
//   ↳ scaled: a_i * i! are its
// coefficients. Converting between the monomial and
//   ↳ falling-factorial bases is exactly the
// STIRLING transform (Stirling numbers of the first/second kind are
//   ↳ the two change-of-basis
// matrices), which is how "sum of i^k" becomes "sum of C(i, j)" and
```

```
↳ telescopes.
// WHICH ROUTINE WHEN
// one value, consecutive samples ..... lagrangeConsecutive, O(n)
// one value, arbitrary samples ..... lagrange, O(n^2)
// coefficients, arbitrary samples ..... interpCoeffs, O(n^2) (or
//   ↳ 04 - MultipointEval)
// coefficients in the binomial basis ... binomialBasis, O(n log n)
```

6.6.6 PRecursive

LINEAR RECURRENCE WITH POLYNOMIAL COEFFICIENTS (a "P-recursive" or holonomic sequence):

```
// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp). MOD must be
//    ↪ PRIME.
//    sum_{i=0..order} P_i(m - i) * a[m - i] = 0    for all large m,
//    ↪ deg P_i <= deg.
// This is the strictly stronger sibling of Berlekamp-Massey. BM only
//    ↪ finds CONSTANT-coefficient
// recurrences, so it fails on n!, C(2n, n), Motzkin, derangements,
//    ↪ Hertzprung, and essentially
// every sequence built from binomials or factorials. Those are all
//    ↪ P-recursive.
// REACH FOR IT in the classic contest workflow: brute-force the
//    ↪ first ~60-100 terms, guess the
// recurrence here, then extend it to n = 1e7 in O(n * order * deg).
//    ↪ Also tells you the answer IS
// holonomic, which is itself a strong hint about the intended
//    ↪ solution.
// COST: guessing is O(n^3 / deg^2)-ish Gaussian elimination on an R
//    ↪ x C matrix; extending is
// linear. You must GUESS deg (try 0, 1, 2, ... until it succeeds) -
//    ↪ order comes out automatically.
// You need roughly (deg + 2) * (order + 1) terms; give it more and
//    ↪ the verification is meaningful.

// coeffs[i][d] = coefficient of m^d in P_i. Returns {} if no such
//    ↪ recurrence fits the data.
vector<vector<Mint>> findPrec(const vector<Mint>& t, int deg) {
    int n = t.size(), B = (n + 2) / (deg + 2), C = B * (deg + 1), R =
    ↪ n - (B - 1);
    if (B < 2 || R < C - 1) return {};
    vector<vector<Mint>> mat(R, vector<Mint>(C));
    for (int y = 0; y < R; y++)
        for (int b = 0; b < B; b++) {
            Mint v = t[y + b];
            for (int d = 0; d <= deg; d++) mat[y][b * (deg + 1) + d]
            ↪ = v, v *= Mint(y + b);
        }
    int rank = 0;
    for (int x = 0; x < C && rank < R; x++) { //
    ↪ RREF
        int piv = -1;
        for (int y = rank; y < R; y++) if (mat[y][x].v) { piv = y;
        ↪ break; }
        if (piv < 0) continue;
        swap(mat[rank], mat[piv]);
        Mint iv = Mint(1) / mat[rank][x];
        for (int j = x; j < C; j++) mat[rank][j] *= iv;
        for (int y = 0; y < R; y++) if (y != rank && mat[y][x].v) {
            Mint c = mat[y][x];
            for (int j = x; j < C; j++) mat[y][j] -= c * mat[rank][j];
        }
        rank++;
    }
    if (rank == C) return {}; //
    ↪ no non-trivial kernel vector
    // column 'rank' is free: set it to 1 and read the pivot columns
    ↪ off (the matrix is in RREF)
    int order = rank / (deg + 1);
    vector<vector<Mint>> r(order + 1, vector<Mint>(deg + 1, 0));
    r[0][rank % (deg + 1)] = 1;
    for (int y = rank - 1; y >= 0; y--)
        r[order - y / (deg + 1)][y % (deg + 1)] = Mint(0) -
    ↪ mat[y][rank];
    return r;
}
// Extend the sequence to index n. 'terms' must hold at least 'order'
//    ↪ correct initial values.
// PRECONDITION: the leading polynomial P_0(m) must be non-zero mod
//    ↪ MOD at every m you generate -
// if it vanishes the recurrence genuinely does not determine a[m] (a
//    ↪ "singularity" of the
// recurrence); shift the start or pick a different MOD.
vector<Mint> extendPrec(int n, const vector<vector<Mint>>& c,
    ↪ vector<Mint> terms) {
    int order = c.size() - 1, deg = c[0].size() - 1;
    vector<Mint> a = terms;
    a.resize(max<int>(n + 1, terms.size()), 0);
    for (int m = terms.size(); m <= n; m++) {
        Mint s = 0;
        for (int i = 1; i <= order; i++) {
            Mint p = 1, k = Mint(m - i);
            for (int d = 0; d <= deg; d++) s += a[m - i] * c[i][d] *
            ↪ p, p *= k;
        }
        Mint den = 0, p = 1;
```

```
        for (int d = 0; d <= deg; d++) den += c[0][d] * p, p *=
    ↪ Mint(m);
        a[m] = (Mint(0) - s) / den;
    }
    return a;
}
// EXAMPLES that come out immediately (deg, order):
// n! -> (1, 1): a[m] - m * a[m-1] = 0. Catalan -> (1, 1).
//    ↪ Motzkin -> (1, 2).
// derangements -> (1, 2). C(2n, n) -> (1, 1). Hertzprung -> (1,
//    ↪ 4). sum C(n, k)^3 -> (2, 2).
// ALWAYS SANITY-CHECK: re-run extendPPrec on the initial terms and
//    ↪ compare against the input; a
// recurrence that fits by accident is the main failure mode when you
//    ↪ feed too few terms.
// THEORY: holonomic sequences are closed under +, Cauchy product,
//    ↪ and Hadamard product, and the
// generating function satisfies a linear ODE with polynomial
//    ↪ coefficients. That is why "sum of
// products of binomials" is essentially always P-recursive
//    ↪ (Zeilberger's algorithm proves it).
// If you also need the k-th term for k = 1e18, P-recursive sequences
//    ↪ admit an O(sqrt(k) log k)
// evaluation by the factorial-style baby-step/giant-step product of
//    ↪ matrices - far more code.
```


6.6.7 ExpPolynomialSum

SUM OF A GEOMETRIC TIMES A POLYNOMIAL: $S(n) = \sum_{i=0}^n a^i \cdot f(i)$ for n up to $1e18$,

```
// Needs: Mint + buildFact (01 - Mint.cpp), lagrangeConsecutive (06 -
//   ↳ Interpolation/01).
// where f is a polynomial of degree <= k given by its values f(0),
//   ↳ f(1), ..., f(k).  $O(k \log \text{MOD})$ .
// REACH FOR IT for "sum i^k r^i", weighted prefix sums in a linear
//   ↳ recurrence, expected values
// where each step is discounted by a factor a, and anything of the
//   ↳ shape  $\sum a^i \cdot (\text{polynomial})$ .
// WHY IT WORKS: for  $a \neq 1$  the answer has the closed form  $S(n) = a^n$ 
//   ↳  $\cdot g(n) + c$  with  $g$  a
// polynomial of degree exactly k and c a constant. Apply the
//   ↳ operator  $(D \cdot h)(n) = h(n+1) - a \cdot h(n)$ :
// it kills  $a^n \cdot g$  one degree at a time, so  $D^k(k+1)$  annihilates the
//   ↳  $a^n \cdot g(n)$  part completely and
// turns the constant into  $c \cdot (1 - a)^{(k+1)}$ . That single equation
//   ↳ gives c; then g is sampled and
// Lagrange-interpolated. For  $a = 1$  the sum is just a polynomial of
//   ↳ degree k+1 in n.
Mint expPolySum(vector<Mint> y, Mint a, ll n) { // y[i] =
    ↳ f(i) for  $i = 0..k$ 
    int k = y.size() - 1;
    if (n < 0) return 0;
    y.push_back(lagrangeConsecutive(y, k + 1)); // need
    ↳ f(k+1) too, so k+2 samples of S
    vector<Mint> S(k + 2);
    Mint p = 1;
    for (int i = 0; i <= k + 1; i++) {
        S[i] = (i ? S[i - 1] : Mint(0)) + p * y[i];
        p *= a;
    }
    if (a == Mint(1)) return lagrangeConsecutive(S, n); // S is a
    ↳ polynomial of degree k+1
    Mint c = 0; // c *
    ↳  $(1-a)^{(k+1)} = D^{(k+1)} S$  at 0
    for (int i = 0; i <= k + 1; i++) {
        Mint t = C(k + 1, i) * (Mint(0) - a).pow(k + 1 - i) * S[i];
        c += t;
    }
    c /= (Mint(1) - a).pow(k + 1);
    vector<Mint> g(k + 1); // g(i) =
    ↳  $(S(i) - c) / a^i$ 
    Mint ia = Mint(1) / a, q = 1;
    for (int i = 0; i <= k; i++) g[i] = (S[i] - c) * q, q *= ia;
    return lagrangeConsecutive(g, n) * a.pow(n) + c;
}
// SPECIAL CASES WORTH KNOWING
// f == 1:  $\sum_{i=0}^n a^i = (a^{n+1} - 1) / (a - 1)$ , and  $n+1$ 
//   ↳ when  $a = 1$ . Do NOT route a
// trivial geometric series through this - and watch a
//   ↳ == 1 mod MOD, which happens
// for  $a = \text{MOD} + 1$  style inputs and makes the closed
//   ↳ form divide by zero.
// a == 1: pure Faulhaber, powerSum in 06 - Interpolation/01.
// INFINITE sum with  $|a| < 1$  formally:  $\sum_{i=0}^{\infty} a^i f(i) =$ 
//   ↳ (applied k+1 times) - or read it off
// the rational generating function, see 03 -
//   ↳ BostanMori.cpp.
// PRECONDITION: MOD prime and larger than k + 2 (we invert
//   ↳ factorials and  $(1 - a)^{(k+1)}$ );  $a \neq 1$ 
// for the second branch, which the code checks. n may be as large as
//   ↳ you like: a.pow(n) and
// lagrangeConsecutive both take a ll argument.
// SAMPLING f: if you are handed the COEFFICIENTS of f instead of its
//   ↳ values, evaluate it at
//  $0..k+1$  by Horner in  $O(k^2)$  or by one multipoint evaluation.
```

6.6.8 PowerSumsAtOnce

ALL POWER SUMS AT ONCE: $s[i] = \sum_{p=0}^t p^i$ for EVERY $i = 0..k$, in $O(k \log k)$.

```
// Needs: multiply / npow (02 - NTT.cpp), Poly / inv / cut (04 -
//   ↳ Polynomial.cpp).
// (Convention  $0^0 = 1$ , so  $s[0] = t + 1$ .) powerSum in 06 -
//   ↳ Interpolation/01 gives one exponent in
//  $O(k)$ ; this gives all k+1 of them for the same asymptotic cost,
//   ↳ which is what you need when the
// answer is "for each k, output  $\sum i^k$ " or when a Newton-identity /
//   ↳ symmetric-function step needs
// the whole vector of power sums.
// EGF IDENTITY:  $\sum_i s[i] x^i / i! = \sum_{p=0}^t e^{p x} = (e^{(n+1)x} - 1) / (e^x - 1)$ 
//   ↳  $= [(e^{(n+1)x} - 1) / x] \cdot [x / (e^x - 1)]$ ,  $n = t + 1$ .
// The right factor is the BERNOULLI generating function,  $B[i] = i! \cdot$ 
//   ↳  $[x^i] x / (e^x - 1)$ , and
//  $x / (e^x - 1) = 1 / \sum_{i=0}^{\infty} x^i / (i+1)! -$  one power-series
//   ↳ inverse,  $O(k \log k)$ . That beats the
//  $O(k^2)$  Bernoulli recurrence in 05 - Combinatorics/11.
Poly bernoulliEGF(int m) { // returns B[i] / i!
    ↳ for  $i = 0..m-1$ 
    Poly d(m);
    d[0] = 1;
    for (int i = 1; i < m; i++) d[i] = d[i - 1] * npow(i + 1, NMOD -
    ↳ 2) % NMOD; // 1 / (i+1)!
    return inv(d, m);
}
Poly powerSums(ll t, int k) { // s[i] =
    ↳  $\sum_{p=0}^t p^i$ ,  $i = 0..k$ 
    Poly B = bernoulliEGF(k + 2), P(k + 2, 0);
    ll n = (t + 1) % NMOD, cur = 1;
    for (int i = 1; i <= k + 1; i++) { //
    ↳  $P[i] = n^i / i!$ 
        cur = cur * n % NMOD * npow(i, NMOD - 2) % NMOD;
        P[i] = cur;
    }
    Poly Q = cut(multiply(P, B), k + 2);
    Poly s(k + 1);
    ll f = 1;
    for (int i = 0; i <= k; i++) { s[i] = Q[i + 1] * f % NMOD; f = f
    ↳ * (i + 1) % NMOD; }
    return s;
}
// RELATED ONE-LINERS BUILT FROM THE SAME PIECES
// SUM OVER A LIST rather than an interval:  $\sum_j a_j \cdot i$  for all  $i$ 
//   ↳  $= 0..k$  in  $O((n + k) \log^2 n)$  -
// build the rational function  $\sum_j 1 / (1 - a_j x)$  by divide and
//   ↳ conquer over the list (keep
// numerator and denominator, merge with  $u = A \cdot D + B \cdot C$ , down =
//   ↳  $B \cdot D$ ), then expand it mod  $x^{(k+1)}$ .
// Those coefficients ARE the power sums; feed them to Newton's
//   ↳ identities (05 - Comb/10)
// to get the elementary symmetric polynomials.
//  $\sum_{p=0}^t e^{p x} = \sum_i p^i \cdot x^i / i!$  for all  $i$ : same shape, use 07 -
//   ↳ ExpPolynomialSum.cpp per i, or convolve
// with the EGF of the geometric series.
// The Faulhaber POLYNOMIAL in n (coefficients rather than a value)
//   ↳ is faulhaberPoly in
// 05 - Combinatorics/11 - BernoulliFaulhaber.cpp.
// PRECONDITION: NMOD prime and  $> k + 2$  (we invert  $1..k+1$  and take a
//   ↳ series inverse whose constant
// term is 1). t may be huge - only  $t + 1 \bmod \text{NMOD}$  enters.
```

6.6.9 ReedsSloane

REEDS-SLOANE: the shortest linear recurrence fitting a sequence MODULO A COMPOSITE m .

```
// Needs: modInverse / extGCD (04 - Number Theory/01), CRT (04 -
// Number Theory/03/01 - CRT.cpp).
// Berlekamp-Massey (06 - Interpolation/02) divides by the
// discrepancy, so it needs a FIELD and
// silently produces garbage whenever a leading coefficient is not
// invertible mod m. This is the
// replacement: it factors m into prime powers, runs a shift-register
// synthesis over each  $Z/p^e$ 
// (tracking, for every 2-adic-style valuation g, the best
// approximation whose discrepancy has that
// valuation), and CRTs the coefficient vectors back together.
// REACH FOR IT when the problem's modulus is  $1e9$ ,  $2^{32}$ ,  $10^k$  or any
// other composite and you still
// want the "brute force a few terms, guess the recurrence,
// extrapolate" attack.  $O(n^2 \log m)$ .
// OUTPUT: A[0..L] with A[0] = 1 and  $\sum_j A[j] * s[i-j] = 0 \pmod m$ ,
// i.e.
//  $s[i] = -(A[1] s[i-1] + \dots + A[L] s[i-L]) \pmod m$ . Feed 2L or
// more terms.
namespace RS {
int deg(const vector<ll>& a) { // -inf encoded as a
    // large negative number
    return (a.size() > 1 || (a.size() == 1 && a[0])) ? (int)a.size()
    // - 1 : -40000;
}
int L(const vector<ll>& a, const vector<ll>& b) { return max(deg(a),
    // deg(b) + 1); }
void ext(vector<ll>& a, size_t d) { if (d > a.size()) a.resize(d, 0);
    // }

// shift-register synthesis modulo  $p^e$  ( $m = p^e$ ); returns the
// connection polynomial
vector<ll> primePower(const vector<ll>& s, ll m, ll p, int e) {
    vector<vector<ll>>> a(e), b(e), an(e), bn(e), ao(e), bo(e);
    vector<ll> t(e), u(e), r(e), to(e, 1), uo(e), pw(e + 1, 1);
    for (int i = 1; i <= e; i++) pw[i] = pw[i - 1] * p;
    for (int i = 0; i < e; i++) {
        a[i] = an[i] = {pw[i]};
        b[i] = {0}, bn[i] = {s[0] * pw[i] % m};
        t[i] = s[0] * pw[i] % m;
        if (!t[i]) t[i] = 1, u[i] = e;
        else for (u[i] = 0; t[i] % p == 0; t[i] /= p) u[i]++;
    }
    for (size_t k = 1; k < s.size(); k++) {
        for (int g = 0; g < e; g++) if (L(an[g], bn[g]) > L(a[g],
        // b[g])) {
            int h = e - 1 - u[g];
            ao[g] = a[h], bo[g] = b[h], to[g] = t[h], uo[g] = u[h],
        // r[g] = k - 1;
        }
        a = an, b = bn;
        for (int o = 0; o < e; o++) {
            ll d = 0;
            for (size_t i = 0; i < a[o].size() && i <= k; i++) d = (d
        // + a[o][i] * s[k - i]) % m;
            if (!d) { t[o] = 1, u[o] = e; continue; }
            for (u[o] = 0, t[o] = d; t[o] % p == 0; t[o] /= p) u[o]++;
            int g = e - 1 - u[o];
            if (L(a[g], b[g]) == 0) { ext(bn[o], k + 1); bn[o][k] =
        // (bn[o][k] + d) % m; continue; }
            ll c = t[o] % m * modInverse(to[g] % m, m) % m * pw[u[o]
        // - uo[g]] % m;
            int sh = k - r[g];
            ext(an[o], ao[g].size() + sh), ext(bn[o], bo[g].size() +
        // sh);
            for (size_t i = 0; i < ao[g].size(); i++)
                an[o][i + sh] = (an[o][i + sh] - c * ao[g][i]) % m;
            for (size_t i = 0; i < bo[g].size(); i++)
                bn[o][i + sh] = (bn[o][i + sh] - c * bo[g][i]) % m;
            for (ll& x : an[o]) x = (x % m + m) % m;
            for (ll& x : bn[o]) x = (x % m + m) % m;
            while (an[o].size() > 1 && !an[o].back())
        // an[o].pop_back();
            while (bn[o].size() > 1 && !bn[o].back())
        // bn[o].pop_back();
        }
    }
    return an[0];
}
}
vector<ll> reedsSloane(vector<ll> s, ll m) {
    vector<array<ll, 3>>> fac; // {p^e, p, e}
    ll t = m;
    for (ll i = 2; i * i <= t; i++) if (t % i == 0) {
```

```
        ll c = 0, q = 1;
        while (t % i == 0) t /= i, c++, q *= i;
        fac.push_back({q, i, c});
    }
    if (t > 1) fac.push_back({t, t, 1});
    vector<vector<ll>>> per;
    size_t n = 0;
    for (auto& f : fac) {
        vector<ll> ss = s;
        for (ll& x : ss) x = (x % f[0] + f[0]) % f[0];
        per.push_back(RS::primePower(ss, f[0], f[1], (int)f[2]));
        n = max(n, per.back().size());
    }
    vector<ll> res(n);
    for (size_t i = 0; i < n; i++) {
        vector<ll> c(fac.size(), md(fac.size()));
        for (size_t j = 0; j < fac.size(); j++)
            md[j] = fac[j][0], c[j] = i < per[j].size() ? per[j][i] :
        // 0;
        res[i] = CRT(c, md).first;
    }
    return res; // res[0] == 1; s[i]
    // = -sum_{j>=1} res[j]*s[i-j]
}
// AFTER YOU HAVE THE RECURRENCE: extending it is trivial (the
// leading coefficient is 1, so no
// inversion is ever needed) - just iterate  $s[i] = -(\sum_{j>=1} A[j]
// s[i-j]) \pmod m$ .
// For the k-th term with  $k = 1e18 \pmod a$  COMPOSITE, do Kitamasa-style
// polynomial exponentiation
// mod the connection polynomial: it only multiplies, never divides,
// so a composite modulus is fine.
// CHEAPER FIRST TRIES, in order:
// - if m is prime, use Berlekamp-Massey; this file is strictly
// heavier.
// - if m is squarefree, run BM under each prime factor separately
// and CRT. That is 10 lines and
// covers  $m = 1e9+6$ -style inputs; the prime-POWER case is the only
// one that really needs this.
// - if the sequence is not constant-coefficient at all (factorials,
// binomials), you want
// 06 - Precursive.cpp instead, and it also works over composite
// moduli when the leading
// polynomial happens to be invertible.
// PRECONDITION:  $m \geq 1$  and the factorisation loop is trial division,
// so m up to  $\sim 1e18$  needs
// Pollard rho (04 - Number Theory/02/06) instead of the loop above.
```

6.7 Numerical

6.7.1 Integration

ADAPTIVE SIMPSON - numeric integral of a black-box f on $[a,b]$ to a requested absolute error.

```
// Simpson approximates f by a parabola through the two ends and the
//   ↳ midpoint; "adaptive" means it
// splits an interval only where the parabola does not yet agree with
//   ↳ the two half-parabolas.
// Reach for it when the answer is an integral you cannot do in
//   ↳ closed form: area of a union of
// circles/ellipses cut by a line, area swept by a moving shape,
//   ↳ probability over a continuous
// parameter, arc lengths. Cost is proportional to how wiggly f is,
//   ↳ not to (b-a).
template <class F> ld simpson(F f, ld a, ld b) {
    return (b - a) * (f(a) + 4 * f((a + b) / 2) + f(b)) / 6;
}
template <class F> ld asr(F f, ld a, ld b, ld eps, ld S, int dep) {
    ld c = (a + b) / 2, L = simpson(f, a, c), R = simpson(f, c, b);
    if (dep <= 0 || fabs(L + R - S) <= 15 * eps) return L + R + (L +
    ↳ R - S) / 15;
    return asr(f, a, c, eps / 2, L, dep - 1) + asr(f, c, b, eps / 2,
    ↳ R, dep - 1);
}
template <class F> ld integrate(F f, ld a, ld b, ld eps = 1e-9) {
    return asr(f, a, b, eps, simpson(f, a, b), 50);
}
// TRAPS. (1) If f has a KINK or a jump, split the interval at that
//   ↳ point by hand and integrate the
// pieces - adaptive Simpson will burn its depth budget trying to
//   ↳ resolve it. (2) The (L+R-S)/15
// term is Richardson extrapolation; keeping it makes the result much
//   ↳ more accurate for free.
// (3) A too-loose eps on a nearly flat region can stop early and
//   ↳ miss a spike: if f can be zero on
// most of the range and huge somewhere, integrate the pieces
//   ↳ separately.
//
// OTHER NUMERICAL TOOLS WORTH THE PAGE:
// TERNARY SEARCH on a strictly unimodal f over reals: 100 iterations
//   ↳ of
//   m1 = l + (r-l)/3, m2 = r - (r-l)/3; f(m1) < f(m2) ? r = m2 : l
//   ↳ = m1; (minimising)
//   fixes the answer to ~1e-30 relative. On INTEGERS use binary
//   ↳ search on the difference
//   f(m+1) - f(m) instead - the three-way split has off-by-one traps
//   ↳ at the last two points.
// NEWTON'S METHOD for a root of g: x == g(x) / g'(x). Doubles the
//   ↳ correct digits per step but
//   needs a good start; bisection first if the sign changes, then
//   ↳ Newton.
// BISECTION on a monotone predicate: 100 iterations of (l+r)/2 is
//   ↳ always safe and never diverges.
// GOLDEN-SECTION SEARCH: like ternary search but one function
//   ↳ evaluation per step instead of two -
//   worth it only when f is expensive.
// SOLVING f(x) = 0 WHERE f IS A POLYNOMIAL: find all real roots by
//   ↳ recursing on the derivative
//   (roots of f' split the line into monotone pieces; bisection inside
//   ↳ each).
// SIMULATED ANNEALING / HILL CLIMBING when the objective is not
//   ↳ unimodal: start from several
//   random points, take a step of size T, accept worse solutions
//   ↳ with probability exp(-dE/T),
//   cool T geometrically (T *= 0.995). Good for geometric-median /
//   ↳ facility-location style
//   problems where an exact method is out of reach; always seed the
//   ↳ RNG from the clock.
// WEISZFELD for the geometric median (minimise sum of Euclidean
//   ↳ distances): iterate
//   x <- (sum p_i / |x - p_i|) / (sum 1 / |x - p_i|), guarding the
//   ↳ case x == p_i. Converges
//   linearly; nested ternary search is the safer contest answer (the
//   ↳ objective is convex).
```

6.7.2 LinearSolvers

SPECIAL-SHAPE LINEAR SOLVERS. Gaussian elimination is $O(n^3)$; these are the shapes where the

```
// structure buys you a factor of n or removes a precondition.

// --- THOMAS ALGORITHM: TRIDIAGONAL system in  $O(n)$ . ---
//   a[i] x[i-1] + b[i] x[i] + c[i] x[i+1] = d[i], with a[0] =
//   ↳ c[n-1] = 0.
// This is THE closer for random walks on a path ("expected steps to
//   ↳ escape [0, n]"), 1-D heat /
// diffusion DPs, and any DP whose state depends on its two
//   ↳ neighbours. Gaussian elimination on
// the same system is  $O(n^3)$  and will TLE at  $n = 1e5$ ; this is 8 lines.
// STABLE without pivoting when the system is diagonally dominant
//   ↳ ( $|b| > |a| + |c|$ ), which every
// hitting-time system is.
vector<ld> thomas(vector<ld> a, vector<ld> b, vector<ld> c,
    ↳ vector<ld> d) {
    int n = b.size();
    for (int i = 1; i < n; i++) {
        ld m = a[i] / b[i - 1];
        b[i] -= m * c[i - 1], d[i] -= m * d[i - 1];
    }
    vector<ld> x(n);
    x[n - 1] = d[n - 1] / b[n - 1];
    for (int i = n - 2; i >= 0; i--) x[i] = (d[i] - c[i] * x[i + 1])
    ↳ / b[i];
    return x;
}
// BANDED systems of half-width w (state depends on w neighbours)
//   ↳ generalise this at  $O(n \cdot w^2)$  -
// the shape of "robot on a grid with m columns", where  $w = m$ .
// CYCLIC tridiagonal (x[0] and x[n-1] also coupled):
//   ↳ Sherman-Morrison - solve two tridiagonal
// systems and combine, still  $O(n)$ .

// --- DETERMINANT MODULO AN ARBITRARY m (composite allowed),  $O(n^3)$ 
//   ↳ log m). ---
// Gaussian elimination inverts the pivot, so it needs a PRIME
//   ↳ modulus. This version never
// divides: it runs the Euclidean algorithm between two rows,
//   ↳ swapping them, until the lower
// entry is 0. Use it when the modulus is  $2^k$ ,  $1e9$ , or anything you
//   ↳ cannot invert in.
ll detMod(vector<vector<ll>> a, ll m) {
    int n = a.size();
    ll det = 1 % m;
    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++)
            while (a[j][i]) { //
                // Euclid on the two rows
                ll t = a[i][i] / a[j][i];
                for (int k = i; k < n; k++) a[i][k] = (a[i][k] - t *
                ↳ a[j][k]) % m;
                swap(a[i], a[j]), det = -det;
            }
        det = det * a[i][i] % m;
        if (!a[i][i]) return 0;
    }
    return (det % m + m) % m;
}
// MATRIX RANK over a field: the usual elimination, counting non-zero
//   ↳ pivot rows.
// RANK over GF(2): use bitsets -  $O(n^3 / 64)$ , see 04 - Linear
//   ↳ Algebra/03.
// INTEGER determinant with no modulus: compute it modulo several
//   ↳ primes and CRT. Size it with
// HADAMARD'S BOUND |det| <= prod of the row 2-norms <=  $n^{(n/2)}$  *
//   ↳ max|a_ij|^n, and take enough
// primes that the product exceeds  $2x$  that.
// SPARSE determinant ( $n = 1e5$  with few non-zeros): Wiedemann - pick
//   ↳ random u, v, feed the Krylov
// sequence  $u^T A^k v$  to Berlekamp-Massey to get the minimal
//   ↳ polynomial, whose constant term is
// +-det after a random diagonal preconditioner.  $O(n \cdot \text{nnz})$ .
// CHARACTERISTIC POLYNOMIAL in  $O(n^3)$ : reduce to upper HESSENBERG
//   ↳ form by similarity transforms
// (so  $\det(xI - H)$  satisfies a short recurrence in the leading
//   ↳ principal minors), then read the
// coefficients off. With Cayley-Hamilton that gives  $A^k$  in  $O(n^3 +$ 
//   ↳  $n^2 \log k)$  instead of
//  $O(n^3 \log k)$  - worth it only for large k and small n.
```

6.7.3 Romberg

ROMBERG INTEGRATION - the other numerical quadrature. Repeatedly halve the step (trapezoid rule)

```
// and RICHARDSON-EXTRAPOLATE the error away: T(h) has an error
// expansion in h^2 only, so combining
// two step sizes cancels the h^2 term, three cancel h^4, and so on.
// Row k of the table is accurate
// to O(h^(2k)). Cost: 2^k function evaluations, each sample reused
// across the whole table.
// REACH FOR IT over adaptive Simpson (07 - Numerical/01) when the
// integrand is SMOOTH and cheap
// and you want many correct digits: Romberg reaches 1e-12 in ~10
// rows where Simpson needs a deep
// adaptive recursion. Use adaptive Simpson instead when the function
// has kinks, spikes, or a
// singular endpoint - Romberg assumes the error expansion exists and
// quietly lies when it does not.
template <class F> ld romberg(F f, ld a, ld b, ld tol = 1e-10, int
    maxIter = 25) {
    vector<ld> t;
    ld h = b - a;
    t.push_back(h * (f(a) + f(b)) / 2); // T(h), one
    // trapezoid
    for (int k = 1, m = 1; k <= maxIter; k++, m <= 1) {
        ld s = 0, x = a + h / 2;
        for (int j = 0; j < m; j++) s += f(x), x += h; // only the
        // NEW midpoints
        h /= 2;
        ld cur = t[0] / 2 + h * s; // refined
        // trapezoid, previous row's T
        ld prev = t.back();
        for (int j = 0; j < k; j++) { //
            // Richardson: t[j] holds the old row
            ld nxt = (powl(4, j + 1) * cur - t[j]) / (powl(4, j + 1)
            // - 1);
            t[j] = cur, cur = nxt;
        }
        t.push_back(cur);
        if (k > 3 && fabsl(prev - cur) < tol) break;
    }
    return t.back();
}
// THE TABLE: R[k][0] = trapezoid with 2^k intervals, R[k][j] = (4^j
// R[k][j-1] - R[k-1][j-1])
// / (4^j - 1). R[k][1] is exactly Simpson's rule; R[k][2] is
// Boole's. The code above keeps only
// one row, which is all you need.
// PRACTICAL NOTES
// - convergence check on |R[k][k] - R[k-1][k-1]| is a heuristic;
// require a few rows first (the
// k > 3 guard) or a flat integrand will make it stop after one
// step.
// - INFINITE range: substitute x = t/(1-t) or integrate to a large
// cut-off and bound the tail.
// - ENDPOINT SINGULARITY (1/sqrt(x) at 0): substitute it away, or
// use the tanh-sinh (double
// exponential) rule, which handles endpoint singularities and
// converges even faster.
// - OSCILLATORY integrands: split at the zeros of the oscillating
// factor and sum, or Romberg will
// happily converge to nonsense.
// - the same Richardson idea accelerates ANY sequence with a known
// error expansion, e.g.
// numerical derivatives (central differences) and extrapolating a
// DP in a step size.
```

6.8 Linear Programming

6.8.1 Simplex

TWO-PHASE SIMPLEX. maximise c.x subject to $A \cdot x \leq b$, $x \geq 0$.

```
// Returns the optimum, or +inf if unbounded, or -inf if infeasible;
// x is filled with a solution.
// Exponential in theory, O(n*m) per pivot and instant at contest
// sizes (n, m up to a few hundred).
// REACH FOR IT when the problem is "optimise a linear objective
// under linear constraints" and no
// combinatorial structure presents itself - resource allocation,
// blending, fractional covering,
// zero-sum matrix games, and any min-cost-flow-with-side-constraints
// that flow cannot express.
typedef vector<ld> vd;
struct Simplex {
    int m, n;
    vector<int> B, N;
    vector<vd> D;
    Simplex(const vector<vd>& A, const vd& b, const vd& c)
        : m(b.size()), n(c.size()), B(m), N(n + 1), D(m + 2, vd(n +
        // 2, 0)) {
        for (int i = 0; i < m; i++) for (int j = 0; j < n; j++)
        // D[i][j] = A[i][j];
        for (int i = 0; i < m; i++) B[i] = n + i, D[i][n] = -1,
        // D[i][n + 1] = b[i];
        for (int j = 0; j < n; j++) N[j] = j, D[m][j] = -c[j];
        N[n] = -1, D[m + 1][n] = 1;
    }
    void pivot(int r, int s) {
        ld inv = 1 / D[r][s];
        for (int i = 0; i < m + 2; i++) if (i != r && fabsl(D[i][s])
        // > 1e-9) {
            ld f = D[i][s] * inv; // the
            // pivot COLUMN is fixed up by
            for (int j = 0; j < n + 2; j++) if (j != s) D[i][j] -= f
            // * D[r][j]; // the loop below
        }
        for (int j = 0; j < n + 2; j++) if (j != s) D[r][j] *= inv;
        for (int i = 0; i < m + 2; i++) if (i != r) D[i][s] *= -inv;
        D[r][s] = inv;
        swap(B[r], N[s]);
    }
    bool simplex(int phase) {
        int x = m + phase - 1;
        for (;;) {
            int s = -1;
            for (int j = 0; j <= n; j++)
                if (N[j] != -phase && (s < 0 || make_pair(D[x][j],
                // N[j]) < make_pair(D[x][s], N[s]))) s = j;
            if (D[x][s] >= -1e-9) return true;
            int r = -1;
            for (int i = 0; i < m; i++) {
                if (D[i][s] <= 1e-9) continue;
                if (r < 0 || make_pair(D[i][n + 1] / D[i][s], B[i]) <
                // make_pair(D[r][n + 1] / D[r][s], B[r])) r = i;
            }
            if (r < 0) return false; //
            // unbounded
            pivot(r, s);
        }
    }
    ld solve(vd& x) {
        int r = 0;
        for (int i = 1; i < m; i++) if (D[i][n + 1] < D[r][n + 1]) r
        // = i;
        if (D[r][n + 1] < -1e-9) { //
            // phase 1: find any feasible point
            pivot(r, n);
            if (!simplex(2) || D[m + 1][n + 1] < -1e-9) return -1.0L
            // / 0.0L; // infeasible
            for (int i = 0; i < m; i++) if (B[i] == -1) {
                int s = 0;
                for (int j = 1; j <= n; j++)
                    if (make_pair(D[i][j], N[j]) < make_pair(D[i][s],
                    // N[s])) s = j;
                pivot(i, s);
            }
        }
        bool ok = simplex(1);
        x.assign(n, 0);
        for (int i = 0; i < m; i++) if (B[i] < n) x[B[i]] = D[i][n +
        // 1];
        return ok ? D[m][n + 1] : 1.0L / 0.0L; //
        // +inf = unbounded
    }
};
/* MODELLING NOTES - the part that actually costs you time
MINIMISE instead: negate c and negate the answer.
```

$\geq$ CONSTRAINT: multiply the row by -1 . EQUALITY: add it twice, as $\hookrightarrow \leq$ and as $\geq$.

FREE VARIABLE (no $x \geq 0$): substitute $x = u - v$ with $u, v \geq 0$. $\hookrightarrow$ Doubles that column.

UPPER BOUND $x_i \leq U$: just another row; simplex has no special $\hookrightarrow$ handling and does not need any.

LINEAR-FRACTIONAL objective $(c \cdot x + a) / (d \cdot x + b)$: CHARNES-COOPER - $\hookrightarrow$ substitute $y = x / (d \cdot x + b)$, $t = 1 / (d \cdot x + b)$ and maximise $c \cdot y + a \cdot t$ subject to $A \cdot y \leq b \cdot t$, $d \cdot y + b \cdot t = 1$, $t \geq 0$. Linear again.

INTEGER answers are NOT guaranteed. If the constraint matrix is $\hookrightarrow$ TOTALLY UNIMODULAR (network matrices, interval matrices, bipartite incidence) the optimum IS $\hookrightarrow$ integral - that is exactly why flow problems have integral optima. Otherwise you need ILP, which $\hookrightarrow$ is NP-hard.

ZERO-SUM MATRIX GAME: shift A so every entry is > 0 , then min sum(y) $\hookrightarrow$ s.t. $A \cdot y \geq 1$, $y \geq 0$ gives value $v = 1 / \text{sum}(y)$ and the column player's mixed strategy $y \cdot v$; the $\hookrightarrow$ DUAL gives the row player's.

Von Neumann: $\max_x \min_y x^T A \cdot y = \min_y \max_x x^T A \cdot y$.

DUALITY, which is often the actual solution: $\max c \cdot x$, $Ax \leq b$, $x \geq 0$ is dual to $\min b \cdot y$, $A^T y \geq c$, $y \geq 0$; the two optima are equal. $\hookrightarrow$ COMPLEMENTARY SLACKNESS at the optimum: $x_j > 0 \Rightarrow$ the j -th dual constraint is tight, and $y_i > 0 \Rightarrow$ the i -th primal row is tight.

The combinatorial specialisations you already know are $\hookrightarrow$ max-flow/min-cut and König.

PRECISION: this is floating point. If the data is integral and $\hookrightarrow$ small, and you need an exact answer, prefer a combinatorial model (flow, matching) over $\hookrightarrow$ simplex. */

7 Strings

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7.1 String Matching

7.1.1 KMP

KMP. $\text{pi}[i]$ = length of the longest proper border of $s[0..i]$ (prefix that is also a suffix).

```
// Gives  $O(n+m)$  substring search, the period of a prefix ( $i+1 \hookrightarrow \text{pi}[i]$ ), and the automaton for DP.
vector<int> compute_pi(const string &s) {
    int n = s.size();
    vector<int> pi(n);
    for (int i = 1; i < n; i++) {
        int j = pi[i - 1];
        while (j > 0 && s[i] != s[j])
            j = pi[j - 1];
        if (s[i] == s[j])
            j++;
        pi[i] = j;
    }
    return pi;
}

vector<int> find_matches(const string &text, const string &pat) {
    int n = pat.length(), m = text.length();
    string s = pat + "#" + text;
    vector<int> pi = compute_pi(s);
    for (int i = n + 1; i <= n + m; i++) { /* n + 1 is where the text
        starts */
        if (pi[i] == n) {
            ans.push_back(i - 2 * n); /* i - (n - 1) - (n + 1) */
        }
    }
    return ans;
}

void compute_automaton(string s, vector<vector<int>> &aut) {
    s += '#';
    int n = s.size();
    vector<int> pi = compute_pi(s);
    aut.assign(n, vector<int>(26));
    for (int i = 0; i < n; i++) {
        for (int c = 0; c < 26; c++) {
            if (i > 0 && 'a' + c != s[i])
                aut[i][c] = aut[pi[i - 1]][c];
            else
                aut[i][c] = i + ('a' + c == s[i]);
        }
    }
}
```

7.1.2 RabinKarp

RABIN-KARP - rolling-hash substring search in $O(n+m)$ expected. Prefer 05 - Hashing for anything

$\hookrightarrow$ serious (this uses a FIXED base, which is hackable); keep this for $\hookrightarrow$ one-off matching.

```
vector<int> rabin_karp(string const &s, string const &t) {
    const int p = 31;
    const int m = 1e9 + 9;
    int S = s.size(), T = t.size();

    vector<ll> p_pow(max({S, T, 1})); // max(0,0) would
    // leave it empty
    p_pow[0] = 1;
    for (int i = 1; i < (int) p_pow.size(); i++)
        p_pow[i] = (p_pow[i - 1] * p) % m;

    vector<ll> h(T + 1, 0);
    for (int i = 0; i < T; i++)
        h[i + 1] = (h[i] + (t[i] - 'a' + 1) * p_pow[i]) % m;
    ll h_s = 0;
    for (int i = 0; i < S; i++)
        h_s = (h_s + (s[i] - 'a' + 1) * p_pow[i]) % m;

    vector<int> occurrences;
    for (int i = 0; i + S - 1 < T; i++) {
        ll cur_h = (h[i + S] + m - h[i]) % m;
        if (cur_h == h_s * p_pow[i] % m)
            occurrences.push_back(i);
    }
    return occurrences;
}
```

7.1.3 ZFunction

$z[i]$ is the length of the longest substring starting from $s[i]$ which is also a prefix of s

```
vector<int> z_function(string s) {
    int n = s.length();
    vector<int> z(n);
    int l = 0, r = 0;
    for (int i = 1; i < n; i++) {
        if (i < r) {
            z[i] = min(r - i, z[i - l]);
        }
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) {
            z[i]++;
        }
        if (i + z[i] > r) {
            l = i;
            r = i + z[i];
        }
    }
    return z;
}
```


7.1.4 WildcardMatching

STRING MATCHING WITH WILDCARDS in $O(n \log n)$. A wildcard matches any character, on EITHER side.

```
// Needs: conv (06 - Math/05 - Convolution/01 - FFT.cpp).
// Map every real character to a positive value and every wildcard to
  ↳ 0. Then position i matches iff
//      sum_j (p[j] - t[i+j])^2 * p[j] * t[i+j] == 0,
// because each term is >= 0 and vanishes exactly when the characters
  ↳ agree or one of them is 0.
// Expanding gives three correlations: sum p^3 t - 2 sum p^2 t^2 +
  ↳ sum p t^3, and a correlation is a
// convolution with the pattern reversed.
vector<int> wildcardMatch(const string& t, const string& p, char wild
  ↳ = '?') {
    int n = t.size(), m = p.size();
    if (m > n) return {};
    // The identity needs values >= 0 and wildcard == 0. Compress the
  ↳ characters that actually
    // occur to 1..k - with a raw 'c - 'a' + 1' any character below
  ↳ 'a' is NEGATIVE and the terms
    // can cancel, which reports matches that do not exist (e.g. text
  ↳ "b" against pattern "Z").
    ll id[256] = {}, k = 0; // a
  ↳ flat table, not a map: the
    for (char c : t) if (c != wild) id[(unsigned char)c] = 1; //
  ↳ conversion is O(n), not O(n log n)
    for (char c : p) if (c != wild) id[(unsigned char)c] = 1;
    for (int c = 0; c < 256; c++) if (id[c]) id[c] = ++k;
    auto val = [&](char c) { return c == wild ? 0LL : id[(unsigned
  ↳ char)c]; };
    vector<ll> a(m), b(n);
    for (int i = 0; i < m; i++) a[i] = val(p[m - 1 - i]); //
  ↳ reversed pattern
    for (int i = 0; i < n; i++) b[i] = val(t[i]);
    auto pw = [](vector<ll> v, int k) { for (ll& x : v) { ll y = 1;
  ↳ for (int i = 0; i < k; i++) y *= x; x = y; } return v; };
    auto c1 = conv(pw(a, 3), b), c2 = conv(pw(a, 2), pw(b, 2)), c3 =
  ↳ conv(a, pw(b, 3));
    vector<int> res;
    for (int i = 0; i + m <= n; i++)
        if (c1[i + m - 1] - 2 * c2[i + m - 1] + c3[i + m - 1] == 0)
  ↳ res.push_back(i);
    return res; //
  ↳ 0-based start positions
}
// PRECISION: the largest intermediate is about k^4 * m, where k is
  ↳ the number of DISTINCT real
// characters (which is why compressing is free). With k = 26 and m =
  ↳ 1e5 that is
// ~4.6e10, comfortable for double FFT. For a big alphabet, compress
  ↳ the characters that actually
// occur first, or run the three convolutions under two NTT primes
  ↳ and CRT.
// COUNT MISMATCHES at every shift (no wildcards, small alphabet):
  ↳ for each letter c convolve the
// 0/1 indicator of c in p (reversed) with the indicator of c in t;
  ↳ summing gives the number of
// MATCHES per shift, and mismatches = m - matches. O(|sigma| n log
  ↳ n).
// k-MISMATCH: same counts, keep shifts with matches >= m - k.
// SUBSET / "at most one character class" matching: encode each
  ↳ position as a bitmask over the
// alphabet and convolve the complement indicators; a shift is valid
  ↳ iff the total is 0.
```

7.1.5 PrefixFunctionApplications

WHAT THE PREFIX FUNCTION ANSWERS once you have pi[] - every one of these is $O(n)$ and every one

```
// Needs: 07 - Strings/01 - String Matching/01 - KMP.cpp
  ↳ (compute_pi)
// of them is a contest problem on its own. INDEX CONVENTION: s is
  ↳ 0-indexed and
//      pi[i] = length of the longest PROPER border of the prefix
  ↳ s[0..i] (so pi[0] = 0).
// A "border" is a string that is both a proper prefix and a proper
  ↳ suffix.
// WHEN: any question about periodicity, repetition, "shortest string
  ↳ whose k-th power is s",
// "how many times does every prefix occur", "which prefixes are also
  ↳ suffixes".

// --- ALL BORDERS of the whole string, longest first. The border of
  ↳ a border is a border, so the
// chain pi[n-1] -> pi[pi[n-1]-1] -> ... enumerates them ALL, in
  ↳ O(number of borders) <= O(n).
vector<int> allBorders(const vector<int>& pi) {
    vector<int> b;
    for (int k = pi.empty() ? 0 : pi.back(); k > 0; k = pi[k - 1])
  ↳ b.push_back(k);
    return b; //
  ↳ lengths, decreasing
}
// --- PERIODS. p is a period of s (s[i] == s[i+p] wherever both
  ↳ exist) IFF n - p is a border.
// So the periods are exactly { n - b : b a border }, and the
  ↳ SMALLEST period is n - pi[n-1].
int smallestPeriod(int n, const vector<int>& pi) { return n - (n ?
  ↳ pi[n - 1] : 0); }
// s is a perfect power t^k IFF n % smallestPeriod == 0, and then k =
  ↳ n / smallestPeriod.
// (Careful: "aabaab" has period 3 but 5 % 3 != 0, so it is NOT a
  ↳ power - the test is required.)

// --- OCCURRENCE COUNT OF EVERY PREFIX inside s. cnt[len] = number
  ↳ of occurrences of s[0..len-1].
// Every position ends the prefix of length i+1, and an occurrence of
  ↳ a prefix implies an
// occurrence of all of ITS borders - so push the counts down the pi
  ↳ chain in reverse order.
vector<int> prefixOccurrences(const string& s, const vector<int>& pi)
  ↳ {
    int n = s.size();
    vector<int> cnt(n + 1, 0);
    for (int i = 0; i < n; i++) cnt[pi[i]]++;
    for (int i = n - 1; i > 0; i--) cnt[pi[i - 1]] += cnt[i];
    for (int i = 0; i <= n; i++) cnt[i]++; // the
  ↳ prefix itself
    cnt[0] = 0;
    return cnt;
}
// The same routine on p + '#' + t counts occurrences of every
  ↳ prefix of p inside t: read
// cnt[|p|] for the number of matches, and cnt[k] - (contribution
  ↳ inside p) for shorter prefixes.

// --- BORDER TREE: parent(i) = pi[i-1] over nodes 0..n (node i = the
  ↳ prefix of length i). Then
// "the borders of the prefix of length i" = the ancestors of i, so
// is u a border of v? -> u is an ancestor of v
  ↳ (Euler-tour interval test, O(1))
// longest common border of two -> LCA
// count borders of prefix i -> depth(i)
vector<int> borderTree(const vector<int>& pi) {
    int n = pi.size();
    vector<int> par(n + 1, 0);
    par[0] = -1;
    for (int i = 1; i <= n; i++) par[i] = pi[i - 1];
    return par;
}
/* MORE PREFIX-FUNCTION FACTS worth having on the page
pi vs Z: z[i] = length of the lcp of s and s[i..]. Either converts
  ↳ to the other in O(n), but pi
  ↳ is the one that extends ONLINE (append a character, update in O(1)
  ↳ amortised) and the one that
  ↳ turns into an automaton; z is the one that answers "match at
  ↳ offset i" directly.
AUTOMATON aut[i][c] (see 01 - KMP.cpp) makes the DP "count strings
  ↳ of length L avoiding p"
  ↳ trivial; without it that DP is O(L n |A|) with the pi chain walked
  ↳ per step.
DISTINCT SUBSTRINGS incrementally: append c to s, run the prefix
  ↳ function of the REVERSED new
```


string; the new distinct substrings number $(i+1) - \max(z\text{-values}) - \hookrightarrow O(n^2)$ total, and the right answer whenever the input arrives online and a suffix automaton is $\hookrightarrow$ overkill.

FINE AND WILF: if p and q are periods of s and $p + q - \gcd(p, q) \leq n$, then $\gcd(p, q)$ is a period.

This is why the border lengths of any prefix form $O(\log n)$

$\hookrightarrow$ ARITHMETIC PROGRESSIONS - the fact that makes "sum over all borders" DPs $O(n \log n)$ instead of $O(n^2)$.

COUNTING STRINGS with a given pi array, or with a given set of $\hookrightarrow$ borders, is Fine and Wilf plus inclusion-exclusion over the progressions; do NOT try to enumerate $\hookrightarrow$ borders one by one. $\ast/$

7.2 Hashing

7.2.1 StringHashing

Double-mod polynomial hashing. RANDOM bases $\Rightarrow$ not anti-hackable.

```
// No modular inverses: h(l,r) is compared already scaled, so get()
// is 2 mults.
ll HB1, HB2;
struct Hash {
    static const ll M1 = 1000000007, M2 = 998244353;
    int n;
    vector<ll> h1, h2, p1, p2;

    Hash(const string& s) : n(s.size()), h1(n + 1, 0), h2(n + 1, 0),
        p1(n + 1, 1), p2(n + 1, 1) {
        if (!HB1) {
            mt19937_64
            rng(chrono::steady_clock::now().time_since_epoch().count());
            HB1 = 256 + rng() % (M1 - 300), HB2 = 256 + rng() % (M2 -
            300); // FULL range: a base drawn
            // from only 1e6 values gives collision probability n/1e6 per
            // modulus (0.1 at n = 1e5)
        }
        for (int i = 0; i < n; i++) {
            h1[i + 1] = (h1[i] * HB1 + s[i]) % M1, p1[i + 1] = p1[i]
            * HB1 % M1;
            h2[i + 1] = (h2[i] * HB2 + s[i]) % M2, p2[i + 1] = p2[i]
            * HB2 % M2;
        }
        pair<ll, ll> get(int l, int r) const { // [l, r]
            // inclusive, 0-based
            ll a = (h1[r + 1] - h1[l] * p1[r - l + 1]) % M1;
            ll b = (h2[r + 1] - h2[l] * p2[r - l + 1]) % M2;
            return {a < 0 ? a + M1 : a, b < 0 ? b + M2 : b};
        }
        bool eq(int l1, int r1, int l2, int r2) const {
            return r1 - l1 == r2 - l2 && get(l1, r1) == get(l2, r2);
        }
        // concat(A over lenA, B): h = hA * base^lenB + hB (works per
        // modulus)
    };
    // USES: substring equality  $O(1)$  * longest common prefix of two
    //  $\hookrightarrow$  suffixes by binary search
    // * dedup of substrings * 2D grid patterns (hash rows, then
    //  $\hookrightarrow$  hash the row-hashes)
    // * tree isomorphism (hash the sorted multiset of child hashes)
```

7.2.2 Hashing2D

2D HASHING - $O(1)$ hash of any submatrix, for grid pattern matching.

```
// 2D prefix hash with random bases b1 (down) and b2 (right), single
//  $\hookrightarrow$  62-bit modulus via __int128.
ll HB2D1 = 0, HB2D2 = 0; // ONE base pair
// per RUN, at file scope.
struct Hash2D { // Per-object
    // bases would make two grids
    static const ll M = (ll)1e18 + 9; // incomparable
    // - and "find this pattern in
    int n, m; // that grid"
    // needs exactly two objects.
    ll b1, b2;
    vector<ll> P1, P2;
    vector<vector<ll>> h;

    static ll mul(ll a, ll b) { return (ll)((__int128)a * b % M); }
    Hash2D(const vector<string>& g) : n(g.size()), m(n ? g[0].size()
    : 0),
        P1(n + 1, 1), P2(m + 1, 1), h(n
        + 1, vector<ll>(m + 1, 0)) {
        if (!HB2D1) {
            mt19937_64
            rng(chrono::steady_clock::now().time_since_epoch().count());
            HB2D1 = 256 + rng() % (M - 300), HB2D2 = 256 + rng() % (M
            - 300);
        }
        b1 = HB2D1, b2 = HB2D2;
        for (int i = 1; i <= n; i++) P1[i] = mul(P1[i - 1], b1);
        for (int j = 1; j <= m; j++) P2[j] = mul(P2[j - 1], b2);
        for (int i = 1; i <= n; i++)
            for (int j = 1; j <= m; j++) {
                __int128 v = (__int128)mul(h[i - 1][j], b1) +
                mul(h[i][j - 1], b2)
                - mul(mul(h[i - 1][j - 1], b1), b2) + g[i
                - 1][j - 1];
                h[i][j] = (ll)((v % M + M) % M);
            }
        // hash of the submatrix with top-left (r, c) and size dr x dc,
        // 0-based
        ll get(int r, int c, int dr, int dc) const {
            int r2 = r + dr, c2 = c + dc;
            __int128 v = (__int128)h[r2][c2] - mul(h[r][c2], P1[dr]) -
            mul(h[r2][c], P2[dc])
            + mul(mul(h[r][c], P1[dr]), P2[dc]);
            return (ll)((v % M + M) % M);
        }
    };
    // USES: count occurrences of a pattern grid * largest common square
    //  $\hookrightarrow$  submatrix (binary search the
    // side, hash all windows of both grids) * count distinct  $k \times k$ 
    //  $\hookrightarrow$  submatrices.
    // CHEAPER when the pattern is fixed: 1D-hash each row of the
    //  $\hookrightarrow$  pattern, then run KMP /
    // Aho-Corasick over the rows treated as single symbols.
```

7.2.3 HashingWithUpdates

HASHING UNDER POINT UPDATES, forward AND backward - the structure that answers "is s[l..r] a

```
// palindrome" while characters keep changing. Static prefix hashes
    ↪ cannot survive an update, and
// Manacher / eertree cannot be updated at all, so this is the only
    ↪ tool for that query shape.
// Two Fenwick trees over the modular values s[i]*p^i and
    ↪ s[i]*p^(n+1-i): point update O(log n),
// substring hash O(log n), palindrome test O(log n), longest
    ↪ palindrome at a centre O(log^2 n).
// MODULUS 2^61 - 1 with 128-bit multiplication: one modulus of that
    ↪ size is safer than two
// 32-bit ones (birthday bound ~ q^2 / 2^61) and shorter to write.
// INDEX CONVENTION: 1-INDEXED, s[1..n], and every l, r below is
    ↪ inclusive 1-based. get(l, r) is
// scaled so that equal substrings of EQUAL LENGTH compare equal -
    ↪ never compare across lengths.
struct DynHash {
    static const unsigned long long MD = (1ULL << 61) - 1;
    static unsigned long long mul(unsigned long long a, unsigned long
    ↪ long b) {
        __int128 c = (__int128)a * b;
        unsigned long long lo = (unsigned long long)(c & MD), hi =
    ↪ (unsigned long long)(c >> 61);
        lo += hi;
        return lo >= MD ? lo - MD : lo;
    }
    static unsigned long long ad(unsigned long long a, unsigned long
    ↪ long b) {
        return a + b >= MD ? a + b - MD : a + b;
    }
    static unsigned long long sb(unsigned long long a, unsigned long
    ↪ long b) {
        return a >= b ? a - b : a + MD - b;
    }
    static unsigned long long pw(unsigned long long b, unsigned long
    ↪ long e) {
        unsigned long long r = 1;
        for (; e >= 1, b = mul(b, b)) if (e & 1) r = mul(r, b);
        return r;
    }
    int n;
    string s;
    ↪ s[1..n]
    vector<unsigned long long> P, IP, ft[2];
    DynHash(const string& in) {
        n = in.size(), s = string(n + 1, 0);
        ↪
    ↪ s[0] unused; filled by upd below
        unsigned long long base = 131 + (unsigned long
    ↪ long)chrono::steady_clock::now()

    ↪ .time_since_epoch().count() % 1000000;
        P.resize(n + 2), IP.resize(n + 2), ft[0].assign(n + 1, 0),
    ↪ ft[1].assign(n + 1, 0);
        P[0] = IP[0] = 1;
        unsigned long long ib = pw(base, MD - 2);
        for (int i = 1; i <= n + 1; i++) P[i] = mul(P[i - 1], base),
    ↪ IP[i] = mul(IP[i - 1], ib);
        for (int i = 1; i <= n; i++) upd(i, in[i - 1]);
    }
    void add(int k, int i, unsigned long long v) {
        for (; i <= n; i += i & -i) ft[k][i] = ad(ft[k][i], v);
    }
    unsigned long long qr(int k, int i) {
        unsigned long long r = 0;
        for (; i > 0; i -= i & -i) r = ad(r, ft[k][i]);
        return r;
    }
    unsigned long long rng(int k, int l, int r) { return sb(qr(k, r),
    ↪ qr(k, l - 1)); }
    void upd(int i, char c) {
        ↪ set
    ↪ s[i] = c
        unsigned long long d = sb(c, s[i]);
        add(0, i, mul(d, P[i])), add(1, i, mul(d, P[n + 1 - i]));
        s[i] = c;
    }
    unsigned long long get(int l, int r) { return mul(rng(0, l, r),
    ↪ IP[l]); }
    unsigned long long rev(int l, int r) { return mul(rng(1, l, r),
    ↪ IP[n + 1 - r]); }
    bool isPal(int l, int r) { return get(l, r) == rev(l, r); }
    int oddRadius(int c) {
        ↪
    ↪ longest s[c-k..c+k] palindrome
        int lo = 0, hi = min(c - 1, n - c), res = 0;
        while (lo <= hi) {
            int m = (lo + hi) / 2;
```

```
        isPal(c - m, c + m) ? (res = m, lo = m + 1) : hi = m - 1;
        }
        return res;
        ↪ length 2 * res + 1
    }
    int evenRadius(int c) {
        ↪
    ↪ longest s[c-k+1..c+k] palindrome
        int lo = 1, hi = min(c, n - c), res = 0;
        while (lo <= hi) {
            int m = (lo + hi) / 2;
            isPal(c - m + 1, c + m) ? (res = m, lo = m + 1) : hi = m
        ↪ - 1;
        }
        return res;
        ↪ length 2 * res
    }
};
/* NOTES
THE SCALING IS THE WHOLE GAME. Forward stores s[i] * p^i, so the raw
    ↪ range sum of [l, r] is
    p^l times the hash of the substring - multiply by p^-l. Backward
    ↪ stores s[i] * p^(n+1-i), whose
    range sum is p^(n+1-r) times the reversed substring's hash. Get
    ↪ one exponent wrong and every
    answer is silently false; test isPal on a known palindrome first.
A SEGMENT TREE does the same with range ASSIGN ("set s[l..r] to c")
    ↪ lazily, which a Fenwick
    cannot: keep (hash, length) per node and lazily set hash = c *
    ↪ (p^len - 1) / (p - 1).
SUBSTRING COMPARISON / SORTING under updates: binary search the
    ↪ first differing position with
    get(), O(log^2 n) per comparison - the standard "compare two
    ↪ suffixes with updates".
LONGEST COMMON EXTENSION under updates is the same binary search,
    ↪ and is what makes this a
    drop-in replacement for a suffix array in dynamic problems.
ANTI-HASH: the base is randomised at construction. Never hard-code a
    ↪ base in an open-hacking
    contest; with a fixed base and mod 2^61-1 the string is still
    ↪ forgeable by a prepared test. */
```

7.3 Aho Corasick

7.3.1 AhoCorasick

AHO-CORASICK - a trie of all patterns plus suffix (fail) links, so one pass over the text finds

```
// every occurrence of every pattern in  $O(|\text{text}| + \text{total pattern length} + \text{\#matches})$ .
// The goto-automaton form (nxt filled in by BFS) is also the
// transition table for DP over strings.
struct Mapper {
    int operator()(char c) const { return c - 'a'; }
};

template <int ALPHABET, typename Mapper>
struct Aho {
    struct Node {
        array<int, ALPHABET> nxt;
        int link = 0;
        int terminal_idx = -1; // -1 if not a terminal, else idx of
        // string insertion
        int exit_link = 0; // Points to the nearest terminal suffix
        int depth = 0; // Size of the prefix the node represents
        Node() { nxt.fill(0); }
    };

    vector<Node> t;
    vector<int> bfs_order;
    Mapper get_idx; // Instance of the mapping function
    int word_count = 0; // Tracks the order of inserted strings

    // Constructor optionally accepts a custom mapper instance
    Aho(Mapper mapper = Mapper()) : get_idx(mapper) {
        t.emplace_back();
    }

    int insert(const string& p) {
        int u = 0;
        for (char c : p) {
            int v = get_idx(c);
            if (!t[u].nxt[v]) {
                t[u].nxt[v] = t.size();
                t.emplace_back();
                t.back().depth = t[u].depth + 1; // New node's depth
                // is parent's depth + 1
            }
            u = t[u].nxt[v];
        }

        // If this exact string hasn't been inserted before, assign
        // it the current index
        if (t[u].terminal_idx == -1) {
            t[u].terminal_idx = word_count;
        }
        word_count++; // Increment for the next insertion

        return u;
    }

    void build() {
        queue<int> q;
        for (int c = 0; c < ALPHABET; ++c) {
            if (t[0].nxt[c]) {
                q.push(t[0].nxt[c]);
            }
        }

        while (!q.empty()) {
            int u = q.front();
            q.pop();
            bfs_order.push_back(u);

            for (int c = 0; c < ALPHABET; ++c) {
                int v = t[u].nxt[c];
                if (v) {
                    t[v].link = t[t[u].link].nxt[c];

                    // Compute the exit link:
                    // If the immediate suffix link is a terminal,
                    // point to it.
                    // Otherwise, copy its exit link.
                    t[v].exit_link = (t[t[v].link].terminal_idx !=
                    // -1) ? t[t[v].link].exit_link;

                    q.push(v);
                }
                else {
                    t[u].nxt[c] = t[t[u].link].nxt[c];
                }
            }
        }
    }
};
```

```
    }
}

int advance(int u, char c) {
    return t[u].nxt[get_idx(c)];
}

};
```

7.3.2 AhoCorasickDP

DP OVER THE AHO-CORASICK AUTOMATON - the high-value use of Aho, more than matching itself.

```
// Needs: Mint (06 - Math/01) and AhoCorasick (01 - AhoCorasick.cpp).
// After build(), nxt[] is a total transition function, so the
// automaton is a DFA over states
// 0..S-1 and you can run any DP on (position, state, extra).
// "bad" marks states that contain a forbidden pattern as a suffix;
// propagate it along exit links.
struct AhoDP {
    static const int A = 26; // trie fan-out
    int alpha = 26; // alphabet the DP
    // counts over (set smaller!)
    vector<array<int, A>> nxt;
    vector<int> link, term;
    vector<char> bad;
    vector<int> order; // BFS order
    // of states

    AhoDP() { newNode(); }
    int newNode() { nxt.push_back({}); nxt.back().fill(0);
    // link.push_back(0); term.push_back(0); bad.push_back(0); return
    // nxt.size() - 1; }
    void insert(const string& p, int id = 1) {
        int u = 0;
        for (char c : p) {
            int v = c - 'a';
            if (!nxt[u][v]) nxt[u][v] = newNode();
            u = nxt[u][v];
        }
        term[u] = id, bad[u] = 1;
    }
    void build() {
        queue<int> q;
        for (int c = 0; c < A; c++) if (nxt[0][c]) q.push(nxt[0][c]);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            order.push_back(u);
            bad[u] = bad[u] || bad[link[u]]; // a state is
            // bad if any SUFFIX is a pattern
            for (int c = 0; c < A; c++) {
                int v = nxt[u][c];
                if (v) link[v] = nxt[link[u]][c], q.push(v);
                else nxt[u][c] = nxt[link[u]][c]; // total
            }
            // transition function
        }
    }
    // #strings of length L over the first 'alpha' letters containing
    // NO pattern. O(L*S*alpha).
    Mint countAvoiding(int L) {
        int S = nxt.size();
        vector<Mint> dp(S, 0), nd;
        dp[0] = 1;
        for (int i = 0; i < L; i++) {
            nd.assign(S, 0);
            for (int u = 0; u < S; u++) if (!bad[u] && dp[u].v)
                for (int c = 0; c < alpha; c++) {
                    int v = nxt[u][c];
                    if (!bad[v]) nd[v] += dp[u];
                }
            dp = nd;
        }
        Mint r = 0;
        for (int u = 0; u < S; u++) if (!bad[u]) r += dp[u];
        return r;
    }
    // #occurrences of every pattern in a text: walk the text, count
    // visits per state, then push
    // the counts DOWN the suffix-link tree (reverse BFS order).
    vector<ll> occurrences(const string& s) {
        vector<ll> cnt(nxt.size(), 0);
        int u = 0;
        for (char c : s) u = nxt[u][c - 'a'], cnt[u]++;
        for (int i = order.size() - 1; i >= 0; i--)
            cnt[link[order[i]]] += cnt[order[i]];
        return cnt; // cnt[state]
    }
    // of pattern p = #occurrences
};
// SHORTEST / LEXICOGRAPHICALLY SMALLEST string avoiding all
// patterns: BFS over (state) with
// the alphabet in order. If a cycle among good states is reachable,
// arbitrarily long strings
// exist - detect it with a cycle check on the good-state subgraph.
```

7.3.3 DynamicAhoCorasick

DYNAMIC (ONLINE) AHO-CORASICK - the pattern set grows while queries arrive. A plain

```
// Aho-Corasick must be rebuilt from scratch after every insertion,
// which is O(total length) per
// insert; here the patterns are kept in O(log P) STATIC automata of
// sizes 1, 2, 4, 8, ... and
// inserting merges the small ones the way binary counter increments
// carry.
// AMORTISED O(|p| log P) per insert, O(|s| log P) per query (log P
// <= ~20 automata to feed).
// WHEN: "add a word, then count occurrences of all words in this
// text", online dictionary
// problems (CF 710F), and any offline-impossible interleaving of
// insert and query.
// DELETION: keep a SECOND identical structure of deleted patterns
// and subtract its answer.
// INDEX CONVENTION: node 0 is the root; cnt[v] is already the SUM
// over the whole suffix-link
// chain, so a query adds cnt[cur] once per text position and never
// walks exit links.
struct StaticAho { //
    // rebuilt wholesale, never patched
    vector<array<int, 26>> nxt;
    vector<int> link, cnt;
    StaticAho() { clear(); }
    void clear() { nxt.assign(1, {}), nxt[0].fill(0), link.assign(1,
    // 0), cnt.assign(1, 0); }
    void insert(const string& p) {
        int u = 0;
        for (char ch : p) {
            int c = ch - 'a';
            if (!nxt[u][c]) {
                nxt[u][c] = nxt.size();
                nxt.push_back({}), nxt.back().fill(0),
            }
            u = nxt[u][c];
        }
        cnt[u]++;
    }
    void build() { //
        // goto-automaton + cumulative cnt
        queue<int> q;
        for (int c = 0; c < 26; c++) if (nxt[0][c]) q.push(nxt[0][c]);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            cnt[u] += cnt[link[u]];
            for (int c = 0; c < 26; c++) {
                int v = nxt[u][c];
                if (v) link[v] = nxt[link[u]][c], q.push(v);
                else nxt[u][c] = nxt[link[u]][c];
            }
        }
    }
    ll count(const string& s) const { //
        // total occurrences of all patterns
        ll r = 0;
        int u = 0;
        for (char ch : s) u = nxt[u][ch - 'a'], r += cnt[u];
        return r;
    }
};
struct DynamicAho { //
    static const int B = 20;
    // supports 2^20 patterns
    vector<string> words[B];
    StaticAho aho[B];
    void insert(const string& p) {
        int lvl = 0;
        while (lvl < B && !words[lvl].empty()) lvl++;
        // the carry of a binary counter
        words[lvl].push_back(p);
        for (int i = 0; i < lvl; i++) {
            for (auto& w : words[i]) words[lvl].push_back(w);
            words[i].clear(), aho[i].clear();
        }
        aho[lvl].clear();
        for (auto& w : words[lvl]) aho[lvl].insert(w);
        aho[lvl].build();
    }
    ll count(const string& s) { //
        // occurrences of all patterns in s
        ll r = 0;
        for (int i = 0; i < B; i++) if (!words[i].empty()) r +=
        // aho[i].count(s);
        return r;
    }
};
```

```

    }
};
/* NOTES AND VARIANTS
WHY THE COST IS AMORTISED: a pattern is copied into a bigger
    ↪ automaton at most log P times, so
    the total rebuild work is  $O(\sum |p| * \log P)$  - exactly the same
    ↪ argument as binary lifting on
    a knapsack, or as the "logarithmic method" for making any static
    ↪ structure insert-only.
THE SAME WRAPPER makes ANY static structure dynamic (suffix arrays,
    ↪ convex hulls, kd-trees):
    keep buckets of size  $2^i$  and merge on carry. That is the reusable
    ↪ idea here.
PER-PATTERN counts instead of a total: store an id per terminal and
    ↪ walk the exit-link chain, or
    put the query positions on the fail tree of each bucket - the
    ↪ total stays  $O(\text{matches} \log P)$ .
MEMORY: 26 ints per node in every bucket. With big alphabets use a
    ↪ hash map per node or the
    sorted-children trick, otherwise 20 rebuilt copies will not fit.
IF ALL PATTERNS ARE KNOWN OFFLINE, do not use this - build one
    ↪ static automaton. */

```

7.3.4 PatternContainmentPoset

ALL-PAIR OCCURRENCE RELATION between the patterns themselves: for every ordered pair (i, j) ,

```

// does pattern i occur as a SUBSTRING of pattern j? That relation is
    ↪ a partial order (a poset),
// and problems then ask for its longest chain, its maximum antichain
    ↪ (Dilworth = minimum path
// cover = k - maximum bipartite matching, see 03 - Graph Theory), or
    ↪ its transitive reduction.
// CF 590E "Birthday" is the canonical use: pick the largest subset
    ↪ with no containments.
// NAIVE is  $O(k^2 * L)$  with a matcher per pair. This is  $O(\sum |p| * A)$ 
    ↪ for the automaton plus
//  $O(k^3 / 64)$  for the closure, and the closure is the whole trick:
// during one pass over  $p_j$  you record ONLY the NEAREST terminal
    ↪ suffix at each position (the
// exit link). Everything else follows by TRANSITIVITY, because
    ↪ "occurs in" is transitive.
// INDEX CONVENTION: patterns are 0-indexed in insertion order and
    ↪ must be DISTINCT (dedup first,
// otherwise  $p_i$  occurs in  $p_j$  and  $p_j$  in  $p_i$  and the poset
    ↪ degenerates).
// Needs: 07 - Strings/03 - Aho Corasick/01 - AhoCorasick.cpp
template <int K>                                // K =
    ↪ compile-time bound on the #patterns
vector<bitset<K>> containment(const vector<string>& p) {
    Aho<26, Mapper> ac;
    vector<int> node(p.size());
    for (size_t i = 0; i < p.size(); i++) node[i] = ac.insert(p[i]);
    ac.build();
    int k = p.size();
    vector<bitset<K>> in(k);                                //
    ↪ in[j][i] =  $p_i$  occurs in  $p_j$ 
    for (int j = 0; j < k; j++) {
        int u = 0;
        for (char c : p[j]) {
            u = ac.advance(u, c);
            // the nearest terminal suffix - skipping  $p_j$  itself,
            ↪ which ends at its own last
            // position and would otherwise hide the shorter terminal
            ↪ suffixes underneath it
            int tu = ac.t[u].terminal_idx;
            int v = (tu != -1 && tu != j) ? u : ac.t[u].exit_link;
            if (v && ac.t[v].terminal_idx != -1 &&
            ↪ ac.t[v].terminal_idx != j)
                in[j][ac.t[v].terminal_idx] = 1;
        }
    }
    vector<int> ord(k);                                //
    ↪ shorter patterns first: a proper
    iota(ord.begin(), ord.end(), 0);                    //
    ↪ substring is strictly shorter
    sort(ord.begin(), ord.end(), [&](int a, int b) { return
    ↪ p[a].size() < p[b].size(); });
    for (int j : ord) {                                //
        ↪ transitive closure
        auto cur = in[j];
        for (int i = 0; i < k; i++) if (cur[i] in[j] != in[i];
        }
    return in;
}
/* USING THE POSET
LARGEST SUBSET WITH NO CONTAINMENT (maximum antichain) = k -
    ↪ (minimum chain cover), and by
    Dilworth on the TRANSITIVELY CLOSED relation, minimum chain cover
    ↪ = k - maximum matching in
    the bipartite graph "i on the left, j on the right, edge if  $p_i$ 
    ↪ occurs in  $p_j$ ". Recover the
    actual antichain from the Konig vertex cover: keep the vertices
    ↪ NOT in the cover.
LONGEST CHAIN = longest path in the DAG = the usual  $O(k^2)$  DP on the
    ↪ closed relation.
COUNTING occurrences instead of deciding: replace the bitset with a
    ↪ fail-tree subtree sum -
    push +1 at every visited node of  $p_j$  and read the subtree sum of
    ↪ each terminal,  $O((L + k) \log)$ .
MEMORY:  $k * K / 8$  bytes.  $k = 750$  (the CF 590E bound) is 70 KB;  $k =$ 
    ↪  $1e5$  is not viable, and at
    that size the question is always something the fail tree answers
    ↪ directly.
IF THE PATTERNS REPEAT, dedup and remember the multiplicity; the
    ↪ poset is on DISTINCT strings. */

```

7.4 Suffix Structures

7.4.1 SuffixArray

SUFFIX ARRAY + LCP + sparse table for $O(1)$ LCE. Build is $O(n \log n)$ (prefix doubling with a radix

```
// sort), Kasai gives the LCP array in  $O(n)$ . Everything you can ask
    ↳ about substrings reduces to a
// range on this array - see 03 - SuffixArrayApplications.cpp.
struct SuffixArray {
    string S;
    // sa is the suffix array with the empty suffix being sa[0]
    // lcp[i] holds the lcp between sa[i], sa[i - 1]
    vector<int> logs, sa, lcp, rank;
    vector<vector<int>> table;

    SuffixArray() {};

    SuffixArray(string &s, int lim = 256) {
        S = s;
        int n = s.size() + 1, k = 0, a, b;
        vector<int> c(s.begin(), s.end() + 1), tmp(n), frq(max(n,
    ↳ lim));
        c.back() = 0; // 0 is less than any character
        sa = lcp = rank = tmp, iota(sa.begin(), sa.end(), 0);
        for (int j = 0, p = 0; p < n; j = max(1, j * 2), lim = p) {
            p = j, iota(tmp.begin(), tmp.end(), n - j);
            for (int i = 0; i < n; i++) {
                if (sa[i] >= j)
                    tmp[p++] = sa[i] - j;
            }

            fill(frq.begin(), frq.end(), 0);

            for (int i = 0; i < n; i++) frq[c[i]]++;
            for (int i = 1; i < lim; i++) frq[i] += frq[i - 1];
            for (int i = n; i--;) sa[--frq[c[tmp[i]]]] = tmp[i];

            swap(c, tmp), p = 1, c[sa[0]] = 0;
            for (int i = 1; i < n; i++)
                a = sa[i - 1], b = sa[i], c[b] = (tmp[a] == tmp[b] &&
    ↳ tmp[a + j] == tmp[b + j]) ? p - 1 : p++;
        }

        for (int i = 1; i < n; i++) rank[sa[i]] = i;
        for (int i = 0, j; i < n - 1; lcp[rank[i++]] = k)
            for (k && k--, j = sa[rank[i] - 1];
                s[i + k] == s[j + k];
                k++);

        void preLcp() {
            int n = S.size() + 1;
            logs = vector<int>(n + 5);
            for (int i = 2; i < n + 5; ++i) {
                logs[i] = logs[i / 2] + 1;
            }
            // level-major: sized by logs[n], not a hard-coded 20 (which
    ↳ is OUT OF BOUNDS at
            // |s| >= 2^20 - and Codeforces routinely gives |s| up to
    ↳ 1e6), and no dead columns.
            table = vector<vector<int>>(logs[n] + 1, vector<int>(n));
            for (int i = 0; i < n; ++i) table[0][i] = lcp[i];
            for (int j = 1; j <= logs[n]; ++j)
                for (int i = 0; i + (1 << j) <= n; ++i)
                    table[j][i] = min(table[j - 1][i], table[j - 1][i +
    ↳ (1 << (j - 1))]);
        }

        // ARGUMENTS ARE SUFFIX-ARRAY INDICES, not string positions. For
    ↳ two string positions
        // p and q use lce(p, q) below, which is what every application
    ↳ actually wants.
        int queryLcp(int i, int j) {
            if (i == j) return (int)S.size() - sa[i];
            if (i > j) swap(i, j);
            i++;
            int len = logs[j - i + 1];
            return min(table[len][i], table[len][j - (1 << len) + 1]);
        }

        int lce(int p, int q) { return queryLcp(rank[p], rank[q]); } //
    ↳ longest common extension
    };
```

7.4.2 SuffixAutomaton

SUFFIX AUTOMATON - $O(n)$ states/transitions, accepts exactly the substrings of s.

```
// st[v].len = longest string in v's endpos class link =
    ↳ suffix link (v's parent)
// cnt[v] = |endpos(v)| = #occurrences of every string in v
// distinct substrings = sum over v != root of (len[v] - len[link[v]])
struct SAM {
    struct S { int len = 0, link = -1; array<int, 26> nxt; S() {
        ↳ nxt.fill(-1); } };
    vector<S> st;
    vector<ll> cnt; // occurrence
    ↳ counts (after countOcc)
    vector<int> firstPos; // end position
    ↳ of the first occurrence
    int last = 0;
    bool counted = false; // countOcc() is
    ↳ idempotent

    SAM(int reserve = 0) { st.reserve(2 * reserve),
    ↳ st.push_back(S()), cnt.push_back(0), firstPos.push_back(0); }

    void extend(char ch) {
        int c = ch - 'a', cur = st.size();
        st.push_back(S()), cnt.push_back(1), firstPos.push_back(0);
        st[cur].len = st[last].len + 1, firstPos[cur] = st[cur].len -
    ↳ 1;
        int p = last;
        while (p != -1 && st[p].nxt[c] == -1) st[p].nxt[c] = cur, p =
    ↳ st[p].link;
        if (p == -1) st[cur].link = 0;
        else {
            int q = st[p].nxt[c];
            if (st[p].len + 1 == st[q].len) st[cur].link = q;
            else {
                int cl = st.size();
                st.push_back(st[q]), cnt.push_back(0),
    ↳ firstPos.push_back(firstPos[q]);
                st[cl].len = st[p].len + 1;
                while (p != -1 && st[p].nxt[c] == q) st[p].nxt[c] =
    ↳ cl, p = st[p].link;
                st[q].link = st[cur].link = cl;
            }
            last = cur;
        }
    }

    void build(const string& s) { for (char c : s) extend(c); }

    vector<int> byLen() const { // states sorted
    ↳ by len (counting sort)
        int n = st.size(), mx = st[0].len;
        for (auto& x : st) mx = max(mx, x.len);
        vector<int> c(mx + 2, 0), ord(n);
        for (auto& x : st) c[x.len]++;
        for (int i = 1; i <= mx; i++) c[i] += c[i - 1];
        for (int v = n - 1; v >= 0; v--) ord[--c[st[v].len]] = v;
        return ord; // increasing
    ↳ len; reverse for link-tree order
    }

    void countOcc() { // push counts
    ↳ down the suffix-link tree
        if (counted) return; // NOT
    ↳ idempotent without this guard
        counted = true;
        auto ord = byLen();
        for (int i = (int)ord.size() - 1; i > 0; i--) {
            int v = ord[i];
            if (st[v].link >= 0) cnt[st[v].link] += cnt[v];
        }
        cnt[0] = 0;
    }

    ll distinctSubstrings() const { // number of
    ↳ DISTINCT non-empty substrings
        ll r = 0;
        for (int v = 1; v < (int)st.size(); v++) r += st[v].len -
    ↳ st[st[v].link].len;
        return r;
    }

    ll totalOccurrences() { // sum over
    ↳ distinct substrings of #occurrences
        countOcc();
        ll r = 0;
        for (int v = 1; v < (int)st.size(); v++) r += cnt[v] *
    ↳ (st[v].len - st[st[v].link].len);
        return r;
    }

    int countOf(const string& p) { // #occurrences
    ↳ of p in s (call countOcc first)
```



```

    int v = 0;
    for (char ch : p) { v = st[v].nxt[ch - 'a']; if (v < 0)
    ↪ return 0; }
    return cnt[v];
}
// K-th SMALLEST distinct substring (k is 1-indexed). Needs
↪ 'paths' = #substrings from v.
vector<ll> paths;
void buildPaths() {
    paths.assign(st.size(), 1); // 1 counts the
    ↪ empty continuation
    auto ord = byLen();
    for (int i = (int)ord.size() - 1; i >= 0; i--) {
        int v = ord[i];
        paths[v] = 1;
        for (int c = 0; c < 26; c++) if (st[v].nxt[c] >= 0)
    ↪ paths[v] += paths[st[v].nxt[c]];
    }
    string kthSubstring(ll k) { // "" if fewer
    ↪ than k distinct substrings
        string r;
        int v = 0;
        while (k > 0) {
            int c = 0;
            for (; c < 26; c++) {
                int u = st[v].nxt[c];
                if (u < 0) continue;
                if (k <= paths[u]) { r += char('a' + c), v = u, k--;
    ↪ break; }
            }
            k -= paths[v];
        }
        if (c == 26) break;
    }
    return r;
}
};
// LONGEST COMMON SUBSTRING of s and t: build the SAM of s, then walk
↪ t keeping the current
// length; on a missing transition follow suffix links and clamp the
↪ length to st[v].len.
int longestCommonSubstring(SAM& A, const string& t) {
    int v = 0, l = 0, best = 0;
    for (char ch : t) {
        int c = ch - 'a';
        while (v && A.st[v].nxt[c] < 0) v = A.st[v].link, l =
    ↪ A.st[v].len;
        if (A.st[v].nxt[c] >= 0) v = A.st[v].nxt[c], l++;
        best = max(best, l);
    }
    return best;
}
// GENERALISED SAM (several strings): reset 'last = 0' before each
↪ string and, in extend(),
// if a transition already exists reuse/clone it instead of creating
↪ a new state.

```

7.4.3 SuffixArrayApplications

WHAT YOU ACTUALLY EXTRACT FROM A SUFFIX ARRAY.

```

// Convention of 01 - SuffixArray.cpp: sa has n+1 entries (sa[0] =
↪ the empty suffix),
// and lcp[i] = LCP(sa[i], sa[i-1]). Everything below assumes that.

// 1) NUMBER OF DISTINCT SUBSTRINGS = sum over i of (len of suffix i)
↪ - lcp with the previous.
ll distinctSubstrings(const SuffixArray& S) {
    ll n = S.S.size(), r = 0;
    for (ll i = 1; i <= n; i++) r += (n - S.sa[i]) - S.lcp[i];
    return r;
}

// 2) LONGEST REPEATED SUBSTRING = max lcp; the substring starts at
↪ sa[argmax].
pair<int, int> longestRepeated(const SuffixArray& S) { //
    ↪ {length, start position}
    int n = S.S.size(), bi = 0, best = 0;
    for (int i = 1; i <= n; i++) if (S.lcp[i] > best) best =
    ↪ S.lcp[i], bi = S.sa[i];
    return {best, bi};
}

// 3) OCCURRENCES of a pattern = the contiguous SA block whose
↪ suffixes start with it.
// Two binary searches, O(|P| log n).
pair<int, int> patternRange(const SuffixArray& S, const string& p) {
    ↪ // [lo, hi], empty if lo>hi
    int n = S.S.size();
    auto cmp = [&](int idx, const string& q) { return
    ↪ S.S.compare(idx, q.size(), q) < 0; };
    int lo = 1, hi = n, L = n + 1, R = n;
    while (lo <= hi) { int m = (lo + hi) / 2; cmp(S.sa[m], p) ? lo =
    ↪ m + 1 : (L = m, hi = m - 1); }
    lo = L, hi = n, R = L - 1; // R
    ↪ MUST be reset here:
    //
    ↪ otherwise an absent pattern
    //
    ↪ returns the range [L, n]
    while (lo <= hi) {
        int m = (lo + hi) / 2;
        S.S.compare(S.sa[m], p.size(), p) == 0 ? (R = m, lo = m + 1)
    ↪ : hi = m - 1;
    }
    return {L, R}; //
    ↪ count = R - L + 1
}

// 4) LONGEST COMMON SUBSTRING of two strings (named
↪ lcsViaSuffixArray to avoid clashing
// with the SAM version in 02 - SuffixAutomaton.cpp): build SA of
↪ a + sep + b (sep smaller than every
// real character), then take the max lcp over ADJACENT pairs
↪ coming from different sides.
int lcsViaSuffixArray(const string& a, const string& b) {
    string t = a + char(1) + b;
    SuffixArray S(t);
    int n = t.size(), best = 0, cut = a.size();
    auto side = [&](int p) { return p < cut ? 0 : (p == cut ? -1 :
    ↪ 1); };
    for (int i = 2; i <= n; i++) {
        int s1 = side(S.sa[i]), s2 = side(S.sa[i - 1]);
        if (s1 >= 0 && s2 >= 0 && s1 != s2) best = max(best,
    ↪ S.lcp[i]);
    }
    return best;
}

// 5) K-TH SMALLEST DISTINCT SUBSTRING: SA entry i contributes (n -
↪ sa[i]) - lcp[i] new
// substrings, all sharing the prefix of length lcp[i]. Prefix-sum
↪ those counts and binary
// search for k; inside the block the answer is a prefix of suffix
↪ sa[i].
// 6) LCS OF K STRINGS: concatenate with distinct separators, then
↪ slide a window over the SA
// that contains at least one suffix of every string; the answer
↪ is the max over windows of
// the minimum lcp inside (monotonic deque), O(n) after the SA.
// 7) COMPARE TWO SUBSTRINGS in O(1): let l = S.lce(i, j) (STRING
↪ positions - queryLcp takes
// SUFFIX-ARRAY indices, do not mix them up); if l >= both
// lengths they are equal, otherwise compare s[i+l] with s[j+l].

```

7.4.4 GeneralizedSAM

GENERALIZED SUFFIX AUTOMATON - one automaton accepting every substring of a SET of strings.

```
// Same size bound O(total length). The only change vs a plain SAM:
// ↳ when extending, the transition
// ↳ may ALREADY exist (a previous string walked here), so handle that
// ↳ case instead of blindly adding.
// Feeding strings one after another into a plain SAM is WRONG - it
// ↳ creates junk states.
struct GenSAM {
    struct S { int len = 0, link = -1; array<int, 26> nxt; S() {
        ↳ nxt.fill(-1); } };
    vector<S> st;
    GenSAM() { st.push_back(S()); }
    int clone(int q, int len) { int c = st.size();
        ↳ st.push_back(st[q]), st[c].len = len; return c; }
    int extend(int last, int c) {
        ↳ returns the new "last"
        if (st[last].nxt[c] != -1) {
            ↳ edge is already there
            int q = st[last].nxt[c];
            if (st[q].len == st[last].len + 1) return q;
            int cl = clone(q, st[last].len + 1);
            for (int p = last; p != -1 && st[p].nxt[c] == q; p =
        ↳ st[p].link) st[p].nxt[c] = cl;
            st[q].link = cl;
            return cl;
        }
        int cur = st.size();
        st.push_back(S()), st[cur].len = st[last].len + 1;
        int p = last;
        while (p != -1 && st[p].nxt[c] == -1) st[p].nxt[c] = cur, p =
        ↳ st[p].link;
        if (p == -1) st[cur].link = 0;
        else {
            int q = st[p].nxt[c];
            if (st[p].len + 1 == st[q].len) st[cur].link = q;
            else {
                int cl = clone(q, st[p].len + 1);
                for (; p != -1 && st[p].nxt[c] == q; p = st[p].link)
        ↳ st[p].nxt[c] = cl;
                st[q].link = st[cur].link = cl;
            }
        }
        return cur;
    }
    void add(const string& s) { int last = 0; for (char ch : s) last
        ↳ = extend(last, ch - 'a'); }
    vector<int> byLen() const {
        ↳ states in increasing len
        int n = st.size(), mx = 0;
        for (auto& x : st) mx = max(mx, x.len);
        vector<int> c(mx + 2, 0), ord(n);
        for (auto& x : st) c[x.len]++;
        for (int i = 1; i <= mx; i++) c[i] += c[i - 1];
        for (int v = n - 1; v >= 0; v--) ord[--c[st[v].len]] = v;
        return ord;
    }
    // HOW MANY OF THE k STRINGS CONTAIN each substring: mark the
    ↳ states each string walks through,
    // then push the marks up the link tree. k <= 64 here; for larger
    ↳ k use small-to-large sets, or
    // a bitset<K>, or process the strings in batches of 64 and add
    ↳ up the popcounts.
    vector<int> countStringsContaining(const vector<string>& v) {
        ↳ // requires v.size() <= 64
        vector<unsigned long long> mask(st.size(), 0);
        for (int i = 0; i < (int)v.size(); i++)
            for (int p = 0, j = 0; j < (int)v[i].size(); j++)
                p = st[p].nxt[v[i][j] - 'a'], mask[p] |= 1ULL << i;
        auto ord = byLen();
        for (int i = ord.size() - 1; i > 0; i--)
            mask[st[ord[i]].link] |= mask[ord[i]];
        vector<int> r(st.size());
        for (size_t i = 0; i < st.size(); i++) r[i] =
        ↳ __builtin_popcountll(mask[i]);
        return r;
        ↳ r[v] for every state v
    }
};
// LONGEST COMMON SUBSTRING OF k STRINGS: build the generalized SAM,
// ↳ take the deepest state with
// count == k; its longest string is len[v] (walk suffix links for
// ↳ the actual string).
// DISTINCT SUBSTRINGS OF THE UNION = sum over v != 0 of (len[v] -
// ↳ len[link[v]]), same formula.
// SUBSTRING PRESENT IN EXACTLY ONE STRING / IN AT LEAST k: filter by
```

↳ the same counts.
 // ALTERNATIVE when the strings are few: concatenate with distinct
 ↳ separators and use one SUFFIX
 // ARRAY + a sliding window over lcp. Shorter to write, $O(n \log n)$,
 ↳ but no online extension.

7.4.5 SuffixArraySAIS

SUFFIX ARRAY IN $O(n)$ - SA-IS. Use it instead of 01 - SuffixArray.cpp ($O(n \log n)$ doubling)

```
// when n is 1e6 or more, when the input is a vector<int> with a huge
// ↪ alphabet (doubling needs a
// ↪ radix sort per round), or when the doubling version is the
// ↪ bottleneck of a heavy solution.
// It is ~4x faster than doubling at n = 1e6 and needs no sorting of
// ↪ pairs at all.
// INDEX CONVENTION - DIFFERENT FROM 01 - SuffixArray.cpp, and mixing
// ↪ them up is a wrong answer:
// sa has EXACTLY n entries (no empty suffix); sa[i] = start of the
// ↪ i-th smallest suffix.
// lcp has n entries with lcp[0] = 0 and lcp[i] = LCP(sa[i-1],
// ↪ sa[i]).
// HOW IT WORKS (enough to debug it): classify every position as
// ↪ S-type (its suffix is smaller
// ↪ than the next one) or L-type. The LMS positions (an S right after
// ↪ an L) split the string into
// ↪ blocks; sort those blocks by recursing on a shorter string, then
// ↪ INDUCE the whole order from
// ↪ them with two linear scans. The recursion halves the length, so
// ↪ the total is  $O(n)$ .
vector<int> saIs(const vector<int>& s, int up) {
    // ↪ values in [0, up], up >= 0
    int n = s.size();
    if (n == 0) return {};
    if (n == 1) return {0};
    if (n == 2) return s[0] < s[1] ? vector<int>{0, 1} :
    // ↪ vector<int>{1, 0};
    vector<int> sa(n);
    vector<char> ls(n, 0);
    // ↪ 1 =
    // ↪ S-type
    for (int i = n - 2; i >= 0; i--) ls[i] = s[i] == s[i + 1] ? ls[i]
    // ↪ + 1 : s[i] < s[i + 1];
    vector<int> sumL(up + 2, 0), sumS(up + 2, 0);
    for (int i = 0; i < n; i++) ls[i] ? sumL[s[i] + 1]++ :
    // ↪ sumS[s[i]]++;
    for (int i = 0; i <= up; i++) { sumS[i] += sumL[i]; if (i < up)
    // ↪ sumL[i + 1] += sumS[i]; }
    auto induce = [&](const vector<int>& lms) {
        fill(sa.begin(), sa.end(), -1);
        vector<int> buf = sumS;
        for (int d : lms) if (d != n) sa[buf[s[d]]++] = d;
        buf = sumL;
        sa[buf[s[n - 1]]++] = n - 1;
        for (int i = 0; i < n; i++) {
            int v = sa[i];
            if (v >= 1 && !ls[v - 1]) sa[buf[s[v - 1]]++] = v - 1;
        }
        buf = sumL;
        for (int i = n - 1; i >= 0; i--) {
            int v = sa[i];
            if (v >= 1 && ls[v - 1]) sa[--buf[s[v - 1] + 1]] = v - 1;
        }
    };
    vector<int> lmsMap(n + 1, -1), lms;
    int m = 0;
    for (int i = 1; i < n; i++) if (!ls[i - 1] && ls[i]) lmsMap[i] =
    // ↪ m++, lms.push_back(i);
    induce(lms);
    if (m) {
        vector<int> sorted;
        for (int v : sa) if (lmsMap[v] != -1) sorted.push_back(v);
        vector<int> rec(m);
        int upRec = 0;
        rec[lmsMap[sorted[0]]] = 0;
        for (int i = 1; i < m; i++) {
            int l = sorted[i - 1], r = sorted[i];
            int el = lmsMap[l] + 1 < m ? lms[lmsMap[l] + 1] : n;
            int er = lmsMap[r] + 1 < m ? lms[lmsMap[r] + 1] : n;
            bool same = el - l == er - r;
            if (same) {
                while (l < el && s[l] == s[r]) l++, r++;
                if (l == n || s[l] != s[r]) same = false;
            }
            if (!same) upRec++;
            rec[lmsMap[sorted[i]]] = upRec;
        }
        vector<int> recSa = saIs(rec, upRec);
        for (int i = 0; i < m; i++) sorted[i] = lms[recSa[i]];
        induce(sorted);
    }
    return sa;
}
vector<int> suffixArray(const string& s) {
    // ↪ n
    // ↪ entries, no empty suffix
```

```
vector<int> a(s.begin(), s.end());
for (int& x : a) x = (unsigned char)x;
return saIs(a, 255);
}
// KASAI: lcp[i] = LCP(suffix sa[i-1], suffix sa[i]), lcp[0] = 0.
// ↪ 0(n).
vector<int> kasai(const string& s, const vector<int>& sa) {
    int n = s.size(), k = 0;
    vector<int> rnk(n), lcp(n, 0);
    for (int i = 0; i < n; i++) rnk[sa[i]] = i;
    for (int i = 0; i < n; i++) {
        if (!rnk[i]) { k = 0; continue; }
        int j = sa[rnk[i] - 1];
        while (i + k < n && j + k < n && s[i + k] == s[j + k]) k++;
        lcp[rnk[i]] = k;
        if (k) k--;
    }
    return lcp;
}
/* NOTES
GENERAL ALPHABETS: call saIs directly on a vector<int> after
    ↪ coordinate-compressing the values
    to [0, up]; every value must be >= 0 and the LAST element does NOT
    ↪ have to be a sentinel -
    this implementation handles the end of string internally.
CONCATENATING MANY STRINGS (generalised suffix array): join with
    ↪ distinct separators SMALLER
    than every real character, or - better with SA-IS - compress to
    ↪ ints and use 0, 1, 2, ... as
    separators, shifting real characters up.
EVERYTHING ELSE (distinct substrings, longest repeated, pattern
    ↪ range, LCE sparse table) is in
    03 - SuffixArrayApplications.cpp; shift its formulas by one
    ↪ because that file's sa[0] is the
    empty suffix and this one has no empty suffix.
MEMORY is ~3n ints at the top level plus the recursion, which sums
    ↪ to < 2x that. Doubling needs
    5 arrays of n and a radix sort - SA-IS also wins on memory.
DC3 / SKEW is the other linear construction; it is shorter to write
    ↪ but slower in practice and
    needs 3n extra memory, so SA-IS is the one to carry. */
```

7.4.6 IsomorphicSuffixArray

PARAMETERIZED / ISOMORPHIC SUFFIX ARRAY. Two strings are ISOMORPHIC (p-match) when some

```
// Needs: 07 - Strings/04 - Suffix Structures/05 -
  ↳ SuffixArraySAIS.cpp (saIs, kasai)
// BIJECTION of the alphabet turns one into the other: "abab" ~
  ↳ "cdcd" ~ "baba", but not "aabb".
// WHEN: "how many substrings of s are isomorphic to s[l..r]",
  ↳ detecting renamed patterns,
// register-allocation style matching. Plain hashing cannot do this -
  ↳ the equality is not on
// characters but on the PATTERN OF REPEATS.
// THE ENCODING that makes it work: a[i] = (i - previous position of
  ↳ s[i]) + 1, or 1 when s[i] is
// the first occurrence of its character. Two substrings are
  ↳ isomorphic iff their encodings match
// AFTER CAPPING: at offset L inside a substring the value is a[x] if
  ↳ a[x] <= L + 1, else 1 (a
// reference that points outside the window becomes "first
  ↳ occurrence"). The cap is why this is
// not just an ordinary suffix array on a[.].
// INDEX CONVENTION: sa has n entries, no empty suffix; sa[i] = start
  ↳ of the i-th smallest suffix
// UNDER ISOMORPHISM, and lcp[i] = longest common ISOMORPHIC prefix
  ↳ of sa[i] and sa[i+1] (note:
// indexed by the LEFT of the pair, unlike 01 - SuffixArray.cpp - one
  ↳ entry per adjacent pair).
// COMPLEXITY: O(n log n * A) worst case (A = alphabet), because one
  ↳ comparison may restart at
// every distinct character; in practice each comparison is a couple
  ↳ of jumps.
struct IsoSuffixArray {
    int n, LG;
    vector<int> a, sa, rnk, lcp, hsa, hrnk, hlcp, lg;
    vector<vector<int>> sp; //
    ↳ sparse table over hlcp
    IsoSuffixArray(const string& s, int A = 26, char base = 'a') {
        n = s.size(), a.resize(n);
        vector<int> last(A, -1);
        for (int i = 0; i < n; i++)
            a[i] = last[s[i] - base] < 0 ? 1 : i - last[s[i] - base]
    ↳ + 1, last[s[i] - base] = i;
        hsa = saIs(a, n + 1); //
    ↳ helper: plain SA of the encoding
        hrnk.resize(n);
        for (int i = 0; i < n; i++) hrnk[hsa[i]] = i;
        hlcp.assign(n, 0);
        for (int i = 0, k = 0; i < n; i++) { //
    ↳ Kasai on the integer array
            if (!hrnk[i]) { k = 0; continue; }
            int j = hsa[hrnk[i] - 1];
            while (i + k < n && j + k < n && a[i + k] == a[j + k])
    ↳ k++;
                hlcp[hrnk[i]] = k;
                if (k) k--;
            }
            lg.assign(n + 1, 0);
            for (int i = 2; i <= n; i++) lg[i] = lg[i / 2] + 1;
            LG = lg[n] + 1;
            sp.assign(LG, vector<int>(n, 0));
            sp[0] = hlcp;
            for (int k = 1; k < LG; k++)
                for (int i = 0; i + (1 << k) <= n; i++)
                    sp[k][i] = min(sp[k - 1][i], sp[k - 1][i + (1 << (k -
    ↳ 1))]);
            sa.resize(n, iota(sa.begin(), sa.end(), 0));
            sort(sa.begin(), sa.end(), [&](int i, int j) { return cmp(i,
    ↳ j); });
            rnk.resize(n);
            for (int i = 0; i < n; i++) rnk[sa[i]] = i;
            lcp.assign(max(0, n - 1), 0);
            for (int i = 0; i + 1 < n; i++) lcp[i] = lce(sa[i], sa[i +
    ↳ 1]);
        }
        int rawLce(int i, int j) { // lcp
    ↳ of a[i..] and a[j..]
            if (i == j) return n - i;
            int l = hrnk[i], r = hrnk[j];
            if (l > r) swap(l, r);
            int k = lg[r - l];
            return min(sp[k][l + 1], sp[k][r - (1 << k) + 1]);
        }
        int lce(int i, int j) { //
    ↳ longest common ISOMORPHIC prefix
            for (int L = 0;;) {
                int x = i + L, y = j + L;
                if (x >= n || y >= n) return L;

```

```
int vx = a[x] > L + 1 ? 1 : a[x], vy = a[y] > L + 1 ? 1 :
    ↳ a[y];
    if (vx != vy) return L;
    L += max(1, rawLce(x, y)); // raw
    ↳ equality implies iso equality
    }
}
bool cmp(int i, int j) {
    int L = lce(i, j);
    if (i + L >= n) return true;
    if (j + L >= n) return false;
    int vx = a[i + L] > L + 1 ? 1 : a[i + L], vy = a[j + L] > L +
    ↳ 1 ? 1 : a[j + L];
    return vx < vy;
}
// Number of substrings isomorphic to s[p .. p+len-1]: the SA
    ↳ block of suffixes whose
    ↳ isomorphic prefix of that length matches. O(log^2 n) with the
    ↳ lcp array as a min-sparse
    ↳ table; the linear scan below is fine when len is large or
    ↳ queries are few.
pii occurrences(int p, int len) {
    int lo = rnk[p], hi = rnk[p];
    while (lo > 0 && lcp[lo - 1] >= len) lo--;
    while (hi + 1 < n && lcp[hi] >= len) hi++;
    return {lo, hi}; //
    ↳ count = hi - lo + 1
}
};
/* NOTES
THE OTHER ENCODINGS you may meet: "previous occurrence index" (0 if
    ↳ none) is the same idea;
    ↳ "order of first appearance" (abcbab -> 01201) is NOT enough for
    ↳ substrings, because a substring
    ↳ restarts the numbering - that is exactly what the cap fixes here.
HASHING ALTERNATIVE: hash the capped encoding with a segment tree
    ↳ over "distance to previous
    ↳ occurrence" and answer isomorphism of two substrings in O(log n) -
    ↳ shorter if you only need
    ↳ EQUALITY tests and not an order.
STRUCTURAL / ORDER-ISOMORPHISM (the same RELATIVE ORDER of numbers,
    ↳ not equality pattern) is a
    ↳ different problem: encode each position by (rank among the
    ↳ previous window) and use a similar
    ↳ suffix structure, or hash the pair (previous smaller, previous
    ↳ larger) positions.
THE COMPARATOR IS NOT TRANSITIVE-SAFE if the cap is wrong; test on
    ↳ "aabb" vs "bbaa" (isomorphic)
    ↳ and "aabb" vs "abab" (not) before trusting a modified version. */

```

7.4.7 DistinctSubstringsInRange

NUMBER OF DISTINCT SUBSTRINGS OF $s[l..r]$, for q queries, OFFLINE in $O((n + q) \log n)$.

```
// Needs: 07 - Strings/04 - Suffix Structures/02 - SuffixAutomaton.cpp
// For the whole string this is one line off a suffix automaton or a
//   ↳ suffix array; for an
// arbitrary WINDOW it is a genuinely hard problem, and this is the
//   ↳ standard solution
// (HackerRank "How Many Substrings", CF gym 102129I).
// INDEX CONVENTION: 0-based inclusive queries  $[l, r]$ ; internally
//   ↳ everything is 1-based over
// positions  $1..n$ , and LCT node id = SAM state id + 1 (id 0 is the
//   ↳ null node).
// THE INVARIANT: sweep  $r$  from left to right and keep, in a Fenwick
//   ↳ with range add and range sum,
//   val[p] = number of distinct substrings whose LAST occurrence
//   ↳ in  $s[1..r]$  STARTS at p.
// Then the answer for  $[l, r]$  is sum of val[l..r] - every distinct
//   ↳ substring of the window is
// counted at the start position of its last occurrence, which is
//   ↳ inside the window exactly when
// the substring occurs in the window at all.
// MAINTAINING IT is the trick: appending  $s[r]$  creates  $r$  new
//   ↳ occurrences (the suffixes of the
// prefix), so add +1 to every position  $1..r$ , and then REMOVE the
//   ↳ previous last-occurrence entries
// of exactly those substrings. Those substrings are the ones on the
//   ↳ suffix-link path from the
// state of prefix  $r$  to the root - and their old "last occurrence"
//   ↳ positions are constant along
// each PREFERRED PATH of a link-cut tree. So maintain the
//   ↳ suffix-link tree in an LCT where
// access(state, r) PAINTS the path with the colour  $r$ : every
//   ↳ preferred-child change reports one
// maximal group (a range of lengths that shared an old colour),
//   ↳ which is one range update.
// That "access as painting" pattern is reusable: any "assign a
//   ↳ colour to a root path, and pay
// only for the  $O(\log n)$  amortised colour groups you destroy" problem
//   ↳ is this code.
struct DistinctSubstringsInRange {
    int n = 0, sz = 0;
    vector<int> par, lz, col, len;
    vector<array<int, 2>> ch;
    vector<pii> mods; //
    ↳ {segment top len, old colour}
    vector<ll> b1, b2; //
    ↳ range-add range-sum Fenwick
    void bAdd(vector<ll>& b, int i, ll v) { for (; i <= n; i += i &
    ↳ -i) b[i] += v; }
    ll bSum(const vector<ll>& b, int i) const {
        ll r = 0;
        for (; i > 0; i -= i & -i) r += b[i];
        return r;
    }
    void rangeAdd(int l, int r, ll v) { // add
    ↳ v to positions [l, r]
        if (l > r) return;
        bAdd(b1, l, v), bAdd(b2, l, v * (l - 1));
        if (r + 1 <= n) bAdd(b1, r + 1, -v), bAdd(b2, r + 1, -v * r);
    }
    ll pref(int i) const { return bSum(b1, i) * i - bSum(b2, i); }
    ll rangeSum(int l, int r) const { return l > r ? 0 : pref(r) -
    ↳ pref(l - 1); }
    void mark(int x, int v) { lz[x] = col[x] = v; }
    void push(int x) {
        if (!lz[x]) return;
        if (ch[x][0]) mark(ch[x][0], lz[x]);
        if (ch[x][1]) mark(ch[x][1], lz[x]);
        lz[x] = 0;
    }
    bool isRoot(int x) { return ch[par[x]][0] != x && ch[par[x]][1]
    ↳ != x; }
    void rotate(int x) {
        int y = par[x], z = par[y], k = ch[y][1] == x, w = ch[x][!k];
        if (!isRoot(y)) ch[z][ch[z][1] == y] = x;
        ch[x][!k] = y, ch[y][k] = w;
        if (w) par[w] = y;
        par[y] = x, par[x] = z;
    }
    void splay(int x) {
        static vector<int> st;
        st.clear(), st.push_back(x);
        for (int i = x; !isRoot(i); i = par[i]) st.push_back(par[i]);
        while (!st.empty()) push(st.back()), st.pop_back();
        while (!isRoot(x)) {
            int y = par[x], z = par[y];
```

```
        if (!isRoot(y)) rotate((ch[y][1] == x) == (ch[z][1] == y)
    ↳ ? y : x);
        rotate(x);
    }
    void access(int x, int v) { //
    ↳ paint root..x with colour v
        mods.clear();
        for (int z = 0; x; z = x, x = par[x]) {
            splay(x);
            mods.push_back({len[x] - 1, col[x]}); //
    ↳ this preferred group dies now
            ch[x][1] = z;
            mark(x, v);
        }
    }
    vector<ll> solve(const string& s, const vector<pii>& queries) {
        n = s.size();
        SAM sam(n);
        vector<int> pos(n + 1, 1);
        for (int i = 0; i < n; i++) sam.extend(s[i]), pos[i + 1] =
    ↳ sam.last + 1;
        sz = sam.st.size();
        par.assign(sz + 1, 0), lz.assign(sz + 1, 0), col.assign(sz +
    ↳ 1, 0);
        ch.assign(sz + 1, {0, 0}), len.assign(sz, 0);
        for (int i = 0; i < sz; i++) len[i] = sam.st[i].len;
        for (int i = 1; i <= sz; i++) par[i] = sam.st[i - 1].link + 1;
        b1.assign(n + 2, 0), b2.assign(n + 2, 0);
        vector<vector<pii>> byR(n + 1); // {l,
    ↳ query id}, 1-based r
        for (int i = 0; i < (int)queries.size(); i++)
            byR[queries[i].second + 1].push_back({queries[i].first +
    ↳ 1, i});
        vector<ll> ans(queries.size(), 0);
        for (int r = 1; r <= n; r++) {
            rangeAdd(1, r, 1); //
    ↳ every suffix of s[1..r] is new
            access(pos[r], r);
            int prevLen = 0;
            for (int j = (int)mods.size() - 1; j >= 1; j--) { //
    ↳ root-most group first
                auto [topLen, oldCol] = mods[j];
                if (!topLen) continue; // the
    ↳ root state (empty string)
                if (oldCol) rangeAdd(oldCol - topLen + 1, oldCol -
    ↳ prevLen, -1);
                prevLen = topLen;
            }
            for (auto [l, id] : byR[r]) ans[id] = rangeSum(l, r);
        }
        return ans;
    }
};
/* NOTES
WHY THE LENGTHS LINE UP: a preferred group is a set of
    ↳ suffix-link-tree nodes whose strings all
    had the same last-occurrence end position 'oldCol', and the group
    ↳ covers exactly the lengths
    (prevLen, topLen]. Those substrings therefore started at positions
    ↳ oldCol - topLen + 1 ..
    oldCol - prevLen, which is the single range you cancel. Get the
    ↳ direction of prevLen wrong and
    the answers drift slowly - test against a brute force on  $n \leq 12$ 
    ↳ before using it.
THE SAM IS BUILT FIRST, in full, and the LCT is built over the FINAL
    ↳ suffix-link tree; no links
    or cuts ever happen, only access(). Clones created later do not
    ↳ break the sweep because the
    path from the state of prefix  $r$  to the root always lists exactly
    ↳ the suffixes of that prefix.
COMPLEXITY:  $O(n \log n)$  amortised for the accesses (the number of
    ↳ preferred-child changes is
     $O(n \log n)$ ), each producing one  $O(\log n)$  Fenwick range update.
ONLINE variant: replace the Fenwick with a PERSISTENT one, keeping a
    ↳ root per  $r$ ; queries then
    need no sorting. Memory becomes  $O(n \log^2 n)$ .
RELATED, MUCH EASIER: distinct substrings of the WHOLE string (sum
    ↳ of len - len[link] over SAM
    states), and distinct VALUES in a range (the classic
    ↳ BIT-over-last-occurrence sweep, which is
    this same idea with one position per value instead of a whole
    ↳ path). */
```

7.5 Palindromes

7.5.1 Manacher

MANACHER - the radius of the longest palindrome centred at every position, in $O(n)$.

```
// Answers "is s[l..r] a palindrome" in  $O(1)$  and counts all
// palindromic substrings in one pass.
struct Manacher {
    vector<int> p;

    // Returns a vector where p[i] is the radius of the longest
    // palindrome around center i
    // Centers are interleaved: 0 is before string, 1 is first char,
    // 2 is between 1st & 2nd, etc.
    Manacher(string s) {
        string t = "#";
        for (char c : s) {
            t += c;
            t += "#";
        }
        int n = t.size();
        p.assign(n, 0);
        int c = 0, r = 0;
        for (int i = 0; i < n; i++) {
            int mirror = 2 * c - i;
            if (i < r) p[i] = min(r - i, p[mirror]);
            while (i - 1 - p[i] >= 0 && i + 1 + p[i] < n && t[i - 1 -
                p[i]] == t[i + 1 + p[i]]) {
                p[i]++;
            }
            if (i + p[i] > r) {
                c = i;
                r = i + p[i];
            }
        }

        // Check if substring [l, r] (0-based) is a palindrome in  $O(1)$ 
        bool is_palindrome(int l, int r) {
            int len = r - l + 1;
            int center = l + r + 1;
            return p[center] >= len;
        }
    };
};
```

7.5.2 PalindromicTree

PALINDROMIC TREE (eertree) - $O(n)$ nodes, one per DISTINCT palindromic substring, built online in

```
//  $O(n)$ . Gives the count of distinct palindromes, occurrence counts,
// and the longest palindromic
// suffix at every prefix - which is what palindromic-factorization
// DP needs.
struct PalindromeTree {
    int n, id, cur, tot;
    vector<array<int, 26>> go;

    // Core structural links
    vector<int> suflink, len, cnt;

    // Extensions for Hard DP (Palindrome Partitioning)
    vector<int> diff; // Difference in length to the suffix link
    vector<int> slink; // Series link (skips arithmetic progressions
    // of suffix links)

    // Extensions for counting
    vector<int> num; // Number of palindromic suffixes ending at
    // this node
    long long total_overlapping; // Running sum of total palindromes

    PalindromeTree() {}

    PalindromeTree(const string &s) {
        build(s);
    }

    void init(int max_len) {
        n = max_len;
        go.assign(n + 2, {});
        suflink.assign(n + 2, 0);
        len.assign(n + 2, 0);
        cnt.assign(n + 2, 0);
        diff.assign(n + 2, 0);
        slink.assign(n + 2, 0);
        num.assign(n + 2, 0);

        suflink[0] = suflink[1] = 1;
        len[1] = -1;
        id = 2;
        cur = 0;
        tot = 0;
        total_overlapping = 0;
    }

    int get(const string &s, int i, int v) {
        while (i - len[v] - 1 < 0 || s[i - len[v] - 1] != s[i]) {
            v = suflink[v];
        }
        return v;
    }

    void add(const string &s, int i) {
        int ch = s[i] - 'a';
        cur = get(s, i, cur);

        if (go[cur][ch] == 0) {
            len[id] = 2 + len[cur];
            suflink[id] = go[get(s, i, suflink[cur])][ch];

            // 1. Number of palindromes ending exactly here
            num[id] = num[suflink[id]] + 1;

            // 2. Arithmetic Progression / Series Links for DP
            diff[id] = len[id] - len[suflink[id]];
            if (diff[id] == diff[suflink[id]]) {
                slink[id] = slink[suflink[id]];
            } else {
                slink[id] = suflink[id];
            }

            tot++;
            go[cur][ch] = id++;
        }
        cur = go[cur][ch];
        cnt[cur]++;
        total_overlapping += num[cur];
    }

    // Standard build
    void build(const string &s) {
        init(s.length());
        for (int i = 0; i < n; i++) {
            add(s, i);
        }
    }
};
```



```

    }
    countAll();
}

// Call this to push frequencies down the suffix links
void countAll() {
    for (int i = id - 1; i >= 2; --i) {
        cnt[suflink[i]] += cnt[i];
    }
}

// --- Black-Box Solvers ---

// 1. Returns number of unique palindromes
int cntDistinct() {
    return tot;
}

// 2. Returns total number of palindromes (including
    ↪ overlaps/duplicates)
long long cntTotal() {
    return total_overlapping;
}

// 3. Solves Minimum Palindrome Partitioning in O(N log N)
// Rebuilds the tree internally to compute the DP concurrently.
int min_partitions(const string &s) {
    init(s.length());
    const int INF = 1e9;
    vector<int> dp(s.length() + 1, INF);
    vector<int> series_ans(s.length() + 2, INF);

    dp[0] = 0;

    for (int i = 0; i < s.length(); ++i) {
        add(s, i);

        for (int v = cur; v > 1; v = slink[v]) {
            series_ans[v] = dp[i + 1 - (len[slink[v]] + diff[v])];

            if (diff[v] == diff[suflink[v]]) {
                series_ans[v] = min(series_ans[v],
                    ↪ series_ans[suflink[v]]);
            }

            dp[i + 1] = min(dp[i + 1], series_ans[v] + 1);
        }
        return dp[s.length()];
    }

// 4. Returns an array of frequencies for string B's palindromes
    ↪ inside this tree (String A)
// Note: You must 'build()' the tree with string A before calling
    ↪ this with B.
vector<int> count_string(const string& B) {
    vector<int> out_cnt(id, 0);
    int curr = 0;

    for (int i = 0; i < B.length(); ++i) {
        int ch = B[i] - 'a';

        while (true) {
            if (i - len[curr] - 1 >= 0 && B[i - len[curr] - 1] ==
                ↪ B[i]) {
                if (go[curr][ch]) {
                    curr = go[curr][ch];
                    break;
                }
            }
            if (curr == 1) {
                curr = 0;
                break;
            }
            curr = suflink[curr];
        }
        out_cnt[curr]++;

        for (int i = id - 1; i >= 2; --i) {
            out_cnt[suflink[i]] += out_cnt[i];
        }
        return out_cnt;
    }
};

```

7.5.3 PalindromicTreeRollback

PALINDROMIC TREE WITH ROLLBACK - same structure, but every extend() can be undone, so you can DFS

```

// NOTE: 02 - PalindromicTree.cpp declares the same 'struct
    ↪ PalindromeTree'. Paste one.
// a trie/tree of characters and maintain the palindromic structure
    ↪ of the current root-to-node string.
struct PalindromeTree {
    int n, id, cur, tot;
    vector<array<int, 26>> go;

    // Core structural links
    vector<int> suflink, len, cnt;

    // Extensions for Hard DP (Palindrome Partitioning)
    vector<int> diff; // Length difference to suffix link
    vector<int> slink; // Series link

    // Extensions for counting
    vector<int> num; // Number of palindromic suffixes ending at
    ↪ this node
    long long total_overlapping; // Running total of all palindromic
    ↪ substrings

    // --- ROLLBACK EXTENSIONS ---
    string S; // Maintains current active string
    struct History {
        int old_cur;
        int added_node; // ID of created node, or 0 if edge already
        ↪ existed
        int parent_node; // Node where go[parent_node][ch] was updated
        int ch; // Character index (0..25)
    };
    vector<History> history;

    PalindromeTree() {}

    PalindromeTree(int max_len) {
        init(max_len);
    }

    void init(int max_len) {
        n = max_len;
        go.assign(n + 2, {});
        suflink.assign(n + 2, 0);
        len.assign(n + 2, 0);
        cnt.assign(n + 2, 0);
        diff.assign(n + 2, 0);
        slink.assign(n + 2, 0);
        num.assign(n + 2, 0);

        suflink[0] = suflink[1] = 1;
        len[1] = -1;
        id = 2;
        cur = 0;
        tot = 0;
        total_overlapping = 0;

        S.clear();
        history.clear();
    }

    int get(int i, int v) {
        while (i - len[v] - 1 < 0 || S[i - len[v] - 1] != S[i]) {
            v = suflink[v];
        }
        return v;
    }

    // 1. PUSH / ADD CHARACTER (WITH ROLLBACK HISTORY)
    void push_back(char c) {
        S.push_back(c);
        int i = (int)S.length() - 1;
        int ch = c - 'a';

        int old_cur = cur;
        cur = get(i, cur);

        int added_node = 0;
        int parent_node = 0;

        if (go[cur][ch] == 0) {
            added_node = id;
            parent_node = cur;

            len[id] = 2 + len[cur];
            suflink[id] = go[get(i, suflink[cur])][ch];

```

```

    num[id] = num[suflink[id]] + 1;

    diff[id] = len[id] - len[suflink[id]];
    if (diff[id] == diff[suflink[id]]) {
        slink[id] = slink[suflink[id]];
    } else {
        slink[id] = suflink[id];
    }

    tot++;
    go[cur][ch] = id++;
}

cur = go[cur][ch];
cnt[cur]++;
total_overlapping += num[cur];

// Save state transition into history stack
history.push_back({old_cur, added_node, parent_node, ch});
}

// 2. POP / ROLLBACK LAST CHARACTER
void pop_back() {
    if (history.empty()) return;

    History h = history.back();
    history.pop_back();

    // Revert occurrence count & running palindrome count
    total_overlapping -= num[cur];
    cnt[cur]--;

    // If a new node was created during this push_back, delete it
    if (h.added_node != 0) {
        int u = h.added_node;
        go[h.parent_node][h.ch] = 0; // Disconnect child edge

        // Reset node u's properties
        go[u].fill(0);
        suflink[u] = len[u] = cnt[u] = diff[u] = slink[u] =
    } num[u] = 0;

        id--;
        tot--;
    }

    // Restore previous active node and string length
    cur = h.old_cur;
    S.pop_back();
}

// --- Black-Box Solvers ---

int cntDistinct() {
    return tot;
}

long long cntTotal() {
    return total_overlapping;
}

// Call this if you need to push end-occurrence counts down
// suffix links
void countAll() {
    for (int i = id - 1; i >= 2; --i) {
        cnt[suflink[i]] += cnt[i];
    }
}
};

```

7.5.4 PalindromicFactorization

COUNTING PALINDROMIC FACTORISATIONS - the number of ways to cut s into palindromic pieces,

```

// Needs: 07 - Strings/05 - Palindromes/02 - PalindromicTree.cpp
// in  $O(n \log n)$ . The tree file already has min_partitions (the
    // MINIMUM number of pieces); this is
// the same series-link machinery with (min, +1) replaced by (sum,
    // *1), plus the even-length
// variant that CF 932G needs. Doing it naively is  $O(n^2)$ : "for each
    //  $i$ , for each palindromic
// suffix" - and a string like  $a^n$  has  $n$  palindromic suffixes at
    // every position.
// WHY IT IS  $O(n \log n)$ : the palindromic suffixes of any prefix fall
    // into  $O(\log n)$  ARITHMETIC
// PROGRESSIONS by length (Fine and Wilf).  $\text{diff}[v] = \text{len}[v] -$ 
    //  $\text{len}[\text{slink}[v]]$  is the common
// difference of  $v$ 's progression and  $\text{slink}[v]$  jumps to the previous
    // progression, so walking
//  $v = \text{cur}$ ,  $\text{slink}[v]$ ,  $\text{slink}[\text{slink}[v]]$ , ... visits  $O(\log n)$  nodes.
    //  $\text{series}[v]$  aggregates the whole
// progression of  $v$  in  $O(1)$  by reusing  $\text{series}[\text{slink}[v]]$  from the
    // PREVIOUS position - that reuse is
// the entire trick and is why  $\text{series}[]$  must be indexed by NODE and
    // not by position.
// INDEX CONVENTION:  $\text{dp}$  is 1-indexed over prefixes,  $\text{dp}[i] = \text{answer}$ 
    // for  $s[0..i-1]$ ,  $\text{dp}[0] = 1$ .
ll countPalindromicFactorizations(const string& s, bool
    // evenLengthPiecesOnly = false) {
    int n = s.size();
    PalindromeTree t;
    t.init(n);
    vector<ll> dp(n + 1, 0), series(n + 2, 0);
    dp[0] = 1;
    for (int i = 0; i < n; i++) {
        t.add(s, i);
        for (int v = t.cur; t.len[v] > 0; v = t.slink[v]) {
            series[v] = dp[i + 1 - (t.len[t.slink[v]] + t.diff[v])];
            int sl = t.suflink[v];
            if (t.diff[v] == t.diff[sl]) series[v] = (series[v] +
    // series[sl]) % MOD;
                dp[i + 1] = (dp[i + 1] + series[v]) % MOD;
            }
            if (evenLengthPiecesOnly && (i + 1) % 2) dp[i + 1] = 0; //
    // an odd prefix cannot be filled
        }
    }
    return dp[n];
}

/* THE VARIANTS THIS UNLOCKS - all the same loop, only the semiring
    // changes
MINIMUM number of pieces: (min, +1) - already in 02 -
    // PalindromicTree.cpp as min_partitions.
NUMBER of factorisations: (sum, *1) - the code above.
FACTORISATIONS INTO AN EVEN NUMBER OF PIECES, or with a parity
    // constraint: carry two dp arrays
    //  $\text{dp}[i][0]$ ,  $\text{dp}[i][1]$  and one  $\text{series}[]$  per parity, exactly as
    // min_partition does for (even, odd).
CF 932G "PALINDROME PARTITION" (split  $s$  into  $k$  pieces with  $p_1 =$ 
    //  $p_k$ ,  $p_2 = p_{k-1}$ , ...):
    // build  $t = s[0] s[n-1] s[1] s[n-2] \dots$  and count factorisations of
    //  $t$  into EVEN-LENGTH
    // palindromes - that is the call with evenLengthPiecesOnly = true.  $n$ 
    // must be even.
WEIGHTED pieces (cost per palindrome length): the series aggregate
    // must then be a min/sum over
    // the progression WITH the weight, which only stays  $O(1)$  if the
    // weight is affine in the length -
    // otherwise you are back to  $O(n^2)$  and should look for a different
    // decomposition.
SANITY: countPalindromicFactorizations("aaa") = 4 (aaa | a+aa | aa+a
    // | a+a+a), and any string of
    // distinct characters gives exactly 1. */

```

7.5.5 PalindromesInRange

HOW MANY PALINDROMIC SUBSTRINGS LIE ENTIRELY INSIDE $s[l..r]$? Offline, $O((n+q)\log n)$.

```
// Needs: 07 - Strings/05 - Palindromes/01 - Manacher.cpp
// The whole-string count is one Manacher pass; the RANGE version is
//   ↳ the hard part, because a
//   ↳ centre's contribution is CLIPPED by both ends of the query.
// INDEX CONVENTION (0-based, inclusive): with the interleaved radii
//   ↳ p[] of 01 - Manacher.cpp,
//   ↳ d1[i] = (p[2i + 1] + 1) / 2 = number of odd palindromes centred
//   ↳ AT i (longest = 2*d1-1)
//   ↳ d2[i] = p[2i] / 2 = number of even palindromes centred
//   ↳ BETWEEN i-1 and i
// Then the answer for [l, r] is
//   ↳ sum over i in [l, r] of min(d1[i], i - l + 1, r - i + 1)
//   ↳ + sum over i in [l+1, r] of min(d2[i], i - l, r - i + 1).
// SPLIT AT THE MIDPOINT so that only ONE of the two clips can bind:
//   ↳ on the left half the binding
// clip is the distance to l, on the right half it is the distance to
//   ↳ r. Each half becomes
//   ↳ sum over i in a range of ( key[i] >= T ? affine(i) : v[i] )
// with key[i] = v[i] - i (left) or v[i] + i (right) - a threshold
//   ↳ sweep with three Fenwicks.
struct PalinRange {
    int n;
    vector<ll> d1, d2;
    PalinRange(const string& s) {
        Manacher M(s);
        n = s.size(), d1.assign(n, 0), d2.assign(n, 0);
        for (int i = 0; i < n; i++) d1[i] = (M.p[2 * i + 1] + 1) / 2,
        ↳ d2[i] = M.p[2 * i] / 2;
    }
    struct BIT {
        vector<ll> f;
        BIT(int n = 0) : f(n + 1, 0) {}
        void add(int i, ll v) { for (i++; i < (int)f.size(); i += i &
        ↳ -i) f[i] += v; }
        ll qr(int i) { ll r = 0; for (i++; i > 0; i -= i & -i) r +=
        ↳ f[i]; return r; }
        ll rng(int l, int r) { return l > r ? 0 : qr(r) - qr(l - 1); }
    };
    // sum over i in [ql, qr] of ( key[i] >= T ? c0 + c1 * i : v[i]
    ↳ ), for many queries at once.
    vector<ll> sweep(const vector<ll>& v, const vector<ll>& key,
        const vector<array<ll, 5>>& qs) { //
    ↳ {ql, qr, T, c0, c1}
        int q = qs.size();
        vector<ll> res(q, 0), pre(n + 1, 0);
        for (int i = 0; i < n; i++) pre[i + 1] = pre[i] + v[i];
        vector<int> ord(n), qo(q);
        iota(ord.begin(), ord.end(), 0), iota(qo.begin(), qo.end(),
        ↳ 0);
        sort(ord.begin(), ord.end(), [&](int a, int b) { return
        ↳ key[a] > key[b]; });
        sort(qo.begin(), qo.end(), [&](int a, int b) { return
        ↳ qs[a][2] > qs[b][2]; });
        BIT cnt(n), sIdx(n), sVal(n);
        int ptr = 0;
        for (int id : qo) {
            auto [ql, qr, T, c0, c1] = qs[id];
            while (ptr < n && key[ord[ptr]] >= T) {
                int i = ord[ptr++];
                cnt.add(i, 1), sIdx.add(i, i), sVal.add(i, v[i]);
            }
            if (ql > qr) continue;
            ll c = cnt.rng(ql, qr), si = sIdx.rng(ql, qr), sv =
            ↳ sVal.rng(ql, qr);
            res[id] = c0 * c + c1 * si + (pre[qr + 1] - pre[ql] - sv);
        }
        return res;
    }
    vector<ll> solve(const vector<pii>& queries) { //
    ↳ {l, r} 0-based inclusive
        int q = queries.size();
        vector<ll> km(n), kp(n), res(q, 0);
        vector<array<ll, 5>> a(q), b(q), c(q), d(q);
        for (int i = 0; i < n; i++) km[i] = d1[i] - i, kp[i] = d1[i]
        ↳ + i;
        for (int j = 0; j < q; j++) {
            ll l = queries[j].first, r = queries[j].second, m = (l +
            ↳ r) / 2;
            a[j] = {l, m, 1 - l, 1 - l, 1}; //
            ↳ odd, left half: i - l + 1
            b[j] = {m + 1, r, r + 1, r + 1, -1}; //
            ↳ odd, right half: r - i + 1
        }
        auto ra = sweep(d1, km, a), rb = sweep(d1, kp, b);
```

```
        for (int i = 0; i < n; i++) km[i] = d2[i] - i, kp[i] = d2[i]
        ↳ + i;
        for (int j = 0; j < q; j++) {
            ll l = queries[j].first, r = queries[j].second, m = (l +
            ↳ r + 1) / 2;
            c[j] = {l + 1, m, -l, -l, 1}; //
            ↳ even, left half: i - l
            d[j] = {m + 1, r, r + 1, r + 1, -1}; //
            ↳ even, right half: r - i + 1
        }
        auto rc = sweep(d2, km, c), rd = sweep(d2, kp, d);
        for (int j = 0; j < q; j++) res[j] = ra[j] + rb[j] + rc[j] +
        ↳ rd[j];
        return res;
    }
};
/* NOTES
WHOLE-STRING COUNT is just sum(d1) + sum(d2) - no sweep needed, and
↳ the number of DISTINCT
palindromic substrings is the node count of an eertree (05 -
↳ Palindromes/02).
ONLINE queries: replace the sweep with a WAVELET TREE (02 - DS/13)
↳ that also stores prefix SUMS
per node, or with a merge-sort tree - same formulas,  $O(\log^2 n)$ 
↳ per query, no sorting of
queries. The offline version above is shorter and faster; use it
↳ unless the queries are forced
to be online.
PALINDROMES STARTING IN [l, r] (rather than contained in it) is a
↳ different and easier sweep:
each centre contributes a contiguous range of start positions.
COUNTING DISTINCT palindromes inside a range is much harder (offline
↳ eertree + BIT over the
"last occurrence" trick, like distinct values in a range); do not
↳ confuse the two.
CHECK THE SPLIT POINT: the odd half uses  $m = (l+r)/2$  and the even
↳ half  $m = (l+r+1)/2$ . Both come
from solving  $i - l + 1 \leq r - i + 1$  and  $i - l \leq r - i + 1$ ; an
↳ off-by-one there double counts
the centre and is the only bug this code ever has. */
```

7.6 Rotations

7.6.1 LyndonAndRotations

LYNDON FACTORISATION (Duval) and MINIMAL CYCLIC ROTATION (Duval), both $O(n)$, $O(1)$ memory.

```
// A Lyndon word is strictly smaller than all of its proper suffixes.
↳ Every string factors
// UNIQUELY as  $w_1 \geq w_2 \geq \dots \geq w_k$  with each  $w_i$  Lyndon.
vector<string> duval(const string& s) {
    int n = s.size(), i = 0;
    vector<string> f;
    while (i < n) {
        int j = i + 1, k = i;
        while (j < n && s[k] <= s[j]) s[k] < s[j] ? k = i : k++, j++;
        while (i <= k) f.push_back(s.substr(i, j - k)), i += j - k;
    }
    return f;
}
// Smallest cyclic rotation: run Duval on s+s and take the factor
↳ starting before n.
int minRotation(string s) {
    int n = s.size();
    s += s;
    int i = 0, ans = 0;
    while (i < n) {
        ans = i;
        int j = i + 1, k = i;
        while (j < (int)s.size() && s[k] <= s[j]) s[k] < s[j] ? k = i : k++, j++;
        ↳ : k++, j++;
        while (i <= k) i += j - k;
    }
    return ans; // index in the
    ↳ ORIGINAL string
}
// USES: canonical form of a cyclic string (necklace dedup -> pairs
↳ with Burnside);
// "is A a rotation of B" (compare minRotation forms, or search
↳ A in B+B);
// lexicographically smallest result of rotating; runs /
↳ repetitions via Lyndon roots.
```

7.6.2 RunsAndTandemRepeats

ALL RUNS (maximal repetitions) of s in $O(n \log n)$ - $O(n)$ if the LCE structure is linear.

```
// Needs: 07 - Strings/04 - Suffix Structures/05 -
  ↳ SuffixArraySAIS.cpp (saIs, kasai)
// A RUN is a triple (l, r, p): s[l..r] has period p, r - l + 1 >=
  ↳ 2p, and it cannot be extended
// left or right with the same period. Every tandem repeat, every
  ↳ square, every "x repeated k
// times" substring lives inside exactly one run, so runs are THE
  ↳ compact encoding of all
// repetitions - and by the RUNS THEOREM there are FEWER THAN n of
  ↳ them, with total exponent
// sum(r - l + 1)/p < 3n. Enumerating squares directly is quadratic;
  ↳ enumerating runs is not.
// WHEN: count/locate squares or k-th powers, "longest substring that
  ↳ is a repetition", periodic
// structure of a string, and as the input to Lempel-Ziv style
  ↳ factorisations.
// INDEX CONVENTION: 0-based, l and r INCLUSIVE, p is the SMALLEST
  ↳ period of s[l..r].
// HOW (Bannai et al.): a run's period is a LYNDON ROOT - for one of
  ↳ the two alphabet orders, the
// run contains a Lyndon word of length p starting at some i. So take
  ↳ every i, let l[i] be the
// longest Lyndon prefix of the suffix i (that is "next smaller
  ↳ suffix" minus i), and try to
// extend the pair (i, i + l[i]) both ways with LCE. Do it for BOTH
  ↳ orders (< and its reverse).
struct RunsFinder {
    struct LceSA { //
        ↳ suffix array + O(1) LCE
        int n, LG;
        vector<int> sa, rnk, lcp, lg;
        vector<vector<int>> sp;
        void build(const string& s) {
            n = s.size(), sa = suffixArray(s), lcp = kasai(s, sa),
            ↳ rnk.assign(n, 0);
            for (int i = 0; i < n; i++) rnk[sa[i]] = i;
            lg.assign(n + 1, 0);
            for (int i = 2; i <= n; i++) lg[i] = lg[i / 2] + 1;
            LG = lg[n] + 1, sp.assign(LG, vector<int>(n, 0)), sp[0] =
            ↳ lcp;
            for (int k = 1; k < LG; k++)
                for (int i = 0; i < n; i + (1 << k) <= n; i++)
                    sp[k][i] = min(sp[k - 1][i], sp[k - 1][i + (1 <<
                    ↳ (k - 1))]);
        }
        int lce(int i, int j) { // lcp
            ↳ of the suffixes i and j
            if (i >= n || j >= n) return 0;
            if (i == j) return n - i;
            int l = rnk[i], r = rnk[j];
            if (l > r) swap(l, r);
            int k = lg[r - l];
            return min(sp[k][l + 1], sp[k][r - (1 << k) + 1]);
        }
    };
    int n;
    LceSA F, B; //
    ↳ forward, and on the reversal
    vector<int> crnk; //
    ↳ suffix runs under the FLIPPED
    RunsFinder(const string& s) { //
        ↳ alphabet order
        n = s.size();
        string r(s.rbegin(), s.rend()), c = s;
        for (char& ch : c) ch = (char)(255 - (unsigned char)ch);
        F.build(s), B.build(r);
        vector<int> csa = suffixArray(c);
        crnk.assign(n, 0);
        for (int i = 0; i < n; i++) crnk[csa[i]] = i;
        // NOT the reverse of F.rnk: flipping the alphabet flips
        ↳ every comparison EXCEPT the
        // prefix rule ("abc" < "abcd" in both orders), so the second
        ↳ order needs its own array.
    }
    int lceF(int i, int j) { return F.lce(i, j); } //
    ↳ common prefix of s[i..], s[j..]
    int lceB(int i, int j) { //
        ↳ common suffix of s[..i], s[..j]
        if (i < 0 || j < 0) return 0;
        return B.lce(n - 1 - i, n - 1 - j);
    }
    vector<array<int, 3>> runs() {
        vector<array<int, 3>> res;
        for (int ord = 0; ord < 2; ord++) { //
            ↳ both alphabet orders
```

```
const vector<int>& rk = ord ? crnk : F.rnk;
vector<int> nxt(n, n), st; //
↳ next suffix smaller in this order
for (int i = n - 1; i >= 0; i--) {
    while (!st.empty() && rk[st.back()] > rk[i])
        ↳ st.pop_back();
    nxt[i] = st.empty() ? n : st.back();
    st.push_back(i);
}
for (int i = 0; i < n; i++) {
    int p = nxt[i] - i, j = i + p; // p =
    ↳ Lyndon root length
    if (j > n) continue;
    int b = j < n ? lceF(i, j) : 0, a = lceB(i - 1, j -
    ↳ 1);
    if (a + b < p || p == 0) continue;
    int l = i - a, r = j + b - 1;
    if (r - l + 1 >= 2 * p) res.push_back({l, r, p});
}
sort(res.begin(), res.end()); // (l,
↳ r, p) with p increasing
vector<array<int, 3>> out; //
↳ keep the SMALLEST p per (l, r)
for (auto& x : res)
    if (out.empty() || out.back()[0] != x[0] || out.back()[1]
    ↳ != x[1]) out.push_back(x);
return out; // < n
↳ runs
}
};
/* WHAT TO DO WITH THE RUNS
NUMBER OF SQUARES (as occurrences, x x with |x| = q): a run (l, r,
↳ p) with length L = r - l + 1
contains, for every multiple q of ... no: for every q with p <= q
↳ and 2q <= L AND q a multiple
of p, exactly L - 2q + 1 occurrences of a square of half-length q.
↳ Summing over runs is O(n)
amortised by the runs theorem (total exponent < 3n).
NUMBER OF DISTINCT SQUARES is at most n (the "three squares lemma");
↳ get them by taking, per
run, the primitively rooted squares and deduplicating with the
↳ run's root.
K-TH POWERS: a run (l, r, p) contains a k-th power iff L >= k p, and
↳ then L - k p + 1 of them.
LONGEST REPETITION = the run maximising L / p, read straight off the
↳ list.
MAIN-LORENTZ is the alternative: divide & conquer with z-functions,
↳ O(n log n), ~40 lines, and
it reports tandem repeats as O(n log n) grouped triples instead of
↳ < n runs. Prefer runs.
PRIMITIVITY of s[l..r] (is it a proper power?) is a prefix-function
↳ question, not a runs one:
see 01 - String Matching/05, "n % (n - pi) == 0".
TESTING: on a^n there is exactly ONE run (0, n-1, 1); on "abababa"
↳ exactly one, (0, 6, 2). If
you get duplicates with different p, the Lyndon-root loop ran with
↳ only one order. */
```

7.7 Sequences

7.7.1 DeBruijnAndBitap

DE BRUIJN SEQUENCES and BIT-PARALLEL MATCHING.

```
// --- DE BRUIJN SEQUENCE B(k, n): a CYCLIC string over an alphabet
//   ↳ of size k, of length k^n, in
//   ↳ which every one of the k^n words of length n appears exactly once
//   ↳ as a (cyclic) substring.
// USES: "open the safe with the fewest keypresses" (the classic),
//   ↳ minimum-length probe sequences,
// and hashing tricks (a de Bruijn constant is how you find the index
//   ↳ of the lowest set bit).
// CONSTRUCTION (FKM / Lyndon): concatenate, in lexicographic order,
//   ↳ every Lyndon word whose
// length DIVIDES n. That is exactly what this O(k^n) recursion emits.
string deBruijn(int k, int n) {
    string s;
    vector<int> a(k * n, 0);
    function<void(int, int)> db = [&](int t, int p) {
        if (t > n) {
            if (n % p == 0) for (int i = 1; i <= p; i++) s +=
↳ char('0' + a[i]);
        } else {
            a[t] = a[t - p];
            db(t + 1, p);
            for (int j = a[t - p] + 1; j < k; j++) a[t] = j, db(t +
↳ 1, t);
        }
    };
    db(1, 1);
    return s;
    //
    ↳ length k^n, read cyclically
}
// The same object is an EULERIAN CIRCUIT in the de Bruijn graph
//   ↳ (vertices = words of length n-1,
// edges = words of length n), so Hierholzer on that graph is the
//   ↳ alternative construction - and
// the BEST theorem then counts how many distinct de Bruijn sequences
//   ↳ exist: (k!)^(k^(n-1)) / k^n.

// --- BITAP / SHIFT-AND: exact matching in O(n*m/64) with bitsets,
//   ↳ and the natural home for
// approximate matching. Build one bitmask per character marking
//   ↳ where it occurs in the pattern;
// then one pass over the text ANDs shifted copies together.
// Beats KMP only when you need the extra flexibility (k mismatches,
//   ↳ wildcards, "any of these
// patterns"), or when you want ALL occurrences as a bitset to
//   ↳ intersect with a range.
vector<int> bitap(const string& t, const string& p) {
    int n = t.size(), m = p.size();
    if (!m || m > n) return {};
    vector<unsigned long long> mask(256, 0);
    // Works for m <= 64 with a single word; for longer patterns use
    ↳ bitset<M> and shift it.
    if (m > 64) return {};
    for (int i = 0; i < m; i++) mask[(unsigned char)p[i]] |= 1ULL <<
    ↳ i;
    unsigned long long st = 0, hit = 1ULL << (m - 1);
    vector<int> res;
    for (int i = 0; i < n; i++) {
        st = ((st << 1) | 1) & mask[(unsigned char)t[i]];
        if (st & hit) res.push_back(i - m + 1);
    }
    return res;
}
/* WHAT BITAP GENERALISES TO, which is the reason to carry it
k MISMATCHES: keep k+1 state words R0..Rk, where Rj = "matched a
↳ prefix with at most j errors".
Rj_new = ((Rj << 1) & mask[c]) | Rj-1 | (Rj-1 << 1) | (Rj-1_old <<
↳ 1) | 1
covering substitution, insertion and deletion respectively - that
↳ is Wu-Manber, O(k n m / 64).
WILDCARDS in the PATTERN: set mask[c] bits for every character the
↳ class accepts. A '?' just
means "every mask has that bit". This is free - no FFT needed -
↳ and is the right tool when the
alphabet is small and the pattern is short. Use the FFT method (01
↳ - String Matching/04) when
the pattern is long or wildcards appear in the TEXT too.
MULTIPLE PATTERNS of the same length: OR their masks and check which
↳ one actually matched.
OCCURRENCES INSIDE [l, r]: keep the whole occurrence set as a bitset
↳ and popcount a range - that
is what makes "how many times does p occur in t[l..r]" O(n/64) per
↳ query with no suffix
structure at all.
MYERS' BIT-VECTOR ALGORITHM computes full edit distance in O(n m /
```

↳ 64) with the same idea
 (carry-save arithmetic on the DP's difference vectors); it is the
 ↳ fastest practical exact edit
 distance and the escape hatch when the O(nm) DP is 1e10. */

7.7.2 EditDistance

EDIT DISTANCE (Levenshtein): fewest single-character insertions, deletions and substitutions

```
// turning a into b. dp[i][j] = distance between the prefixes
//   ↪ a[0..i-1] and b[0..j-1], so the
// table is (n+1) x (m+1) and the answer is dp[n][m]. O(n m) time,
//   ↪ O(min(n,m)) memory here.
// WHEN: fuzzy matching, "minimum operations to make equal",
//   ↪ DNA-style alignment. If you only need
// insert+delete (no substitution) the answer is n + m - 2*LCS(a, b)
//   ↪ - use the LCS DP instead.
int editDistance(const string& a, const string& b) {
    int n = a.size(), m = b.size();
    vector<int> prv(m + 1), cur(m + 1);
    iota(prv.begin(), prv.end(), 0);
    for (int i = 1; i <= n; i++) {
        cur[0] = i;
        for (int j = 1; j <= m; j++)
            cur[j] = min({prv[j] + 1, cur[j - 1] + 1, prv[j - 1] +
                ↪ (a[i - 1] != b[j - 1])});
        swap(prv, cur);
    }
    return prv[m];
}

// --- BANDED: when you only care whether the distance is <= k. Cells
//   ↪ off the diagonal by more
// than k can never be on an optimal path (each step off the diagonal
//   ↪ costs >= 1), so evaluate
// only |i - j| <= k. O(n k). Returns k + 1 as "more than k".
int editDistanceAtMost(const string& a, const string& b, int k) {
    int n = a.size(), m = b.size(), BIG = k + 1;
    if (abs(n - m) > k) return BIG;
    vector<int> prv(m + 2, BIG), cur(m + 2, BIG);
    for (int j = 0; j <= min(m, k); j++) prv[j] = j; //
    ↪ row 0
    for (int i = 1; i <= n; i++) {
        int lo = max(0, i - k), hi = min(m, i + k);
        cur[lo] = lo ? min({prv[lo] + 1, prv[lo - 1] + (a[i - 1] !=
            ↪ b[lo - 1]), BIG}) : min(i, BIG);
        for (int j = lo + 1; j <= hi; j++) //
            ↪ cur[lo-1] is OUTSIDE the band
            cur[j] = min({prv[j] + 1, cur[j - 1] + 1, prv[j - 1] +
                ↪ (a[i - 1] != b[j - 1]), BIG});
        if (hi + 1 <= m) cur[hi + 1] = BIG; //
        ↪ the cell the next row reads
        swap(prv, cur);
    }
    return min(prv[m], BIG);
}

// --- DAMERAU-LEVENSHTEIN, RESTRICTED (optimal string alignment):
//   ↪ also allows swapping two
// ADJACENT characters at cost 1, but forbids editing a substring
//   ↪ twice. This is the version
// contests mean; the unrestricted one needs the "last row/column per
//   ↪ character" table.
int damerau(const string& a, const string& b) {
    int n = a.size(), m = b.size();
    vector<vector<int>> dp(n + 1, vector<int>(m + 1));
    for (int i = 0; i <= n; i++) dp[i][0] = i;
    for (int j = 0; j <= m; j++) dp[0][j] = j;
    for (int i = 1; i <= n; i++)
        for (int j = 1; j <= m; j++) {
            dp[i][j] = min({dp[i - 1][j] + 1, dp[i][j - 1] + 1,
                ↪ dp[i - 1][j - 1] + (a[i - 1] != b[j -
                    ↪ 1])});
            if (i > 1 && j > 1 && a[i - 1] == b[j - 2] && a[i - 2] ==
                ↪ b[j - 1])
                dp[i][j] = min(dp[i][j], dp[i - 2][j - 2] + 1);
        }
    return dp[n][m];
}

// --- MYERS' BIT-VECTOR ALGORITHM: exact Levenshtein in O(n * m /
//   ↪ 64), the escape hatch when the
// O(n m) table is 1e10. It stores the VERTICAL differences dp[i][j]
//   ↪ - dp[i-1][j] (always in
// {-1,0,+1}) as two bitmasks Pv/Mv over the rows, and advances a
//   ↪ whole column with 12 word ops.
// Here a is the "pattern" and must satisfy |a| <= 64; loop over
//   ↪ blocks of 64 rows for longer a.
int myersEditDistance(const string& a, const string& b) {
    int m = a.size();
    if (!m) return b.size();
    unsigned long long peq[256] = {};
    for (int i = 0; i < m; i++) peq[(unsigned char)a[i]] |= 1ULL << i;
    unsigned long long Pv = ~0ULL, Mv = 0, high = 1ULL << (m - 1);
    int score = m;
    for (char ch : b) {
```

```
        unsigned long long Eq = peq[(unsigned char)ch];
        unsigned long long Xv = Eq | Mv;
        unsigned long long Xh = (((Eq & Pv) + Pv) ^ Pv) | Eq;
        unsigned long long Ph = Mv | ~(Xh | Pv), Mh = Pv & Xh;
        if (Ph & high) score++;
        if (Mh & high) score--;
        Ph = (Ph << 1) | 1; //
        ↪ dp[0][j] = j (GLOBAL alignment)
        Mh <<= 1;
        Pv = Mh | ~(Xv | Ph);
        Mv = Ph & Xv;
    }
    return score;
}

/* VARIANTS AND TRAPS
SHIFT IN 0 INSTEAD OF 1 ('Ph <= 1') and Myers computes, for every
    ↪ j, the distance from a to the
    BEST-MATCHING substring of b ending at j - i.e. approximate
    ↪ pattern matching, k-differences.
RECONSTRUCTING the alignment needs the table; with O(n) memory use
    ↪ Hirschberg
    (08 - DP/04 - Classics), which is the same divide & conquer for
    ↪ any such 2-string DP.
WEIGHTED costs (ins != del != sub) work in the plain DP but break
    ↪ Myers and the LCS identity.
LONGEST COMMON SUBSTRING (contiguous) is NOT this DP: use hashing +
    ↪ binary search, a suffix
    automaton, or dp[i][j] = dp[i-1][j-1] + 1 with the answer read as
    ↪ a max over all cells.
EDIT DISTANCE TO A REGULAR LANGUAGE / to any string of a trie: same
    ↪ DP with the automaton state
    as the second dimension, O(n * states).
LOWER BOUND: |n - m| <= dist <= max(n, m), and dist >= (number of
    ↪ characters whose counts
    differ) / 2 style bounds - cheap filters before running the real
    ↪ DP on many pairs. */
```


7.7.3 BitParallelLCS

BIT-PARALLEL LCS - the length of the longest common subsequence of a and b in $O(n \cdot m / 64)$,

```
// where  $n = |a|$ . The classic DP is  $O(n \cdot m)$  and dies at  $n = m = 5e4$ ;
// this survives  $n = m = 1e5$ .
// IDEA: the LCS row is monotone and its DIFFERENCE vector is a bit
// string over the positions of
//  $a$  (bit  $i = \text{"the LCS value increases at position } i \text{ of } a"$ ). One row
// update is then a handful of
// word operations, of which the carry-propagating ADD is what does
// the actual work.
// PRECONDITION: none - this is exact, for arbitrary alphabets (map
// characters to  $0..A-1$ ).
// The answer is the POPCOUNT of the final vector; the bit positions
// themselves are NOT an LCS.
struct BitLCS {
    int n, W; // W =
    // number of 64-bit words
    vector<vector<unsigned long long>> match; //
    // match[c] = positions of c in a
    BitLCS(const string& a, int A = 26, char base = 'a') {
        n = a.size(), W = n / 64 + 1;
        match.assign(A, vector<unsigned long long>(W, 0));
        for (int i = 0; i < n; i++) match[a[i] - base][i >> 6] |=
        1ULL << (i & 63);
    }
    int lcs(const string& b, char base = 'a') {
        vector<unsigned long long> row(W, 0), x(W);
        for (char ch : b) {
            const auto& mc = match[ch - base];
            for (int i = 0; i < W; i++) x[i] = row[i] | mc[i]; // x
            // = row | match[c]
            unsigned long long carry = 1; //
            row = (row << 1) | 1
            for (int i = 0; i < W; i++) {
                unsigned long long nc = row[i] >> 63;
                row[i] = (row[i] << 1) | carry;
                carry = nc;
            }
            unsigned long long borrow = 0; //
            row = x - row
            for (int i = 0; i < W; i++) {
                unsigned long long d = x[i] - row[i] - borrow;
                borrow = (x[i] < row[i] + borrow) || (borrow &&
                row[i] == ~0ULL);
                row[i] = d;
            }
            for (int i = 0; i < W; i++) row[i] = x[i] & ~row[i]; //
            row = x & ~(x - row)
            int res = 0;
            for (int i = 0; i < W; i++) res +=
            __builtin_popcountll(row[i]);
            return res;
        }
    };
    // USAGE: build once on the SHORTER string (that is the one that
    // costs memory and words),
    // then call lcs() for every other string - the build is
    // reused.
    // BitLCS L(a); int k = L.lcs(b);
    /* NOTES
    WHY POPCOUNT: after processing all of b, bit i of 'row' is set iff
    LCS(a[0..i], b) is strictly
    greater than LCS(a[0..i-1], b) - the difference vector of a
    monotone row of the DP table.
    EDIT DISTANCE has the same treatment (Myers, 07 - Sequences/02) with
    two vectors instead of one,
    because its differences live in  $\{-1, 0, +1\}$  while the LCS row only
    ever steps by 0 or 1.
    LCS ITSELF (the subsequence, not its length) still needs the  $O(n \cdot m)$ 
    table or Hirschberg
    (08 - DP/04 - Classics) - bit-parallelism gives you the LENGTH
    only.
    LCS OF PREFIXES: read the popcount after each character of b to get
    LCS(a, b[0..j]) for all j at
    no extra cost. That is what makes this the practical answer to
    "all-prefix" LCS queries.
    SPARSE ALTERNATIVE: when the number of matching pairs r is small,
    the Hunt-Szymanski algorithm
    is  $O(r \log n)$  - better on near-distinct alphabets, worse on binary
    strings.
    ALPHABET: 'match' is  $A * W$  words. For a large alphabet, map to the
    distinct characters of a
    first (everything else can never match and can be dropped from b).
    */
```

7.7.4 AllSubstringLCS

ALL-SUBSTRING LCS - after ONE $O(n \cdot m)$ pass you can answer $\text{lcs}(s, t[i..j])$ for EVERY one of the

```
//  $m^2$  substrings of t. The classic DP gives one pair per  $O(n \cdot m)$ ;
// this gives all of them at once.
// INDEX CONVENTION: s and t are 1-indexed below; the returned array
// F is indexed 1..m and
//  $\text{lcs}(s, t[i..j]) = \# \{ k \text{ in } [i, j] : F[k] < i \}$ .
// So one array of m integers encodes all  $m^2$  answers, and a single
// query is a RANGE COUNT of
// values below a threshold - BIT offline, wavelet tree, or
// merge-sort tree,  $O(\log m)$  each.
// WHEN: "for every substring of t, how similar is it to s",
// edit-distance-to-a-window sweeps,
// and problems asking for the shortest/longest window of t whose LCS
// with s reaches a target.
// WHY IT WORKS: for a fixed left end i the row of the LCS table is
// monotone with steps of 0/1;
// F[k] is the largest left end for which position k still
// contributes a step. f/g below track
// the row and column "critical points" of the table - that is the
// Alves-Caceres-Song theory.
vector<int> allSubstringLCS(const string& s, const string& t) {
    int n = s.size(), m = t.size();
    vector<int> f(m + 1), g(m + 1, 0), nf(m + 1, 0);
    for (int j = 0; j <= m; j++) f[j] = j; // row
    // 0 of the table
    for (int i = 1; i <= n; i++) {
        nf[0] = 0;
        for (int j = 1; j <= m; j++) {
            int gp = g[j - 1], fp = f[j]; //
            // g[i][j-1] and f[i-1][j]
            if (s[i - 1] == t[j - 1]) nf[j] = gp, g[j] = fp;
            else nf[j] = max(fp, gp), g[j] = min(gp, fp);
        }
        swap(f, nf);
    }
    return f; //
    // F[1..m], F[0] is unused
}
// DIRECT USE - all answers for one left end i in  $O(m)$ :
// int cnt = 0; for (int j = i; j <= m; j++) { if (F[j] < i) cnt++;
// /*  $\text{lcs}(s, t[i..j]) = \text{cnt}$  */ }
// MANY (i, j) QUERIES: sort by i, sweep i from m down to 1 adding
// position k when  $F[k] < i$  to a
// BIT, then the answer is a prefix-sum query on [i, j].  $O((m + q) \log m)$  total.
/* RELATED SHAPES
ALL-SUBSTRING LCS OF s AGAINST ITS OWN SUBSTRINGS is the same call
with t = s.
THE TRANSPOSE (all substrings of s against the whole t) is the same
routine with the arguments
swapped - the construction is NOT symmetric, so swap deliberately.
SEMI-LOCAL LCS is the full theory this is a corner of: it also
answers " $s[a..b]$  vs  $t[i..j]$ "
using a permutation matrix and range counting, in  $O(n \cdot m \log)$ 
preprocessing. Rarely needed.
EDIT DISTANCE has no such compact all-substring encoding; only the
+1/-1 step structure of LCS
makes the answers a range count.
SANITY CHECK:  $\text{lcs}(s, t[i..i])$  is 1 exactly when  $t[i]$  appears in s,
and  $F[k] < i$  is then the test
"the first position of  $t[k]$ 's character in s is early enough" -
verify on a tiny case before
trusting the direction of the inequality. */
```

7.7.5 CyclicLCS

CYCLIC LCS - max over all n cyclic rotations of s of $\text{lcs}(\text{rotation}, t)$, in $O(n \cdot m)$ TOTAL.

```
// The naive answer is  $n$  separate LCS computations,  $O(n^2 \cdot m)$ . WHEN:
//   ↳ the two sequences are
// necklaces / circular arrangements and the cut point is yours to
//   ↳ choose.
// HOW: build the LCS table of  $s+s$  against  $t$  once, then RE-ROOT it  $n$ 
//   ↳ times. Deleting the first
// row of the table only ever cancels ONE unit of the answer, and the
//   ↳ path that loses it can be
// followed in  $O(m)$  - so all  $n$  rotations cost  $O(n \cdot m)$  together, not
//   ↳  $O(n^2 \cdot m)$ .
// INDEX CONVENTION: rows  $1..2n$  of  $dp$  read the character  $s[(row - 1) \cdot n]$ ;  $dp[i][j]$  is the LCS of
// the first  $i$  characters of the doubled  $s$  with  $t[0..j-1]$ . Rotation  $r$ 
//   ↳ reads  $dp[r + n][m]$ .
// from[i][j] records WHICH transition was taken: 0 = came from the
//   ↳ left, 1 = diagonal match,
// 2 = came from above. The re-rooting walk needs it, so it is not
//   ↳ optional bookkeeping.
int cyclicLCS(const string& s, const string& t) {
    int n = s.size(), m = t.size();
    if (!n || !m) return 0;
    vector<vector<int>>> dp(2 * n + 1, vector<int>(m + 1, 0));
    vector<vector<char>>> from(2 * n + 1, vector<char>(m + 1, 0));
    auto eq = [&](int i, int j) { return s[(i - 1) % n] == t[j - 1]; };
    for (int i = 0; i <= 2 * n; i++)
        for (int j = 0; j <= m; j++) {
            dp[i][j] = 0, from[i][j] = 0;
            if (j && dp[i][j - 1] > dp[i][j]) dp[i][j] = dp[i][j - 1], from[i][j] = 0;
            if (i && j && eq(i, j) && dp[i - 1][j - 1] + 1 > dp[i][j])
                dp[i][j] = dp[i - 1][j - 1] + 1, from[i][j] = 1;
            if (i && dp[i - 1][j] > dp[i][j]) dp[i][j] = dp[i - 1][j], from[i][j] = 2;
        }
    int res = 0;
    for (int r = 0; r < n; r++) {
        res = max(res, dp[r + n][m]);
        // rotation starting at s[r]
        int x = r + 1, y = 0;
        // now
        while (y <= m && from[x][y] == 0) y++;
        // find where that row is used
        for (; y <= m && x <= 2 * n; x++) {
            from[x][y] = 0, dp[x][m]--;
            if (x == 2 * n) break;
            for (; y <= m; y++) {
                // follow the path downward
                if (from[x + 1][y] == 2) break;
                if (y + 1 <= m && from[x + 1][y + 1] == 1) { y++;
            }
        }
    }
    return res;
}
/* WHAT THIS DOES AND DOES NOT COMPUTE
IT ROTATES ONLY s. That is the standard definition of CLCS and it is
↳ enough: rotating BOTH
strings is a strictly larger maximum in general, and needs  $n \cdot m$ 
↳ separate cut pairs - if a
problem really wants both, rotate  $t$  in an outer loop ( $O(n \cdot m^2)$ ) or
↳ think again.
MEMORY is  $O(n \cdot m)$  for two tables - at  $n = m = 2000$  that is ~32 MB for
↳  $dp$  plus ~8 MB for  $from$ .
Make  $s$  the SHORTER string; the doubled dimension is the expensive
↳ one.
THE ANSWER IS NOT MONOTONE in the rotation, so binary searching the
↳ cut point is wrong.
CYCLIC EDIT DISTANCE is the same re-rooting idea (Maes' algorithm)
↳ but is  $O(n \cdot m \log n)$  via
divide & conquer on non-crossing optimal paths - do not expect
↳ this code to adapt.
SANITY: cyclicLCS(s, s) == |s| and cyclicLCS(s, t) >= lcs(s, t)
↳ always. */
```

7.7.6 SubsequenceAutomaton

SUBSEQUENCE AUTOMATON (a.k.a. the "next occurrence" table). $O(n \cdot A)$ build, $O(1)$ transition.

```
// INDEX CONVENTION:  $\text{nxt}[i][c]$  = the SMALLEST  $j \geq i$  with  $s[j] == c$ ,
//   ↳ or  $n$  if there is none.
// There are  $n + 1$  rows,  $i = 0..n$ ; row  $i$  is the state "we have
//   ↳ consumed  $s[0..i-1]$ ".
// After matching a character at  $j$  you move to row  $j + 1$  - never to
//   ↳ row  $j$ .
// WHEN: everything of the form "is  $t$  a subsequence of  $s$ ", counting
//   ↳ or ranking DISTINCT
// subsequences, and shortest strings that are NOT subsequences.
//   ↳ Build it from the RIGHT.
vector<array<int, 26>>> subseqAutomaton(const string& s) {
    int n = s.size();
    vector<array<int, 26>>> nxt(n + 1);
    for (int c = 0; c < 26; c++) nxt[n][c] = n;
    for (int i = n - 1; i >= 0; i--) nxt[i] = nxt[i + 1], nxt[i][s[i]
        ↳ - 'a'] = i;
    return nxt;
}
// --- IS t A SUBSEQUENCE OF s?  $O(|t|)$  per query after the build (the
//   ↳ reason to build at all:
// the two-pointer scan is  $O(|s| + |t|)$  per query, which loses the
//   ↳ moment there are many queries).
bool isSubsequence(const string& t, const vector<array<int, 26>>&
    ↳ nxt) {
    int n = nxt.size() - 1, i = 0;
    for (char ch : t) {
        int j = nxt[i][ch - 'a'];
        if (j == n) return false;
        i = j + 1;
    }
    return true;
}
// --- NUMBER OF DISTINCT NON-EMPTY SUBSEQUENCES,  $O(n \cdot A)$ . Each
//   ↳ distinct subsequence is read by
// exactly ONE path in the automaton (always take the earliest
//   ↳ occurrence), so just count paths.
//  $f[i]$  = number of distinct non-empty subsequences of  $s[i..]$ ,
//   ↳ counted as "first letter is  $c$ ".
ll distinctSubsequences(const string& s, const vector<array<int,
    ↳ 26>>& nxt, ll mod = MOD) {
    int n = s.size();
    vector<ll> f(n + 2, 0);
    for (int i = n; i >= 0; i--) {
        f[i] = 0;
        for (int c = 0; c < 26; c++) if (nxt[i][c] < n) f[i] = (f[i]
            ↳ + f[nxt[i][c] + 1] + 1) % mod;
        return f[0];
    }
}
// --- K-TH DISTINCT SUBSEQUENCE in lexicographic order (1-indexed,
//   ↳ non-empty). Use the UNCAPPED
// counts  $f[i]$  above (saturating at  $k$ , not modular - a modular count
//   ↳ cannot be compared).
string kthSubsequence(const string& s, const vector<array<int, 26>>&
    ↳ nxt, ll k) {
    int n = s.size();
    vector<ll> f(n + 2, 0);
    // f[i] saturated at  $2e18$ 
    for (int i = n; i >= 0; i--)
        for (int c = 0; c < 26; c++) if (nxt[i][c] < n)
            f[i] = min((ll)2e18, f[i] + f[nxt[i][c] + 1] + 1);
    if (k > f[0]) return "";
    // fewer than  $k$  subsequences
    string res;
    int i = 0;
    while (k > 0) {
        for (int c = 0; c < 26; c++) {
            int j = nxt[i][c];
            if (j == n) continue;
            ll cur = f[j + 1] + 1;
            // subsequences starting with  $c$ 
            if (k <= cur) { res += char('a' + c), i = j + 1, k--;
            ↳ break; }
            k -= cur;
        }
    }
    return res;
}
// --- SHORTEST STRING OVER THE ALPHABET THAT IS NOT A SUBSEQUENCE OF
//   ↳  $s$ ,  $O(n \cdot A)$ .
//  $g[i]$  = that length for  $s[i..]$ ; a missing letter gives 1, otherwise
//   ↳ pay one letter and recurse.
int shortestNonSubsequence(const string& s, const vector<array<int,
    ↳ 26>>& nxt, int A = 26) {
```

```

int n = s.size();
vector<int> g(n + 2, 0);
for (int i = n; i >= 0; i--) {
    g[i] = inf;
    for (int c = 0; c < A; c++)
        g[i] = min(g[i], nxt[i][c] == n ? 1 : 1 + g[nxt[i][c] +
    1]);
}
return g[0];
// reconstruct by replaying the min
}
/* VARIANTS
DISTINCT COMMON SUBSEQUENCES of s and t: run both automata in
    ↳ lockstep,
    f[i][j] = sum over c of (f[nxt1[i][c] + 1][nxt2[j][c] + 1] + 1),
    ↳ O(n m A).
COUNTING SUBSEQUENCES EQUAL TO t (occurrences, not distinct) is the
    ↳ other classic DP,
    dp[i][j] = dp[i-1][j] + (s[i] == t[j]) * dp[i-1][j-1], O(n m) -
    ↳ the automaton does not help.
DISTINCT SUBSEQUENCES OF LENGTH EXACTLY k: add the length as a
    ↳ dimension, O(n k A).
LARGE ALPHABET: store nxt as a vector of sorted position lists per
    ↳ character and binary search;
    that is O(n + A) memory and O(log n) per transition, which is the
    ↳ right trade when A is 1e5.
DO NOT CONFUSE with the SUFFIX automaton (04 - Suffix
    ↳ Structures/02), which recognises
    SUBSTRINGS. This one recognises SUBSEQUENCES and is a DAG with no
    ↳ suffix links. */

```

7.8 Theorems

7.8.1 String Theorems

STRING THEOREMS - the facts that turn an $O(n^2)$ idea into an $O(n)$ one. No code: these are the

```

/* STRING THEOREMS - the facts that turn an O(n^2) idea into an O(n)
    ↳ one. No code: these are the
    statements you need to REMEMBER, because none of them is derivable
    ↳ at the table.

```

PERIODICITY

BORDER-PERIOD DUALITY: p is a period of s ($s[i] = s[i+p]$) whenever

↳ both exist) IFF $n - p$ is the length of a border. So the periods and the borders are the same
 ↳ set read backwards, and the prefix function gives all of them (07 - Strings/01/05).

FINE AND WILF: if p and q are both periods of s and $p + q - \gcd(p, q) \leq n$, then $\gcd(p, q)$ is

also a period. The bound is TIGHT (Fibonacci words). Consequences:
 - the borders of any string form $O(\log n)$ ARITHMETIC PROGRESSIONS
 ↳ (each progression is one

"period class"); this is what makes palindromic-factorisation
 ↳ and border-sum DPs $O(n \log n)$

instead of $O(n^2)$ - see 05 - Palindromes/04.
 - two occurrences of a pattern P closer than $|P| / 2$ force P to be

↳ periodic; hence in any text window of length $2|P|$ a non-periodic pattern occurs at most
 ↳ twice (the basis of constant-space

matching and of "the occurrences form $O(\log)$ arithmetic
 ↳ progressions").

CRITICAL FACTORISATION: every string s with period p has a position

↳ $i < p$ such that the local period at i equals the global period. Splitting there makes the
 ↳ two halves comparable independently - the basis of the two-way (Crochemore-Perrin) $O(n)$
 ↳ constant-space matcher.

PRIMITIVITY: s is a power t^k with $k \geq 2$ IFF $n \% (n - \text{pi}[n-1]) == 0$

↳ and $n - \text{pi}[n-1] < n$. A

primitive word has exactly n distinct rotations; a non-primitive
 ↳ one has n / k .

THE CYCLE LEMMA / rotations: s and t are rotations of each other IFF

↳ t is a substring of $s + s$
 IFF $\text{minRotation}(s) == \text{minRotation}(t)$ (06 - Rotations/01).

REPETITIONS

RUNS THEOREM (Bannai et al. 2015): a string of length n has FEWER

↳ THAN n runs (maximal repetitions), and the sum of their exponents is $< 3n$. Both bounds
 ↳ are what make "enumerate all repetitions" linear rather than quadratic. Construction: 06 -
 ↳ Rotations/02.

THREE SQUARES LEMMA: if three squares start at the same position and

↳ u is a prefix of v is a prefix of w , then $|u| + |v| \leq |w|$. Hence at most $O(\log n)$ squares
 ↳ start at any position, and a string of length n contains fewer than n DISTINCT squares.

CHEN-FOX-LYNDON: every string factors uniquely as $w_1 \geq w_2 \geq \dots \geq$

↳ w_k with each w_i Lyndon, and Duval computes it in $O(n)$ with $O(1)$ memory. The last factor is the
 ↳ smallest suffix; the first is the longest Lyndon prefix. Lyndon roots are the pivot of the
 ↳ runs algorithm.

A LYNDON WORD is strictly smaller than all of its proper suffixes;

↳ equivalently it is the unique minimum among its own rotations, and it is primitive. There are
 ↳ $(1/n) * \sum_{d|n} \mu(d) * A^{(n/d)}$ Lyndon words of length exactly n over an alphabet
 ↳ of size A (Witt's formula).

SUBSTRINGS AND SUFFIX STRUCTURES

ENDPOS EQUIVALENCE (suffix automaton): two substrings are equivalent

↳ when they occur at exactly the same set of END positions. The classes form a TREE under "is a
 ↳ suffix of" (the suffix-link tree), there are at most $2n - 1$ states and $3n - 4$ transitions, and
 ↳ one class contains exactly the strings of lengths $(\text{len}[\text{link}[v]], \text{len}[v])$ - so a class
 ↳ contributes $\text{len}[v] - \text{len}[\text{link}[v]]$ distinct substrings. Summing that is the count of distinct
 ↳ substrings.

THE SUFFIX-LINK TREE of a SAM built on the REVERSED string IS the

↳ suffix tree of the string.

A STRING OF LENGTH n HAS AT MOST n DISTINCT PALINDROMIC SUBSTRINGS

↳ (eertree node count), and the palindromic suffixes of any prefix again form $O(\log n)$ arithmetic
 ↳ progressions.

DISTINCT SUBSTRINGS = $n(n+1)/2$ - sum of the LCP array; adding one

↳ character to the end changes that by (new length) - (longest suffix that already occurred),

Dynamic Programming (DP)

↪ which is the SAM's `len[last]`.

MATCHING BOUNDS

KNUTH-MORRIS-PRATT does at most $2n$ character comparisons;

↪ Boyer-Moore is sublinear on average
but $O(nm)$ in the worst case without Galil's rule; Z and prefix
↪ functions are exactly linear.

A PATTERN WITH WILDCARDS is not solvable by automata in linear time;

↪ the FFT convolution
 $\sum (p_i - t_j)^2 p_i t_j$ is the standard $O(n \log n)$ route (01 -
↪ String Matching/04).

EDIT DISTANCE cannot be computed in $O(n^{2-\epsilon})$ unless SETH fails -

↪ so the $O(nm / 64)$
bit-parallel version (07 - Sequences/02) is essentially the last
↪ word.

LCS OF k STRINGS is NP-hard in k ; for $k = 2$ it is $O(nm / 64)$ and no

↪ truly subquadratic
algorithm is known (same SETH barrier).

COUNTING

NUMBER OF DISTINCT NECKLACES (rotation classes) of length n over A

↪ letters:
 $(1/n) * \sum \text{over } d \mid n \text{ of } \phi(d) * A^{(n/d)}$
↪ (Burnside)

NUMBER OF BRACELETS adds the reflections; see 05 - Combinatorics/02

↪ - BurnsidePolya.cpp.

DE BRUIJN SEQUENCES $B(A, n)$: there are $(A!)^{(A^{(n-1)})} / A^n$ distinct

↪ ones (07 - Sequences/01). */

8 Dynamic Programming (DP)

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8.1 DP Optimizations

8.1.1 DivideAndConquerDP

Divide & Conquer DP Optimization - Time: $O(K * N \log N)$, Space: $O(N)$

// Applies when DP transition is $dp[k][i] = \min_{j < i} \{ dp[k-1][j] +$
↪ $cost(j+1, i) \}$

// Condition: $opt(i, j) \leq opt(i, j+1)$ (Monotonicity of optimal

↪ split points)

const int N = 200005; // ALSO in 01 -

↪ Template.cpp: keep one copy

int a[N];
ll curCost;
int curL = 0, curR = -1, freq[N];
pair<**int**, **ll**> pre[N], cur[N];
// Modify at will :
int lo(**int** ind) { **return** 1; }
int hi(**int** ind) { **return** ind; }

void add(**int** val) {
 curCost += freq[val]++;
}

void remove(**int** val) {
 curCost -= --freq[val];
}

ll Cost(**int** l, **int** r) {
 while (curR < r)
 add(a[++curR]);
 while (curL > l)
 add(a[--curL]);
 while (curR > r)
 remove(a[curR--]);
 while (curL < l)
 remove(a[curL++]);
 return curCost;
}

ll f(**int** ind, **int** k) { **return** pre[k-1].second + Cost(k, ind); }
void store(**int** ind, **int** k, **ll** v) { cur[ind] = **pair**(k, v); }

void rec(**int** L, **int** R, **int** LO, **int** HI) {
 if (L > R) **return**;
 int mid = (L + R) >> 1;
 pair best(llinf, LO);
 for (**int** k = max(LO, lo(mid)); k <= min(HI, hi(mid)); ++k)
 best = min(best, **pair**(f(mid, k), k));
 store(mid, best.second, best.first);
 rec(L, mid-1, LO, best.second);
 rec(mid+1, R, best.second, HI);
}

void solve(**int** L, **int** R) { rec(L, R, -inf, inf); }

8.1.2 CHT

DYNAMIC CONVEX HULL TRICK (UPPER hull -> MAXIMUM of $m*x + b$). Lines may be inserted in ANY order

// and queried at any x , $O(\log n)$ each. If your slopes AND queries
↪ are both monotone, the deque

// version at the bottom of this file is $O(1)$ amortised and much

↪ shorter - prefer it.

// For a MINIMUM, insert $(-m, -b)$ and negate the answer.

// To maximise over all x in $[l, r]$, query only l and r : the upper

↪ envelope is convex.

```
struct Line {  
    ll m, b;  
    mutable function<const Line *(>> succ;  
  
    bool operator<(const Line &other) const {  
        return m < other.m;  
    }  
  
    bool operator<(const ll &x) const {  
        const Line *s = succ();  
        if (!s)  
            return 0;  
        return (lll)(b - s->b) < (lll)(s->m - m) * x;    // __int128:  
        ↪ |dm|*|x| overflows ll  
    }  
};
```

// will maintain upper hull for maximum

```
struct HullDynamic : public multiset<Line, less<>> {  
    bool bad(iterator y) {  
        auto z = next(y);  
        if (y == begin()) {  
            if (z == end())  
                return 0;  
            return y->m == z->m && y->b <= z->b;  
        }  
        auto x = prev(y);  
        if (z == end())  
            return y->m == x->m && y->b <= x->b;  
        // __int128, NOT long double: each factor reaches ~2e18, the  
        ↪ product needs ~123 bits,  
        // and this test is consulted exactly when three lines are  
        ↪ nearly collinear.  
        return (lll)(x->b - y->b) * (z->m - y->m) >= (lll)(y->b -  
        ↪ z->b) * (y->m - x->m);  
    }  
  
    void insert_line(ll m, ll b) {  
        // for minimum  
        // m *= -1;  
        // b *= -1;  
        auto y = insert({m, b});  
        y->succ = [=] { return next(y) == end() ? 0 : &*next(y); };  
        if (bad(y)) {  
            erase(y);  
            return;  
        }  
        while (next(y) != end() && bad(next(y)))  
            erase(next(y));  
        while (y != begin() && bad(prev(y)))  
            erase(prev(y));  
    }  
  
    ll query(ll x) { // UB if empty -  
        ↪ insert at least one line first  
        auto l = *lower_bound(x);  
        // for minimum  
        // return -(l.m * x + l.b);  
        return l.m * x + l.b;  
    }  
};
```

↪ -----
// MONOTONE CHT (deque) - the version you actually want when the DP
↪ feeds lines in slope order.

// MINIMISES $m*x + b$. Requires slopes added in DECREASING order;

↪ queries in increasing x are $O(1)$

// amortised (move the head), arbitrary x is $O(\log n)$ by binary

↪ search on the hull.

// For a MAXIMUM, negate m and b on the way in and negate the answer
↪ on the way out.

```
struct MonoCHT {  
    vector<ll> M, B;  
    int head = 0;  
    bool bad(int a, int b, int c) { // is line b  
        ↪ unnecessary between a and c?  
        return (lll)(B[c] - B[a]) * (M[a] - M[b]) <= (lll)(B[b] -
```

```

    ↪ B[a]) * (M[a] - M[c]);
    }
    void add(ll m, ll b) { // m must be <=
    ↪ the last slope added
        M.push_back(m), B.push_back(b);
        while (M.size() >= 3 && bad(M.size() - 3, M.size() - 2,
    ↪ M.size() - 1))
            M.erase(M.end() - 2), B.erase(B.end() - 2);
        head = min(head, (int)M.size() - 1);
    }
    ll f(int i, ll x) { return M[i] * x + B[i]; }
    ll queryInc(ll x) { // x
    ↪ non-decreasing across calls
        while (head + 1 < (int)M.size() && f(head + 1, x) <= f(head,
    ↪ x)) head++;
        return f(head, x);
    }
    ll query(ll x) { // any x, O(log
    ↪ n)
        int lo = 0, hi = M.size() - 1;
        while (lo < hi) { int m = (lo + hi) / 2; f(m + 1, x) <= f(m,
    ↪ x) ? lo = m + 1 : hi = m; }
        return f(lo, x);
    }
};
// WHEN EACH ONE: monotone slopes + monotone queries ->
    ↪ MonoCHT::queryInc, O(n).
//     monotone slopes, arbitrary queries ->
    ↪ MonoCHT::query, O(n log n).
//     arbitrary slopes -> HullDynamic
    ↪ above, or Li Chao (04),
//     which needs
    ↪ no ordering at all.
//     lines inserted and removed -> CHT with
    ↪ rollback (03) or Li Chao merge.

```

8.1.3 CHTRollback

Rollback Convex Hull Trick - Space: $O(N)$, Add: $O(\log N)$, Query: $O(\log N)$, Rollback: $O(1)$

```

// Supports inserting monotonic lines with history stack rollback for
    ↪ tree/dsu DP
// Template parameters:
// IS_MAX: Set to true for Maximum queries, false for Minimum queries.
// IS_INC: Set to true if slopes added are strictly Increasing, false
    ↪ for Decreasing.
template<bool IS_MAX = false, bool IS_INC = false>
struct RollbackCHT {
    struct Line {
        ll m, c;
        Line() : m(0), c(llinf) {}
        Line(ll m, ll c) : m(m), c(c) {}
        ll eval(ll x) const { return m * x + c; }
    };

    struct History {
        int pos;
        int sz;
        Line old_line;
    };

    vector<Line> st;
    vector<History> hist;
    int sz;

    RollbackCHT(int max_nodes) {
        st.resize(max_nodes);
        sz = 0;
    }

    bool redundant(Line l1, Line l2, Line l3) {
        __int128 dx1 = l1.m - l2.m;
        __int128 dc1 = l2.c - l1.c;
        __int128 dx2 = l2.m - l3.m;
        __int128 dc2 = l3.c - l2.c;
        return dc1 * dx2 >= dc2 * dx1;
    }

    void add(ll m, ll c) {
        // The magic happens here: automatically formats internal
    ↪ state
        ll m_int = IS_INC ? -m : m;
        ll c_int = IS_MAX ? -c : c;

        Line l_new(m_int, c_int);
        int pos = sz;
        int low = 1, high = sz - 1;

        while (low <= high) {
            int mid = low + (high - low) / 2;
            if (redundant(st[mid - 1], l_new, st[mid])) {
                pos = mid;
                high = mid - 1;
            } else {
                low = mid + 1;
            }
        }

        hist.push_back({pos, sz, st[pos]});
        st[pos] = l_new;
        sz = pos + 1;
    }

    void rollback() {
        if (hist.empty()) return;
        History h = hist.back();
        hist.pop_back();
        sz = h.sz;
        st[h.pos] = h.old_line;
    }

    ll query(ll x) {
        if (sz == 0) return IS_MAX ? -llinf : llinf;

        // The magic happens here: matches x to the internal geometry
        ll x_int = (IS_INC != IS_MAX) ? -x : x;

        int low = 0, high = sz - 2;
        int best = sz - 1;

        while (low <= high) {
            int mid = low + (high - low) / 2;
            if (st[mid].eval(x_int) <= st[mid + 1].eval(x_int)) {
                best = mid;
            }
        }
    }
};

```

```

        high = mid - 1;
    } else {
        low = mid + 1;
    }
}

ll ans = st[best].eval(x_int);
return IS_MAX ? -ans : ans;
};
}
};

```

8.1.4 LiChaoTree

LI CHAO TREE - MAXIMUM of a set of lines $y = m \cdot x + b$ at any query x , with NO order requirement

```

// on the slopes or on the queries (that is the whole point; CHT
//  ↳ needs monotone slopes).
// O(log X) per line insert and per query, O(log^2 X) to insert a
//  ↳ line only on a SEGMENT [lx, rx].
// Space O(n log X). Works for any family of functions where two
//  ↳ members cross AT MOST ONCE -
// lines, and also  $a \cdot x^2 + b \cdot x + c$  with a FIXED  $a$ .
// EXACT INTEGER ARITHMETIC: never compute the crossing abscissa.
//  ↳ Compare the two candidates at
// ns, mid and ne only - a division in double is off by  $\sim 1e3$  in  $x$ 
//  ↳ when the coefficients reach  $1e18$ ,
// which silently sends the loser down the wrong half.
// SET the domain [X0, X1] before use; it must cover every  $x$  you will
//  ↳ ever query.
struct LiChao {
    struct Line { ll m = 0, b = LLONG_MIN / 4; }; //
    ↳ neutral = -infinity
    ll X0, X1;
    vector<Line> t;
    vector<int> lf, rt;
    LiChao(ll X0, ll X1) : X0(X0), X1(X1) { newNode(); }
    int newNode() { t.push_back({}); lf.push_back(-1),
    ↳ rt.push_back(-1); return t.size() - 1; }
    static ll f(const Line& L, ll x) { return L.m == 0 && L.b ==
    ↳ LLONG_MIN / 4 ? L.b : L.m * x + L.b; }
    void add(Line nw, int v, ll l, ll r) { //
    ↳ insert on the whole node range
        while (true) {
            ll m = l + (r - l) / 2; // NOT
            ↳ (l+r)/2: negative l+r truncates up
            bool lef = f(nw, l) > f(t[v], l), mid = f(nw, m) >
            ↳ f(t[v], m);
            if (mid) swap(t[v], nw);
            if (l == r) return;
            if (lef != mid) {
                if (lf[v] < 0) lf[v] = newNode();
                v = lf[v], r = m;
            } else {
                if (f(nw, r) > f(t[v], r)) {
                    if (rt[v] < 0) rt[v] = newNode();
                    v = rt[v], l = m + 1;
                } else return;
            }
        }
    }
    void addLine(ll m, ll b) { add({m, b}, 0, X0, X1); }
    void addSegment(ll m, ll b, ll lx, ll rx, int v = 0, ll l =
    ↳ LLONG_MIN, ll r = LLONG_MIN) {
        if (l == LLONG_MIN) l = X0, r = X1;
        if (rx < l || r < lx) return;
        if (lx <= l && r <= rx) { add({m, b}, v, l, r); return; }
        ll md = l + (r - l) / 2;
        if (lf[v] < 0) lf[v] = newNode();
        if (rt[v] < 0) rt[v] = newNode();
        addSegment(m, b, lx, rx, lf[v], l, md), addSegment(m, b, lx,
        ↳ rx, rt[v], md + 1, r);
    }
    ll query(ll x) {
        ll r = LLONG_MIN, l = X0, rr = X1;
        for (int v = 0; v >= 0; ) {
            r = max(r, f(t[v], x));
            if (l == rr) break;
            ll m = l + (rr - l) / 2;
            if (x <= m) v = lf[v], rr = m;
            else v = rt[v], l = m + 1;
        }
        return r; //
    }
    ↳ LLONG_MIN/4 if no line covers x
};
// FOR A MINIMUM: insert (-m, -b) and negate the answer.
// PERSISTENT / MERGEABLE: see 05 - PersistentLiChaoTree.cpp. MERGING
//  ↳ two Li Chao trees costs
// O(n log X) amortised in total (push the loser line down into the
//  ↳ other subtree, exactly like
// segment-tree merging) - that is how you do convex-hull DP on a
//  ↳ tree without small-to-large.

```


8.1.5 PersistentLiChaoTree

Persistent Li Chao Tree - Space: $O(N \log X)$, Insert Line: $O(\log X)$, Query: $O(\log X)$

```
// Version-persistent Li Chao Segment Tree for tree / history DP
↳ queries
const ll maxN = 1e7 + 5;

struct Line {
    ll m, c;

    Line() : m(0), c(1llinf) {}

    Line(ll m, ll c) : m(m), c(c) {}
};

ll sub(ll x, Line l) {
    return x * l.m + l.c;
}

// Persistent Li Chao
struct Node {
    // range I am responsible for
    Line line;
    Node *left, *right;

    Node() {
        left = right = NULL;
    }

    Node(ll m, ll c) {
        line = Line(m, c);
        left = right = NULL;
    }

    void extend(int l, int r) {
        if (left == NULL && l != r) {
            left = new Node();
            right = new Node();
        }

        Node *copy(Node *node) {
            Node *newNode = new Node;
            newNode->left = node->left;
            newNode->right = node->right;
            newNode->line = node->line;
            return newNode;
        }

        Node *add(Line toAdd, int l, int r) {
            int mid = l + (r - l) / 2; // NOT (l+r)/2: that truncates
            ↳ toward ZERO, so on a negative // interval the left child
            ↳ equals the parent -> infinite recursion
            Node *cur = copy(this);
            if (l == r) {
                if (sub(l, toAdd) < sub(l, cur->line))
                    swap(toAdd, cur->line);
                return cur;
            }
            bool lef = sub(l, toAdd) < sub(l, cur->line);
            bool midE = sub(mid + 1, toAdd) < sub(mid + 1, cur->line);
            if (midE)
                swap(cur->line, toAdd);
            cur->extend(l, r);
            if (lef != midE) {
                cur->left = cur->left->add(toAdd, l, mid);
            } else {
                if (mid + 1 > r)
                    return cur;
                cur->right = cur->right->add(toAdd, mid + 1, r);
            }
            return cur;
        }

        Node *add(Line toAdd) {
            return add(toAdd, -maxN + 1, maxN - 1);
        }
    }

    ll query(ll x, int l, int r) {
        int mid = l + (r - l) / 2; // NOT (l+r)/2: that truncates
        ↳ toward ZERO, so on a negative // interval the left child
        ↳ equals the parent -> infinite recursion
        if (l == r || left == NULL)
            return sub(x, line);
        extend(l, r);
        if (x <= mid)
```

```
        return min(sub(x, line), left->query(x, l, mid));
    } else
        return min(sub(x, line), right->query(x, mid + 1, r));
}

ll query(ll x) {
    return query(x, -maxN + 1, maxN - 1);
}

void clear() {
    if (left != NULL) {
        left->clear();
        right->clear();
    }
    delete this;
}

};

const int N = 200005; // #versions; also
↳ in 01 - Template.cpp
Node *tree[N];
```

8.1.6 KnuthDP

KNUTH OPTIMIZATION - interval DP $dp[i][j] = \min_{i \leq k < j} \{dp[i][k] + dp[k+1][j]\} + \text{cost}(i, j)$

```
// from  $O(n^3)$  to  $O(n^2)$ , by restricting k to  $[\text{opt}[i][j-1],$ 
↳  $\text{opt}[i+1][j]]$ .
// THE HYPOTHESIS IS ON cost, NOT ON opt. 'opt[i][j-1] <= opt[i][j]'
↳  $\leq \text{opt}[i+1][j]$  is the
// CONCLUSION - you cannot test it before running. What you must
↳ check is BOTH of:
// (1) QUADRANGLE INEQUALITY (Monge):  $\text{cost}(a, c) + \text{cost}(b, d) \leq$ 
↳  $\text{cost}(a, d) + \text{cost}(b, c)$ 
// for all  $a \leq b \leq c \leq d$ . Equivalent local form, and the
↳ only one checkable by hand:
//  $\text{cost}(a, c) + \text{cost}(a+1, c+1) \leq \text{cost}(a+1, c) + \text{cost}(a, c+1)$ .
// (2) MONOTONE ON INTERVALS:  $a \leq b \leq c \leq d \Rightarrow \text{cost}(b, c) \leq$ 
↳  $\text{cost}(a, d)$ .
// Both hold for: sum of a non-negative array over  $[i, j]$ ;  $(S_j -$ 
↳  $S_i)^2$ ;  $f(S_j - S_i)$  with f convex
// and  $a[] \geq 0$ ; "number of equal pairs inside  $[i, j]$ ". They FAIL for
↳ anything built from max, and
// monotonicity fails as soon as  $a[]$  can be negative (QI can still
↳ hold - D&C needs only QI,
// Knuth needs both). Applied without them there is no crash and no
↳ TLE, just a wrong answer.
// Classic uses: optimal BST, merging n piles of stones, CF
↳ 1101F-style interval costs.
```

```
struct KnuthDP {
    int n;
    vector<vector<ll>> dp;
    vector<vector<int>> opt;

    KnuthDP(int n = 0) : n(n), dp(n + 2, vector<ll>(n + 2, 0)), opt(n
↳ + 2, vector<int>(n + 2, 0)) {}

    // User-defined cost function cost(i, j)
    ll solve(function<ll(int, int)> cost) {
        for (int i = 1; i <= n; i++) {
            opt[i][i] = i;
        }

        for (int len = 2; len <= n; len++) {
            for (int i = 1; i + len - 1 <= n; i++) {
                int j = i + len - 1;
                dp[i][j] = 1llinf;
                int L = opt[i][j - 1], R = opt[i + 1][j];
                for (int k = L; k <= min(R, j - 1); k++) {
                    ll cur = dp[i][k] + dp[k + 1][j] + cost(i, j);
                    if (cur < dp[i][j]) {
                        dp[i][j] = cur;
                        opt[i][j] = k;
                    }
                }
            }
        }
        return dp[1][n];
    }
};
```

8.1.7 AliensTrick

ALIENS TRICK (Lagrangian relaxation / wqs binary search) - $O(n \log(\text{range}))$.

```
// Drops an "exactly k pieces" constraint: charge lambda per piece,
// ↳ solve the UNCONSTRAINED DP,
// and binary search lambda until the optimum uses k pieces. Answer =
// ↳ cost(lambda) - k*lambda.
// PRECONDITION: f(k), the optimum with exactly k pieces, must be
// ↳ CONVEX in k (for a minimum;
// ↳ concave for a maximum). NOT "strictly" - strictness is never
// ↳ needed and rarely true.
// THREE THINGS THAT MAKE IT CORRECT, all easy to get wrong:
// (1) dp_solver MUST return the LARGEST optimal count for that
// ↳ lambda. With f convex, the
// difference d(k) = f(k) - f(k-1) is non-decreasing; the search
// ↳ below finds the largest
// lambda with count >= k, and then d(k) <= -lambda <= d(k+1),
// ↳ so k really is optimal at
// that lambda and cost - k*lambda = f(k). Any other tie-break
// ↳ makes the last line wrong.
// (2) An integer lambda suffices ONLY if all costs are integers.
// (3) [min_penalty, max_penalty] must straddle every slope of f;
// ↳ assert it.
// CONVEX families: "k pairwise non-adjacent items", "exactly k
// ↳ segments with a Monge cost",
// "k disjoint intervals", "min-cost flow of value k" (flow cost is
// ↳ always convex in the value),
// matroid-rank-constrained optima. TRAPS: two coupled constraints,
// ↳ or a cost rewarding a parity
// of k. CHECK IT: brute-force f(k) for n <= 60 and assert
// ↳ f(k+1)-f(k) >= f(k)-f(k-1).
struct AliensTrick {
    // Binary searches optimal penalty lambda to choose exactly K
    // ↳ items
    // Returns minimum cost to choose exactly K items
    static ll solve(int n, int k, ll min_penalty, ll max_penalty,
    // ↳ function<pair<ll, int>>(ll) dp_solver) {
        ll l = min_penalty, r = max_penalty, best_lambda = 0;

        while (l <= r) {
            ll mid = l + (r - l) / 2;
            auto [cost, count] = dp_solver(mid);
            if (count >= k) {
                best_lambda = mid;
                l = mid + 1; // Increase penalty if we used too many
            } else {
                r = mid - 1; // Decrease penalty if we used too few
            }
        }

        // True cost = cost_with_penalty - (k * best_lambda)
        auto [cost, count] = dp_solver(best_lambda);
        return cost - (ll)k * best_lambda;
    }
};
```

8.1.8 AliensTrickExample

ALIENS TRICK (Lagrangian / wqs binary search) - drops the "exactly k pieces" constraint by charging

```
// a penalty lambda per piece, solving the unconstrained DP, and
// ↳ binary searching lambda until the
// optimal solution uses k. REQUIRES the optimum as a function of k
// ↳ to be CONVEX; check that first.
// 1. DP State Structure
struct DPState {
    long long cost;
    long long count; // Number of items/segments picked

    // TIE-BREAKER LOGIC:
    // If costs are equal, we maximize the count to handle collinear
    // ↳ points on the convex hull.
    // If you are writing a MINIMIZATION problem with min(), change
    // ↳ this to:
    // if (cost != other.cost) return cost > other.cost; (since
    // ↳ priority queue / max naturally takes the "biggest")
    // Actually, when using standard >, < operators for min/max DP:
    bool operator<(const DPState& other) const {
        if (cost != other.cost) return cost < other.cost;
        return count < other.count; // Always prefer taking MORE
    }
};

void solve_alien() {
    int n, k;
    cin >> n >> k;
    // Read ABC 218 H input: A array is of size N-1
    vector<long long> A(n);
    for(int i = 1; i < n; i++) cin >> A[i];
    // If K > N / 2, swap colors so K <= N / 2 (the score is
    // ↳ symmetric)
    if (k > n / 2) k = n - k;
    // Construct 1-indexed values array 'a' where a[i] = A[i-1] + A[i]
    vector<long long> a(n + 1);
    for (int i = 1; i <= n; i++) {
        a[i] = A[i - 1] + (i < n ? A[i] : 0);
    }

    // 2. The Unconstrained DP
    // Returns the optimal {cost, count} for a given penalty.
    auto check_dp = [&](long long penalty) -> DPState {
        // dp[i][0]: Max score in prefix i, ending completely outside
        // ↳ a segment
        // dp[i][1]: Max score in prefix i, ending inside an active
        // ↳ segment
        vector<vector<DPState>> dp(n + 1, vector<DPState>(2, {0, 0}));

        for (int i = 1; i <= n; i++) {
            // Transition 0: We don't pick item i
            dp[i][0] = max(dp[i-1][0], dp[i-1][1]);

            // Transition 1: We pick item i
            // Rule: ONLY apply penalty when STARTING a new
            // ↳ pick/segment!
            // Type B logic: To ensure non-adjacency, we must
            // ↳ transition from dp[i-1][0].
            // --- IF MAXIMIZING PROFIT ---
            DPState start_new = {dp[i-1][0].cost + a[i] - penalty,
            // ↳ dp[i-1][0].count + 1};

            // Replaced max(start_new, continue_old) with just
            // ↳ start_new
            dp[i][1] = start_new;
        }
        return max(dp[n][0], dp[n][1]);
    };

    // 3. WQS Binary Search
    // The bounds must be large enough to cover the maximum possible
    // ↳ penalty.
    // FIXED: Dropped bounds to 2e9 to prevent k * mid_penalty from
    // ↳ overflowing long long
    long long low = -2e9 - 7, high = 2e9 + 7;
    long long best_ans = 0;

    while (low <= high) {
        long long mid_penalty = low + (high - low) / 2;

        DPState res = check_dp(mid_penalty);

        // We want EXACTLY k items.
        // If we picked >= k items, it means the penalty wasn't harsh
        // ↳ enough to stop us.
    }
}
```

```

// Even if res.count > k, we save the answer because of
↳ collinear points on the hull.
if (res.count >= k) {

    // --- IF MAXIMIZING PROFIT ---
    // We subtracted the penalty K times, so we add it back.
    best_ans = res.cost + (1LL * k * mid_penalty);

    // --- IF MINIMIZING COST ---
    // best_ans = res.cost - (1LL * k * mid_penalty);

    // Penalty was too weak, increase it to force picking
    ↳ fewer items next time
    low = mid_penalty + 1;
} else {
    // We picked too few items. The penalty was too harsh.
    high = mid_penalty - 1;
}

// Output the final answer
cout << best_ans << "\n";
}

```

8.1.9 MaxAverageSubarray

UPPER hull (maximum) and this one is the LOWER hull (minimum) - they are NOT

```

// NOTE: 02 - CHT.cpp also declares 'struct Line' / 'struct
↳ HullDynamic', but that one is the
// UPPER hull (maximum) and this one is the LOWER hull
↳ (minimum) - they are NOT
// interchangeable. If you need both, rename one pair to
↳ UpperHull / LowerHull.
/*
↳ =====
* DYNAMIC CONVEX HULL TRICK (CHT) - AVERAGE SUBARRAY TEMPLATE
*
↳ =====
*
* HOW TO USE THE 4 VARIATIONS:
*
* 1. MAX Average, LONGEST length (Default)
* - Code: Use 'lhs < rhs' in Query::operator<
* - Call: hull.insert_line(i, S[i]); hull.query_tangent_index(i,
↳ S[i]);
*
* 2. MAX Average, SHORTEST length
* - Code: Use 'lhs <= rhs' in Query::operator<
* - Call: hull.insert_line(i, S[i]); hull.query_tangent_index(i,
↳ S[i]);
*
* 3. MIN Average, LONGEST length
* - Code: Use 'lhs < rhs' in Query::operator<
* - Call: hull.insert_line(i, -S[i]); hull.query_tangent_index(i,
↳ -S[i]); // Negated
*
* 4. MIN Average, SHORTEST length
* - Code: Use 'lhs <= rhs' in Query::operator<
* - Call: hull.insert_line(i, -S[i]); hull.query_tangent_index(i,
↳ -S[i]); // Negated
*
↳ =====
↳ */

```

```

struct Query {
    ll i, Si;
};

```

```

struct Line {
    ll m, b;
    mutable function<const Line *(> succ;

    bool operator<(const Line &other) const {
        return m < other.m;
    }

    bool operator<(const Query &q) const {
        const Line *s = succ();
        if (!s) return false;

        // CORRECTED ALGEBRA: Direct cross-multiplication of slopes
        __int128_t lhs = (__int128_t)(q.Si - b) * (q.i - s->m);
        __int128_t rhs = (__int128_t)(q.Si - s->b) * (q.i - m);

        // --- TIE-BREAKING CONTROL ---
        // Change this return statement based on length requirements:
        // return lhs < rhs; // <-- Use for LONGEST subarray
        // return lhs <= rhs; // <-- Use for SHORTEST subarray
        return lhs < rhs;
    }
};

```

```

struct HullDynamic : public multiset<Line, less<>> {
    bool bad(iterator y) {
        auto z = next(y);
        if (y == begin()) {
            if (z == end()) return false;
            // CORRECTED: Lower hull keeps smallest y
            return y->m == z->m && y->b >= z->b;
        }
        auto x = prev(y);
        if (z == end()) return y->m == x->m && y->b >= x->b;

        return (__int128_t)(x->b - y->b) * (z->m - y->m) <=
↳ (__int128_t)(y->b - z->b) * (y->m - x->m);
    }

    void insert_line(ll m, ll b) {
        auto y = insert({m, b});
        y->succ = [=] { return next(y) == end() ? 0 : &*next(y); };
        if (bad(y)) {
            erase(y);
        }
    }
};

```

```

        return;
    }
    while (next(y) != end() && bad(next(y))) erase(next(y));
    while (y != begin() && bad(prev(y))) erase(prev(y));
}

int query_tangent_index(ll i, ll Si) {
    auto l = *lower_bound(Query{i, Si});
    return l.m;
}

};

// USAGE (longest subarray with average >= 0, after subtracting the
//   target from every element):
//   S[0..n] = prefix sums; the answer for each i is i - (tangent
//   index from the hull of (j, S[j])).
//   HullDynamic hull; hull.insert_line(0, 0);
//   for (int i = 1; i <= n; i++) {
//       int k = hull.query_tangent_index(i, S[i]);          // best
//       j < i
//       ans[i] = i - k;
//       hull.insert_line(i, S[i]);
//   }
// WHY IT WORKS: avg(j+1..i) >= 0 <=> S[i] - S[j] >= 0, and among
//   all valid j you want the
//   SMALLEST, which is the point of tangency from (i, S[i]) to the
//   lower hull of the points (j, S[j]).
// Binary search on the answer + prefix sums is the shorter O(n log
//   n) alternative; this is O(n).
// THE FOUR VARIATIONS, all from the same hull - only the tie-break
//   and the sign change:
//   MAX average, LONGEST segment: insert(i, S[i]), tangent(i,
//   S[i]), strict '<' in the compare
//   MAX average, SHORTEST segment: insert(i, S[i]), tangent(i,
//   S[i]), non-strict '<='
//   MIN average, LONGEST segment: insert(i, -S[i]), tangent(i,
//   -S[i]), strict '<'
//   MIN average, SHORTEST segment: insert(i, -S[i]), tangent(i,
//   -S[i]), non-strict '<='
// (Minimising is just maximising the negated prefix sums; the
//   tie-break decides which of two
//   equally-good segments you keep, which is what "longest" versus
//   "shortest" means here.)

```

8.1.10 SlopeTrick

SLOPE TRICK - maintain a CONVEX piecewise-linear $f(x)$ as two heaps of its break-points.

```

// L = left slopes (max-heap), R = right slopes (min-heap), minf =
//   current minimum value.
// Every operation below is O(log n). Classic use: "minimum total
//   |change| to make an array
//   non-decreasing", scheduling with earliness/tardiness penalties,
//   and any DP where
//   dp_i(x) = min over y<=x of dp_{i-1}(y) + |x - a_i|.
struct SlopeTrick {
    priority_queue<ll> L; // left
    // breakpoints (max-heap)
    priority_queue<ll, vector<ll>, greater<ll>> R; // right
    // breakpoints (min-heap)
    ll minf = 0, addL = 0, addR = 0; // lazy
    // shifts of each side

    ll topL() { return L.top() + addL; }
    ll topR() { return R.top() + addR; }
    void pushL(ll x) { L.push(x - addL); }
    void pushR(ll x) { R.push(x - addR); }

    void addConst(ll c) { minf += c; }
    void addRampR(ll a) { // f(x)
        += max(0, x - a)
        if (!L.empty()) minf += max(0LL, topL() - a);
        pushL(a);
        pushR(topL()); L.pop();
    }
    void addRampL(ll a) { // f(x)
        += max(0, a - x)
        if (!R.empty()) minf += max(0LL, a - topR());
        pushR(a);
        pushL(topR()); R.pop();
    }
    void addAbs(ll a) { addRampL(a), addRampR(a); } // f(x)
    += |x - a|
    void prefixMin() { while (!R.empty()) R.pop(); } // f(x)
    = min_{y<=x} f(y)
    void suffixMin() { while (!L.empty()) L.pop(); } // f(x)
    = min_{y>=x} f(y)
    void shift(ll dl, ll dr) { addL += dl, addR += dr; } // f(x)
    -> min over [x-dr, x-dl]
    ll minVal() const { return minf; }
};

// RECIPE for "make a[] non-decreasing with min sum |b_i - a_i|":
//   for each a_i: st.prefixMin(); st.addAbs(a_i); answer =
//   st.minVal()
// For "strictly increasing", first replace a_i by a_i - i.

```

8.1.11 MinPlusConvolution

MIN-PLUS / MAX-PLUS CONVOLUTION: $c[k] = \min_i (a[i] + b[k-i])$.

```
// In general this is quadratic and CONJECTURED to stay that way (the
  ↳ MinConv hypothesis:
// MinConv, MaxConv, unbounded knapsack and 0/1 knapsack are all
  ↳ equivalent under subquadratic
// reductions). So the ONLY fast versions rely on convexity.

// BOTH CONVEX -> O(n + m). The difference sequence of the answer is
  ↳ the sorted MERGE of the two
// difference sequences: greedily always take the smaller next slope.
vector<ll> minPlusConvexConvex(const vector<ll>& a, const vector<ll>&
  ↳ b) {
    int n = a.size(), m = b.size();
    vector<ll> c(n + m - 1);
    c[0] = a[0] + b[0];
    int i = 0, j = 0;
    for (int k = 1; k < n + m - 1; k++) {
        ll da = (i + 1 < n) ? a[i + 1] - a[i] : LLONG_MAX;
        ll db = (j + 1 < m) ? b[j + 1] - b[j] : LLONG_MAX;
        if (da <= db) c[k] = c[k - 1] + da, i++;
        else c[k] = c[k - 1] + db, j++;
    }
    return c;
}

// MAX-PLUS with both CONCAVE: same code, take the LARGER slope first
  ↳ (or negate and reuse).
vector<ll> maxPlusConcaveConcave(vector<ll> a, vector<ll> b) {
    for (ll& x : a) x = -x;
    for (ll& x : b) x = -x;
    auto c = minPlusConvexConvex(a, b);
    for (ll& x : c) x = -x;
    return c;
}

// ONLY ONE CONVEX -> the matrix  $X[k][i] = a[i] + b[k-i]$  is totally
  ↳ monotone (Monge), so the row
// minima can be found with SMAWK in  $O(n + m)$ , or with divide &
  ↳ conquer in  $O(n \log n)$ :
// solve(kl, kr, il, ir): take k = mid, scan i in [il, ir] for the
  ↳ best, recurse with the
// split at the optimum. Same skeleton as DivideAndConquerDP.
// 'valid(k)' must return the legal range of i for output index k -
  ↳ for the convolution above
// that is  $[\max(0, k-m+1), \min(n-1, k)]$ . Intersecting with it is what
  ↳ keeps 'best' well defined;
// without it an empty candidate set leaves best = -1 and poisons
  ↳ both recursive calls.
template <class F, class V>
void dcOpt(int kl, int kr, int il, int ir, vector<ll>& c, F cost, V
  ↳ valid) {
    if (kl > kr) return;
    int km = (kl + kr) / 2, best = -1;
    ll bv = LLONG_MAX;
    auto [lo, hi] = valid(km);
    for (int i = max(il, lo); i <= min(ir, hi); i++) {
        ll v = cost(km, i);
        if (v < bv) bv = v, best = i;
    }
    c[km] = bv;
    if (best < 0) best = max(il, lo); // no candidate:
    ↳ keep the bounds sane
    dcOpt(kl, km - 1, il, best, c, cost, valid);
    dcOpt(km + 1, kr, best, ir, c, cost, valid);
}

// NEITHER CONVEX ->  $O(n*m)$  is the best you should plan for, and that
  ↳ is not laziness: min-plus
// convolution, 0/1 knapsack, unbounded knapsack and tree knapsack
  ↳ are subquadratic-EQUIVALENT,
// so a general fast merge would break all four. Look for convexity,
  ↳ a small value range, or a
// bitset instead of a cleverer convolution.
// WHERE THIS APPEARS: merging two "cost as a function of how many
  ↳ you take" curves - group
// knapsack, tree DP where each child contributes a convex cost
  ↳ curve, and combining the
// per-part curves produced by Alien's trick.
```

8.1.12 OneDOneD

1D/1D DP OPTIMISATION - the self-referential case that divide & conquer structurally cannot do.

```
// dp[i] = min over j < i of ( dp[j] + w(j, i) )
// D&C optimisation needs the LAYERED form  $dp[k][i] = \min(dp[k-1][j]$ 
  ↳  $+ w)$  so that the whole
// previous layer is already known. Here  $dp[j]$  is produced by the
  ↳ very loop that consumes it, so
// you must decide  $dp[i]$  before you may use it. That is what this
  ↳ file solves.
// PRECONDITION: w satisfies the QUADRANGLE INEQUALITY (Monge)
//  $w(a,c) + w(b,d) \leq w(a,d) + w(b,c)$  for all  $a \leq b \leq c \leq d$ 
// checkable in the local form  $w(a,c) + w(a+1,c+1) \leq w(a+1,c) +$ 
  ↳  $w(a,c+1)$ .
// Then the optimal j for i is NON-DECREASING in i, so the array of
  ↳ "who is currently best for
// each future i" is a sequence of INTERVALS, each owned by one j -
  ↳ and finalising  $dp[i]$  can only
// append one new interval and overwrite a suffix of the old ones.
  ↳ Keep them on a stack.
//  $O(n \log n)$  (the binary search for the crossover);  $O(n)$  if the
  ↳ crossover is  $O(1)$  to find.
template <class F>
vector<ll> onedOned(int n, F w) {
    ↳  $w(j, i)$  for  $j < i$ 
    vector<ll> dp(n + 1, llinf);
    dp[0] = 0;
    vector<array<int, 3>> st;
    ↳ {owner j, from, to} intervals
    st.push_back({0, 1, n});
    int head = 0;
    for (int i = 1; i <= n; i++) {
        while (head < (int)st.size() && st[head][2] < i) head++;
        dp[i] = dp[st[head][0]] + w(st[head][0], i);
        ↳ the owner of position i
        st[head][1] = i;
        ↳ now insert i as a candidate owner for the suffix it wins
        auto better = [&](int a, int b, int x) { // is
            ↳ a better than b at x?
            return dp[a] + w(a, x) <= dp[b] + w(b, x);
        };
        while (!st.empty() && st.back()[1] > i && better(i,
        ↳ st.back()[0], st.back()[1])) st.pop_back();
        if (st.empty() || head >= (int)st.size()) st.push_back({i, i
        ↳ + 1, n});
        else {
            auto [j, l, r] = st.back();
            if (better(i, j, r)) { // i
                ↳ wins part of this interval
                int lo = max(l, i + 1), hi = r, pos = r + 1;
                while (lo <= hi) { //
                    ↳ first x where i beats j
                    int m = (lo + hi) / 2;
                    better(i, j, m) ? (pos = m, hi = m - 1) : lo = m
                    ↳ + 1;
                }
                st.back()[2] = pos - 1;
                if (pos <= n) st.push_back({i, pos, n});
            } else if (r < n) st.push_back({i, r + 1, n});
        }
    }
    return dp;
}

/* WHEN YOU NEED WHICH OPTIMISATION - the whole family on one card
dp[i] = min_j (dp[j] + w(j,i)), w Monge, SELF-REFERENTIAL
  ↳ -> THIS FILE,  $O(n \log n)$ 
dp[k][i] = min_j (dp[k-1][j] + w(j,i)), w Monge, LAYERED
  ↳ -> D&C opt (01),  $O(n k \log n)$ 
dp[i][j] = min_k (dp[i][k] + dp[k+1][j]) + w(i,j), w Monge AND
  ↳ monotone on intervals
  ↳ -> Knuth (06),  $O(n^2)$ 
dp[i] = min_j (m_j * x_i + b_j)
  ↳ -> CHT (02) / Li Chao (04)
"exactly k pieces", optimum convex in k
  ↳ -> Aliens (07)
min-plus convolution of a CONVEX sequence with anything
  ↳ -> SMAWK / D&C (11)
the query point moves monotonically and lines are range-updated
  ↳ -> kinetic segment tree
VERIFY the quadrangle inequality numerically on random quadruples
  ↳ before trusting any of them:
a DP optimisation applied without its precondition is a silent
  ↳ wrong answer, never a crash.
COMMON MONGE COSTS:  $(S_j - S_i)^2$ ;  $f(S_j - S_i)$  with f convex and
  ↳  $a[] \geq 0$ ;  $c * (j - i)$ ;
"number of equal pairs inside [i,j]";  $a_i * b_j$  with a
  ↳ non-increasing and b non-decreasing.
```

NOT MONGE: anything built from max, and anything with negative a[]
 ↪ where you also need the
 interval-monotonicity half (D&C only needs QI; Knuth needs both).
 ↪ */

8.1.13 KineticSegmentTree

KINETIC SEGMENT TREE ("segment tree beats" for lines). Each position i owns a LINE

```
// f_i(x) = a_i * x + b_i and a current parameter x; the tree supports
// addX(l, r, d) : x += d for every position in [l, r] (d >=
  ↪ 0, MONOTONE)
// queryMax(l, r) : max over the range of a_i * x_i + b_i
// setLine(i, a, b): replace one line
// in AMORTISED O(log^2 n) per operation (O(n log^2 n) total), and
  ↪ that amortisation is the point:
// a Li Chao tree answers "many lines, one query point" while THIS
  ↪ answers "one line per position,
// query points that drift". 12 - OneDOneD.cpp lists it as the tool
  ↪ for "the query point moves
// monotonically and the lines are range-updated" - this is that tool.
// PRECONDITION FOR THE COMPLEXITY (not for correctness): x only ever
  ↪ INCREASES. Each node keeps
// 'melt' = how much further x may advance before the argmax inside
  ↪ it changes; a decrease would
// invalidate every melt and destroy the potential-function argument
  ↪ (the total number of argmax
// changes over the whole run is O(n log^2 n) only for monotone x).
// INDEX CONVENTION: 0-based positions, inclusive ranges, x starts at
  ↪ 0 for every position.
struct KineticSegTree {
    struct Node { ll val, slope, melt; }; //
    ↪ melt = INF when nothing can flip
    int n;
    vector<Node> t;
    vector<ll> lz; //
    ↪ pending x-advance for a subtree
    KineticSegTree(const vector<ll>& a, const vector<ll>& b) {
        n = a.size(), t.assign(4 * n, {0, 0, 0}), lz.assign(4 * n, 0);
        build(1, 0, n - 1, a, b);
    }
    Node merge(const Node& L, const Node& R) {
        const Node &w = L.val >= R.val ? L : R, &o = L.val >= R.val ?
    ↪ R : L;
        ll cross = llinf;
        if (o.slope > w.slope) cross = (w.val - o.val) / (o.slope -
    ↪ w.slope) + 1;
        return {w.val, w.slope, min({L.melt, R.melt, cross})};
    }
    void build(int v, int l, int r, const vector<ll>& a, const
    ↪ vector<ll>& b) {
        if (l == r) { t[v] = {b[l], a[l], llinf}; return; }
        int m = (l + r) / 2;
        build(2 * v, l, m, a, b), build(2 * v + 1, m + 1, r, a, b);
        t[v] = merge(t[2 * v], t[2 * v + 1]);
    }
    void apply(int v, ll d) { //
    ↪ legal only when d < t[v].melt
        t[v].val += t[v].slope * d;
        if (t[v].melt != llinf) t[v].melt -= d;
        lz[v] += d;
    }
    void push(int v) {
        if (!lz[v]) return;
        apply(2 * v, lz[v]), apply(2 * v + 1, lz[v]), lz[v] = 0;
    }
    void addX(int v, int l, int r, int ql, int qr, ll d) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr && d < t[v].melt) { apply(v, d);
    ↪ return; }
        if (l == r) { t[v].val += t[v].slope * d; return; } // a
    ↪ leaf never "melts"
        push(v);
        int m = (l + r) / 2;
        addX(2 * v, l, m, ql, qr, d), addX(2 * v + 1, m + 1, r, ql,
    ↪ qr, d);
        t[v] = merge(t[2 * v], t[2 * v + 1]); // the
    ↪ RECURSION is the beats step
    }
    ll query(int v, int l, int r, int ql, int qr) {
        if (qr < l || r < ql) return LLONG_MIN;
        if (ql <= l && r <= qr) return t[v].val;
        push(v);
        int m = (l + r) / 2;
        return max(query(2 * v, l, m, ql, qr), query(2 * v + 1, m +
    ↪ 1, r, ql, qr));
    }
    void setLine(int v, int l, int r, int i, ll a, ll b, ll x) {
        if (l == r) { t[v] = {a * x + b, a, llinf}; return; }
        push(v);
        int m = (l + r) / 2;
        i <= m ? setLine(2 * v, l, m, i, a, b, x) : setLine(2 * v +
    ↪ 1, m + 1, r, i, a, b, x);
    }
```



```

    t[v] = merge(t[2 * v], t[2 * v + 1]);
}
void addX(int l, int r, ll d) { addX(1, 0, n - 1, l, r, d); }
ll queryMax(int l, int r) { return query(1, 0, n - 1, l, r); }
void setLine(int i, ll a, ll b, ll x) { setLine(1, 0, n - 1, i,
    ↪ a, b, x); }
};
/* USING IT
THE MODEL PROBLEM: process items left to right; item i contributes
    ↪ a_i * (how long it has been
    open) + b_i, and you repeatedly ask for the best open item. Every
    ↪ "time step" is one range
    addX, every answer one queryMax.
SLOPE TRICK / APERIODIC DPs of the form dp[i] = max_j (dp[j] + w_j *
    ↪ (i - j)) become: insert a
    line at position j when it becomes available, addX(0, n-1, 1) each
    ↪ step, query the max. That
    is the self-referential case D&C optimisation cannot do (see 12 -
    ↪ OneDOneD.cpp).
THE MELT FORMULA is integer division; with real-valued slopes use
    ↪ long double and a small
    epsilon, or scale to integers. 'cross' must be the FIRST x at
    ↪ which the loser strictly wins,
    which is why it is (diff / dslope) + 1 - dropping the +1 makes the
    ↪ tree do nothing and the
    answers silently stale.
DECREASING x, or arbitrary jumps, still give CORRECT answers (the
    ↪ recursion just triggers more
    often) but the complexity degrades to O(n) per operation in the
    ↪ worst case.
SIMPLER ALTERNATIVES FIRST: if all lines are inserted once and
    ↪ queries are offline-sortable, use
    CHT (02) or Li Chao (04). Reach for this only when positions are
    ↪ updated in RANGES. */

```

8.2 Bitmask DP

8.2.1 SOS_DP

Computes the sum of all subsets for every bitmask in $O(N * 2^N)$

```

vector<long long> SOS_DP(int n_bits, const vector<long long>& a) {
    int max_mask = 1 << n_bits;
    vector<long long> dp = a; // dp[mask] initially holds exactly the
    ↪ value for mask

    for (int i = 0; i < n_bits; ++i) {
        for (int mask = 0; mask < max_mask; ++mask) {
            if (mask & (1 << i)) {
                dp[mask] += dp[mask ^ (1 << i)];
            }
        }
    }
    return dp; // dp[mask] now contains sum of all submasks of mask
}

```

8.2.2 BitmaskDP

BITMASK DP - the $n \leq 20$ toolkit.

```

// MEMORY: vector<vector<ll>> dp(1<<20, vector<ll>(20)) is ~193 MB
    ↪ (24 B of vector header per row).
// Flatten to ONE vector<ll> of size (1<<n)*n (168 MB) or vector<int>
    ↪ (84 MB), or iterate masks in
// increasing popcount and keep only two layers. At n = 20, 2^n * n^2
    ↪ = 4e8 - usually not intended.

// TSP: shortest Hamiltonian cycle from 0 back to 0.  $O(2^n * n^2)$ .
ll tsp(const vector<vector<ll>>& d) {
    int n = d.size(), N = 1 << n;
    vector<vector<ll>> dp(N, vector<ll>(n, llinf));
    dp[1][0] = 0;
    for (int m = 1; m < N; m++)
        for (int u = 0; u < n; u++) if ((m >> u & 1) && dp[m][u] <
            ↪ llinf)
            for (int v = 0; v < n; v++) if (!(m >> v & 1) && d[u][v]
                ↪ < llinf)
                dp[m | 1 << v][v] = min(dp[m | 1 << v][v], dp[m][u] +
                    ↪ d[u][v]);
    ll best = llinf;
    for (int u = 1; u < n; u++) if (dp[N - 1][u] < llinf && d[u][0] <
        ↪ llinf)
        best = min(best, dp[N - 1][u] + d[u][0]);
    return n == 1 ? 0 : best; // drop the
    ↪ +d[u][0] for a Hamiltonian PATH
}

// MINIMUM SET COVER / partition into the fewest valid groups.  $O(3^n)$ 
    ↪ via submask enumeration.
// ok[m] = true if the set m is a legal single group.
int minPartition(int n, const vector<char>& ok) {
    int N = 1 << n;
    vector<int> dp(N, inf);
    dp[0] = 0;
    for (int m = 1; m < N; m++) {
        int low = m & -m; // fix the
        ↪ lowest element: kills the n! factor
        for (int s = m; s; s = (s - 1) & m) if ((s & low) && ok[s])
            dp[m] = min(dp[m], dp[m ^ s] + 1);
    }
    return dp[N - 1];
}

// COUNTING the partitions instead: same loop with dp[m] += dp[m ^ s].
// If groups are UNORDERED, fixing the lowest element (above) already
    ↪ prevents double counting.

// ASSIGNMENT (n tasks to n workers, minimum cost),  $O(2^n * n)$ . dp[m]
    ↪ = best cost using the
// first popcount(m) workers on exactly the task set m.
ll assignment(const vector<vector<ll>>& c) {
    int n = c.size(), N = 1 << n;
    vector<ll> dp(N, llinf);
    dp[0] = 0;
    for (int m = 0; m < N - 1; m++) if (dp[m] < llinf) {
        int i = __builtin_popcount(m); // next worker
        for (int j = 0; j < n; j++) if (!(m >> j & 1))
            dp[m | 1 << j] = min(dp[m | 1 << j], dp[m] + c[i][j]);
    }
    return dp[N - 1]; // n > 20 -> use
    ↪ the Hungarian algorithm
}

// COUNTING HAMILTONIAN PATHS / permutations with adjacency
    ↪ constraints: same shape as tsp but
// dp[m][u] += dp[m ^ (1<<u)][v] over allowed v.
// SUBSET-SUM STYLE: when the state is "which items used" AND the
    ↪ order does not matter, prefer
// dp over masks; when only a total matters, use a knapsack - 2^n is
    ↪ not always necessary.
// MEET IN THE MIDDLE: n <= 40 -> split in half, enumerate 2^20 each
    ↪ side, sort + two pointers.

```

8.2.3 SubsetConvolution

ZETA / MOBIUS OVER THE SUBSET LATTICE, and SUBSET CONVOLUTION. 01 - SOS_DP.cpp has the plain

```
// "sum over submasks"; this file adds the INVERSE (which is what
↳ turns a sum-over-subsets answer
// back into an exact-set answer) and the convolution that requires
↳ the parts to be DISJOINT.
// OR-convolution c[k] = sum over i | j = k of a[i] b[j] ->
↳ zeta, multiply, mobius
// AND-convolution c[k] = sum over i & j = k of a[i] b[j] ->
↳ superZeta, mul, superMobius
// XOR-convolution c[k] = sum over i ^ j = k of a[i] b[j] ->
↳ FWHT, see 06 - Math/05/03
// SUBSET conv c[k] = sum over i | j = k AND i & j = 0 ->
↳ THIS FILE, O(2^n n^2)
// WHEN: "partition a set into groups" DPs (count ways to split into
↳ k connected pieces, minimum
// cost exact cover, chromatic number), where the naive "for each
↳ mask, for each submask" is
// O(3^n) - subset convolution replaces that with O(2^n n^2), which
↳ at n = 20 is 4e8 vs 3.5e9.
void zeta(vector<ll>& f, int n) {
↳ g[s] = sum over t subset of s
for (int i = 0; i < n; i++)
for (int s = 0; s < (1 << n); s++)
if (s >> i & 1) f[s] = (f[s] + f[s ^ (1 << i)]) % MOD;
}
void mobius(vector<ll>& f, int n) {
↳ exact inverse of zeta
for (int i = 0; i < n; i++)
for (int s = 0; s < (1 << n); s++)
if (s >> i & 1) f[s] = (f[s] - f[s ^ (1 << i)]) % MOD +
↳ MOD) % MOD;
}
void superZeta(vector<ll>& f, int n) {
↳ g[s] = sum over t superset of s
for (int i = 0; i < n; i++)
for (int s = 0; s < (1 << n); s++)
if (!(s >> i & 1)) f[s] = (f[s] + f[s | (1 << i)]) % MOD;
}
void superMobius(vector<ll>& f, int n) {
for (int i = 0; i < n; i++)
for (int s = 0; s < (1 << n); s++)
if (!(s >> i & 1)) f[s] = (f[s] - f[s | (1 << i)]) % MOD +
↳ MOD) % MOD;
}
// SUBSET CONVOLUTION. Rank by POPCOUNT: after the zeta transform, a
↳ product of the rank-i and
// rank-j layers counts pairs with i | j SUBSET of s and |i| + |j| =
↳ k; reading the answer at
// k = popcount(s) forces i | j = s and |i| + |j| = |s|, which for a
↳ union is exactly i & j = 0.
// That ranking is the entire idea - without it, disjointness is not
↳ expressible as a transform.
vector<ll> subsetConvolution(const vector<ll>& A, const vector<ll>&
↳ B, int n) {
int N = 1 << n;
vector<vector<ll>> f(n + 1, vector<ll>(N, 0)), g = f, h = f;
for (int s = 0; s < N; s++) {
int p = __builtin_popcount(s);
f[p][s] = A[s], g[p][s] = B[s];
}
for (int k = 0; k <= n; k++) zeta(f[k], n), zeta(g[k], n);
for (int k = 0; k <= n; k++)
for (int i = 0; i <= k; i++)
for (int s = 0; s < N; s++) h[k][s] = (h[k][s] + f[i][s]
↳ * g[k - i][s]) % MOD;
for (int k = 0; k <= n; k++) mobius(h[k], n);
vector<ll> c(N);
for (int s = 0; s < N; s++) c[s] = h[__builtin_popcount(s)][s];
return c;
}
/* USING IT
SET PARTITION / EXACT COVER: let f[s] = 1 if s is a legal piece. The
↳ number of ways to cover the
full set with exactly k pieces is the k-th subset-convolution
↳ POWER of f at s = full; dividing
by k! makes the pieces unordered. Chromatic number = smallest k
↳ with (independent sets)^k > 0.
SUBSET EXP / LOG exist too (O(2^n n^2)) and turn "connected pieces"
↳ into "all pieces" and back -
the standard route for counting CONNECTED labelled structures on
↳ subsets.
MIN-PLUS version (minimum cost exact cover) does NOT work: min-plus
↳ has no inverse, so the
mobius step is invalid. Use the O(3^n) submask enumeration for
↳ min/max objectives.
```

O(3^n) SUBMASK ENUMERATION is 'for (int t = s; t; t = (t - 1) & s)'
↳ and remains the right answer
for n <= 15 or when the operation is not a ring - it is shorter
↳ and has no modulus.
XOR needs FWHT, not this; AND/OR need only the transform pair with
↳ no ranking.
OVERFLOW: everything here is mod MOD. Without a modulus use __int128
↳ or accept unsigned wraparound
for OR/AND (the mobius step is exact in Z, so wraparound cancels).
↳ */

8.2.4 DynamicSubmaskCount

DYNAMIC SUBMASK COUNTING - a multiset of masks with INSERT, ERASE and the query

```
// "how many stored masks a satisfy a & s == a (a is a submask
//   ↳ of s)"
// in  $O(2^{B/2})$  per operation, B = number of bits. SOS DP answers
//   ↳ this in  $O(1)$  but only for a
// STATIC multiset (any insert costs a full  $O(2^B)$  rebuild);
//   ↳ enumerating submasks of s per query
// is  $O(2^B)$ . This is the meet-in-the-middle between the two, and the
//   ↳ technique generalises to any
// "precompute forwards for heavy items, backwards for light items"
//   ↳ split.
// THE SPLIT: call a mask HEAVY if popcount > B/2, LIGHT otherwise.
// LIGHT masks are stored twice: once at their own index, and once
//   ↳ at EVERY SUBMASK of theirs
// ( $\leq 2^{B/2}$  of them) in sup[], so sup[t] = #light masks that
//   ↳ CONTAIN t.
// HEAVY masks are stored at every SUPERMASK of theirs ( $\leq 2^{B/2}$ )
//   ↳ of them) in sub[],
// so sub[t] = #heavy masks that are CONTAINED in t.
// A query with a light s only needs light answers (a submask of s
//   ↳ has popcount  $\leq$  popcount(s)),
// and is a submask enumeration over s,  $\leq 2^{B/2}$  steps. A query
//   ↳ with a heavy s reads sub[s] for
// the heavy part and inclusion-exclusion over the ZERO bits of s
//   ↳ (there are  $\leq B/2$  of them) for
// the light part: #light masks meeting  $\sim s$  = sum over non-empty T in
//   ↳  $\sim s$  of  $(-1)^{|T|+1}$  sup[T].
template <int B>
struct SubmaskCounter {
    static const int SZ = 1 << B, H = B / 2;
    vector<int> own, sup, sub; //
    ↳ own[m] = #light copies of m
    int lightTotal = 0;
    SubmaskCounter() : own(SZ, 0), sup(SZ, 0), sub(SZ, 0) {}
    void modify(int m, int d) { // d =
    ↳ +1 to insert, -1 to erase
        if (__builtin_popcount(m) <= H) {
            own[m] += d, lightTotal += d;
            for (int t = m;; t = (t - 1) & m) { sup[t] += d; if (!t)
    ↳ break; }
        } else {
            int zeros = ((SZ - 1) ^ m);
            for (int t = zeros;; t = (t - 1) & zeros) { sub[m | t] +=
    ↳ d; if (!t) break; }
        }
    }
    void insert(int m) { modify(m, 1); }
    void erase(int m) { modify(m, -1); }
    int count(int s) { //
    ↳ #stored masks that are submasks
        if (__builtin_popcount(s) <= H) {
            int r = 0;
            for (int t = s;; t = (t - 1) & s) { r += own[t]; if (!t)
    ↳ break; }
            return r;
        }
        int zeros = ((SZ - 1) ^ s), bad = 0;
        for (int t = zeros; t; t = (t - 1) & zeros) //
    ↳ inclusion-exclusion, T non-empty
            bad += (__builtin_popcount(t) & 1) ? sup[t] : -sup[t];
        return sub[s] + lightTotal - bad;
    }
};
/* NOTES
THE THRESHOLD H = B/2 balances  $2^H$  against  $2^{(B-H)}$ ; with skewed
↳ workloads (many more queries
than inserts) move it - all four loops stay correct for any H.
MEMORY is  $3 * 2^B$  ints: B = 20 is 12 MB, B = 22 is 50 MB. Above
↳ that, drop 'own' (it equals the
light part of a separate map) or switch to a hash map for the
↳ sparse arrays.
SUPERMASK COUNTING ("how many stored a with a & s == s") is the same
↳ code with the roles of sup
and sub swapped, or just complement everything: a superset of s is
↳ a submask of  $\sim s$  complemented.
EXACTLY-EQUAL, OR "how many a with a & s == 0" (disjoint) =
↳ count( $\sim s$ ) - both fall out for free.
STATIC ALTERNATIVE: if all inserts precede all queries, run SOS DP
↳ once and answer in  $O(1)$ . Check
that first; this structure is only for genuinely interleaved
↳ updates and queries. */
```

8.3 Knapsack

8.3.1 BoundedKnapsackBitset

BOUNDED KNAPSACK, FEASIBILITY ONLY, with a bitset: $O(n W / 64)$. Binary-group the counts

```
// (1,2,4,...) so "cnt copies of w" becomes  $O(\log \text{cnt})$  shift-ors. Use
//   ↳ when you only need which sums
// are reachable, not the best value.
const int MAX_W = 100005;
bitset<MAX_W> dp;

void multiset_bitset_dp() {
    // Example: We have 13 items, each with weight 5
    int count = 13;
    int weight = 5;

    vector<int> bundled_weights;

    // Binary grouping logic
    int current_power = 1;
    while (count >= current_power) {
        bundled_weights.push_back(current_power * weight);
        count -= current_power;
        current_power *= 2;
    }
    // Add whatever is left over
    if (count > 0) {
        bundled_weights.push_back(count * weight);
    }

    // Now run the ultra-fast Bitset DP on the bundles
    dp[0] = 1;
    for (int bundle_w : bundled_weights) {
        dp |= (dp << bundle_w);
    }
}
```

8.3.2 KnapsackFamily

KNAPSACK FAMILY. W = capacity. All $O(n*W)$ unless noted.

```
// 0/1: iterate capacity DOWNWARD so each item is used once.
ll knap01(const vector<int>& w, const vector<ll>& v, int W) {
    vector<ll> dp(W + 1, 0);
    for (size_t i = 0; i < w.size(); i++)
        for (int c = W; c >= w[i]; c--) dp[c] = max(dp[c], dp[c -
    ↳ w[i]] + v[i]);
    return dp[W];
}
// UNBOUNDED: iterate capacity UPWARD.
ll knapInf(const vector<int>& w, const vector<ll>& v, int W) {
    vector<ll> dp(W + 1, 0);
    for (size_t i = 0; i < w.size(); i++)
        for (int c = w[i]; c <= W; c++) dp[c] = max(dp[c], dp[c -
    ↳ w[i]] + v[i]);
    return dp[W];
}
// BOUNDED (item i available cnt[i] times) via BINARY GROUPING: split
↳ cnt into 1,2,4,...,rest
// and run 0/1 on the groups.  $O(W * \sum \log \text{cnt})$ .
ll knapBounded(const vector<int>& w, const vector<ll>& v, const
    ↳ vector<int>& cnt, int W) {
    vector<int> W2; vector<ll> V2;
    for (size_t i = 0; i < w.size(); i++) {
        int c = cnt[i];
        for (int k = 1; c > 0; k <= 1) {
            int t = min(k, c); c -= t;
            W2.push_back(w[i] * t), V2.push_back(v[i] * t);
        }
    }
    return knap01(W2, V2, W);
}
// FEASIBILITY ONLY (which sums are reachable) -> bitset,  $O(n*W/64)$ .
↳ Massively faster.
bitset<100001> reachableSums(const vector<int>& w) {
    bitset<100001> dp;
    dp[0] = 1;
    for (int x : w) dp |= dp << x;
    return dp;
}
// COUNTING ways (mod p): same loops, dp[c] += dp[c - w[i]].
// Order matters: item-outer/capacity-inner counts COMBINATIONS,
// capacity-outer/item-inner counts PERMUTATIONS
↳ (ordered sequences).
```

8.3.3 BoundedKnapsackDeque

BOUNDED KNAPSACK IN $O(n * W)$ - item i may be used at most $cnt[i]$ times. The binary-grouping

```
// version in 02 - KnapsackFamily.cpp is  $O(W * \sum \log cnt)$ , which is
    ↳ the one to write by default;
// reach for this when sum log cnt is too big (huge counts, or W
    ↳ already at the limit).
// THE REWRITE: fix item  $i$  and split the capacities by RESIDUE mod
    ↳  $w[i]$ . Along one residue class,
//  $dp'[t] = \max \text{ over } k \text{ in } [0, cnt] \text{ of } dp[t - k] + k * val$ 
//  $= (\max \text{ over } j \text{ in } [t - cnt, t] \text{ of } dp[j] - j * val) + t$ 
    ↳  $* val$ 
// which is a SLIDING WINDOW MAXIMUM of width  $cnt + 1$  - a monotone
    ↳ deque,  $O(1)$  amortised per cell.
// PRECONDITION: none beyond  $w[i] > 1$ ; this is an exact rewrite, not
    ↳ an approximation. (The
// subtracted term  $j * val$  is what makes the window comparable - do
    ↳ not forget to add it back.)
ll boundedKnapsackDeque(const vector<int>& w, const vector<ll>& val,
    const vector<int>& cnt, int W) {
    int n = w.size();
    vector<ll> dp(W + 1, 0), nd(W + 1, 0);
    for (int i = 0; i < n; i++) {
        int wi = w[i], m = cnt[i];
        ll v = val[i];
        if (wi <= 0 || m <= 0) continue;
        ↳ weight 0 items: add  $m * v$  to all
        if (m > W / wi) m = W / wi;
        ↳ more copies than fit
        for (int s = 0; s < wi && s <= W; s++) {
            deque<pair<int, ll>> dq;
            ↳ {t,
            dp[s + t*wi] - t*v}
            for (int t = 0; s + (ll)t * wi <= W; t++) {
                ll cur = dp[s + t * wi] - (ll)t * v;
                while (!dq.empty() && dq.back().second <= cur)
                    dq.pop_back();
                dq.push_back({t, cur});
                while (dq.front().first < t - m) dq.pop_front();
                nd[s + t * wi] = dq.front().second + (ll)t * v;
            }
            dp.swap(nd);
        }
        return dp[W];
    }
    ↳ dp[c] = best with weight <= c
}
/* THE FAMILY, so you pick the right one under time pressure
cnt[i] = 1 -> 0/1 knapsack,  $O(n W)$ 
    ↳ (02 - KnapsackFamily)
cnt[i] = infinity -> unbounded,  $O(n W)$ , forward loop
    ↳ (02 - KnapsackFamily)
general cnt[i], small sum log -> binary grouping,  $O(W \sum \log cnt)$ 
    ↳ (02 - KnapsackFamily)
general cnt[i], W large -> THIS FILE,  $O(n W)$ 
FEASIBILITY only (no values) -> bitset,  $O(n W / 64)$ 
    ↳ (02 - KnapsackFamily)
feasibility, many equal weights ->  $O(S \sqrt{S})$  by distinct value
    ↳ (04 - SubsetSumSqrt)
COUNTING the number of ways with bounded multiplicity: the deque
    ↳ does NOT apply (sums, not
    ↳ maxima) - use prefix sums along the same residue classes instead,
    ↳  $dp'[t] = P[t] - P[t - cnt - 1]$ ,
    ↳ which is the same  $O(n W)$  with the same residue decomposition.
RECONSTRUCTION: store which  $t$  the deque front came from, or re-run
    ↳ the DP layer by layer keeping
    ↳ all  $n$  rows ( $O(n W)$  memory) and walk back.
IF THE VALUES ARE THE WEIGHTS (subset sum), the reachability bitset
    ↳ beats everything by a factor
    ↳ of 64 - check that first, it is one line. */
```

8.3.4 SubsetSumSqrt

SUBSET SUM IN $O(S \sqrt{S})$ WITH RECONSTRUCTION, where S is the sum of all values (or the target,

```
// whichever is smaller). The bitset DP is  $O(n S / 64)$  and usually
    ↳ wins, but it does NOT
// reconstruct the subset and it degrades when  $n$  is large; this one
    ↳ is  $O(S \sqrt{S})$  REGARDLESS of
//  $n$  and hands back the item indices.
// THE COUNTING ARGUMENT that gives sqrt: only DISTINCT values
    ↳ matter, and if the values sum to  $S$ 
// there are at most  $O(\sqrt{S})$  distinct ones. For a fixed value  $v$ 
    ↳ with multiplicity  $c$ , "reachable
// with at most  $c$  copies of  $v$ " is computed in ONE  $O(S)$  pass: walk
    ↳ each residue class mod  $v$  and
// carry a counter of how many copies are still available - a
    ↳ reachable cell refills the counter
// to  $c$ , an unreachable cell consumes one copy if any remain. That is
    ↳ the bounded-knapsack
// feasibility trick without a deque.
// PRECONDITION: all values > 0 (a zero value makes the residue walk
    ↳ spin).
// RETURNS the indices of a subset summing to exactly 'target', or an
    ↳ empty vector if impossible.
vector<int> subsetSumSqrt(const vector<int>& a, int target) {
    if (target < 0) return {};
    vector<char> dp(target + 1, 0);
    vector<int> last(target + 1, -1), cnt(target + 1, 0);
    vector<vector<int>> id(target + 1);
    for (int i = 0; i < (int)a.size(); i++)
        if (a[i] > 0 && a[i] <= target) id[a[i]].push_back(i),
        ↳ cnt[a[i]]++;
    dp[0] = 1;
    for (int v = 1; v <= target; v++) {
        if (!cnt[v]) continue;
        for (int r = 0; r < v; r++) {
            int left = 0;
            ↳ copies of  $v$  still usable
            for (int k = r; k <= target; k += v) {
                if (dp[k]) left = cnt[v];
                ↳ reachable without any copy here
                else if (left) dp[k] = 1, left--, last[k] =
                ↳ id[v][left];
            }
        }
        if (!dp[target]) return {};
    }
    vector<int> res;
    for (int s = target; s > 0; s -= a[last[s]])
        ↳ res.push_back(last[s]);
    return res;
}
/* WHEN TO USE WHAT
n small, S small -> plain  $O(n S)$  DP, easiest to modify.
feasibility only -> bitset  $dp |= dp << a[i]$ ,  $O(n S /$ 
    ↳  $64)$ . Fastest by far.
need the SUBSET back -> THIS FILE, or the bitset plus  $O(n)$ 
    ↳ "peel one item at a time" with
    ↳ a per-item bitset snapshot ( $O(n S /$ 
    ↳  $64)$  memory - usually too much).
many equal values -> THIS FILE is optimal: it is  $O(S)$  per
    ↳ DISTINCT value.
counting the number of subsets, not feasibility -> the residue walk
    ↳ does not apply; use prefix
    ↳ sums per residue class (bounded) or
    ↳ plain  $O(n S)$  (unbounded).
CLASSIC USES: "split the array into two halves with a given sum",
    ↳ partitioning tree levels
    ↳ (CF 1481F), and any "choose a subset of the multiset of subtree
    ↳ sizes" argument - those all
    ↳ have sum of values =  $n$ , hence  $O(n \sqrt{n})$ .
THE SAME sqrt ARGUMENT gives the  $O(n \sqrt{n} / 64)$  bound for "subset
    ↳ sums of an array whose
    ↳ values sum to  $n$ " with bitsets - group equal values, binary-group
    ↳ each group, and the total
    ↳ number of groups is  $O(\sqrt{n} \log n)$ . */
```

8.4 Classics

8.4.1 DigitDP

DIGIT DP - count numbers in [L, R] with a digit property. $f(R) - f(L-1)$.

```
// State: (position, tight, started, problem state). 'started'
// ↳ handles leading zeros.
// THIS CODE IS BASE 10 (to_string, digit cap 9, memo[20] positions).
// ↳ For another base, write the
// digits yourself, change the 9, and size memo by the digit count
// ↳ (base 2 needs 64 positions).
// L - 1 IS NOT ALWAYS AVAILABLE: if the bounds are given as huge
// ↳ decimal STRINGS you cannot
// subtract one, so carry two tight flags (tightLo, tightHi) and
// ↳ count [L, R] in a single pass.
string DIG;
ll memo[20][2][2][200];
bool vis[20][2][2][200];

ll go(int p, int tight, int started, int st) {
    if (p == (int)DIG.size()) return started ? /* ACCEPT? */ (st ==
        ↳ 0) : 0; // ↳ edit
    if (vis[p][tight][started][st]) return
        ↳ memo[p][tight][started][st];
    vis[p][tight][started][st] = true;
    ll res = 0;
    int hi = tight ? DIG[p] - '0' : 9;
    for (int d = 0; d <= hi; d++) {
        int nStarted = started || d > 0;
        int nSt = nStarted ? (st + d) % 10 : 0;
        ↳ // ↳ edit
        res += go(p + 1, tight && d == hi, nStarted, nSt);
    }
    return memo[p][tight][started][st] = res;
}

bool qualifiesZero = 1; // does the number 0
// ↳ satisfy the predicate?
ll countUpTo(ll x) {
    if (x < 0) return 0;
    DIG = to_string(x);
    memset(vis, 0, sizeof vis);
    return go(0, 1, 0, 0) + qualifiesZero; // set qualifiesZero
    ↳ = 1 only if 0 itself counts
}

ll countRange(ll L, ll R) { return countUpTo(R) - countUpTo(L - 1); }
// VARIANTS: track (sum of digits), (value mod k), (last digit), (a
// ↳ bitmask of used digits),
// (whether we already saw "13"), (count so far) - anything with a
// ↳ small state.
// For "k-th number with the property", binary search on countUpTo.
```

8.4.2 LIS

LIS in $O(n \log n)$, with reconstruction. STRICT increasing by default.

```
// Non-decreasing: change lower_bound -> upper_bound.
// Longest DEcreasing: negate the array. Longest non-increasing:
// ↳ negate + upper_bound.
vector<int> lis(const vector<ll>& a) {
    int n = a.size();
    vector<ll> tail; // tail[k] =
    ↳ smallest possible tail of an LIS of length k+1
    vector<int> pos(n, -1), idx;
    for (int i = 0; i < n; i++) {
        int k = lower_bound(all(tail), a[i]) - tail.begin();
        if (k == (int)tail.size()) tail.push_back(a[i]),
        ↳ idx.push_back(i);
        else tail[k] = a[i], idx[k] = i;
        pos[i] = k;
        prv[i] = k ? idx[k - 1] : -1;
    }
    vector<int> res;
    for (int i = idx.empty() ? -1 : idx.back(); i >= 0; i = prv[i])
        ↳ res.push_back(i);
    reverse(all(res));
    return res; // indices of one
    ↳ LIS; length = tail.size()
}

// SAME SHAPE SOLVES: longest chain of pairs (sort by x, LIS on y);
// ↳ maximum non-overlapping
// intervals; "minimum number of non-increasing subsequences covering
// ↳ the array" = LIS length
// (Dilworth). For LIS with an extra weight, replace 'tail' with a
// ↳ max-BIT over compressed values.
```

8.4.3 IntervalDP

INTERVAL DP - dp over [l, r], processed by increasing length. $O(n^3)$ ($O(n^2)$ with Knuth).

```
// This is the base pattern that Knuth's optimisation speeds up;
// ↳ write this first, optimise later.

// 1) SPLIT form: dp[l][r] = min over k of dp[l][k] + dp[k+1][r] +
// ↳ cost(l, r)
// ↳ matrix chain multiplication, merging stones, optimal BST, "cost
// ↳ to combine a range"
template <class Cost> ll intervalSplit(int n, Cost cost) {
    vector<vector<ll>> dp(n, vector<ll>(n, 0));
    for (int len = 2; len <= n; len++)
        for (int l = 0, r = len - 1; r < n; l++, r++) {
            dp[l][r] = llinf;
            for (int k = l; k < r; k++) dp[l][r] = min(dp[l][r],
                ↳ dp[l][k] + dp[k + 1][r]);
            dp[l][r] += cost(l, r);
        }
    return dp[0][n - 1];
}

// 2) ENDPOINT form: decide about a[l] or a[r] and shrink.
// ↳ Palindromes, "remove from either end".
// Minimum insertions to make s a palindrome:
int minInsertPalindrome(const string& s) {
    int n = s.size();
    vector<vector<int>> dp(n, vector<int>(n, 0));
    for (int len = 2; len <= n; len++)
        for (int l = 0, r = len - 1; r < n; l++, r++)
            dp[l][r] = s[l] == s[r] ? dp[l + 1][r - 1] : 1 + min(dp[l
                ↳ + 1][r], dp[l][r - 1]);
    return n ? dp[0][n - 1] : 0;
}

// 3) "BURST BALLOONS" form: choose the LAST element removed in the
// ↳ range, so the neighbours are
// ↳ the (still intact) boundaries l-1 and r+1.
ll burst(vector<ll> a) {
    int n = a.size();
    a.insert(a.begin(), 1), a.push_back(1);
    vector<vector<ll>> dp(n + 2, vector<ll>(n + 2, 0));
    for (int len = 1; len <= n; len++)
        for (int l = 1, r = len; r <= n; l++, r++)
            for (int k = l; k <= r; k++)
                dp[l][r] = max(dp[l][r], dp[l][k - 1] + a[l - 1] *
                    ↳ a[k] * a[r + 1] + dp[k + 1][r]);
    return dp[1][n];
}

// WHEN KNUTH APPLIES (form 1): cost is monotone on intervals and
// ↳ satisfies the quadrangle
// inequality; then  $opt[l][r-1] \leq opt[l][r] \leq opt[l+1][r]$  and the
// ↳ k-loop becomes amortised  $O(1)$ .
// COUNTING variants (e.g. #ways to parenthesise) replace min with +
// ↳ and cost with 1.
```


8.4.4 BrokenProfileDP

BROKEN PROFILE DP - sweep the grid cell by cell carrying a bitmask of the "broken" frontier.

```
// O(n * m * 2^m); always make m the SMALLER dimension.
// mask bit j = "cell (i, j) is already covered by something placed
  ⇨ earlier".

// Count tilings of an n x m grid by 1x2 dominoes.
ll dominoTilings(int n, int m) {
    if (m > n) swap(n, m);
    if (((ll)n * m) & 1) return 0;
    int M = 1 << m;
    vector<ll> cur(M, 0), nxt;
    cur[0] = 1;
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++) {
            nxt.assign(M, 0);
            for (int mask = 0; mask < M; mask++) if (cur[mask]) {
                if (mask >> j & 1) nxt[mask ^ (1 << j)] += cur[mask];
                ⇨ // already covered
            } else {
                if (i + 1 < n) nxt[mask | (1 << j)] += cur[mask];
                ⇨ // vertical domino
                if (j + 1 < m && !(mask >> (j + 1) & 1))
                    nxt[mask | (1 << (j + 1))] += cur[mask];
                ⇨ // horizontal domino
            }
        }
    cur = nxt;
    return cur[0];
}

// BLOCKED CELLS: keep a grid 'bad[i][j]'; a blocked cell must arrive
  ⇨ already "covered"
// (skip it by clearing bit j and forbidding any placement on it).
// OTHER PIECES: add one transition per shape; the mask must record
  ⇨ every cell the shape
// occupies in a LATER position of the sweep.
// COUNTING INDEPENDENT SETS / no-two-adjacent colourings: mask =
  ⇨ which cells of the previous
// row are used; transition over masks with (a & b) == 0 and no two
  ⇨ adjacent bits in b.
// AS WRITTEN this is O(n * 4^m) (nested mask loops), not O(n * m *
  ⇨ 2^m): at m = 14 that is 2.7e8
// per row. To get O(n * 2^m), note sum over a with (a & b) == 0 of
  ⇨ cur[a] = zetaSub(cur)[~b],
// so one subset-zeta pass per row replaces the inner loop (see 05 -
  ⇨ Combinatorics/07).
ll independentSets(int n, int m) { // #ways to
    ⇨ pick cells, none orthogonally adjacent
    if (m > n) swap(n, m);
    int M = 1 << m;
    vector<ll> cur(M, 0), nxt;
    for (int b = 0; b < M; b++) if (!(b & (b << 1))) cur[b] = 1;
    for (int i = 1; i < n; i++) {
        nxt.assign(M, 0);
        for (int a = 0; a < M; a++) if (cur[a])
            for (int b = 0; b < M; b++) if (!(b & (b << 1)) && !(a &
                ⇨ b)) nxt[b] += cur[a];
        cur = nxt;
    }
    ll r = 0;
    for (ll x : cur) r += x;
    return r;
}

// CONNECTED-COMPONENT DP ("plug DP") is the heavy version: the state
  ⇨ stores, for each frontier
// cell, WHICH connected component its plug belongs to (a
  ⇨ bracket-like labelling). Use it for
// "count Hamiltonian cycles / single-closed-loop tilings"; it is
  ⇨ long - only write it if the
// problem clearly needs connectivity, not just coverage.
```

8.4.5 MaximalRectangle

LARGEST ALL-ZERO (or all-one) RECTANGLE in a binary grid, O(n*m).

```
// Row by row, keep the histogram of consecutive zeros ending at this
  ⇨ row, then run the
// largest-rectangle-in-histogram monotonic stack.
ll maximalRectangle(const vector<vector<int>>& g, int target = 0) {
    int n = g.size(), m = n ? g[0].size() : 0;
    vector<int> h(m, 0);
    ll best = 0;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) h[j] = (g[i][j] == target) ? h[j]
            ⇨ + 1 : 0;
        vector<int> st; // indices,
            ⇨ increasing height
        for (int j = 0; j <= m; j++) {
            int cur = (j == m) ? -1 : h[j];
            while (!st.empty() && h[st.back()] >= cur) {
                int ht = h[st.back()]; st.pop_back();
                int left = st.empty() ? -1 : st.back();
                best = max(best, (ll)ht * (j - left - 1));
            }
            st.push_back(j);
        }
        st.clear();
    }
    return best;
}

// LARGEST SQUARE instead: dp[i][j] = min(dp[i-1][j], dp[i][j-1],
  ⇨ dp[i-1][j-1]) + 1 when free.
// COUNTING all all-zero submatrices: for each row, for each column j
  ⇨ keep 'run[j]' = the number
// of rectangles ending at (i, j); with a monotonic stack, run[j] =
  ⇨ run[prevSmaller] + h[j] *
// (j - prevSmaller), and the total is the sum of run[j] over all
  ⇨ cells. O(n*m).
ll countAllZeroSubmatrices(const vector<vector<int>>& g) {
    int n = g.size(), m = n ? g[0].size() : 0;
    vector<ll> h(m, 0), run(m, 0);
    ll total = 0;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) h[j] = g[i][j] == 0 ? h[j] + 1 :
            ⇨ 0;
        vector<int> st;
        for (int j = 0; j < m; j++) {
            while (!st.empty() && h[st.back()] >= h[j]) st.pop_back();
            int p = st.empty() ? -1 : st.back();
            run[j] = (p < 0 ? 0 : run[p]) + h[j] * (j - p);
            total += run[j];
            st.push_back(j);
        }
    }
    return total;
}
```


8.4.6 Hirschberg

HIRSCHBERG - reconstruct the ANSWER of a two-sequence DP (LCS, alignment, edit script) in

```
// O(n m) time but only O(min(n, m)) memory instead of the O(n m)
//   ↳ table. WHEN: n = m = 1e5 and you
// need the actual subsequence, not its length - the table would be
//   ↳ 1e10 cells.
// THE IDEA (divide & conquer on the ROW, not on the value): the last
//   ↳ row of the DP for the top
// half of a, and the last row of the REVERSED DP for the bottom
//   ↳ half, meet at the middle row.
// The column k maximising L[k] + R[m - k] is a point the optimal
//   ↳ path passes through, so the
// problem splits into (top half of a vs b[0..k]) and (bottom half of
//   ↳ a vs b[k..m]).
// COST: T(n, m) = O(n m) + T(n/2, k) + T(n/2, m - k) = O(n m) total
//   ↳ - the halves of the second
// level together cost half of the first level, so the series is 2 n
//   ↳ m.
// PRECONDITION: the DP must be reversible - dp computed
//   ↳ left-to-right on the reversed strings
// must equal the suffix DP. True for LCS, edit distance and
//   ↳ affine-gap alignment; NOT true for
// DPs whose transition depends on the direction (e.g. "no two chosen
//   ↳ positions adjacent" is fine,
// but anything reading a prefix STATE such as a running remainder is
//   ↳ not).
vector<int> lcsLastRow(const string& a, const string& b) { //
    ↳ O(|b|) memory
    int m = b.size();
    vector<int> prv(m + 1, 0), cur(m + 1, 0);
    for (char ca : a) {
        for (int j = 1; j <= m; j++)
            cur[j] = ca == b[j - 1] ? prv[j - 1] + 1 : max(prv[j],
    ↳ cur[j - 1]);
        swap(prv, cur);
    }
    return prv;
}
string hirschbergLCS(const string& a, const string& b) {
    int n = a.size(), m = b.size();
    if (n == 0 || m == 0) return "";
    if (n == 1) return b.find(a[0]) != string::npos ? string(1, a[0])
    ↳ : "";
    int i = n / 2;
    string a1 = a.substr(0, i), a2 = a.substr(i);
    string ra(a2.rbegin(), a2.rend()), rb(b.rbegin(), b.rend());
    vector<int> L = lcsLastRow(a1, b), R = lcsLastRow(ra, rb);
    int best = -1, k = 0;
    for (int j = 0; j <= m; j++) if (L[j] + R[m - j] > best) best =
    ↳ L[j] + R[m - j], k = j;
    return hirschbergLCS(a1, b.substr(0, k)) + hirschbergLCS(a2,
    ↳ b.substr(k));
}
/* ADAPTING IT
EDIT DISTANCE / ALIGNMENT: replace lcsLastRow with the edit-distance
    ↳ row (initialised to
    0,1,2,... instead of zeros) and take the MINIMUM of L[j] + R[m-j]
    ↳ instead of the maximum. The
    split column is then a cell the optimal alignment passes through;
    ↳ emit the operations.
AFFINE GAP COSTS need three rows (match / gap-in-a / gap-in-b) and
    ↳ the split must record WHICH
    of the three states the path is in at the middle row - otherwise
    ↳ the two halves disagree.
MEMORY: O(m) for the rows plus O(m log n) for the substrings taken
    ↳ at each recursion level; pass
    index ranges instead of substrings if that matters.
RECURSION DEPTH is O(log n) - safe.
IF YOU ONLY NEED THE LENGTH, do not use this: one rolling row is
    ↳ enough (and for LCS, the
    bit-parallel version in 07 - Strings/07 - Sequences/03 is O(n m /
    ↳ 64)).
THE SAME TRICK reconstructs paths in any layered DAG DP where a
    ↳ middle layer is cheap to
    evaluate from both ends - it is the DP analogue of "meet in the
    ↳ middle". */
```

8.4.7 MaximumSubarray

MAXIMUM SUBARRAY (Kadane) and its dimensional relatives. All exact, all one pass per dimension.

```
// KADANE: the best subarray ending at i either extends the best one
//   ↳ ending at i-1 or restarts.
// O(n). Returns {sum, l, r} with the segment NON-EMPTY (if every
//   ↳ element is negative the answer
// is the largest single element - the empty-segment convention gives
//   ↳ 0 instead, so decide which
// one the problem wants BEFORE writing the loop; that is the only
//   ↳ bug this ever has).
array<ll, 3> maxSubarray(const vector<ll>& a) {
    int n = a.size();
    ll best = LLONG_MIN, cur = 0;
    int bl = 0, br = 0, start = 0;
    for (int i = 0; i < n; i++) {
        if (cur <= 0) cur = a[i], start = i; //
        ↳ restart: the prefix only hurts
        else cur += a[i];
        if (cur > best) best = cur, bl = start, br = i;
    }
    return {best, bl, br};
}
// CIRCULAR (the array wraps): the answer is either an ordinary
//   ↳ subarray, or the complement of the
// MINIMUM subarray. Guard the all-negative case - "total - min"
//   ↳ would return the empty segment.
ll maxCircularSubarray(const vector<ll>& a) {
    ll total = 0, mx = LLONG_MIN, mn = LLONG_MAX, cx = 0, cn = 0;
    for (ll x : a) {
        total += x;
        cx = max(x, cx + x), mx = max(mx, cx);
        cn = min(x, cn + x), mn = min(mn, cn);
    }
    return mx < 0 ? mx : max(mx, total - mn);
}
// MAXIMUM SUM SUBMATRIX, O(rows^2 * cols): fix the top and bottom
//   ↳ rows, compress the columns
// between them into a 1D array by prefix sums, and run Kadane on it.
//   ↳ Make ROWS the smaller side.
ll maxSubmatrix(const vector<vector<ll>>& g) {
    int n = g.size(), m = n ? g[0].size() : 0;
    ll best = LLONG_MIN;
    for (int top = 0; top < n; top++) {
        vector<ll> col(m, 0);
        for (int bot = top; bot < n; bot++) {
            for (int j = 0; j < m; j++) col[j] += g[bot][j];
            best = max(best, maxSubarray(col)[0]);
        }
    }
    return best;
}
/* THE VARIANTS, and which structure each one needs
LENGTH AT LEAST k: prefix sums S, answer = max over i >= k of S[i] -
    ↳ min(S[0..i-k]) - one running
    minimum, O(n).
LENGTH AT MOST k: S[i] - min over a SLIDING WINDOW of S, so a
    ↳ monotone deque, O(n).
AT MOST k NEGATIVE ELEMENTS / at most k "skips": dp[i][j] = best
    ↳ ending at i having used j skips.
MAXIMUM AVERAGE subarray (with a length constraint): binary search
    ↳ the average and run the
    "length at least k" version on a[i] - mid, or the tangent trick in
    ↳ 01 - DP Opt/09.
MAXIMUM PRODUCT subarray: carry BOTH the running max and the running
    ↳ min (a negative times a
    negative flips them), O(n).
K-TH LARGEST subarray sum: binary search the value plus a
    ↳ two-pointer/BIT count of pairs.
MAXIMUM SUBARRAY WITH UPDATES / on a range: segment tree storing
    ↳ (total, best prefix, best
    suffix, best inside) - see 02 - Data Structures; this file is the
    ↳ static case.
ALL-NEGATIVE ARRAYS are the standard trap in every one of these. */
```

8.4.8 ConnectedComponentDP

CONNECTED-COMPONENT DP (a.k.a. the "broken interval" or profile-free connectivity DP).

```
// SHAPE OF THE PROBLEM: you must build one object out of n items
//   ↳ processed in SORTED order, the
// cost depends on how far apart consecutive items end up, and the
//   ↳ final object must be CONNECTED
// (one block, one path, one permutation with a cyclic condition).
//   ↳ The state is NOT "which items
// are used" (2^n) but "how many disjoint fragments exist so far",
//   ↳ because the fragments are
// interchangeable - only their COUNT matters, and every fragment can
//   ↳ be glued later.
// STATE: dp[i][c][cost][ends] = first i items placed, c open
//   ↳ fragments, accumulated cost, and
// 'ends' = how many of the two outer boundaries have been fixed (0,
//   ↳ 1 or 2) when the object is a
// PATH rather than a cycle. Drop 'ends' for objects with no
//   ↳ distinguished endpoints.
// THE COST TRICK, which is the whole point: when moving from item i
//   ↳ to item i+1, EVERY fragment
// boundary that is still open must eventually cross the gap (a[i+1]
//   ↳ - a[i]). There are
// (2c - ends) such open boundaries, so add (a[i+1] - a[i]) * (2c -
//   ↳ ends) to the cost NOW, once,
// instead of trying to know later which items are adjacent. That is
//   ↳ what removes the 2^n.
// COMPLEXITY: O(n^2 * K) states with O(1) transitions, K = cost
//   ↳ budget. The classic instance is
// JOI Skyscraper: count permutations of a[] with sum |a[i+1] - a[i]|
//   ↳ <= K.
// PRECONDITION: sort a[] first - the cost accounting above is only
//   ↳ valid for sorted items.
ll skyscraper(vector<ll> a, int K) {
    int n = a.size();
    sort(a.begin(), a.end());
    if (n == 1) return 1;
    // dp[c][cost][ends]
    vector<vector<array<ll, 3>>> dp(n + 2, vector<array<ll, 3>>>(K +
    ↳ 1, {0, 0, 0}));
    dp[0][0][0] = 1;
    for (int i = 0; i < n; i++) {
        vector<vector<array<ll, 3>>> nd(n + 2, vector<array<ll, 3>>>(K
    ↳ + 1, {0, 0, 0}));
        ll gap = i ? a[i] - a[i - 1] : 0;
        for (int c = 0; c <= i; c++)
            for (int cost = 0; cost <= K; cost++)
                for (int e = 0; e < 3; e++) {
                    ll v = dp[c][cost][e];
                    if (!v) continue;
                    ll nc = cost + gap * (2 * c - e); //
    ↳ pay for every open boundary
                    if (nc > K) continue;
                    auto add = [&](int c2, int e2, ll ways) {
                        // after placing an item at least one
                        // guard the state (c = 0, ends = 1) survives
                        if (c2 < 1 || c2 > n || ways <= 0) return;
                        nd[c2][nc][e2] = (nd[c2][nc][e2] + v % MOD *
    ↳ (ways % MOD)) % MOD;
                    };
                    add(c + 1, e, c + 1 - e); //
    ↳ new fragment, interior
                    add(c + 1, e + 1, e < 2 ? 2 - e : 0); //
    ↳ new fragment, at an end
                    add(c, e, 2 * c - e); //
    ↳ append to one fragment's side
                    add(c, e + 1, e < 2 ? 2 - e : 0); //
    ↳ append and make it an end
                    if (c >= 2) add(c - 1, e, c - 1); //
    ↳ merge two fragments
                }
        dp = nd;
    }
    ll res = 0;
    for (int cost = 0; cost <= K; cost++) res = (res +
    ↳ dp[1][cost][2]) % MOD;
    return res; // one
    ↳ fragment, both ends fixed
}
/* READING THE TRANSITIONS (they are always these five, in every
   ↳ problem of this family)
NEW FRAGMENT: the item starts a block of its own. Interior blocks
   ↳ can be created in
(c + 1 - ends) ways (the gaps between existing blocks); if it is
   ↳ to become an outer end of the
```

final object, that is one of the (2 - ends) remaining end slots.
 EXTEND: glue the item to one of the (2c - ends) open sides of an
 ↳ existing block, or spend it as
 an outer end of that block (again 2 - ends ways).
 MERGE: join two adjacent blocks with this item, (c - 1) ordered
 ↳ choices of which pair.
 ACCEPT at the end only c == 1 (connected!) and ends == 2 (both
 ↳ boundaries closed).
 OTHER PROBLEMS IN THIS FAMILY: "arrange n numbers in a row
 ↳ minimising the sum of adjacent
 differences with constraints", counting linear extensions with gap
 ↳ costs, "cut a rod into
 pieces so that ..." and any permutation problem whose cost
 ↳ telescopes over sorted values.
 IF THE OBJECT IS A CYCLE, delete the 'ends' dimension entirely and
 ↳ accept c == 1.
 WATCH THE OVERCOUNTING: 'add(c, e + 1, 2 - e)' and 'add(c + 1, e +
 ↳ 1, 2 - e)' both consume an end
 slot - forgetting one of them, or letting e exceed 2, silently
 ↳ changes the answer. */

8.4.9 PlugDP

PLUG DP WITH CONNECTIVITY (the bracket / minimum-representation profile DP). This is the

```
// heavyweight cousin of 04 - BrokenProfileDP.cpp: broken profile
// only remembers WHICH frontier
// cells are occupied, which is enough for tilings; here the frontier
// must also remember WHICH
// PARTIAL PATHS ARE CONNECTED TO WHICH, so that closing a loop early
// can be forbidden.
// ENCODING: the frontier has m + 1 "plugs". Each plug is 0
// (nothing), 1 (a LEFT bracket) or
// 2 (a RIGHT bracket), packed two bits per plug. Because the paths
// live in a planar grid they
// can never cross, so the connection pattern is a balanced bracket
// sequence - matching brackets
// = "these two plug ends belong to the same partial path". THAT is
// why 3 states per plug suffice
// instead of a general set partition of the frontier.
// COMPLEXITY: O(n * m * S) where S = number of reachable states,
// empirically a few thousand for
// m <= 12 (the Motzkin-like count, far below 3^(m+1)); keep the
// states in a hash map.
// INDEX CONVENTION (get this wrong and nothing works): cells are
// 0-based and scanned row by row.
// BEFORE processing cell (i, j) the state holds, in plug index order
// 0..m:
// indices < j : the BOTTOM edges of the cells (i, 0..j-1) already
// placed in this row
// index j : the LEFT edge of (i, j) <- p below
// index j+1 : the TOP edge of (i, j) <- q below
// indices > j+1: the TOP edges of (i, j+1..m-1), inherited from
// the previous row
// AFTER processing, plug j becomes the BOTTOM edge of (i, j) and
// plug j+1 its RIGHT edge, which
// is exactly the invariant for cell (i, j+1). At the end of a row
// plug m must be 0 (no path may
// leave the grid) and the state is SHIFTED LEFT by one plug. Plug k
// uses bits 2k and 2k+1.
// THE PROBLEM SOLVED HERE: count HAMILTONIAN CYCLES in a grid with
// blocked cells (URAL 1519
// "Formula 1"). Every other plug-DP problem is this loop with
// different accept/reject rules.
struct PlugDP {
    int n, m;
    vector<string> g; // '. '
    // free, '*' blocked
    PlugDP(const vector<string>& grid) : g(grid) { n = g.size(), m =
    n ? g[0].size() : 0; }
    static int get(ll s, int i) { return (s >> (2 * i)) & 3; }
    static ll set(ll s, int i, int v) { return (s & ~(3LL << (2 *
    i))) | ((ll)v << (2 * i)); }
    int matchRight(ll s, int i) { //
    // find the ')' matching '(' at i
    int cnt = 0;
    for (int j = i; j <= m; j++) {
        int v = get(s, j);
        if (v == 1) cnt++;
        else if (v == 2 && --cnt == 0) return j;
    }
    return -1;
    }
    int matchLeft(ll s, int i) { //
    // find the '(' matching ')' at i
    int cnt = 0;
    for (int j = i; j >= 0; j--) {
        int v = get(s, j);
        if (v == 2) cnt++;
        else if (v == 1 && --cnt == 0) return j;
    }
    return -1;
    }
    ll solve() {
        int li = -1, lj = -1; // the
        // LAST free cell
        for (int i = 0; i < n; i++)
            for (int j = 0; j < m; j++) if (g[i][j] == '. ') li = i,
        lj = j;
        if (li < 0) return 0;
        ll ans = 0;
        map<ll, ll> cur, nxt;
        cur[0] = 1;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) {
                nxt.clear();
                for (auto& [s, w] : cur) {
                    int p = get(s, j), q = get(s, j + 1);
                    if (g[i][j] == '*') {

```

```

        // blocked: no plug may touch it
        if (!p && !q) nxt[s] += w;
        continue;
    }
    if (!p && !q) { //
        // open a new path: down and right
        if (i + 1 < n && j + 1 < m) nxt[set(set(s, j,
        1), j + 1, 2)] += w;
    } else if (!p || !q) { // one
        // plug: continue down or right
        int v = p ? p : q;
        if (i + 1 < n) nxt[set(set(s, j, v), j + 1,
        0)] += w; // go down
        if (j + 1 < m) nxt[set(set(s, j, 0), j + 1,
        v)] += w; // go right
    } else if (p == 1 && q == 1) { // two
        // '(' : merge, retag the match
        int k = matchRight(s, j + 1);
        nxt[set(set(set(s, k, 1), j, 0), j + 1, 0)]
        += w;
    } else if (p == 2 && q == 2) { // two
        // ')' : merge, retag the match
        int k = matchLeft(s, j);
        nxt[set(set(set(s, k, 2), j, 0), j + 1, 0)]
        += w;
    } else if (p == 2 && q == 1) { //
        // different paths: just join them
        nxt[set(set(s, j, 0), j + 1, 0)] += w;
    } else if (i == li && j == lj) { // p
        // == 1 && q == 2: closes a loop
        ll t = set(set(s, j, 0), j + 1, 0); //
        // legal ONLY at the last free cell
        if (t == 0) ans += w;
    }
    }
    cur.swap(nxt);
    }
    nxt.clear(); // end
    // of row: shift the profile
    for (auto& [s, w] : cur) if (!get(s, m)) nxt[s << 2] += w;
    cur.swap(nxt);
    }
    return ans;
}
};
/* ADAPTING IT - the five cases above never change, only what you do
// with them
COUNT HAMILTONIAN PATHS (not cycles): allow the "close" case at any
// cell but require exactly one
open path at the end, or add a virtual start/end cell.
SIMPLE CYCLE THAT NEED NOT COVER EVERY CELL (max weight cycle): let
// a free cell be SKIPPED (the
// 'p && !q' case also allows keeping the state unchanged), and
// accept the closure at any cell,
// taking the max instead of the sum.
PARTITION INTO CLOSED LOOPS: accept every closure and multiply, no
// "last cell" restriction.
THE THREE-STATE ENCODING ONLY WORKS FOR NON-CROSSING (planar)
// connections. For general
connectivity (Steiner-tree profile DPs, spanning forests) you need
// MINIMUM REPRESENTATION:
store, per frontier cell, the id of its component normalised to
// 0,1,2,... left to right - same
loop, bigger state, and a canonicalisation step after every move.
ALWAYS make m the SMALLER dimension (transpose if needed) - the
// state count is exponential in m.
DEBUGGING: a 4x4 empty grid has 6 Hamiltonian cycles, 6x6 has 1072,
// 8x8 has 425; a grid with an
odd number of free cells has none. Check those before running
// anything bigger. */

```

8.4.10 DPOverDivisors

DP OVER THE DIVISOR LATTICE OF ONE NUMBER - the exact analogue of SOS DP, with "submask"

```
// replaced by "divisor". The divisors of n form a lattice isomorphic
// to a product of chains (one
// per prime, of length e_p), so the same "one dimension at a time"
// trick works:
// for each prime p: for divisors in increasing order: g[d] +=
//   g[d / p]
// gives g[d] = sum over e | d of f[e] in O(D * omega(n)) instead of
//   O(D^2) pair testing, where
// D = number of divisors (<= 1344 below 1e9) and omega = number of
//   distinct primes (<= 9).
// WHEN: "for every divisor d of n, count the array elements whose
//   gcd with n is exactly d" (the
//   mobius direction), "sum over all divisors of a divisor",
//   multiplicative-function DPs restricted
// to one n, and the CF-style "count subsets whose gcd is exactly d".
// INDEX CONVENTION: divs is SORTED ASCENDING and f is indexed by
//   POSITION in that array; pos maps
// a divisor value to its index. Sorted order guarantees d / p comes
//   before d, which is what makes
// the in-place update legal - do not shuffle divs.
struct DivisorLattice {
    vector<ll> divs;
    vector<ll> primes;
    map<ll, int> pos;
    DivisorLattice(ll n, const vector<ll>& primeFactors) :
    { primes(primeFactors) {
        for (ll i = 1; i * i <= n; i++)
            if (n % i == 0) { divs.push_back(i); if (i != n / i)
    divs.push_back(n / i); }
        sort(divs.begin(), divs.end());
        for (int i = 0; i < (int)divs.size(); i++) pos[divs[i]] = i;
    }
    void zeta(vector<ll>& f) { //
    { g[d] = sum over e | d of f[e]
        for (ll p : primes)
            for (int i = 0; i < (int)divs.size(); i++)
                if (divs[i] % p == 0) f[i] += f[pos[divs[i] / p]];
    }
    void mobius(vector<ll>& f) { //
    { exact inverse of zeta
        for (ll p : primes)
            for (int i = (int)divs.size() - 1; i >= 0; i--)
                if (divs[i] % p == 0) f[i] -= f[pos[divs[i] / p]];
    }
    void superZeta(vector<ll>& f) { //
    { g[d] = sum over d | e of f[e]
        for (ll p : primes)
            for (int i = (int)divs.size() - 1; i >= 0; i--)
                if (divs[i] % p == 0) f[pos[divs[i] / p]] += f[i];
    }
    void superMobius(vector<ll>& f) { //
    { exact inverse of superZeta
        for (ll p : primes)
            for (int i = 0; i < (int)divs.size(); i++)
                if (divs[i] % p == 0) f[pos[divs[i] / p]] -= f[i];
    }
};
/* THE CANONICAL USE
cnt[d] = number of array elements divisible by d (easy: for each
    element, add 1 at gcd(a, n));
then superZeta turns "exactly d" counts into "divisible by d" counts
    and superMobius turns them
back. Counting subsets with gcd exactly d is then: ways[d] =
    2^(divisible by d) - 1, followed by
superMobius over the divisor lattice.
OVER 1..N INSTEAD OF THE DIVISORS OF ONE n: the same transforms are
    the harmonic loops
    for (d = 1..N) for (m = 2d, 3d, ...) f[m] += f[d] (divisor zeta,
    O(N log N))
    for (d = N..1) for (m = 2d, 3d, ...) f[d] += f[m] (multiple
    zeta, O(N log N))
and their mobius inverses reverse the inner direction. See 04 -
    Number Theory/04/02.
GCD CONVOLUTION c[k] = sum over gcd(i,j) = k of a[i] b[j]:
    multipleZeta both, multiply pointwise,
multipleMobius the result. LCM convolution is the same with the
    divisor transforms.
WHY NOT JUST TEST ALL PAIRS: D^2 is 1.8e6 for D = 1344, fine once -
    but inside a loop over
queries or over n it is not, and the lattice version is 12000
    operations.
THE PRIME LIST must be the DISTINCT primes of n (factorise with 04 -
    NT/02/06 Pollard). Passing a
prime twice double-counts and silently corrupts every value. */
```

8.4.11 XorEquationCount

COUNTING SOLUTIONS OF $x_1 \text{ xor } x_2 \text{ xor } \dots \text{ xor } x_n = X$ with $0 \leq x_i \leq a_i$, in $O(n * B)$.

```
// The bounds are DIFFERENT per variable, which is what makes this
//   hard: with a common bound
// 2^B - 1 the answer is just 2^(B(n-1)) [X < 2^B]. Digit DP over the
//   bits with an "is this
// variable still tight" flag per variable would need 2^n states -
//   the trick below avoids that.
// THE TRICK: classify a solution by the HIGHEST bit k at which some
//   bits with an "is this
// Above bit k every variable equals a_i exactly, so that prefix is
//   forced. At bit k, at least one
// variable with a_i's bit set chooses 0 and becomes FREE below k;
//   every other variable is either
// still tight (its low bits are then bounded by a_i's low bits) or
//   already free (2^k choices).
// One designated free variable absorbs the xor constraint on the low
//   bits - it has 2^k choices
// and exactly ONE of them fixes the remaining xor - which is why the
//   first freed variable
// contributes a factor of 1 and every later one a factor of 2^k.
// If NOBODY drops at bit k, the parity of the k-th bits is forced to
//   popcount-parity, so the
// recursion continues to bit k-1 only when that parity matches X's
//   k-th bit.
// PRECONDITION: a_i >= 0 and X >= 0; B must cover every a_i and X
//   (bit B-1 is the top).
ll countXorSolutions(const vector<ll>& a, ll X, int B = 60) {
    int n = a.size();
    ll res = 0;
    for (int k = B - 1; k >= 0; k--) {
        ll low = (1LL << k) % MOD;
        // dp[freed][parity of the k-th bits chosen so far]
        ll dp[2][2] = {{1, 0}, {0, 0}};
        for (int i = 0; i < n; i++) {
            ll lo = a[i] & ((1LL << (k + 1)) - 1); //
            // bits k..0 of a_i
            ll nd[2][2] = {{0, 0}, {0, 0}};
            if (a[i] >> k & 1) {
                ll keep = (lo - (1LL << k) + 1) % MOD; // x_i
            }
            // bit k = 1, low bits <= a_i
            for (int f = 0; f < 2; f++)
                for (int p = 0; p < 2; p++) {
                    if (!dp[f][p]) continue;
                    nd[f][p ^ 1] = (nd[f][p ^ 1] + dp[f][p] *
            keep) % MOD; // stay tight
                    ll fac = f ? low : 1; // the
                }
            // FIRST freed one absorbs xor
            nd[1][p] = (nd[1][p] + dp[f][p] * fac) % MOD;
            // drop to 0 here
        }
        // drop to 0 here
    }
    // else {
        ll ch = (lo + 1) % MOD; // x_i
        // bit k = 0, low bits <= a_i
        for (int f = 0; f < 2; f++)
            for (int p = 0; p < 2; p++) nd[f][p] = dp[f][p] *
        ch % MOD;
    }
    memcpy(dp, nd, sizeof dp);
    res = (res + dp[1][(X >> k) & 1]) % MOD;
    int par = 0;
    // nobody dropped at bit k
    for (int i = 0; i < n; i++) par ^= (a[i] >> k) & 1;
    if (par != ((X >> k) & 1)) return res; // the
    // forced prefix already fails
    return (res + 1) % MOD; // the
}
/* VARIANTS
SUM OF THE SOLUTIONS, or counting with x_i in [lo_i, hi_i]:
    inclusion-exclusion over the lower
    bounds (subtract the count with x_i <= lo_i - 1) - 2^n terms, so
    only for tiny n; otherwise
    carry a second "already below lo_i" flag, which doubles the state,
    not the exponent.
AND / OR CONSTRAINTS instead of xor: the same "first variable that
    drops" decomposition works,
    with the parity condition replaced by the AND/OR of the k-th bits.
XOR EQUAL TO AT MOST X: sum the answers over the prefixes of X, i.e.
    for each bit where X has a
    1, count solutions matching X above that bit and 0 there - one
    extra outer loop.
LINEAR ALGEBRA ALTERNATIVE: if the bounds are all 2^B - 1 (or the
    variables range over a
```

subspace), this is a rank computation over GF(2), not a DP - see
 ↳ 06 - Math/07 XOR basis.
 SANITY: $n = 1$ gives $[X \leq a_1]$; all $a_i = 2^B - 1$ gives $2^{(B(n-1))}$.
 ↳ */

8.4.12 TransferMatrixDP

TRANSFER-MATRIX DP - when the DP has a HUGE time dimension (n up to $1e18$) but few states.

```
// Needs: 06 - Math/03 - Matrices/01 - MatrixExponentiation.cpp
↳ (Mat, operator*, mpow)
// RECOGNITION TEST: dp[t] is a vector over S states, and dp[t+1] = M
↳ * dp[t] with M INDEPENDENT
// of t. Then dp[n] = M^n * dp[0], computed in  $O(S^3 \log n)$ . If M
↳ depends on t only through a
// small period p, multiply the p matrices once and exponentiate the
↳ product.
// WHEN: count strings of length  $1e18$  avoiding a pattern, count walks
↳ of a given length in a
// graph, count tilings of a  $4 \times 1e18$  strip, any linear recurrence
↳ with n astronomically large.
// PRECONDITIONS, all checkable: (1) the transition is LINEAR in the
↳ dp values (sums of products
// with CONSTANTS - "max" is not linear, see min-plus below); (2) the
↳ state set does not grow with
// t; (3) the same matrix applies at every step (or with a fixed
↳ period).
// Build the matrix by RUNNING ONE STEP of the DP from each unit
↳ vector - that is mechanical and
// far less error-prone than deriving the entries by hand.
template <class Step>
Mat transferMatrix(int S, Step step) { //
↳ step(from, add) enumerates edges
  Mat M(S, vector<Mint>(S, 0));
  for (int i = 0; i < S; i++) step(i, [&](int j, Mint w) { M[j][i]
↳ += w; });
  return M; //
↳ COLUMN convention:  $dp' = M * dp$ 
}
// EXAMPLE - count strings of length n over an alphabet of size A
↳ that avoid a set of patterns:
// states = the nodes of the Aho-Corasick automaton with terminal
↳ nodes deleted, and the step is
// for each node u, for each character c: v = advance(u, c); if v
↳ is not terminal, add 1 edge.
// dp[0] = e_root, and the answer is the sum of the entries of M^n *
↳ dp[0].  $O(S^3 \log n)$ 
/* THE VARIANTS
MIN-PLUS ("shortest walk with exactly n edges", "minimum cost after
↳ n steps"): replace (+, *)
with (min, +) and exponentiate the same way - see 06 - Math/03/07
↳ - MinPlusMatrixPower.cpp.
The matrix product is still associative, which is ALL that
↳ exponentiation needs; only the
 $O(S^3)$  per multiply survives, there is no Strassen and no inverse.
KITAMASA / BOSTAN-MORI (06 - Math/06/03) computes the n-th term of a
↳ linear recurrence of order
k in  $O(k \log k \log n)$  instead of  $O(k^3 \log n)$ . If your matrix is a
↳ companion matrix - i.e. the
DP is a plain linear recurrence - use that instead; k = 1000 is
↳ hopeless for matrices and easy
for Bostan-Mori.
BERLEKAMP-MASSEY finds the recurrence for you: generate 2k terms
↳ with the slow DP, recover the
order-k recurrence, then jump to n. This is the practical route
↳ when the state space is
structured but you cannot describe the matrix cleanly.
PERIODIC / TIME-DEPENDENT transitions: if M_t has period p,
↳ exponentiate (M_{p-1} ... M_0).
PROFILE DPs on a k x n grid become transfer matrices over the 2^k
↳ profiles - that is how "count
tilings of a 5 x 1e18 board" is solved (and 2^k must stay under
↳ ~200 for the cube to fit).
EIGENVALUES / CLOSED FORMS are almost never worth it in a contest:
↳ the matrix power is  $O(S^3 \log n)$  with no numerical risk, and a closed form needs distinct
↳ eigenvalues you cannot verify.
SANITY: check M^1 and M^2 against the slow DP for small n before
↳ trusting M^1e18. */
```

8.5 Probability

8.5.1 ExpectedValueDP

EXPECTED VALUE / PROBABILITY DP.

```
// Needs: Mint (06 - Math/01) and solveLinear (06 - Math/04 - Linear
↳ Algebra/02 - GaussianElimination.cpp).
// LINEARITY OF EXPECTATION FIRST: E[sum of indicators] = sum of
↳ P(indicator). Most "expected
// number of X" problems collapse to summing per-element
↳ probabilities with no DP at all.
// Only when states depend on each other CYCLICALLY do you need a
↳ linear system.

// E[i] = 1 + sum_j P[i][j] * E[j], with E[absorbing] = 0.
// Move the unknowns left: E[i] - sum_j P[i][j] E[j] = 1 -> (I -
↳ P) E = 1.  $O(n^3)$ .
// Needs solveLinear from GaussianElimination.cpp (values are Mint,
↳ so probabilities must be
// given as modular fractions p * inv(q)).
vector<Mint> hittingTimes(const vector<vector<Mint>>& P, const
↳ vector<char>& absorbing) {
  int n = P.size();
  vector<vector<Mint>> A(n, vector<Mint>(n, 0));
  vector<Mint> b(n, 0);
  for (int i = 0; i < n; i++) {
    if (absorbing[i]) { A[i][i] = 1, b[i] = 0; continue; }
    A[i][i] = 1;
    for (int j = 0; j < n; j++) A[i][j] -= P[i][j];
    b[i] = 1;
  }
  return solveLinear(A, b);
}
/* ABSORBING MARKOV CHAIN, canonical form P = [Q R; 0 I]:
fundamental matrix N = (I - Q)^-1; expected steps before
↳ absorption = N * 1
absorption probabilities B = N * R (B[i][j] = P(absorbed in j |
↳ started in i))

TRIDIAGONAL SYSTEMS (grid random walks that only move left/right
↳ within a row) -  $O(n)$  with
the Thomas algorithm instead of  $O(n^3)$ :
a_i x_{i-1} + b_i x_i + c_i x_{i+1} = d_i, a_1 = c_n = 0
forward: c'_i = c_i / (b_i - a_i c'_{i-1}), d'_i = (d_i - a_i
↳ d'_{i-1}) / (b_i - a_i c'_{i-1})
back: x_n = d'_n, x_i = d'_i - c'_i x_{i+1}
Diagonally dominant (always true for hitting times), so it is
↳ stable.

TREES: substitute E_v = A_v * E_parent + B_v, push A and B up in
↳ one DFS, then resolve at the
root and push back down.  $O(n)$ . (Folklore - no canonical reference;
↳ derive it, do not trust it.)

COMMON TRAPS
- "expected value of a maximum" is NOT the maximum of expected
↳ values; sum over thresholds:
E[max] = sum_{t>=1} P(max >= t).
- Probabilities mod p: store p*inv(q); never mix doubles and
↳ Mint.
- Infinite processes: E exists only if absorption has
↳ probability 1; check reachability first.
- Conditional expectation: E[X] = sum_y P(Y=y) * E[X | Y=y] -
↳ condition on the FIRST step. */
```

9 Trees

Lowest Common Ancestor (LCA) 200 · DSU on Tree 202 · Heavy Light Decomposition (HLD) 203 · Centroid Decomposition 203 · Tree Diameter 205 · Euler Tour 205 · Rerooting 206 · Virtual Tree 206 · Isomorphism 207 · Tree DP 207 · Dynamic Trees 208 · Long Path Decomposition 209 · Path Operations 210

9.1 Lowest Common Ancestor (LCA)

9.1.1 LCA

LCA by Euler tour + sparse table: $O(n \log n)$ build, $O(1)$ query. 0-based, rooted at 'root'.

```
// For k-th ancestor / "jump while predicate" use BinaryLifting
// instead (02 - BinaryLifting.cpp).
struct LCA {
    static const int B = 21;
    vector<array<int, 2>> tour;
    vector<array<array<int, 2>, B>> T;
    vector<int> d, in, lg;

    template <class G>
    LCA(int n, const G& adj, int root = 0) : d(n), in(n), lg(2 * n +
    // 1) {
        tour.reserve(2 * n);
        dfs(root, -1, adj);
        int m = tour.size();           // == 2n - 1
        T.resize(m);
        for (int i = 2; i <= 2 * n; i++) lg[i] = lg[i / 2] + 1;
        for (int i = 0; i < m; i++) T[i][0] = tour[i];
        for (int j = 1; j < B; j++)
            for (int i = 0; i + (1 << j) - 1 < m; i++)
                T[i][j] = min(T[i][j - 1], T[i + (1 << (j - 1))][j -
    // 1]);
    }
    template <class G>
    void dfs(int u, int p, const G& adj) {
        in[u] = tour.size();
        tour.push_back({d[u], u});
        for (int v : adj[u]) if (v != p) {
            d[v] = d[u] + 1;
            dfs(v, u, adj);
            tour.push_back({d[v], v});
        }
    }
    int get(int u, int v) const {
        int l = min(in[u], in[v]), r = max(in[u], in[v]), j = lg[r -
    // l + 1];
        return min(T[l][j], T[r - (1 << j) + 1][j])[1];
    }
    int dist(int u, int v) const { return d[u] + d[v] - 2 * d[get(u,
    // v)]; }
};
```

9.1.2 BinaryLifting

BINARY LIFTING - k-th ancestor, LCA, and "jump while a predicate holds", all $O(\log n)$.

```
// This is NOT only a tree tool: the same table jumps  $2^k$  steps in
// ANY functional graph
// (i -> f(i)), which is how you do "apply the operation  $1e18$  times",
// permutation powers,
// and "min lamps to cover [L,R]" (jump by farthest-reach).
struct Lift {
    int n, B;
    vector<vector<int>> up;           // up[k][v] =  $2^k$ -th
    // ancestor (-1 above the root)
    vector<int> d;

    template <class G>
    Lift(int n, const G& adj, int root = 0) : n(n), B(1), d(n, 0) {
        while ((1 << B) < n) B++;
        up.assign(B, vector<int>(n, -1));
        dfs(root, -1, adj);
        for (int k = 1; k < B; k++)
            for (int v = 0; v < n; v++)
                up[k][v] = up[k - 1][v] < 0 ? -1 : up[k - 1][up[k -
    // 1][v]];
    }
    template <class G>
    void dfs(int u, int p, const G& adj) {
        up[0][u] = p;
        for (int v : adj[u]) if (v != p) d[v] = d[u] + 1, dfs(v, u,
    // adj);
    }
    int kth(int v, ll k) const {           // k-th ancestor, -1
    // if it does not exist
        for (int i = 0; i < B && v >= 0; i++) if (k >> i & 1) v =
    // up[i][v];
        return k >> B ? -1 : v;
    }
    int lca(int u, int v) const {
        if (d[u] < d[v]) swap(u, v);
        u = kth(u, d[u] - d[v]);
        if (u == v) return u;
        for (int i = B - 1; i >= 0; i--) if (up[i][u] != up[i][v]) u
    // = up[i][u], v = up[i][v];
        return up[0][u];
    }
    int dist(int u, int v) const { return d[u] + d[v] - 2 * d[lca(u,
    // v)]; }
    int onPath(int u, int v, int k) const {           // k-th vertex on the
    // path u -> v (k = 0 gives u)
        int a = lca(u, v), du = d[u] - d[a];
        return k <= du ? kth(u, k) : kth(v, d[u] + d[v] - 2 * d[a] -
    // k);
    }
    // highest ancestor of v still satisfying pred (monotone: true
    // near v, false near the root)
    template <class F>
    int jumpWhile(int v, F pred) const {
        for (int i = B - 1; i >= 0; i--) if (up[i][v] >= 0 &&
    // pred(up[i][v])) v = up[i][v];
        return v;
    }
};
// STANDALONE version for a functional graph f (no tree needed):
// up[0][v] = f(v); up[k][v] = up[k-1][up[k-1][v]]; then apply k
// steps by the bits of k.
```


9.1.3 LCAConstantTime

LCA IN $O(1)$ after $O(n \log n)$ preprocessing - Euler tour + sparse table on DEPTH.

```
// Binary lifting (02) is  $O(\log n)$  per query and needs  $O(n \log n)$ 
// memory too, so reach for this
// when the query count dominates:  $O(1)$  LCA turns "distance between
// every pair in a list" and
// "is u an ancestor of v" from  $n \log n$  into  $n$ . It is also the
// primitive behind  $O(1)$  LCE on a
// suffix tree and behind virtual trees built in bulk.
// The tour has  $2n-1$  entries; the LCA of  $u$  and  $v$  is the SHALLOWEST
// vertex between their first
// occurrences, and since consecutive tour depths differ by exactly
//  $\pm 1$  a plain sparse table over
// (depth, vertex) pairs is enough - min over an overlapping range is
// idempotent.
struct LCA {
    int n, LOG;
    vector<int> first, depth, tour;
    vector<vector<pair<int, int>>> sp; //
    // (depth, vertex)
    LCA(int n, const vector<vector<int>>& adj, int root = 0)
        : n(n), first(n, -1), depth(n, 0) {
        tour.reserve(2 * n);
        vector<pair<int, int>> st{{root, -1}}; //
        // iterative: no stack overflow
        vector<int> it(n, 0);
        depth[root] = 0;
        while (!st.empty()) {
            auto& [u, p] = st.back();
            if (first[u] < 0) first[u] = tour.size();
            if (it[u] < (int)adj[u].size()) {
                int v = adj[u][it[u]++];
                if (v == p) continue;
                tour.push_back(u), depth[v] = depth[u] + 1;
                st.push_back({v, u});
            } else {
                tour.push_back(u);
                st.pop_back();
            }
        }
        int m = tour.size();
        LOG = 1;
        while ((1 << LOG) <= m) LOG++;
        sp.assign(LOG, vector<pair<int, int>>(m));
        for (int i = 0; i < m; i++) sp[0][i] = {depth[tour[i]],
        // tour[i]};
        for (int k = 1; k < LOG; k++)
            for (int i = 0; i + (1 << k) <= m; i++)
                sp[k][i] = min(sp[k-1][i], sp[k-1][i + (1 << (k-1))]);
        // }
        int lca(int u, int v) const {
            int a = first[u], b = first[v];
            if (a > b) swap(a, b);
            int k = 31 - __builtin_clz(b - a + 1);
            return min(sp[k][a], sp[k][b - (1 << k) + 1]).second;
        }
        int dist(int u, int v) const { return depth[u] + depth[v] - 2 *
        // depth[lca(u, v)]; }
        bool isAncestor(int u, int v) const { return lca(u, v) == u; }
    };
    // TRUE  $O(n)$  PREPROCESSING is possible (Farach-Colton-Bender): split
    // the  $\pm 1$  depth array into
    // blocks of size  $\log(n)/2$ , precompute every possible block pattern
    // (there are only  $\sqrt{n}$  of
    // them), and put a sparse table over the block minima. It is  $\sim 60$ 
    // lines and only worth it at
    //  $n \geq 5e5$ , where the  $O(n \log n)$  table's memory is the real problem.

    // --- SUBTREE HASHING: are these two subtrees the same shape?  $O(n)$ 
    //  $\log n$ . ---
    // Canonical form: sort the children's ids and look the sorted vector
    // up in a map. Two subtrees
    // get the same id iff they are isomorphic as ROOTED trees.
    // Shorter than AHU (09 - Isomorphism) and merges naturally into
    // other tree DPs; AHU is still the
    // one to use when you need the UNROOTED comparison (root at the
    // centres and take the minimum).
    map<vector<int>, int> subtreeId;
    vector<int> subHash;
    int hashSubtree(int u, int p, const vector<vector<int>>& adj) {
        vector<int> ch;
        for (int v : adj[u]) if (v != p) ch.push_back(hashSubtree(v, u,
        // adj));
        sort(all(ch));
        auto it = subtreeId.find(ch);
        if (it == subtreeId.end()) it = subtreeId.emplace(ch,
```

```
    // subtreeId.size()).first;
    return subHash[u] = it->second;
}
// RANDOMISED VARIANT,  $O(n)$  and mergeable:  $h(u)$  = XOR over children
// of  $f(h(\text{child}))$  with  $f$  a fixed
// random 64-bit mixing function (splitmix64). No map, no sorting,
// and it composes under rerooting
// - which is what you want when you need the hash of every subtree
// AND of every "rest of tree".

// --- MERGING TWO TREE PATHS. A segment tree node stores the MINIMAL
// PATH CONTAINING all the
// vertices in its range, as that path's two endpoints ( $\{-1, -1\}$  =
// empty,  $\{-2, -2\}$  = they do not fit
// on any path). Merging is  $O(1)$  given an  $O(1)$  LCA: the covering path
// of two paths, if it exists,
// has its endpoints among the four given ones, so test each of the
// four candidate pairs ( $u, v$ )
// with  $\text{dist}(u, v) == \text{dist}(u, w) + \text{dist}(w, v)$  for every endpoint  $w$ .
// NOTE THE SEMANTIC: this is the smallest path CONTAINING both, not
// their union - two disjoint
// pieces of one long path merge into that long path, which is what a
// covering query wants.
// USES: "is this set of vertices contained in one path", "the
// smallest path covering a range of
// queries", and segment-tree-over-queries problems on trees.
struct PathNode { int x = -1, y = -1; };
PathNode mergePaths(PathNode a, PathNode b, const LCA& L) {
    if (a.x < 0) return b;
    if (b.x < 0) return a;
    int e[4] = {a.x, a.y, b.x, b.y}; // ALL
    // SIX pairs, not just four:
    for (int i = 0; i < 4; i++) for (int j = i; j < 4; j++) { //
        // (a.x, b.x) and (a.y, b.y) are
        int u = e[i], v = e[j]; //
        // spanning pairs too
        bool ok = true;
        for (int w : e) if (L.dist(u, v) != L.dist(u, w) + L.dist(w,
        // v)) ok = false;
        if (ok) return {u, v};
    }
    return {-2, -2}; // not
    // a path: mark it invalid
}
```

9.2 DSU on Tree

9.2.1 DSUOnTree

DSU ON TREE (small to large) - answers offline subtree queries in $O(n \log n)$ by keeping the heavy

// child's data and re-adding only the light subtrees. Each vertex is
 ↳ added $O(\log n)$ times.
 // Use it for "most frequent colour in each subtree" style problems
 ↳ where merging maps would be $O(n^2)$.

```
class DSUOnTree {
private:
    struct Query {
        int idx, x;
    };

    int n;
    vector<vector<int>> adj;
    vector<int> sz, depth;
    vector<bool> big;
    vector<vector<Query>> queries;
    vector<int> ans;

    // =====
    // PROBLEM SPECIFIC STATE VARIABLES
    // =====
    vector<char> c; // Example: Array of characters per node

    // Example state variables
    // int distinct_count = 0;
    // vector<int> freq;

    void add(int u, int p) {
        // --- ADD LOGIC HERE ---
        // Example: freq[c[u] - 'a']++;

        for (int v : adj[u]) {
            if (v == p || big[v]) continue; // Skip parent and heavy
            ↳ child
            add(v, u);
        }
    }

    void remove(int u, int p) {
        // --- REMOVE LOGIC HERE ---
        // Example: freq[c[u] - 'a']--;

        for (int v : adj[u]) {
            if (v == p || big[v]) continue; // Skip parent and heavy
            ↳ child
            remove(v, u);
        }
    }

    int get_answer(int u, int x = 0) {
        // --- QUERY ANSWER LOGIC HERE ---
        return 0;
    }
    // =====

    void pre(int u, int p = 0) {
        sz[u] = 1;
        depth[u] = depth[p] + 1;
        for (int v : adj[u]) {
            if (v == p) continue;
            pre(v, u);
            sz[u] += sz[v]; // Compute subtree size
        }
    }

    void dfs(int u, int p = 0, bool keep = false) {
        int mx = -1, bigChild = -1;

        // Find the heavy child
        for (int v : adj[u]) {
            if (v == p) continue;
            if (sz[v] > mx) {
                mx = sz[v];
                bigChild = v;
            }
        }

        // Process light children and clear them
        for (int v : adj[u]) {
            if (v == p || v == bigChild) continue;
            dfs(v, u, false);
        }

        // Process heavy child and keep it
        if (bigChild != -1) {
```

```
        dfs(bigChild, u, true);
        big[bigChild] = true;
    }

    // Add current node and its light children
    add(u, p);

    // Answer all queries for the current node
    for (const auto& q : queries[u]) {
        ans[q.idx] = get_answer(u, q.x);
    }

    // Reset the big child flag
    if (bigChild != -1) {
        big[bigChild] = false;
    }

    // Clear state if this node is not meant to be kept
    if (!keep) {
        remove(u, p);
    }
}

public:
    // Initialize with 1-based indexing in mind
    DSUOnTree(int nodes, const vector<char>& chars) : n(nodes),
    ↳ c(chars) {
        adj.assign(n + 1, vector<int>());
        sz.assign(n + 1, 0);
        depth.assign(n + 1, 0);
        big.assign(n + 1, false);
        queries.assign(n + 1, vector<Query>());
        // Initialize problem-specific arrays here (e.g.,
        ↳ freq.assign(26, 0);
    }

    void add_edge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    void add_query(int node, int query_idx, int x = 0) {
        queries[node].push_back({query_idx, x});
    }

    vector<int> solve(int root, int num_queries) {
        ans.assign(num_queries, 0);
        pre(root, 0); // Precalculate sizes and depths
        dfs(root, 0, false); // Run the sack algorithm
        return ans;
    }
};
```

9.3 Heavy Light Decomposition (HLD)

9.3.1 HLD

HEAVY-LIGHT DECOMPOSITION - splits the tree into chains so any root-to-node path crosses $O(\log n)$

// of them; combined with a segment tree it gives path update/query
 ↳ in $O(\log^2 n)$.
 // Subtrees are contiguous in tin[], so subtree queries are a single
 ↳ range on the same tree.

```
class HLD {
public:
    vector<int> par, sz, head, tin, tout, who, depth;

    int dfs1(int u, vector<vector<int>> &adj) {
        for (int &v: adj[u]) {
            if (v == par[u]) continue;
            depth[v] = depth[u] + 1;
            par[v] = u;
            sz[u] += dfs1(v, adj);
            if (sz[v] > sz[adj[u][0]] || adj[u][0] == par[u]) swap(v,
↳ adj[u][0]);
        }
        return sz[u];
    }

    void dfs2(int u, int &timer, const vector<vector<int>> &adj) {
        tin[u] = timer++;
        for (int v: adj[u]) {
            if (v == par[u]) continue;
            head[v] = (timer == tin[u] + 1 ? head[u] : v);
            dfs2(v, timer, adj);
        }
        tout[u] = timer - 1;
    }

    HLD(vector<vector<int>> adj, int r = 0)
        : par(adj.size(), -1), sz(adj.size(), 1), head(adj.size(),
↳ r), tin(adj.size()), tout(adj.size()),
        who(adj.size()),
        depth(adj.size()) {
        dfs1(r, adj);
        int x = 0;
        dfs2(r, x, adj);
        for (int i = 0; i < adj.size(); ++i) who[tin[i]] = i;
    }

    vector<pair<int, int>> path(int u, int v) {
        vector<pair<int, int>> res;
        for (; v = par[head[v]]) {
            if (depth[head[u]] > depth[head[v]]) swap(u, v);
            if (head[u] != head[v]) {
                res.emplace_back(tin[head[v]], tin[v]);
            } else {
                if (depth[u] > depth[v]) swap(u, v);
                res.emplace_back(tin[u], tin[v]);
                return res;
            }
        }
    }

    pair<int, int> subtree(int u) {
        return {tin[u], tout[u]};
    }

    int dist(int u, int v) {
        return depth[u] + depth[v] - 2 * depth[lca(u, v)];
    }

    int lca(int u, int v) {
        for (; v = par[head[v]]) {
            if (depth[head[u]] > depth[head[v]]) swap(u, v);
            if (head[u] == head[v]) {
                if (depth[u] > depth[v]) swap(u, v);
                return u;
            }
        }
    }

    bool isAncestor(int u, int v) {
        return tin[u] <= tin[v] && tout[u] >= tout[v];
    }
};
```

9.4 Centroid Decomposition

9.4.1 Centroid

CENTROID DECOMPOSITION - recursively remove the centroid (the vertex whose largest remaining

// component is $\leq n/2$), giving a decomposition tree of depth $O(\log$
 ↳ $n)$. Every path in the original
 // tree passes through the centroid of some level, which turns
 ↳ path-counting problems into
 // "for each centroid, combine the distance lists of its components"
 ↳ in $O(n \log n)$.

```
class CentroidDecomposition {
private:
    int n;
    vector<vector<int>> adj;
    vector<int> sz, par, nodes;
    vector<bool> vis;

    int dfsSz(int u, int p) {
        sz[u] = 1;
        for (int v : adj[u]) {
            if (vis[v] || v == p) continue;
            sz[u] += dfsSz(v, u);
        }
        return sz[u];
    }

    int getCentroid(int u, int p, int total_nodes) {
        for (int v : adj[u]) {
            if (vis[v] || v == p) continue;
            if (sz[v] * 2 > total_nodes) {
                return getCentroid(v, u, total_nodes);
            }
        }
        return u;
    }

    void collect(int u, int p) {
        nodes.push_back(u);
        for (int v : adj[u]) {
            if (vis[v] || v == p) continue;
            collect(v, u);
        }
    }

    void build(int u, int p) {
        int total_nodes = dfsSz(u, p);
        int centroid = getCentroid(u, p, total_nodes);

        if (p != -1) {
            par[centroid] = p;
        } else {
            par[centroid] = centroid;
        }

        vis[centroid] = true;

        // =====
        // PROBLEM SPECIFIC PROCESSING
        // =====
        // Process paths going through the 'centroid' here
        for (int v : adj[centroid]) {
            if (vis[v]) continue;
            nodes.clear();
            collect(v, centroid);

            // e.g., Update answers using the nodes collected from
            ↳ this subtree branch
        }
        // =====

        // Recursively decompose the remaining components
        for (int v : adj[centroid]) {
            if (vis[v]) continue;
            build(v, centroid);
        }
    }

public:
    // Initialize with 1-based indexing
    CentroidDecomposition(int nodes_count) : n(nodes_count) {
        adj.assign(n + 1, vector<int>());
        sz.assign(n + 1, 0);
        par.assign(n + 1, 0);
        vis.assign(n + 1, false);
    }

    void add_edge(int u, int v) {
        adj[u].push_back(v);
    }
};
```

```

    adj[v].push_back(u);
}

// Start decomposition, usually from node 1
void decompose(int root = 1) {
    build(root, -1);
}

// Returns the parent of a node in the centroid tree
int get_centroid_parent(int u) const {
    return par[u];
}
};

```

9.4.2 PathLengthCounts

NUMBER OF PATHS OF EVERY LENGTH IN A TREE - $\text{cnt}[L]$ = how many unordered pairs $\{u, v\}$ have

```

// Needs: conv (06 - Math/05 - Convolution/01 - FFT.cpp).
// dist(u, v) == L, for ALL L at once, in  $O(n \log^2 n)$ . 0-indexed,
//    ↳ unweighted tree.
// The naive "for each length, count" is  $O(n^2)$ ; a plain centroid
//    ↳ decomposition answers "paths of
// length  $\leq K$ " in  $O(n \log n)$  but not all lengths simultaneously. The
//    ↳ fix is to convolve the depth
// HISTOGRAMS instead of merging them pairwise: at each centroid,
//    ↳ square the histogram of all
// distances and subtract the square of each child component's
//    ↳ histogram (inclusion-exclusion for
// the pairs that never actually cross the centroid).
// Reach for it for "for every k, how many pairs are at distance k",
//    ↳ "sum over pairs of f(dist)"
// when f is arbitrary, and distance-distribution / diameter-profile
//    ↳ problems.
vector<ll> pathLengthCounts(int n, const vector<vector<int>>& adj) {
    vector<ll> cnt(n + 1, 0);
    vector<int> sub_(n, 0), dep;
    vector<char> done(n, 0);
    int tot = 0;
    function<void(int, int)> calcSz = [&](int u, int p) {
        tot++, sub_[u] = 1;
        for (int v : adj[u]) if (v != p && !done[v]) calcSz(v, u),
        ↳ sub_[u] += sub_[v];
    };
    function<int(int, int)> centroid = [&](int u, int p) {
        for (int v : adj[u]) if (v != p && !done[v] && sub_[v] * 2 >
        ↳ tot) return centroid(v, u);
        return u;
    };
    function<void(int, int, int)> gather = [&](int u, int p, int d) {
        dep.push_back(d);
        for (int v : adj[u]) if (v != p && !done[v]) gather(v, u, d +
        ↳ 1);
    };
    function<void(int, int, int, int)> add = [&](int u, int p, int
    ↳ d0, int sign) {
        dep.clear(), gather(u, p, d0);
        vector<ll> h(*max_element(all(dep)) + 1, 0);
        for (int d : dep) h[d]++;
        vector<ll> r = conv(h, h);
        ↳ ordered pairs, diagonal included
        for (int i = 0; i < (int)r.size() && i <= n; i++) cnt[i] +=
        ↳ sign * r[i];
    };
    function<void(int, int)> decomp = [&](int u, int p) {
        tot = 0, calcSz(u, p);
        int c = centroid(u, p);
        done[c] = 1;
        add(c, p, 0, 1);
        ↳ pairs inside this component
        for (int v : adj[c]) if (!done[v]) add(v, c, 1, -1);
        ↳ minus the ones inside one child
        for (int v : adj[c]) if (!done[v]) decomp(v, c);
    };
    if (n) decomp(0, -1);
    cnt[0] = 0;
    for (ll x : cnt) x /= 2;
    ↳ ordered -> unordered
    return cnt;
    ↳ cnt[L] for L = 1..n-1
}

// WEIGHTED trees: the same decomposition works, but the histogram is
//    ↳ indexed by weight, so it is
// only practical when the total weight is small. Otherwise sort the
//    ↳ distance lists and use two
// pointers - that answers "pairs with dist  $\leq K$ " but not all lengths
//    ↳ at once.
// SUM OVER PAIRS OF f(dist): convolve as above and dot with f. Any f
//    ↳ works, which is the whole
// point of getting the full distribution.
// ONE LENGTH ONLY, or "count paths with dist  $\leq K$ ": plain centroid
//    ↳ decomposition with sorting,
//  $O(n \log^2 n)$  and no FFT at all - see 01 - Centroid.

```

9.5 Tree Diameter

9.5.1 TreeDiameter

Tree Diameter via 2-BFS - Time: $O(V + E)$, Space: $O(V + E)$

```
// Computes diameter length, endpoints (u, v), and the full diameter
↳ path
struct TreeDiameter {
    int n;
    vector<vector<int>> adj;
    vector<int> dist, par;

    TreeDiameter(int n = 0) : n(n), adj(n + 1), dist(n + 1), par(n + 1) {}

    void add_edge(int u, int v) {
        adj[u].pb(v);
        adj[v].pb(u);
    }

    int bfs(int src) {
        fill(all(dist), -1);
        queue<int> q;
        q.push(src);
        dist[src] = 0;
        par[src] = -1;

        int farthest = src;
        while (!q.empty()) {
            int u = q.front();
            q.pop();
            if (dist[u] > dist[farthest]) farthest = u;

            for (int v : adj[u]) {
                if (dist[v] == -1) {
                    dist[v] = dist[u] + 1;
                    par[v] = u;
                    q.push(v);
                }
            }
        }
        return farthest;
    }

    // Returns {diameter_length, {endpoint_u, endpoint_v}}
    pair<int, pair<int, int>> get_diameter() {
        int u = bfs(1);
        int v = bfs(u);
        return {dist[v], {u, v}};
    }

    // Returns the sequence of vertices on the diameter path from u
    ↳ to v
    vector<int> get_path() {
        auto [len, endpoints] = get_diameter();
        auto [u, v] = endpoints;
        vector<int> path;
        for (int curr = v; curr != -1; curr = par[curr]) {
            path.pb(curr);
        }
        reverse(all(path));
        return path;
    }
};
```

9.6 Euler Tour

9.6.1 EulerTour

Euler Tour Technique (Tree Flattening into 1D Array Intervals) - $O(V + E)$

```
// Subtree of node u corresponds to contiguous range [in_time[u],
↳ out_time[u]] in flat_tree
struct EulerTour {
    int n, timer = 0;
    vector<vector<int>> adj;
    vector<int> in_time, out_time, flat_tree;

    EulerTour(int n = 0) : n(n), adj(n + 1), in_time(n + 1),
↳ out_time(n + 1) {}

    void add_edge(int u, int v) {
        adj[u].pb(v);
        adj[v].pb(u);
    }

    void dfs(int u, int p = -1) {
        in_time[u] = ++timer;
        flat_tree.pb(u);
        for (int v : adj[u]) {
            if (v != p) dfs(v, u);
        }
        out_time[u] = timer;
    }

    void build(int root = 1) {
        timer = 0;
        flat_tree.clear();
        dfs(root);
    }

    // Returns [L, R] 1-based index range for subtree queries on node
    ↳ u
    pair<int, int> get_subtree_range(int u) const {
        return {in_time[u], out_time[u]};
    }
};

/* THE EULER-TOUR TOOLBOX - four different tours, each answering a
↳ different query family.
CONVENTION HERE: in_time is 1-based (in_time[u] = ++timer) but
↳ flat_tree is a 0-based vector,
so flat_tree[in_time[u] - 1] == u. Keep that straight when you index
↳ a BIT.

1) SUBTREE = a contiguous range [in[v], out[v]].
    point update, subtree query -> BIT over in[]: add at in[u],
↳ query [in[v], out[v]]
    subtree update, point query -> DIFFERENCE BIT: +x at in[v], -x
↳ at out[v]+1,
                                then the value at u is the
↳ prefix sum up to in[u]
2) PATH TO THE ROOT, using the SAME difference trick in the other
↳ direction:
    subtree update, path-to-root query <-> path update, point
↳ query (they are duals).
    PATH UPDATE u..v by +x, point query: +x at in[u], +x at in[v],
↳ -x at in[lca],
                                -x at in[parent(lca)]
    PATH QUERY u..v, point update: query(in[u]) +
↳ query(in[v]) - 2*query(in[lca])
                                (+ the LCA's own value if
↳ vertices carry weights)
3) THE 2n TOUR (+1 on entry, -1 on exit) turns ancestor tests and
↳ LCA into RMQ:
    u is an ancestor of v iff in[u] <= in[v] && out[v] <= out[u]
    LCA(u,v) = the vertex of minimum DEPTH in the tour range
↳ [in[u], in[v]] - and consecutive
    depths differ by exactly 1, which is what the O(n)/O(1) +-1 RMQ
↳ exploits.
4) THE EDGE TOUR (each edge appears twice) is what Mo's-on-trees
↳ uses: a vertex appearing TWICE
    inside the range cancels, so a path becomes one contiguous
↳ range plus the LCA by hand.
    See 02 - Data Structures/06 - Square Root Decomposition/03 -
↳ MoOnPaths.cpp.
```

EDGE WEIGHTS instead of vertex weights: push each edge's weight onto
↳ its DEEPER endpoint, then
every path formula above drops the LCA term (the LCA's incoming edge
↳ is not on the path). */

9.7 Rerooting

9.7.1 Rerooting

REROOTING DP - compute $f(v \text{ as root})$ for EVERY v in $O(n)$.

```
// Four hooks: E() identity of merge * self(v) contribution of v
//      ↳ itself *
//      merge(a,b) associative+commutative * lift(a, child)
//      ↳ push a child's value
//      across one edge into the parent's frame.
// Shipped example: ans[v].first = sum of distances from v to every
//      ↳ other node.
struct Reroot {
    typedef pair<ll, ll> T; // {sum of
    ↳ distances, #nodes}
    static T E() { return {0, 0}; }
    static T self(int v) { return {0, 1}; }
    static T merge(T a, T b) { return {a.first + b.first, a.second +
    ↳ b.second}; }
    static T lift(T a, int child) { return {a.first + a.second,
    ↳ a.second}; }

    int n;
    vector<vector<int>> adj;
    vector<T> down, up, ans;
    vector<int> par, ord;

    Reroot(int n) : n(n), adj(n), down(n, E()), up(n, E()), ans(n,
    ↳ E()), par(n, -1) {}
    void add_edge(int u, int v) { adj[u].push_back(v),
    ↳ adj[v].push_back(u); }

    void build(int root = 0) {
        ord.clear(), ord.reserve(n);
        vector<int> st = {root};
        par[root] = -1;
        while (!st.empty()) { // iterative order:
            ↳ no recursion depth limit
                int u = st.back(); st.pop_back(); ord.push_back(u);
                for (int v : adj[u]) if (v != par[u]) par[v] = u,
            ↳ st.push_back(v);
        }
        for (int i = (int)ord.size() - 1; i >= 0; i--) { // pass 1:
            ↳ leaves -> root
                int u = ord[i];
                down[u] = self(u);
                for (int v : adj[u]) if (v != par[u]) down[u] =
            ↳ merge(down[u], lift(down[v], v));
        }
        up[root] = E();
        for (int u : ord) { // pass 2: root ->
            ↳ leaves
                vector<int> ch;
                for (int v : adj[u]) if (v != par[u]) ch.push_back(v);
                int k = ch.size();
                vector<T> pre(k + 1, E()), suf(k + 1, E());
                for (int i = 0; i < k; i++) pre[i + 1] = merge(pre[i],
            ↳ lift(down[ch[i]], ch[i]));
                for (int i = k - 1; i >= 0; i--) suf[i] = merge(suf[i +
            ↳ 1], lift(down[ch[i]], ch[i]));
                ans[u] = merge(down[u], up[u]);
                for (int i = 0; i < k; i++)
                    up[ch[i]] = lift(merge(self(u), merge(up[u],
            ↳ merge(pre[i], suf[i + 1])), u);
        }
    };
    // The prefix/suffix trick ("aggregate over all children EXCEPT one")
    ↳ is reusable on its own:
    // use it whenever a DP needs "everything but this branch" and merge
    ↳ has no inverse.
    // OTHER INSTANTIATIONS: max depth from each node (T = ll, lift =
    ↳ a+1, self = 0);
    // count of nodes at even distance; product of subtree answers; tree
    ↳ diameter through each node.
```

9.8 Virtual Tree

9.8.1 VirtualTree

VIRTUAL (AUXILIARY) TREE - compress k marked vertices plus their pairwise LCAs into a tree

```
// with at most  $2k-1$  nodes, in  $O(k \log k)$ . Run the real DP on that
//      ↳ instead of on all  $n$  nodes.
// THE pattern for "q queries, each with  $k_i$  marked vertices, sum of
//      ↳  $k_i$  bounded".
// Needs tin/tout from a DFS and an LCA (binary lifting or
//      ↳ Euler+sparse table).
struct VirtualTree {
    int n;
    vector<int> tin, tout, dep;
    vector<vector<int>> adj, vadj; // vadj = the
    ↳ compressed tree
    int timer = 0;

    VirtualTree(int n) : n(n), tin(n), tout(n), dep(n, 0), adj(n),
    ↳ vadj(n) {}
    void add_edge(int u, int v) { adj[u].push_back(v),
    ↳ adj[v].push_back(u); }
    void dfs(int u, int p) {
        tin[u] = timer++;
        for (int v : adj[u]) if (v != p) dep[v] = dep[u] + 1, dfs(v,
    ↳ u);
        tout[u] = timer++;
    }
    bool isAnc(int u, int v) const { return tin[u] <= tin[v] &&
    ↳ tout[v] <= tout[u]; }

    // 'lca' must be a working LCA callable. Returns the roots' node
    ↳ list; edges land in vadj.
    template <class LCA>
    vector<int> build(vector<int> key, LCA lca) {
        if (key.empty()) return {};
        sort(all(key), [&](int a, int b) { return tin[a] < tin[b]; });
        key.erase(unique(all(key)), key.end());
        int k = key.size();
        for (int i = 0; i + 1 < k; i++) key.push_back(lca(key[i],
    ↳ key[i + 1]));
        sort(all(key), [&](int a, int b) { return tin[a] < tin[b]; });
        key.erase(unique(all(key)), key.end());
        for (int u : key) vadj[u].clear();
        vector<int> st = {key[0]};
        for (size_t i = 1; i < key.size(); i++) {
            int u = key[i];
            while (!isAnc(st.back(), u)) st.pop_back();
            vadj[st.back()].push_back(u); // parent ->
            ↳ child (directed is enough)
            st.push_back(u);
        }
        return key; // key[0] is
    ↳ the root of the virtual tree
    };
    // ALWAYS clear vadj for exactly the nodes you used (done above) -
    ↳ never clear all  $n$  per query,
    // or the total cost becomes  $O(q \cdot n)$  and defeats the whole point.
    // The compressed edge  $u \rightarrow v$  represents the real path between them;
    ↳ its "weight" is usually
    //  $dep[v] - dep[u]$ , or a path aggregate you can query in  $O(1)$  with
    ↳ prefix sums / HLD.
```

9.9 Isomorphism

9.9.1 Treelsomorphism

TREE ISOMORPHISM (AHU canonical form). $O(n \log n)$ with a map, $O(n)$ with per-level bucketing.

```
// Gives every rooted subtree a canonical ID: two subtrees are
// ↳ isomorphic iff their IDs match.
// Exact - unlike random subtree hashing, which can collide.
struct TreeIso {
    map<vector<int>, int> id; // canonical
    ↳ child-multiset -> id
    // canonical id of the subtree rooted at u
    int rootedId(int u, int p, const vector<vector<int>>& adj) {
        vector<int> ch;
        for (int v : adj[u]) if (v != p) ch.push_back(rootedId(v, u,
        ↳ adj));
        sort(all(ch));
        auto it = id.find(ch);
        if (it != id.end()) return it->second;
        int nid = id.size();
        return id[ch] = nid;
    }
    // CENTERS of a tree: 1 or 2 vertices (Jordan). Root there to
    ↳ canonicalise an UNROOTED tree.
    vector<int> centers(int n, const vector<vector<int>>& adj) {
        if (n == 1) return {0};
        vector<int> deg(n), leaves;
        for (int i = 0; i < n; i++) if ((deg[i] = adj[i].size()) <=
        ↳ 1) leaves.push_back(i);
        int removed = leaves.size();
        while (removed < n) {
            if (leaves.empty()) break; // guard: not
            ↳ a tree / disconnected
            vector<int> nxt;
            for (int u : leaves)
                for (int v : adj[u]) if (--deg[v] == 1)
                    ↳ nxt.push_back(v);
            leaves = nxt, removed += leaves.size();
        }
        return leaves; // 1 or 2
    }
    ↳ vertices, adjacent if 2
    // Canonical id of an UNROOTED tree: min over its centers.
    int unrootedId(int n, const vector<vector<int>>& adj) {
        auto c = centers(n, adj);
        int best = rootedId(c[0], -1, adj);
        if (c.size() == 2) best = min(best, rootedId(c[1], -1, adj));
        return best;
    }
};
// Two unrooted trees are isomorphic iff unrootedId is equal (use ONE
// ↳ shared TreeIso object so
// the id space is common). Two rooted trees: compare rootedId at
// ↳ their roots.
// USES: "how many distinct tree shapes", dedupe trees, count
// ↳ isomorphism classes, group
// identical subtrees for a DP over shapes.
// JORDAN: a tree has exactly one or two centers; if two, they are
// ↳ adjacent.
```

9.10 Tree DP

9.10.1 TreeDP

TREE DP - the most common tree shape in a contest. Root the tree, compute a value per subtree

```
// bottom-up, and combine children. If the answer must be "for every
// ↳ root", see 07 - Rerooting.

// --- MAXIMUM WEIGHT INDEPENDENT SET / MINIMUM VERTEX COVER on a
// ↳ tree,  $O(n)$ . ---
// dp[v][0] = best for v's subtree with v NOT taken, dp[v][1] = with
// ↳ v taken.
// A tree is bipartite, so alpha = n - nu and (Konig) min vertex
// ↳ cover = maximum matching - but
// this DP is shorter than building a matching, and it handles
// ↳ weights, which Konig does not.
pair<ll, ll> misTree(int v, int p, const vector<vector<int>>& adj,
    ↳ const vector<ll>& w) {
    ll take = w[v], skip = 0;
    for (int u : adj[v]) if (u != p) {
        auto [s, t] = misTree(u, v, adj, w);
        skip += max(s, t), take += s; // if
        ↳ v is taken, no child may be
    }
    return {skip, take}; //
    ↳ answer = max(skip, take)
}
// MINIMUM VERTEX COVER (unweighted) = n - MIS. MAXIMUM MATCHING on a
// ↳ tree = the greedy "match
// every leaf to its parent, delete both, repeat" - and that greedy
// ↳ is optimal (exchange argument).

// --- TREE KNAPSACK,  $O(n^2)$  - and the one character that decides  $n^2$ 
// ↳ vs  $n^3$ . ---
// dp[v][j] = best value using exactly j vertices of v's subtree.
// ↳ Merge child by child, capping
// BOTH loops by the ALREADY ACCUMULATED size, not by n. Each
// ↳ unordered pair (a, b) is then
// charged exactly once, at their LCA, giving  $O(n^2)$  total. Writing
// ↳ 'u <= n' instead of
// 'u <= acc' is  $O(n^3)$  and looks identical.
vector<ll> knapTree(int v, int p, const vector<vector<int>>& adj,
    ↳ const vector<ll>& w) {
    vector<ll> dp = {0, w[v]}; //
    ↳ dp[0] = take nothing at all
    for (int u : adj[v]) if (u != p) {
        vector<ll> c = knapTree(u, v, adj, w); //
        ↳ c[0] = skip this child entirely
        vector<ll> nd(dp.size() + c.size() - 1, LLONG_MIN);
        nd[0] = 0;
        for (size_t i = 1; i < dp.size(); i++) // i
            ↳ >= 1: v IS taken, so a child
            for (size_t j = 0; j < c.size(); j++) // may
                ↳ if (dp[i] != LLONG_MIN && c[j] != LLONG_MIN) nd[i +
                ↳ j] = max(nd[i + j], dp[i] + c[j]);
        dp = nd;
    }
    return dp; //
    ↳ dp[j] = best CONNECTED set of j
}
// ↳ vertices containing v (dp[0] = 0)
// Dropping the 'i >= 1' restriction gives the UNCONSTRAINED version:
// ↳ best j vertices anywhere in
// the subtree, no connectivity - that is the
// ↳ group-knapsack-on-a-tree variant.
// WITH A CAP k: allocate min(size, k) + 1 slots instead, giving  $O(n$ 
// ↳ * k).

// --- DIAMETER / LONGEST PATH by DP,  $O(n)$ . The version you reuse
// ↳ inside other tree DPs. ---
// down1[v], down2[v] = the two longest downward paths from v through
// ↳ DIFFERENT children.
// The answer is max over v of down1[v] + down2[v]; the two-BFS trick
// ↳ gives the same on an
// unweighted or POSITIVELY weighted tree, but only this DP survives
// ↳ negative weights.

/* MORE TREE-DP SHAPES, all  $O(n)$  unless noted
    COUNT SUBTREES / connected subgraphs containing v: prod over
    ↳ children of (1 + f(child)).
    COUNT MATCHINGS / independent sets: same [0/1] split, sum instead
    ↳ of max.
    SUM OF DISTANCES from every vertex: rerooting with (subtree size,
    ↳ subtree distance sum).
    K-COLOURINGS with adjacent-different:  $k * (k-1)^{(n-1)}$ . With a
    ↳ colour list per vertex, DP
    dp[v][c] and take the product over children of (sum over c' != c
    ↳ of dp[u][c']) - keep the
```

total per child and subtract the forbidden term to stay $O(n * k)$.
 LONGEST PATH WITH AT MOST K EDGES / paths of length exactly L:
 → dp[v][d] merged small-to-large
 over the depth arrays, or long-path decomposition for $O(n)$.
 BINARY LIFTING ON A DP: when the transition is associative you can
 → jump 2^k ancestors at once
 (max edge on a path, "the first ancestor with property P").
 TREE + BITMASK: dp[v][mask] for $k \leq 20$ marked vertices (Steiner
 → tree on a tree).
 REROOTING is the answer to "compute f(v) for EVERY root" - do not
 → run the DP n times.
 AUXILIARY / VIRTUAL TREE (09 - Trees/08) collapses q marked
 → vertices into an $O(q)$ tree, so a
 per-query tree DP costs $O(q \log q)$ instead of $O(n)$.
 DSU ON TREE (09 - Trees/02) is the answer when the per-subtree
 → state is a MULTISET you cannot
 merge algebraically (most frequent colour, k-th value), $O(n \log$
 → $n)$.
 SEGMENT TREE MERGING (02 - DS/16) does the same in $O(n \log n)$
 → while keeping the state
 queryable, at the cost of memory.
 RECURSION DEPTH: a path-shaped tree with $n = 2e5$ overflows the
 → default stack once frames get
 fat. Move big locals out of the recursive function or convert to
 → the iterative order used
 in 07 - Rerooting.cpp. */

9.11 Dynamic Trees

9.11.1 LinkCutTree

LINK-CUT TREE - a FOREST that changes shape, with path queries. $O(\log n)$ amortised per op.

```
// This is dynamic HLD: link, cut, re-root, and aggregate over the
    → path u..v, all online.
// If the tree is STATIC use HLD (09 - Trees/03) - it is faster and
    → far shorter. Reach for an LCT
// only when edges are added/removed, or when you need makeRoot.
// HOW IT WORKS: each vertex has at most one "preferred" child;
    → preferred paths are stored as
// splay trees keyed by DEPTH; a non-preferred edge is a path-parent
    → pointer the splay does not
// know about. access(v) makes root..v one preferred path, and the
    → same heavy/light potential that
// bounds HLD bounds the number of preferred-child changes, hence the
    →  $O(\log n)$  amortised.
struct LCT {
    struct Node { int ch[2] = {0, 0}, p = 0; ll val = 0, sum = 0;
        → bool rev = 0; };
    vector<Node> t;
    LCT(int n) : t(n + 1) {} //
    → 1-indexed; node 0 is the null
    bool isRoot(int x) { return t[t[x].p].ch[0] != x &&
        → t[t[x].p].ch[1] != x; }
    void pull(int x) { t[x].sum = t[t[x].ch[0]].sum ^ t[x].val ^
        → t[t[x].ch[1]].sum; }
    void flip(int x) { if (x) swap(t[x].ch[0], t[x].ch[1]), t[x].rev
        → ^= 1; }
    void push(int x) { if (t[x].rev) flip(t[x].ch[0]),
        → flip(t[x].ch[1]), t[x].rev = 0; }
    void rot(int x) {
        int p = t[x].p, g = t[p].p, d = t[p].ch[1] == x;
        if (!isRoot(p)) t[g].ch[t[g].ch[1] == p] = x;
        t[x].p = g;
        t[p].ch[d] = t[x].ch[!d], t[t[x].ch[!d]].p = p;
        t[x].ch[!d] = p, t[p].p = x;
        pull(p), pull(x);
    }
    void splay(int x) {
        vector<int> st{x};
        for (int y = x; !isRoot(y); y = t[y].p) st.push_back(t[y].p);
        for (int i = st.size() - 1; i >= 0; i--) push(st[i]);
        while (!isRoot(x)) {
            int p = t[x].p, g = t[p].p;
            if (!isRoot(p)) rot((t[g].ch[1] == p) == (t[p].ch[1] ==
        → x) ? p : x);
            rot(x);
        }
        int access(int x) { //
        → root..x becomes one splay tree
            int last = 0;
            for (int y = x; y; y = t[y].p) splay(y), t[y].ch[1] = last,
        → pull(y), last = y;
            splay(x);
            return last; // =
        → the LCA on the previous access
        }
        void makeRoot(int x) { access(x), flip(x), push(x); }
        int findRoot(int x) { access(x); while (t[x].ch[0]) push(x), x =
        → t[x].ch[0]; splay(x); return x; }
        bool connected(int x, int y) { return findRoot(x) == findRoot(y);
        → }
        void link(int x, int y) { makeRoot(x); if (findRoot(y) != x)
        → t[x].p = y; }
        void cut(int x, int y) { //
        → no-op if the edge is absent
            makeRoot(x);
            if (findRoot(y) == x && t[y].p == x && !t[y].ch[0]) t[y].p =
        → t[x].ch[1] = 0, pull(x);
        }
        ll query(int x, int y) { makeRoot(x), access(y); return t[y].sum;
        → } // path aggregate
        void update(int x, ll v) { access(x), t[x].val = v, pull(x); }
        int lca(int r, int x, int y) { makeRoot(r), access(x); return
        → access(y); }
};
// THE AGGREGATE is XOR here purely because it is self-inverse and
    → needs no reversal handling.
// For SUM/MIN/MAX just change pull(); for a NON-COMMUTATIVE
    → aggregate (matrix product, max
// prefix sum) you must store BOTH directions in the node and swap
    → them inside flip().
// EDGE WEIGHTS: give each edge its own vertex (n + edgeId) with the
    → weight, and link both ends
// to it. Vertex values then stay 0 and every path query still works.
// SUBTREE aggregates need "virtual subtree" bookkeeping: keep a
```

```

    ↪ second sum over the non-preferred
// children, updated in access() when you swap t[y].ch[1]. That
    ↪ doubles the code - if the tree is
// static, an Euler tour (09 - Trees/06) is  $O(\log n)$  and ten lines.
/* WHAT LCT UNLOCKS
    ONLINE DYNAMIC CONNECTIVITY ON A FOREST: connected() in  $O(\log n)$ .
    ↪ For general graphs (cycles)
        you need the offline segment-tree-on-time trick (02 - Data
    ↪ Structures/14) or Holm-de
        Lichtenberg-Thorup at  $O(\log^2 n)$ .
    DYNAMIC MST: to add edge (u,v,w), if u and v are connected find
    ↪ the MAXIMUM edge on the path;
        if it exceeds w, cut it and link the new edge.  $O(\log n)$  per
    ↪ update.
    MAXIMUM FLOW (Dinic with link-cut) improves the blocking-flow step
    ↪ to  $O(VE \log V)$  - almost
        never needed at contest sizes.
    OFFLINE "PAINT THE ROOT PATH": access() itself is the operation
    ↪ "recolour root..v with a new
        colour"; the number of colour changes is  $O(n \log n)$  total, so if
    ↪ each change costs a segment
        tree update you get  $O(n \log^2 n)$ . This is the standard trick for
    ↪ "count distinct substrings
        over all substrings of s" and for tree-path-colouring problems.
    ↪ */

```

9.12 Long Path Decomposition

9.12.1 LongPathDecomposition

LONG PATH DECOMPOSITION - split the tree into vertical paths by always following the child with

```

// the DEEPEST subtree (HLD follows the child with the BIGGEST
    ↪ subtree instead). 0-indexed, rooted.
// TWO THINGS IT BUYS YOU THAT HLD DOES NOT.
// 1.  $O(1)$  K-TH ANCESTOR after  $O(n \log n)$  preprocessing. Binary
    ↪ lifting is  $O(\log n)$  per query;
// this is the standard constant-time answer, and it is only ~30
    ↪ lines.
// 2. DEPTH-INDEXED TREE DP IN  $O(n)$  TOTAL ("dp[v][d] = something
    ↪ about the vertices at depth d
// below v"). Give each path head an array of its length and let
    ↪ every vertex on the path SHARE
// that array with an offset - the heavy child inherits the
    ↪ parent's array for free, and only
// the  $O(\sqrt{n})$ -ish path heads allocate. The classic use is "for
    ↪ each v, the most common depth
// in its subtree" and "count vertices at distance k below v" for
    ↪ all v.
// The number of paths equals the number of LEAVES, and the crucial
    ↪ property is: the path containing
// v has length at least (height of v), so jumping to the head of v's
    ↪ path already covers a lot.
// K-TH ANCESTOR IN  $O(1)$ . For every path head h of length L, store
    ↪ the L ancestors above h as well
// as the L vertices of the path itself ("up" and "down" tables of
    ↪ size 2L, total  $O(n)$ ). To answer
// kth(v, k): let j = highest power of two  $\leq k$ , jump v to its  $2^j$ -th
    ↪ ancestor with a lifting table
// (one step), and the remaining  $k - 2^j < 2^j \leq$  (height of the new
    ↪ vertex), so the rest of the
// jump lands inside the new vertex's own path table - one array
    ↪ lookup.
struct LongPath {
    int n, B;
    vector<int> par, dep, hgt, heavy, head;
    vector<vector<int>> jmp, tab; //
    ↪ tab[h] = ladder of head h
    vector<int> posInTab, headOf, logs;
    LongPath(int n, const vector<vector<int>>& adj, int root = 0)
        : n(n), B(1), par(n, -1), dep(n, 0), hgt(n, 0), heavy(n, -1),
    ↪ head(n, -1),
        posInTab(n, 0), headOf(n, -1), logs(n + 1, 0) {
        while ((1 << B) <= n) B++;
        for (int i = 2; i <= n; i++) logs[i] = logs[i / 2] + 1;
        vector<int> ord; //
    ↪ iterative DFS -> preorder
        vector<int> st{root};
        par[root] = -1;
        while (!st.empty()) {
            int u = st.back(); st.pop_back();
            ord.push_back(u);
            for (int v : adj[u]) if (v != par[u]) par[v] = u, dep[v]
    ↪ = dep[u] + 1, st.push_back(v);
        }
        for (int i = n - 1; i >= 0; i--) { //
    ↪ heights, bottom up
            int u = ord[i];
            for (int v : adj[u]) if (v != par[u] && hgt[v] + 1 >
    ↪ hgt[u])
                hgt[u] = hgt[v] + 1, heavy[u] = v;
        }
        jmp.assign(B, vector<int>(n, -1));
        for (int u = 0; u < n; u++) jmp[0][u] = par[u];
        for (int k = 1; k < B; k++) for (int u = 0; u < n; u++)
            jmp[k][u] = jmp[k - 1][u] < 0 ? -1 : jmp[k - 1][jmp[k -
    ↪ 1][u]];
        for (int u : ord) head[u] = (par[u] >= 0 && heavy[par[u]] ==
    ↪ u) ? head[par[u]] : u;
        tab.assign(n, {});
        for (int h = 0; h < n; h++) if (head[h] == h) { //
    ↪ build the ladder of this path
            int L = hgt[h] + 1;
            vector<int> down;
            for (int v = h; v >= 0; v = heavy[v]) down.push_back(v);
            vector<int> upv; // L
    ↪ ancestors above h, if they exist
            for (int x = par[h], c = 0; x >= 0 && c < L; x = par[x],
    ↪ c++) upv.push_back(x);
            reverse(all(upv));
            for (int v : down) upv.push_back(v); //
    ↪ [far ancestors ... h ... deepest]
            tab[h] = upv;
            int off = (int)tab[h].size() - (int)down.size();
            for (int i = 0; i < (int)down.size(); i++)
                posInTab[down[i]] = off + i, headOf[down[i]] = h;

```

```

    }
}
int kth(int v, int k) const { //
    ↪ k-th ancestor, -1 if too high
    if (k == 0) return v;
    if (k > dep[v]) return -1;
    int j = logs[k];
    v = jmp[j][v], k -= 1 << j; // one
    ↪ lifting step
    int h = headOf[v];
    return tab[h][posInTab[v] - k]; // the
    ↪ rest is inside the ladder
}
};
// WHY THE LADDER IS LONG ENOUGH: after the single 2^j jump, the
    ↪ remainder k' < 2^j and the vertex
// we are on has height >= 2^j - 1, so its path (and hence its
    ↪ ladder's upper half) covers k'.
// MEMORY is O(n log n) for the lifting table and O(n) for the
    ↪ ladders (each path of length L keeps
// 2L entries, and the path lengths sum to n).
// LEVEL ANCESTOR can also be done in O(n) preprocessing (Euler tour
    ↪ + the ladder trick without the
// lifting table), but the version above is the one you can write
    ↪ from memory.
// SMALL-TO-LARGE ALTERNATIVE for depth-indexed DP: merge
    ↪ maps/multisets child by child, O(n log^2)
// - shorter but slower; use it when the DP state is not a plain
    ↪ array indexed by depth.

```

9.13 Path Operations

9.13.1 PathIntersectionAndUnion

INTERSECTION AND UNION OF TWO TREE PATHS, $O(\log n)$ each. 0-indexed, rooted tree.

```

// Needs: Lift (09 - Trees/01 - Lowest Common Ancestor (LCA)/02 -
    ↪ BinaryLifting.cpp).
// A path is the pair of its endpoints {a, b} (unordered); {-1, -1}
    ↪ is the EMPTY path and
// {-2, -2} is "these two paths do not union into a path".
// Reach for it for "how many of these q paths pass through a common
    ↪ vertex", "the longest prefix
// of the queries whose paths all intersect" (the intersection is
    ↪ associative, so put it in a
// segment tree or binary-search it), "merge the constraints u..v
    ↪ must be inside one path", and
// road/route-overlap problems.
// KEY DECOMPOSITION: a path a..b splits at l = lca(a, b) into two
    ↪ VERTICAL chains l..a and l..b.
// The intersection of two paths is the union of the (at most four)
    ↪ pairwise chain intersections,
// and each of those is again a vertical chain that is trivial to
    ↪ compute from depths and lca.
struct PathOps {
    const Lift& L;
    typedef pair<int, int> P;
    PathOps(const Lift& L) : L(L) {}
    bool isAnc(int u, int a) const { // is
        ↪ a an ancestor of u (or == u)?
        return L.d[u] >= L.d[a] && L.kth(u, L.d[u] - L.d[a]) == a;
    }
    int len(P p) const { return p.first < 0 ? 0 : L.dist(p.first,
        ↪ p.second) + 1; } // #vertices
    // The common part of the vertical chains u..a and v..c, where u
        ↪ is an ancestor of a and v of c.
    P chainMeet(int u, int a, int v, int c) const { // v
        ↪ must lie on the chain u..a
        if (!isAnc(a, v)) return {-1, -1};
        int x = L.lca(a, c);
        if (L.d[v] < L.d[u]) { if (isAnc(x, u)) return {u, x}; }
        else if (isAnc(x, v)) return {v, x};
        return {-1, -1};
    }
    P joinChains(P x, P y) const { //
        ↪ union, ASSUMED to be a path
        if (x.first < 0) return y;
        if (y.first < 0) return x;
        int can[4] = {x.first, x.second, y.first, y.second}, a =
        ↪ can[0], b = -1;
        for (int u : can) if (L.d[u] > L.d[a]) a = u; //
        ↪ deepest endpoint
        for (int u : can) if (!isAnc(a, u) && (b < 0 || L.d[b] <
        ↪ L.d[u])) b = u;
        if (b < 0) { b = can[0]; for (int u : can) if (L.d[u] <
        ↪ L.d[b]) b = u; }
        return {a, b};
    }
    P intersect(P p, P q) const { //
        ↪ {-1,-1} when they are disjoint
        if (p.first < 0 || q.first < 0) return {-1, -1};
        int u = L.lca(p.first, p.second), v = L.lca(q.first,
        ↪ q.second);
        P r1 = joinChains(chainMeet(u, p.first, v, q.first),
        ↪ chainMeet(u, p.first, v, q.second));
        P r2 = joinChains(chainMeet(u, p.second, v, q.first),
        ↪ chainMeet(u, p.second, v, q.second));
        return joinChains(r1, r2);
    }
    P unite(P x, P y) const { //
        ↪ {-2,-2} if the union is not a path
        if (x.first == -2 || y.first == -2) return {-2, -2};
        if (x.first < 0) return y;
        if (y.first < 0) return x;
        int e[4] = {x.first, x.second, y.first, y.second};
        int need = len(x) + len(y) - len(intersect(x, y)); //
        ↪ |union| by inclusion-exclusion
        for (int i = 0; i < 4; i++) for (int j = i; j < 4; j++) {
            P r(e[i], e[j]); //
            ↪ candidate covering path
            bool ok = len(r) == need; // no
            ↪ branch and no gap
            for (int w : e)
                if (L.dist(r.first, w) + L.dist(w, r.second) !=
                ↪ L.dist(r.first, r.second)) ok = 0;
            if (ok) return r;
        }
        return {-2, -2};
    }
};

```

```
// BOTH OPERATIONS ARE ASSOCIATIVE, so they drop straight into a
    ↳ segment tree over an array of
// paths: "the intersection of a range of queries" or "the longest
    ↳ prefix whose union is a path"
// (descend the tree greedily, which is exactly the classic "sort a
    ↳ permutation along a tree" task).
// SMALLEST PATH CONTAINING two given paths is a THIRD operation, and
    ↳ it is not the union - see
// mergePaths in 01 - LCA/03 - LCAConstantTime, which returns the
    ↳ minimal COVERING path.
// DOES VERTEX w LIE ON PATH a..b? dist(a,w) + dist(w,b) ==
    ↳ dist(a,b). That one-liner replaces this
// whole file when the query is that simple.
// PATH INTERSECTION LENGTH is len(intersect(p, q)) in VERTICES;
    ↳ subtract one for edges.
```

10 Game Theory

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NamedGames 214 · NimbersAndSums 215 · Hackenbush 217

10.1 GrundyAndNim

SPRAGUE-GRUNDY. Every FINITE, ACYCLIC (progressively bounded) IMPARTIAL game under normal

```
/* SPRAGUE-GRUNDY. Every FINITE, ACYCLIC (progressively bounded)
    ↳ IMPARTIAL game under normal
    play (last to move wins) is equivalent to a nim pile of size
    ↳ Grundy(position).
    ACYCLIC IS A HYPOTHESIS, NOT A DETAIL: on a graph with cycles the
    ↳ mex recursion is not
    well-founded and Grundy() below returns silently wrong values -
    ↳ use retrograde BFS instead.
    Grundy(p) = mex{ Grundy(q) : q reachable from p }.
    A position is LOSING for the player to move iff Grundy = 0.
    SUM of independent games: Grundy = XOR of the parts.
    ↳ */
int mex(const vector<int>& v) {
    vector<bool> seen(v.size() + 1, false);
    for (int x : v) if (x >= 0 && x <= (int)v.size()) seen[x] = true;
    int m = 0;
    while (m < (int)seen.size() && seen[m]) m++;
    return m;
}
// Grundy over a DAG of at most n positions, memoised.
vector<int> g_;
int Grundy(int u, const vector<vector<int>>& nxt) {
    if (g_[u] != -1) return g_[u];
    g_[u] = 0; // guard (DAG
    ↳ => no real cycles)
    vector<int> s;
    for (int v : nxt[u]) s.push_back(Grundy(v, nxt));
    return g_[u] = mex(s);
}
/* NIM AND FRIENDS
    NIM (take any number from one pile). Losing iff XOR of pile sizes
    ↳ == 0.
    Winning move: find a pile with (x ^ X) < x where X is the total
    ↳ xor; reduce it to x ^ X.
    MISERE NIM (last to move LOSES). If every pile is size 1: first
    ↳ player wins iff #piles is EVEN.
    Otherwise identical to normal nim (win iff xor != 0).
    NIM_k / MOORE'S NIM (may take from up to k piles). Losing iff, for
    ↳ every bit, the number of
    piles with that bit set is divisible by (k+1).
    STAIRCASE NIM. INDEX 0 IS THE GROUND - coins there are DEAD and
    ↳ have no move.
    Losing iff XOR of the counts at ODD indices == 0. (If instead a
    ↳ coin on index 0 may still
    slide OFF the board, the ground is a virtual index -1 and you
    ↳ must XOR the EVEN indices.
    Getting this backwards is wrong on 96 of the 256 four-position
    ↳ states.)
    Recognise it as "move something toward a sink, and moving it to
    ↳ an even slot is reversible".
    SUBTRACTION GAME, allowed set S: Grundy is eventually PERIODIC.
    ↳ Compute a prefix, find the
    period, extrapolate - the standard route for n up to 1e9.
    S = {1..k}: Grundy(n) = n mod (k+1).
    WYTHOFF (two piles, take from one or equally from both). Losing
    ↳ positions are
    (floor(k*phi), floor(k*phi^2)), phi = (1+sqrt(5))/2, k >= 0.
    FIBONACCI NIM (first move any amount < pile; then at most 2x the
    ↳ previous move).
    First player loses iff the pile size is a Fibonacci number.
    TURNING / COIN GAMES: Grundy of a set of coins = XOR of the Grundy
    ↳ of each single coin.
    GREEN HACKENBUSH (all edges neutral): Grundy of a path of length n
    ↳ is n; FUSION principle -
    a cycle of odd length fuses to a single edge, an even cycle to
    ↳ nothing; sum branches by XOR.
    COLON PRINCIPLE (trees): Grundy of a node = XOR over children of
    ↳ (Grundy(child) + 1). */

// RETROGRADE ANALYSIS - when the game graph HAS CYCLES (draws
    ↳ possible) or the game is partisan,
// Grundy does not apply. BFS backwards from terminal positions
    ↳ counting out-degrees.
// win[u] = 1 first player to move wins, 0 loses, -1 draw.
vector<int> retrograde(int n, const vector<vector<int>>& radj,
    ↳ vector<int> deg,
    const vector<int>& terminalLose) {
    vector<int> res(n, -1);
    queue<int> q;
    for (int u : terminalLose) res[u] = 0, q.push(u);
    while (!q.empty()) {
        int v = q.front(); q.pop();
        for (int u : radj[v]) {
            if (res[u] != -1) continue;
```

```

    if (res[v] == 0) res[u] = 1, q.push(u);    // can move
    ↪ to a losing position
    else if (--deg[u] == 0) res[u] = 0, q.push(u); // all
    ↪ moves lead to wins
    }
    return res;                                // remaining
    ↪ -1 are draws
}

```

10.2 GameAlgorithms

Concrete game solvers. All impartial games under NORMAL play (last move wins) unless stated.

```

// SUBTRACTION GAME with allowed move set S: Grundy is eventually
    ↪ PERIODIC. Compute a prefix,
// detect the period, extrapolate to n up to 1e18. This is the
    ↪ standard route for huge n.
struct Periodic {
    vector<int> g;
    int start = -1, per = -1;
    Periodic(const vector<int>& S, int prefix = 2000) {
        g.assign(prefix, 0);
        for (int i = 1; i < prefix; i++) {
            vector<bool> seen(S.size() + 2, false);
            for (int s : S) if (s <= i && g[i - s] <
                ↪ (int)seen.size()) seen[g[i - s]] = true;
                while (g[i] < (int)seen.size() && seen[g[i]]) g[i]++;
            }
            detect();
        }
        void detect() {                                // smallest
            ↪ (start, per) that fits
            int n = g.size();
            for (int p = 1; p <= n / 3; p++) {
                bool ok = true;
                for (int i = n / 3; i + p < n; i++) if (g[i] != g[i + p])
                    ↪ { ok = false; break; }
                    if (!ok) continue;
                int st = 0;                                // the LAST
                ↪ break, not the first match:
                for (int i = n - p - 1; i >= 0; i--)        //
                    ↪ periodicity must hold from start ON
                    if (g[i] != g[i + p]) { st = i + 1; break; }
                    if (n - p - st < 2 * p + 20) continue; // demand
                ↪ real evidence before believing
                start = st, per = p;
                return;
            }
        }
        int at(ll n) const {
            if (n < (ll)g.size()) return g[n];
            assert(per > 0);                                // no
            ↪ period found: do NOT extrapolate
            return g[start + (n - start) % per];
        }
    };
// S = {1..k} => Grundy(n) = n mod (k+1) (no need to compute
    ↪ anything)

// WYTHOFF'S GAME (two piles; take any amount from one pile, or the
    ↪ SAME amount from both).
// Losing (P-)positions: (floor(k*phi), floor(k*phi^2)) for k >= 0,
    ↪ phi = (1+sqrt(5))/2.
// Since phi^2 - phi = 1, the two coordinates differ by EXACTLY k, so
    ↪ k = b - a.
// Tested exactly, no floating point: a == floor(k*phi) <=> d <=
    ↪ k*sqrt(5) < d+2, d = 2a-k.
bool wythoffLosing(ll a, ll b) {
    if (a > b) swap(a, b);
    ll k = b - a, d = 2 * a - k;
    return d >= 0 && (__int128)d * d <= (__int128)5 * k * k
        && (__int128)5 * k * k < (__int128)(d + 2) * (d +
        ↪ 2);
}
// P-positions: (0,0) (1,2) (3,5) (4,7) (6,10) (8,13) (9,15) (11,18)
    ↪ (12,20) (14,23) ...

// MOORE'S NIM_k: you may take from up to k piles in one move.
// Losing iff, for EVERY bit, the number of piles having that bit set
    ↪ is divisible by (k+1).
bool mooreLosing(const vector<ll>& piles, int k) {
    for (int b = 0; b < 63; b++) {
        int c = 0;
        for (ll x : piles) c += (x >> b) & 1;
        if (c % (k + 1)) return false;
    }
    return true;
}

// STAIRCASE NIM: coins on positions 0..n-1, a move slides some coins
    ↪ from i to i-1.
// INDEX 0 IS THE GROUND: coins there are DEAD and cannot move. Then
    ↪ only ODD indices matter
// (a move from an even index is reversible, so it is worth nothing).
// If your problem instead lets a coin at index 0 slide OFF the
    ↪ board, the ground is a virtual

```



```
// index -1 and you must XOR the EVEN indices. Brute force over 4
    ↪ positions: the wrong convention
// is wrong on 96 of 256 states, e.g. (0,0,0,1).
bool staircaseLosing(const vector<ll>& cnt) {
    ll x = 0;
    for (size_t i = 1; i < cnt.size(); i += 2) x ^= cnt[i];
    return x == 0;
}

// GREEN HACKENBUSH (all edges neutral), Grundy of a rooted tree =
    ↪ COLON PRINCIPLE:
// g(v) = XOR over children c of (g(c) + 1)
int hackenbushTree(int u, int p, const vector<vector<int>>& adj) {
    int r = 0;
    for (int v : adj[u]) if (v != p) r ^= hackenbushTree(v, u, adj) +
    ↪ 1;
    return r;
}

// FUSION PRINCIPLE for graphs: all vertices of a cycle can be fused
    ↪ into one; an ODD cycle
// leaves a single loop (= one edge, Grundy 1), an EVEN cycle leaves
    ↪ nothing (Grundy 0).

// COIN TURNING GAMES: a position is a set of "heads". The Grundy
    ↪ value of the whole position is
// the XOR of the Grundy values of each single head considered alone
    ↪ (Mock Turtles, Ruler, ...).

// MISERE NIM (last to move LOSES): if EVERY pile is 1, the first
    ↪ player wins iff #piles is even;
// otherwise the normal-play rule (xor != 0 => win) applies unchanged.
```

10.3 MinimaxAndMatching

Games that Sprague-Grundy does NOT cover: partisan games, scored games, draws.

// Needs: Blossom (03 - Graph Theory/07 - Matching/04 - Blossom.cpp)
 ↪ for the matching part.

// MINIMAX with ALPHA-BETA pruning. Returns the value in PLAYER 0's
 ↪ units (a fixed sign), not
 // "the value for the player to move" - with maxing = false the
 ↪ result is still player-0 scored.
 // 'moves(state)' lists successors; 'eval(state)' scores
 ↪ terminal/leaf states from player 0's view.

```
template <class S, class Moves, class Eval>
ll alphabeta(const S& s, int depth, ll alpha, ll beta, bool maxing,
    ↪ Moves moves, Eval eval) {
    auto nxt = moves(s);
    if (!depth || nxt.empty()) return eval(s);
    if (maxing) {
        ll best = LLONG_MIN;
        for (auto& t : nxt) {
            best = max(best, alphabeta(t, depth - 1, alpha, beta,
    ↪ false, moves, eval));
            alpha = max(alpha, best);
            if (beta <= alpha) break; // beta
    ↪ cutoff
        }
        return best;
    }
    ll best = LLONG_MAX;
    for (auto& t : nxt) {
        best = min(best, alphabeta(t, depth - 1, alpha, beta, true,
    ↪ moves, eval));
        beta = min(beta, best);
        if (beta <= alpha) break; // alpha
    ↪ cutoff
    }
    return best;
}
```

// Order moves best-first: pruning is only good if you try strong
 ↪ moves early.

// Memoise on (state) when states repeat; use iterative deepening
 ↪ when depth is unbounded.

// MATCHING GAME ON A GRAPH: a token sits on vertex s; players
 ↪ alternately move it along an
 // edge to an unvisited vertex; a player who cannot move loses.
 // THEOREM: the FIRST player wins iff s is in EVERY maximum matching,
 ↪ i.e. iff removing s
 // decreases the size of the maximum matching.
 // This works on ANY graph, and you get EVERY start vertex at once
 ↪ for the price of one matching.
 // After a maximum matching is fixed, the vertices left exposed by
 ↪ SOME maximum matching are
 // exactly those reachable from an already-exposed vertex by an
 ↪ EVEN-length alternating path -
 // which is precisely the set an unsuccessful blossom search paints
 ↪ as OUTER. So run the matching,
 // then one failing search from each exposed vertex, and the
 ↪ unpainted vertices are the wins.
 // O(V^3) for the whole table, versus O(V^4) for "recompute the
 ↪ matching without s" per vertex.
 // Needs Blossom (03 - Graph Theory/07 - Matching/04 - Blossom.cpp) -
 ↪ see the wrapper below.

```
vector<char> matchingGameWins(Blossom& B) { // B
    ↪ already has all edges added
    B.solve();
    vector<char> miss(B.n, 0);
    for (int v = 0; v < B.n; v++) if (B.match[v] < 0) {
        B.augment(v); // always
    ↪ fails; it is here for used[]
        for (int u = 0; u < B.n; u++) miss[u] |= B.used[u];
    }
    for (int v = 0; v < B.n; v++) miss[v] = !miss[v]; // in every
    ↪ maximum matching -> P1 wins
    return miss;
}
```

// A vertex with NO edges is exposed in every matching, so the first
 ↪ player loses there - which is
 // right, they cannot move. ISOLATED vertices therefore need no
 ↪ special case.

// GAMES WITH SCORES (not just win/lose): plain minimax DP over
 ↪ states,
 // dp[state] = best over moves of (gain + (-dp[next])) for a
 ↪ zero-sum symmetric game,
 // or keep two arrays (dp0, dp1) when the players' objectives
 ↪ differ.

// GAMES WITH DRAWS / CYCLES: retrograde BFS (see 01 -
 ↪ GrundyAndNim.cpp), never Grundy.

10.4 NamedGames

NAMED GAMES: the values you cannot re-derive during a contest =====

```
/* ===== NAMED GAMES: the values you cannot re-derive during a
  ↳ contest =====
Two different families live here: TAKE-AND-BREAK (octal) games, and
  ↳ COIN-TURNING games.
They have different theory - octal games are the splitting rule
  ↳ below, coin-turning games are
XOR of single-head values and, in 2-D, nim MULTIPLICATION. Do not
  ↳ mix them up.
Every value on this card was brute-force verified.
All are impartial, NORMAL play. Grundy of a sum = XOR of parts.
```

SPLITTING RULE (the general take-and-break move):

```
g(v) = mex{ g(a1) XOR g(a2) XOR ... : v -> (a1, a2, ...) }
```

LASKER'S NIM (take any amount from a pile, OR split a pile of size

↳ ≥ 2 into two non-empty)

```
g(4k+1) = 4k+1      g(4k+2) = 4k+2      g(4k+3) = 4k+4      g(4k+4) =
↳ 4k+3
```

KAYLES (remove 1 or 2 adjacent pins, possibly splitting the row)

Periodic with period 12 from $n = 71$ onward (Winning Ways says 72;
↳ 71 is tight).

block for $n \geq 71$: 7 4 1 2 8 1 4 7 2 1 8 2

14 exceptions (n, G):

```
↳ (0,0)(3,3)(6,3)(9,4)(11,6)(15,7)(18,3)(21,4)(22,6)(28,5)(34,6)
      (39,3)(57,4)(70,6)      last exception
```

↳ $n = 70$

DAWSON'S CHESS (octal game .137)

Period 34 from $n = 52$ onward.

block for $n \geq 52$: 3 3 0 1 1 3 0 2 1 1 0 4 5 3 7 4 8 1 1 2 0 3 1

↳ 1 0 3 3 2 2 4 4 5 5 9

7 exceptions (n, G): (0,0)(14,0)(16,2)(17,2)(31,2)(34,0)(51,2)

↳ last exception $n = 51$

DAWSON'S KAYLES (.07) period 34 from $n = 53$, same block, 8

↳ exceptions, last at $n = 52$.

GRUNDY'S GAME (split a pile into two UNEQUAL non-empty piles)

Periodicity is an OPEN PROBLEM - more than 2×10^{10} values computed
↳ and no period found.

Do NOT assume a period here; compute the prefix you need.

GUY-SMITH PERIODICITY THEOREM (how to PROVE a period, not just guess
↳ it)

For an octal game whose largest non-zero code digit has index k : if

$G(n + p) = G(n)$ for all n with $n_0 \leq n < 2 \cdot n_0 + p + k$

then the period holds for ALL $n \geq n_0$. So checking up to $2 \cdot n_0 + p$

↳ $+ k$ is a proof.

TURNING GAMES (a position is a set of heads; the whole position's

↳ Grundy value is the XOR of

the values of each single head considered alone)

Twins (turn exactly 2 coins), 0-indexed: $g(x) = x$

↳ [plain nim]

Mock Turtles(turn 1, 2 or 3 coins), 0-indexed: $g(x) =$ the x -th

↳ ODIIOUS number

(odious = odd popcount): 1, 2, 4, 7, 8, 11, 13, 14,

↳ 16, ...

$g(x) = 2x$ if $2x$ is odious, else $2x + 1$

Ruler (turn any number of CONSECUTIVE coins), 1-indexed:

$g(x) =$ largest power of two dividing $x = x \& (-x) \rightarrow$

↳ 1,2,1,4,1,2,1,8,...

TRAP: Ruler is 1-INDEXED while Twins and Mock Turtles are

↳ 0-INDEXED. Mixing them up gives a

silently wrong answer.

NIM WITH ONE PASS (Guy's model: the pass may be used once, at any
↳ NON-terminal position)

1 heap: $G(0) = 0$, $G(n) = n+1$ for n odd, $G(n) = n-1$ for n even

↳ ≥ 2 .

2 heaps: P-positions are exactly (0,0) and (a, a+1) with a ODD.

3+ heaps: OPEN. No pattern; 34 P-positions with $a \leq b \leq c \leq 13$ and

↳ neither $xor == 0$ nor $xor == 1$

comes close. Never guess.

If the pass is ALSO legal at a terminal position (a weaker rule),

↳ the game collapses:

$G = (XOR \text{ of piles}) XOR 1$, so P-positions are exactly $XOR == 1$.

↳ */

```
bool odious(ll x) { return __builtin_parityll(x); }
```

```
ll mockTurtles(ll x) { return odious(2 * x) ? 2 * x : 2 * x + 1; }
```

↳ // 0-indexed

```
ll ruler(ll x) { return x & -x; }
```

↳ // 1-indexed

```
ll lasker(ll n) { // g(0) = 0, NOT
```

↳ -1

```
ll r = n % 4;
```

```
if (r == 0) return n ? n - 1 : 0;
```

```
return r == 3 ? n + 1 : n;
```

}

// Grundy of a sum of turning-game heads = XOR of the single-head

↳ values.

```
ll turningGameValue(const vector<ll>& heads, ll (*g)(ll)) {
```

```
ll r = 0;
```

```
for (ll h : heads) r ^= g(h);
```

```
return r;
```

}

10.5 NimbersAndSums

NIMBERS, OCTAL GAMES, AND THE SUM VARIANTS. Everything on this page was brute-force verified.

```
// --- NIM PRODUCT. [0, 2^(2^k)) is a FIELD under (XOR, nimMul). ---
// Definition: a (*) b = mex{ a'(*)b ^ a(*)b' ^ a'(*)b' : a' < a, b'
    < b }.
// Computed with the two identities: for a Fermat 2-power F = 2^(2^k),
// x (*) F = x * F for x < F, and F (*) F = 3F/2.
// USES: Turning Corners (turn the 4 corners of a rectangle) has
    < g(x,y) = x (*) y, and the TARTAN
// THEOREM says the 2-D product of two coin-turning games has g(x,y)
    < = g1(x) (*) g2(y). Also
// CF 1310F, which asks you to solve a^(*)x = b in the nimber field
    < of size 2^64.
typedef unsigned long long u64; // numbers
    < are UNSIGNED, always
map<pair<u64, u64>, u64> nmMemo;
u64 nimMul(u64 a, u64 b) {
    if (a < b) swap(a, b);
    if (b == 0) return 0;
    if (b == 1) return a;
    auto it = nmMemo.find({a, b});
    if (it != nmMemo.end()) return it->second;
    int L = 0;
    while (L < 6 && (a >> (1u << L))) L++;
    u64 F = 1ULL << (1u << (L - 1)); //
    < largest Fermat 2-power <= a
    u64 ah = a / F, al = a % F, bh = b / F, bl = b % F;
    u64 A = nimMul(ah, bh), B = nimMul(ah, bl), C = nimMul(al, bh), D
    < = nimMul(al, bl);
    return nmMemo[{a, b}] = ((A ^ B ^ C) * F) ^ D ^ nimMul(A, F / 2);
}

// --- OCTAL (TAKE-AND-BREAK) GAME GRUNDY GENERATOR. ---
// The code is ".d1d2d3...": digit d_k governs "remove exactly k
    < tokens from one heap".
// bit 1 (value 1): you may take the WHOLE heap (i.e. leave
    < zero heaps)
// bit 2 (value 2): you may leave ONE non-empty heap
// bit 4 (value 4): you may SPLIT the remainder into TWO non-empty
    < heaps
// So .333... = Nim, .77 = Kayles, .07 = Dawson's Kayles, .137 =
    < Dawson's chess.
// Being able to read a statement into a code string turns "invent a
    < Grundy recurrence" into a
// function call. O(N^2 * digits) because of the split term; O(N *
    < digits) without it.
vector<int> octal(const string& code, int N) { //
    < code is the digits AFTER the dot
    vector<int> d;
    for (char c : code) d.push_back(isdigit(c) ? c - '0' : c - 'a' +
    < 10);
    vector<int> g(N + 1, 0);
    for (int n = 1; n <= N; n++) {
        set<int> s;
        for (int i = 0; i < (int)d.size(); i++) {
            int k = i + 1;
            if (k > n) break;
            int rem = n - k;
            if ((d[i] & 1) && rem == 0) s.insert(0);
            if ((d[i] & 2) && rem >= 1) s.insert(g[rem]);
            if (d[i] & 4) for (int a = 1; a < rem; a++) s.insert(g[a]
    < ^ g[rem - a]);
            int m = 0;
            while (s.count(m)) m++;
            g[n] = m;
        }
    }
    return g;
}

// GUY-SMITH PERIODICITY THEOREM turns a guess into a PROOF: with k =
    < the index of the last
// non-zero code digit, if G(n + p) = G(n) for all n0 <= n < 2*n0 + p
    < + k, the period holds for
// ALL n >= n0. See 02 - GameAlgorithms.cpp for the (fixed) period
    < detector.

// --- EUCLID'S GAME. (a, b); subtract a positive multiple of the
    < smaller from the larger. ---
// FIRST PLAYER WINS iff a == b OR max/min > phi. The usual phrasing
    < ">= phi" is WRONG on the
// whole diagonal: (a, a) is always an immediate win.
bool euclidWin(ll a, ll b) {
    if (a < b) swap(a, b);
    if (b == 0) return false;
    if (a == b || a >= 2 * b) return true;
```

```
    return !euclidWin(b, a - b);
}

// --- ANTI-SG / the SJ theorem: MISERE play on a SUM of impartial
    < games. ---
// PRECONDITION, and it is not optional: in every component, SG(u) ==
    < 0 must imply u is TERMINAL.
// (Without it this rule is wrong on ~1.3% of random impartial games;
    < with it, exact.)
// Misere nim satisfies it, which is why the classic misere-nim rule
    < looks so clean.
bool antiSGFirstWins(const vector<int>& g) {
    int x = 0, mx = 0;
    for (int v : g) x ^= v, mx = max(mx, v);
    return (x && mx > 1) || (!x && mx <= 1);
}

// --- EVERY-SG: you must move in EVERY unfinished component. Not a
    < function of the XOR at all. ---
// step(v) = 0 if v is terminal
// step(v) = 1 + max{ step(u) : SG(u) == 0 } if SG(v) > 0
    < (drag it out - you will win it)
// step(v) = 1 + min{ step(u) } if SG(v) == 0 (end
    < it fast - you will lose it)
// First player wins iff max over components of step(root) is ODD.
// The mirrored max/min assignment is WRONG; this orientation is the
    < verified one.
void everySGStep(const vector<vector<int>>& succ, const vector<int>&
    < sg, vector<int>& step) {
    int n = succ.size();
    step.assign(n, 0);
    for (int u = 0; u < n; u++) { // u
        < in REVERSE topological order
        if (succ[u].empty()) { step[u] = 0; continue; }
        if (sg[u] > 0) {
            int b = 0;
            for (int v : succ[u]) if (!sg[v]) b = max(b, step[v] + 1);
            step[u] = b;
        } else {
            int b = INT_MAX;
            for (int v : succ[u]) b = min(b, step[v] + 1);
            step[u] = b;
        }
    }
}

/* ===== VERIFIED CLOSED FORMS AND FACTS =====
COIN-TURNING FAMILY (a position is a set of heads; the whole
    < position = XOR of single-head
values). THE INDEXING IS LOAD-BEARING - mixing 0- and 1-indexed
    < values is silently wrong.
Twins turn exactly 2 0-idx g(x) = x
TurningTurtles turn 1 or 2 1-idx g(x) = x
MockTurtles turn 1, 2 or 3 0-idx g(x) = 2x or 2x+1,
    < whichever is ODDIOUS
( = the x-th
odious number: 1 2 4 7 8 11 13 )
Ruler any CONSECUTIVE run 1-idx g(x) = x & -x
Triplets turn exactly 3 0-idx g(x) = 0 for x <
    < 2, MockTurtles(x-2) otherwise
Mogul turn 1..5 NO closed form (1
    < 2 4 8 16 31 32 64 103 ...)
TurningCorners turn 4 rectangle corners g(x, y) = x (*) y
    < [nim product]
TARTAN THEOREM: for the 2-D product of two coin-turning games,
    < g(x,y) = g1(x) (*) g2(y).

k-WYTHOFF (Fraenkel): take from one pile, or k1, k2 > 0 from both
    < with |k1 - k2| < k.
P-positions: a_n = mex{a_i, b_i : i < n}, b_n = a_n + k*n.
Closed form: (floor(n*alpha), floor(n*beta)) with alpha = (2 - k +
    < sqrt(k*k + 4)) / 2,
    beta = alpha + k. k = 1 is ordinary Wythoff.

SILVER DOLLAR (coins at p_1 < ... < p_k on a strip, slide one left
    < onto an empty square, no
jumping): with gaps g_i = p_i - p_{i-1} - 1 and p_0 = -1, the
    < GRUNDY VALUE is
g_k XOR g_{k-2} XOR g_{k-4} XOR ... - staircase nim on alternate
    < gaps counted FROM THE RIGHT.

CHOMP: every m x n board is a FIRST-PLAYER WIN except 1x1, by
    < strategy stealing - and there is
NO known constructive winning move. Two-row Chomp: P-positions are
    < exactly (b+1, b).
Strategy stealing is a pure EXISTENCE proof; if the problem wants
    < the move, it is useless.
```

BLUE-RED HACKENBUSH (partisan!): a string of edges from the ground
 ↪ has a NUMBER value, not a number. The first run of m equal colours contributes $\pm m$ (Blue ↪ positive), and every later edge contributes $\pm 1/2, \pm 1/4, \dots$ halving. This is the cleanest ↪ concrete model of surreal numbers.
 Partisan values in general: 0 = second player wins, $*$ = $\{0|0\}$ = ↪ first player wins,
 $up = \{0|*\}$ is positive but smaller than every positive number, a ↪ SWITCH $\{a|b\}$ with $a > b$ is "hot" (both players want to move there), and the TEMPERATURE is ↪ what it costs the opponent to let you move first. SIMPLICITY RULE: if some number lies strictly ↪ between the Left and Right options, the game IS the simplest such number. NUMBER AVOIDANCE: ↪ never move in a number component while a non-number component exists.

MISERE THEORY, honestly: misere nim's clean rule is an ACCIDENT. In ↪ general the misere outcome of a sum is NOT a function of the Grundy values; the correct ↪ invariant is the GENUS, and only "tame" games (genus matching nim's) obey nim-like rules. Anti-SG ↪ above is exactly the tame case, which is why it carries a precondition.

COMPLEXITY LANDSCAPE - when to stop looking for a formula:
 UNDIRECTED vertex geography is in P (the maximum-matching theorem ↪ in 03 - MinimaxAndMatching).
 DIRECTED vertex geography, Node Kayles, generalized geography and ↪ generalized Hex are all PSPACE-complete; generalized chess and Go are EXPTIME-complete. So ↪ a $1e5$ -sized instance of directed geography MUST have extra structure - find it instead of ↪ searching.

BOGUS NIM / REVERSIBLE MOVES: adding a move that the opponent can ↪ immediately undo does not change the Grundy value. That is precisely why the even indices in ↪ staircase nim are worth nothing, and it is the impartial case of the "bypass a reversible ↪ option" simplification. */

10.6 Hackenbush

BLUE-RED HACKENBUSH ON A TREE - the one PARTISAN game you can actually compute. The board is a

```
// tree rooted at the ground; every edge is Blue (only Left may
  ↪ delete it) or Red (only Right may
  ↪ delete it), and deleting an edge also deletes everything no longer
  ↪ connected to the root. The
// value of such a position is a DYADIC RATIONAL (a surreal NUMBER,
  ↪ never a number), and the usual
// number rules then decide everything: value > 0 means Left wins
  ↪ whoever moves, value < 0 means
// Right wins whoever moves, value == 0 means the SECOND player wins.
// THE RECURSION. For a stalk (a path from the ground) the value is
  ↪ easy and worth remembering:
// the first run of  $m$  equal colours is worth  $\pm m$ , and every edge
  ↪ after it is worth  $\pm 1/2, \pm 1/4,$ 
// halving each time. For a tree, evaluate each subtree, then push
  ↪ its value  $v$  through the edge
// that joins it to its parent: Blue edge  $\rightarrow (v + p) / 2^{(p-1)}$  with
  ↪  $p$  the smallest positive
// integer making  $v + p > 1$ ; Red edge  $\rightarrow (v - p) / 2^{(p-1)}$  with  $p$ 
  ↪ smallest making  $v - p < -1$ .
// Sum the results over the children. That is the "one edge above a
  ↪ game" simplification, and it
// is what makes the whole tree computable in near-linear time.
// WHEN: exactly the CodeChef GERALD08 shape - a tree of black/white
  ↪ edges, two players delete
// their own colour, print who wins moving first and who wins moving
  ↪ second. More generally, any
// partisan game whose components are trees of forced-order moves.
// COMPLEXITY:  $O(n \log^2 n)$  with small-to-large merging of the
  ↪ fraction maps. The value needs
//  $\Theta(n)$  bits in the worst case, which is why it is stored as an
  ↪ integer part plus a map of
// set fraction bits rather than as a long double - a double loses
  ↪ the answer at depth 53.
// DEGENERATE: a single vertex (no edges) has value 0, so the second
  ↪ player wins. All-Blue trees
// have value equal to the number of edges' contribution and are a
  ↪ Left win; the routine handles
// them with no special case. EXACT - all arithmetic is integer bit
  ↪ shuffling.
struct Surreal { // value =
  ↪ val + sum of frac bits, in  $[0,1)$ 
  int val = 0, w = 0; // absolute
  ↪ bit position = key + w
  map<int, int> frac; // only
  ↪ positions < 0 survive normalisation
  void clear() { frac.clear(), val = w = 0; }
  void add(int x) { // set bit at
    ↪ key x, then carry upward
    frac[x]++;
    auto it = frac.find(x);
    while (it != frac.end() && it->se != 1) {
      if (it->se > 1) frac[it->fi + 1] += it->se >> 1, it->se
    }
    ↪ &= 1;
    auto d = it++;
    if (!d->se) frac.erase(d);
  }
  void divide(int x) { // value /=
    ↪  $2^x$ 
    for (int i = 0; i < x; i++) {
      if (!val) break;
      if (val & 1) add(i - w); // bit i of
    }
    ↪ val lands at position i - x
    val >>= 1;
    w -= x;
  }
  void operator+=(Surreal& o) {
    val += o.val;
    for (auto& [k, b] : o.frac) if (b) add(k + o.w - w);
    while (!frac.empty() && frac.rbegin()->fi + w >= 0) {
    ↪ // promote carries into val
      if (frac.rbegin()->se) val += 1 << (frac.rbegin()->fi +
    ↪ w);
      frac.erase(prev(frac.end()));
    }
    void blueEdge() { // v  $\rightarrow (v +$ 
    ↪  $p) / 2^{(p-1)}$ 
    int p = max(1, 1 - val);
    if (frac.empty() && p + val == 1) p++; // v + p > 1
    ↪ is STRICT
    val += p, divide(p - 1);
  }
```

```

void redEdge() { // v -> (v -
    ↪ p) / 2^(p-1)
    int p = max(1, 1 + val);
    if (val - p == -1) p++;
    val -= p, divide(p - 1);
}
int sign() { return val > 0 ? 1 : val < 0 ? -1 : !frac.empty(); }
};
// g[u] = {v, colour}, colour 0 = Blue (Left / first player's edges),
    ↪ 1 = Red. Tree rooted at 0.
// Returns the sign of the value: +1 Left wins always, -1 Right wins
    ↪ always, 0 second player wins.
int hackenbush(int n, const vector<vector<pii>>& g) {
    vector<Surreal> a(n);
    vector<int> id(n), vis(n, 0);
    iota(all(id), 0);
    function<void(int)> dfs = [&](int u) {
        vis[u] = 1;
        for (auto& [v, c] : g[u]) {
            if (vis[v]) continue;
            dfs(v);
            c ? a[id[v]].redEdge() : a[id[v]].blueEdge();
            if (sz(a[id[v]].frac) > sz(a[id[u]].frac)) swap(id[v],
    ↪ id[u]); // small to large
            a[id[u]] += a[id[v]], a[id[v]].clear();
        }
    };
    dfs(0);
    return a[id[0]].sign();
}
// READING THE ANSWER: sign > 0 -> Left wins moving first AND second;
    ↪ sign < 0 -> Right wins both;
// sign == 0 -> whoever moves second wins. There is no "first player
    ↪ wins whoever they are" case,
// because a Hackenbush tree value is always a number and numbers are
    ↪ never fuzzy with 0.
// GREEN HACKENBUSH (every edge deletable by both) is impartial and
    ↪ is a NIMBER instead: the
// colon principle, Grundy(u) = XOR over children of (Grundy(child) +
    ↪ 1) - see 02 - GameAlgorithms.
// A tree with edges of all three colours has no closed form; you are
    ↪ back to computing surreal
// values of general games, which is exponential.
// THE PARTISAN TOOLKIT this file is the concrete instance of (05 -
    ↪ NimbersAndSums has the rest):
// 0 = second player wins, * = {0|0} = first player wins, up = {0|*}
    ↪ beats every 0 but loses to
// every positive number, a SWITCH {a|b} with a > b is hot,
    ↪ TEMPERATURE measures how much the move
// is worth, SIMPLICITY RULE picks the simplest number strictly
    ↪ between the Left and Right options,
// and NUMBER AVOIDANCE says never move in a number component while a
    ↪ non-number one exists.

```

11 Geometry

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11.1 Geometry 2D

11.1.1 Core

Point / vector core. $P \ll l$ = exact (use for all predicates), $P \ll d$ = real answers.

```

// cross(a,b) > 0 <=> b is counter-clockwise from a.
// a.cross(b,c) > 0 <=> c lies strictly LEFT of the directed line
    ↪ a->b. [= orient(a,b,c)]
int sgn(ll x) { return (x > 0) - (x < 0); }
int sgn(ld x) { return (x > eps) - (x < -eps); }
int sgn(int x) { return (x > 0) - (x < 0); } // without these
    ↪ two overloads a bare int or
int sgn(double x) { return sgn((ld)x); } // double
    ↪ argument is AMBIGUOUS: hard error
ll gcdll(ll a, ll b) { return b ? gcdll(b, a % b) : a; } //
    ↪ std::__gcd is libstdc++-only

template <class T>
struct P {
    typedef P Pt; T x = 0, y = 0;
    P(T x = 0, T y = 0) : x(x), y(y) {}
    Pt operator+(Pt p) const { return {x + p.x, y + p.y}; }
    Pt operator-(Pt p) const { return {x - p.x, y - p.y}; }
    Pt operator*(T d) const { return {x * d, y * d}; }
    Pt operator/(T d) const { return {x / d, y / d}; }
    T dot(Pt p) const { return x * p.x + y * p.y; }
    T cross(Pt p) const { return x * p.y - y * p.x; }
    T cross(Pt a, Pt b) const { return (a - *this).cross(b - *this); }
    ↪ } // orient
    T dist2() const { return x * x + y * y; }
    ld dist() const { return sqrtl((ld)dist2()); }
    ld angle() const { return atan2l(y, x); } // (-pi, pi]
    Pt perp() const { return {-y, x}; } // rotate
    ↪ +90 deg
    Pt unit() const { return *this / dist(); }
    Pt rot(ld a) const { return {x * cosl(a) - y * sinl(a), x *
    ↪ sinl(a) + y * cosl(a)}; }
    bool operator<(Pt p) const { return tie(x, y) < tie(p.x, p.y); }
    bool operator==(Pt p) const { return tie(x, y) == tie(p.x, p.y); }
    friend istream& operator>>(istream& i, Pt& p) { return i >> p.x
    ↪ >> p.y; }
    friend ostream& operator<<(ostream& o, Pt p) { return o << p.x <<
    ↪ ' ' << p.y; }
};
typedef P<ll> Pi;
typedef P<ld> Pd;

// Signed angle from a to b in (-pi, pi]. Sign tells you the rotation
    ↪ direction.
template <class T> ld angleTo(P<T> a, P<T> b) { return
    ↪ atan2l(a.cross(b), a.dot(b)); }
// Angle a-0-b in [0, 2pi).
template <class T> ld angleAt(P<T> a, P<T> 0, P<T> b) {
    ld r = angleTo(a - 0, b - 0);
    return r < 0 ? r + 2 * PI : r;
}

// EXACT polar sort, no atan2. Orders by angle in [0, 2pi) starting
    ↪ at +x axis.
// sort(all(v), angCmp<ll>); ties (same direction) keep input order.
template <class T> bool half(P<T> p) { return p.y < 0 || (p.y == 0 &&
    ↪ p.x < 0); }
template <class T> bool angCmp(P<T> a, P<T> b) {
    return half(a) != half(b) ? half(a) < half(b) : a.cross(b) > 0;
}

// OVERFLOW: |coord| <= 1e9 => cross fits in ll (max ~4e18, ll max
    ↪ 9.2e18).
// dist2 of a difference fits too. Anything squared twice
    ↪ needs __int128.

```


11.1.2 Lines

Lines, segments, rays. "line ab" = infinite line through a and b.

```
// Predicates are exact on P<ll>; anything returning a point needs
  ⇨ P<ld>.

template <class P> ld lineDist(P p, P a, P b) {           // distance
  ⇨ point -> line
  return fabsl((ld)(b - a).cross(p - a)) / (b - a).dist();
}
template <class P> P proj(P p, P a, P b) {               // foot of
  ⇨ perpendicular
  return a + (b - a) * ((ld)(p - a).dot(b - a) / (b - a).dist2());
}
template <class P> P refl(P p, P a, P b) { return proj(p, a, b) * 2 -
  ⇨ p; }

template <class P> bool onSeg(P p, P a, P b) {           // inclusive
  ⇨ of endpoints
  return a.cross(b, p) == 0 && (a - p).dot(b - p) <= 0;
}
template <class P> ld segDist(P p, P a, P b) {           // distance
  ⇨ point -> segment (exact-safe)
  if (a == b) return (p - a).dist();
  ld d = (ld)(b - a).dist2(), t = (ld)(p - a).dot(b - a);
  if (t <= 0) return (p - a).dist();
  if (t >= d) return (p - b).dist();
  return fabsl((ld)(b - a).cross(p - a)) / sqrtl(d);
}
template <class P> P closestOnSeg(P p, P a, P b) {
  if ((p - a).dot(b - a) <= 0) return a;
  if ((p - b).dot(b - a) >= 0) return b;
  return proj(p, a, b);
}

// LINE x LINE. first = 1 unique, 0 parallel-disjoint, -1 same line.
template <class P> pair<int, P> lineInter(P a, P b, P c, P d) {
  auto x = (b - a).cross(d - c);
  if (sgn(x) == 0) return {-(a.cross(b, c) == 0), P()};
  auto p = c.cross(b, d), q = c.cross(d, a);           // p + q ==
  ⇨ x (affine weights)
  return {1, (a * p + b * q) / x};                    // P must
  ⇨ be floating
}

// SEGMENT x SEGMENT. Returns 0, 1, or 2 points (2 = collinear
  ⇨ overlap endpoints).
// Exact on P<ll> when the answer has 0 or 2 points. The 1-point case
  ⇨ DIVIDES and the numerator
// is ~2*C^3, so it is only exact for |coordinate| <= ~1.6e6. Above
  ⇨ that use segCross for the
// boolean, or Frac (01 - Miscellaneous/13) for an exact rational
  ⇨ point.
template <class P> vector<P> segInter(P a, P b, P c, P d) {
  auto oa = c.cross(d, a), ob = c.cross(d, b), oc = a.cross(b, c),
  ⇨ od = a.cross(b, d);
  if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0)
    return {(a * ob - b * oa) / (ob - oa)};
  set<P> s;
  if (onSeg(c, a, b)) s.insert(c);
  if (onSeg(d, a, b)) s.insert(d);
  if (onSeg(a, c, d)) s.insert(a);
  if (onSeg(b, c, d)) s.insert(b);
  return {all(s)};
}
// Just "do they touch?" - fully exact, NO division and no
  ⇨ allocation, so it is both safe at
// |c| = 1e9 and ~30x faster than asking segInter. Use this whenever
  ⇨ you only need a bool.
template <class P> bool segCross(P a, P b, P c, P d) {
  auto oa = c.cross(d, a), ob = c.cross(d, b), oc = a.cross(b, c),
  ⇨ od = a.cross(b, d);
  if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0) return true;
  return onSeg(c, a, b) || onSeg(d, a, b) || onSeg(a, c, d) ||
  ⇨ onSeg(b, c, d);
}

template <class P> ld segSegDist(P a, P b, P c, P d) {
  if (segCross(a, b, c, d)) return 0;
  return min({segDist(a, c, d), segDist(b, c, d), segDist(c, a, b),
  ⇨ segDist(d, a, b)});
}

// RAY from s through e.
template <class P> bool onRay(P p, P s, P e) {
  return s.cross(e, p) == 0 && (p - s).dot(e - s) >= 0;
}
template <class P> ld rayDist(P p, P s, P e) {
```

```
  return (p - s).dot(e - s) <= 0 ? (p - s).dist() : lineDist(p, s,
  ⇨ e);
}

// Perpendicular bisector of ab, returned as two points on it.
// FLOATING P ONLY: (a+b)/2 truncates on P<ll> and returns a
  ⇨ DIFFERENT line (for a=(0,0),
// b=(1,1) it gives y = -x instead of y = -x + 1). For an exact
  ⇨ integer version work on the
// doubled lattice: the line through (a+b) and (a+b) + (b-a).perp().
template <class P> pair<P, P> perpBisector(P a, P b) {
  P m = (a + b) / 2;
  return {m, m + (b - a).perp()};
}
// Interior angle bisector direction at B for angle A-B-C (unit
  ⇨ vectors summed).
template <class P> P bisectorDir(P a, P b, P c) { return (a -
  ⇨ b).unit() + (c - b).unit(); }

// Line as a*x + b*y = c -> two points on it (needs floating P).
template <class P> pair<P, P> fromABC(ld A, ld B, ld C) {
  P n(A, B), p = n * (C / n.dist2());
  return {p, p + n.perp()};
}
```


11.1.3 Polygon

Polygons. Vertices in order (CW or CCW); convex routines want CCW, no collinear points.

```
template <class P> auto area2(const vector<P>& v) { // SIGNED
    ↪ 2*area. >0 <=> CCW.
    decltype(v[0].cross(v[0])) a = 0; // exact on
    ↪ P<ll>; safe on empty
    for (int i = 0, n = v.size(); i < n; i++) a += v[i].cross(v[(i +
    ↪ 1) % n]);
    return a;
}

template <class P> ld area(const vector<P>& v) { return
    ↪ fabsl((ld)area2(v)) / 2; }

template <class P> ld perimeter(const vector<P>& v) {
    ld r = 0;
    for (int i = 0, n = v.size(); i < n; i++) r += (v[i] - v[(i + 1)
    ↪ % n]).dist();
    return r;
}

template <class P> P centroid(const vector<P>& v) { // area
    ↪ centroid (needs floating P)
    P r; ld A = 0;
    for (int i = 0, j = v.size() - 1, n = v.size(); i < n; j = i++) {
        auto c = v[j].cross(v[i]);
        r = r + (v[i] + v[j]) * c; A += c;
    }
    return r / A / 3;
}

// A completely collinear (zero-area) polygon reports TRUE here -
    ↪ guard with area2 != 0 if that
// matters. Bow-ties and spikes are correctly rejected.
template <class P> bool isConvex(const vector<P>& v) {
    int n = v.size(), pos = 0, neg = 0;
    for (int i = 0; i < n; i++) {
        int s = sgn(v[i].cross(v[(i + 1) % n], v[(i + 2) % n]));
        s > 0 ? pos++ : s < 0 ? neg++ : 0;
    }
    return !pos || !neg;
}

// Point in ANY simple polygon, O(n), exact. strict=true -> boundary
    ↪ counts as outside.
template <class P> bool inPoly(const vector<P>& v, P a, bool strict =
    ↪ true) {
    int cnt = 0, n = v.size();
    for (int i = 0; i < n; i++) {
        P q = v[(i + 1) % n];
        if (onSeg(a, v[i], q)) return !strict;
        cnt ^= ((a.y < v[i].y) - (a.y < q.y)) * a.cross(v[i], q) > 0;
    }
    return cnt;
}

// Convex hull, CCW, strictly convex (collinear points dropped). O(n
    ↪ log n).
// Change <= 0 to < 0 in the pop test to KEEP collinear points on the
    ↪ hull.
template <class P> vector<P> hull(vector<P> p) {
    if (p.size() <= 1) return p;
    sort(all(p));
    vector<P> h(p.size() + 1);
    int s = 0, t = 0;
    for (int it = 2; it--; s = --t, reverse(all(p)))
        for (P q : p) {
            while (t >= s + 2 && h[t - 2].cross(h[t - 1], q) <= 0)
                ↪ t--;
            h[t++] = q;
        }
    return {h.begin(), h.begin() + t - (t == 2 && h[0] == h[1])};
}

// Point in CONVEX polygon (CCW, strict), O(log n). 1 inside, 0 on
    ↪ boundary, -1 outside.
// h must be a strictly convex CCW hull with n >= 1. Returns 1
    ↪ inside, 0 on the boundary,
// -1 outside.
template <class P> int inConvex(const vector<P>& h, P p) {
    if (h.empty()) return -1;
    int n = h.size();
    if (n == 1) return p == h[0] ? 0 : -1;
    if (n == 2) return onSeg(p, h[0], h[1]) ? 0 : -1;
    if (h[0].cross(h[1], p) < 0 || h[0].cross(h[n - 1], p) > 0)
        ↪ return -1;
    int l = 1, r = n - 1;
    while (r - l > 1) { int m = (l + r) / 2; (h[0].cross(h[m], p) >=
        ↪ 0 ? l : r) = m; }
}
```

```
int s = sgn(h[l].cross(h[l + 1], p));
if (s < 0) return -1;
if (s == 0) return 0;
return (h[0].cross(h[1], p) == 0 || h[0].cross(h[n - 1], p) == 0)
    ↪ ? 0 : 1;
}

// Index of the hull vertex furthest in direction 'dir', O(log n).
    ↪ CCW hull, no collinear.
// Use to get tangents from a far-away direction, or max/min dot
    ↪ product.
#define _cmp(i, j) sgn(dir.perp().cross(h[(i) % n] - h[(j) % n]))
#define _ext(i) (_cmp(i + 1, i) >= 0 && _cmp(i, i - 1 + n) < 0)
template <class P> int extreme(const vector<P>& h, P dir) {
    int n = h.size(), lo = 0, hi = n;
    if (_ext(0)) return 0;
    while (lo + 1 < hi) {
        int m = (lo + hi) / 2;
        if (_ext(m)) return m;
        int ls = _cmp(lo + 1, lo), ms = _cmp(m + 1, m);
        (ls < ms || (ls == ms && ls == _cmp(lo, m)) ? hi : lo) = m;
    }
    return lo;
}

// Cut polygon by the line s->e, KEEP the part on the LEFT (boundary
    ↪ included). O(n).
// Needs floating P. Boundary-inclusive matters: with a strict test,
    ↪ vertices lying exactly on
// the line are dropped from BOTH halves and you lose area. Repeat
    ↪ over many lines = half-plane
// intersection. cut(v,e,s) gives the other half; the two areas sum
    ↪ to area(v).
// Keeps the part LEFT of the directed line s->e, boundary INCLUSIVE.
    ↪ Vertices exactly on the
// line are kept, so a line through a vertex produces a DUPLICATE
    ↪ point - fine for area, but
// dedup before feeding the result to hull / isConvex / minkowski /
    ↪ diameter2.
template <class P> vector<P> cut(const vector<P>& v, P s, P e) {
    vector<P> r;
    for (int i = 0, n = v.size(); i < n; i++) {
        P cur = v[i], prv = v[(i + n - 1) % n];
        bool side = sgn(s.cross(e, cur)) >= 0;
        if (side != (sgn(s.cross(e, prv)) >= 0))
            ↪ r.push_back(lineInter(s, e, cur, prv).second);
        if (side) r.push_back(cur);
    }
    return r;
}

// Minkowski sum of two CONVEX CCW polygons. O(n+m). Result is convex
    ↪ CCW.
// BOTH polygons must be CONVEX and COUNTER-CLOCKWISE. On clockwise
    ↪ input the merge below makes
// no progress and loops forever (an unexplained TLE/MLE, not a wrong
    ↪ answer) - so normalise.
template <class P> vector<P> minkowski(vector<P> A, vector<P> B) {
    if (area2(A) < 0) reverse(all(A));
    if (area2(B) < 0) reverse(all(B));
    auto low = [](vector<P>& v) {
        auto lo = [](P a, P b) { return tie(a.y, a.x) < tie(b.y,
        ↪ b.x); };
        rotate(v.begin(), min_element(all(v), lo), v.end());
    };
    low(A); low(B);
    A.push_back(A[0]); A.push_back(A[1]);
    B.push_back(B[0]); B.push_back(B[1]);
    vector<P> r; size_t i = 0, j = 0;
    while (i + 2 < A.size() || j + 2 < B.size()) {
        r.push_back(A[i] + B[j]);
        auto c = (A[i + 1] - A[i]).cross(B[j + 1] - B[j]);
        if (c >= 0 && i + 2 < A.size()) i++;
        if (c <= 0 && j + 2 < B.size()) j++;
    }
    return r;
}

// Rotating calipers. Hull must be CCW strictly convex.
template <class P> auto diameter2(const vector<P>& h) { // max
    ↪ squared distance
    int n = h.size(), j = n < 2 ? 0 : 1;
    decltype(h[0].dist2()) r = 0;
    for (int i = 0; i < j; i++)
        for (; j = (j + 1) % n) {
            r = max(r, (h[i] - h[j]).dist2());
            if ((h[(j + 1) % n] - h[j]).cross(h[i + 1] - h[i]) >= 0)
                ↪ continue;
        }
}
```

```

    ↪ break;
    }
    return r;
}
template <class P> ld width(const vector<P>& h) {          // min
    ↪ distance between parallel supports
    int n = h.size(); if (n <= 2) return 0;
    ld r = 1e18; int j = 1;
    for (int i = 0; i < n; i++) {
        while ((h[(i + 1) % n] - h[i]).cross(h[(j + 1) % n] - h[j])
    ↪ >= 0) j = (j + 1) % n;
        r = min(r, lineDist(h[j], h[i], h[(i + 1) % n]));
    }
    return r;
}
// Minimum-area enclosing rectangle. One side always lies on a hull
    ↪ edge (that's the theorem),
// so try every edge. O(n^2) on the HULL only - fine, and immune to
    ↪ caliper tie bugs.
template <class P> ld minRectArea(const vector<P>& h) {
    int n = h.size(); if (n < 3) return 0;
    ld best = 1e18;
    for (int i = 0; i < n; i++) {
        P d = h[(i + 1) % n] - h[i];
        ld L = d.dist(), lo = 1e18, hi = -1e18, ht = 0;
        for (P q : h) {
            ld u = (ld)d.dot(q - h[i]) / L, v = (ld)d.cross(q - h[i])
    ↪ / L;
            lo = min(lo, u); hi = max(hi, u); ht = max(ht, fabsL(v));
        }
        best = min(best, (hi - lo) * ht);
    }
    return best;
}

// PICK'S THEOREM. Lattice polygon: A = I + B/2 - 1.
template <class P> ll boundaryPts(const vector<P>& v) {    // B
    ll b = 0;
    for (int i = 0, n = v.size(); i < n; i++) {
        P d = v[(i + 1) % n] - v[i];
        b += gcdll(llabs(d.x), llabs(d.y));
    }
    return b;
}
// PICK'S THEOREM, and it needs a SIMPLE polygon with integer
    ↪ vertices. On a self-intersecting
// or degenerate (zero-area) polygon the result is meaningless, not
    ↪ merely approximate.
template <class P> ll interiorPts(const vector<P>& v) {    // I = A -
    ↪ B/2 + 1
    return (llabs(area2(v)) - boundaryPts(v)) / 2 + 1;
}

```

11.1.4 Circles

Circles. All of these want a floating P (Pd).

```

// --- circle x line: 0, 1 (tangent) or 2 points on segment/line ab
    ↪ ---
template <class P> vector<P> circleLine(P o, ld r, P a, P b) {
    P p = proj(o, a, b);
    ld h2 = r * r - (p - o).dist2();
    if (h2 < -eps) return {};
    if (h2 < eps) return {p};
    P h = (b - a).unit() * sqrtL(h2);
    return {p - h, p + h};
}
template <class P> vector<P> circleSeg(P o, ld r, P a, P b) {
    vector<P> out;
    for (P p : circleLine(o, r, a, b)) if (segDist(p, a, b) < eps)
    ↪ out.push_back(p);
    return out;
}

// --- circle x circle: the 0/1/2 intersection points ---
template <class P> vector<P> circleCircle(P a, ld r1, P b, ld r2) {
    P d = b - a; ld d2 = d.dist2();
    if (d2 < eps) return {}; //
    ↪ concentric (equal => infinite)
    ld p = (d2 + r1 * r1 - r2 * r2) / (2 * d2), h2 = r1 * r1 - p * p
    ↪ * d2;
    if (r1 + r2 < sqrtL(d2) - eps || fabsL(r1 - r2) > sqrtL(d2) +
    ↪ eps) return {};
    P mid = a + d * p, per = d.perp() * sqrtL(max((ld)0, h2) / d2);
    if (h2 < eps) return {mid};
    return {mid + per, mid - per};
}

// --- AREA of the lens (intersection region) of two circles ---
template <class P> ld circleInterArea(P a, ld r1, P b, ld r2) {
    ld d = (a - b).dist();
    if (d >= r1 + r2) return 0;
    if (d + min(r1, r2) <= max(r1, r2)) return PI * min(r1, r2) *
    ↪ min(r1, r2);
    ld A = acosL((d * d + r1 * r1 - r2 * r2) / (2 * d * r1));
    ld B = acosL((d * d + r2 * r2 - r1 * r1) / (2 * d * r2));
    return r1 * r1 * (A - sinL(2 * A) / 2) + r2 * r2 * (B - sinL(2 *
    ↪ B) / 2);
}

// --- TANGENTS. Each result is a pair (touch point on c1, touch
    ↪ point on c2).
// inner=false -> the 2 external tangents; inner=true -> the 2
    ↪ internal (crossing) ones.
// Tangent lines from a POINT p to circle (c,r): tangents(p, 0, c,
    ↪ r, false).
template <class P> vector<pair<P, P>> tangents(P c1, ld r1, P c2, ld
    ↪ r2, bool inner) {
    if (inner) r2 = -r2;
    P d = c2 - c1;
    ld dr = r1 - r2, d2 = d.dist2(), h2 = d2 - dr * dr;
    if (d2 < eps || h2 < -eps) return {};
    vector<pair<P, P>> out;
    for (ld s : {-1.0L, 1.0L}) {
        P v = (d * dr + d.perp() * sqrtL(max((ld)0, h2)) * s) / d2;
        out.push_back({c1 + v * r1, c2 + v * r2});
    }
    if (h2 < eps) out.pop_back();
    return out;
}

// --- circumcircle / incircle of a triangle ---
template <class P> P circumcenter(P a, P b, P c) {
    P B = c - a, C = b - a;
    return a + (B * C.dist2() - C * B.dist2()).perp() / B.cross(C) /
    ↪ 2;
}
template <class P> ld circumradius(P a, P b, P c) {
    return (b - a).dist() * (c - b).dist() * (a - c).dist() / (2 *
    ↪ fabsL((ld)(b - a).cross(c - a)));
}
template <class P> P incenter(P a, P b, P c) {
    ld A = (b - c).dist(), B = (c - a).dist(), C = (a - b).dist();
    return (a * A + b * B + c * C) / (A + B + C);
}
template <class P> ld inradius(P a, P b, P c) {
    ld s = ((b - a).dist() + (c - b).dist() + (a - c).dist()) / 2;
    return fabsL((ld)(b - a).cross(c - a)) / 2 / s;
}
template <class P> P orthocenter(P a, P b, P c) { return a + b + c -
    ↪ circumcenter(a, b, c) * 2; }

```

```
// --- MINIMUM ENCLOSING CIRCLE, expected O(n) (Welzl, incremental)
↳ ---
template <class P> pair<P, ld> mec(vector<P> p) {
    if (p.empty()) return {P(), 0};
    shuffle(all(p),
    mt19937(chrono::steady_clock::now().time_since_epoch().count()));
    P o = p[0]; ld r = 0, E = 1 + 1e-9;
    for (int i = 0; i < (int)p.size(); i++) if ((o - p[i]).dist() > r
    ↳ * E) {
        o = p[i], r = 0;
        for (int j = 0; j < i; j++) if ((o - p[j]).dist() > r * E) {
            o = (p[i] + p[j]) / 2, r = (o - p[i]).dist();
            for (int k = 0; k < j; k++) if ((o - p[k]).dist() > r * E)
                o = circumcenter(p[i], p[j], p[k]), r = (o -
            ↳ p[i]).dist();
        }
    }
    return {o, r};
}

// --- AREA of (circle centered at c, radius r) INTERSECT polygon ps
↳ ---
template <class P> ld circlePolyArea(P c, ld r, vector<P> ps) {
    auto arg = [](P p, P q) { return atan2l(p.cross(q), p.dot(q)); };
    auto tri = [&](P p, P q) -> ld {
        ld r2 = r * r / 2;
        P d = q - p;
        ld a = d.dot(p) / d.dist2(), b = (p.dist2() - r * r) /
    ↳ d.dist2(), det = a * a - b;
        if (det <= 0) return arg(p, q) * r2;
        ld s = max((ld)0, -a - sqrtl(det)), t = min((ld)1, -a +
    ↳ sqrtl(det));
        if (t < 0 || 1 <= s) return arg(p, q) * r2;
        P u = p + d * s, v = p + d * t;
        return arg(p, u) * r2 + u.cross(v) / 2 + arg(v, q) * r2;
    };
    ld sum = 0;
    for (int i = 0, n = ps.size(); i < n; i++) sum += tri(ps[i] - c,
    ↳ ps[(i + 1) % n] - c);
    return fabsl(sum);
}

// --- power of a point / radical axis ---
// pow(p) = |p-c|^2 - r^2 : <0 inside, 0 on circle, >0 outside;
↳ equals (tangent length)^2.
template <class P> ld power(P p, P c, ld r) { return (p - c).dist2()
    ↳ - r * r; }

// Radical axis of two non-concentric circles, as two points on it.
template <class P> pair<P, P> radicalAxis(P a, ld r1, P b, ld r2) {
    P d = b - a; ld t = ((r1 * r1 - r2 * r2) / d.dist2() + 1) / 2;
    P m = a + d * t;
    return {m, m + d.perp()};
}
```

11.1.5 Sweep

Global / sweep-line geometry: problems about a whole SET of objects.

```
// --- CLOSEST PAIR of points, O(n log n). Returns the two points.
↳ Exact on P<ll>. ---
template <class P> pair<P, P> closestPair(vector<P> v) {
    assert(v.size() > 1); // n < 2
    ↳ has no pair to return
    set<P> S; // ordered
    ↳ by (x, y)
    sort(all(v), [](P a, P b) { return a.y < b.y; });
    pair<ll, pair<P, P>> best{LLONG_MAX, {P(), P()}};
    int j = 0;
    for (P p : v) {
        P d{1 + (ll)sqrtl((ld)best.first), 0};
        while (v[j].y <= p.y - d.x) S.erase(v[j++]);
        auto lo = S.lower_bound(p - d), hi = S.upper_bound(p + d);
        for (; lo != hi; ++lo) best = min(best, ((*lo - p).dist2(),
    ↳ {*lo, p}));
        S.insert(p);
    }
    return best.second;
}

// --- AREA OF UNION OF AXIS-ALIGNED RECTANGLES, O(n log n). ---
// rects: {x1, y1, x2, y2} half-open [x1,x2) x [y1,y2). Sweep x,
↳ segment tree counts covered y.
struct CoverTree { // "count">0
    ↳ length" segment tree
    int n; vector<int> cnt; vector<ll> len; vector<ll> ys;
    CoverTree(vector<ll> y) : ys(y) {
        n = ys.size() - 1, cnt.assign(4 * n, 0), len.assign(4 * n, 0);
    }
    void upd(int v, int l, int r, int ql, int qr, int d) {
        if (qr <= l || r <= ql) return;
        if (ql <= l && r <= qr) cnt[v] += d;
        else {
            int m = (l + r) / 2;
            upd(2 * v, l, m, ql, qr, d); upd(2 * v + 1, m, r, ql, qr,
    ↳ d);
        }
        len[v] = cnt[v] ? ys[r] - ys[l] : (r - l == 1 ? 0 : len[2 *
    ↳ v] + len[2 * v + 1]);
    }
    void upd(int l, int r, int d) { upd(1, 0, n, l, r, d); }
    ll get() { return len[1]; }
};

ll rectUnionArea(vector<array<ll, 4>> r) {
    vector<ll> ys;
    for (auto& q : r) ys.push_back(q[1]), ys.push_back(q[3]);
    sort(all(ys)); ys.erase(unique(all(ys)), ys.end());
    if (ys.size() < 2) return 0;
    auto id = [&](ll y) { return (int)(lower_bound(all(ys), y) -
    ↳ ys.begin()); };
    vector<array<ll, 4>> ev; // {x, +-1,
    ↳ y1, y2}
    for (auto& q : r) ev.push_back({q[0], 1, q[1], q[3]}),
    ↳ ev.push_back({q[2], -1, q[1], q[3]});
    sort(all(ev));
    CoverTree T(ys);
    ll area = 0, px = ev.empty() ? 0 : ev[0][0];
    for (auto& e : ev) {
        area += T.get() * (e[0] - px); px = e[0];
        T.upd(id(e[2]), id(e[3]), (int)e[1]);
    }
    return area;
}

// PERIMETER of the union: run the same sweep twice (once per axis)
↳ accumulating
// |len(after) - len(before)| at each event x, and add the vertical
↳ contributions.

// --- MAX POINTS ON A LINE, O(n^2 log n) ---
template <class P> int maxOnLine(vector<P> p) {
    int n = p.size(), best = min(n, 1);
    for (int i = 0; i < n; i++) {
        map<pair<ll, ll>, int> c; int dup = 0;
        for (int j = 0; j < n; j++) if (j != i) {
            ll dx = p[j].x - p[i].x, dy = p[j].y - p[i].y;
            if (!dx && !dy) { dup++; continue; } //
    ↳ duplicate points
            ll g = gcdll(llabs(dx), llabs(dy)); dx /= g; dy /= g;
            if (dx < 0 || (dx == 0 && dy < 0)) dx = -dx, dy = -dy;
            c[{dx, dy}]++;
        }
        best = max(best, dup + 1);
        for (auto& [k, v] : c) best = max(best, v + dup + 1);
    }
}
```

```

    }
    return best;
}

// --- COUNT PAIRS OF INTERSECTING SEGMENTS,  $O(n^2)$  exact. ---
// For "do ANY two of them cross", use Shamos-Hoey (17), which is  $O(n \log n)$  and returns the pair.
// For  $O((n+k) \log n)$  reporting of ALL of them use Bentley-Ottmann;
// almost never needed.
template <class P> ll countSegInter(vector<pair<P, P>> s) {
    ll c = 0;
    for (int i = 0; i < (int)s.size(); i++)
        for (int j = i + 1; j < (int)s.size(); j++)
            c += segCross(s[i].first, s[i].second, s[j].first, s[j].second);
    return c;
}

// --- CONVEX LAYERS (onion peeling): repeatedly strip the hull.
//  $O(n^2 \log n)$  naive. ---
// Fine to  $n \sim 3000$ ; past that use the decremental hull in 22 -
// OnionDecomposition.cpp,
//  $O(n \log^2 n)$ . Note this version DROPS points lying inside a hull
// edge, 22 keeps them.
template <class P> vector<vector<P>> onion(vector<P> p) {
    vector<vector<P>> L;
    while (!p.empty()) {
        auto h = hull(p);
        L.push_back(h);
        for (P q : h) p.erase(find(all(p), q));
    }
    return L;
}

```

11.1.6 HalfPlaneIntersection

HALF-PLANE INTERSECTION - $O(n \log n)$. A half-plane is "everything LEFT of the directed

```

// line a -> b". Returns the CCW vertices of the intersection, or {}
// if it is empty.
// A big bounding box is added, so the result is always bounded.
// 2D LINEAR PROGRAMMING is exactly this: constraints -> half-planes,
// then optimise over the
// resulting convex polygon (ternary search on the boundary, or check
// all vertices).
struct HP {
    Pd p, d; // keep the side LEFT
    // of p -> p + d
    HP() {}
    HP(Pd a, Pd b) : p(a), d(b - a) {}
    bool out(Pd r) const { return d.cross(r - p) < -eps; }
};

Pd hpInter(const HP& a, const HP& b) { // assumes a.d and b.d are not parallel
    return a.p + a.d * ((b.p - a.p).cross(b.d) / a.d.cross(b.d));
}

vector<Pd> halfPlaneInter(vector<HP> h, ld BOX = 1e9) {
    Pd box[4] = {Pd(-BOX, -BOX), Pd(BOX, -BOX), Pd(BOX, BOX), Pd(-BOX, BOX)};
    for (int i = 0; i < 4; i++) h.push_back(HP(box[i], box[(i + 1) % 4]));
    sort(all(h), [](const HP& a, const HP& b) {
        if (half(a.d) != half(b.d)) return half(a.d) < half(b.d);
        ld c = a.d.cross(b.d);
        if (fabs(c) > eps) return c > 0;
        return a.out(b.p); // same direction:
    });
    deque<HP> dq;
    for (auto& x : h) {
        if (!dq.empty() && fabs(dq.back().d.cross(x.d)) <= eps && dq.back().d.dot(x.d) > 0) continue;
        while (dq.size() > 1 && x.out(hpInter(dq[dq.size() - 1], dq[dq.size() - 2]))) dq.pop_back();
        while (dq.size() > 1 && x.out(hpInter(dq[0], dq[1]))) dq.pop_front();
        dq.push_back(x);
    }
    while (dq.size() > 2 && dq[0].out(hpInter(dq[dq.size() - 1], dq[dq.size() - 2]))) dq.pop_back();
    while (dq.size() > 2 && dq.back().out(hpInter(dq[0], dq[1]))) dq.pop_front();
    if (dq.size() < 3) return {};
    vector<Pd> r;
    for (size_t i = 0; i < dq.size(); i++) r.push_back(hpInter(dq[i], dq[(i + 1) % dq.size()]));
    return r;
}

// If the constraints arrive ONLINE, or you must report after each
// one, use 21 - DynamicHalfPlane.
// If you only have a handful of half-planes, repeated cut() (03 -
// Polygon.cpp) on a big box is
// shorter to retype and just as correct -  $O(n^2)$  but n is small.

```

11.1.7 Unions

AREA OF UNIONS. The three shapes that actually show up.

```
// --- UNION OF POLYGONS (any simple polygons, may overlap),  $O(n^2 \log n)$ . ---
// Works by, for every edge, measuring how much of it is "on the
// boundary of the union" and
// accumulating the shoelace contribution with the right multiplicity.
ld edgeRatio(Pd a, Pd b) { return sgn(b.x) ? a.x / b.x : a.y / b.y; }
ld polyUnion(vector<vector<Pd>>& ps) { // every
    // polygon must be CCW
    ld ret = 0;
    for (size_t i = 0; i < ps.size(); i++)
        for (size_t v = 0; v < ps[i].size(); v++) {
            Pd A = ps[i][v], B = ps[i][(v + 1) % ps[i].size()];
            vector<pair<ld, int>> seg = {{0, 0}, {1, 0}};
            for (size_t j = 0; j < ps.size(); j++) {
                if (i == j) continue;
                for (size_t u = 0; u < ps[j].size(); u++) {
                    Pd C = ps[j][u], D = ps[j][(u + 1) %
                ps[j].size()];
                    int sc = sgn(A.cross(B, C)), sd = sgn(A.cross(B,
                    D));
                    if (sc != sd) {
                        ld sa = C.cross(D, A), sb = C.cross(D, B);
                        if (min(sc, sd) < 0) seg.push_back({sa / (sa
                        - sb), sc > sd ? 1 : -1});
                    } else if (!sc && !sd && j < i && sgn((B -
                    A).dot(D - C)) > 0) {
                        seg.push_back({edgeRatio(C - A, B - A), 1});
                        seg.push_back({edgeRatio(D - A, B - A), -1});
                    }
                }
            }
            sort(all(seg));
            for (auto& s : seg) s.first = min(max(s.first, (ld)0),
            (ld)1);
            ld sum = 0;
            int cnt = seg[0].second;
            for (size_t j = 1; j < seg.size(); j++) { // fraction
                // of AB on the union boundary
                if (!cnt) sum += seg[j].first - seg[j - 1].first;
                cnt += seg[j].second;
            }
            ret += A.cross(B) * sum;
        }
    return ret / 2;
}

// --- UNION OF CIRCLES,  $O(n^2 \log n)$  by integrating each arc that is
// on the outer boundary. ---
ld circleUnion(vector<Pd> c, vector<ld> r) {
    int n = c.size();
    ld total = 0;
    for (int i = 0; i < n; i++) {
        vector<pair<ld, int>> ev;
        int cover = 0;
        for (int j = 0; j < n; j++) {
            if (i == j) continue;
            ld d = (c[i] - c[j]).dist();
            if (d + r[i] <= r[j] + eps &&
                (r[i] < r[j] || (fabsl(r[i] - r[j]) < eps && j < i)))
                { cover = -1; break; }
            if (d >= r[i] + r[j] - eps || d + r[j] <= r[i] + eps)
                continue;
            ld a = (c[j] - c[i]).angle();
            ld w = acosl((d * d + r[i] * r[i] - r[j] * r[j]) / (2 * d
            * r[i]));
            ld L = a - w, R = a + w;
            if (L < -PI) L += 2 * PI;
            if (R > PI) R -= 2 * PI;
            if (L > R) ev.push_back({L, 1}), ev.push_back({PI, -1}),
                ev.push_back({-PI, 1}), ev.push_back({R, -1});
            else ev.push_back({L, 1}), ev.push_back({R, -1});
        }
        if (cover < 0) continue;
        ev.push_back({-PI, 0}), ev.push_back({PI, 0});
        sort(all(ev));
        int cnt = 0; ld last = -PI;
        for (auto [a, t] : ev) {
            if (!cnt && a > last) { // arc
                [last, a] is on the boundary
                Pd p1 = c[i] + Pd(cosl(last), sinl(last)) * r[i];
                Pd p2 = c[i] + Pd(cosl(a), sinl(a)) * r[i];
                total += p1.cross(p2) / 2 + r[i] * r[i] * ((a - last)
                - sinl(a - last)) / 2;
            }
        }
    }
}
```

```
        cnt += t;
        last = max(last, a);
    }
    return total;
}

// --- PERIMETER OF A UNION OF AXIS-ALIGNED RECTANGLES ---
// Run the rectUnionArea sweep TWICE (once per axis). At each event
// x, the vertical contribution
// is |coveredLen(after) - coveredLen(before)|; sum those, then
// repeat with x/y swapped.
```

11.1.8 Transforms

COORDINATE TRANSFORMS - the tricks that turn a hard metric into an easy one.

```
// CHEBYSHEV <-> MANHATTAN.  rot45(p) = (x+y, x-y).
// Manhattan distance in the original = Chebyshev distance after
//   ↪ rot45 (and vice versa).
// Use it for: "max Manhattan distance over a set" (becomes max over
//   ↪ 4 coordinate extremes),
// "count pairs within Manhattan distance k", chessboard-king
//   ↪ problems.
template <class P> P rot45(P p) { return P(p.x + p.y, p.x - p.y); }
template <class P> P unrot45(P p) { return P((p.x + p.y) / 2, (p.x -
//   ↪ p.y) / 2); }
// Max Manhattan distance between any two of the points, O(n).
ll maxManhattan(const vector<P>& v) {
    ll best = 0;
    for (int s = 0; s < 2; s++) { // the 2^1
        // sign patterns for (x +- y)
        ll mn = LLONG_MAX, mx = LLONG_MIN;
        for (auto& p : v) { ll t = s ? p.x - p.y : p.x + p.y; mn =
        //   ↪ min(mn, t), mx = max(mx, t); }
        best = max(best, mx - mn);
    }
    return best;
}
// In d dimensions the same idea uses all 2^(d-1) sign patterns.

// MANHATTAN MST - the MST under L1 distance, O(n log n) instead of
//   ↪ O(n^2).
// Key fact: every point only needs edges to its nearest neighbour in
//   ↪ each of 8 octants; by
// symmetry 4 sweeps suffice (rotate/reflect the plane between
//   ↪ sweeps).
vector<array<ll, 3>> manhattanEdges(vector<P> ps) { // {w, i,
//   ↪ j}; indices stay ORIGINAL
    int n = ps.size();
    vector<int> id(n);
    iota(all(id), 0);
    vector<array<ll, 3>> e;
    for (int s = 0; s < 4; s++) {
        sort(all(id), [&](int a, int b) { return ps[a].x + ps[a].y <
        //   ↪ ps[b].x + ps[b].y; });
        map<ll, int> act; // key = -y,
        //   ↪ value = index
        for (int i : id) {
            for (auto it = act.lower_bound(-ps[i].y); it !=
            //   ↪ act.end(); act.erase(it++)) {
                int j = it->second;
                P d = ps[i] - ps[j];
                if (d.x > d.y) break;
                e.push_back({d.x + d.y, (ll)i, (ll)j});
            }
            act[-ps[i].y] = i;
        }
        for (auto& p : ps) { if (s & 1) p.x = -p.x; else swap(p.x,
        //   ↪ p.y); } // next octant pair
    }
    return e; // feed
//   ↪ these to Kruskal
}

// LINEAR TRANSFORMATION mapping p0->q0 and p1->q1 (rotation +
//   ↪ scaling + translation).
Pd linTrans(Pd p0, Pd p1, Pd q0, Pd q1, Pd r) {
    Pd dp = p1 - p0, dq = q1 - q0, num(dp.cross(dq), dp.dot(dq));
    return q0 + Pd((r - p0).cross(num), (r - p0).dot(num)) /
    //   ↪ dp.dist2();
}

// POINT <-> LINE DUALITY: point (a, b) <-> line y = a*x - b.
// "max points on a common line" becomes "max lines through a
//   ↪ common point"
// convex hull of points <-> upper/lower envelope of lines (this
//   ↪ IS the CHT correspondence)
// incidences are preserved, above/below is preserved.
```

11.1.9 KDTree

KD-TREE over 2D points: nearest-neighbour and "count points in a rectangle".

```
// Build O(n log n); NN query O(log n) expected, O(n) worst. The
//   ↪ O(sqrt n) bound is for
// RECTANGLE queries; a rectangle count costs O(sqrt n + k).
// Use when coordinates are large/sparse and a grid or sort-sweep
//   ↪ will not do.
struct KD {
    struct Node {
        P p; // the
        //   ↪ splitting point
        ll x0 = LLONG_MAX, x1 = LLONG_MIN, y0 = LLONG_MAX, y1 =
        //   ↪ LLONG_MIN; // bounding box
        int l = -1, r = -1, cnt = 0;
    };
    vector<Node> t;
    vector<P> pts;

    KD(vector<P> v) : pts(v) { if (!v.empty()) build(0, v.size(),
    //   ↪ false); }

    int build(int lo, int hi, bool divX) {
        if (lo >= hi) return -1;
        int id = t.size();
        t.push_back({});
        int mid = (lo + hi) / 2;
        nth_element(pts.begin() + lo, pts.begin() + mid, pts.begin()
        //   ↪ + hi,
        //   ↪ [&](P a, P b) { return divX ? a.x < b.x : a.y <
        //   ↪ b.y; });
        Node nd;
        nd.p = pts[mid], nd.cnt = hi - lo;
        for (int i = lo; i < hi; i++)
            nd.x0 = min(nd.x0, pts[i].x), nd.x1 = max(nd.x1,
            //   ↪ pts[i].x),
            nd.y0 = min(nd.y0, pts[i].y), nd.y1 = max(nd.y1,
            //   ↪ pts[i].y);
        t[id] = nd;
        int L = build(lo, mid, !divX), R = build(mid + 1, hi, !divX);
        t[id].l = L, t[id].r = R;
        return id;
    }
    ll boxDist(int i, P q) const { // squared
        //   ↪ distance from q to the box
        const Node& n = t[i];
        ll dx = max({(ll)0, n.x0 - q.x, q.x - n.x1}), dy =
        //   ↪ max({(ll)0, n.y0 - q.y, q.y - n.y1});
        return dx * dx + dy * dy;
    }
    void nn(int i, P q, ll& best, P& bp, bool skipSelf) const {
        if (i < 0 || boxDist(i, q) >= best) return;
        ll d = (t[i].p - q).dist2();
        if (d < best && !(skipSelf && d == 0)) best = d, bp = t[i].p;
        int a = t[i].l, b = t[i].r;
        if (a >= 0 && b >= 0 && boxDist(b, q) < boxDist(a, q))
            swap(a, b);
        nn(a, q, best, bp, skipSelf), nn(b, q, best, bp, skipSelf);
    }
    // skipSelf skips EVERY point at distance 0, not just q itself -
    //   ↪ with duplicate input points
    // that is wrong for "nearest OTHER point". If duplicates are
    //   ↪ possible, count multiplicity
    // separately and treat a repeated point as its own nearest
    //   ↪ neighbour at distance 0.
    pair<ll, P> nearest(P q, bool skipSelf = false) const {
        if (t.empty()) return {llinf, P{}};
        ll best = LLONG_MAX; P bp;
        nn(0, q, best, bp, skipSelf);
        return {best, bp}; // {squared
        //   ↪ distance, point}
    }
    int count(int i, ll x0, ll y0, ll x1, ll y1) const { // #points
        //   ↪ in [x0,x1] x [y0,y1]
        if (i < 0 || t[i].x1 < x0 || t[i].x0 > x1 || t[i].y1 < y0 ||
        //   ↪ t[i].y0 > y1) return 0;
        if (x0 <= t[i].x0 && t[i].x1 <= x1 && y0 <= t[i].y0 &&
        //   ↪ t[i].y1 <= y1) return t[i].cnt;
        int r = (x0 <= t[i].p.x && t[i].p.x <= x1 && y0 <= t[i].p.y
        //   ↪ && t[i].p.y <= y1);
        return r + count(t[i].l, x0, y0, x1, y1) + count(t[i].r, x0,
        //   ↪ y0, x1, y1);
    }
    int count(ll x0, ll y0, ll x1, ll y1) const { return t.empty() ?
    //   ↪ 0 : count(0, x0, y0, x1, y1); }
};
```


11.1.10 Triangulation

EAR CLIPPING - triangulates any SIMPLE polygon (convex or not, no self-intersections, no holes)

```
// Needs: Core.cpp, Polygon.cpp (area2).
// into n-2 triangles,  $O(n^2)$ . Exact with P<ll>. Returns index
//   ↳ triples into the ORIGINAL vector.
// An "ear" is a convex vertex whose triangle contains no other
//   ↳ vertex of the polygon.
// USES: area/integral of anything over a polygon, rendering,
//   ↳ splitting a region for a sweep,
// point-location preprocessing, computing a polygon's centre of mass
//   ↳ with a weight function.
template <class P>
vector<array<int, 3>> triangulate(const vector<P>& poly) {
    int n = poly.size();
    vector<array<int, 3>> tri;
    if (n < 3) return tri;
    vector<int> id(n);
    iota(all(id), 0);
    if (area2(poly) < 0) reverse(all(id));
    ↳ work counter-clockwise
    auto inTri = [&](int a, int b, int c, int q) {
        ↳ closed triangle
        return poly[a].cross(poly[b], poly[q]) >= 0 &&
        poly[b].cross(poly[c], poly[q]) >= 0 &&
        poly[c].cross(poly[a], poly[q]) >= 0;
    };
    while (id.size() > 2) {
        int m = id.size(), cut = -1, flat = -1;
        for (int i = 0; i < m && cut < 0; i++) {
            int a = id[(i + m - 1) % m], b = id[i], c = id[(i + 1) %
        ↳ m];
            auto o = poly[a].cross(poly[b], poly[c]);
            if (o == 0) { flat = i; continue; }
            ↳ collinear vertex: just drop it
            if (o < 0) continue;
            ↳ reflex
            bool ok = true;
            for (int j = 0; j < m && ok; j++)
                if (id[j] != a && id[j] != b && id[j] != c &&
                inTri(a, b, c, id[j])) ok = false;
            if (ok) cut = i, tri.push_back({a, b, c});
        }
        if (cut < 0) cut = flat;
        if (cut < 0) break;
        ↳ not a simple polygon
        id.erase(id.begin() + cut);
    }
    return tri;
}
// FASTER: monotone decomposition (sweep,  $O(n \log n)$ ) then
//   ↳ triangulate each monotone piece in  $O(n)$ .
// Only worth writing for n beyond ~2000; ear clipping handles
//   ↳ everything a contest throws at you.
// DELAUNAY TRIANGULATION (13 - Delaunay.cpp,  $O(n \log n)$ ) maximises
//   ↳ the minimum angle instead, and
// its dual is the VORONOI DIAGRAM (14 - Voronoi.cpp). Cheap
//   ↳ derivation to remember: lift each
// point to the paraboloid  $(x, y, x^2 + y^2)$  and take the LOWER hull of
//   ↳ the 3D convex hull - its
// faces project to the Delaunay triangles.
// Delaunay contains: the Euclidean minimum spanning tree, the
//   ↳ nearest-neighbour graph, and the
// closest pair. So "EMST of points in the plane" = Delaunay ( $O(n$ 
//   ↳  $\log n$ ) edges) + Kruskal.
// In-circle test (a,b,c counter-clockwise): d is strictly inside
//   ↳ the circumcircle of a,b,c iff
// the 3x3 determinant of rows (a-d, |a-d|^2), (b-d, |b-d|^2),
//   ↳ (c-d, |c-d|^2) is > 0. [...]128]
// ART GALLERY THEOREM: floor(n/3) guards always suffice for a simple
//   ↳ n-gon, and are sometimes
// necessary (comb polygon). Proof = triangulate, 3-colour the
//   ↳ triangulation graph, take the
// smallest colour class. The triangulation above is exactly what
//   ↳ that construction needs.
// POLYGON WITH HOLES: connect each hole to the outer boundary with a
//   ↳ "bridge" edge (pick the hole's
// rightmost vertex, shoot a ray right, split the hit edge) to get
//   ↳ one simple polygon, then clip.
```

11.1.11 PolygonOps

POLYGON AGAINST POLYGON: clipping, intersection, containment, distance, and the $O(\log n)$

```
// Needs: Core.cpp, Lines.cpp, Polygon.cpp.
// queries you get once one side is a convex hull.

// --- IS THIS POLYGON SIMPLE?  $O(n^2)$ , exact on P<ll>. ---
// Every polygon routine (inPoly, area, triangulate, Pick, polyUnion)
//   ↳ silently returns nonsense on
// a self-intersecting polygon, so this is also the oracle you
//   ↳ stress-test them with.
template <class P> bool isSimple(const vector<P>& v) {
    int n = v.size();
    if (n < 3) return false;
    for (int i = 0; i < n; i++) {
        if (v[i] == v[(i + 1) % n]) return false;
        ↳ zero-length edge
        for (int j = i + 1; j < n; j++) {
            int a = i, b = (i + 1) % n, c = j, d = (j + 1) % n;
            if (a == c || a == d || b == c || b == d) {
                ↳ adjacent edges: they may only
                if (a == d || b == c) {
                    ↳ share the one common vertex
                    P sh = (a == d) ? v[a] : v[b];
                    for (P q : {v[a], v[b], v[c], v[d]})
                        if (!(q == sh) && onSeg(q, v[a], v[b]) &&
                        onSeg(q, v[c], v[d])) return false;
                }
                continue;
            }
            if (segCross(v[a], v[b], v[c], v[d])) return false;
        }
    }
    return true;
}
// --- WINDING NUMBER. inPoly uses the EVEN-ODD rule, which is wrong
//   ↳ for a self-intersecting
// outline and for "polygon with holes given as one vertex list". The
//   ↳ nonzero rule uses this. ---
template <class P> int windingNumber(const vector<P>& v, P p) {
    ↳ 0 = outside (nonzero rule)
    int n = v.size(), w = 0;
    for (int i = 0; i < n; i++) {
        P a = v[i], b = v[(i + 1) % n];
        if (onSeg(p, a, b)) return 0;
        ↳ on
        ↳ the boundary: caller decides
        if (a.y <= p.y) { if (b.y > p.y && a.cross(b, p) > 0) w++; }
        else if (b.y <= p.y && a.cross(b, p) < 0) w--;
    }
    return w;
}
// --- SUTHERLAND-HODGMAN: clip any polygon by a CONVEX window,  $O(n *$ 
//   ↳  $m)$ . ---
// The window must be convex and counter-clockwise; the subject may
//   ↳ be concave.
// This is the general "intersect with a convex region" and it is
//   ↳ just 'cut' in a loop.
template <class P> vector<P> clipByConvex(vector<P> subject, const
vector<P>& win) {
    int m = win.size();
    for (int i = 0; i < m && !subject.empty(); i++)
        subject = cut(subject, win[i], win[(i + 1) % m]);
    return subject;
}
// --- CONVEX intersect CONVEX in  $O(n + m)$  is the classic O'Rourke
//   ↳ walk, but clipByConvex is
//  $O(nm)$  and fits on one line - at contest sizes (n, m <= a few
//   ↳ thousand) prefer it. Use
// halfPlaneInter (06) when the polygons come as half-plane
//   ↳ constraints, not vertex lists.

// --- MINIMUM DISTANCE BETWEEN TWO CONVEX POLYGONS,  $O(n * m)$  as
//   ↳ written. ---
// 0 if they touch or overlap. The  $O(n+m)$  rotating-calipers version
//   ↳ exists but this is 6 lines and
// the distance is a floating answer anyway. Equivalent trick:
//   ↳ dist(origin, minkowski(A, -B)).
template <class P> ld convexDist(const vector<P>& A, const vector<P>&
B) {
    for (P p : A) if (inPoly(B, p, false)) return 0;
    for (P p : B) if (inPoly(A, p, false)) return 0;
    ld best = 1e18;
    int n = A.size(), m = B.size();
    for (int i = 0; i < n; i++) for (int j = 0; j < m; j++) {
        P a = A[i], b = A[(i + 1) % n], c = B[j], d = B[(j + 1) % m];
        if (segCross(a, b, c, d)) return 0;
        best = min(best, segSegDist(a, b, c, d));
    }
}
```

```

    }
    return best;
}
// --- TANGENTS FROM A POINT TO A CONVEX POLYGON, O(n) here (O(log n)
//   ↳ with a binary search). ---
// Returns the two vertex indices l, r such that the whole polygon
//   ↳ lies to one side of p->h[l]
// and of p->h[r]: the visible arc is h[r], h[r+1], ..., h[l]. p must
//   ↳ be strictly OUTSIDE.
// USES: visibility ("how much of the wall can I see"),
//   ↳ shadow/silhouette, and "add p to the hull".
template <class P> pair<int, int> pointTangents(const vector<P>& h, P
//   ↳ p) {
    int n = h.size(), l = 0, r = 0;
    for (int i = 1; i < n; i++) {
        if (p.cross(h[i], h[l]) > 0) l = i;           //
//   ↳ h[l] is the CCW-most
        if (p.cross(h[i], h[r]) < 0) r = i;           //
//   ↳ h[r] is the CW-most
    }
    return {l, r};
}
// --- LINE x CONVEX POLYGON in O(log n): which two edges does the
//   ↳ line a->b cross? ---
// Returns {-1,-1} for no hit; {i,-1} for a single touched corner i;
//   ↳ {i,j} when the line crosses
// the edges (i,i+1) and (j,j+1). h must be a strictly convex CCW
//   ↳ hull (no collinear vertices).
// USES: "length of the segment inside the polygon" per query, in
//   ↳ O(log n) instead of O(n).
template <class P> pair<int, int> lineHull(const vector<P>& h, P a, P
//   ↳ b) {
    int n = h.size();
    int endA = extreme(h, (a - b).perp()), endB = extreme(h, (b -
//   ↳ a).perp());
    auto cml = [&](int i) { return sgn(a.cross(h[i], b)); };
    if (cml(endA) < 0 || cml(endB) > 0) return {-1, -1}; //
//   ↳ whole hull on one side
    auto bs = [&](int lo, int hi) { //
//   ↳ last index still >= 0
        int step = (hi - lo + n) % n;
        for (int s = step; s > 0; s /= 2)
            while (s && cml((lo + s) % n) >= 0) lo = (lo + s) % n,
//   ↳ step -= s;
            return lo;
    };
    int i = bs(endB, endA), j = bs(endA, endB);
    if (i == j) return {i, -1};
    return {i, j};
}
/* MORE POLYGON FACTS AND RECIPES
AREA OF INTERSECTION of two CONVEX polygons: clipByConvex then
//   ↳ area(). For two ARBITRARY simple
//   ↳ polygons, triangulate one (10 - Triangulation.cpp) and clip each
//   ↳ triangle by the other.
AREA OF THE UNION: |A| + |B| - |A n B|, or polyUnion (07 -
//   ↳ Unions.cpp) for many polygons at once.
SYMMETRIC DIFFERENCE: |A| + |B| - 2|A n B|.
POINT IN A POLYGON WITH HOLES: outer boundary CCW, every hole CW,
//   ↳ then the sign of the total
//   ↳ winding number is the answer (this is exactly why windingNumber
//   ↳ exists).
CONVEX POLYGON CONTAINS CONVEX POLYGON: every vertex of B is in A
//   ↳ (inConvex, O(m log n)).
CENTROID OF THE PERIMETER (not of the area): sum of edge midpoints
//   ↳ weighted by edge LENGTH.
The two differ - "balance the wire frame" vs "balance the plate".
POLYGON DIAMETER / WIDTH / MIN RECTANGLE: 03 - Polygon.cpp (rotating
//   ↳ calipers).
REGULAR n-GON of circumradius R centred at c: c + R*(cos(2*pi*k/n +
//   ↳ t), sin(2*pi*k/n + t)).
Its area is (n/2) R^2 sin(2 pi/n); with INRADIUS r it is n r^2
//   ↳ tan(pi/n).
SPLIT A POLYGON BY A LINE into both halves: cut(v, s, e) and cut(v,
//   ↳ e, s).
LARGEST INSCRIBED / SMALLEST ENCLOSING: smallest enclosing circle is
//   ↳ mec (04 - Circles.cpp);
smallest enclosing rectangle is minRectArea; largest inscribed
//   ↳ CIRCLE in a convex polygon is
the Chebyshev centre, 18 - InscribedCircle.cpp (a nested ternary
//   ↳ search, no LP needed);
largest inscribed AXIS-ALIGNED rectangle needs a sweep and is
//   ↳ genuinely hard - do not improvise.
OFFSET / BUFFER a convex polygon by d: push every edge outward by d
//   ↳ along its normal and
//   ↳ intersect the half-planes (06 - HalfPlaneIntersection.cpp). The
//   ↳ exact offset of a NON-convex

```

polygon has circular arcs at the reflex corners - approximate them.
 TRIANGLE CONTAINS POINT, exact: the three orientations all have the
 ↳ same sign (or one is 0 for
 the boundary). Faster than inPoly for n = 3 and it is the inner
 ↳ loop of triangulate. */

11.1.12 Primitives

THE SMALL-FUNCTION LAYER. Individually trivial, collectively the thing you actually spend

```
// Needs: Core.cpp, Lines.cpp, Circles.cpp.
// contest minutes re-deriving. Everything here is one to six lines.

// --- ANGLE BISECTORS OF TWO LINES. Two lines have TWO bisectors,
//   ↳ perpendicular to each other.
// Direction of the internal one = the sum of the unit direction
//   ↳ vectors; external = difference.
// (Take the pair whose sign matches the side you want; if the lines
//   ↳ are parallel the bisector is
// the parallel line halfway between them.)
template <class P> pair<P, P> lineBisectors(P d1, P d2) {
    ↳ returns the two DIRECTIONS
    P u = d1.unit(), v = d2.unit();
    return {u + v, u - v};
}
// Internal bisector of the angle at b in triangle a-b-c, as a
//   ↳ direction from b:
template <class P> P angleBisector(P a, P b, P c) { return (a -
    ↳ b).unit() + (c - b).unit(); }
// The internal bisector from A meets BC at the point dividing it in
//   ↳ ratio AB : AC.
template <class P> P bisectorFoot(P a, P b, P c) {
    ld x = (a - b).dist(), y = (a - c).dist();
    return (b * y + c * x) / (x + y);
}
// --- ROTATE / SCALE ABOUT AN ARBITRARY CENTRE (rot() in Core is
//   ↳ about the origin). ---
template <class P> P rotAbout(P p, P c, ld a) { return c + (p -
    ↳ c).rot(a); }
template <class P> P scaleAbout(P p, P c, ld k) { return c + (p - c)
    ↳ * k; }
// Reflect p across the POINT c (a 180-degree rotation):
template <class P> P reflPoint(P p, P c) { return c * 2 - p; }
// --- SIGNED AREA OF A TRIANGLE, and the barycentric coordinates of
//   ↳ p in a-b-c. ---
template <class P> auto triArea2(P a, P b, P c) { return a.cross(b,
    ↳ c); } // 2 * signed
template <class P> array<ld, 3> barycentric(P a, P b, P c, P p) {
    ld s = a.cross(b, c);
    return {(ld)p.cross(b, c) / s, (ld)a.cross(p, c) / s,
    ↳ (ld)a.cross(b, p) / s};
}
// p is inside the triangle iff all three coordinates are >= 0 (they
//   ↳ always sum to 1).
template <class P> bool inTriangle(P a, P b, P c, P p) {
    ↳ exact, boundary included
    int s1 = sgn(a.cross(b, p)), s2 = sgn(b.cross(c, p)), s3 =
    ↳ sgn(c.cross(a, p));
    return (s1 >= 0 && s2 >= 0 && s3 >= 0) || (s1 <= 0 && s2 <= 0 &&
    ↳ s3 <= 0);
}
// --- SEGMENT x CIRCLE: clip the segment a-b to the disc, returning
//   ↳ the surviving piece. ---
template <class P> vector<P> segCircle(P a, P b, P c, ld r) {
    auto v = circleLine(c, r, a, b);
    ↳ 0, 1 or 2 points
    vector<P> res;
    for (P p : v) if (onSeg(p, a, b)) res.push_back(p);
    return res;
}
// --- CIRCLE THROUGH THREE POINTS = circumcenter (04 - Circles.cpp);
//   ↳ its radius = circumradius.
// CIRCLE THROUGH TWO POINTS WITH A GIVEN RADIUS: the two centres are
//   ↳ m +- h * (b-a).perp().unit()
// where m is the midpoint and h = sqrt(r^2 - |b-a|^2/4). None if 2r
//   ↳ < |b-a|.
template <class P> vector<P> circlesThrough(P a, P b, ld r) {
    ld d2 = (b - a).dist2(), h2 = r * r - d2 / 4;
    if (d2 < eps || h2 < 0) return {};
    ↳ a == b: infinitely many
    P m = (a + b) / 2, dir = (b - a).perp().unit() * sqrt(h2);
    return h2 < eps ? vector<P>{m} : vector<P>{m - dir, m + dir};
}
// --- CIRCLE OF RADIUS r THROUGH p AND TANGENT TO THE LINE a-b. Its
//   ↳ centre is at distance r from
// the line, so on one of the two PARALLELS at offset r, and at
//   ↳ distance r from p, so on the
// circle (p, r). Intersect the two: 0, 1 or 2 centres. With d =
//   ↳ distance from p to the line,
// there are 2 for d < 2r (both, tangentially, when d == 0), 1 when d
//   ↳ == 2r, and none beyond.
template <class P> vector<P> circleTangentLine(P a, P b, P p, ld r) {
    vector<P> res;
    P n = (b - a).perp().unit() * r;
```

```
    for (int s = -1; s <= 1; s += 2)
        for (P c : circleLine(p, r, a + n * s, b + n * s))
            ↳ res.push_back(c);
    return res;
}
// --- APOLLONIUS CIRCLE: the locus of p with |pa| / |pb| = k (k !=
//   ↳ 1) is a circle whose diameter
// endpoints are the two points dividing ab internally and externally
//   ↳ in ratio k : 1.
template <class P> pair<P, ld> apollonius(P a, P b, ld k) {
    P u = (b * k + a) / (k + 1), v = (b * k - a) / (k - 1);
    ↳ internal, external
    return {(u + v) / 2, (u - v).dist() / 2};
}
// --- IS THE QUADRILATERAL a,b,c,d CONCYCLIC? Exact on P<ll> with
//   ↳ __int128 (in-circle test). ---
// > 0: d strictly inside the circumcircle of a CCW triangle a,b,c;
//   ↳ == 0: concyclic.
template <class P> int inCircle(P a, P b, P c, P d) {
    auto f = [&](P p) { return (__int128)(p - d).dist2(); };
    __int128 det = (__int128)(a.x - d.x) * ((b.y - d.y) * f(c) - (c.y
    ↳ - d.y) * f(b))
        - (__int128)(a.y - d.y) * ((b.x - d.x) * f(c) - (c.x
    ↳ - d.x) * f(b))
        + f(a) * ((__int128)(b.x - d.x) * (c.y - d.y)
        - (__int128)(c.x - d.x) * (b.y - d.y));
    return (det > 0) - (det < 0);
}
// --- MAXIMUM POINTS COVERED BY A CIRCLE OF RADIUS r, 0(n^2 log n).
//   ↳ ---
// For each i, every j within 2r gives an ARC of centre-angles that
//   ↳ cover both; sweep the arcs.
template <class P> int maxCovered(const vector<P>& p, ld r) {
    int n = p.size(), best = 1;
    for (int i = 0; i < n; i++) {
        vector<pair<ld, int>> ev;
        int base = 1;
        ↳ p[i], plus its duplicates
        for (int j = 0; j < n; j++) if (j != i) {
            ld d = (p[j] - p[i]).dist();
            if (d > 2 * r + eps) continue;
            if (d < eps) { base++; continue; }
            ↳ // DUPLICATE of p[i]: the
            ↳ // division below would be 0/0
            ↳ // and it is always covered
            ld th = (p[j] - p[i]).angle(), ph = acosl(min((ld)1,
            ↳ max((ld)-1, d / (2 * r))));
            ld s = th - ph, e = th + ph;
            ↳ the arc of valid centres
            while (s < -PI) s += 2 * PI;
            ↳ normalise, then SPLIT the
            e = s + 2 * ph;
            ↳ arcs that wrap past +pi -
            // 0 = arc opens, 1 = arc closes. Pair-sorting then puts
            ↳ the OPEN first at an equal
            // angle, which is what makes a TANGENTIAL arc (d == 2r,
            ↳ so the arc is a single point)
            // still count - with {+1}/{-1} the close sorts first and
            ↳ the point is lost.
            if (e <= PI) ev.push_back({s, 0}), ev.push_back({e, 1});
            else {
                ev.push_back({s, 0}), ev.push_back({PI, 1});
                ev.push_back({-PI, 0}), ev.push_back({e - 2 * PI, 1});
            }
        }
        sort(all(ev));
        int cur = base;
        best = max(best, cur);
        for (auto& [a, t] : ev) best = max(best, cur += t ? -1 : 1);
    }
    return best;
    ↳ the circle passes through p[i]
}
// --- GEOMETRIC MEDIAN (minimise the SUM of distances). Nested
//   ↳ ternary search - the objective is
// convex, so this is exact to any eps and cannot get stuck.
// Weiszfeld iteration is faster but
// breaks when the iterate lands exactly on a sample point.
template <class P> P geoMedian(const vector<P>& p) {
    auto f = [&](ld x, ld y) { ld s = 0; for (auto& q : p) s += (P{x,
    ↳ y} - q).dist(); return s; };
    auto fy = [&](ld x) { ld lo = -2e9, hi = 2e9;
        for (int it = 0; it < 100; it++) { ld a = lo + (hi - lo) / 3,
        ↳ b = hi - (hi - lo) / 3;
            f(x, a) < f(x, b) ? hi = b : lo = a; } return (lo + hi) /
```

```

    ↪ 2; };
    ld lo = -2e9, hi = 2e9;
    for (int it = 0; it < 100; it++) {
        ld a = lo + (hi - lo) / 3, b = hi - (hi - lo) / 3;
        f(a, fy(a)) < f(b, fy(b)) ? hi = b : lo = a;
    }
    ld x = (lo + hi) / 2;
    return {x, fy(x)};
}
/* THE REST OF THE SMALL LAYER, as one-liners you can reconstruct
    ↪ instantly
COLLINEAR a, b, c          a.cross(b, c) == 0
PARALLEL / PERPENDICULAR  d1.cross(d2) == 0 / d1.dot(d2) == 0
SAME SIDE OF A LINE        sgn(a.cross(b, p)) == sgn(a.cross(b, q))
POINT ON A RAY / SEGMENT   onRay, onSeg (02 - Lines.cpp)
DISTANCE point-line/segment lineDist, segDist ray: rayDist
    ↪ seg-seg: segSegDist
PROJECT / REFLECT over a line proj, refl (02 - Lines.cpp)
ANGLE between two vectors atan2(cross, dot) -> signed, in (-pi,
    ↪ pi]. NEVER acos(dot/|a||b|):
        it loses all precision near 0 and pi
    ↪ and cannot give you a sign.
ROTATE 90 DEGREES          p.perp() = (-y, x). Twice = negation.
    ↪ This is the whole of "turn
        left"; you almost never need a general
    ↪ rot() for a right angle.
SORT BY ANGLE              angCmp (01 - Core.cpp), NOT atan2 -
    ↪ exact and 5x faster.
AREA OF A TRIANGLE         |cross| / 2. Heron only if you are
    ↪ given the three SIDES, and then
        use the sorted stable form: sort a >= b
    ↪ >= c and
        A =
    ↪ sqrt((a+(b+c))(c-(a-b))(c+(a-b))(a+(b-c))) / 4.
TRIANGLE CENTRES           centroid (a+b+c)/3, circumcenter,
    ↪ incenter, orthocenter (04).
        EULER LINE: O, G, N, H are collinear
    ↪ with OG:GN:NH = 2:1:3, and
        H = A + B + C - 2O. Nine-point centre =
    ↪ midpoint of OH, radius R/2.
TRIANGLE IDENTITIES        A = rs = abc/(4R); R = abc/(4A); r =
    ↪ A/s; r_a = A/(s-a)
        law of cosines c^2 = a^2 + b^2 - 2ab
    ↪ cos C
        law of sines a/sin A = 2R
        OI^2 = R(R - 2r) => EULER'S
    ↪ INEQUALITY R >= 2r
        Stewart (cevian d to BC split m + n):
    ↪ b^2 m + c^2 n = a(d^2 + mn)
        median m_a = sqrt(2b^2 + 2c^2 - a^2)/2
CYCLIC QUADRILATERAL      PTOLEMY: AC*BD = AB*CD + BC*AD (in
    ↪ cyclic order). For ANY four
        points AC*BD <= AB*CD + BC*AD, equality
    ↪ iff concyclic in that order.
        BRAHMAGUPTA area =
    ↪ sqrt((s-a)(s-b)(s-c)(s-d)).
        General quadrilateral (BRETSCHNEIDER,
    ↪ angle-free):
        A = sqrt(4p^2q^2 -
    ↪ (b^2+d^2-a^2-c^2)^2)/4 with p, q the diagonals.
POWER OF A POINT           pow(p) = |pc|^2 - r^2; for ANY line
    ↪ through p meeting the circle at
        A and B the signed product PA*PB equals
    ↪ it. Outside: = tangent^2.
        RADICAL AXIS = equal power, a line
    ↪ perpendicular to the centre line.
        RADICAL CENTRE of three circles with
    ↪ non-collinear centres: the
        three pairwise axes are concurrent.
DESCARTES CIRCLE THEOREM  four mutually tangent circles,
    ↪ curvatures k_i = +-1/r_i (negative
        for the enclosing one):
    ↪ (k1+k2+k3+k4)^2 = 2(k1^2+k2^2+k3^2+k4^2),
        so k4 = k1+k2+k3 +- 2 sqrt(k1k2 + k2k3
    ↪ + k3k1).
LATTICE POINTS            on a segment: gcd(|dx|, |dy|) + 1
    ↪ endpoints included.
        inside a polygon: PICK, I = A - B/2 + 1
    ↪ (simple polygons only).
        under a line, in O(log): floorSum (04 -
    ↪ Number Theory/01/03).
HELLY (2D)                if every THREE of a family of convex
    ↪ sets meet, all of them meet.
RADON (2D)                any FOUR points split into two sets
    ↪ whose hulls intersect.
CARATHEODORY (2D)        a point in the hull of S is in the hull
    ↪ of at most THREE points of S

```

```

- which is exactly why the minimum
    ↪ enclosing circle needs only 3.
ART GALLERY                floor(n/3) guards always suffice for a
    ↪ simple n-gon (triangulate,
        3-colour, take the smallest class).
    ↪ Orthogonal polygon: floor(n/4).
EULER, PLANAR              V - E + F = 1 + C. n lines in general
    ↪ position cut the plane into
        1 + n + C(n,2) regions, C(n-1,2) of
    ↪ them bounded; n circles in
        general position give n^2 - n + 2
    ↪ regions.
DUALITY                    point (a,b) <-> line y = ax - b.
    ↪ Incidence and above/below are
        preserved, so "max collinear points"
    ↪ becomes "max concurrent
        lines"
        and a convex hull becomes an upper
    ↪ envelope. */

```

11.1.13 Delaunay

DELAUNAY TRIANGULATION, $O(n \log n)$, divide and conquer over the quad-edge structure.

```
// Needs: Core.cpp.
// Returns index triples into the input, each one COUNTER-CLOCKWISE.
//   ↳ The defining property, and
//   ↳ the only one you ever use: the circumcircle of every returned
//   ↳ triangle is EMPTY of input points.
// WHEN: EUCLIDEAN MST of points in the plane - the EMST is a
//   ↳ subgraph of the Delaunay graph, so
// Delaunay (0(n) edges) + Kruskal replaces the 0(n^2) complete
//   ↳ graph. Also the nearest-neighbour
// graph, the closest pair, the largest empty circle, "maximise the
//   ↳ minimum angle" meshing, and
// the VORONOI DIAGRAM, which is exactly the dual (14 - Voronoi.cpp).
// COMPLEXITY:  $O(n \log n)$  time. At most  $2n - 2 - h$  triangles and  $3n -$ 
//   ↳  $3 - h$  edges,  $h$  = hull size.
// Allocates quad-edges with new and never frees them; budget ~200
//   ↳ bytes per point.
// DEGENERATE: DUPLICATE POINTS ARE NOT ALLOWED - dedupe before
//   ↳ calling. Fewer than three points,
// or ALL POINTS COLLINEAR, returns {} - there is genuinely no
//   ↳ triangle, handle it upstream.
// Four or more CONCYCLIC points make the triangulation non-unique;
//   ↳ you get one legal answer.
// EXACT on P<ll> for |coord| <= 1e9: orient fits in ll (<= 8e18) and
//   ↳ inCircle is __int128.
typedef struct QuadEdge* QE;
Pi qArb(LLONG_MAX, LLONG_MAX); // a sentinel
// unequal to any real point
struct QuadEdge {
    QE rot, o; Pi p = qArb; bool mark = 0;
    Pi& F() { return r()->p; } // the far
    // endpoint of this directed edge
    QE& r() { return rot->rot; } // the same edge
    // reversed
    QE prev() { return rot->o->rot; }
    QE next() { return r()->prev(); }
};
bool inCirc(Pi p, Pi a, Pi b, Pi c) { // is p strictly
    // inside circumcircle(a, b, c)?
    __int128 p2 = p.dist2(), A = a.dist2() - p2, B = b.dist2() - p2,
    // C = c.dist2() - p2;
    // return p.cross(a, b) * C + p.cross(b, c) * A + p.cross(c, a) * B
    // > 0;
}
QE qeEdge(Pi orig, Pi dest) {
    QE q[] = {new QuadEdge{0, 0, orig}, new QuadEdge{0, 0, qArb}, new
    // QuadEdge{0, 0, dest},
    // new QuadEdge{0, 0, qArb}};
    for (int i = 0; i < 4; i++) q[i]->o = q[-i & 3], q[i]->rot = q[(i
    // + 1) & 3];
    return *q;
}
void qeSplice(QE a, QE b) { swap(a->o->rot->o, b->o->rot->o),
    // swap(a->o, b->o); }
QE qeConnect(QE a, QE b) {
    QE q = qeEdge(a->F(), b->p);
    qeSplice(q, a->next()), qeSplice(q->r(), b);
    return q;
}
QE dNext(QE e, bool lf) { return lf ? e->o : e->prev(); } //
    // ↳ walk left / right of the base
bool dValid(QE e, QE base) { return e->F().cross(base->F(), base->p)
    // > 0; }
QE dDel(QE init, QE base, bool lf) { // drop edges
    // ↳ that the new base invalidates
    QE e = dNext(init, lf);
    if (dValid(e, base))
        while (inCirc(dNext(e, lf)->F(), base->F(), base->p, e->F()))
            {
                QE t = dNext(e, lf);
                qeSplice(e, e->prev()), qeSplice(e->r(), e->r()->prev());
                e = t;
            }
    return e;
}
pair<QE, QE> dRec(const vector<Pi>& s) { // {ccw hull edge at
    // ↳ left end, cw at right}
    if (s.size() <= 3) {
        QE a = qeEdge(s[0], s[1]), b = qeEdge(s[1], s.back());
        if (s.size() == 2) return {a, a->r()};
        qeSplice(a->r(), b);
        auto side = s[0].cross(s[1], s[2]);
        QE c = side ? qeConnect(b, a) : 0; // side == 0:
        // ↳ three collinear points, no face
        return {side < 0 ? c->r() : a, side < 0 ? c : b->r()};
    }
}
```

```
}
QE A, B, ra, rb;
int half = s.size() / 2;
tie(ra, A) = dRec({s.begin(), s.end() - half});
tie(B, rb) = dRec({s.begin() + (s.size() - half), s.end()});
while ((B->p.cross(A->F(), A->p) < 0 && (A = A->next())) ||
    // ↳ find the lower common tangent
    (A->p.cross(B->F(), B->p) > 0 && (B = B->r()->o)));
QE base = qeConnect(B->r(), A);
if (A->p == ra->p) ra = base->r();
if (B->p == rb->p) rb = base;
for (;;) { // zip the two
    // ↳ hulls together, bottom to top
    QE lc = dDel(base->r(), base, 1), rc = dDel(base, base, 0);
    if (!dValid(lc, base) && !dValid(rc, base)) break;
    if (!dValid(lc, base) || (dValid(rc, base) && inCirc(rc->F(),
    // ↳ rc->p, lc->F(), lc->p)))
        base = qeConnect(rc, base->r());
    else base = qeConnect(base->r(), lc->r());
}
return {ra, rb};
}
vector<array<int, 3>> delaunay(const vector<Pi>& pts) {
    vector<Pi> p = pts;
    sort(all(p));
    vector<array<int, 3>> res;
    if (p.size() < 3) return res;
    bool line = true; // guard
    // ↳ collinear input: dRec spins on it
    for (int i = 2; i < sz(p) && line; i++) line = p[0].cross(p[1],
    // ↳ p[i]) == 0;
    if (line) return res;
    QE e = dRec(p).first;
    vector<QE> q = {e};
    while (e->o->F().cross(e->F(), e->p) < 0) e = e->o;
    vector<Pi> out;
    auto walk = [&](QE e) { // emit one
        // ↳ face, queue its neighbours
        QE c = e;
        do { c->mark = 1, out.pb(c->p), q.pb(c->r()), c = c->next();
        } while (c != e);
    };
    walk(e), out.clear(); // the first face
    // ↳ walked is the OUTER one
    for (int i = 0; i < sz(q); i++) if (!(e = q[i])->mark) walk(e);
    map<pair<ll, ll>, int> id;
    for (int i = 0; i < sz(pts); i++) id[{pts[i].x, pts[i].y}] = i;
    for (int i = 0; i + 2 < sz(out); i += 3)
        res.pb({id[{out[i].x, out[i].y}], id[{out[i + 1].x, out[i +
    // ↳ 1].y}],
        // id[{out[i + 2].x, out[i + 2].y}]});
    return res;
}
// EUCLIDEAN MST: collect the 3 edges of every triangle, dedupe, sort
// ↳ by squared length, Kruskal.
// vector<array<ll, 3>> E; for (auto& t : delaunay(p)) for (int k =
// ↳ 0; k < 3; k++)
//     E.pb({p[t[k]] - p[t[(k+1)%3]].dist2(), t[k], t[(k+1)%3]});
// LARGEST EMPTY CIRCLE whose centre is inside the hull: its centre
// ↳ is a Delaunay circumcentre or
// a point on the hull boundary - check all 0(n) circumcentres, then
// ↳ the hull edges.
// The DELAUNAY GRAPH also contains the nearest-neighbour graph and
// ↳ the Gabriel graph, so any
// "for each point, its closest other point" question is answered by
// ↳ scanning its Delaunay edges.
// FLIP ALGORITHM (any triangulation -> Delaunay by flipping illegal
// ↳ edges) is  $O(n^2)$  and simple,
// but you already have this; the paraboloid lift  $(x, y, x^2+y^2) +$ 
// ↳ LOWER 3D hull is the other
// derivation and is the one to remember when you need the connection
// ↳ to convexity.
```


11.1.14 Voronoi

VORONOI DIAGRAM. The cell of site i is the set of points closer to site i than to any other

```
// Needs: Core.cpp, HalfPlaneIntersection.cpp (Delaunay.cpp for the
// fast variant).
// site; it is a convex polygon (unbounded for sites on the convex
// hull, so everything here is
// clipped to a box of half-width BOX - make BOX comfortably larger
// than any coordinate you care
// about, and remember that a vertex of the clipped cell may be an
// artefact of the box).
// The cell is just the intersection of the half-planes "closer to i
// than to j" over all  $j \neq i$ ,
// and each of those is the side of the perpendicular bisector of  $s_i$ 
//  $s_j$  containing  $s_i$ .
// WHEN: nearest-site queries with a planar point location structure
// on top; largest empty circle;
// "which post office serves this house"; any facility-location or
// growth/flood simulation where
// regions expand at equal speed. If you only need nearest
// neighbours, a KD-tree (09) is shorter.
// COMPLEXITY:  $O(n^2 \log n)$  for all  $n$  cells as written ( $O(n \log n)$ 
// per cell). That is fine up to
//  $n$  around 2000. For  $O(n \log n)$  total, build the DELAUNAY dual
// instead - see the tail.
// DEGENERATE: duplicate sites give two identical cells that are both
// empty ({} is returned);
// dedupe first.  $n == 1$  gives the whole box. Collinear sites give
// slab cells, handled correctly.
// FLOATING POINT: the bisectors are exact only if you keep sites
// integral, but halfPlaneInter is
// an eps algorithm, so the cell vertices carry the usual  $1e-9$ 
// relative error. Four or more
// cocircular sites make a Voronoi vertex of degree  $> 3$  and the cell
// may pick up a zero-length
// edge; strip edges shorter than eps if you are counting them.
vector<Pd> voronoiCell(const vector<Pd>& s, int i, ld BOX = 1e9) {
    vector<HP> h;
    for (int j = 0; j < sz(s); j++) if (j != i) {
        Pd m = (s[i] + s[j]) / 2; // the
        // perpendicular bisector, directed so
        h.pb(HP(m, m + (s[j] - s[i]).perp())); // that the
        // LEFT side contains s[i]
    }
    return halfPlaneInter(h, BOX);
}

vector<vector<Pd>> voronoi(const vector<Pd>& s, ld BOX = 1e9) {
    vector<vector<Pd>> c;
    for (int i = 0; i < sz(s); i++) c.pb(voronoiCell(s, i, BOX));
    return c;
}

//  $O(n \log n)$  VIA THE DELAUNAY DUAL (13 - Delaunay.cpp). One Voronoi
// vertex per Delaunay triangle,
// namely its CIRCUMCENTRE; one Voronoi edge per Delaunay edge,
// joining the circumcentres of the
// two triangles that share it. Recipe: run delaunay(p); for every
// triangle t store
// circumcenter(p[t[0]], p[t[1]], p[t[2]]); for each site collect the
// circumcentres of the
// triangles incident to it and sort them by angle around the site -
// that is its cell, in CCW
// order. A site on the CONVEX HULL has an unbounded cell: its two
// hull edges contribute rays
// along their outward perpendicular bisectors, so clip them to the
// box by hand.
// NEAREST SITE for a query point = the cell containing it: build the
// diagram, then run planar
// point location (15). For a handful of queries just scan all sites.
// LARGEST EMPTY CIRCLE with centre inside the hull: its centre is a
// Voronoi vertex (= a Delaunay
// circumcentre) or a point where a Voronoi edge crosses the hull
// boundary. Check both families.
// DELAUNAY EDGE <=> VORONOI EDGE, and the MST / nearest-neighbour
// consequences, are all in 13.
// WEIGHTED (POWER / LAGUERRE) DIAGRAM: replace the bisector by the
// RADICAL AXIS of the two
// circles (04 - Circles.cpp) - the cells stay convex polygons, so
// this exact code works with the
// bisector swapped out. The MULTIPLICATIVELY weighted (Apollonius)
// diagram has circular arcs for
// boundaries and none of this applies.
```

11.1.15 PointLocation

PLANAR POINT LOCATION, offline, by a left-to-right sweep. Input: a set of segments that are

```
// Needs: Core.cpp.
// pairwise NON-CROSSING (they may share endpoints) - i.e. the edges
// of a planar subdivision - and
// a list of query points. Output, for each query, the index of the
// segment DIRECTLY BELOW it, or
// -1 if the query is below everything. That single answer is the
// whole primitive: label every
// segment with the id of the face lying immediately ABOVE it and the
// query's face falls out.
// WHEN: "n disjoint polygons, q points, which polygon contains each"
// with  $n$  and  $q$  both  $1e5$ , so
// the  $O(nq)$  scan is dead. Also "which cell of an arrangement", and
// nearest-site queries once you
// have built the Voronoi diagram (14). For ONE convex polygon use
// inConvex (03) instead; for a
// handful of polygons just call inPoly (03) on each.
// COMPLEXITY:  $O((n + q) \log n)$  time,  $O(n + q)$  memory. Everything is
// offline - all queries must be
// known up front. (Online needs a persistent segment tree over the
// vertical slabs, which is
//  $O(n \log n)$  memory and rarely worth typing.)
// DEGENERATE: NO VERTICAL SEGMENTS - the comparator below assumes
// every segment points strictly
// rightwards. If you have them, shear the plane with  $(x, y) \rightarrow (x *
// (2C + 1) + y, y)$ , which is
// exact on integers and makes all  $x$  distinct, but squares your
// coordinates (use __int128 cross).
// A query exactly ON a segment reports that segment (endpoints
// included), because segments
// starting at  $x$  are inserted before the queries at  $x$  and segments
// ending at  $x$  are removed after.
// Zero-length segments must be filtered out first.
// EXACT on P<ll> with |coord|  $\leq 1e9$ ; the comparator is pure
// orientation signs, no division.
vector<int> locateBelow(vector<pair<Pi, Pi>> s, const vector<Pi>& qs)
{
    int n = sz(s), m = sz(qs);
    for (auto& e : s) if (e.second < e.first) swap(e.first, e.second);
    s.pb(Pi(), Pi()); // slot n:
    // the current query, as a point
    // Two non-crossing segments are comparable wherever their
    // x-ranges overlap: ask which side of
    // a's line BOTH endpoints of b fall on, and if that is
    // inconclusive (shared endpoint) ask the
    // mirror question. Only ever consulted while both are in the
    // sweep set, so this is a valid
    // strict weak ordering at every instant even though it changes
    // as the sweep advances.
    auto below = [&](int i, int j) {
        auto& [a, b] = s[i]; auto& [c, d] = s[j];
        int v = sgn(a.cross(b, c)) + sgn(a.cross(b, d));
        if (v) return v > 0;
        return sgn(c.cross(d, a)) + sgn(c.cross(d, b)) < 0;
    };
    set<int, decltype(below)> t(below);
    vector<int> add(n), del(n), qi(m), res(m, -1);
    iota(all(add), 0), iota(all(del), 0), iota(all(qi), 0);
    sort(all(add), [&](int i, int j) { return s[i].first.x <
    s[j].first.x; });
    sort(all(del), [&](int i, int j) { return s[i].second.x <
    s[j].second.x; });
    sort(all(qi), [&](int i, int j) { return qs[i].x < qs[j].x; });
    int ia = 0, id = 0, iq = 0;
    while (ia < n || id < n || iq < m) { // sweep
        // EVERY event x, not just query x:
        ll x = LLONG_MAX; // the set
        // order is only valid while all
        if (ia < n) x = min(x, s[add[ia]].first.x); // members
        // still span the sweep line
        if (id < n) x = min(x, s[del[id]].second.x);
        if (iq < m) x = min(x, qs[iq].x);
        while (ia < n && s[add[ia]].first.x == x)
            t.insert(add[ia++]); // open,
        while (id < n && s[del[id]].x == x) {
            // then answer,
            int k = qi[iq++];
            s[n] = {qs[k], qs[k]}; // a
            // zero-length segment = the query point
            auto it = t.lower_bound(n);
            if (it != t.begin()) res[k] = *prev(it);
        }
        while (id < n && s[del[id]].second.x == x)
            t.erase(del[id++]); // then close
    }
}
```



```

    return res;
}
// ATTACHING FACES. Build the subdivision as half-edges: every
//   ↳ segment carries the face id above
// it and the face id below it (the unbounded face is -1). Then:
// f = locateBelow(...)[k]; answer = f < 0 ? -1 : faceAbove[f];
// and the query is ON the boundary iff qs[k].cross(s[f].first,
//   ↳ s[f].second) == 0.
// For DISJOINT SIMPLE POLYGONS the labelling is free: walk each
//   ↳ polygon in CCW order and give
// every edge whose direction points LEFT (a.x > b.x) the polygon's
//   ↳ id as its "above" face, and
// every rightward edge the id of whatever the polygon sits in (-1
//   ↳ for a top-level polygon).
// ARRANGEMENT OF LINES: n lines cut the plane into 1 + n + C(n,2)
//   ↳ faces; build the arrangement by
// sorting the intersections along each line, then run this on the
//   ↳ resulting O(n^2) segments.
// A different, shorter answer to "point in one of many disjoint
//   ↳ polygons": if the polygons are
// convex and you only need containment, sort by x-range and binary
//   ↳ search, or build the union
// hull. Reach for point location only when the subdivision is
//   ↳ genuinely a subdivision.

```

11.1.16 DynamicHull

DYNAMIC CONVEX HULL - insert points ONLINE, ask "is this point inside the hull so far" in

```

// Needs: Core.cpp.
// O(log n), amortised O(log n) per insert. The structure below is
//   ↳ the UPPER hull only, stored as
// a map x -> max y; run a second copy fed the negated points for the
//   ↳ lower hull and you have the
// full hull. Storing one y per x is what makes this short: the upper
//   ↳ chain has strictly
// increasing x by definition, and if two points share an x only the
//   ↳ higher one can be on it.
// WHEN: points arrive one at a time and you must answer containment
//   ↳ / support / area queries
// between insertions; "add point, print hull area"; maximising a
//   ↳ linear function over a growing
// set (that special case is CHT / Li Chao, 08 - DP Optimizations,
//   ↳ and is simpler - use it if the
// query is always a dot product). Deletions are NOT supported; for
//   ↳ those, use offline divide and
// conquer over time, or Kinetic / segment-tree-on-time rebuilding.
// COMPLEXITY: O(log n) amortised insert (each point is erased at
//   ↳ most once), O(log n) query,
// O(n) memory.
// DEGENERATE: COLLINEAR points are dropped (the chain keeps only
//   ↳ strict turns), so a set of
// collinear points collapses to its two extremes and under() still
//   ↳ answers correctly. Inserting a
// point already inside is a no-op. An empty structure answers false
//   ↳ to everything.
// EXACT on integers: the only arithmetic is a cross product, so
//   ↳ |coord| <= 1e9 keeps it in ll.
struct UpHull { // the upper
    ↳ chain, left to right
    map<ll, ll> h;
    typedef map<ll, ll>::iterator it_t;
    Pi at(it_t i) { return Pi(i->fi, i->se); }
    bool under(ll x, ll y) { // is (x, y)
        ↳ on or below the chain?
        auto it = h.lower_bound(x);
        if (it == h.end() || (it == h.begin() && it->fi != x)) return
        ↳ false; // outside the span
        if (it->fi == x) return y <= it->se;
        return at(prev(it)).cross(at(it), Pi(x, y)) <= 0;
        ↳ // not strictly left
    }
    bool bad(it_t it) { // does *it
        ↳ sit on or below its neighbours?
        if (it == h.begin() || next(it) == h.end()) return false;
        return at(prev(it)).cross(at(next(it)), at(it)) <= 0;
    }
    void add(ll x, ll y) {
        if (under(x, y)) return;
        h[x] = y;
        auto it = h.find(x);
        while (next(it) != h.end() && bad(next(it)))
        ↳ h.erase(next(it));
        while (it != h.begin() && bad(prev(it))) h.erase(prev(it));
    }
};
// FULL HULL: UpHull up, dn; insert p as up.add(p.x, p.y) and
//   ↳ dn.add(-p.x, -p.y);
// inside(p) = up.under(p.x, p.y) && dn.under(-p.x, -p.y).
// EXTREME POINT in direction d, O(log n): binary search the upper
//   ↳ chain for the first edge whose
// direction turns from d-positive to d-negative, i.e. the last
//   ↳ vertex with
// dot(next - cur, d) > 0. With a map you have to walk, so keep the
//   ↳ chain in a
// vector-backed balanced BST (or an ordered set with order_of_key)
//   ↳ if you need this a lot.
// AREA UNDER INSERTIONS, the usual companion query: keep ll s2 = sum
//   ↳ of cross(p_i, p_{i+1}) over
// consecutive chain vertices. Every erase of a middle vertex b
//   ↳ between a and c does
// s2 -= cross(a,b) + cross(b,c) - cross(a,c);
// and every insert of b between a and c does the same with a plus
//   ↳ sign; erases and inserts at an
// end touch only the single adjacent term. The hull's area is |s2_up
//   ↳ + s2_down| / 2 with the
// lower chain's points negated, since negating both coordinates
//   ↳ leaves every cross product alone.
// OFFLINE ALTERNATIVE, almost always easier: if you know all the
//   ↳ points up front and the queries
// are interleaved, sort by time and rebuild the hull in O(n log n)
//   ↳ once per sqrt-block, or run
// divide and conquer on the time axis. Only type this file when the

```

↪ input is genuinely online.

11.1.17 ShamosHoeY

SHAMOS-HOEY: do ANY TWO of n segments intersect? $O(n \log n)$, and it returns the witness pair.

```
// Needs: Core.cpp, Lines.cpp (segCross).
// The sweep keeps the active segments in a set ordered by their y at
  ↪ the sweep line. Two segments
// can only be the FIRST crossing pair if they are adjacent in that
  ↪ order immediately before they
// cross, so it is enough to test a segment against its two
  ↪ neighbours when it is inserted, and to
// test the two segments that become neighbours when one is removed.
// WHEN: "is this polygon simple", "do these n roads/walls cross",
  ↪ "is this a planar graph
// drawing" - anything where you only need YES/NO or one example,
  ↪ with n up to 1e5 so the  $O(n^2)$ 
// double loop (05 - Sweep.cpp, countSegInter) is out. isSimple (11 -
  ↪ PolygonOps.cpp) is the
// polygon-shaped special case and is  $O(n^2)$ ; replace it with this
  ↪ when n is large.
// COMPLEXITY:  $O(n \log n)$  time,  $O(n)$  memory.
// DEGENERATE: touching at a shared endpoint COUNTS as an
  ↪ intersection (segCross is inclusive), so
// for "is this polygon simple" you must skip the adjacent pairs
  ↪ yourself - easiest is to shrink
// every polygon edge by a hair, or to special-case the pair (i, i+1
  ↪ mod n) in the answer.
// Overlapping collinear segments are reported. Zero-length segments
  ↪ are reported against anything
// they touch. Vertical segments are fine here (unlike point
  ↪ location).
// The ANSWER (which pair) is exact - segCross is pure integer
  ↪ orientation. Only the SET ORDER
// uses floating point, and only to decide adjacency, so an eps slip
  ↪ can at worst make the sweep
// miss a crossing whose two segments are within  $1e-9$  of each other
  ↪ in y; with |coord|  $\leq 1e9$  and
// ld that does not happen for integer input.
struct SSeg {
    Pi a, b; int id; // normalised
    ↪ so that a.x  $\leq$  b.x
    ld yAt(ld x) const {
        return a.x == b.x ? (ld)a.y : a.y + (ld)(b.y - a.y) * (x -
        ↪ a.x) / (b.x - a.x);
    }
    bool operator<(const SSeg& o) const { // compare
        ↪ where both are already alive
        ld x = max((ld)a.x, (ld)o.a.x);
        return yAt(x) < o.yAt(x) - eps;
    }
};
pair<int, int> anyIntersect(const vector<pair<Pi, Pi>>& in) {
    int n = sz(in);
    vector<SSeg> s(n);
    vector<array<ll, 3>> ev; // {x, +1
    ↪ open / -1 close, id}
    for (int i = 0; i < n; i++) {
        s[i] = {in[i].fi, in[i].se, i};
        if (s[i].b < s[i].a) swap(s[i].a, s[i].b);
        ev.pb({s[i].a.x, 1, i}), ev.pb({s[i].b.x, -1, i});
    }
    sort(all(ev), [](const array<ll, 3>& p, const array<ll, 3>& q) {
        return p[0] != q[0] ? p[0] < q[0] : p[1] > q[1];
    });
    ↪ // open before close
    set<SSeg> t;
    vector<set<SSeg>::iterator> pos(n);
    typedef set<SSeg>::iterator sit;
    auto hit = [&](sit x, sit y) {
        return x != t.end() && y != t.end() && segCross(x->a, x->b,
        ↪ y->a, y->b);
    };
    for (auto& e : ev) {
        int i = (int)e[2];
        if (e[1] > 0) {
            sit nx = t.lower_bound(s[i]), pv = nx == t.begin() ?
            ↪ t.end() : prev(nx);
            sit me = t.insert(nx, s[i]); // insert
            ↪ first, then test both neighbours
            if (hit(nx, me)) return {nx->id, i};
            if (hit(pv, me)) return {pv->id, i};
            pos[i] = me;
        } else {
            sit nx = next(pos[i]), pv = pos[i] == t.begin() ? t.end()
            ↪ : prev(pos[i]);
            if (hit(nx, pv)) return {pv->id, nx->id}; // they
            ↪ become adjacent once i leaves
            t.erase(pos[i]);
        }
    }
}
```

```

    }
}
return {-1, -1};
}
// COUNT or REPORT ALL k intersections: BENTLEY-OTTMANN, O((n + k)
//   ↳ log n). Same sweep, but the
// event queue is a priority queue that also receives the
//   ↳ intersection points themselves; when two
// segments swap order at a crossing you re-test each against its new
//   ↳ neighbour. It is long,
// fiddly with degeneracies (three segments through one point), and
//   ↳ almost never what a problem
// wants - if you need a COUNT of crossing pairs of ARBITRARY
//   ↳ segments and n is 1e5, look for
// extra structure (all horizontal/vertical -> BIT sweep; all chords
//   ↳ of a circle -> inversions).
// IS A POLYGON SIMPLE, in O(n log n): shrink every edge towards its
//   ↳ midpoint by a factor
// (1 - 1e-9) so consecutive edges no longer share their endpoint,
//   ↳ then this returns {-1, -1}
// exactly for a simple polygon. Shrinking is the honest fix;
//   ↳ filtering "adjacent indices" out of
// the answer is not, because the sweep stops at the FIRST adjacent
//   ↳ pair it meets.

```

11.1.18 InscribedCircle

LARGEST INSCRIBED CIRCLE IN A CONVEX POLYGON - the CHEBYSHEV CENTRE. Returns {centre, radius}.

```

// Needs: Core.cpp, Polygon.cpp (area2).
// The trick is that f(p) = min over edges of the signed distance
//   ↳ from p to that edge's LINE is a
// minimum of affine functions, hence CONCAVE on the whole plane, and
//   ↳ it is exactly the distance
// to the boundary for p inside. So a nested ternary search on (x, y)
//   ↳ finds its maximum with no
// case analysis at all: no medial axis, no half-plane shrinking, no
//   ↳ binary search on r.
// WHEN: "place a disc of maximum radius inside this region", "the
//   ↳ point of a convex region
// farthest from its boundary", robot clearance, and 2D LP with an
//   ↳ objective that is a min of
// linear functions (this IS that LP - maximise r subject to n linear
//   ↳ constraints).
// COMPLEXITY: O(n * it^2) with it = 100 iterations per axis; about
//   ↳ 1e4 * n operations, so keep n
// below a few thousand. If n is large, run the same search on the
//   ↳ O(n) half-plane form or binary
// search r and test emptiness with halfPlaneInter (06) at O(n log n
//   ↳ log(1/eps)).
// DEGENERATE: the polygon must be CONVEX and given in order; the
//   ↳ code reverses a CW input for
// you. A degenerate (zero-area) polygon returns radius 0 at some
//   ↳ point of it. Fewer than 3
// vertices returns radius 0. For a NON-CONVEX polygon f is not
//   ↳ concave and this silently returns
// a local optimum - the correct answer there lives on the medial
//   ↳ axis and is genuinely hard.
// FLOATING POINT, always: the answer is irrational. 100 ternary
//   ↳ iterations over a 2e9-wide box
// leave the centre accurate to ~1e-9 * box, which is the best ld can
//   ↳ do anyway.
pair<Pd, ld> inscribedCircle(vector<Pd> v) {
    int n = sz(v);
    if (n < 3) return {n ? v[0] : Pd(), 0};
    if (area2(v) < 0) reverse(all(v)); // f must be
    // POSITIVE inside
    auto f = [&](ld x, ld y) { // distance
        // to the nearest edge LINE
        ld r = 1e18;
        for (int i = 0; i < n; i++)
            r = min(r, v[i].cross(v[(i + 1) % n], Pd(x, y)) / (v[(i +
        // 1) % n] - v[i]).dist());
        return r;
    };
    ld lo = 1e18, hi = -1e18, bo = 1e18, bi = -1e18;
    for (Pd& p : v) lo = min(lo, p.x), hi = max(hi, p.x), bo =
    // min(bo, p.y), bi = max(bi, p.y);
    auto best = [&](ld x) { // max over y
        // for a fixed x, still concave
        ld a = bo, b = bi;
        for (int it = 0; it < 100; it++) {
            ld m1 = a + (b - a) / 3, m2 = b - (b - a) / 3;
            f(x, m1) < f(x, m2) ? a = m1 : b = m2;
        }
        return (a + b) / 2;
    };
    for (int it = 0; it < 100; it++) {
        ld m1 = lo + (hi - lo) / 3, m2 = hi - (hi - lo) / 3;
        f(m1, best(m1)) < f(m2, best(m2)) ? lo = m1 : hi = m2;
    }
    ld x = (lo + hi) / 2, y = best(x);
    return {Pd(x, y), max((ld)0, f(x, y))};
}
// WHY THE NESTED SEARCH IS LEGAL: g(x) = max over y of f(x, y) is
//   ↳ the partial maximisation of a
// concave function, and that is concave in x. The same argument
//   ↳ licenses geoMedian (12) and every
// other nested ternary search - check concavity of the PARTIAL max,
//   ↳ not just of f.
// LP FORM, if you prefer it or need n large: maximise r subject to
//   ↳ cross(b_i - a_i, p - a_i) / |b_i - a_i| >= r for every edge,
// which is n constraints in three variables (x, y, r) - Seidel's
//   ↳ randomised LP solves it in O(n)
// expected. Equivalently, binary search r, push every edge line
//   ↳ inward by r, and ask whether the
// half-plane intersection is non-empty.
// LARGEST INSCRIBED CIRCLE CENTRED ON A GIVEN LINE: one ternary
//   ↳ search along the line.
// LARGEST INSCRIBED AXIS-ALIGNED RECTANGLE is a different and much
//   ↳ harder problem - it needs a
// sweep with a stack and is not a two-line variation of this. Do not
//   ↳ improvise it under time.

```

```
// MINIMUM ENCLOSING CIRCLE is mec (04 - Circles.cpp); minimum
    ↳ enclosing rectangle is minRectArea
// (03 - Polygon.cpp) by rotating calipers.
```

11.1.19 LineClip

TOTAL LENGTH OF A LINE INSIDE A POLYGON, including NON-CONVEX and self-intersecting ones, in

```
// Needs: Core.cpp.
// O(n log n). The line a -> b generally enters and leaves the region
    ↳ many times; this sums all of
// the pieces. Works by collecting the parameters t at which the line
    ↳ crosses each edge, tagging
// each crossing with the direction the boundary sweeps past it,
    ↳ sorting, and integrating the
// stretches where the running WINDING NUMBER is non-zero. cut() (03
    ↳ - Polygon.cpp) only handles
// the convex case and only gives you one side; this gives you the
    ↳ measure.
// WHEN: "how much of this laser / road / sight line is inside the
    ↳ region", area of a polygon
// swept by a moving line, and any integral over a region done by
    ↳ integrating over parallel lines
// (fix a direction, sweep the offset, and this is the inner
    ↳ integrand).
// COMPLEXITY: O(n log n) for the sort, O(n) memory.
// DEGENERATE: a vertex lying exactly ON the line is handled by
    ↳ treating a zero orientation as
// POSITIVE, which is the same as nudging the line infinitesimally
    ↳ toward its right; that is
// consistent, so a vertex merely touching the line contributes
    ↳ nothing and a vertex the boundary
// genuinely passes through contributes once. An EDGE lying along the
    ↳ line is skipped for the same
// reason - if you need boundary-along-the-line to count, add its
    ↳ length yourself.
// SELF-INTERSECTING input is fine and uses the non-zero winding rule
    ↳ (windingNumber, 11), so a
// doubly-wound region counts once, not twice. Orientation (CW or
    ↳ CCW) does not matter.
// The crossing TEST is exact on P<ll> (lcoord| <= 1e9); only the
    ↳ parameter t is floating point,
// so nearly-tangent edges can shift a boundary by ~1e-9 relative.
template <class P> ld lineInPoly(const vector<P>& v, P a, P b) {
    P d = b - a;
    int n = sz(v);
    vector<pair<ld, int>> ev;
    for (int i = 0; i < n; i++) {
        P p = v[i], q = v[(i + 1) % n];
        int s1 = sgn(d.cross(p - a)), s2 = sgn(d.cross(q - a));
        if (!s1) s1 = 1; // a point ON
        ↳ the line counts as LEFT
        if (!s2) s2 = 1;
        if (s1 == s2) continue;
        ev.pb({(ld)(p - a).cross(q - p) / d.cross(q - p), s1 > s2 ? 1
        ↳ : -1});
    }
    sort(all(ev));
    ld len = 0;
    int c = 0;
    for (int i = 0; i + 1 < sz(ev); i++)
        if ((c += ev[i].second)) len += ev[i + 1].fi - ev[i].fi;
    ↳ // inside <=> winding != 0
    return len * d.dist();
}
// SEGMENT instead of a line: same events, but clamp the accumulated
    ↳ interval to [0, 1] before
// adding, i.e. len += max((ld)0, min((ld)1, ev[i+1].fi) - max((ld)0,
    ↳ ev[i].fi)).
// RAY: clamp to [0, infinity) the same way.
// THE PIECES THEMSELVES, not just their total length: emit the
    ↳ interval [ev[i].fi, ev[i+1].fi]
// whenever the running count is non-zero; a + d * t gives the
    ↳ endpoints.
// AREA OF THE POLYGON, as a sanity check on this file: integrate
    ↳ lineInPoly over a family of
// parallel lines. That is also how you get "the area of the part of
    ↳ a polygon on one side of an
// arbitrary curve" when the curve is not a line and cut() does not
    ↳ apply.
// LINE x CONVEX POLYGON in O(log n): lineHull (11 - PolygonOps.cpp)
    ↳ gives the two crossed edges
// directly; there is exactly one interval and you do not need this
    ↳ file.
// LINE x CIRCLE is circleLine (04); LINE x DISC intersected with a
    ↳ polygon: clip against both.
```

11.1.20 MaxAreaPolygon

MAXIMUM-AREA POLYGON WITH GIVEN SIDE LENGTHS. Given n side lengths in any order, the polygon

```
// of largest area that has exactly those sides is CYCLIC - all its
    ↳ vertices lie on one circle
// (and the order of the sides does not matter, which is the
    ↳ surprising half of the theorem). So
// the whole problem is "find the circumradius R", and each side a_i
    ↳ subtends a central angle
// 2 * asin(a_i / (2R)). Two cases: if the centre is INSIDE, the
    ↳ angles sum to 2 * pi; if the
// longest side is long enough to push the centre OUTSIDE, its angle
    ↳ is subtracted instead and the
// signed sum is 0. Both are monotone in R, so binary search.
// WHEN: "arrange these n sticks into the polygon of largest area";
    ↳ the n = 3 case is Heron and
// the n = 4 case is Brahmagupta, and this is the general answer for
    ↳ both. Also the standing proof
// that a regular polygon maximises area among equilateral ones.
// COMPLEXITY: O(n * 100) - one binary search with 100 iterations,
    ↳ each O(n).
// DEGENERATE: no polygon exists at all unless the longest side is
    ↳ strictly shorter than the sum
// of the others (the polygon inequality); the routine returns -1 for
    ↳ that, and for n < 3. Sides
// of length 0 are harmless. All sides equal gives the regular n-gon,
    ↳ area n * r^2 * sin(2 pi / n) / 2
// with r = a / (2 sin(pi / n)) - use that to check the code.
// FLOATING POINT, necessarily: R is irrational. asin loses precision
    ↳ when a_i / (2R) is near 1,
// which is exactly the degenerate "longest side is a diameter" case,
    ↳ so clamp the argument.
ld maxAreaPolygon(vector<ld> a) {
    int n = sz(a);
    if (n < 3) return -1;
    int m = max_element(all(a)) - a.begin();
    ld s = 0;
    for (int i = 0; i < n; i++) if (i != m) s += a[i];
    if (a[m] >= s) return -1; // the
    ↳ polygon inequality fails
    auto ang = [&](ld R, int sg) { // sum of
        ↳ central angles, a[m] signed
        ld t = 0;
        for (int i = 0; i < n; i++) {
            ld u = 2 * asinl(min((ld)1, a[i] / (2 * R)));
            t += (i == m ? sg * u : u);
        }
        return t;
    };
    ld lo = a[m] / 2, hi = 1e18; // R >= a[m]
    ↳ / 2 always
    int sg = ang(lo, 1) > 2 * PI ? 1 : -1; // centre
    ↳ inside (+1) or outside (-1)
    ld tgt = sg > 0 ? 2 * PI : 0; // ang(.) is
    ↳ DECREASING in R for sg = +1,
    for (int it = 0; it < 200; it++) { // INCREASING
        ↳ for sg = -1
        ld mid = (lo + hi) / 2;
        ((ang(mid, sg) > tgt) == (sg > 0) ? lo : hi) = mid;
    }
    ld R = (lo + hi) / 2, area = 0;
    for (int i = 0; i < n; i++) {
        ld u = 2 * asinl(min((ld)1, a[i] / (2 * R)));
        area += (i == m ? sg : 1) * R * R * sinl(u) / 2;
    }
    return area;
}
// THE VERTICES, if the problem wants them and not just the area:
    ↳ walk the angles. Start at
// (R, 0), and after each side rotate by its central angle (negated
    ↳ for the long side when the
// centre is outside). p_0 = R * (cos 0, sin 0), theta_{k+1} =
    ↳ theta_k + u_k, p_k = R * (cos, sin).
// TRIANGLE (n = 3), closed form, no search: HERON with the stable
    ↳ sorted form,
// sort a >= b >= c, A = sqrt((a+b+c)(c-a-b)(c+a-b)(a+b-c))
    ↳ / 4.
// CYCLIC QUADRILATERAL (n = 4): BRAHMAGUPTA, A =
    ↳ sqrt((s-a)(s-b)(s-c)(s-d)) with s the
// semiperimeter - and by this theorem that IS the maximum over all
    ↳ quadrilaterals with those
// sides. General quadrilateral with diagonals p, q (BRETSCHNEIDER):
// A = sqrt(4 p^2 q^2 - (b^2 + d^2 - a^2 - c^2)^2) / 4.
// MAXIMUM AREA WITH GIVEN PERIMETER, no side constraints: the
    ↳ circle. Among n-gons: the regular
// one. This file is the constrained version of the same
    ↳ isoperimetric fact.
```

```
// A DIFFERENT PROBLEM often confused with this one: pick the
    ↳ triangle of maximum area from a set
// of given points - that is rotating calipers on the hull,  $O(n^2)$ ,
    ↳ and is not here.
```

11.1.21 DynamicHalfPlane

DYNAMIC HALF-PLANE INTERSECTION - a convex region you can CUT online, in $O(\log n)$ amortised per

```
// Needs: Core.cpp (angCmp).
// cut, while asking for its area or its extreme point in any
    ↳ direction between cuts. The static
// routine (06 - HalfPlaneIntersection.cpp) needs every constraint up
    ↳ front and re-sorts; this one
// does not, so it is what you want when the constraints arrive one
    ↳ at a time, or when you must
// report an answer after each one.
// REPRESENTATION: the region's boundary, as a multimap from EDGE
    ↳ DIRECTION to the vertex where
// that edge begins, in counter-clockwise angular order. That single
    ↳ choice is the whole trick -
// the edge a new cut attacks is found by one lower_bound on its
    ↳ direction, and every vertex is
// erased at most once, so the amortised cost is a lookup plus the
    ↳ erases you paid to insert.
// It starts as a huge box, so the region is always bounded; make B
    ↳ larger than any coordinate you
// care about and remember that a surviving box corner is an
    ↳ artefact, not an answer.
// WHEN: "after each of q constraints, print the area of the feasible
    ↳ region"; online 2D LP;
// incremental visibility / lit-region problems; kernel of a polygon
    ↳ built edge by edge.
// COMPLEXITY:  $O(\log n)$  amortised per cut,  $O(\log n)$  per extreme-point
    ↳ query,  $O(n)$  per area query
// (keep a running sum of cross products if you need area after every
    ↳ cut - update it in the same
// places the code erases and inserts).
// DEGENERATE: cutting with a line that misses the region entirely is
    ↳ a no-op; cutting everything
// away leaves the map EMPTY and every later cut returns immediately,
    ↳ so check empty() for
// infeasible. Duplicate directions are allowed (that is why it is a
    ↳ multimap). A cut exactly
// along an existing edge removes the now-redundant vertex.
// FLOATING POINT: directions and vertices are ld and compared with
    ↳ the usual eps through sgn.
// Feed it integer-valued Pd if you can; a cut that is within eps of
    ↳ tangent may keep or drop a
// zero-length edge, so filter edges shorter than eps before counting
    ↳ them.
struct DirLess { bool operator()(Pd a, Pd b) const { return
    ↳ angCmp<ld>(a, b); } };
struct HPSet {
    typedef multimap<Pd, Pd, DirLess> M;
    typedef M::iterator it_t;
    M m; // edge
    ↳ direction -> start vertex of edge
    HPSet(ld B = 1e9) {
        m.insert({Pd(1, 0), Pd(-B, -B)}), m.insert({Pd(0, 1), Pd(B,
    ↳ -B)});
        m.insert({Pd(-1, 0), Pd(B, B)}), m.insert({Pd(0, -1), Pd(-B,
    ↳ B)});
    }
    it_t nx(it_t i) { return next(i) == m.end() ? m.begin() :
    ↳ next(i); }
    it_t pv(it_t i) { return i == m.begin() ? prev(m.end()) :
    ↳ prev(i); }
    it_t fix(it_t i) { return i == m.end() ? m.begin() : i; }
    void cut(Pd a, Pd b) { // keep only
    ↳ the part LEFT of a -> b
        if (m.empty()) return;
        int old = sz(m);
        Pd d = b - a;
        auto ev = [&](it_t i) { return sgn(a.cross(b, i->second)); };
        auto ix = [&](it_t i) { // where i's
    ↳ edge meets the line a -> b
            return i->second + i->first * (d.cross(a - i->second) /
    ↳ d.cross(i->first));
        };
        it_t it = fix(m.lower_bound(d)); // the vertex
    ↳ farthest RIGHT of d
        if (ev(it) >= 0) return; // it
    ↳ survives, so everything does
        while (sz(m) && ev(pv(it)) < 0) m.erase(pv(it));
        while (sz(m) && ev(nx(it)) < 0) it = fix(m.erase(it));
        if (m.empty()) return;
        if (ev(nx(it)) > 0) it->second = ix(it); // pull this
    ↳ vertex onto the cutting line
        else it = fix(m.erase(it));
        if (old <= 2) return;
        it = pv(it);
        m.insert(it, {d, ix(it)}); // the new
```

```

    ↪ edge starts where we re-entered
        if (ev(it) == 0) m.erase(it); // the old
    ↪ vertex is now redundant
    }
    Pd extreme(Pd dir) { // vertex
    ↪ maximising dot(v, dir), O(log n)
        return fix(m.lower_bound(dir.perp()))->second;
    }
    ld area() {
        if (sz(m) <= 2) return 0;
        ld s = 0;
        for (it_t i = m.begin(); i != m.end(); ++i) s +=
    ↪ i->second.cross(nx(i)->second);
        return fabs(s) / 2;
    }
    vector<Pd> poly() { // the region
    ↪ as a CCW vertex list
        vector<Pd> v;
        for (auto& e : m) v.pb(e.second);
        return v;
    }
};
// USAGE: HPSet h; then h.cut(p, q) for each constraint "left of the
    ↪ directed line p -> q", which
// is the same convention as HP in 06. To cut with the half-plane {x
    ↪ : n.dot(x) <= c}, take a
// point p on the boundary line and cut(p, p + n.perp()).
// AREA AFTER EVERY CUT in O(log n): keep ld s = sum of cross(v_i,
    ↪ v_{i+1}). Every erase of a
// vertex b between a and c does s -= cross(a,b) + cross(b,c) -
    ↪ cross(a,c), every insert adds it,
// and moving a vertex is an erase plus an insert. Same bookkeeping
    ↪ as 16 - DynamicHull.cpp.
// 2D LP ONLINE: after each constraint the optimum of a linear
    ↪ objective c is extreme(c) - one
// lower_bound, no scan. That is the reason to prefer this over
    ↪ rebuilding with 06.
// DELETIONS are not supported. If constraints are added AND removed,
    ↪ go offline: segment tree on
// time, and rebuild with the static routine in each node.

```

11.1.22 OnionDecomposition

ONION DECOMPOSITION (convex layers) in $O(n \log^2 n)$: strip the convex hull, then the hull of

```

// Needs: Core.cpp.
// what remains, and so on, and report which layer each point lands
    ↪ in. The naive loop in
// 05 - Sweep.cpp rebuilds a hull per layer and is  $O(n^2 \log n)$ ,
    ↪ which dies at  $n = 1e5$  on the
// worst input (points on nested circles give  $\Theta(n^{2/3})$  layers,
    ↪ and a spiral gives  $\Theta(n)$ ).
// THE STRUCTURE: a DECREMENTAL hull - it only supports DELETE, which
    ↪ is exactly what peeling
// needs. A segment tree over the points sorted by y stores, in every
    ↪ node, the two endpoints of
// the BRIDGE joining the hulls of its two children; pull() finds
    ↪ that bridge by the classic
// simultaneous descent in  $O(\log n)$ , so a delete costs  $O(\log^2 n)$  and
    ↪ reading off the current hull
// costs  $O(\text{hull size})$ . Two copies handle the two sides: the LEFT hull
    ↪ of the points, and the left
// hull of the negated-and-reversed points, which is the right hull.
// WHEN: "how many peels until the set is empty", "which layer is
    ↪ point i on", k-th layer queries,
// and the classic "repeatedly remove the hull" simulation. Also the
    ↪ fastest route to the DEPTH of
// a point set (the number of layers), which some
    ↪ rotating-calipers-free arguments need.
// COMPLEXITY:  $O(n \log^2 n)$  time,  $O(n)$  memory. Beats the naive
    ↪ version from about  $n = 3000$ .
// DEGENERATE: ALL POINTS MUST BE DISTINCT - dedupe first. COLLINEAR
    ↪ points on a hull edge are
// KEPT on that layer (the bridge search uses strict turns, so a
    ↪ point in the middle of an edge is
// still reported), which differs from hull() in 03 - Polygon.cpp; if
    ↪ you need them dropped,
// filter each layer afterwards.  $n \leq 2$  puts everything on layer 1.
// EXACT on P<ll>: cross fits in ll for  $|\text{coord}| \leq 1e9$ , and the one
    ↪ product of a cross with a
// coordinate inside pull() is widened to __int128, so nothing here
    ↪ overflows at that budget.
struct LeftHull { // points
    ↪ must be sorted by (y, x)
    struct Nd { int bl, br, L, R, lc, rc; };
    vector<Pi> ps;
    vector<Nd> t;
    int root = 0;
    bool leaf(int w) { return t[w].lc < 0 && t[w].rc < 0; }
    void pull(int w) { // find the
    ↪ bridge between the two children
        int l = t[w].lc, r = t[w].rc;
        ll sy = ps[t[r].L].y;
        while (!leaf(l) || !leaf(r)) {
            int a = t[l].bl, b = t[l].br, c = t[r].bl, d = t[r].br;
            if (a != b && ps[a].cross(ps[b], ps[c]) > 0) l = t[l].lc;
            else if (c != d && ps[b].cross(ps[c], ps[d]) > 0) r =
    ↪ t[r].rc;
            else if (a == b) r = t[r].lc;
            else if (c == d) l = t[l].rc;
            else {
                ll s1 = ps[a].cross(ps[b], ps[c]), s2 =
    ↪ ps[b].cross(ps[a], ps[d]);
                if (s1 + s2 == 0 || (__int128)s1 * ps[d].y +
    ↪ (__int128)s2 * ps[c].y <
                    (__int128)sy * (s1 + s2)) l =
    ↪ t[l].rc;
                else r = t[r].lc;
            }
        }
        t[w].bl = t[l].L, t[w].br = t[r].L;
    }
    void build(int w, int L, int R) {
        t[w].L = L, t[w].R = R;
        if (R - L == 1) { t[w].lc = t[w].rc = -1, t[w].bl = t[w].br =
    ↪ L; return; }
        int M = (L + R) / 2;
        t[w].lc = w + 1, t[w].rc = w + 2 * (M - L);
        build(t[w].lc, L, M), build(t[w].rc, M, R), pull(w);
    }
    int erase(int w, int L, int R) {
        if (R <= t[w].L || L >= t[w].R) return w;
        if (L <= t[w].L && R >= t[w].R) return -1;
        t[w].lc = erase(t[w].lc, L, R), t[w].rc = erase(t[w].rc, L,
    ↪ R);
        if (t[w].lc < 0) return t[w].rc;
        if (t[w].rc < 0) return t[w].lc;
        return pull(w), w;
    }
}

```



```

void walk(int w, int l, int r, vector<int>& res) {
    if (leaf(w)) res.pb(t[w].L);
    else if (r <= t[w].bl) walk(t[w].lc, l, r, res);
    else if (l >= t[w].br) walk(t[w].rc, l, r, res);
    else walk(t[w].lc, l, t[w].bl, res), walk(t[w].rc, t[w].br,
    ↪ r, res);
}
LeftHull(const vector<Pi>& p) : ps(p), t(max<size_t>(2, p.size()
    ↪ * 2)) { build(0, 0, sz(p)); }
void erase(int i) { root = erase(root, i, i + 1); }
vector<int> hull() {
    vector<int> res;
    if (root >= 0) walk(root, 0, sz(ps) - 1, res);
    return res;
}
};
// layer[i] = 1-based convex-layer index of the ORIGINAL point i. All
    ↪ points must be distinct.
vector<int> onionLayers(const vector<Pi>& p) {
    int n = sz(p);
    if (!n) return {};
    vector<int> ord(n), lay(n), res(n);
    iota(all(ord), 0);
    sort(all(ord), [&](int i, int j) { return tie(p[i].y, p[i].x) <
    ↪ tie(p[j].y, p[j].x); });
    vector<Pi> a(n), b(n);
    for (int i = 0; i < n; i++) a[i] = p[ord[i]];
    for (int i = 0; i < n; i++) b[i] = Pi(-a[n - 1 - i].x, -a[n - 1 -
    ↪ i].y); // the right hull
    LeftHull L(a), R(b);
    for (int id = 1, cnt = 0; cnt < n; id++) {
        set<int> h;
        for (int i : L.hull()) h.insert(i);
        for (int i : R.hull()) h.insert(n - 1 - i);
        for (int i : h) cnt++, lay[i] = id, L.erase(i), R.erase(n - 1
    ↪ i);
    }
    for (int i = 0; i < n; i++) res[ord[i]] = lay[i];
    return res;
}
// The same structure answers "hull of the set after these deletions"
    ↪ at any moment, so it also
// does OFFLINE DELETIONS on a hull: process the deletes in order and
    ↪ read hull() in between.
// INSERTIONS instead of deletions are 16 - DynamicHull.cpp; you
    ↪ cannot have both here.
// LAYER COUNT bounds worth knowing: n points in convex position give
    ↪ 1 layer, a spiral gives n/3,
// and a uniform random square gives Theta(n^(2/3)) - so "peel until
    ↪ empty" is NOT O(sqrt n).

```

11.2 Geometry 3D

11.2.1 Geometry3D

3D point / vector, planes, and the few 3D things that actually appear.

```

template <class T>
struct P3 {
    typedef P3 Pt; T x = 0, y = 0, z = 0;
    P3(T x = 0, T y = 0, T z = 0) : x(x), y(y), z(z) {}
    Pt operator+(Pt p) const { return {x + p.x, y + p.y, z + p.z}; }
    Pt operator-(Pt p) const { return {x - p.x, y - p.y, z - p.z}; }
    Pt operator*(T d) const { return {x * d, y * d, z * d}; }
    Pt operator/(T d) const { return {x / d, y / d, z / d}; }
    T dot(Pt p) const { return x * p.x + y * p.y + z * p.z; }
    Pt cross(Pt p) const { return {y * p.z - z * p.y, z * p.x - x *
    ↪ p.z, x * p.y - y * p.x}; }
    T dist2() const { return x * x + y * y + z * z; }
    ld dist() const { return sqrtl((ld)dist2()); }
    Pt unit() const { return *this / dist(); }
    ld phi() const { return atan2l(y, x); } //
    ↪ longitude, (-pi, pi]
    ld theta() const { return atan2l(sqrtl((ld)(x * x + y * y)), z);
    ↪ } // colatitude, [0, pi]
    Pt rot(ld a, Pt ax) const { //
    ↪ Rodrigues, ccw around ax
        Pt u = ax.unit(); ld s = sinl(a), c = cosl(a);
        return u * dot(u) * (1 - c) + (*this) * c + u.cross(*this) *
    ↪ s;
    }
    bool operator<(Pt p) const { return tie(x, y, z) < tie(p.x, p.y,
    ↪ p.z); }
    bool operator==(Pt p) const { return tie(x, y, z) == tie(p.x,
    ↪ p.y, p.z); }
    friend istream& operator>>(istream& i, Pt& p) { return i >> p.x
    ↪ >> p.y >> p.z; }
    friend ostream& operator<<(ostream& o, Pt p) { return o << p.x <<
    ↪ ' ' << p.y << ' ' << p.z; }
};
typedef P3<ll> Pi3;
typedef P3<ld> Pd3;

// PLANE n.dot(X) = d. Build from 3 points, or from a normal and a
    ↪ point.
template <class P> struct Plane {
    P n; ld d;
    Plane(P n = P(), ld d = 0) : n(n), d(d) {}
    Plane(P a, P b, P c) : n((b - a).cross(c - a)), d(n.dot(a)) {}
    ld side(P p) const { return n.dot(p) - d; } //
    ↪ sign = which side
    ld dist(P p) const { return fabsl(side(p)) / n.dist(); }
    P proj(P p) const { return p - n * (side(p) / n.dist2()); }
    P refl(P p) const { return p - n * (2 * side(p) / n.dist2()); }
};
// LINE a->b vs plane. {0 parallel-disjoint, 1 unique, -1 line lies
    ↪ in plane}
template <class P> pair<int, P> planeLine(const Plane<P>& pl, P a, P
    ↪ b) {
    ld da = pl.side(a), db = pl.side(b);
    if (fabsl(da - db) < eps) return {fabsl(da) < eps ? -1 : 0, P()};
    return {1, (b * da - a * db) / (da - db)};
}
// PLANE x PLANE -> a line, given as {point on it, direction}. dir==0
    ↪ means parallel.
template <class P> pair<P, P> planePlane(const Plane<P>& A, const
    ↪ Plane<P>& B) {
    P dir = A.n.cross(B.n);
    if (dir.dist2() < eps) return {P(), P()};
    P p = (B.n * A.d - A.n * B.d).cross(dir) / dir.dist2();
    return {p, dir};
}

// VOLUME of the tetrahedron abcd, signed. The x6 form is exact on
    ↪ P3<ll> only up to
// |coordinate| ~ 5.7e5, since |vol6| <= 48 * C^3; above that make
    ↪ the return type __int128.
template <class P> auto vol6(P a, P b, P c, P d) { return (b -
    ↪ a).cross(c - a).dot(d - a); }
template <class P> ld tetraVolume(P a, P b, P c, P d) { return
    ↪ fabsl((ld)vol6(a, b, c, d)) / 6; }

// SPHERE x LINE: points where line a->b meets the sphere (o, r).
template <class P> vector<P> sphereLine(P o, ld r, P a, P b) {
    P d = b - a, f = a - o;
    ld A = d.dist2(), B = 2 * f.dot(d), C = f.dist2() - r * r, D = B
    ↪ * B - 4 * A * C;
    if (D < -eps) return {};
    D = sqrtl(max((ld)0, D));
    if (D < eps) return {a + d * (-B / (2 * A))};
    return {a + d * ((-B - D) / (2 * A)), a + d * ((-B + D) / (2 *

```

```

    ↪ A));
}

// GREAT-CIRCLE distance on a sphere of radius R, inputs in DEGREES
    ↪ (haversine).
ld greatCircle(ld lat1, ld lon1, ld lat2, ld lon2, ld R) {
    auto rad = [](ld d) { return d * PI / 180; };
    ld dl = rad(lat2 - lat1) / 2, dg = rad(lon2 - lon1) / 2;
    ld h = sinl(dl) * sinl(dl) + cosl(rad(lat1)) * cosl(rad(lat2)) *
    ↪ sinl(dg) * sinl(dg);
    return 2 * R * asinl(sqrtl(min((ld)1, h)));
}

```

11.2.2 Lines3D

LINES, SEGMENTS AND TRIANGLES IN 3D. L3 is the line $\{o + t*d\}$ through two points, carrying

```

// Needs: Geometry3D.cpp, Core.cpp (sgn).
// distance / projection / reflection of a point. On top of it the
    ↪ two things everyone re-derives
// under pressure: the SMALLEST DISTANCE BETWEEN TWO LINES (project
    ↪ the gap onto the common normal
// d1 x d2) and the PAIR OF CLOSEST POINTS realising it, one on each
    ↪ line. Then point-segment,
// segment-segment and point-triangle distance, and SEGMENT x
    ↪ TRIANGLE intersection (does segment
// d-e cross triangle abc, and where). Everything is O(1) time and
    ↪ memory.
// You reach for this the moment the input stops being planar: two 3D
    ↪ lines generically MISS, so
// lineInter is meaningless and "distance between the lines" replaces
    ↪ it; and mesh problems
// (visibility, is this ray blocked, closest approach of two moving
    ↪ points) are exactly segTri and
// segSegDist3.
// DEGENERATE: parallel lines fall back to point-line, and
    ↪ lineClosest then returns {a.o, its foot
// on b} - one of infinitely many optimal pairs. A zero-length
    ↪ segment degrades to its point. A
// collinear triangle makes triDist3 return the min over its three
    ↪ edges and makes segTri report
// -1. segTri with d == e reports 0 (or -1 if that point is in the
    ↪ plane of abc), never 1.
// EXACTNESS: the predicates are sgn(vol6), so segTri is EXACT on
    ↪ P3<ll> up to |coord| <= 5.7e5
// (the vol6 budget of Geometry3D.cpp; wrap vol6 in __int128 for
    ↪ 1e9). triDist3's inside test and
// L3::dist2 are DEGREE-4 forms (((v-u) x (p-u)).n and |d x
    ↪ (p-o)|^2), so they hold in ll only for
// |coord| <= 1.5e4 - again __int128 lifts that to 9e8. Anything
    ↪ returning a distance or a point
// divides, so it comes back in ld whatever you passed in; and note
// proj / refl / lineClosest silently TRUNCATE on P3<ll>. The eps
    ↪ tests below are absolute, so
// scale your input to O(1)..O(1e6) if you care about near-degenerate
    ↪ cases.
template <class P> struct L3 {
    P o, d; // {o +
    ↪ t*d}; d must be nonzero
    L3(P a = P(), P b = P()) : o(a), d(b - a) {} // point
    ↪ + DIRECTION: L3(o, o + dir)
    ld dist2(P p) const { return (ld)d.cross(p - o).dist2() /
    ↪ d.dist2(); }
    ld dist(P p) const { return sqrtl(dist2(p)); } // ratio
    ↪ of two exact integer forms
    P proj(P p) const { return o + d * ((ld)d.dot(p - o) /
    ↪ d.dist2()); }
    P refl(P p) const { return proj(p) * 2 - p; }
    bool cmpProj(P p, P q) const { return d.dot(p) < d.dot(q); } //
    ↪ sort points ALONG the line
};
// SMALLEST DISTANCE BETWEEN TWO LINES: the gap vector measured along
    ↪ the common normal.
template <class P> ld lineLineDist(L3<P> a, L3<P> b) {
    P n = a.d.cross(b.d);
    if (n == P()) return a.dist(b.o); //
    ↪ parallel: any point of b will do
    return fabsl((ld)(b.o - a.o).dot(n)) / n.dist();
}
// The pair of points realising it, {on a, on b}. The point on a is
    ↪ where a pierces the plane
// spanned by b and the common normal, so the "closest point on a"
    ↪ needs only one division.
template <class P> pair<P, P> lineClosest(L3<P> a, L3<P> b) {
    P n = a.d.cross(b.d);
    if (n == P()) return {a.o, b.proj(a.o)}; //
    ↪ parallel: infinitely many pairs
    P na = b.d.cross(n), nb = a.d.cross(n);
    return {a.o + a.d * ((ld)(b.o - a.o).dot(na) / a.d.dot(na)),
    ↪ b.o + b.d * ((ld)(a.o - b.o).dot(nb) / b.d.dot(nb))};
}
// POINT -> SEGMENT. Clamp the line parameter to [0, 1] and expand
    ↪ the norm, so no point of type
// P is ever constructed: this stays correct when you hand it P3<ll>.
template <class P> ld segDist3(P p, P a, P b) {
    P u = b - a, w = p - a;
    ld L = (ld)u.dist2(), uw = (ld)u.dot(w);
    if (L < eps) return w.dist(); // a == b
    ld t = min((ld)1, max((ld)0, uw / L));
    return sqrtl(max((ld)0, (ld)w.dist2() - t * (2 * uw - t * L)));
}

```

```
// SEGMENT -> SEGMENT. The minimum is either at an endpoint or at the
    ↳ unconstrained closest pair
// of the two LINES, and that one only counts when both parameters
    ↳ land strictly inside (0, 1).
// den = |u|^2|v|^2 - (u.v)^2 loses absolute precision at huge
    ↳ coordinates, but a garbage (s, t)
// still names two REAL points of the segments, so it can only fail
    ↳ to improve the answer.
template <class P> ld segSegDist3(P a, P b, P c, P d) {
    P u = b - a, v = d - c, w = a - c;
    ld A = (ld)u.dist2(), B = (ld)u.dot(v), C = (ld)v.dist2();
    ld D = (ld)u.dot(w), E = (ld)v.dot(w), W = (ld)w.dist2(), den = A
    ↳ * C - B * B;
    ld r = min({segDist3(a, c, d), segDist3(b, c, d), segDist3(c, a,
    ↳ b), segDist3(d, a, b)});
    if (den > eps) {
        ld s = (B * E - C * D) / den, t = (A * E - B * D) / den;
        if (s > 0 && s < 1 && t > 0 && t < 1) // |w +
    ↳ s*u - t*v|^2, expanded
            r = min(r, sqrtl(max((ld)0, W + s * s * A + t * t * C
                + 2 * s * D - 2 * t * E - 2 * s
    ↳ * t * B)));
    }
    return r; //
    ↳ parallel / collinear: endpoints win
}
// POINT -> TRIANGLE (the solid triangle). Project onto the plane; if
    ↳ the projection is inside,
// the plane distance is the answer, else the min over the three
    ↳ edges. The inside test uses p
// itself, NOT the projection: p and its projection differ by a
    ↳ multiple of n, and ((v-u) x n).n
// is 0, so the three orientations are unchanged - which keeps the
    ↳ test exact on integers.
template <class P> ld triDist3(P p, P a, P b, P c) {
    P n = (b - a).cross(c - a);
    ld e = min({segDist3(p, a, b), segDist3(p, b, c), segDist3(p, c,
    ↳ a)});
    if (n.dist2() < eps) return e; //
    ↳ collinear abc: no plane to hit
    auto f = [&](P u, P v) { return (v - u).cross(p - u).dot(n); };
    auto s1 = f(a, b), s2 = f(b, c), s3 = f(c, a);
    bool in = (s1 >= 0 && s2 >= 0 && s3 >= 0) || (s1 <= 0 && s2 <= 0
    ↳ && s3 <= 0);
    return in ? fabsl((ld)n.dot(p - a)) / n.dist() : e; // no
    ↳ eps: the two branches AGREE on
} // the
    ↳ boundary, so either is right
// SEGMENT d-e x TRIANGLE a-b-c, by four orientation signs and
    ↳ nothing else.
// {1, p} they meet in the single point p (grazing a vertex or an
    ↳ edge counts); {0, _} disjoint;
// {-1, _} d-e is COPLANAR with abc, where the meet can be empty, a
    ↳ point or a whole segment -
// map both to 2D with 03 - CoplanarFrame.cpp and clip there.
template <class P> pair<int, Pd3> segTri(P a, P b, P c, P d, P e) {
    auto va = vol6(a, b, c, d), vb = vol6(a, b, c, e);
    int sa = sgn(va), sb = sgn(vb);
    if (!sa && !sb) return {-1, Pd3()}; // (also
    ↳ fires when abc is collinear)
    if (sa == sb) return {0, Pd3()}; //
    ↳ segment strictly on one side
    int t1 = sgn(vol6(d, e, a, b)), t2 = sgn(vol6(d, e, b, c)), t3 =
    ↳ sgn(vol6(d, e, c, a));
    // The LINE d-e pierces the triangle iff it turns the same way
    ↳ around all three edges.
    if (!((t1 >= 0 && t2 >= 0 && t3 >= 0) || (t1 <= 0 && t2 <= 0 &&
    ↳ t3 <= 0))) return {0, Pd3()};
    Pd3 D((ld)d.x, (ld)d.y, (ld)d.z), E((ld)e.x, (ld)e.y, (ld)e.z);
    return {1, (E * (ld)va - D * (ld)vb) / ((ld)va - (ld)vb)}; //
    ↳ = planeLine, in vol6 terms
}
/* RELATED
SEGMENT x PLANE planeLine (01) then check the parameter
    ↳ lies in [0, 1].
TRIANGLE -> TRIANGLE min over the 6 (edge, triangle) pairs of
    ↳ segment-to-triangle distance;
    ↳ segment parameter - the distance
    ↳ that one is a ternary search over the
    ↳ from a point to a convex set is convex, so
    ↳ it is safe.
SEGMENT -> TRIANGLE same ternary search; 0 exactly when segTri
    ↳ says 1.
RAY x TRIANGLE same four signs, but replace the sa != sb
    ↳ test with sa != 0 and sa
    ↳ opposite to the side the ray points at
    ↳ (Moller-Trumbore is this test
```

```
with the divisions folded in - no reason to
    ↳ type it).
TWO SEGMENTS MEET in 3D only if coplanar: vol6(a, b, c, d) ==
    ↳ 0, then use the 2D
    ↳ segInter inside a coplanar frame.
    ↳ segSegDist3 == 0 is the cheap test.
CLOSEST APPROACH of two points moving linearly = segSegDist3 of the
    ↳ two trajectories with a
    ↳ shared parameter; if the speeds differ,
    ↳ minimise |w + t(v1 - v2)|^2
    ↳ directly, t* = -w.(v1-v2) / |v1-v2|^2
    ↳ clamped to the time window. */
```

11.2.3 CoplanarFrame

THE COPLANAR FRAME - the single most valuable file in 3D geometry, because it deletes 3D.

```
// Needs: Geometry3D.cpp, Core.cpp, Polygon.cpp.
// Given points known to lie on one plane, it builds a frame (o, dx,
//   ↳ dy, dz) with dx, dy in the
// plane and dz along the normal, and maps each point to a 2D point:
//   ↳ pos2d. From there EVERY 2D
// routine in the notebook (hull, area2, inPoly, inConvex, rotating
//   ↳ calipers, cut, minkowski,
// Pick, ...) applies UNCHANGED, and to3d maps the answer back. pos3d
//   ↳ keeps the third coordinate,
// which is the signed offset off the plane - 0 exactly for the
//   ↳ coplanar points.
// You reach for it whenever a statement says "a polygon in space",
//   ↳ "the points lie on a common
// plane", or you have just cut a polyhedron by a plane and hold the
//   ↳ cross-section as a bag of
// unordered 3D points. Building it is O(1); mapping is O(1) per
//   ↳ point.
//
// TWO FRAMES, AND THE CHOICE IS THE WHOLE POINT.
// ORTHONORMAL (ortho = true, the default): dx, dy, dz are unit and
//   ↳ mutually perpendicular, so the
// map is a rigid motion - DISTANCES, ANGLES and AREAS are
//   ↳ preserved exactly as they are in 3D.
// It normalises, so it needs P3<ld> and inherits float error.
// AFFINE (ortho = false): dx = q - p and dy = r - p as they came in,
//   ↳ no normalisation, so it
// stays on P3<ll> and every 2D predicate downstream stays EXACT.
//   ↳ It does NOT preserve distances
// or angles - a square can come out as a parallelogram - but it is
//   ↳ an invertible linear map of
// the plane, so it preserves COLLINEARITY, CONVEXITY, ORIENTATION,
//   ↳ ratios along a line and
// inside/outside. That is exactly enough for hull, inPoly,
//   ↳ inConvex, isConvex, orientation
// tests and point location. Areas are multiplied by the constant
//   ↳ |dx x dy| (the 2D area2 comes
// out |dx x dy|^2 times the area in plane coordinates, and the
//   ↳ true 3D area is the 2D area
// divided by |dx x dy|, so ratios of areas survive too. Never
//   ↳ feed the affine map to anything
// metric: closest pair, minimum enclosing circle, width,
//   ↳ perimeter, angle sorting.
// DEGENERATE: p, q, r must not be collinear (both constructors
//   ↳ produce dz = 0 otherwise, and the
// orthonormal one divides by 0). pos2d does not check that its
//   ↳ argument is on the plane - an
// off-plane point is silently flattened onto it, so test (p -
//   ↳ o).dot(dz) == 0 first if you are
// not sure. to3d is the inverse of pos2d for the ORTHONORMAL frame
//   ↳ only; for the affine frame
// carry the input INDEX through the 2D routine instead of mapping
//   ↳ back (see hullCoplanar), or
// solve the 2x2 Gram system: with A = dx.dx, B = dx.dy, C = dy.dy,
//   ↳ det = AC - B^2,
// a = (C*X - B*Y)/det, b = (A*Y - B*X)/det and the point is o + dx*a
//   ↳ + dy*b.
// EXACTNESS: affine frame on P3<ll> is exact; pos2d is a dot
//   ↳ product, so |coord| <= 1e9 keeps it
// in ll, and a downstream area2 squares that again - drop to |coord|
//   ↳ <= 3e4 or use __int128
// there. Orthonormal frame is floating throughout.
template <class Q> struct Frame {
    typedef decltype(Q().dot(Q())) T; // ll
    Q o, dx, dy, dz; // dz is
    ↳ the plane NORMAL
    Frame(Q p, Q q, Q r, bool ortho = true) : o(p), dx(q - p), dy(r -
    ↳ p) {
        dz = dx.cross(dy); // exact
    ↳ normal, before any scaling
        if (ortho) { dx = dx.unit(); dz = dz.unit(); dy =
        ↳ dz.cross(dx); }
    }
    P<T> pos2d(Q p) const { return {(p - o).dot(dx), (p -
    ↳ o).dot(dy)}; }
    Q pos3d(Q p) const { return {(p - o).dot(dx), (p - o).dot(dy), (p
    ↳ - o).dot(dz)}; }
    Q to3d(P<T> p) const { return o + dx * p.x + dy * p.y; } //
    ↳ ORTHONORMAL frames only
};
// CONVEX HULL OF COPLANAR 3D POINTS, returned as 3D points, ccw when
//   ↳ seen from the dz side.
// Exact on P3<ll>: it uses the affine frame and carries the point
//   ↳ itself through a map, so
```

```
// nothing is ever mapped back. O(n log n).
template <class Q> vector<Q> hullCoplanar(vector<Q> p) {
    sort(all(p)); p.erase(unique(all(p)), p.end());
    int n = sz(p), j = 2;
    if (n < 3) return p;
    while (j < n && (p[1] - p[0]).cross(p[j] - p[0]) == Q()) j++;
    if (j == n) return {p[0], p[n - 1]}; // all
    ↳ collinear: the two extremes
    Frame<Q> F(p[0], p[1], p[j], false);
    vector<decltype(F.pos2d(p[0]))> q;
    map<decltype(F.pos2d(p[0])), Q> back; // pos2d
    ↳ is injective on the plane
    for (Q x : p) q.push_back(F.pos2d(x)), back[q.back()] = x;
    vector<Q> res;
    for (auto& y : hull(q)) res.push_back(back[y]);
    return res;
}
// IS A 3D POINT INSIDE A COPLANAR POLYGON? Exact on P3<ll>. The
//   ↳ plane test comes FIRST: without
// it a point floating above the polygon projects straight into it
//   ↳ and reports true.
template <class Q> bool inCoplanar(const vector<Q>& v, Q p, bool
    ↳ strict = true) {
    int n = sz(v), j = 2;
    while (j < n && (v[1] - v[0]).cross(v[j] - v[0]) == Q()) j++;
    if (j == n) return false; //
    ↳ degenerate polygon: no plane
    Frame<Q> F(v[0], v[1], v[j], false);
    if (sgn((p - F.o).dot(F.dz)) != 0) return false; // off
    ↳ the plane
    vector<decltype(F.pos2d(p))> q;
    for (Q x : v) q.push_back(F.pos2d(x));
    return inPoly(q, F.pos2d(p), strict);
}
/* RECIPES
AREA OF A PLANAR 3D POLYGON no frame needed: |sum p_i x p_{i+1}| /
    ↳ 2, the vector area
    ↳ (faceArea in 06 - Polyhedron.cpp).
    ↳ Through the frame you get the
    ↳ same number as area(pos2d(.)) for an
    ↳ ORTHONORMAL frame, and
    ↳ area(pos2d(.)) / |dx x dy| for the
    ↳ affine one - use that identity
    ↳ as your sanity check that the frame is
    ↳ built right.
IS A SET OF 3D POINTS COPLANAR - fix the first three non-collinear
    ↳ ones, then vol6 == 0 for the
    ↳ rest. O(n), exact.
CUT A CONVEX POLYHEDRON BY A PLANE - clip each face, collect every
    ↳ new point on the cutting
    ↳ plane, and close the hole with
    ↳ hullCoplanar of those points.
SORT COPLANAR POINTS AROUND THEIR CENTROID - map with the affine
    ↳ frame and use angCmp (01 -
    ↳ Core.cpp) - exact, and gives the ccw
    ↳ face order a mesh wants.
POLYGON x POLYGON IN SPACE only meaningful if they are coplanar
    ↳ (planePlane first); then map
    ↳ both with ONE frame and run the 2D
    ↳ clip.
A 3D ROTATION MATRIX from the frame: the rows dx, dy, dz of an
    ↳ orthonormal frame ARE the matrix
    ↳ that takes the plane to z = 0, and its
    ↳ transpose takes it back. */
```

11.2.4 ConvexHull3D

3D CONVEX HULL, incremental. Input: n points (duplicates allowed, any order). Output: the hull

```
// Needs: Geometry3D.cpp.
// surface as TRIANGULAR faces, each an index triple {i, j, k} into
//   the input vector, ordered so
//   that (p[j]-p[i]) x (p[k]-p[i]) points OUTWARD - equivalently
//   vol6(p[i], p[j], p[k], x) <= 0 for
// every input point x. Coplanar facets come back cut into triangles
//   by an arbitrary
// triangulation, and points strictly inside a facet are simply
//   absent from the output.
// Reach for it for "volume/surface of the hull", "is this point
//   inside the hull", "diameter of a
// 3D point set", "smallest enclosing box", or any time a 2D hull
//   idea has to survive one more
// coordinate. O(n^2) faces touched (O(n^2 log n) as written, the log
//   being the sort of the
// horizon edges - swap in an n x n matrix to drop it; it has never
//   mattered).
// Memory O(n) for the face list, O(n) for the horizon.
// DEGENERATE, and this is the part people get wrong: n < 4, ALL
//   POINTS EQUAL, ALL POINTS
// COLLINEAR and ALL POINTS COPLANAR each return an EMPTY face list -
//   a flat hull has no volume
// and no consistently outward orientation, so it cannot be expressed
//   in this format. Detect it
// (the return is empty) and rerun the problem in 2D through 03 -
//   CoplanarFrame.cpp, which is
// where a coplanar input belongs. Duplicate points are harmless: a
//   repeat never strictly sees a
// face, so it is skipped and only one copy can appear in the output.
// EXACTNESS: no divisions anywhere - only vol6 signs - so on P3<ll>
//   the hull is EXACT with
// |coord| <= 3.6e5 (the orientation test sums four vol6, 192*C^3 <=
//   9.2e18; make vol6 return
// __int128 to reach 1e9). The three seeding tests convert to ld to
//   compare against eps, which is
// what makes the same code safe on P3<ld>; there eps is absolute, so
//   scale your input to O(1).
template <class P> vector<array<int, 3>> hull3d(const vector<P>& p) {
    int n = sz(p), k;
    if (n < 4) return {}; // no
    // solid hull exists
    vector<array<int, 3>> F;
    vector<int> id(n);
    iota(all(id), 0);
    auto nz = [](P v) { return max({fabsl((ld)v.x), fabsl((ld)v.y),
    // fabsl((ld)v.z)) > eps; };
    for (k = 1; k < n && !nz(p[id[k]] - p[id[0]]); k++); // a
    // second DISTINCT point
    if (k == n) return {};
    swap(id[1], id[k]);
    for (k = 2; k < n && !nz((p[id[1]] - p[id[0]]).cross(p[id[k]] -
    // p[id[0]])); k++);
    if (k == n) return {}; // every
    // point on one line
    swap(id[2], id[k]);
    for (k = 3; k < n && fabsl((ld)vol6(p[id[0]], p[id[1]], p[id[2]],
    // p[id[k]])) < eps; k++);
    if (k == n) return {}; // every
    // point on one plane
    swap(id[3], id[k]);
    // Orient against the centroid of the seed tetrahedron, which is
    // strictly inside it and so
    // stays strictly inside every later hull. Summing the four vol6
    // IS testing that centroid,
    // scaled by 4, so the test never ties and never divides.
    auto face = [&](int a, int b, int c) {
        auto s = vol6(p[a], p[b], p[c], p[id[0]]) + vol6(p[a], p[b],
    // p[c], p[id[1]])
    // + vol6(p[a], p[b], p[c], p[id[2]]) + vol6(p[a], p[b],
    // p[c], p[id[3]]);
    // F.push_back(s > 0 ? array<int, 3>{a, c, b} : array<int, 3>{a,
    // b, c});
    };
    for (int i = 0; i < 4; i++) face(id[(i + 1) % 4], id[(i + 2) %
    // 4], id[(i + 3) % 4]);
    for (int t = 4; t < n; t++) {
        int u = id[t];
        vector<array<int, 3>> keep;
        vector<pii> E; //
    // directed edges of the dying faces
        for (auto& f : F)
            if (vol6(p[f[0]], p[f[1]], p[f[2]], p[u]) > eps) {
                for (int j = 0; j < 3; j++) E.push_back({f[j], f[(j +
```

```
    // 1) % 3}]);
        } else keep.push_back(f);
        if (E.empty()) continue; // u is
    // inside the hull: nothing to do
        F = keep;
        sort(all(E));
        // The HORIZON is the directed edges whose reverse did not
    // also die: those are the ones
        // whose other face survives. Cone them to u.
        for (auto& [a, b] : E) if (!binary_search(all(E), pii(b, a)))
            face(a, b, u);
        }
    return F;
}
/* RELATED
VOLUME / AREA / CENTROID of the result: build the face list and call
    // 06 - Polyhedron.cpp -
    vector<vector<P>> fs; for (auto& f : hull3d(p))
    // fs.push_back({p[f[0]], p[f[1]], p[f[2]]});
    // the faces are already consistently outward, so polyVol6 comes
    // out POSITIVE.
IS A POINT INSIDE THE HULL    vol6(p[f[0]], p[f[1]], p[f[2]], q) <= 0
    // for every face, O(F).
EXTREME POINT in a direction d: max dot over the hull VERTICES (the
    // indices that appear in F).
DIAMETER of a 3D point set: no rotating calipers in 3D worth typing
    // - take the hull vertices
    // and brute force the pairs, it is O(h^2) with a tiny h.
O(n log n) HULL exists (divide and conquer, or randomised
    // incremental with a conflict graph).
    // It is 150 lines of merge bookkeeping for a constant factor you
    // do not need: n = 2000 here is
    // 4e6 orientation tests, a few milliseconds.
DELAUNAY TRIANGULATION of 2D points: lift each (x, y) to (x, y, x^2
    // + y^2) on the paraboloid,
    // take the 3D hull, and keep the faces whose outward normal has
    // NEGATIVE z (the LOWER hull) -
    // those triangles, projected back, are exactly the Delaunay
    // triangulation. The upper hull is
    // the farthest-point Delaunay. Lifting squares the coordinates, so
    // |coord| <= 1e4 to stay in
    // ll here; the Voronoi diagram is the dual of what comes out.
HALFSPACE INTERSECTION in 3D is the dual problem: map each plane n.x
    // = d (d > 0, so that the
    // origin is interior) to the point n/d, take the hull of those,
    // and dualise the faces back. */
```


11.2.5 Spherical

SPHERICAL GEOMETRY: solid angles, spherical triangles and polygons, caps, the lens of two

```
// Needs: Geometry3D.cpp.
// spheres, the circumsphere of four points and the smallest
//   ↪ enclosing sphere of a point set.
// Points "on the sphere" are always passed as vectors FROM ITS
//   ↪ CENTRE and only their DIRECTION is
// read, so you may hand in unnormalised vectors, and a sphere
//   ↪ centred elsewhere is handled by
// subtracting the centre first. Angles are radians, areas are on a
//   ↪ sphere of radius R.
// You reach for it for radar/antenna coverage, "what fraction of the
//   ↪ sky", area of a country on
// a globe, overlapping-ball volumes, and the two enclosing-object
//   ↪ questions. All O(1) except
// sphPolyArea (O(n)) and minSphere (O(n) per iteration).
// DEGENERATE: solidAngleTet returns 0 when a, b, c are coplanar with
//   ↪ the origin (a flat corner)
// and is unaffected by their lengths. sphAngle is undefined when a
//   ↪ is parallel to b or c (a
// zero-length spherical segment) and returns 0 there. sphPolyArea
//   ↪ needs a SIMPLE polygon, ccw
// seen from outside, with no two consecutive vertices antipodal (the
//   ↪ arc between them is not
// unique); it returns the area of the complement if you feed it cw.
//   ↪ circumsphere needs four
// points in general position - vol6 == 0 makes it divide by zero.
//   ↪ minSphere needs a non-empty
// input and returns {p[0], 0} for a single point.
// EXACTNESS: this whole file is floating point - every routine
//   ↪ either normalises, calls a trig
// function, or divides. The predicates it rests on (vol6, dot,
//   ↪ cross) are still exact on P3<ll>,
// so use integer input where you can and only the final value
//   ↪ carries error. atan2 is used
// everywhere in place of acos: acos(dot/|a||b|) loses half its
//   ↪ digits near 0 and pi, which is
// exactly the regime these formulas live in. eps is absolute here,
//   ↪ so normalise your radii to
// O(1)..O(1e6).
// SOLID ANGLE of a cone of half-angle th (steradians): the full
//   ↪ sphere is 4*pi.
ld solidAngleCone(ld th) { return 2 * PI * (1 - cosl(th)); }
// SOLID ANGLE of the tetrahedron corner at the ORIGIN spanned by a,
//   ↪ b, c (VAN OOSTEROM-STRACKEE):
// tan(0m/2) = |a.(b x c)| / (|a||b||c| + (a.b)|c| + (a.c)|b| +
//   ↪ (b.c)|a|).
// atan2 takes the correct branch by itself, so this is right on the
//   ↪ whole range [0, 2*pi] with no
// sign patching, and it stays accurate for the needle-thin corners
//   ↪ that Girard's formula ruins.
template <class P> ld solidAngleTet(P a, P b, P c) {
    ld A = a.dist(), B = b.dist(), C = c.dist();
    ld num = fabsl((ld)a.dot(b.cross(c)));
    ld den = A * B * C + (ld)a.dot(b) * C + (ld)a.dot(c) * B +
    ↪ (ld)b.dot(c) * A;
    return 2 * atan2l(num, den);
}
// ANGLE at a of the spherical triangle abc = the dihedral angle
//   ↪ between the planes Oab and Oac,
// in [0, pi]. The ORIENTED version measures how far ccw you turn
//   ↪ from arc a->b to arc a->c, in
// [0, 2*pi), which is what a signed area needs.
template <class P> ld sphAngle(P a, P b, P c) {
    P u = a.cross(b), v = a.cross(c);
    return atan2l(u.cross(v).dist(), (ld)u.dot(v));
}
template <class P> ld sphAngleOr(P a, P b, P c) {
    ld t = sphAngle(a, b, c);
    return a.cross(b).dot(c) >= 0 ? t : 2 * PI - t;
}
// SPHERICAL TRIANGLE AREA on a sphere of radius R. GIRARD: R^2 * (A
//   ↪ + B + C - pi), the spherical
// EXCESS - a spherical triangle's angles overshoot pi by exactly its
//   ↪ area. Computed through the
// solid angle instead, which is the same number without the
//   ↪ catastrophic cancellation of summing
// three angles that are each close to pi/2 and subtracting pi.
template <class P> ld sphTriArea(P a, P b, P c, ld R) { return R * R
//   ↪ * solidAngleTet(a, b, c); }
// SPHERICAL POLYGON AREA, vertices in ccw order seen from OUTSIDE:
//   ↪ R^2 * (sum of the interior
// angles - (n-2)*pi), the same excess with n corners. Handles
//   ↪ non-convex polygons.
template <class P> ld sphPolyArea(const vector<P>& p, ld R) {
    int n = sz(p);
```

```
ld s = -(n - 2) * PI;
for (int i = 0; i < n; i++) s += sphAngleOr(p[(i + 1) % n], p[(i
↪ + 2) % n], p[i]);
return R * R * s;
}
// SPHERICAL CAP of height h on a sphere of radius R (a plane at
//   ↪ distance R - h from the centre;
// from the polar half-angle th instead, h = R * (1 - cos th) and the
//   ↪ base radius is R * sin th).
// capArea is the CURVED part only - add pi*a^2 for the flat disc,
//   ↪ a^2 = h * (2R - h).
ld capArea(ld R, ld h) { return 2 * PI * R * h; }
ld capVolume(ld R, ld h) { return PI * h * h * (3 * R - h) / 3; }
// SPHERICAL SEGMENT / zone: the slab between two parallel planes,
//   ↪ base radii a and b, height h.
ld zoneArea(ld R, ld h) { return 2 * PI * R * h; } //
//   ↪ curved part; independent of WHERE
ld zoneVolume(ld a, ld b, ld h) { return PI * h * (3 * a * a + 3 * b
//   ↪ * b + h * h) / 6; }
// SPHERE x SPHERE, centres d apart: {surface area, volume} of the
//   ↪ lens they share. The lens is
// two caps glued on the plane of the intersection circle.
//   ↪ CONTAINMENT is checked FIRST - there
// the cap formula silently returns nonsense, and the answer is just
//   ↪ the smaller ball.
pair<ld, ld> sphereSphere(ld R, ld r, ld d) {
    if (R < r) swap(R, r);
    if (d >= R + r) return {0, 0}; //
//   ↪ disjoint (tangent counts)
    if (d + r <= R) return {4 * PI * r * r, 4 * PI * r * r * r / 3};
//   ↪ small one swallowed
    ld x = (R * R - r * r + d * d) / (2 * d); // plane
//   ↪ of the circle, from centre R
    ld h1 = R - x, h2 = r - (d - x); // the
//   ↪ two cap heights
    return {capArea(R, h1) + capArea(r, h2), capVolume(R, h1) +
//   ↪ capVolume(r, h2)};
}
// The intersection CIRCLE itself has radius sqrt(R^2 - x^2) and sits
//   ↪ at x along the centre line.
// CIRCUMSPHERE of four points, {centre, radius}. Needs P3<ld>; the
//   ↪ denominator IS 2*vol6, so
// coplanar input divides by zero - test vol6 first.
template <class P> pair<P, ld> circumsphere(P a, P b, P c, P d) {
    P B = b - a, C = c - a, D = d - a;
    ld den = 2 * (ld)B.dot(C.cross(D));
    P o = a + (C.cross(D) * (ld)B.dist2() + D.cross(B) * (ld)C.dist2()
//   ↪ + B.cross(C) * (ld)D.dist2()) / den;
    return {o, (o - a).dist()};
}
// SMALLEST ENCLOSEING SPHERE, {centre, radius}, by shrinking steps
//   ↪ toward the current farthest
// point. Start at the centroid and step a fraction q toward the
//   ↪ farthest point, with q decaying
// geometrically: the centre converges to the true one and the step
//   ↪ budget bounds the error by
// (initial spread) * 0.999^30000 ~ 1e-13 of the diameter - fine for
//   ↪ 1e-6 answers, NOT a proof.
// O(n) per iteration. Welzl with a 4-point basis is the exact O(n)
//   ↪ expected version and is 40
// lines; this is 8 and has never lost a contest problem.
template <class P> pair<P, ld> minSphere(const vector<P>& p) {
    P c;
    for (P q : p) c = c + q;
    c = c / sz(p);
    for (ld q = 0.1; q > 1e-13; q *= 0.999) {
        int f = 0;
        for (int i = 1; i < sz(p); i++) if ((c - p[i]).dist2() > (c -
//   ↪ p[f]).dist2()) f = i;
        c = c + (p[f] - c) * q;
    }
    ld r = 0;
    for (P x : p) r = max(r, (c - x).dist());
    return {c, r};
}
// GREAT CIRCLE, the part 01 - Geometry3D.cpp does not cover: its
//   ↪ greatCircle takes lat/lon in
// DEGREES, these take 3D vectors. Distance along the surface between
//   ↪ two points of the sphere,
// and the lat/lon -> vector conversion (latitude from the equator,
//   ↪ longitude east, degrees).
template <class P> ld sphDist(P a, P b, ld R) {
    return R * atan2l(a.cross(b).dist(), (ld)a.dot(b));
}
Pd3 fromLatLon(ld R, ld lat, ld lon) {
    lat *= PI / 180, lon *= PI / 180;
    return {R * cosl(lat) * cosl(lon), R * cosl(lat) * sinl(lon), R *
```



```

    ↪ sinl(lat));
}
/* RELATED
TANGENT CONE from an external point at distance d: half-angle acos(R
    ↪ / d), and the cap it cuts
    off the sphere has height R * (1 - R / d). d < R means the point
    ↪ is inside: no tangent.
SPHERE x PLANE    a circle of radius sqrt(R^2 - dist(centre,
    ↪ plane)^2), centred at the projection.
SPHERE x LINE / SEGMENT    sphereLine (01 - Geometry3D.cpp), then
    ↪ clamp the parameters.
VOLUME OF A UNION OF BALLS - no closed form beyond two; for three or
    ↪ more integrate over one axis
    numerically, or use inclusion-exclusion only when the statement
    ↪ promises few overlaps.
SPHERICAL SEGMENT x SEGMENT - two great-circle arcs ab and cd cross
    ↪ where the plane normals a x b
    and c x d cross: the candidate directions are +--((a x b) x (c x
    ↪ d)); accept the one lying
    inside both arcs, which is the exact 3D test sgn((a x b).c) !=
    ↪ sgn((a x b).d) and vice versa.
SPHERICAL CONVEX HULL of directions = the 3D convex hull of the unit
    ↪ vectors (04), as long as
    they fit in a hemisphere.
SOLID ANGLE OF A POLYGONAL CONE - triangulate it from one vertex and
    ↪ sum solidAngleTet - that is
    exactly what sphPolyArea does with R = 1. */

```

11.2.6 Polyhedron

POLYHEDRA GIVEN AS A FACE LIST: fs[i] is one face, a vector of its vertices in order around it,

```

// Needs: Geometry3D.cpp, Spherical.cpp (winding3 only).
// coplanar, no repeat of the first vertex at the end. Faces may be
    ↪ any polygon, not just
// triangles. Everything here wants the list CLOSED (every edge
    ↪ shared by exactly two faces) and
// CONSISTENTLY oriented (all normals outward, or all inward);
    ↪ reorient produces the consistency
// from a face list that has it in neither direction, and the sign of
    ↪ polyVol6 tells you which of
// the two you ended up with. Computes: signed 6*volume, surface
    ↪ area, volume centroid, the
// orientation fixup, and the 3D winding number that answers "is the
    ↪ origin inside this thing".
// You reach for this file after a 3D hull, after cutting a solid by
    ↪ planes, or when the statement
// hands you a mesh (an .obj-shaped input) and asks for volume, mass
    ↪ centre, or containment.
// COMPLEXITY: volume, area, centroid O(total vertices). reorient O(E
    ↪ log E) for the edge map,
// O(E) memory. winding3 O(total vertices) with a few trig calls each.
// DEGENERATE: an empty face list gives volume 0, area 0 and centroid
    ↪ 0/0 - guard it. A face with
// fewer than 3 vertices contributes nothing (its fan is empty). A
    ↪ polyhedron that is NOT closed
// still returns a number from polyVol6, and that number is
    ↪ meaningless - there is no cheap check,
// so verify that every directed edge appears exactly once (which is
    ↪ what reorient's map does).
// reorient BFSes the face adjacency graph, so a DISCONNECTED surface
    ↪ (two separate shells) comes
// back with only the component of face 0 fixed; run it per
    ↪ component. Non-manifold edges (three
// faces meeting) break the map, which keeps only the first face per
    ↪ directed edge.
// EXACTNESS: polyVol6 and faceArea's vector sum are exact on P3<ll>
    ↪ (|coord| <= 5.7e5 for vol6);
// faceArea then takes one sqrt, and centroid divides, so those
    ↪ return ld / need P3<ld>. reorient
// compares vertices for EQUALITY, so on P3<ld> two faces only share
    ↪ an edge if the coordinates
// are bit-identical - with float input, or any input where the same
    ↪ vertex was computed twice,
// index your vertices and run reorient on vector<vector<int>> with
    ↪ the map keyed by pii instead.
// SIGNED 6*VOLUME, by the divergence theorem: fan every face from
    ↪ the ORIGIN and sum the
// tetrahedra. The parts outside the solid cancel exactly. Positive
    ↪ iff the faces are OUTWARD.
template <class P> auto polyVol6(const vector<vector<P>>& fs) {
    decltype(fs[0][0].dot(fs[0][0])) s = 0;
    for (auto& f : fs) for (int i = 1; i + 1 < sz(f); i++) s +=
        ↪ vol6(P(), f[0], f[i], f[i + 1]);
    return s;
}
template <class P> ld polyVolume(const vector<vector<P>>& fs) {
    return fabsll((ld)polyVol6(fs)) / 6;
}
// AREA of one planar face, from its VECTOR AREA sum p_i x p_(i+1)
    ↪ (which is 2A times the unit
// normal, and is independent of where the origin sits because the
    ↪ face is closed and planar).
template <class P> ld faceArea(const vector<P>& f) {
    P s;
    for (int i = 0, m = sz(f); i < m; i++) s = s + f[i].cross(f[(i +
        ↪ 1) % m]);
    return s.dist() / 2;
}
template <class P> ld surfaceArea(const vector<vector<P>>& fs) {
    ld a = 0;
    for (auto& f : fs) a += faceArea(f);
    return a;
}
// CENTROID of the SOLID (not of the vertices, not of the surface):
    ↪ the volume-weighted average of
// the tetrahedron centroids of the same fan. Needs P3<ld>. Signs
    ↪ cancel exactly as in polyVol6.
template <class P> P polyCentroid(const vector<vector<P>>& fs) {
    P c; ld V = 0;
    for (auto& f : fs) for (int i = 1; i + 1 < sz(f); i++) {
        ld v = (ld)vol6(P(), f[0], f[i], f[i + 1]); // 6 *
        ↪ signed volume of 0-f0-fi-fi+1
        c = c + (f[0] + f[i] + f[i + 1]) * (v / 4); // its
        ↪ centroid is (0+a+b+c)/4
        V += v;
    }
}

```

```

    }
    return c / V;
}
// FACE-ORIENTATION FIXUP. Flip a subset of the faces so that all
    ↪ normals agree (either all out or
// all in - which one you get depends on face 0; check sgn(polyVol6)
    ↪ afterwards and reverse every
// face if you wanted the other). THE RULE: two faces sharing an edge
    ↪ agree exactly when they
// traverse that edge in OPPOSITE directions, so seeing the SAME
    ↪ directed edge twice means exactly
// one of the two faces must flip. BFS that parity over the face
    ↪ adjacency graph.
template <class P> vector<vector<P>> reorient(vector<vector<P>> fs) {
    int n = sz(fs);
    if (!n) return fs;
    vector<vector<pii>> g(n); //
    ↪ {neighbour, same direction?}
    map<pair<P, P>, int> es;
    for (int u = 0; u < n; u++)
        for (int i = 0, m = sz(fs[u]); i < m; i++) {
            P a = fs[u][i], b = fs[u][(i + 1) % m];
            if (es.count({b, a})) { int v = es[{b, a}]; g[u].pb({v,
    ↪ 0}); g[v].pb({u, 0}); }
            else if (es.count({a, b})) { int v = es[{a, b}];
    ↪ g[u].pb({v, 1}); g[v].pb({u, 1}); }
            else es[{a, b}] = u;
        }
    vector<int> vis(n, 0), flip(n, 0), q = {0};
    vis[0] = 1; // face
    ↪ 0 defines the common direction
    for (int i = 0; i < sz(q); i++)
        for (auto& [v, same] : g[q[i]])
            if (!vis[v]) vis[v] = 1, flip[v] = flip[q[i]] ^ same,
    ↪ q.push_back(v);
        for (int u = 0; u < n; u++) if (flip[u] reverse(all(fs[u]));
        return fs;
}
// 3D WINDING NUMBER about the ORIGIN: how many times the surface
    ↪ wraps around it. 0 = the origin
// is OUTSIDE; +1 = inside with the faces oriented outward; -1 =
    ↪ inside with them oriented inward.
// Each face is projected onto the unit sphere, where it covers a
    ↪ signed solid angle; the total is
// 4*pi per wrap. remainder folds each face's contribution into
    ↪ (-2pi, 2pi] so that faces covering
// more than a hemisphere still add up right. To test a point other
    ↪ than the origin, subtract it
// from every vertex first. The origin exactly ON the surface makes a
    ↪ face's solid angle
// undefined - handle that case before calling (segTri / inCoplanar
    ↪ in 02 and 03).
template <class P> int winding3(const vector<vector<P>>& fs) {
    ld s = 0;
    for (auto& f : fs) s += remainderl(sphPolyArea(f, 1), 4 * PI);
    return (int)llroundl(s / (4 * PI));
}
/* RELATED
IS A POINT INSIDE A CONVEX POLYHEDRON - do not call winding3: test
    ↪ sgn of the side of every face
    plane, O(F), exact on integers. Convex + outward faces => inside
    ↪ iff all sides <= 0.
EULER    V - E + F = 2 for any closed surface of genus 0; a
    ↪ triangulated one has E = 3F/2, so
    F = 2V - 4 and E = 3V - 6. Use it as a free assertion on the
    ↪ output of a 3D hull.
SURFACE OF A NON-PLANAR "face" - faceArea silently returns the area
    ↪ of its projection onto the
    best-fit plane. Triangulate first if the input can be skew.
MOMENTS / INERTIA - the same fan works: sum the tetrahedron inertia
    ↪ tensors weighted by vol6.
CUT BY A PLANE - clip every face (keep the vertices with side <= 0
    ↪ plus the crossing points), then
    close the hole with hullCoplanar of the new points (03 -
    ↪ CoplanarFrame.cpp).
MINKOWSKI SUM of two convex polyhedra: hull of the pairwise vertex
    ↪ sums, O((nm)^2) but n and m
    are tiny in practice.
VOLUME OF A UNION of polyhedra has no cheap formula - sweep a plane
    ↪ and integrate the
    cross-section area, or inclusion-exclusion on convex pieces. */

```

12 Problem Solving

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12.1 Reframing

REFRAMING: CHANGE THE PROBLEM, NOT THE EFFORT
=====

```

/* ===== REFRAMING: CHANGE THE PROBLEM, NOT THE EFFORT
    ↪ =====
This chapter has no algorithms - it has the moves you make BEFORE
    ↪ you know which template to
paste. Stuck for 10 minutes? Do not re-read the statement a fourth
    ↪ time. Walk the six questions,
then walk the TRIGGERS below and try each against your problem.

```

THE SIX QUESTIONS

1. What object am I counting / optimising over? Say it in one
↪ sentence with no input format.
2. What is INVARIANT under the allowed operation?
3. What does the ANSWER look like (a prefix? one of the inputs? a
↪ threshold? always <= 2?)
4. Can I CHECK a candidate faster than I can find one? ->
↪ binary search the answer
5. What do the constraints forbid, and what does the leftover
↪ budget exactly equal?
6. What would the brute force print for n = 1..8?

--- REVERSE THE PROCESS ---

Run time backwards: deletions become insertions, splits become
↪ merges. Every structure is
add-friendly and delete-hostile (DSU above all), so reversing
↪ turns impossible into trivial.
TRIGGER: things are removed one at a time and you must answer
↪ after each removal.
EXAMPLE: CF 722C Destroying Array - process destructions in
↪ reverse, union with restored
neighbours. Also CSES 1163 Traffic Lights.
TRAP: emit answers in reverse too, and check the off-by-one at
↪ both ends with n = 1.

--- WORK BACKWARDS FROM THE GOAL ---

Define the answer at terminal states and propagate backwards.
↪ Standard for games and for
"what must have been true one step earlier".
TRIGGER: few terminal states, huge forward branching, small
↪ backward branching.
TRAP: on a CYCLIC game graph this is NOT a plain DP - you need
↪ retrograde BFS with out-degree
counters (a position is losing only once ALL successors are
↪ known winning). See 10 - Game
Theory. Writing win[v] = OR over children on a cyclic graph
↪ silently mislabels draws.

--- COMPLEMENTARY COUNTING ---

Count the bad ones and subtract; count $P(X \leq x)$ instead of $P(X =$
↪ $x)$.
TRIGGER: "at least one", "there exists", "not all equal", "some
↪ pair".
EXAMPLE: $E[\max] = \text{sum over } x \text{ of } P(\max \geq x)$, with $P(\max \leq x) =$
↪ $(x/n)^k$.
TRAP: the complement is easy for AT MOST / AT LEAST, not for
↪ EXACTLY. Getting from one to the
other is binomial inversion: $\text{exactly}_k = \text{sum}_{\{j \geq k\}} (-1)^{(j-k)}$
↪ $C(j, k) \text{ atleast}_j$.

--- THE CONTRIBUTION TECHNIQUE ---

Instead of "for each subarray, compute f", ask "for each element,
↪ in how many objects does it
contribute, and how much". Turns $O(n^2)$ enumeration into $O(n)$.
TRIGGER: "sum over all subarrays / pairs / subsets of [min | max |
↪ gcd | #distinct | bottleneck]".
EXAMPLE: sum over subarrays of the minimum -> monotonic stack
↪ gives each element's span.
Sum over pairs of the bottleneck -> Kruskal, each merge
↪ contributes $w * |A| * |B|$ pairs.
TRAP: ties. With equal values the span must be strict on ONE side
↪ and non-strict on the other,
or every duplicated minimum is counted twice. Decide which side
↪ BEFORE coding.

--- SWAP THE ORDER OF SUMMATION ---

$\text{sum}_i \text{sum}_{\{d \mid i\}} f(d) = \text{sum}_d f(d) * \text{floor}(n/d)$. Move the outer
↪ loop to the variable whose
inner count has a closed form.
TRIGGER: a double sum whose inner range depends on the outer
↪ index; divisors, multiples,
ancestors, intervals containing a point.
EXAMPLE: sum of $\sigma(i)$ for $i \leq n = \text{sum}_d d * \text{floor}(n/d)$, then
↪ divisor blocks -> $O(\sqrt{n})$.

--- LINEARITY OF EXPECTATION ---

$E[\sum X_i] = \sum E[X_i]$, with NO independence needed. Decompose
 ↳ into indicators.
 TRIGGER: "expected number of ...", or an expectation over a
 ↳ process you cannot untangle.
 TRAP: linearity does not survive max, min, or division. $E[\max] \neq \max E$, $E[1/X] \neq 1/E[X]$.
 For an expected maximum use the complementary-counting tail sum
 ↳ above.

--- FIX THE EXTREMUM ---
 Add an artificial outer loop "for each choice of the maximum / the
 ↳ leftmost / the last move".
 Everything else becomes bounded relative to the fixed thing.
 TRIGGER: the statement mentions max or min over a range; a
 ↳ Cartesian-tree recursion is available.
 EXAMPLE: CF 1156E - fix the position of the maximum, then
 ↳ enumerate the SMALLER side.
 TRAP: always iterate the smaller side (small-to-large, $O(n \log n)$)
 ↳ total). Iterating the left
 side unconditionally is $O(n^2)$ on a sorted permutation - and it
 ↳ passes the samples.

--- DUALIZE ---
 max-flow = min-cut; max bipartite matching = min vertex cover
 ↳ (Konig); min path cover of a DAG
 = $n - \text{matching}$; longest antichain = min chain cover (Dilworth); "a
 ↳ perfect matching exists" =
 "no Hall violator".
 TRIGGER: the statement asks for a MINIMUM removal / blocking set /
 ↳ number of chains - that is
 almost always the dual of a maximum you can compute.
 TRAP: Konig gives the SIZE for free, but recovering the cover
 ↳ needs the alternating-BFS
 construction $(L \setminus Z) \cup (R \text{ and } Z)$. And Konig is BIPARTITE
 ↳ ONLY.

--- MODEL IT AS A GRAPH ---
 Objects -> nodes, relations -> edges. States can be tuples: (cell,
 ↳ fuel), (vertex, parity),
 (position, mask).
 TRIGGER: "can I get from A to B", "minimum number of operations",
 ↳ "are these constraints
 consistent", "each x points at exactly one y" (functional graph).
 TRAP: compute $|V| * |E|$ explicitly before coding. In a layered
 ↳ graph the answer is usually
 "any layer", not layer 0.

--- MODEL IT AS INTERVALS / A TIMELINE ---
 Recast objects as [start, end] and use a sweep,
 ↳ greedy-by-endpoint, or a difference array.
 Tree subtrees become intervals via the Euler tour.
 TRIGGER: "from time a to time b", "covers l..r", "is available
 ↳ during", subtree aggregates.
 TRAP: decide ONCE whether [1,3] and [3,5] overlap, and encode it
 ↳ in exactly one place.

--- MODEL IT AS A POLYNOMIAL ---
 "For every possible sum, count the ways" is a convolution.
 ↳ Multiplying GFs = combining
 independent choices; x^k = "used k of the budget".
 TRIGGER: modulus 998244353 (NTT is intended); independent parts
 ↳ whose sizes add.
 TRAP: convolving an array with itself double-counts $i \neq j$ and
 ↳ mis-counts $i = j$.

--- MODEL IT AS LINEAR ALGEBRA OVER $GF(2)$ ---
 Each element is a bit-vector; "can I make X" = "is X in the span";
 ↳ the number of subsets giving
 a representable Y is $2^{(n - \text{rank})}$.
 TRIGGER: XOR anywhere; "product is a perfect square" (exponent
 ↳ parity vector over primes);
 switch-toggling puzzles; "each constraint satisfied an even
 ↳ number of times".
 TRAP: $2^{(n - \text{rank})}$ counts solutions only if the system is CONSISTENT
 ↳ - check first. And subtract
 the empty subset when the problem wants a non-empty one.

--- MODEL IT AS THE SUBSET LATTICE ---
 Bitmask DP plus zeta/Mobius (SOS) to aggregate over submasks in
 ↳ $O(2^n n)$ instead of $O(3^n)$.
 TRIGGER: $n \leq 20$ with a "set of chosen items" state, or a $\leq$
 ↳ 22-bit value domain with queries
 "over all y contained in x".
 TRAP: the bit loop goes OUTSIDE the mask loop. Swapping them gives
 ↳ a wrong-but-plausible array.
 Submask enumeration must be for $(s = m; ; s = (s-1) \& m) \{ \dots;$
 ↳ if (!s) break; }.

--- THINK ABOUT WHAT THE ANSWER LOOKS LIKE ---

Characterise the optimum's SHAPE first: it is a prefix; it is one
 ↳ of the inputs; it is at most
 2; it has a threshold structure. Then search only that shape.
 TRIGGER: the output is one small number regardless of n; the
 ↳ problem screams "constructive".
 TRAP: "the answer is at most k" needs a matching lower-bound
 ↳ example, or you print k-1.

--- CHANGE OF VARIABLES ---

Difference array (range add -> two point updates; "make all equal"
 ↳ -> zero the differences),
 prefix sums (subarray -> pair of points), Chebyshev <-> Manhattan
 ↳ via $(x+y, x-y)$, logs
 (products -> sums).
 TRIGGER: operations act on ranges; the objective is $\max(|dx|, |dy|)$
 ↳ or $|dx| + |dy|$.
 TRAP: C++ % is negative for negative operands - use $((x \% n) + n)$
 ↳ % n. Seed the empty prefix.

--- PER-BIT / PER-PRIME INDEPENDENCE ---

If the objective decomposes across bits (XOR/AND/OR sums, popcount
 ↳ weights) or across primes
 (gcd/lcm exponent-wise), solve 30 tiny problems and recombine.
 EXAMPLE: sum over pairs of $(a_i \text{ XOR } a_j) = \text{sum}_b 2^b * \text{cnt0}_b *$
 ↳ cnt1_b .
 TRAP: bit-independence holds for SUMS, not for COMPARISONS. "Count
 ↳ pairs with $a_i * a_j < k$ " is
 not separable - that needs a binary trie walked high bit first.

--- REDUCE TO SOMETHING YOU ALREADY HAVE ---

"minimum adjacent swaps to sort" = inversions; "maximum
 ↳ pairwise-compatible set" = matching;
 "minimise the maximum edge on a path" = MST bottleneck / Kruskal
 ↳ reconstruction tree; "longest
 chain of pairs" = LIS; "pairwise gap constraints" = difference
 ↳ constraints + Bellman-Ford;
 "each person points at one other" = functional graph.
 TRIGGER: you can state the problem without the story. Do that,
 ↳ then match it.
 TRAP: the disguise is usually ALMOST the classic - one extra
 ↳ constraint breaks the reduction.
 Check the reduction on the samples by hand before trusting it.

=====

↳ */

12.2 SearchAndProof

SEARCH: TURN "FIND" INTO "CHECK" =====

/* ===== SEARCH: TURN "FIND" INTO "CHECK" =====

BINARY SEARCH ON THE ANSWER. If feasible(x) is monotone, stop
 ↳ optimising and find the boundary.
 TRIGGER: "minimise the maximum", "maximise the minimum", "minimum within k". Also: the answer is an integer in a huge range and no
 ↳ formula presents itself.
 TRAPS: (1) the check itself overflows - break early once a running
 ↳ count passes the target;
 (2) 'hi' must be PROVABLY feasible - seed it with the trivial
 ↳ answer, not 1e9;
 (3) if the predicate is not monotone you get a plausible wrong
 ↳ answer and no crash.

BINARY SEARCH THE VALUE, COUNT WITH A SECOND SEARCH. For "k-th
 ↳ smallest X" over a set you cannot
 materialise (pairs, products, distances, subarray sums): binary
 ↳ search x, count how many are
 ≤ x. The counting routine is the real problem.
 TRAP: use firstTrue(count(x) ≥ k) so the returned x actually
 ↳ occurs.

PARAMETRIC SEARCH / LAGRANGIAN (Aliens, wqs). Replace "exactly k" by
 ↳ a penalty per item and
 binary search the penalty. Requires the optimum to be CONVEX in k.
 ↳ See 08 - DP/01/07.
 TRAP: verify convexity by brute-forcing f(k) for small n; and the
 ↳ inner DP must break ties
 toward the largest count or the final subtraction is off by an
 ↳ arbitrary amount.

TERNARY SEARCH. Unimodal f: cut a third each step. Reals: fixed ~200
 ↳ iterations (never an eps
 loop). Integers: while (hi - lo > 2), then scan the survivors.
 TRAP: PLATEAUS KILL IT. f(m1) == f(m2) on a flat region can
 ↳ discard the true optimum. For any
 discretely convex f use binary search on the difference instead:
 argmin = firstTrue(lo, hi-1, [&](ll x){ return f(x+1) ≥ f(x);
 ↳ });

TWO POINTERS - and why it is legal. Legal precisely when validity is
 ↳ MONOTONE IN l for fixed r:
 if [l, r] is valid then so is [l', r] for every l' ≥ l.
 TRAP: "subarray with sum ≤ S" is two-pointerable with positive
 ↳ values and WRONG with negative
 ones. Every time you reach for two pointers, say out loud
 ↳ "shrinking can only help". If it
 cannot, stop.

SWEEP LINE. Sort events by one coordinate, maintain a structure over
 ↳ the other.
 TRAP: event ordering at TIES is the whole problem - for closed
 ↳ intervals process opens first;
 for touching rectangles process closes first. Getting it wrong
 ↳ is off-by-exactly-the-ties.

COORDINATE COMPRESSION. Only the order matters; map ≤ 2n values to
 ↳ 0..m-1.
 TRAP: if the problem cares about GAPS (lengths, areas, "how many
 ↳ integers strictly between"),
 keep the original values to weight each compressed cell. Insert
 ↳ both l and r+1.

DIVIDE AND CONQUER ON TIME. Put each object's lifetime on a segment
 ↳ tree over the timeline, DFS
 it with a ROLLBACK structure, answer at the leaves. This is how
 ↳ you delete from a DSU.
 TRAP: rollback DSU must use union by size only - path compression
 ↳ makes undo impossible, so
 find is O(log n), not alpha. Budget accordingly.

MEET IN THE MIDDLE. Split in half, enumerate 2^(n/2) per side,
 ↳ recombine by sorting + binary
 search / two pointers / hash map.
 TRIGGER: n ≤ 40..44 with a subset-flavoured question.
 TRAP: memory (2²⁰ ll = 8 MB per array); and in "mod m" variants
 ↳ you must also consider the
 wrap-around pair. Put the extra element of an odd n in the
 ↳ CHEAPER half.

SQRT DECOMPOSITION OF THE PROBLEM (not the data structure). Two
 ↳ complementary bad algorithms,
 O(x) and O(n/x): threshold at sqrt and use each where it wins.
 ↳ Heavy vs light vertices, small

vs large divisors, big vs small queries, "sum a_i ≤ 1e5 so at
 ↳ most sqrt distinct values".
 TRAP: tune the threshold to the ACTUAL costs; if updates are 10x
 ↳ cheaper the optimum is far
 from sqrt(n). Measure.

OFFLINE REORDERING. If nothing forces you online, sort the queries:
 ↳ by right endpoint (+ BIT),
 by threshold (+ DSU), by Mo's ordering, by time. The highest-yield
 ↳ contest move for "q queries
 on a static array".
 TRIGGER: no "xor with the previous answer" marker; all queries
 ↳ readable up front.
 TRAP: store answers by the ORIGINAL query id and print in input
 ↳ order.

SMALL TO LARGE. Always insert the smaller container into the larger:
 ↳ each element moves
 O(log n) times. SWAP THE HANDLES, never copy - copying the big one
 ↳ in is O(n²) and looks
 identical.

AMORTIZATION / POTENTIAL. An occasionally-expensive operation is
 ↳ cheap on average if each
 expensive step DESTROYS something expensive to create. Justifies
 ↳ monotonic stacks, ODT, beats.
 TRAP: ODT's amortization needs assign-heavy input; it is O(n) per
 ↳ op in the worst case.
 Monotonic-stack amortization dies if you ever re-push an index.

ADD ONE ELEMENT AT A TIME. Solve for prefixes and maintain the
 ↳ answer incrementally. Turns "for
 every prefix, output X" from O(n²) into O(n log n) and usually
 ↳ reveals the structure.
 TRAP: LIS - lower_bound gives strictly increasing, upper_bound
 ↳ gives non-decreasing. The tails
 array is NOT the subsequence; reconstruct with parent pointers.

PARALLEL BINARY SEARCH. q independent binary searches over one
 ↳ shared timeline: bucket queries
 by their current mid, sweep the timeline once, resolve one bit for
 ↳ every query.
 TRAP: reset or rebuild the shared structure between LEVELS.
 ↳ Forgetting produces monotone-
 looking garbage.

===== PROOF TOOLS: THE TEN MINUTES THAT SAVE THE SUBMIT
 ↳ =====

EXCHANGE ARGUMENT (for greedy). Take any optimum; show that swapping
 ↳ two ADJACENT elements
 toward your greedy order never makes it worse. For a sorting
 ↳ greedy this DERIVES the
 comparator: compare cost(a then b) against cost(b then a).
 TRAP: the derived comparator must be a STRICT WEAK ORDERING. Ties
 ↳ returning true, or an
 overflowing cross-multiplication, makes std::sort run off the
 ↳ array - an RE that looks like
 a logic bug. Test with all-equal input, and cross-multiply in
 ↳ __int128.

INVARIANTS. A quantity unchanged by every legal operation. If start
 ↳ and target differ on it the
 answer is "impossible" - and you have a proof instead of a guess.
 TRAP: an invariant proves IMPOSSIBILITY only. You still owe a
 ↳ construction for the other side.

MONOVARIANTS. A quantity that strictly decreases and is bounded:
 ↳ proves termination and bounds
 the number of operations. (Euclid: a+b strictly decreases.)
 TRAP: on reals a monovariant proves nothing - you need integrality
 ↳ or a floor.

PARITY AND COLOURING. Colour the board so every legal move changes
 ↳ the colour counts in a fixed
 way, then count. Try in order: checkerboard, columns mod k, (i+j)
 ↳ mod 3, a weight function.
 EXAMPLE: the mutilated chessboard has 32/30 of the two colours, so
 ↳ no domino tiling exists.
 TRAP: if no colouring produces an imbalance in five minutes, the
 ↳ obstruction is not parity.

THE EXTREMAL PRINCIPLE. Argue about the largest / smallest / deepest
 ↳ object - extremes have
 neighbours on one side only, which collapses the cases.
 EXAMPLE: two-BFS tree diameter is correct because the farthest
 ↳ vertex from ANY start is an

endpoint of SOME diameter.
 TRAP: that theorem is TREE-ONLY. It is false on general graphs and
 ↪ on negative weights.

PIGEONHOLE. $n+1$ objects in n boxes. In practice: $n+1$ prefix sums but
 ↪ only n residues, so some
 subarray sum is divisible by n .
 TRAP: the constructive version needs the FIRST repeat - store the
 ↪ earliest index per residue.

SYMMETRY AND WLOG. If the problem is invariant under swapping two
 ↪ roles, assume an order and
 divide by the symmetry factor at the end.
 TRAP: handle the diagonal $i == j$ separately: $(\text{total} - \text{diagonal})/2$
 ↪ + diagonal. Under a modulus,
 "divide by 2" means multiply by $\text{inv}2$.

STRATEGY STEALING. In a symmetric game where an extra move never
 ↪ hurts, the first player wins or
 draws: if the second player had a winning strategy the first could
 ↪ steal it.
 TRAP: pure EXISTENCE. It gives you no winning move, and it fails
 ↪ under misere play.

ADVERSARY ARGUMENTS / QUERY LOWER BOUNDS. q binary queries
 ↪ distinguish at most 2^q inputs, so
 $q \geq \log_2(\#\text{inputs})$; comparison sorting needs $\log_2(n!) \sim n \log n$.
 TRAP: assume the interactor is ADAPTIVE unless told otherwise - it
 ↪ fixes nothing in advance,
 only stays consistent. Strategies relying on "the answer was
 ↪ fixed before I started" lose.

===== CONSTRAINTS ARE A MESSAGE FROM THE SETTER =====
 $n \leq 10$ permutations $n!$, or backtracking with pruning
 $n \leq 20$ 2^n bitmask DP, SOS, 3^n submask partition
 $n \leq 24..28$ 2^n with a small factor
 $n \leq 40..44$ MEET IN THE MIDDLE, $2^{(n/2)}$
 $n \leq 100$ $O(n^3)/O(n^4)$: Floyd, Gauss, interval DP, min-cost
 ↪ flow, Hungarian
 $n \leq 500$ $O(n^3)$ or $O(n^2 \log n)$; flows on n^2 edges
 $n \leq 5000$ $O(n^2)$: LCS, tree knapsack, or $O(n^2/64)$ with a
 ↪ bitset
 $n \leq 1e5$, $\log^2$ slack segtree of segtrees, merge-sort tree,
 ↪ parallel binary search, HLD
 $n \leq 2e5$ $O(n \log n)$. This is the default and it means "find
 ↪ the $n \log n$ "
 $n \leq 1e6$ $O(n)$ or $O(n \log n)$ with a SMALL constant: no map,
 ↪ no set, fast IO mandatory
 $n, q \leq 1e5$, offline Mo's ($n \sqrt{q}$) is in budget, $\sim 3e7$ pointer
 ↪ moves
 $\text{value} \leq 1e9$, n small coordinate compression, or binary search
 ↪ the value
 $a_i \leq 1e18$ Miller-Rabin + Pollard rho, $__int128$, no sieve
 $\text{sum } n \leq 2e5$ over tests per-test $O(n \log n)$; NEVER memset a MAXN
 ↪ global per test
 $t \leq 1e4$, no sum bound the intended solution is $O(1)$ or $O(\log)$
 ↪ per test
 $\text{mod } 998244353$ NTT / polynomial methods intended
 $\text{mod } 1e9+7$ counting DP / combinatorics, NOT NTT
 real output, $1e-6$ geometry, ternary/binary search on reals,
 ↪ or expectation
 $\leq 20-30$ oracle queries binary search ($2^{20} > 1e6$) or bit-by-bit
 ↪ determination
 TL 3-5 s at $n \leq 1e5$ the intended solution really is $n \sqrt{n}$
 ↪ or $n \log^2 n$
 ML 32-64 MB offline/streaming, rolling DP arrays, no persistence

THE /64 ARGUMENT. An "impossible" $O(n^2)$ or $O(n^3)$ over BOOLEANS
 ↪ becomes $n^2/64$ - fine up to
 $\sim 1e10$ raw ops. Reachability, subset sum, LCS, string matching, set
 ↪ intersection.
 TRAP: memory. n bitsets of n bits is $n^2/8$ bytes: at $n = 5e4$ that
 ↪ is 312 MB. Process in
 blocks of ~ 1000 and reuse. bitset needs a compile-time size; a
 ↪ vector<u64> hand-roll is
 $\sim 30\%$ slower and always available.

THE "ANSWER IS SMALL" ARGUMENT. Prove the answer is $\leq$ a tiny
 ↪ constant, then brute force over
 it. Each operation halves something / clears a bit / merges two
 ↪ components; "OR of a range
 equals its SUM" means the values are bit-disjoint so at most 30
 ↪ are nonzero; $\text{sum } a_i \leq 1e5$
 means at most ~ 632 distinct values; $a_i \leq 1e6$ means at most 7
 ↪ distinct primes, 240 divisors;
 gcd over an expanding range takes only $O(\log V)$ distinct values.
 TRAP: prove the bound, do not observe it on samples, and handle

↪ "impossible" explicitly.

↪ */

12.3 ContestCraft

CONSTRUCTIVE, INTERACTIVE, RANDOMISED, EMPIRICAL
=====

```
/* ===== CONSTRUCTIVE, INTERACTIVE, RANDOMISED, EMPIRICAL
   ↳ =====
```

BUILD GREEDILY AND PROVE AS YOU GO. Emit the answer one decision at
↳ a time, always taking the best local choice that KEEPS THE REMAINDER FEASIBLE - that check
↳ is the whole problem.
TRAP: for LEXICOGRAPHICALLY SMALLEST, greedy alone is wrong. You
↳ need "pick the smallest candidate such that the rest is still completable", and the
↳ completability test must be exact. If you cannot write that test, you cannot do
↳ lexicographic greedy.

BUILD RECURSIVELY. Solve n from n/2 (or n-1, or four quadrants) and
↳ glue with a fixed pattern.
TRIGGER: n is a power of two, or the object is self-similar (Gray
↳ code, Hanoi, de Bruijn).
TRAP: base cases are usually n = 1 AND n = 2. Recursive output of
↳ 2^n lines needs buffered output - endl will TLE.

EXTEND FROM A KNOWN SMALL CASE. Brute force n = 1..8, find the
↳ pattern of CONSTRUCTION (not just the count), then prove the inductive step.
TRAP: the small cases are exactly where the pattern fails. Find
↳ the smallest n where it first works and special-case everything below. Nearly every
↳ constructive WA is a small-n exception.

CERTIFICATE CONSTRUCTION (prove the NO). Output the witness: the
↳ negative cycle, the odd cycle, the Hall violator, the parity mismatch. Producing it also
↳ validates your YES logic.
TRAP: after n Bellman-Ford rounds you have a vertex AFFECTED BY a
↳ negative cycle, not one ON it. Walk n parent pointers back first, then follow parents until
↳ a vertex repeats.

INTERACTIVE: BUDGET FIRST, ALGORITHM SECOND. Compare 2^Q against the
↳ number of possibilities.
Q ~ log n -> binary search; Q ~ n -> one query per element; Q ~
↳ n log n -> sorting/merging.
TRAPS: (1) FLUSH after every query (endl, or fflush) or you get
↳ "idleness limit exceeded", which looks nothing like the real bug; (2) count your queries
↳ locally and assert the limit; (3) assume the interactor is ADAPTIVE - it only has to stay
↳ consistent with past answers.

RANDOM SHUFFLE to destroy adversarial order: Kuhn's matching,
↳ quickselect, incremental convex hull, minimum enclosing circle (which is O(n) only after a
↳ shuffle), KD-tree splits.
TRAP: seed from the clock (mt19937_64 with steady_clock), never
↳ srand(time(0)) + rand(). unordered_map needs the splitmix64 custom hash or it is O(n^2)
↳ on a crafted test.

RANDOM SAMPLING. If a target occupies more than a 1/c fraction, k
↳ samples miss it with probability (1 - 1/c)^k.
TRIGGER: "majority element of a range", "a value shared by at
↳ least half".
TRAP: sampling gives a CANDIDATE - you must verify it exactly. And
↳ the failure probability compounds over q queries: with q = 1e5 you need per-query
↳ failure below 1e-12, so ~40 samples, not 20.

RANDOM WEIGHTS FOR UNIQUENESS (Zobrist / XOR hashing). Give each
↳ value a random 64-bit label; then multiset equality, "every value appears a multiple of k
↳ times", and "this subtree shape occurred before" become single-number comparisons.
TRAP: one random assignment has a real false-positive rate -
↳ repeat ~30 times with FRESH randomness. A fixed small set of randomness is breakable by Gaussian
↳ elimination.

HASHING AS O(1) EQUALITY. Polynomial hashing for substrings, 2D for
↳ submatrices.
TRAP: RANDOM base, and either double-mod or one 61-bit Mersenne
↳ mod. Fixed base 31 with mod

1e9+7 is anti-hashed by a Thue-Morse construction. Birthday
↳ bound: comparing m hashes pairwise collides with probability ~m^2/(2*MOD), so m = 1e6
↳ against a single 1e9+7 mod is a coin flip.

THE REPETITION MATH. k independent trials fail with p^k. Budget PER
↳ QUERY, not per test, then multiply by 100 for safety. For primality use the DETERMINISTIC
↳ witness set {2,3,5,7,11,13,17,19,23,29,31,37} (valid below 3.3e24) - zero
↳ failure probability, same cost.

STRESS TESTING AS A THINKING TOOL. gen + brute + a diff loop; then
↳ SHRINK the counterexample until it fits on one line and stare at it.
TRAPS: (1) the generator must produce DEGENERATE cases - all
↳ equal, n = 1, all zero, maximum values, duplicate coordinates. A uniform generator over [1, 1e9]
↳ never produces a tie and will never find your tie bug. (2) If the problem says "print
↳ any", diffing is useless - write a CHECKER. (3) Stress the brute force against the samples
↳ first.

BRUTE FORCE SMALL CASES, THEN LOOK IT UP. Compute n = 1..12 exactly,
↳ then search OEIS, or feed the sequence to Berlekamp-Massey, or fit a polynomial by Lagrange
↳ interpolation.
TRAP: an OEIS match on 4 terms is noise - get 8, then confirm the
↳ 9th. BM needs terms computed under the SAME prime you will use, and at least 2x the order.

PRINT THE DP TABLE AND LOOK FOR STRUCTURE. Dump the n x n table (and
↳ the argmin table) for n = 20 and look for monotone rows, monotone opt[i][j], convex
↳ differences, sparsity. That is how you decide between Knuth, divide & conquer, and Aliens.
TRAP: monotonicity that holds at n = 20 can fail at n = 200. Also
↳ verify the quadrangle inequality numerically on random quadruples.

INSTRUMENT, DO NOT GUESS (for TLE). Print a counter of inner-loop
↳ iterations on your largest local test and compare against 1e8-1e9. That distinguishes "wrong
↳ complexity" from "wrong constant", which need completely different fixes. Compile locally
↳ with -O2.

===== CONTEST CRAFT =====

READING THE SET
Minute 0-15: all three read DIFFERENT problems, front to back.
↳ Nobody codes. On the board, per problem: a one-line restatement, n, and a guess at the family.
Order by (confidence x 1/effort), not by letter. Setters mis-order
↳ ~30% of the time. Use the scoreboard from minute 30 - it is the best difficulty
↳ signal you have. ONE coder at the keyboard, always. A problem is not "started"
↳ until it has a written plan with its complexity and its main data structure.

WHEN TO ABANDON - when any TWO hold:
* 25+ minutes with no NEW idea (the idea count stopped growing,
↳ not the code).
* The complexity is 10x over budget with no plan to close it.
* Two teammates failed to find the bug and 5 minutes of stress
↳ testing found nothing => you are testing the wrong thing or misread the statement.
↳ Re-read it OUT LOUD.
* Another unattempted problem has more solves.
Write the state on paper before switching, and set a hard return
↳ time.

DEBUGGING UNDER TIME PRESSURE - in this order:
1. RE-READ THE STATEMENT: output format, the modulus, 1-based vs
↳ 0-based, multi-test. About a quarter of "impossible" bugs are misreads.
2. Re-read your input parsing (n and m swapped; looping to n over
↳ an array sized m).
3. Run ALL the samples, including the one you skipped.
4. Run n = 1, n = 2, and the all-equal case.
5. Stress test. This beats reading code whenever you have a brute
↳ force.
6. Only then read the code - out loud, to a teammate, for INTENT.
↳ Silent re-reading just

re-hallucinates the same thing.

7. Nothing after 15 minutes: REWRITE the suspicious function from scratch.

PRE-SUBMIT CHECKLIST (40 seconds, every submit)

- [] Overflow: every product of two inputs; inside comparators; inside prefix sums.
- [] Modulus: right constant, +MOD after subtraction, inverse instead of division, non-negative.
- [] $n = 1$, $m = 1$, $k = 0$, single-element array.
- [] Degenerate input: empty string, zero edges, one-node tree, all equal, all zero, negatives.
- [] STATE RESET between test cases: every global, vector, counter, visited array, the answer variable, and any function-local static. Clear only what you touched if sum n is bounded.
- [] Right bound read: MAXN vs the actual limit; the sum over queries can exceed the per-test n .
- [] Uninitialised: dp arrays that need $-\text{INF}$ not 0 ; DSU parent[i] = i.
- [] Recursion depth: DFS on a path with $n = 2e5$ is $2e5$ frames - move big locals out.
- [] Output format: newline vs space, YES vs Yes, -1 vs IMPOSSIBLE, setprecision.
- [] endl removed from any loop running more than $\sim 1e4$ times; debug output removed.
- [] The 'cin >> tc' line matches whether the problem is multi-test.

VERDICT TRIAGE

- WA -> which test? Test 1 means a misread or a format error; a late test means an edge case or overflow. Then: modulus, overflow, ties/duplicates (strict vs non-strict), stress test, unreset state, floating-point comparison.
- TLE -> is the complexity really what you think (count the loops on paper)? endl in a loop / missing sync_with_stdio. map/set in a hot loop.
- Reallocation and pass-by-value. memset of a huge global per test. Infinite loop: a binary search with $l = \text{mid}$.
- Only then change algorithm. A /64 win never fixes a wrong complexity class.
- RE -> array bounds first: rebuild with `-fsanitize=address,undefined -D_GLIBCXX_DEBUG` and run the samples; it finds it in seconds. Then stack overflow, division by zero, a comparator that is not a strict weak ordering (crashes INSIDE `std::sort`), `.back()/top()` on an empty container, 0 - vs 1 -indexed access.
- MLE -> $n \times n$ arrays (5000^2 ints = 100 MB); `vector<vector<int>>` for a graph (5-8x the raw data - use CSR); persistent structures (count the nodes before allocating); recursion frames; go offline and stream instead of storing versions.
- Idleness on an interactive -> a missing flush, or debug output on `stdout`.

TEAM PROTOCOL

- Never submit without the checklist: a rejection costs 20 penalty minutes plus morale.
- One person owns the printed notebook and fetches pages; the coder does not browse.
- Paste, adapt, then read once - the #1 template bug is an index-convention mismatch.
- Two solutions disagree? Trust the brute force. Always.
- Last 30 minutes: stop starting new problems; finish the nearest one and re-check the submits.

THE TEN MOST EXPENSIVE RECURRING BUGS (by contest-time cost)

1. Overflow in an intermediate product, invisible because the final answer is small.
2. State not reset between test cases - passes locally, fails on test 2.
3. Wrong strictness ($<$ vs $<=$) in a comparator, a binary-search predicate, or a tie-break.
4. Two pointers on a non-monotone predicate: plausible output, silently wrong.
5. Binary search bounds: hi not provably feasible, or $l = \text{mid}$ causing an infinite loop.
6. 0 - vs 1 -based off-by-one when pasting a template.
7. endl in an output loop, or a missing `sync_with_stdio`, at $n = 1e6$.
8. Negative modulo in C++ ($-3 \% 5 == -3$).
9. A greedy that was never proved and passes the samples.
10. Recursion depth on a path-shaped tree with $n = 2e5$.

=====
-> */

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